

Modern Control Theory

From Historical Foundations to Practical Implementation

An IEEE-Formatted Textbook with Hands-On Laboratory Experiments

A Collaborative Educational Work

January 14, 2026

Preface

This textbook takes a unique approach to teaching control systems engineering. Rather than presenting theory as abstract mathematics, every concept is motivated by the real-world problem that necessitated its invention. From James Watt’s struggle to regulate steam engine speed to Harold Black’s revolutionary insight about negative feedback in telephone amplifiers, each chapter begins with the historical context that gave birth to the mathematical tools we now take for granted.

Philosophy: We believe that understanding *why* a mathematical technique was developed is just as important as understanding *how* to apply it. When you know that Bode plots were invented to analyze the stability of transcontinental telephone amplifier chains, the mathematics becomes not just a tool, but a solution to a tangible problem.

Practical Focus: Every chapter includes hands-on experiments that can be implemented with accessible equipment:

- 3D-printed mechanical components
- Arduino or similar microcontrollers
- DC motors, servos, and stepper motors
- Basic sensors (encoders, potentiometers, IMUs)
- Simple masses, springs, and dampers

Self-Contained Chapters: Each chapter is designed to stand alone. Instructors may reorder chapters or select specific topics without losing coherence.

Mathematical Rigor with Clarity: We present complete derivations, but prioritize elegance and intuition. Every equation should make physical sense.

Contents

Preface	iii
I Foundations & Mindset	1
1 What is Control?	3
1.1 Historical Motivation: The Birth of Automatic Control	3
1.1.1 The Ktesibios Water Clock (c. 270 BC)	3
1.1.2 Cornelis Drebbel's Thermostat (c. 1620)	4
1.1.3 James Watt's Centrifugal Governor (1788)	5
1.2 Modern Applications: Open-Loop vs. Closed-Loop Systems	6
1.2.1 Open-Loop Control Systems	6
1.2.2 Closed-Loop Control Systems	7
1.3 Mathematical Foundations	8
1.3.1 System Definitions and Notation	8
1.3.2 Block Diagram Representation	9
1.3.3 Open-Loop Control Analysis	9
1.3.4 Closed-Loop Control Analysis	9
1.3.5 Why Feedback Reduces Sensitivity	11
1.3.6 Steady-State Error Analysis	11
1.4 Practical Laboratory: DC Motor Position Control	12
1.4.1 Learning Objectives	12
1.4.2 Required Components	12
1.4.3 Wiring Configuration	13
1.4.4 Software Implementation	17
1.4.5 Experimental Procedure	17
1.4.6 Data Collection and Analysis	17
1.5 Worked Examples	18
1.6 Figures for TikZ Implementation	22
1.6.1 Figure: Watt Centrifugal Governor	22
1.6.2 Figure: Laboratory Setup	22
1.7 Chapter Summary	22
1.8 Exercises	24

2	Modeling Physical Systems	25
2.1	Historical Motivation: The Need to Predict Before Building	25
2.1.1	Newton's Laws of Motion (1687): The Birth of Mathematical Physics . . .	25
2.1.2	Kirchhoff's Circuit Laws (1845): Extending Models to Electricity	26
2.1.3	Lord Kelvin's Mechanical Analog Computers (1870s)	26
2.1.4	Heaviside's Operational Calculus (1880s-1890s)	27
2.1.5	The Cost of Not Modeling: Engineering Disasters	28
2.1.6	The Evolution to Digital Simulation	28
2.2	Modern Applications of Physical System Models	28
2.2.1	Digital Twins in Manufacturing	28
2.2.2	Vehicle Dynamics Simulation	29
2.2.3	HVAC System Modeling	29
2.2.4	Pharmacokinetics: Modeling Drug Dynamics in the Body	29
2.3	Mathematical Foundation: Deriving Differential Equations from Physics	30
2.3.1	Newton's Second Law: The Foundation of Mechanical Systems	30
2.3.2	Electrical Systems: RLC Circuits from Kirchhoff's Laws	32
2.3.3	Thermal Systems: Heat Flow and RC Analogs	32
2.3.4	Rotational Systems	33
2.3.5	Lagrangian Mechanics: An Energy-Based Alternative	33
2.3.6	System Order and State Variables	34
2.4	Practical Experiment: Building and Characterizing a Mass-Spring-Damper System	34
2.4.1	Apparatus Design and Construction	35
2.4.2	Measurement Procedure	35
2.4.3	Data Analysis and Parameter Extraction	37
2.4.4	Comparing Theory and Experiment	38
2.4.5	Sources of Error and Improvements	38
2.5	Worked Examples	38
2.5.1	Example 1: Simple Pendulum	39
2.5.2	Example 2: DC Motor Model	40
2.5.3	Example 3: Finding Natural Frequency and Damping Ratio	40
2.5.4	Example 4: Fluid Tank Level Control	41
2.5.5	Example 5: Coupled Mass-Spring System	42
2.6	Chapter Summary	43
2.7	Exercises	44
2.8	Further Reading	45
3	Transfer Functions and Block Diagrams	47
3.1	Historical Motivation: From Telegraph Cables to Modern Control	47
3.1.1	The Transatlantic Cable Problem	47
3.1.2	Oliver Heaviside and Operational Calculus	48
3.1.3	The Earlier Mathematical Foundations	49
3.1.4	The Telephone Era: Systematizing Transfer Functions	49
3.1.5	The Power of Abstraction	50
3.2	Modern Applications	50
3.2.1	Audio Processing and Equalization	50

3.2.2	Communication Systems	50
3.2.3	Mechanical Vibration Analysis	51
3.2.4	Control System Design Software	51
3.3	Mathematical Foundation	51
3.3.1	The Laplace Transform	51
3.3.2	Key Laplace Transform Pairs	52
3.3.3	Essential Transform Properties	53
3.3.4	The Transfer Function	54
3.3.5	Derivation: Mass-Spring-Damper System	55
3.3.6	Poles and Zeros	55
3.3.7	Block Diagram Algebra	57
3.3.8	Mason's Gain Formula (Simplified)	58
3.4	Practical Experiment: RC Low-Pass Filter	59
3.4.1	Theory and Transfer Function Derivation	59
3.4.2	Hardware Implementation	60
3.4.3	Expected Results and Analysis	61
3.4.4	Alternative: Computational Simulation	61
3.5	Example Problems	63
3.6	Summary	66
3.7	Exercises	68
	Chapter Overview	69
3.8	Historical Development and Motivation	69
3.8.1	The Limitations of Classical Control	69
3.8.2	Rudolf Kalman and the State-Space Revolution	70
3.8.3	The Space Race Imperative	70
3.8.4	Parallel Soviet Developments	71
3.8.5	Connections to Classical Physics	71
3.9	Modern Applications	72
3.9.1	Aerospace Guidance and Navigation	72
3.9.2	Robotic Manipulation	72
3.9.3	Economic and Financial Modeling	72
3.9.4	Power Grid State Estimation	73
3.10	Mathematical Foundation	73
3.10.1	State Variable Definition	73
3.10.2	Standard State-Space Form	73
3.10.3	Converting Differential Equations to State-Space Form	75
3.10.4	Example: Mass-Spring-Damper System	75
3.10.5	Transfer Function to State-Space Conversion	76
3.10.6	Eigenvalues and System Poles	77
3.10.7	State Transition Matrix	78
3.10.8	Complete Solution with Forcing	78
3.11	Laboratory Experiment: Two-Mass System	79
3.11.1	Objective	79
3.11.2	Apparatus	79
3.11.3	System Modeling	79

3.11.4	Transfer Function Comparison	80
3.11.5	Experimental Procedure	80
3.11.6	Expected Results	81
3.11.7	Discussion Questions	81
3.12	Worked Examples	82
3.12.1	Example 1: Converting an ODE to State-Space Form	82
3.12.2	Example 2: State-Space to Transfer Function	83
3.12.3	Example 3: Eigenvalue Analysis and Stability	83
3.12.4	Example 4: DC Motor State-Space Model	84
3.12.5	Example 5: Different State Variable Choices	85
3.13	Chapter Summary	86
3.14	Exercises	87
	Further Reading	89

II System Behavior 91

5 First and Second Order Systems 93

5.1	Historical Motivation: The Quest to Understand Oscillation	93
5.1.1	Lord Rayleigh and the Theory of Sound (1877)	94
5.1.2	Seismograph Development and the Need for Instrument Theory (1880s)	94
5.1.3	Automotive Suspension Design (1920s–1930s)	95
5.1.4	Aircraft Flutter Analysis: When Resonance Kills (1930s)	96
5.1.5	Why These Canonical Forms?	96
5.2	Modern Applications	97
5.2.1	Audio Speaker Design and Room Acoustics	97
5.2.2	Building Earthquake Response	97
5.2.3	Drug Absorption Kinetics	98
5.2.4	Electric Vehicle Battery Thermal Management	98
5.3	Mathematical Foundation	98
5.3.1	First-Order Systems	98
5.3.2	Second-Order Systems	101
5.3.3	Pole Locations and Response Characteristics	104
5.4	Laboratory Experiment: Variable Damping Mass-Spring System	105
5.4.1	Apparatus Design and Construction	106
5.4.2	Experimental Procedure	108
5.4.3	Data Analysis and Parameter Extraction	109
5.4.4	Discussion Questions	110
5.5	Worked Examples	110
5.6	Summary and Key Formulas	116
5.7	Exercises	118
5.8	Further Reading	118

6	Stability: When Systems Blow Up	121
6.1	Historical Motivation: The Quest for Stable Machines	121
6.1.1	James Watt and the Governor Problem	121
6.1.2	Maxwell's "On Governors" (1868)	122
6.1.3	Routh's Criterion (1877)	123
6.1.4	Hurwitz's Independent Discovery (1895)	123
6.1.5	Dramatic Failures: Lessons in Instability	123
6.2	Modern Applications of Stability Theory	124
6.2.1	Power Grid Stability	124
6.2.2	Flight Control: Intentionally Unstable Aircraft	125
6.2.3	Audio Feedback	125
6.2.4	Economic and Financial Stability	125
6.3	Mathematical Foundation of Stability	126
6.3.1	BIBO Stability: Definition	126
6.3.2	Impulse Response and Stability	126
6.3.3	Pole Location Criterion	127
6.3.4	Marginal Stability	128
6.3.5	Closed-Loop Stability	129
6.4	The Routh-Hurwitz Stability Criterion	129
6.4.1	The Routh Array	129
6.4.2	Special Cases	130
6.5	Practical Experiment: Demonstrating Stability and Instability	131
6.5.1	Circuit Description	131
6.5.2	Alternative: Arduino-Based Motor Control Experiment	132
6.5.3	Experimental Procedure	133
6.6	Worked Examples	134
6.7	Summary	139
6.8	Exercises	141
7	Transient vs Steady-State Response	143
7.1	Historical Development: The Quest for Performance Specifications	143
7.1.1	Industrial Process Control: The Foundation (1920s–1940s)	143
7.1.2	Servomechanisms and World War II: Speed Becomes Critical	144
7.1.3	Ziegler and Nichols: The First Systematic Tuning Rules (1942)	145
7.1.4	Telephone Switching: Precision Timing Requirements (1950s)	145
7.1.5	The Fundamental Tradeoff: Speed vs. Accuracy	146
7.2	Modern Applications: Where Performance Specifications Matter	146
7.2.1	Hard Disk Drive Head Positioning	146
7.2.2	CNC Machining: Accuracy vs. Throughput	147
7.2.3	Insulin Pumps: Overshoot is Dangerous	147
7.2.4	Autonomous Vehicle Lane Keeping	147
7.3	Mathematical Foundation: Transient Response Specifications	148
7.3.1	The Standard Second-Order System	148
7.3.2	Unit Step Response Derivation	149
7.3.3	Rise Time	150

7.3.4	Peak Time	150
7.3.5	Percent Overshoot	150
7.3.6	Settling Time	151
7.3.7	Summary: Transient Specifications	152
7.4	Mathematical Foundation: Steady-State Error Analysis	153
7.4.1	The Final Value Theorem	153
7.4.2	Unity Feedback Configuration	153
7.4.3	System Type and Error Constants	154
7.4.4	Steady-State Error for Standard Inputs	154
7.4.5	Steady-State Error Summary by System Type	155
7.4.6	Physical Interpretation	155
7.4.7	The Speed-Accuracy Tradeoff Revisited	156
7.5	Design in the s-Plane: Meeting Multiple Specifications	156
7.5.1	s-Plane Design Regions	156
7.5.2	Natural Frequency Constraints	157
7.6	Practical Experiment: Motor Position Control System	157
7.6.1	Experimental Setup	157
7.6.2	Controller Implementation	158
7.6.3	Experimental Procedure	159
7.6.4	Data Analysis	159
7.6.5	Discussion Questions	160
7.7	Worked Examples	161
7.8	Diagrams and Visual Reference	167
7.9	Summary	168
7.10	Exercises	170
8	Linearization	173
8.1	Historical Motivation: From Newton to Modern Control	173
8.1.1	Newton and the Birth of Local Approximation (1687)	173
8.1.2	Small Oscillation Theory in Physics	174
8.1.3	Poincaré and the Qualitative Theory (1880s)	174
8.1.4	Aircraft Stability Analysis (1900s)	175
8.1.5	Why Linearize? The Practical Argument	176
8.2	Modern Applications of Linearization	176
8.2.1	Aircraft Autopilot Design	176
8.2.2	Chemical Process Control	177
8.2.3	Power Electronics: Small-Signal Analysis	177
8.2.4	Robot Arm Control: Jacobian Linearization	177
8.3	Mathematical Foundation	178
8.3.1	Taylor Series Review	178
8.3.2	The Linearization Procedure	178
8.3.3	Example: The Simple Pendulum	180
8.3.4	Multiple Variables: Jacobian Matrix Construction	183
8.3.5	Validity: When Is Linearization “Good Enough”?	183
8.4	Practical Experiment: Pendulum Linearization	184

8.4.1	Apparatus Design and Construction	184
8.4.2	Experimental Procedure	185
8.4.3	Data Analysis and Expected Results	186
8.4.4	Discussion Questions	186
8.5	Worked Examples	187
8.5.1	Example 3: Finding Equilibrium Points	189
8.5.2	Example 4: Linearization Around Non-Zero Equilibrium	189
8.5.3	Example 5: Linear vs. Nonlinear Simulation	190
8.6	Operating Point Concept	192
8.7	Summary	193
8.8	Exercises	194
8.9	Further Reading	195

III Classical Control Design 197

9 Feedback and Sensitivity 199

9.1	Historical Motivation: The Birth of Negative Feedback	199
9.1.1	The Telephone Amplifier Problem	199
12.6	Root Locus Diagrams: A Visual Reference	202
12.7	Summary	203
12.8	Exercises	204

13 Frequency Response and Bode Plots 205

13.1	Historical Development and Motivation	205
13.1.1	The Bell Labs Challenge	205
13.1.2	Hendrik Bode and the Logarithmic Revolution	206
13.1.3	Harry Nyquist and the Stability Criterion	207
13.1.4	The Fundamental Insight: Sinusoids as Eigenfunctions	207
13.2	Modern Applications	208
13.2.1	Audio System Design	208
13.2.2	Filter Design	208
13.2.3	Control Loop Bandwidth Specification	208
13.2.4	Electromagnetic Compatibility (EMC)	209
13.3	Mathematical Foundation	209
13.3.1	Frequency Response Function	209
13.3.2	Bode Plot Construction	209
13.3.3	First-Order System: $G(s) = \frac{1}{\tau s + 1}$	210
13.3.4	Second-Order System	211
13.3.5	Combining Elements: Poles and Zeros	214
13.3.6	Stability Margins	215
13.4	Laboratory Experiment: Frequency Response Measurement	215
13.4.1	Theoretical Background	216
13.4.2	Circuit Construction	217
13.4.3	Arduino Code for Sine Wave Generation	217

13.4.4	Data Collection and Analysis	217
13.4.5	Plotting the Bode Diagram	217
13.4.6	Alternative Experiment: Motor Transfer Function	219
13.5	Worked Examples	219
13.6	Summary	225
13.7	Exercises	226
14	Lead-Lag Compensation	229
	Chapter Overview	229
14.1	Historical Motivation: From Anti-Aircraft Guns to Modern Servos	229
14.1.1	The Fire Control Problem of World War II	230
14.1.2	The Limitation of Pure PID Control	230
14.1.3	Bode's Revolutionary Contribution	230
14.1.4	Lead Compensation: Filtered Derivative Action	231
14.1.5	Lag Compensation: Stable Integral Action	232
14.1.6	Lead-Lag: The Best of Both Worlds	232
14.2	Modern Applications	232
14.2.1	Hard Disk Drive Servo Systems	232
14.2.2	Antenna Tracking Systems	232
14.2.3	Power System Stabilizers	233
14.2.4	Precision Motion Control	233
14.3	Mathematical Foundation	233
14.3.1	The General First-Order Compensator	233
14.3.2	Lead Compensator Analysis	233
14.3.3	Lag Compensator Analysis	236
14.3.4	Lead-Lag Compensator	237
14.3.5	Root Locus Interpretation	237
14.4	Practical Experiment: Motor Position Control with Lead Compensation	239
14.4.1	Objective	239
14.4.2	Required Equipment	239
14.4.3	System Model	239
14.4.4	Experiment Procedure	240
14.4.5	Expected Results	241
14.4.6	Verifying Phase Margin Improvement	241
14.5	Example Problems	243
14.5.1	Example 14.3: Maximum Phase Lead Calculation	245
14.5.2	Example 14.4: Lead-Lag Design for Combined Specifications	246
14.5.3	Example 14.5: Digital Implementation of Lead Compensator	247
14.5.4	Example 14.6: Phase Margin from Bode Plot	248
14.6	Chapter Summary	249
14.7	Exercises	250
	Further Reading	251

IV State-Space Control 253

15 Controllability and State Feedback 255

15.1	Historical Motivation: The Birth of State-Space Control	255
15.1.1	The Classical Control Dilemma	255
15.1.2	Kalman's Revolutionary Framework	256
15.1.3	The Space Race Context	256
15.1.4	The Controllability Question	256
15.1.5	From Theory to Practice: Ackermann's Contribution	257
15.1.6	The Modern Legacy	257
15.2	Modern Applications	258
15.2.1	Spacecraft Attitude Control	258
15.2.2	Multi-Axis Robot Control	258
15.2.3	Active Suspension Systems	258
15.2.4	Process Control	258
15.3	Mathematical Foundation	259
15.3.1	Definition of Controllability	259
15.3.2	The Controllability Matrix	259
15.3.3	The Controllability Rank Condition	260
15.3.4	The Controllability Gramian	261
15.3.5	State Feedback Control	262
15.3.6	Pole Placement Theorem	263
15.3.7	Ackermann's Formula	263
15.3.8	Bass-Gura Formula	264
15.3.9	Practical Considerations	264
15.4	Practical Experiment: Inverted Pendulum State Feedback Control	265
15.4.1	System Description	265
15.4.2	Hardware Implementation	266
15.4.3	Controller Implementation	267
15.4.4	Experimental Results	267
15.4.5	Extensions and Variations	269
15.5	Example Problems	269
15.6	Summary	273
15.7	Exercises	275

16 Observability and Observers 279

16.1	Historical Motivation: The Hidden State Problem	279
16.1.1	The Fundamental Challenge of State Estimation	279
16.1.2	Kalman's Dual Discovery (1960)	280
16.1.3	Luenberger's Observer (1964)	280
16.1.4	Aerospace Applications: Apollo Navigation	280
16.1.5	Submarine Target Motion Analysis	281
16.1.6	The Key Insight	281
16.2	Modern Applications	282
16.2.1	GPS Receivers	282

16.2.2	Motor Control: Velocity Estimation from Position	282
16.2.3	Battery State-of-Charge Estimation	282
16.2.4	Soft Sensors in Chemical Processes	283
16.3	Mathematical Foundation	283
16.3.1	System Description	283
16.3.2	Definition of Observability	284
16.3.3	The Observability Matrix	284
16.3.4	Duality Between Controllability and Observability	285
16.3.5	The Luenberger Observer	285
16.3.6	Observer Design by Pole Placement	287
16.3.7	The Separation Principle	287
16.3.8	Observer-Based Control Implementation	288
16.3.9	Transfer Function Interpretation	289
16.4	Practical Experiment: Motor Velocity Observer	289
16.4.1	Objective	289
16.4.2	Required Equipment	289
16.4.3	Motor Model	289
16.4.4	Observability Analysis	290
16.4.5	Observer Design	290
16.4.6	Microcontroller Implementation	291
16.4.7	Experimental Procedure	291
16.4.8	Expected Results	291
16.5	Block Diagrams	293
16.5.1	Basic Observer Structure	293
16.5.2	Observer-Based Control System	293
16.5.3	State Estimation Error Convergence	293
16.6	Worked Examples	293
16.6.1	Example 3: Duality Between Controllability and Observability	296
16.6.2	Example 4: Observer-Based Controller Design	297
16.6.3	Example 5: Observer Convergence Rate Analysis	298
16.6.4	Example 6: Unobservable System Analysis	299
16.7	Summary	301
16.7.1	Key Concepts	301
16.7.2	Design Procedure	302
16.7.3	Looking Ahead	302
16.8	Exercises	303
17	LQR: Optimal State Feedback	305
17.1	Historical Motivation: The Quest for Optimal Control	305
17.1.1	The Fundamental Dilemma: Where to Place the Poles?	305
17.1.2	Rudolf Kalman and the Birth of Optimal Control Theory	306
17.1.3	Parallel Developments: Pontryagin and Bellman	306
17.1.4	From Theory to the Moon: LQR in the Apollo Program	307
17.1.5	Why LQR Matters Today	308
17.2	Modern Applications of LQR	308

17.2.1	Autonomous Vehicle Trajectory Planning	308
17.2.2	Quadrotor Attitude Control	308
17.2.3	Economic Policy Optimization	308
17.2.4	Humanoid Robot Balance Control	309
17.2.5	Power Grid Frequency Regulation	309
17.3	Mathematical Foundation	309
17.3.1	Problem Formulation	309
17.3.2	Interpretation of the Cost Function	310
17.3.3	Solution via Calculus of Variations	311
17.3.4	The Algebraic Riccati Equation	312
17.3.5	The Optimal Gain and Closed-Loop System	312
17.3.6	The Optimal Cost	313
17.3.7	Tuning Q and R: The Art Within the Science	313
17.3.8	The Pareto Frontier: Performance vs. Effort	314
17.4	Practical Experiment: LQR Control of an Inverted Pendulum	315
17.4.1	Hardware Setup	315
17.4.2	System Model	315
17.4.3	LQR Design Procedure	316
17.4.4	Microcontroller Implementation	319
17.4.5	Experimental Results	320
17.4.6	Discussion	320
17.5	Example Problems	321
17.5.1	Example 17.3: Applying Bryson's Rule	322
17.5.2	Example 17.4: LQR vs. Pole Placement Comparison	323
17.5.3	Example 17.5: Verifying ARE Solution Stability	324
17.5.4	Example 17.6: Multi-Input LQR	325
17.6	Chapter Summary	326
17.6.1	Key Equations	326
17.6.2	Key Concepts	326
17.7	Exercises	328
18	Kalman Filter: Optimal Estimation	331
18.1	Historical Motivation: The Problem of Noisy Measurements	331
18.1.1	Rudolf Kalman and the Birth of Optimal Filtering	331
18.1.2	Before Kalman: The Wiener Filter	332
18.1.3	Kalman's Revolutionary Insight	332
18.1.4	Stanley Schmidt and the Apollo Program	332
18.1.5	Duality: The Kalman Filter is to Estimation What LQR is to Control	333
18.2	Modern Applications	333
18.2.1	GPS/INS Sensor Fusion	333
18.2.2	Self-Driving Car Localization	334
18.2.3	Weather Prediction	334
18.2.4	Financial Time Series	334
18.3	Mathematical Foundation	334
18.3.1	System Model with Stochastic Noise	334

18.3.2	The Estimation Problem	335
18.3.3	Structure of the Optimal Estimator	335
18.3.4	Derivation of the Kalman Gain	336
18.3.5	The Continuous-Time Riccati Equation	337
18.3.6	Summary: Continuous-Time Kalman Filter	337
18.3.7	Discrete-Time Kalman Filter	337
18.3.8	Understanding the Kalman Gain	338
18.3.9	Tuning Q and R : Model vs. Sensor Confidence	338
18.3.10	Duality with LQR: The Algebraic Riccati Equation	339
18.4	Practical Experiment: IMU Sensor Fusion	339
18.4.1	The Sensor Fusion Problem	339
18.4.2	Hardware Setup	340
18.4.3	Mathematical Model for Angle Estimation	340
18.4.4	Implementation	340
18.4.5	Visualization and Tuning	340
18.5	Worked Examples	342
18.5.1	Example 3: Effect of Q and R on Filter Behavior	344
18.5.2	Example 4: Step-by-Step Discrete Kalman Filter	344
18.5.3	Example 5: Sensor Fusion—Combining Two Noisy Measurements	345
18.6	Diagrams and Visualizations	346
18.6.1	Kalman Filter Block Diagram	346
18.6.2	Prediction-Update Cycle	346
18.6.3	Gaussian Fusion: Prior, Measurement, and Posterior	346
18.6.4	Sensor Comparison: Raw vs. Filtered	346
18.7	Chapter Summary	346
18.8	Exercises	349
18.9	Further Reading	349

V Digital & Discrete Control 353

19 Sampling and Discretization 355

19.1	Historical Motivation: From Telegraph Lines to Digital Control	355
19.1.1	Harry Nyquist and Telegraph Transmission Theory (1928)	355
19.1.2	Claude Shannon and the Sampling Theorem (1949)	356
19.1.3	The Problem That Drove the Mathematics	357
19.1.4	Early Digital Computers and Sampled-Data Systems	358
19.1.5	The Fundamental Question for Control Engineers	359
19.2	Modern Applications of Sampling and Discretization	359
19.2.1	Digital Audio	359
19.2.2	Digital Control in Microcontrollers	359
19.2.3	Anti-Aliasing in Signal Processing	360
19.2.4	Real-Time Embedded Systems	360
19.3	Mathematical Foundations of Sampling	360
19.3.1	The Ideal Sampling Process	360

19.3.2	The Nyquist-Shannon Sampling Theorem	362
19.3.3	Aliasing: When Sampling Fails	363
19.3.4	The Zero-Order Hold (ZOH)	364
19.3.5	Discretization Methods for Controllers	365
19.3.6	Choosing the Sample Rate	367
19.4	Practical Laboratory: Demonstrating Aliasing and Discretization Effects	368
19.4.1	Learning Objectives	368
19.4.2	Required Components	369
19.4.3	Experiment 1: Visualizing Aliasing	369
19.4.4	Experiment 2: Effect of Sample Rate on Control Quality	371
19.4.5	Experiment 3: Comparing Discretization Methods	371
19.5	Worked Examples	373
19.6	Figures for TikZ Implementation	379
19.6.1	Figure: Sampling and Reconstruction Block Diagram	379
19.6.2	Figure: Aliasing Time Domain	379
19.6.3	Figure: ZOH Step Response	379
19.6.4	Figure: s-Plane to z-Plane Mapping	380
19.7	Chapter Summary	380
19.8	Exercises	382
20	Z-Transform and Discrete Stability	383
20.1	Historical Motivation and Development	383
20.1.1	The Problem with Sampled Signals	383
20.1.2	Witold Hurewicz and the Birth of the Z-Transform (1947)	384
20.1.3	Eliahu Jury and the Stability Criterion (1958)	384
20.1.4	The Digital Revolution in Control	385
20.1.5	The Fundamental Mapping: $z = e^{sT}$	385
20.2	Modern Applications	386
20.2.1	Digital Filter Design	386
20.2.2	Embedded Control Systems	386
20.2.3	Digital Signal Processors	387
20.2.4	Discrete-Time Financial Models	387
20.3	Mathematical Foundation	387
20.3.1	Definition of the Z-Transform	387
20.3.2	Z-Transforms of Common Sequences	387
20.3.3	Properties of the Z-Transform	389
20.3.4	Inverse Z-Transform	390
20.3.5	Discrete Transfer Functions	391
20.3.6	S-Plane to Z-Plane Mapping	391
20.3.7	Discrete Stability Criterion	392
20.3.8	The Jury Stability Test	393
20.4	Practical Experiment: Digital Filter on Arduino	395
20.4.1	Objectives	395
20.4.2	Theoretical Background	395
20.4.3	Required Components	395

20.4.4	Microcontroller Implementation	395
20.4.5	Experimental Procedure	397
20.4.6	Expected Results	397
20.4.7	Comparison to Continuous-Time Equivalent	398
20.4.8	Analysis Questions	398
20.5	Worked Examples	398
20.6	Chapter Summary	403
20.7	Exercises	404

VI Modern & AI-Driven Control 407

Chapter Overview	409
20.8 Historical Development and Motivation	409
20.8.1 The Constraint Problem in Process Control	409
20.8.2 Shell Oil and Dynamic Matrix Control (1970s)	410
20.8.3 Jacques Richalet and Model Predictive Heuristic Control (1978)	411
20.8.4 Why Not Earlier? The Computational Barrier	411
20.8.5 The Theoretical Revolution (1980s–1990s)	411
20.8.6 Becoming the Industry Standard	412
20.9 Modern Applications	412
20.9.1 Chemical Process Control	413
20.9.2 Autonomous Vehicles	413
20.9.3 Building HVAC Optimization	413
20.9.4 Quadrotor and UAV Control	414
20.10 Mathematical Foundation	414
20.10.1 The Fundamental Idea	414
20.10.2 System Model: Discrete-Time State-Space	415
20.10.3 Cost Function Formulation	415
20.10.4 Constraint Specification	416
20.10.5 The MPC Optimization Problem	416
20.10.6 Conversion to Quadratic Programming Form	416
20.10.7 The Receding Horizon Algorithm	418
20.10.8 Stability Considerations	418
20.10.9 Relationship to LQR	419
20.11 Practical Laboratory: Tank Level Control with Constraints	419
20.11.1 Objective	419
20.11.2 System Description	420
20.11.3 System Modeling	420
20.11.4 Required Components	421
20.11.5 Software Implementation	422
20.11.6 Experimental Procedure	423
20.11.7 Expected Results	424
20.11.8 Discussion Questions	425
20.12 Worked Examples	425
20.12.1 Example 3: Comparing Unconstrained MPC to LQR	427

20.12.2 Example 4: Effect of Prediction Horizon Length	427
20.12.3 Example 5: MPC for Setpoint Tracking	428
20.13 MPC Block Diagram and Architecture	429
20.14 Chapter Summary	429
20.15 Exercises	431
20.15.1 Conceptual Exercises	431
20.15.2 Computational Exercises	431
20.15.3 Analysis Exercises	431
20.15.4 Implementation Exercises	432
Further Reading	432
21 System Identification	433
21.1 Historical Motivation: Learning Models from Data	433
21.1.1 Gauss and the Birth of Least Squares (1809)	433
21.1.2 From Astronomy to Engineering	434
21.1.3 Lennart Ljung and the Modern Framework (1970s–1980s)	435
21.1.4 “All Models Are Wrong, Some Are Useful”	436
21.1.5 Aerospace Applications: Flutter Testing	436
21.2 Modern Applications	437
21.2.1 Adaptive Control: Identify Then Control	437
21.2.2 Digital Twins from Operational Data	437
21.2.3 Machine Learning for Dynamics	437
21.2.4 Biomedical System Modeling	438
21.3 Mathematical Foundation	438
21.3.1 Problem Formulation	438
21.3.2 Input Signal Design	438
21.3.3 Least Squares Estimation	440
21.3.4 ARX Model Structure	441
21.3.5 Frequency Domain Identification	442
21.3.6 Model Validation	443
21.3.7 Bias-Variance Tradeoff	445
21.4 Practical Experiment: Identifying DC Motor Dynamics	445
21.4.1 Learning Objectives	445
21.4.2 Equipment and Setup	446
21.4.3 System Model	446
21.4.4 Experimental Procedure	446
21.4.5 Comparison with Physics-Based Model	450
21.4.6 Data Recording Table	450
21.5 Worked Examples	450
21.6 System Identification Flowchart	455
21.7 Chapter Summary	455
21.8 Exercises	458

22 Adaptive Control	459
22.1 Historical Motivation: The Need for Adaptation	459
22.1.1 The Aircraft Problem: Where Fixed Controllers Fail	459
22.1.2 The MIT Rule: Birth of Adaptive Control (1958)	460
22.1.3 Lyapunov-Based Adaptation: Mathematical Rigor (1960s)	461
22.1.4 The Robustness Crisis: Rohrs' Counterexamples (1985)	461
22.1.5 Resolution and Modern Perspective	462
22.2 Modern Applications of Adaptive Control	462
22.2.1 Aerospace and Flight Control	463
22.2.2 Robotics and Manufacturing	463
22.2.3 Process Industry	463
22.2.4 Biomedical Applications	463
22.2.5 Automotive Systems	463
22.3 Mathematical Foundations of Adaptive Control	464
22.3.1 The Adaptive Control Problem	464
22.3.2 Model Reference Adaptive Control (MRAC)	464
22.3.3 MIT Rule Derivation	465
22.3.4 Lyapunov-Based Adaptive Control	466
22.3.5 Self-Tuning Regulators	468
22.3.6 Gain Scheduling	469
22.3.7 Robustness Modifications	470
22.4 Practical Experiment: Adaptive Motor Control	471
22.4.1 Experimental Setup	472
22.4.2 Mathematical Model	472
22.4.3 Controller Implementation	473
22.4.4 Arduino Implementation	473
22.4.5 Experimental Procedure	475
22.4.6 Expected Results	475
22.4.7 Discussion Questions	475
22.5 Example Problems	476
22.6 Summary and Key Equations	481
22.6.1 Key Concepts	481
22.6.2 Essential Equations	482
22.6.3 Design Guidelines	482
22.7 Problems	483
22.8 Further Reading	484
24 Introduction to Reinforcement Learning for Control	485
24.1 Historical Motivation	485
24.1.1 Early Foundations: Learning Machines	485
24.1.2 The Temporal Difference Revolution (1980s-1990s)	487
24.1.3 The Deep Learning Revolution (2013-Present)	487
24.1.4 The Control Connection: Optimal Control and RL Duality	488
24.1.5 Why Now? Enabling Technologies	488
24.2 Modern Applications	489

24.2.1	Robotic Manipulation	489
24.2.2	Game Playing as Control	489
24.2.3	Autonomous Vehicle Decision-Making	489
24.2.4	HVAC and Building Energy Optimization	490
24.3	Mathematical Foundation	490
24.3.1	Markov Decision Processes	490
24.3.2	Policies	490
24.3.3	Value Functions	491
24.3.4	The Bellman Equations	492
24.3.5	Q-Learning Algorithm	493
24.3.6	Policy Gradient Methods	493
24.3.7	Connection to LQR: The Linear Quadratic Case	495
24.4	Practical Experiment: CartPole Balancing	495
24.4.1	Problem Description	496
24.4.2	Implementation: Q-Learning with Discretization	496
24.4.3	Learning Curve Analysis	496
24.4.4	Comparison with Hand-Designed Controller	496
24.4.5	Policy Visualization	499
24.5	Worked Examples	499
24.6	Summary	505
25	Neural Networks in the Loop	509
25.1	Historical Motivation: From Biological Neurons to Learning Machines	509
25.1.1	McCulloch and Pitts (1943): The Computational Neuron	509
25.1.2	Rosenblatt (1958): The Perceptron and Learning	510
25.1.3	Minsky and Papert (1969): The First AI Winter	510
25.1.4	Werbos (1974) and Rumelhart et al. (1986): Backpropagation	511
25.1.5	Narendra and Parthasarathy (1990): Neural Networks Meet Control	511
25.1.6	The Second AI Winter and Deep Learning Revolution	512
25.1.7	Why Neural Networks for Control?	512
25.2	Modern Applications of Neural Networks in Control	513
25.2.1	End-to-End Autonomous Driving	513
25.2.2	Neural Network Controllers for Drones	513
25.2.3	Learned Dynamics Models for MPC	514
25.2.4	Industrial Soft Sensors	514
25.3	Mathematical Foundation: Neural Networks for Dynamics and Control	514
25.3.1	The Artificial Neuron	514
25.3.2	Activation Functions	515
25.3.3	Multi-Layer Neural Networks	516
25.3.4	Universal Approximation Theorem	516
25.3.5	Neural Network as Controller	517
25.3.6	Neural Network as Plant Model	518
25.3.7	Training Neural Networks: Loss Functions and Optimization	518
25.3.8	Backpropagation	519
25.3.9	Backpropagation Through Time for Dynamics	520

25.3.10	Challenges in Neural Network Control	520
25.3.11	Hybrid Approaches: Neural Networks + Physics	521
25.4	Practical Experiment: Neural Network Controller for DC Motor	521
25.4.1	Experimental Setup	521
25.4.2	Phase 1: PID Controller and Data Collection	522
25.4.3	Phase 2: Neural Network Training	524
25.4.4	Phase 3: Export and Deployment	524
25.4.5	Phase 4: Evaluation and Comparison	527
25.4.6	Discussion	527
25.5	Worked Examples	529
25.5.1	Example 3: Designing NN Architecture for Control	530
25.5.2	Example 4: Training NN to Approximate a Known Function	531
25.5.3	Example 5: Analyzing NN Controller vs Linear Controller	531
25.5.4	Example 6: Neural Network for System Identification	533
25.6	Chapter Summary	534
25.7	Exercises	535
25.8	Further Reading	536
26	Robustness and Uncertainty	537
26.1	Historical Motivation: Why Models Are Never Enough	537
26.1.1	Bode and the Origins of Robustness Thinking (1940s)	537
26.1.2	The Multivariable Challenge: Doyle and Stein (1981)	539
26.1.3	H_∞ Control: Zames and the Optimization of Robustness (1981)	539
26.1.4	Lessons from Failure: When Models Betray Us	540
26.1.5	The Fundamental Principle	541
26.2	Modern Applications of Robust Control	541
26.2.1	Aerospace Systems	541
26.2.2	Robotics and Automation	542
26.2.3	Power Systems and Renewable Energy	542
26.2.4	Medical Devices	542
26.3	Mathematical Foundations of Robust Control	542
26.3.1	Types of Uncertainty	543
26.3.2	The Small Gain Theorem	544
26.3.3	Sensitivity Functions and Trade-offs	546
26.3.4	The H_∞ Control Problem	547
26.3.5	Robust Performance	549
26.3.6	Introduction to μ -Analysis	549
26.3.7	Practical Robustness Assessment	549
26.4	Practical Laboratory: Robust Motor Control	550
26.4.1	Learning Objectives	550
26.4.2	Required Components	551
26.4.3	System Identification	551
26.4.4	Controller Design	552
26.4.5	Experimental Procedure	552
26.4.6	Data Analysis	554

26.5	Worked Examples	555
26.6	Figures and Diagrams	562
26.6.1	Uncertainty Models	562
26.6.2	Robust Stability in Nyquist Diagram	562
26.6.3	Sensitivity Function Bode Plots	562
26.6.4	Nominal vs Robust Response Comparison	562
26.7	Chapter Summary	562
26.8	Exercises	563

Part I

Foundations & Mindset

Chapter 1

What is Control?

1.1 Historical Motivation: The Birth of Automatic Control

The science of automatic control did not emerge from abstract mathematical inquiry. Rather, it arose from urgent practical necessity—the need to regulate physical processes that humans could not monitor and adjust continuously. Understanding these historical origins provides essential intuition for why feedback mechanisms are structured as they are.

1.1.1 The Ktesibios Water Clock (c. 270 BC)

The First Feedback Controller

The earliest known feedback control device was invented by Ctesibius of Alexandria (also written Ktesibios), a Greek inventor and mathematician working in Ptolemaic Egypt around 270 BC. His ingenious water clock, or *clepsydra*, solved a problem that had plagued time-keeping for millennia. Ktesibios was the son of a barber and initially followed his father's trade before becoming one of the most prolific inventors of the ancient world. His contributions extended beyond the water clock to include the pipe organ (hydraulis), force pump, and various pneumatic devices. The float valve mechanism he invented for the *clepsydra* represents humanity's first documented automatic control system—a remarkable achievement that predates the Industrial Revolution by over two millennia.

The earliest known feedback control device was invented by Ctesibius of Alexandria (also written Ktesibios), a Greek inventor and mathematician working in Ptolemaic Egypt around 270 BC. His ingenious water clock, or *clepsydra*, solved a problem that had plagued timekeeping for millennia.

The Problem: Variable Flow Rate

Prior water clocks operated on a simple principle: water draining from a vessel through an orifice at the bottom would indicate time by the falling water level. However, these devices suffered from a fundamental flaw rooted in fluid dynamics. The flow rate Q through an orifice depends on the

pressure head h above it, following Torricelli's law:

$$Q = C_d A \sqrt{2gh} \quad (1.1)$$

where C_d is the discharge coefficient, A is the orifice area, and g is gravitational acceleration. As water drained and h decreased, the flow rate diminished, causing the clock to run progressively slower. A clock accurate in the morning would lose significant time by evening.

The Solution: Float Valve Regulation

Ktesibios introduced a remarkable solution: a separate reservoir maintained at constant level by a float valve, which then fed the timing vessel. Water entered the regulating reservoir from an external source at a variable rate. A float, constructed from cork or a hollow sphere, rose and fell with the water level. This float was mechanically connected to a conical valve known as the *cock*. As the water level rose, the float lifted and progressively closed the inlet valve, restricting flow. Conversely, as the water level fell, the float dropped and opened the valve to admit more water.

This mechanism maintained a nearly constant head h in the regulating reservoir, ensuring constant flow rate to the timing vessel regardless of variations in the external water supply. The float valve constitutes a true feedback controller: the controlled variable (water level) directly influences the manipulated variable (inlet valve position) through a mechanical linkage.

Why Open-Loop Failed

Consider the alternative: manually setting a fixed inlet flow rate intended to match the outlet flow. Any disturbance—variation in source pressure, partial clogging of the inlet, evaporation, temperature changes affecting viscosity—would cause the level to drift. Without continuous human intervention, accuracy was impossible. The float valve made the system *self-regulating*, automatically compensating for disturbances that the designer could not anticipate or eliminate.

1.1.2 Cornelis Drebbel's Thermostat (c. 1620)

The Dutch inventor Cornelis Drebbel constructed what is considered the first temperature feedback control system around 1620. His application was the regulation of furnace temperature for alchemical operations and, notably, for chicken egg incubators—where maintaining precise temperature was essential for successful hatching.

The Problem: Temperature Fluctuation

Maintaining constant temperature in a furnace or incubator using only fuel rate adjustment is extraordinarily difficult. Heat loss varies with ambient temperature, fuel quality varies, and the thermal mass of the system creates delays between fuel input and temperature response. An operator attempting to maintain temperature by periodic adjustment inevitably produces oscillations: over-correction leads to overheating, followed by under-fueling and cooling, in an endless cycle.

The Mechanism

Drebbel's thermostat employed a mercury-and-alcohol thermometer connected to a damper controlling air flow to the furnace. A glass vessel containing alcohol was placed inside the incubator, where the alcohol expanded or contracted in response to temperature changes. This expansion or contraction moved a column of mercury in a connected tube, and the mercury column was mechanically linked to a damper or flap controlling airflow. When temperature increased, the alcohol expanded, causing the mercury to rise and the damper to close, thereby reducing combustion. When temperature decreased, the alcohol contracted, the mercury fell, and the damper opened to increase combustion.

This represents a proportional feedback controller, where the damper position varied continuously with temperature deviation from the setpoint (determined by the mechanical linkage geometry).

1.1.3 James Watt's Centrifugal Governor (1788)

The device that launched control theory as a mathematical discipline was James Watt's centrifugal governor for steam engines, patented in 1788. While Watt did not invent the centrifugal mechanism (similar devices regulated windmill speeds), his application to steam engines created the technological foundation of the Industrial Revolution—and the theoretical problems that birthed modern control theory.

The Problem: Catastrophic Speed Variation

Early steam engines suffered from dangerous speed fluctuations. The fundamental issue arose from the mismatch between steam power input and mechanical load. Industrial loads such as pumps, looms, and mills varied moment to moment; a loom engaging produced sudden load while thread breaking released it. Steam pressure variations compounded the problem, as boiler output fluctuated with fuel quality, water level, and draft conditions. Furthermore, some loads exhibited positive feedback instabilities, with decreasing resistance at higher speeds creating potential runaway conditions.

Without automatic regulation, a sudden load decrease could cause the engine to “run away,” potentially to mechanical destruction. Conversely, sudden load increases could stall the engine. Human operators could not react quickly enough, and continuous manual throttle adjustment was impractical in factory settings.

The Centrifugal Governor Mechanism

Watt's governor operated on an elegant principle relating rotational speed to centrifugal force. Two heavy balls were mounted on hinged arms connected to a vertical spindle, which was driven by the engine through gearing and rotated proportionally to engine speed. As speed increased, centrifugal force swung the balls outward and upward, and the ball arm geometry converted this outward motion to vertical motion of a sliding collar. The collar was linked to the steam throttle valve, creating the feedback mechanism: higher speed caused the balls to rise, raising the collar and closing the throttle, which admitted less steam and decreased speed. Conversely, lower speed

caused the balls to fall, lowering the collar and opening the throttle, which admitted more steam and increased speed.

The equilibrium condition occurs when centrifugal force exactly balances gravity and spring forces, determining a specific speed at which the throttle position stabilizes.

Mathematical Analysis: Maxwell's 1868 Paper

The Watt governor worked remarkably well in practice, but some installations exhibited puzzling oscillations—the engine speed would hunt continuously about the setpoint rather than settling. This phenomenon prompted James Clerk Maxwell to publish “On Governors” in 1868, founding the mathematical theory of feedback control.

Maxwell modeled the governor-engine system as a set of differential equations and analyzed their stability using linearization. He showed that oscillations arose when certain relationships between governor parameters (ball mass, arm length, spindle friction) and engine parameters (inertia, throttle sensitivity) were violated. The mathematical condition for stability—that all roots of the characteristic equation have negative real parts—was a precursor to the Routh-Hurwitz criterion developed later.

The Fundamental Lesson

The Watt governor illustrates the central principle motivating feedback control:

The Feedback Principle

Feedback control enables a system to regulate itself against disturbances that cannot be predicted, measured, or eliminated at their source. The controller need not “know” why the output deviated—only that it deviated—to take corrective action.

An open-loop alternative—setting the throttle to a fixed position calculated to produce the desired speed—would fail immediately upon any load change, boiler pressure variation, or friction change. The governor succeeds because it continuously measures the actual output (speed) and adjusts the input (steam) accordingly.

1.2 Modern Applications: Open-Loop vs. Closed-Loop Systems

The distinction between open-loop and closed-loop control pervades modern technology, though users rarely consider which paradigm governs the devices they use.

1.2.1 Open-Loop Control Systems

Open-loop systems apply a predetermined input without measuring the output. They rely on accurate system models and the absence of disturbances.

Household Examples

Common household appliances illustrate open-loop operation. A timer-based toaster applies heat for a fixed duration regardless of actual bread browning, and variations in bread moisture, thickness, and initial temperature cause inconsistent results. Similarly, many basic washing machines run fixed cycles assuming standard loads, so overloading or underloading produces suboptimal cleaning. Microwave ovens apply power for a user-specified time without measuring food temperature, though some modern units add temperature probes, thereby converting to closed-loop operation.

Industrial Examples

Industrial settings also employ open-loop control in certain applications. In low-precision stepper motor positioning, motors are commanded a specific number of steps assuming each step produces exact displacement, but missed steps accumulate as position error. Some industrial processes also add chemical reagents on a time schedule rather than measuring concentration, relying on predictable reaction kinetics.

When Open-Loop Suffices

Open-loop control is appropriate under specific conditions. The system model must be accurate and stable, and disturbances must be negligible or repeatable. Additionally, the cost of feedback sensing should exceed the value of improved performance, and the consequences of error must be acceptable. When these conditions are met, the simplicity and lower cost of open-loop control make it the preferred choice.

1.2.2 Closed-Loop Control Systems

Closed-loop systems measure the output and adjust the input to minimize error between the output and desired reference.

Automotive Examples

Modern automobiles employ numerous closed-loop control systems. Cruise control uses a speed sensor to continuously measure vehicle velocity, adjusting the throttle to maintain the setpoint despite hills, wind, and rolling resistance variations. Anti-lock braking systems (ABS) use wheel speed sensors to detect impending lockup and modulate brake pressure to maintain optimal slip ratio. Electronic stability control employs yaw rate and lateral acceleration sensors to detect skids, applying individual wheel braking to restore the intended trajectory.

Industrial Process Control

Industrial processes rely heavily on closed-loop control for safety and quality. In refinery operations, temperature, pressure, flow rate, and composition sensors feed controllers that adjust heaters, valves, and pumps. Power grid frequency control uses generator governors—descendants of Watt's device—along with automatic generation control to maintain 50/60 Hz despite load variations.

Semiconductor fabrication requires sub-nanometer positioning, achieved through continuous feedback from interferometric sensors.

Robotics and Automation

Robotics and automation depend critically on feedback control. Robot arm positioning uses encoder feedback to achieve sub-millimeter repeatability despite payload variations. Drone flight control integrates data from accelerometers, gyroscopes, barometers, and GPS to maintain attitude and position. Self-driving vehicles employ LIDAR, cameras, and radar for perception, with control algorithms maintaining lane position and vehicle following distance.

Biological Feedback Systems

Biological organisms are extraordinarily sophisticated feedback control systems. Thermoregulation exemplifies this: the hypothalamus maintains core body temperature at 37°C through sweating, shivering, and vasodilation or vasoconstriction. Blood glucose regulation employs insulin and glucagon as a dual-controller system maintaining glucose homeostasis. The pupillary light reflex adjusts pupil diameter to maintain appropriate retinal illumination, while the vestibulo-ocular reflex compensates eye position for head rotation, thereby stabilizing the visual field.

1.3 Mathematical Foundations

We now establish the mathematical framework for analyzing control systems, beginning with precise definitions and progressing through the fundamental equations of open-loop and closed-loop operation.

1.3.1 System Definitions and Notation

Definition 1.1 (System). A **system** is a collection of components that act together to perform a function. In control engineering, we model systems as operators that transform input signals into output signals.

Definition 1.2 (Signals). We define the following signals in a control system. The **reference input** $r(t)$ represents the setpoint or desired value. The **control input** $u(t)$ is the manipulated variable or actuator command. The **output** $y(t)$ denotes the controlled variable or measured response. External influences enter the system as the **disturbance** $d(t)$, representing uncontrolled factors, and **noise** $n(t)$, representing measurement corruption. Finally, the **error signal** $e(t)$ is defined as the difference between reference and output: $e(t) = r(t) - y(t)$.

Definition 1.3 (Transfer Function). For a linear time-invariant (LTI) system, the **transfer function** $G(s)$ relates the Laplace transform of the output to the Laplace transform of the input:

$$G(s) = \frac{Y(s)}{U(s)} \quad (1.2)$$

where $Y(s) = \mathcal{L}\{y(t)\}$ and $U(s) = \mathcal{L}\{u(t)\}$.

1.3.2 Block Diagram Representation

Control systems are represented using block diagrams, a graphical notation with several standard elements. Blocks represent transfer functions, with input entering from the left and output exiting to the right. Arrows indicate signal flow direction. Summing junctions, depicted as circles with Σ or $+/-$ symbols, add or subtract signals. Pickoff points, shown as filled circles, duplicate a signal for routing to multiple destinations.

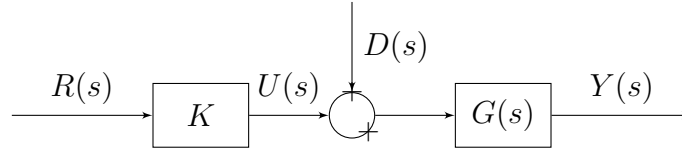


Figure 1.1: Open-loop control system block diagram. The reference input $R(s)$ passes through the controller K to produce the control signal $U(s)$, which combines with the disturbance $D(s)$ at the plant input. No feedback path exists.

1.3.3 Open-Loop Control Analysis

In open-loop control, the controller operates on the reference input without knowledge of the output. The control input is:

$$U(s) = K \cdot R(s) \quad (1.3)$$

where K represents the controller (possibly a constant gain, possibly a transfer function).

The output, including disturbance effects, is:

$$Y(s) = G(s) \cdot U(s) + G(s) \cdot D(s) = G(s)K \cdot R(s) + G(s) \cdot D(s) \quad (1.4)$$

For ideal tracking with no disturbance ($D(s) = 0$), we desire $Y(s) = R(s)$, requiring:

$$G(s)K = 1 \quad \Rightarrow \quad K = \frac{1}{G(s)} = G^{-1}(s) \quad (1.5)$$

This inverse controller approach has severe limitations. The plant $G(s)$ must be exactly known, and the plant must be minimum-phase (having no right-half-plane zeros) for stable inversion. Any model error ΔG causes proportional output error, and disturbances $D(s)$ appear directly at the output, amplified by $G(s)$.

1.3.4 Closed-Loop Control Analysis

In closed-loop control, the error between reference and measured output drives the controller:

The mathematical relationships are:

$$E(s) = R(s) - H(s)Y(s) \quad (1.6)$$

$$U(s) = K(s)E(s) \quad (1.7)$$

$$Y(s) = G(s) [U(s) + D(s)] \quad (1.8)$$

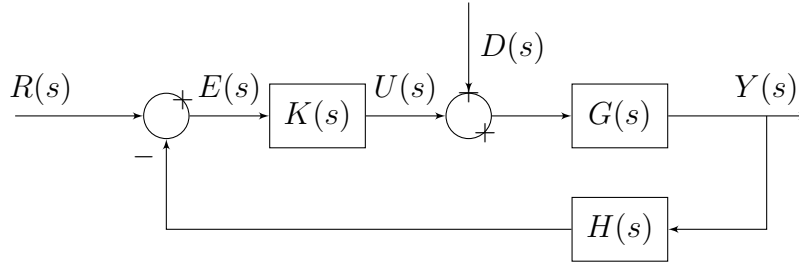


Figure 1.2: Closed-loop control system block diagram. The error $E(s) = R(s) - H(s)Y(s)$ drives the controller, and the feedback path through sensor $H(s)$ enables automatic disturbance rejection.

Substituting (??) into (??) and then into (??):

$$Y(s) = G(s) [K(s) (R(s) - H(s)Y(s)) + D(s)] \quad (1.9)$$

Expanding and collecting terms:

$$Y(s) + G(s)K(s)H(s)Y(s) = G(s)K(s)R(s) + G(s)D(s) \quad (1.10)$$

$$Y(s) [1 + G(s)K(s)H(s)] = G(s)K(s)R(s) + G(s)D(s) \quad (1.11)$$

Solving for $Y(s)$:

$$Y(s) = \frac{G(s)K(s)}{1 + G(s)K(s)H(s)} R(s) + \frac{G(s)}{1 + G(s)K(s)H(s)} D(s) \quad (1.12)$$

Define the **loop transfer function**:

$$L(s) \triangleq G(s)K(s)H(s) \quad (1.13)$$

Then the closed-loop transfer functions are:

Fundamental Closed-Loop Transfer Functions

Reference to Output (Complementary Sensitivity):

$$T(s) = \frac{Y(s)}{R(s)} = \frac{G(s)K(s)}{1 + L(s)} \quad (1.14)$$

Disturbance to Output:

$$\frac{Y(s)}{D(s)} = \frac{G(s)}{1 + L(s)} \quad (1.15)$$

Reference to Error (Sensitivity):

$$S(s) = \frac{E(s)}{R(s)} = \frac{1}{1 + L(s)} \quad (1.16)$$

For unity feedback ($H(s) = 1$), these simplify further. Note the fundamental relationship:

$$S(s) + T(s) = 1 \quad (1.17)$$

1.3.5 Why Feedback Reduces Sensitivity

Consider a plant with nominal transfer function $G_0(s)$ subject to multiplicative uncertainty:

$$G(s) = G_0(s)(1 + \Delta(s)) \quad (1.18)$$

where $\Delta(s)$ represents model error or parameter variation.

Open-Loop Sensitivity

For open-loop control with $K = 1/G_0$ designed for the nominal plant:

$$Y = G \cdot K \cdot R = G_0(1 + \Delta) \cdot \frac{1}{G_0} \cdot R = (1 + \Delta)R \quad (1.19)$$

The output error due to plant variation is:

$$\Delta Y_{OL} = \Delta \cdot R \quad (1.20)$$

The open-loop sensitivity to plant variations equals 1—any plant change appears directly at the output.

Closed-Loop Sensitivity

For closed-loop control with high loop gain $L_0 = G_0KH \gg 1$:

$$T = \frac{GK}{1 + GKH} = \frac{G_0(1 + \Delta)K}{1 + G_0(1 + \Delta)KH} \quad (1.21)$$

For large L_0 :

$$T \approx \frac{G_0(1 + \Delta)K}{G_0(1 + \Delta)KH} = \frac{1}{H} \quad (1.22)$$

The closed-loop transfer function becomes independent of the plant G , depending only on the sensor H ! More precisely, the sensitivity of T to plant variations is:

$$\frac{\partial T}{\partial G} \cdot \frac{G}{T} = \frac{1}{1 + L} = S \quad (1.23)$$

The sensitivity function $S(s)$ quantifies how much plant variations affect the closed-loop response. At frequencies where $|L(j\omega)| \gg 1$, we have $|S(j\omega)| \ll 1$, and the system is insensitive to plant variations.

Theorem 1.1 (Sensitivity Reduction). *Feedback reduces sensitivity to plant variations by the factor $(1 + L)^{-1}$. At frequencies where the loop gain magnitude $|L(j\omega)| = 100$ (40 dB), plant variations are attenuated by a factor of 100 at the output.*

1.3.6 Steady-State Error Analysis

For asymptotic tracking of reference inputs, we analyze the steady-state error using the Final Value Theorem. For a stable closed-loop system:

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s \cdot E(s) = \lim_{s \rightarrow 0} s \cdot S(s) \cdot R(s) \quad (1.24)$$

System Type and Error Constants

The **system type** n is defined as the number of free integrators in the loop transfer function:

$$L(s) = \frac{K_L \prod_i (s + z_i)}{s^n \prod_j (s + p_j)} \quad (1.25)$$

The error constants are defined as:

$$K_p = \lim_{s \rightarrow 0} L(s) \quad (\text{position constant}) \quad (1.26)$$

$$K_v = \lim_{s \rightarrow 0} s \cdot L(s) \quad (\text{velocity constant}) \quad (1.27)$$

$$K_a = \lim_{s \rightarrow 0} s^2 \cdot L(s) \quad (\text{acceleration constant}) \quad (1.28)$$

Table 1.1: Steady-State Error for Different System Types

System Type	Step Input $R(s) = 1/s$	Ramp Input $R(s) = 1/s^2$	Parabolic $R(s) = 1/s^3$
Type 0	$\frac{1}{1 + K_p}$	∞	∞
Type 1	0	$\frac{1}{K_v}$	∞
Type 2	0	0	$\frac{1}{K_a}$

1.4 Practical Laboratory: DC Motor Position Control

This laboratory exercise demonstrates the fundamental difference between open-loop and closed-loop control using readily available components. Students will implement both control strategies on a DC motor position system and quantify the performance improvement achieved through feedback.

1.4.1 Learning Objectives

Upon completion of this laboratory, students will be able to explain why open-loop control fails under disturbances and model uncertainty, implement a basic closed-loop position controller using encoder feedback, measure and compare steady-state error, disturbance rejection, and repeatability, and tune a proportional controller gain while observing its effects on system behavior.

1.4.2 Required Components

For advanced experiments, optional components include a 3D-printed dial with degree markings for the motor shaft, a second potentiometer for setpoint adjustment, and serial plotting software such as Arduino Serial Plotter or Processing.

Table 1.2: Bill of Materials for Motor Control Laboratory

Component	Specification	Qty
Arduino Uno or Nano	ATmega328P microcontroller	1
DC Motor with Encoder	12V, 200+ PPR quadrature encoder	1
Motor Driver	L298N or TB6612FNG H-bridge	1
Potentiometer	10k Ω linear taper	1
Power Supply	12V, 2A DC adapter	1
Breadboard	Full-size	1
Jumper Wires	Male-male, assorted	20
Position Indicator	Pointer or dial attached to motor shaft	1
Reference Scale	Printed protractor or degree markings	1

1.4.3 Wiring Configuration

Table 1.3: Motor Control System Wiring Connections

Category	Source	Destination
Power	12V Supply	Motor Driver VCC
	Motor Driver GND	Common Ground / Arduino GND
	Arduino 5V	Potentiometer Terminal 1
	Arduino GND	Potentiometer Terminal 2
Motor Driver	Driver IN1	Arduino D5 (PWM)
	Driver IN2	Arduino D6 (PWM)
	Driver ENA	Arduino D9 (PWM) or 5V jumper
Encoder	Encoder VCC	Arduino 5V
	Encoder GND	Arduino GND
	Encoder A	Arduino D2 (interrupt capable)
	Encoder B	Arduino D3 (interrupt capable)
Potentiometer	Wiper	Arduino A0

Algorithm 1 Quadrature Encoder Reading with Interrupts

```

1: Initialize:
2:    $encoderCount \leftarrow 0$ 
3:    $lastEncoded \leftarrow 0$ 
4:   Configure encoder pins A and B as inputs with pull-up resistors
5:   Attach interrupt handlers to both encoder channels
6:
7: procedure UPDATEENCODER ▷ Called on any encoder pin change
8:    $MSB \leftarrow \text{read encoder channel A}$ 
9:    $LSB \leftarrow \text{read encoder channel B}$ 
10:   $encoded \leftarrow (MSB \times 2) + LSB$ 
11:   $sum \leftarrow (lastEncoded \times 4) + encoded$ 
12:  if  $sum$  indicates forward rotation then
13:     $encoderCount \leftarrow encoderCount + 1$ 
14:  else if  $sum$  indicates reverse rotation then
15:     $encoderCount \leftarrow encoderCount - 1$ 
16:  end if
17:   $lastEncoded \leftarrow encoded$ 
18: end procedure
19:
20: function GETPOSITIONDEGREES
21:    $COUNTS\_PER\_REV \leftarrow 400$  ▷ Adjust for encoder PPR
22:   return  $(encoderCount / COUNTS\_PER\_REV) \times 360$ 
23: end function

```

Algorithm 2 Open-Loop Position Control (Timed Pulses)

```

1: Constants (students must calibrate):
2:    $CAL\_PWM \leftarrow 150$                                 ▷ Motor speed setting
3:    $CAL\_TIME\_90DEG \leftarrow 500$                         ▷ Milliseconds for 90 degrees
4:
5: Initialize:
6:   Configure motor pins as outputs
7:   Initialize encoder (see Algorithm 1)
8:
9: procedure SETMOTORPWM( $pwm$ )
10:  if  $pwm > 0$  then
11:    Set forward direction with magnitude  $pwm$ 
12:  else if  $pwm < 0$  then
13:    Set reverse direction with magnitude  $|pwm|$ 
14:  else
15:    Stop motor
16:  end if
17: end procedure
18:
19: procedure MOVEOPENLOOP( $targetDegrees$ )
20:    $timeRequired \leftarrow |targetDegrees| \times (CAL\_TIME\_90DEG/90)$ 
21:    $direction \leftarrow \text{sign}(targetDegrees)$ 
22:    $startPos \leftarrow \text{GetPositionDegrees}()$ 
23:   SETMOTORPWM( $direction \times CAL\_PWM$ )
24:   Wait for  $timeRequired$  milliseconds
25:   SETMOTORPWM(0)
26:   Wait 500 ms for motor to settle
27:    $endPos \leftarrow \text{GetPositionDegrees}()$ 
28:    $error \leftarrow targetDegrees - (endPos - startPos)$ 
29:   Report target, actual position, and error
30: end procedure
31:
32: Main Loop:
33: while true do
34:   if user input available then
35:      $target \leftarrow$  read target angle from user
36:     Reset encoder count to zero
37:     MOVEOPENLOOP( $target$ )
38:   end if
39: end while

```

Algorithm 3 Closed-Loop Proportional Position Control

```

1: Parameters (students tune these):
2:    $K_p \leftarrow 2.0$  ▷ Proportional gain
3:    $DEADBAND \leftarrow 1.0$  ▷ Degrees; stop when error below this
4:    $MAX\_PWM \leftarrow 200$ 
5:    $MIN\_PWM \leftarrow 30$  ▷ Minimum to overcome friction
6:
7: State Variables:
8:    $setpoint \leftarrow 0$ 
9:
10: Initialize:
11:   Configure encoder and motor pins
12:
13: Main Control Loop (runs at 100 Hz):
14: while true do
15:   if user command received then
16:     if command is 'S' then
17:        $setpoint \leftarrow$  new setpoint value
18:     else if command is 'K' then
19:        $K_p \leftarrow$  new gain value
20:     end if
21:   end if
22:
23:    $position \leftarrow$  GetPositionDegrees()
24:    $error \leftarrow setpoint - position$ 
25:    $pwmOutput \leftarrow 0$ 
26:
27:   if  $|error| > DEADBAND$  then
28:      $pwmOutput \leftarrow K_p \times error$  ▷ Proportional control
29:      $pwmOutput \leftarrow \text{clamp}(pwmOutput, -MAX\_PWM, MAX\_PWM)$ 
30:     if  $|pwmOutput| < MIN\_PWM$  then
31:        $pwmOutput \leftarrow \text{sign}(error) \times MIN\_PWM$ 
32:     end if
33:   end if
34:
35:   SETMOTORPWM( $pwmOutput$ )
36:   Report setpoint, position, error, and PWM (every 100 ms)
37:   Wait 10 ms
38: end while

```

1.4.4 Software Implementation

Part 1: Encoder Reading (Common to Both Experiments)

Part 2: Open-Loop Control Experiment

Part 3: Closed-Loop Control Experiment

1.4.5 Experimental Procedure

Experiment A: Open-Loop Calibration and Testing

Begin by uploading the open-loop code to the Arduino. For calibration, command 90-degree movements and adjust the `CAL_TIME_90DEG` parameter until the motor consistently moves 90 degrees, then record this calibrated value. Next, conduct a repeatability test by commanding ten 90-degree movements, recording the actual positions achieved, and calculating the mean and standard deviation of the error. For the disturbance test, while the motor is at rest, manually rotate the shaft, then command a 90-degree movement and observe that the system has no knowledge of this disturbance. Finally, perform a load test by attaching a small mass to the shaft and repeating the 90-degree movements; observe the increased error due to the changed dynamics.

Experiment B: Closed-Loop Control

Upload the closed-loop code to the Arduino and start with $K_p = 1.0$, commanding position changes and observing the response. For gain tuning, increase K_p gradually and observe the system behavior: at low K_p values, the response will be sluggish with large steady-state error; at medium K_p values, the response becomes crisp with small error; at high K_p values, oscillation or overshoot will appear. Test disturbance rejection by holding the motor at a position, manually pushing the shaft, and observing the automatic return to setpoint. Conduct a repeatability test by commanding ten 90-degree movements and comparing the error statistics to those from the open-loop experiment. Finally, perform a load test by attaching the same mass used in Experiment A and observing that the closed-loop system compensates automatically.

1.4.6 Data Collection and Analysis

Students should complete the following table:

Table 1.4: Experimental Results Comparison

Metric	Open-Loop	Closed-Loop
Mean error (90° target)		
Std. dev. of error		
Maximum error observed		
Time to reach 2% of target		
Disturbance recovery	None	
Effect of added load		

Discussion Questions

Students should consider and discuss the following questions. First, why does open-loop control fail when load is added, while closed-loop adapts automatically? Second, what limits how high K_p can be set, and what physical phenomena cause oscillation at high gain? Third, how would you modify the controller to eliminate steady-state error entirely? Fourth, what is the role of the deadband, and what happens if it is set too small or too large?

1.5 Worked Examples

Steady-State Error Comparison

A DC motor position system has plant transfer function $G(s) = \frac{10}{s(s+2)}$. Compare the steady-state error to a unit ramp input for (a) open-loop control with $K = 1$, and (b) closed-loop control with proportional controller $K = 5$ and unity feedback.

Solution:

Example 1.1 (Steady-State Error Comparison). A DC motor position system has plant transfer function $G(s) = \frac{10}{s(s+2)}$. Compare the steady-state error to a unit ramp input for (a) open-loop control with $K = 1$, and (b) closed-loop control with proportional controller $K = 5$ and unity feedback.

Solution:

(a) Open-Loop:

For open-loop control, the output is $Y(s) = G(s)KR(s)$ and the error is $E(s) = R(s) - Y(s)$. For a unit ramp, $R(s) = 1/s^2$:

$$Y(s) = \frac{10}{s(s+2)} \cdot 1 \cdot \frac{1}{s^2} = \frac{10}{s^3(s+2)}$$

The error is:

$$E(s) = R(s) - Y(s) = \frac{1}{s^2} - \frac{10}{s^3(s+2)} = \frac{s(s+2) - 10}{s^3(s+2)}$$

Applying the Final Value Theorem:

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot E(s) = \lim_{s \rightarrow 0} \frac{s(s+2) - 10}{s^2(s+2)} = \lim_{s \rightarrow 0} \frac{s^2 + 2s - 10}{s^2(s+2)}$$

As $s \rightarrow 0$, the numerator approaches -10 while the denominator approaches 0, indicating the error grows without bound. The open-loop system cannot track a ramp input.

Alternatively, since the open-loop system GK is Type 1 (one integrator), but tracking requires matching the input, and the system does not inherently invert the input dynamics, ramp tracking fails.

$$e_{ss, \text{open-loop}} = \infty$$

(b) Closed-Loop:

The loop transfer function is:

$$L(s) = G(s)K = \frac{10 \cdot 5}{s(s+2)} = \frac{50}{s(s+2)}$$

The velocity constant is:

$$K_v = \lim_{s \rightarrow 0} s \cdot L(s) = \lim_{s \rightarrow 0} s \cdot \frac{50}{s(s+2)} = \lim_{s \rightarrow 0} \frac{50}{s+2} = \frac{50}{2} = 25$$

For a Type 1 system with ramp input:

$$e_{ss} = \frac{1}{K_v} = \frac{1}{25} = 0.04$$

$$e_{ss, \text{closed-loop}} = 0.04 \text{ (or 4\% of ramp slope)}$$

Conclusion: Feedback reduces the error from unbounded to a finite, small value.

Sensitivity Improvement Through Feedback

This example demonstrates the power of feedback for reducing sensitivity to plant variations. A temperature control system has nominal plant gain $G_0 = 5$ ̊C/V, but due to environmental variations, the actual gain can vary by $\pm 20\%$. We will design a proportional feedback controller to reduce the output sensitivity to plant variations by a factor of 10, showing how feedback can make a system robust to uncertainty.

Example 1.2 (Sensitivity Improvement). A temperature control system has nominal plant gain $G_0 = 5$ ̊C/V. Due to environmental variations, the actual gain can vary by $\pm 20\%$, i.e., $G = G_0(1 + \Delta)$ where $|\Delta| \leq 0.2$.

Design a proportional feedback controller to reduce the output sensitivity to plant variations by a factor of 10.

Solution:

For an open-loop system, a 20% plant variation causes a 20% output variation. We want to reduce this to 2%.

The sensitivity function for closed-loop control is:

$$S = \frac{1}{1+L} = \frac{1}{1+GKH}$$

For unity feedback ($H = 1$) and nominal plant:

$$S_0 = \frac{1}{1+G_0K}$$

We require $|S_0| = 0.1$ (factor of 10 reduction):

$$\frac{1}{1+G_0K} = 0.1$$

$$\begin{aligned}
 1 + G_0 K &= 10 \\
 G_0 K &= 9 \\
 K &= \frac{9}{G_0} = \frac{9}{5} = 1.8 \text{ V/}^\circ\text{C}
 \end{aligned}$$

Verification:

With $K = 1.8$ and $G_0 = 5$, the loop gain is $L_0 = 9$.

If the plant varies by +20%: $G = 6$, $L = 6 \times 1.8 = 10.8$

$$T = \frac{L}{1 + L} = \frac{10.8}{11.8} = 0.915$$

Nominal closed-loop gain: $T_0 = \frac{9}{10} = 0.9$

Variation in T : $\frac{0.915 - 0.9}{0.9} = 1.67\%$

A 20% plant variation now causes only 1.67% output variation, confirming the sensitivity reduction.

$$\boxed{K = 1.8 \text{ V/}^\circ\text{C}}$$

Example 1.3 (Disturbance Rejection Analysis). Consider the motor control system with block diagram shown in Figure 1.2. The plant is $G(s) = \frac{1}{s+1}$, the controller is $K(s) = K_p$ (proportional), and feedback is unity ($H = 1$). A constant disturbance $D(s) = D_0/s$ acts at the plant input. Determine (a) the steady-state output error due to the disturbance for the open-loop system, (b) the steady-state output error due to the disturbance for the closed-loop system with $K_p = 10$, and (c) what value of K_p reduces the disturbance effect by a factor of 100.

Solution:**(a) Open-Loop Disturbance Response:**

For open-loop control, the disturbance-to-output transfer function is simply $G(s)$:

$$Y_d(s) = G(s)D(s) = \frac{1}{s+1} \cdot \frac{D_0}{s}$$

Steady-state (Final Value Theorem):

$$y_{d,ss} = \lim_{s \rightarrow 0} s \cdot Y_d(s) = \lim_{s \rightarrow 0} \frac{D_0}{s+1} = D_0$$

$$\boxed{y_{d,ss}^{\text{OL}} = D_0} \text{ (the full disturbance appears at output)}$$

(b) Closed-Loop Disturbance Response:

From equation (??):

$$\frac{Y(s)}{D(s)} = \frac{G(s)}{1 + G(s)K_p} = \frac{\frac{1}{s+1}}{1 + \frac{K_p}{s+1}} = \frac{1}{s+1+K_p}$$

For step disturbance:

$$Y_d(s) = \frac{1}{s+1+K_p} \cdot \frac{D_0}{s}$$

Steady-state:

$$y_{d,ss} = \lim_{s \rightarrow 0} s \cdot Y_d(s) = \frac{D_0}{1 + K_p} = \frac{D_0}{1 + 10} = \frac{D_0}{11}$$

$$\boxed{y_{d,ss}^{\text{CL}} = D_0/11 \approx 0.091D_0} \text{ (disturbance attenuated by factor of 11)}$$

(c) Design for 100× Attenuation:

We require:

$$\begin{aligned} \frac{y_{d,ss}^{\text{CL}}}{y_{d,ss}^{\text{OL}}} &= \frac{1}{100} \\ \frac{D_0/(1 + K_p)}{D_0} &= \frac{1}{1 + K_p} = \frac{1}{100} \\ 1 + K_p &= 100 \end{aligned}$$

$$\boxed{K_p = 99}$$

Example 1.4 (Complete Closed-Loop Design). Design a feedback control system for a thermal system with the following specifications: the plant transfer function is $G(s) = \frac{2}{5s+1}$ (rC per watt of heater power), the steady-state error to a step change in setpoint must be $\leq 2\%$, and the sensor has transfer function $H(s) = 1$ (ideal thermocouple). Determine the required proportional controller gain K_p .

Solution:

For a unity feedback system with proportional controller, the closed-loop transfer function is:

$$T(s) = \frac{G(s)K_p}{1 + G(s)K_p} = \frac{\frac{2K_p}{5s+1}}{1 + \frac{2K_p}{5s+1}} = \frac{2K_p}{5s + 1 + 2K_p}$$

The DC gain (setting $s = 0$):

$$T(0) = \frac{2K_p}{1 + 2K_p}$$

For a step input of magnitude R_0 , the steady-state output is:

$$y_{ss} = T(0) \cdot R_0 = \frac{2K_p}{1 + 2K_p} R_0$$

The steady-state error is:

$$e_{ss} = R_0 - y_{ss} = R_0 \left(1 - \frac{2K_p}{1 + 2K_p} \right) = R_0 \cdot \frac{1}{1 + 2K_p}$$

The fractional error is:

$$\frac{e_{ss}}{R_0} = \frac{1}{1 + 2K_p} \leq 0.02$$

Solving:

$$1 + 2K_p \geq 50$$

$$K_p \geq 24.5$$

$$K_p \geq 24.5 \text{ W/rC}$$

Verification: With $K_p = 25$:

$$e_{ss}/R_0 = \frac{1}{1+50} = \frac{1}{51} \approx 1.96\% < 2\% \quad \checkmark$$

Note: This analysis assumes the closed-loop system is stable, which should be verified. The characteristic equation is $5s + 1 + 2K_p = 0$, giving pole at $s = -(1 + 2K_p)/5$. For any $K_p > 0$, this pole is in the left half-plane, confirming stability.

1.6 Figures for TikZ Implementation

The following detailed descriptions enable creation of publication-quality figures using TikZ.

1.6.1 Figure: Watt Centrifugal Governor

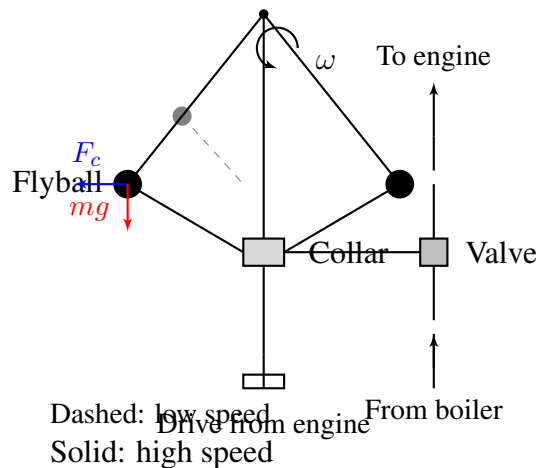


Figure 1.3: Watt centrifugal governor schematic. As rotational speed ω increases, centrifugal force F_c swings the flyballs outward, raising the collar and closing the throttle valve to reduce steam flow.

1.6.2 Figure: Laboratory Setup

1.7 Chapter Summary

This chapter established the fundamental concepts of control systems. From a historical perspective, feedback control emerged from practical necessity—the need to regulate systems against unpredictable disturbances. From Ktesibios’s water clock to Watt’s steam governor, each invention solved problems that open-loop control could not address.

The distinction between open-loop and closed-loop systems is fundamental. Open-loop systems apply predetermined inputs without measuring outputs; they fail when disturbances occur or

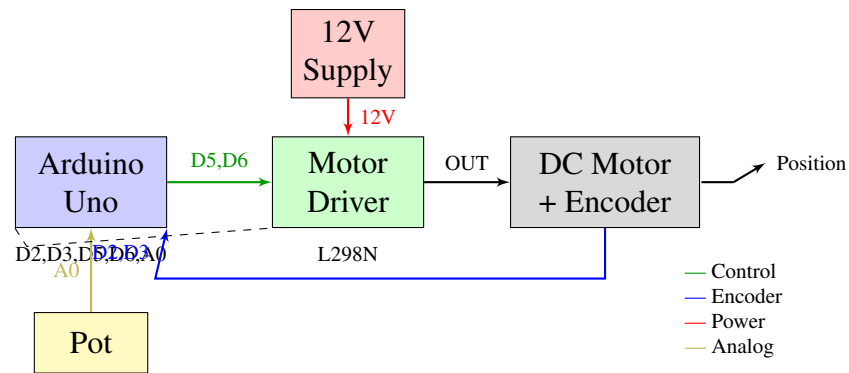


Figure 1.4: Laboratory setup showing the Arduino microcontroller, motor driver, DC motor with encoder, potentiometer for setpoint input, and power supply. Arrows indicate signal flow direction.

models are inaccurate. Closed-loop systems measure output error and adjust input to compensate, providing inherent robustness.

The mathematical framework developed in this chapter shows that the closed-loop transfer function $T(s) = \frac{GK}{1 + GKH}$ differs fundamentally from the open-loop transfer function GK through the $(1 + L)$ denominator, which reduces sensitivity to plant variations and disturbances. Feedback reduces sensitivity to plant variations by the factor $S = 1/(1 + L)$, and system type determines steady-state error characteristics for polynomial inputs.

The laboratory exercises demonstrate these principles using accessible hardware, showing quantitatively how closed-loop control improves accuracy and disturbance rejection compared to open-loop approaches.

Key Takeaway

Feedback control enables systems to regulate themselves against disturbances and uncertainties that cannot be eliminated at their source. The fundamental principle—measure the error, apply correction—is universal across all control applications, from ancient water clocks to modern spacecraft.

1.8 Exercises

Exercise 1.1. (Historical Analysis) Explain why Ktesibios's float valve constitutes a feedback controller. Identify the reference input, controlled variable, manipulated variable, and feedback element.

Exercise 1.2. (Open vs. Closed Loop) A coffee maker heats water for a fixed time (open-loop). Propose a closed-loop alternative and identify what sensor and actuator would be required.

Exercise 1.3. (Steady-State Error) For a system with $G(s) = \frac{5}{s+2}$, $K = 4$, and unity feedback, calculate the steady-state error to (a) a unit step input, and (b) a unit ramp input.

Exercise 1.4. (Sensitivity Analysis) A control system has loop gain $L_0 = 20$ at DC. If the plant gain increases by 10%, what is the percentage change in closed-loop gain?

Exercise 1.5. (Controller Design) Design a proportional controller for $G(s) = \frac{3}{s+5}$ with unity feedback such that the steady-state error to a unit step is less than 5%.

Exercise 1.6. (Disturbance Rejection) For the system in Exercise 5, calculate the steady-state output due to a unit step disturbance at the plant input, both with and without feedback.

Exercise 1.7. (Laboratory Extension) Modify the closed-loop Arduino code to implement proportional-integral (PI) control. What improvement do you expect in steady-state error?

Exercise 1.8. (System Type) Determine the system type and calculate all applicable error constants for the loop transfer function

$$L(s) = \frac{100(s+2)}{s^2(s+5)(s+10)}$$

Exercise 1.9. (Biological Feedback) Human body temperature is regulated at approximately 37°C through sweating, shivering, and vasodilation/vasoconstriction. Draw a block diagram of this thermoregulatory system, identifying the setpoint, sensor, controller, and actuators. What type of controller action (proportional, integral, derivative, or combination) does the body appear to use?

Exercise 1.10. (Closed-Loop Gain Calculation) For the closed-loop system shown in Figure 1.2 with $G(s) = \frac{10}{s(s+5)}$, $K(s) = K_p$, and $H(s) = 1$:

- Derive the closed-loop transfer function $T(s) = Y(s)/R(s)$.
- Find the value of K_p that results in a closed-loop DC gain of 0.9.
- Calculate the closed-loop poles for this value of K_p and determine if the system is stable.

Chapter 2

Modeling Physical Systems

“The purpose of computing is insight, not numbers.” — Richard Hamming

2.1 Historical Motivation: The Need to Predict Before Building

The ability to predict how a physical system will behave—before constructing it—represents one of humanity’s greatest intellectual achievements. This section traces the development of mathematical modeling from Newton’s mechanics through the analog computers of the Victorian era to modern digital simulation.

2.1.1 Newton’s Laws of Motion (1687): The Birth of Mathematical Physics

Newton and the Birth of System Modeling

In 1687, Isaac Newton published *Philosophiæ Naturalis Principia Mathematica*, fundamentally transforming our understanding of the physical world. This work, written in Latin and often simply called the *Principia*, established the mathematical framework that would enable engineers to predict system behavior for centuries to come. Newton developed the calculus simultaneously with Leibniz (leading to a famous priority dispute), and this mathematical tool proved essential for expressing the laws of motion as differential equations. The *Principia* was so influential that it remained the foundation of physics until Einstein’s relativity theory in 1905, and Newtonian mechanics continues to govern most engineering applications today.

In 1687, Isaac Newton published *Philosophiæ Naturalis Principia Mathematica*, fundamentally transforming our understanding of the physical world. Newton’s revolutionary insight was that the motion of all objects—from falling apples to orbiting planets—could be described by three simple mathematical laws.

The Second Law, expressed mathematically as

$$\mathbf{F} = m\mathbf{a} = m \frac{d^2\mathbf{x}}{dt^2}, \quad (2.1)$$

established the differential equation as the language of physics. This was not merely an academic exercise; Newton needed these tools to explain Kepler’s laws of planetary motion and the phenomenon of tides.

The profound implication was this: if one knew the forces acting on a body and its initial conditions (position and velocity), one could predict its future trajectory with mathematical certainty. This was the first “system model”—a mathematical description that captured the essential physics of motion.

Newton’s work raised a compelling question: could this same approach be applied to predict the behavior of engineered systems? The answer, developed over the following centuries, would prove essential to the Industrial Revolution and beyond.

2.1.2 Kirchhoff’s Circuit Laws (1845): Extending Models to Electricity

Gustav Kirchhoff, a German physicist, recognized that electrical circuits followed conservation principles analogous to those in mechanics. In 1845, at just 21 years old, Kirchhoff formulated his circuit laws:

Kirchhoff’s Current Law (KCL): The algebraic sum of currents entering any node equals zero:

$$\sum_{k=1}^n i_k = 0. \quad (2.2)$$

Kirchhoff’s Voltage Law (KVL): The algebraic sum of voltages around any closed loop equals zero:

$$\sum_{k=1}^n v_k = 0. \quad (2.3)$$

These laws, combined with the constitutive relationships for resistors, inductors, and capacitors, enabled the systematic derivation of differential equations describing electrical network behavior. For the first time, engineers could analyze complex electrical systems before building them.

The telegraph industry, rapidly expanding in the 1840s-1860s, provided the economic motivation. Laying submarine cables across the Atlantic was extraordinarily expensive (the first successful transatlantic cable in 1866 cost approximately \$12 million in contemporary dollars). Mathematical modeling allowed engineers to predict signal degradation and design appropriate compensation before committing to construction.

2.1.3 Lord Kelvin’s Mechanical Analog Computers (1870s)

William Thomson (later Lord Kelvin) recognized that differential equations appeared across many domains of physics and engineering. His insight: build mechanical devices that solve differential equations by physical analogy.

In 1876, Kelvin constructed his famous **tide-predicting machine**, a mechanical analog computer using pulleys, gears, and cables to sum harmonic components of tidal motion. The device could predict tide heights for any harbor, years into the future, by physically implementing the

Fourier series:

$$h(t) = h_0 + \sum_{n=1}^N A_n \cos(\omega_n t + \phi_n), \quad (2.4)$$

where each term corresponded to a specific astronomical influence (lunar, solar, etc.).

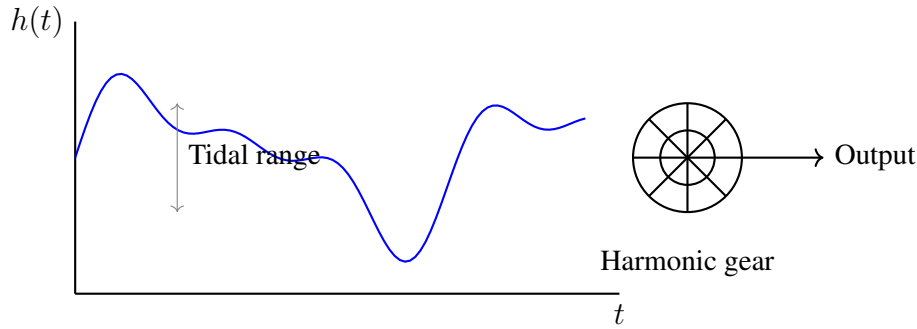


Figure 2.1: Conceptual representation of Kelvin's tide-predicting machine. Gears rotating at different frequencies (corresponding to astronomical periods) are summed mechanically to produce the tide prediction.

This was perhaps the first example of “simulation”—using one physical system (the mechanical computer) to model another (ocean tides). The machine remained in use by the British Admiralty until the 1960s, demonstrating the enduring value of accurate physical models.

2.1.4 Heaviside's Operational Calculus (1880s-1890s)

Oliver Heaviside, a self-taught English electrical engineer, faced a practical problem: analyzing the propagation of electrical signals through long telegraph cables. The partial differential equations governing transmission lines (derived from Maxwell's equations) were notoriously difficult to solve.

Heaviside developed “operational calculus,” treating the differentiation operator d/dt as an algebraic quantity p . Differential equations became algebraic equations:

$$\text{If } p \equiv \frac{d}{dt}, \quad \text{then } L \frac{di}{dt} + Ri = v(t) \quad \Rightarrow \quad (Lp + R)i = v. \quad (2.5)$$

This revolutionary simplification allowed engineers to solve complex circuit problems by algebraic manipulation. Although mathematicians initially dismissed Heaviside's methods as lacking rigor, his techniques worked remarkably well in practice. The formal justification came later through Laplace transform theory, which we shall study in Chapter 3.

Heaviside's work on transmission lines led to the discovery of the “distortionless line” condition:

$$\frac{R}{L} = \frac{G}{C}, \quad (2.6)$$

where R , L , G , and C are the resistance, inductance, conductance, and capacitance per unit length. This insight, derived purely from mathematical modeling, enabled the design of long-distance telephone and telegraph lines that transmitted signals without distortion.

2.1.5 The Cost of Not Modeling: Engineering Disasters

The 19th and early 20th centuries provided tragic demonstrations of what happens when engineers build without adequate mathematical models.

The Tacoma Narrows Bridge (1940): This suspension bridge in Washington State collapsed just four months after opening, due to aeroelastic flutter—a dynamic instability caused by the interaction between wind forces and structural motion. The designers had modeled static loads but not the dynamic, frequency-dependent behavior that proved fatal. The disaster catalyzed the development of aeroelasticity as a rigorous engineering discipline.

The de Havilland Comet (1954): The world’s first commercial jet airliner suffered a series of catastrophic mid-flight breakups. Investigation revealed that the square-cornered windows created stress concentrations, and repeated pressurization cycles caused metal fatigue. The failure to model cyclic stress behavior cost 56 lives and nearly destroyed the British commercial aviation industry.

These disasters underscored a fundamental truth: *mathematical models are not optional luxuries but essential tools for predicting and preventing failure.* Modern engineering ethics require modeling and simulation before any safety-critical system is built.

2.1.6 The Evolution to Digital Simulation

The electronic analog computers of the 1940s-1960s used operational amplifiers, resistors, and capacitors to physically implement differential equations. Each coefficient in the equation corresponded to a component value that could be adjusted. Engineers could “simulate” a system by building an electronic circuit with the same mathematical structure.

The transition to digital computers in the 1970s and beyond enabled more complex simulations and eliminated the analog computer’s sensitivity to component drift. Today, software tools like MATLAB/Simulink, Python with SciPy, and specialized packages enable engineers to simulate systems of virtually unlimited complexity.

Yet the fundamental principle remains unchanged from Newton’s time: derive differential equations from physical laws, solve them (analytically or numerically), and use the solutions to predict system behavior. This is the essence of mathematical modeling.

2.2 Modern Applications of Physical System Models

The mathematical techniques developed centuries ago now underpin virtually every engineered system. This section highlights contemporary applications across diverse fields.

2.2.1 Digital Twins in Manufacturing

A **digital twin** is a virtual replica of a physical system that is continuously updated with real-time sensor data. The concept, pioneered in aerospace (NASA used digital twins for Apollo 13 contingency planning), has now spread to manufacturing, power systems, and infrastructure.

Consider a digital twin of a CNC milling machine. The model includes mechanical dynamics such as motor inertias, gearbox ratios, and structural compliance. Thermal effects are also captured, including spindle heating and thermal expansion of the workpiece. Cutting force models

relate material properties to the forces experienced during machining, while wear models predict tool degradation as a function of cutting time and operating conditions.

By comparing predicted behavior (from the model) with measured behavior (from sensors), deviations can be detected that indicate impending failure or quality problems. This “model-based diagnostics” approach can predict tool breakage hours before it occurs, preventing costly damage and downtime.

2.2.2 Vehicle Dynamics Simulation

Modern vehicles are developed through extensive simulation before physical prototypes are built. A comprehensive vehicle dynamics model encompasses several interconnected subsystems. Chassis dynamics account for sprung and unsprung masses along with suspension geometry. Tire models, such as the Pacejka “magic formula,” relate slip to force generation. Powertrain modeling captures engine torque curves and transmission dynamics, while driver models enable closed-loop evaluation of handling characteristics.

The differential equations governing a vehicle can exceed thousands of states when detailed subsystem models are included. Yet the fundamental approach—derive equations from physics, solve them numerically—remains the same as in Newton’s era.

2.2.3 HVAC System Modeling

Heating, ventilation, and air conditioning (HVAC) systems represent a compelling control problem: maintain comfortable temperature and humidity while minimizing energy consumption. Mathematical models enable several critical capabilities. Load prediction determines how building temperature will evolve given weather forecasts. Optimal control strategies, such as model predictive control (MPC), can pre-cool buildings during off-peak hours to reduce energy costs. Fault detection identifies deviations from model predictions that indicate sensor faults or equipment degradation.

Building thermal models typically use lumped-parameter (RC network) representations, where resistances represent insulation and capacitances represent thermal mass. These models are directly analogous to electrical circuits, demonstrating the power of the electrical-thermal analogy.

2.2.4 Pharmacokinetics: Modeling Drug Dynamics in the Body

Pharmacokinetic models describe how drugs are absorbed, distributed, metabolized, and excreted (ADME) in the body. A simple one-compartment model treats the body as a well-mixed tank:

$$V \frac{dC}{dt} = -k_e V C + R(t), \quad (2.7)$$

where C is drug concentration, V is volume of distribution, k_e is elimination rate constant, and $R(t)$ is the rate of drug administration.

These models are essential for determining dosing schedules that maintain therapeutic concentrations, predicting drug interactions, and designing controlled-release formulations.

The parallel to engineering control systems is striking: the “plant” is the human body, the “input” is drug dosage, and the “output” is blood concentration. Feedback control concepts, including

proportional-integral controllers for automated drug delivery, are increasingly common in medical devices.

2.3 Mathematical Foundation: Deriving Differential Equations from Physics

We now develop the systematic approach to deriving differential equations that describe physical systems. The key insight is that physical laws (conservation of energy, momentum, charge, etc.) combined with constitutive relationships (describing material properties) yield differential equations that govern system dynamics.

2.3.1 Newton's Second Law: The Foundation of Mechanical Systems

Newton's Second Law states that the net force on a body equals its mass times acceleration:

$$\sum F = ma = m \frac{d^2x}{dt^2} = m\ddot{x}. \quad (2.8)$$

We use the overdot notation $\dot{x} \equiv dx/dt$ and $\ddot{x} \equiv d^2x/dt^2$ throughout this text.

The Mass-Spring-Damper System: A Complete Derivation

Consider the canonical mechanical system shown in Fig. 2.2: a mass m connected to a fixed support by a spring (stiffness k) and a viscous damper (damping coefficient b), with an external force $F(t)$ applied.

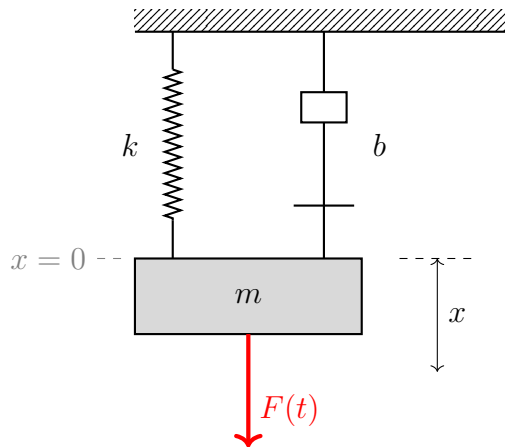


Figure 2.2: Mass-spring-damper system. The position x is measured from the equilibrium position where the spring is at its natural length.

Step 1: Define Coordinates and Sign Conventions

Let x denote the displacement of the mass from its equilibrium position, with positive x defined downward. We measure from equilibrium so that gravity effects are already accounted for by the static spring deflection.

Step 2: Draw the Free Body Diagram

Isolating the mass, we identify all forces acting on it. The spring exerts a restoring force proportional to displacement according to Hooke's Law: $F_{\text{spring}} = -kx$, where the negative sign indicates the force opposes displacement. The viscous damper exerts a force proportional to velocity: $F_{\text{damper}} = -b\dot{x}$, with the negative sign indicating the force opposes motion. Finally, the external applied force $F(t)$ acts on the mass.

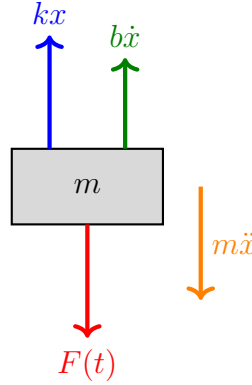
**Free Body Diagram**

Figure 2.3: Free body diagram of the mass showing all forces.

Step 3: Apply Newton's Second Law

Summing forces in the positive x direction:

$$\sum F = ma \quad \Rightarrow \quad F(t) - kx - b\dot{x} = m\ddot{x}. \quad (2.9)$$

Step 4: Rearrange into Standard Form

The standard form places all terms involving the dependent variable on the left and the input on the right:

$$\boxed{m\ddot{x} + b\dot{x} + kx = F(t)} \quad (2.10)$$

This is a second-order, linear, ordinary differential equation (ODE) with constant coefficients. The order of the equation (second) corresponds to the number of energy storage elements in the system: kinetic energy in the mass and potential energy in the spring.

Standard Parameters: Natural Frequency and Damping Ratio

Dividing Eq. (2.9) by m yields:

$$\ddot{x} + \frac{b}{m}\dot{x} + \frac{k}{m}x = \frac{F(t)}{m}. \quad (2.11)$$

We define two fundamental parameters. The **natural frequency** $\omega_n = \sqrt{k/m}$ (in rad/s) characterizes how fast the system oscillates in the absence of damping. The **damping ratio** $\zeta = \frac{b}{2\sqrt{km}} = \frac{b}{2m\omega_n}$ (dimensionless) quantifies the degree to which oscillations are attenuated.

The equation becomes:

$$\boxed{\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \frac{F(t)}{m}} \quad (2.12)$$

This **standard second-order form** appears throughout control systems engineering. The parameters ω_n and ζ completely characterize the system's dynamic behavior, as we shall explore in Chapter ??.

2.3.2 Electrical Systems: RLC Circuits from Kirchhoff's Laws

The electrical analog of the mass-spring-damper system is the series RLC circuit.

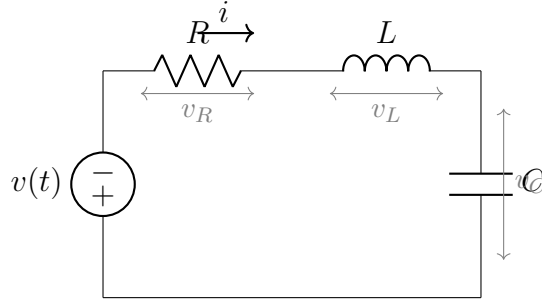


Figure 2.4: Series RLC circuit driven by voltage source $v(t)$.

Constitutive Relations: The resistor obeys Ohm's Law: $v_R = Ri$. The inductor voltage is proportional to the rate of change of current: $v_L = L \frac{di}{dt}$. The capacitor relates current to the rate of change of voltage: $i = C \frac{dv_C}{dt}$, or equivalently $v_C = \frac{1}{C} \int i dt$.

Derivation using KVL:

Applying Kirchhoff's Voltage Law around the loop:

$$v(t) = v_R + v_L + v_C = Ri + L \frac{di}{dt} + v_C. \quad (2.13)$$

Since $i = C \frac{dv_C}{dt}$, we can differentiate the entire equation:

$$\frac{dv}{dt} = R \frac{di}{dt} + L \frac{d^2 i}{dt^2} + \frac{i}{C}. \quad (2.14)$$

Alternatively, expressing everything in terms of the capacitor voltage v_C :

$$\boxed{LC \frac{d^2 v_C}{dt^2} + RC \frac{dv_C}{dt} + v_C = v(t)} \quad (2.15)$$

This has exactly the same form as the mechanical system! The **mechanical-electrical analogy** is summarized in Table 2.1.

2.3.3 Thermal Systems: Heat Flow and RC Analogs

Heat transfer in lumped-parameter systems follows similar principles. Consider a body with thermal capacitance C_{th} (ability to store heat) connected to ambient through thermal resistance R_{th} (resistance to heat flow).

Table 2.1: Force-Voltage Analogy Between Mechanical and Electrical Systems

Property	Mechanical	Electrical
Through variable	Force F	Current i
Across variable	Velocity $v = \dot{x}$	Voltage v
Displacement variable	Position x	Charge q
Inertia/Inductance	Mass m	Inductance L
Dissipation	Damping b	Resistance R
Compliance/Capacitance	$1/k$ (spring)	Capacitance C
Kinetic energy	$\frac{1}{2}mv^2$	$\frac{1}{2}Li^2$
Potential energy	$\frac{1}{2}kx^2$	$\frac{1}{2}Cv^2$

Conservation of Energy:

$$C_{\text{th}} \frac{dT}{dt} = Q_{\text{in}} - \frac{T - T_{\text{ambient}}}{R_{\text{th}}}, \quad (2.16)$$

where T is temperature, and Q_{in} is heat input (e.g., from a heater).

Defining $\theta = T - T_{\text{ambient}}$ as the temperature rise:

$$R_{\text{th}} C_{\text{th}} \frac{d\theta}{dt} + \theta = R_{\text{th}} Q_{\text{in}} \quad (2.17)$$

This is a **first-order system** with time constant $\tau = R_{\text{th}} C_{\text{th}}$. The thermal-electrical analogy provides a direct mapping between domains: temperature corresponds to voltage, heat flow rate corresponds to current, thermal capacitance corresponds to electrical capacitance, and thermal resistance corresponds to electrical resistance.

2.3.4 Rotational Systems

Rotational mechanical systems follow the angular analog of Newton's Second Law:

$$\sum \tau = J\alpha = J \frac{d^2\theta}{dt^2} = J\ddot{\theta}, \quad (2.18)$$

where τ is torque, J is moment of inertia, and θ is angular displacement.

For a rotational system with viscous friction (coefficient b) driven by an applied torque $\tau(t)$:

$$J\ddot{\theta} + b\dot{\theta} = \tau(t) \quad (2.19)$$

This first-order system (when there's no rotational spring) has a time constant $\tau = J/b$.

2.3.5 Lagrangian Mechanics: An Energy-Based Alternative

For complex systems, directly applying Newton's laws becomes cumbersome. The **Lagrangian formulation** provides an elegant alternative based on energy.

Define the Lagrangian as:

$$\mathcal{L} = T - V, \quad (2.20)$$

where T is kinetic energy and V is potential energy. The equations of motion are obtained from the **Euler-Lagrange equation**:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i, \quad (2.21)$$

where q_i are generalized coordinates and Q_i are non-conservative generalized forces.

Example: Mass-Spring-Damper via Lagrangian

$$T = \frac{1}{2}m\dot{x}^2, \quad V = \frac{1}{2}kx^2 \quad (2.22)$$

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2 \quad (2.23)$$

Applying Eq. (??):

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}, \quad \frac{d}{dt}(m\dot{x}) = m\ddot{x} \quad (2.24)$$

$$\frac{\partial \mathcal{L}}{\partial x} = -kx \quad (2.25)$$

The non-conservative force includes damping and the applied force: $Q = F(t) - b\dot{x}$.

Thus:

$$m\ddot{x} - (-kx) = F(t) - b\dot{x} \quad \Rightarrow \quad m\ddot{x} + b\dot{x} + kx = F(t), \quad (2.26)$$

which matches Eq. (??).

2.3.6 System Order and State Variables

The **order** of a system is the highest derivative appearing in its differential equation, which equals the number of independent energy storage elements. A first-order system has one energy storage element, such as an RC circuit or thermal system. A second-order system has two energy storage elements, such as a mass-spring system or RLC circuit. Higher-order systems arise from multiple coupled energy storage elements.

The **state variables** are the minimum set of variables that, together with the input, uniquely determine the future behavior of the system. For a second-order mechanical system, the natural state variables are position x and velocity $\dot{x}$ (or equivalently, position and momentum).

This concept becomes central in state-space analysis (Chapter ??), where we write:

$$\mathbf{x} = \begin{bmatrix} x \\ \dot{x} \end{bmatrix}, \quad \dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} F(t). \quad (2.27)$$

2.4 Practical Experiment: Building and Characterizing a Mass-Spring-Damper System

This section describes a hands-on laboratory exercise where students build a physical mass-spring-damper system, measure its dynamic response, and compare measurements with theoretical predictions.

2.4.1 Apparatus Design and Construction

Components and Bill of Materials

Table 2.2: Bill of Materials for Mass-Spring-Damper Experiment

Item	Description	Qty	Approx. Cost
<i>Structural Components</i>			
3D-printed base	150mm × 100mm × 10mm	1	\$5
3D-printed tower	300mm vertical guide rail	1	\$8
3D-printed carriage	Mass holder with linear bearing	1	\$4
<i>Mechanical Elements</i>			
Extension spring	0.5–2 N/mm stiffness	2	\$3
Rubber bands	Assorted sizes for damping	10	\$2
Steel washers	M8, approx. 15g each	20	\$5
Fishing line	For frictionless guide	2m	\$2
<i>Electronics</i>			
Arduino Uno	Microcontroller	1	\$25
HC-SR04	Ultrasonic distance sensor	1	\$4
MPU-6050	3-axis accelerometer/gyro	1	\$5
Breadboard and wires	Standard prototyping	1 set	\$8

3D Printing Specifications

The apparatus can be printed on any FDM printer with a build volume of at least 150mm × 100mm × 50mm (printing the tower in sections if necessary). Recommended settings include PLA or PETG material with a layer height of 0.2mm. Use 20% infill for the base and tower, and 50% infill for the carriage to ensure adequate strength. Supports are required for bearing housing overhangs.

2.4.2 Measurement Procedure

Static Measurements

Before dynamic testing, characterize the system parameters statically:

1. Spring Constant Measurement:

1. Hang the spring vertically with no load. Measure rest length L_0 .
2. Add known masses incrementally (e.g., 50g, 100g, 150g, 200g).
3. Measure deflection δ for each mass.
4. Plot force $F = mg$ vs. deflection δ .
5. Spring constant $k = \Delta F / \Delta \delta$ (slope of linear fit).

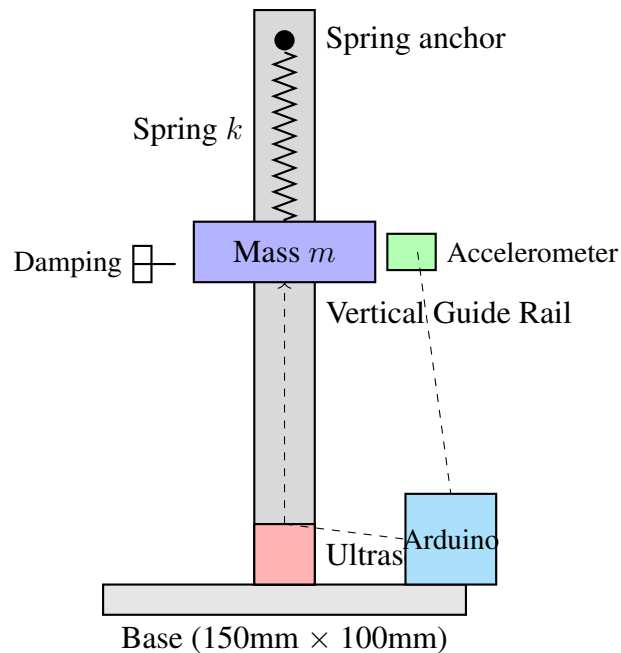


Figure 2.5: Experimental apparatus schematic showing the mass-spring-damper assembly with instrumentation.

2. Total Mass Measurement:

1. Weigh the carriage assembly including accelerometer and wiring.
2. Add washers to achieve desired mass (typically 100–300g total).
3. Record total moving mass m .

Dynamic Measurements: Step Response

The step response reveals the natural frequency and damping ratio:

1. Initialize Arduino data logging at 100+ samples/second.
2. Manually displace the mass downward by approximately 20–30mm.
3. Release the mass and record position or acceleration data for 5–10 seconds.
4. Observe the oscillatory decay (if underdamped) or exponential approach (if overdamped).

Algorithm 4 Step Response Data Acquisition

```

1: Initialize:
2:   Set trigger pin as output, echo pin as input
3:   Initialize serial communication at high baud rate
4:   Record start time  $t_{\text{start}}$ 
5:
6: while data acquisition active do
7:   Trigger ultrasonic pulse:
8:     Set trigger pin LOW, wait  $2 \mu\text{s}$ 
9:     Set trigger pin HIGH, wait  $10 \mu\text{s}$ 
10:    Set trigger pin LOW
11:
12:   Measure echo:
13:      $t_{\text{echo}} \leftarrow$  pulse duration on echo pin
14:      $d \leftarrow t_{\text{echo}} \times 0.034/2$  ▷ Convert to distance in cm
15:
16:   Output data:
17:      $t_{\text{elapsed}} \leftarrow$  current time  $- t_{\text{start}}$ 
18:     Transmit  $(t_{\text{elapsed}}, d)$  via serial
19:
20:   Wait 10 ms ▷ 100 Hz sampling rate
21: end while

```

2.4.3 Data Analysis and Parameter Extraction

Method 1: Logarithmic Decrement

For an underdamped system, the amplitude decays exponentially while oscillating. The **logarithmic decrement** δ is defined as:

$$\delta = \ln \left(\frac{x_n}{x_{n+1}} \right) = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}, \quad (2.28)$$

where x_n and x_{n+1} are successive peak amplitudes.

Solving for damping ratio:

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}. \quad (2.29)$$

The **damped natural frequency** ω_d is measured from the period of oscillation:

$$\omega_d = \frac{2\pi}{T_d}, \quad \omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}}. \quad (2.30)$$

Method 2: Curve Fitting

The theoretical response to an initial displacement x_0 is:

$$x(t) = x_0 e^{-\zeta\omega_n t} \left(\cos \omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t \right). \quad (2.31)$$

Using least-squares curve fitting (in MATLAB, Python, or similar), fit this equation to measured data with ω_n and ζ as free parameters.

2.4.4 Comparing Theory and Experiment

Predicted Parameters:

$$\omega_n^{\text{predicted}} = \sqrt{k/m} \quad (2.32)$$

$$\zeta^{\text{predicted}} = \frac{b}{2\sqrt{km}} \quad (2.33)$$

Note: The damping coefficient b is difficult to measure directly. Friction in the linear guide, air resistance, and internal spring damping all contribute. The experimental measurement of ζ allows back-calculation of the effective b .

Expected Results:

For a typical apparatus with $k \approx 1 \text{ N/mm} = 1000 \text{ N/m}$, $m \approx 0.2 \text{ kg}$:

$$\omega_n = \sqrt{1000/0.2} = 70.7 \text{ rad/s} \approx 11.3 \text{ Hz}. \quad (2.34)$$

With light damping ($\zeta \approx 0.05 - 0.1$), oscillations should persist for 5–10 cycles before decaying to small amplitude.

2.4.5 Sources of Error and Improvements

Several sources of error can affect measurement accuracy. Friction in linear guides introduces Coulomb (dry) friction, which is not captured by the viscous damping model; consider using an air track or magnetic levitation for reduced friction. Sensor noise is another concern: ultrasonic sensors have $\pm 3\text{mm}$ resolution, and accelerometers may exhibit offset drift, though averaging and filtering can improve results. Spring nonlinearity affects measurements when real springs exhibit nonlinear stiffness at large deflections, so oscillation amplitude should be kept small to remain in the linear region. Finally, the mass distribution of the spring itself contributes to the dynamics; approximately 1/3 of the spring mass adds to the effective moving mass.

2.5 Worked Examples

Simple Pendulum Dynamics

The simple pendulum is one of the most fundamental systems in physics and engineering. Despite its apparent simplicity, the pendulum exhibits rich nonlinear behavior and serves as an excellent introduction to linearization techniques. In this example, we derive the equation of motion and show how small-angle approximation leads to simple harmonic motion with a well-defined natural frequency.

2.5.1 Example 1: Simple Pendulum

Problem: Derive the equation of motion for a simple pendulum consisting of a point mass m at the end of a massless rod of length ℓ .

Solution:

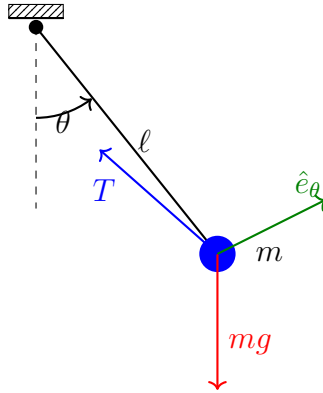


Figure 2.6: Simple pendulum geometry and free body diagram.

Using polar coordinates centered at the pivot, the tangential component of Newton's Second Law gives:

$$m\ell\ddot{\theta} = -mg \sin \theta. \quad (2.35)$$

Thus:

$$\boxed{\ddot{\theta} + \frac{g}{\ell} \sin \theta = 0} \quad (2.36)$$

This is a **nonlinear** differential equation due to the $\sin \theta$ term.

Linearization for Small Angles:

For small angles, $\sin \theta \approx \theta$, yielding:

$$\ddot{\theta} + \frac{g}{\ell} \theta = 0. \quad (2.37)$$

This is simple harmonic motion with natural frequency:

$$\omega_n = \sqrt{g/\ell}. \quad (2.38)$$

For $\ell = 1$ m: $\omega_n = \sqrt{9.81/1} = 3.13$ rad/s, corresponding to period $T = 2\pi/\omega_n = 2.01$ s.

DC Motor: A Multi-Domain System

The DC motor is an excellent example of a system that couples electrical and mechanical domains. The back-EMF creates a feedback mechanism within the motor itself, linking shaft velocity to armature current. Understanding this coupling is essential for motor control applications ranging from robotics to electric vehicles.

2.5.2 Example 2: DC Motor Model

Problem: Derive the transfer function from applied voltage $V(s)$ to motor shaft angular velocity $\Omega(s)$ for a DC motor.

Solution:

A DC motor couples electrical and mechanical domains. The relevant equations are:

Electrical (armature circuit):

$$V = L_a \frac{di_a}{dt} + R_a i_a + e_b, \quad (2.39)$$

where $e_b = K_b \omega$ is the back-EMF (proportional to angular velocity).

Mechanical (rotor):

$$J \frac{d\omega}{dt} + b\omega = \tau_m = K_t i_a, \quad (2.40)$$

where K_t is the torque constant and for a DC motor, $K_t = K_b = K$ (motor constant).

State-Space Form:

Defining states $\mathbf{x} = [i_a, \omega]^T$:

$$\begin{bmatrix} \dot{i}_a \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} -R_a/L_a & -K/L_a \\ K/J & -b/J \end{bmatrix} \begin{bmatrix} i_a \\ \omega \end{bmatrix} + \begin{bmatrix} 1/L_a \\ 0 \end{bmatrix} V. \quad (2.41)$$

Transfer Function:

Taking Laplace transforms (with zero initial conditions):

$$V(s) = (L_a s + R_a) I_a(s) + K \Omega(s) \quad (2.42)$$

$$(Js + b)\Omega(s) = K I_a(s). \quad (2.43)$$

Solving for $\Omega(s)/V(s)$:

$$\frac{\Omega(s)}{V(s)} = \frac{K}{(L_a s + R_a)(Js + b) + K^2} = \frac{K}{L_a J s^2 + (L_a b + R_a J)s + (R_a b + K^2)} \quad (2.44)$$

For many motors, L_a is small enough to neglect, yielding a first-order approximation:

$$\frac{\Omega(s)}{V(s)} \approx \frac{K/R_a}{(JR_a/(R_a b + K^2))s + 1} = \frac{K_m}{\tau_m s + 1}, \quad (2.45)$$

where $K_m = K/(R_a b + K^2)$ is the motor gain and $\tau_m = JR_a/(R_a b + K^2)$ is the mechanical time constant.

2.5.3 Example 3: Finding Natural Frequency and Damping Ratio

Problem: A mass-spring-damper system has $m = 2$ kg, $k = 200$ N/m, and $b = 8$ N·s/m. Find ω_n , ζ , and classify the system response.

Solution:

$$\omega_n = \sqrt{k/m} = \sqrt{200/2} = 10 \text{ rad/s} \quad (2.46)$$

$$\zeta = \frac{b}{2\sqrt{km}} = \frac{8}{2\sqrt{200 \cdot 2}} = \frac{8}{2 \cdot 20} = 0.2 \quad (2.47)$$

Since $0 < \zeta < 1$, the system is **underdamped** and will exhibit oscillatory decay. The damped natural frequency is:

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 10\sqrt{1 - 0.04} = 9.80 \text{ rad/s}. \quad (2.48)$$

The oscillation period is $T_d = 2\pi/\omega_d = 0.641 \text{ s}$.

2.5.4 Example 4: Fluid Tank Level Control

Problem: Derive the differential equation governing the liquid level h in a tank with cross-sectional area A , inlet flow rate $q_{in}(t)$, and outlet flow through a resistance R (where flow is proportional to pressure drop).

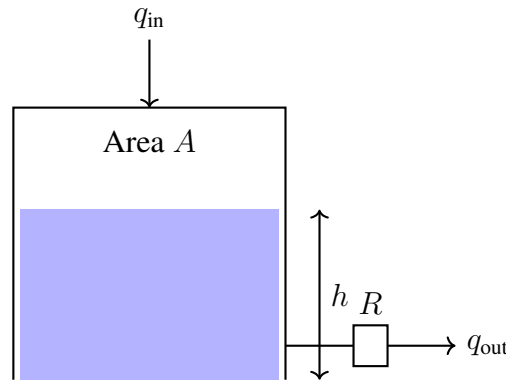


Figure 2.7: Tank level control system schematic.

Solution:

Conservation of Mass:

$$\frac{d(\rho V)}{dt} = \rho q_{in} - \rho q_{out}, \quad (2.49)$$

where $V = Ah$ is the liquid volume. For incompressible fluid:

$$A \frac{dh}{dt} = q_{in} - q_{out}. \quad (2.50)$$

Outlet Flow Relationship:

The outlet flow is driven by hydrostatic pressure ρgh . For a linear resistance:

$$q_{out} = \frac{\rho gh}{R}. \quad (2.51)$$

Final Equation:

$$A \frac{dh}{dt} + \frac{\rho g}{R} h = q_{\text{in}}. \quad (2.52)$$

Defining the time constant $\tau = AR/(\rho g)$:

$$\boxed{\tau \frac{dh}{dt} + h = \frac{R}{\rho g} q_{\text{in}}} \quad (2.53)$$

This is a first-order system. The steady-state level for constant inlet flow $q_{\text{in}} = Q$ is:

$$h_{\text{ss}} = \frac{RQ}{\rho g}. \quad (2.54)$$

2.5.5 Example 5: Coupled Mass-Spring System

Problem: Two masses m_1 and m_2 are connected by springs as shown. Derive the equations of motion.

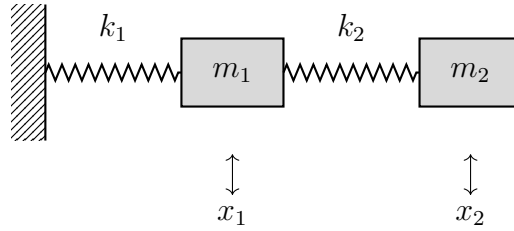


Figure 2.8: Two-mass coupled oscillator system.

Solution:

Free Body Diagram for m_1 : The mass m_1 experiences a restoring force $-k_1 x_1$ from spring k_1 directed toward the wall, and a force $k_2(x_2 - x_1)$ from spring k_2 that depends on the relative displacement between the two masses.

Free Body Diagram for m_2 : The mass m_2 experiences only the force $-k_2(x_2 - x_1)$ from spring k_2 .

Equations of Motion:

$$m_1 \ddot{x}_1 = -k_1 x_1 + k_2(x_2 - x_1) \quad (2.55)$$

$$m_2 \ddot{x}_2 = -k_2(x_2 - x_1) \quad (2.56)$$

Matrix Form:

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (2.57)$$

Or more compactly:

$$\boxed{\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{0}} \quad (2.58)$$

This system has two natural frequencies (normal modes), found by solving the eigenvalue problem:

$$\det(\mathbf{K} - \omega^2 \mathbf{M}) = 0. \quad (2.59)$$

2.6 Chapter Summary

This chapter established the foundation for all subsequent control system analysis: the mathematical model.

From a **historical perspective**, mathematical modeling of physical systems evolved from Newton's mechanics (1687) through Kirchhoff's circuit laws (1845) to Heaviside's operational calculus (1880s), driven by practical needs in astronomy, telegraphy, and engineering.

The modeling process applies physical laws—conservation of energy, momentum, and charge—combined with constitutive relationships such as Hooke's Law and Ohm's Law to derive differential equations that govern system behavior.

The mass-spring-damper paradigm demonstrates that the equation $m\ddot{x} + b\dot{x} + kx = F(t)$ appears across mechanical, electrical, thermal, and other domains through appropriate analogies, providing a unifying framework for dynamic systems analysis.

The **standard form** for second-order systems introduces the natural frequency ω_n and damping ratio ζ , which completely characterize the dynamic response of a system.

System order equals the highest derivative in the differential equation, which corresponds to the number of independent energy storage elements in the physical system.

Finally, **experimental validation** remains essential: physical models must be validated against real measurements, and discrepancies between predicted and observed behavior reveal unmodeled dynamics or parameter uncertainty.

2.7 Exercises

Exercise 2.1. A mass $m = 0.5$ kg is suspended from a spring with stiffness $k = 50$ N/m and viscous damper with coefficient $b = 2$ N·s/m.

- Derive the equation of motion.
- Find the natural frequency and damping ratio.
- Classify the response as overdamped, critically damped, or underdamped.
- If the mass is displaced 5 cm and released, estimate the time for oscillations to decay to 1 cm amplitude.

Exercise 2.2. For the series RLC circuit with $R = 100$ Ω , $L = 0.1$ H, and $C = 10$ μ F:

- Derive the differential equation for capacitor voltage.
- Find the natural frequency and damping ratio.
- Sketch the expected response to a step input voltage.

Exercise 2.3. A thermal system consists of a heated metal block (mass 1 kg, specific heat 500 J/(kg·K)) in an insulated enclosure with a small air gap providing thermal resistance $R_{th} = 2$ K/W to ambient.

- Derive the differential equation relating block temperature to heater power.
- Find the thermal time constant.
- If the heater provides 50 W and ambient is 20°C, what is the steady-state block temperature?

Exercise 2.4. Design an experiment to measure the moment of inertia of a flywheel using only a stopwatch and known applied torque. Derive the relevant equations and describe the procedure.

Exercise 2.5. For the tank level system of Example 4, the tank has cross-sectional area $A = 1$ m², and water exits through an orifice with effective resistance such that $q_{out} = 0.1\sqrt{h}$ m³/s (nonlinear).

- Linearize the outlet flow relationship about an operating point $h_0 = 1$ m.
- Find the linearized time constant.
- Compare the linear and nonlinear responses to a small step change in inlet flow.

Exercise 2.6. Using the Lagrangian method, derive the equations of motion for a double pendulum (two point masses connected by two massless rods). You need not solve the resulting equations.

Exercise 2.7. A DC motor has the following parameters: $R_a = 2$ Ω , $L_a = 5$ mH, $K = 0.1$ V·s/rad = 0.1 N·m/A, $J = 0.01$ kg·m², $b = 0.001$ N·m·s/rad.

- Write the state-space model with states $[i_a, \omega]^T$.

- (b) Find the transfer function from voltage to angular velocity.
- (c) Determine if the electrical time constant is small enough to justify the first-order approximation.

Exercise 2.8. A vehicle suspension can be modeled as a quarter-car model with sprung mass $m_s = 300$ kg (body), unsprung mass $m_u = 50$ kg (wheel/tire assembly), suspension spring $k_s = 20000$ N/m, tire stiffness $k_t = 200000$ N/m, and suspension damper $b_s = 1500$ N·s/m.

- (a) Draw a free body diagram for both masses.
- (b) Derive the equations of motion for this two-degree-of-freedom system.
- (c) Express the system in standard second-order form and identify the natural frequencies.

Exercise 2.9. An electrical heating element has resistance $R = 50 \Omega$ and is used to heat water in an insulated tank. The water has mass 10 kg and specific heat 4186 J/(kg·K). The tank loses heat to the environment through walls with thermal resistance $R_{th} = 0.5$ K/W.

- (a) Derive the differential equation relating water temperature to the voltage applied to the heater.
- (b) Find the thermal time constant of the system.
- (c) If the ambient temperature is 20°C and a constant 120 V is applied, what is the steady-state water temperature?

Exercise 2.10. A pendulum clock uses a pendulum of length $L = 1$ m with a bob of mass $m = 0.5$ kg. Air resistance provides viscous damping with coefficient $b = 0.01$ N·s/m.

- (a) Derive the linearized equation of motion for small oscillations.
- (b) Calculate the natural frequency and period of oscillation.
- (c) Estimate how many oscillations occur before the amplitude decays to 37% ($1/e$) of its initial value.
- (d) If the pendulum is shortened by 1 mm, by how much does the clock gain or lose per day?

2.8 Further Reading

For deeper exploration of the topics in this chapter, several classic texts are recommended. Cannon's *Dynamics of Physical Systems* (McGraw-Hill, 1967) provides excellent coverage of deriving equations of motion from physical principles. Shearer, Murphy, and Richardson's *Introduction to System Dynamics* (Addison-Wesley, 1967) offers an outstanding treatment of analogies between physical domains. Ogata's *System Dynamics*, 4th ed. (Pearson, 2003), provides comprehensive coverage with many worked examples suitable for self-study. For historical interest, Heaviside's original *Electromagnetic Theory* (1893) presents the source material for operational calculus, though it is mathematically challenging by modern standards.

Chapter 3

Transfer Functions and Block Diagrams

“Mathematics is an experimental science, and definitions do not come first, but later on.”

— Oliver Heaviside (1850–1925)

The analysis of dynamic systems underwent a profound transformation in the late nineteenth and early twentieth centuries. What began as a pragmatic need to solve differential equations for electrical networks evolved into one of the most powerful abstractions in engineering: the transfer function. This chapter traces the historical development of these ideas, establishes the rigorous mathematical foundation, and demonstrates their application through both theoretical analysis and practical experimentation.

3.1 Historical Motivation: From Telegraph Cables to Modern Control

3.1.1 The Transatlantic Cable Problem

In the summer of 1866, the successful laying of the first permanent transatlantic telegraph cable represented one of the greatest engineering achievements of the Victorian era. Yet the triumph was accompanied by a profound technical mystery: signals transmitted across the 2,000-mile cable emerged distorted, delayed, and barely recognizable. The sharp “dots” and “dashes” of Morse code spread into overlapping pulses, limiting transmission to a painfully slow 8 words per minute.

The mathematical description of signal propagation along submarine cables had been established by William Thomson (later Lord Kelvin) in 1855. The *telegrapher’s equations* that governed voltage $V(x, t)$ and current $I(x, t)$ along the cable took the form of coupled partial differential equations:

$$\frac{\partial V}{\partial x} = -L \frac{\partial I}{\partial t} - RI \tag{3.1}$$

$$\frac{\partial I}{\partial x} = -C \frac{\partial V}{\partial t} - GV \tag{3.2}$$

where R , L , C , and G represent the resistance, inductance, capacitance, and conductance per unit length, respectively.

Solving these equations for practical cable designs required considerable mathematical sophistication. Engineers needed solutions for various input signals and cable parameters, and each new configuration demanded fresh calculations. The situation called for a fundamentally new approach.

3.1.2 Oliver Heaviside and Operational Calculus

Oliver Heaviside: The Maverick Engineer

Oliver Heaviside (1850–1925) was one of the most remarkable figures in the history of engineering mathematics. Largely self-taught and working in isolation after leaving his position as a telegraph operator due to increasing deafness, Heaviside developed mathematical techniques that were decades ahead of their time. His operational calculus initially faced fierce opposition from mathematicians who objected to his lack of rigorous foundations. When challenged about the validity of his methods, Heaviside famously replied: “Shall I refuse my dinner because I do not fully understand the process of digestion?” His work also included the reformulation of Maxwell’s equations into their modern vector form, the prediction of the Kennelly–Heaviside layer in the atmosphere, and the concept of impedance. Despite his profound contributions, Heaviside died in poverty, largely unrecognized during his lifetime.

Enter Oliver Heaviside (1850–1925), a self-taught English mathematician and electrical engineer whose unconventional methods would revolutionize the analysis of electrical systems. Working in isolation after leaving his position as a telegraph operator, Heaviside developed what he called *operational calculus*—a symbolic method that treated the differential operator d/dt as an algebraic quantity.

Heaviside introduced the operator $p = d/dt$ and manipulated it according to algebraic rules. For instance, to solve the differential equation

$$L \frac{di}{dt} + Ri = v(t) \quad (3.3)$$

Heaviside would write $(Lp + R)i = v$, and “solve” for i as

$$i = \frac{1}{Lp + R}v = \frac{1}{R} \cdot \frac{1}{1 + (L/R)p}v \quad (3.4)$$

He then expanded this in a series or used tables to find the time-domain solution. The expression $1/(Lp + R)$ is recognizable today as what we call a *transfer function*.

Controversy and Vindication

Heaviside’s methods were controversial. Prominent mathematicians objected that his manipulations lacked rigorous justification. The operational calculus produced correct answers, but *why* it worked remained unclear. Heaviside himself was unapologetic, famously retorting: “Shall I refuse my dinner because I do not fully understand the process of digestion?”

The rigorous foundation came later, through the work of Thomas John I'Anson Bromwich (1875–1929) and others who connected Heaviside's operators to the contour integrals of complex analysis. The crucial link was the *Laplace transform*, a mathematical tool that had been developed nearly a century earlier but whose engineering applications were only then becoming apparent.

3.1.3 The Earlier Mathematical Foundations

The Laplace transform takes its name from Pierre-Simon Laplace (1749–1827), who used similar integral transforms in his work on probability theory and celestial mechanics. The specific form we use today:

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt \quad (3.5)$$

converts functions of time into functions of a complex variable s .

Joseph Fourier (1768–1830) had developed related transform methods for studying heat conduction. Fourier's transform, using $j\omega$ instead of s , remains essential for frequency-domain analysis. The Laplace transform can be viewed as a generalization that handles a broader class of functions, including those with exponential growth.

The connection between these mathematical tools and Heaviside's operational methods became clear in the early twentieth century: Heaviside's operator p corresponds to the complex variable s in the Laplace transform. His intuitive manipulations had been, in effect, transform-domain algebra conducted without explicit acknowledgment of the underlying integral transformation.

3.1.4 The Telephone Era: Systematizing Transfer Functions

The development of telephone networks in the 1920s and 1930s brought new challenges that accelerated the formalization of transfer function methods. Engineers at Bell Telephone Laboratories, including Hendrik Bode, Harry Nyquist, and others, needed to design amplifiers, filters, and feedback systems for long-distance telephony.

These engineers faced the problem of *interconnection*: how to predict the behavior of complex systems built from simpler components. The transfer function provided the answer. By characterizing each component by its transfer function $G(s)$, engineers could analyze cascaded systems by multiplying transfer functions, parallel systems by adding them, and feedback systems by applying simple algebraic formulas.

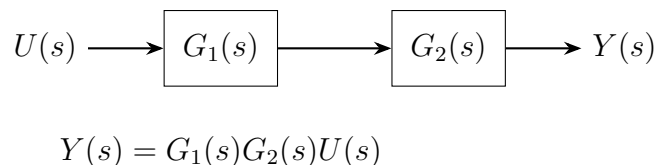


Figure 3.1: The cascade connection of two systems. The overall transfer function is simply the product of the individual transfer functions—an elegant result that enables modular system design.

3.1.5 The Power of Abstraction

The transfer function embodies a fundamental engineering principle: *abstraction*. By representing a system as a transfer function, we treat it as a “black box” characterized solely by its input-output relationship. The internal details—whether the system is electrical, mechanical, thermal, or hybrid—become irrelevant for interconnection analysis.

This abstraction enables several powerful capabilities. First, it facilitates modular design, allowing systems to be designed independently and combined predictably. Second, it provides analysis simplification by transforming differential equations into algebraic equations. Third, setting $s = j\omega$ yields frequency-domain insight, revealing the steady-state frequency response directly. Finally, pole locations determine stability without requiring explicit solution of the time-domain equations, greatly simplifying stability analysis.

The journey from Heaviside’s pragmatic operators to the modern transfer function illustrates how engineering necessity drives mathematical development. Today, transfer functions form the foundation of control system analysis, and the block diagram has become the universal language for describing interconnected dynamic systems.

3.2 Modern Applications

The transfer function concept, born from nineteenth-century telegraph engineering, has become indispensable across virtually every domain of modern technology. We highlight several representative applications.

3.2.1 Audio Processing and Equalization

Digital and analog audio systems rely extensively on transfer function design. An audio equalizer, whether in a music production studio or a smartphone, consists of filters characterized by their transfer functions. A parametric equalizer band, for example, might implement a second-order transfer function:

$$H(s) = \frac{s^2 + (A/Q)\omega_0 s + \omega_0^2}{s^2 + (1/Q)\omega_0 s + \omega_0^2} \quad (3.6)$$

where ω_0 is the center frequency, Q is the quality factor, and A determines the boost or cut amount.

Professional audio software displays these transfer functions as frequency response curves, allowing engineers to shape sound precisely. The underlying mathematics—pole-zero placement and frequency response calculation—remains exactly as developed for control systems.

3.2.2 Communication Systems

Every communication channel—wireless, fiber optic, or copper—can be characterized by a transfer function that describes how it affects transmitted signals. The channel transfer function $H(f)$ captures three essential characteristics: frequency-dependent attenuation through its magnitude response, signal delay and dispersion through its phase response, and the inherent bandwidth limitations of the physical medium.

Modern communication systems use *equalization*—the design of inverse transfer functions—to compensate for channel distortion. The principles of cascade systems (Section 3.3.7) apply directly: if the channel has transfer function $H_c(s)$ and the equalizer has $H_e(s)$, the cascade $H_e(s)H_c(s)$ should approximate unity for distortion-free transmission.

3.2.3 Mechanical Vibration Analysis

The mass-spring-damper system we derive in Section 3.3.5 represents the fundamental model for mechanical vibration. Transfer functions enable engineers to predict resonance frequencies and amplitudes, design vibration isolators for sensitive equipment, analyze vehicle suspension systems, and study structural dynamics in buildings and bridges.

The same mathematical framework applies whether analyzing a microscopic MEMS accelerometer or the vibration modes of a skyscraper.

3.2.4 Control System Design Software

Modern engineering relies heavily on computational tools that implement transfer function methods. MATLAB's Control System Toolbox and Simulink provide transfer function representation and manipulation, automated block diagram reduction, pole-zero analysis and plotting, frequency response computation including Bode plots and Nyquist diagrams, as well as time response simulation.

Python's control systems library (`python-control`) offers similar capabilities in an open-source environment. These tools automate the calculations developed in this chapter, but understanding the underlying theory remains essential for effective use.

3.3 Mathematical Foundation

We now develop the rigorous mathematical framework for transfer function analysis, beginning with the Laplace transform and progressing through transfer function derivation, pole-zero analysis, and block diagram algebra.

3.3.1 The Laplace Transform

Definition 3.1 (Laplace Transform). The *Laplace transform* of a function $f(t)$ defined for $t \geq 0$ is

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t)e^{-st} dt \quad (3.7)$$

where $s = \sigma + j\omega$ is a complex variable. The transform exists for $\text{Re}(s) > \sigma_0$, where σ_0 depends on the growth rate of $f(t)$.

The inverse Laplace transform recovers $f(t)$ from $F(s)$:

$$\mathcal{L}^{-1}\{F(s)\} = f(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} F(s)e^{st} ds \quad (3.8)$$

where $c > \sigma_0$. In practice, we rarely evaluate this integral directly, instead using transform tables and partial fraction expansion.

3.3.2 Key Laplace Transform Pairs

We derive the transforms of the most important functions for control systems analysis.

Unit Step Function

The unit step function is defined as:

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases} \quad (3.9)$$

Applying the definition:

$$\begin{aligned} \mathcal{L}\{u(t)\} &= \int_0^{\infty} 1 \cdot e^{-st} dt = \left[-\frac{1}{s} e^{-st} \right]_0^{\infty} \\ &= 0 - \left(-\frac{1}{s} \right) = \frac{1}{s} \quad \text{for } \operatorname{Re}(s) > 0 \end{aligned} \quad (3.10)$$

Ramp Function

For $f(t) = t \cdot u(t)$:

$$\mathcal{L}\{t\} = \int_0^{\infty} t e^{-st} dt \quad (3.11)$$

Using integration by parts with $u = t$ and $dv = e^{-st} dt$:

$$\begin{aligned} \mathcal{L}\{t\} &= \left[-\frac{t}{s} e^{-st} \right]_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-st} dt \\ &= 0 + \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2} \end{aligned} \quad (3.12)$$

Exponential Function

For $f(t) = e^{-at} u(t)$:

$$\begin{aligned} \mathcal{L}\{e^{-at}\} &= \int_0^{\infty} e^{-at} e^{-st} dt = \int_0^{\infty} e^{-(s+a)t} dt \\ &= \left[-\frac{1}{s+a} e^{-(s+a)t} \right]_0^{\infty} = \frac{1}{s+a} \end{aligned} \quad (3.13)$$

valid for $\operatorname{Re}(s) > -a$.

Sinusoidal Functions

Using Euler's formula, $\cos(\omega t) = \frac{1}{2}(e^{j\omega t} + e^{-j\omega t})$:

$$\begin{aligned}\mathcal{L}\{\cos(\omega t)\} &= \frac{1}{2} \left(\frac{1}{s - j\omega} + \frac{1}{s + j\omega} \right) \\ &= \frac{1}{2} \cdot \frac{2s}{s^2 + \omega^2} = \frac{s}{s^2 + \omega^2}\end{aligned}\tag{3.14}$$

Similarly:

$$\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2}\tag{3.15}$$

3.3.3 Essential Transform Properties

Property 3.1 (Linearity). *For constants a and b :*

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)\tag{3.16}$$

Property 3.2 (Differentiation in Time).

$$\mathcal{L}\left\{\frac{df}{dt}\right\} = sF(s) - f(0^-)\tag{3.17}$$

where $f(0^-)$ denotes the initial value just before $t = 0$.

Proof. Using integration by parts with the definition (??):

$$\begin{aligned}\mathcal{L}\left\{\frac{df}{dt}\right\} &= \int_0^\infty \frac{df}{dt} e^{-st} dt \\ &= [f(t)e^{-st}]_0^\infty + s \int_0^\infty f(t)e^{-st} dt \\ &= 0 - f(0^-) + sF(s) = sF(s) - f(0^-)\end{aligned}\tag{3.18}$$

assuming $f(t)$ grows slower than $e^{\sigma t}$ for some σ . □

Applying this property twice:

$$\mathcal{L}\left\{\frac{d^2f}{dt^2}\right\} = s^2F(s) - sf(0^-) - \dot{f}(0^-)\tag{3.19}$$

Property 3.3 (Integration in Time).

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}\tag{3.20}$$

Property 3.4 (Frequency Shift).

$$\mathcal{L}\{e^{-at}f(t)\} = F(s + a)\tag{3.21}$$

Property 3.5 (Time Delay).

$$\mathcal{L}\{f(t - \tau)u(t - \tau)\} = e^{-\tau s} F(s) \quad (3.22)$$

Property 3.6 (Final Value Theorem). *If $\lim_{t \rightarrow \infty} f(t)$ exists and is finite:*

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s) \quad (3.23)$$

Property 3.7 (Initial Value Theorem).

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s) \quad (3.24)$$

Table 3.1: Common Laplace Transform Pairs

Time Function $f(t)$, $t \geq 0$	Laplace Transform $F(s)$
$\delta(t)$ (unit impulse)	1
$u(t)$ (unit step)	$\frac{1}{s}$
t (ramp)	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$
e^{-at}	$\frac{1}{s + a}$
te^{-at}	$\frac{1}{(s + a)^2}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{-at} \sin(\omega t)$	$\frac{\omega}{(s + a)^2 + \omega^2}$
$e^{-at} \cos(\omega t)$	$\frac{s + a}{(s + a)^2 + \omega^2}$

3.3.4 The Transfer Function

Definition 3.2 (Transfer Function). The *transfer function* of a linear time-invariant (LTI) system is the ratio of the Laplace transform of the output to the Laplace transform of the input, assuming all initial conditions are zero:

$$G(s) = \left. \frac{Y(s)}{U(s)} \right|_{\text{zero ICs}} \quad (3.25)$$

The transfer function completely characterizes the input-output behavior of an LTI system. Given any input $U(s)$, the output is simply:

$$Y(s) = G(s)U(s) \quad (3.26)$$

3.3.5 Derivation: Mass-Spring-Damper System

Consider the canonical mechanical system shown in Figure 3.2: a mass m connected to a spring with stiffness k and a damper with coefficient b . An external force $f(t)$ is applied, and we seek the displacement $x(t)$.

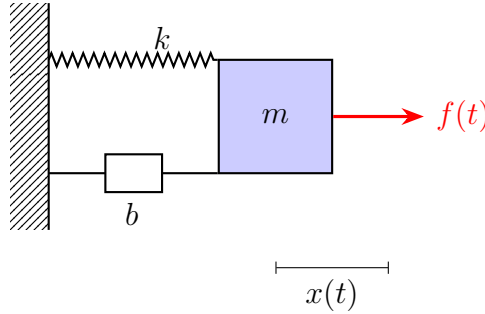


Figure 3.2: Mass-spring-damper system. The displacement $x(t)$ is measured from the equilibrium position.

Newton's second law gives:

$$m\ddot{x} + b\dot{x} + kx = f(t) \quad (3.27)$$

Taking the Laplace transform with zero initial conditions ($x(0) = \dot{x}(0) = 0$):

$$m[s^2 X(s)] + b[sX(s)] + k[X(s)] = F(s) \quad (3.28)$$

Factoring out $X(s)$:

$$(ms^2 + bs + k)X(s) = F(s) \quad (3.29)$$

The transfer function from force to displacement is:

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k} = \frac{1/m}{s^2 + (b/m)s + k/m} \quad (3.30)$$

Introducing the standard second-order parameters:

$$\omega_n = \sqrt{\frac{k}{m}} \quad (\text{natural frequency}), \quad \zeta = \frac{b}{2\sqrt{km}} \quad (\text{damping ratio}) \quad (3.31)$$

we can write:

$$G(s) = \frac{\omega_n^2/k}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (3.32)$$

3.3.6 Poles and Zeros

A transfer function can be written in factored form:

$$G(s) = K \frac{(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)} = K \frac{\prod_{i=1}^m (s - z_i)}{\prod_{j=1}^n (s - p_j)} \quad (3.33)$$

Definition 3.3 (Poles and Zeros). The *zeros* $z_1, z_2, \dots, z_m$ are the roots of the numerator polynomial; they are values of s where $G(s) = 0$. The *poles* $p_1, p_2, \dots, p_n$ are the roots of the denominator polynomial; they are values of s where $G(s) \rightarrow \infty$.

Physical Meaning of Poles

The poles of a system determine its natural response—the behavior when excited by an impulse or released from initial conditions. Each pole p_i contributes a term $e^{p_i t}$ to the time response.

Pole Type	Time Response Contribution	Stability
Real negative ($p = -a, a > 0$)	e^{-at} (decaying exponential)	Stable
Real positive ($p = a, a > 0$)	e^{at} (growing exponential)	Unstable
Complex conjugate ($p = -\sigma \pm j\omega_d$)	$e^{-\sigma t}[\cos(\omega_d t) + \sin(\omega_d t)]$	Stable if $\sigma > 0$
Imaginary axis ($p = \pm j\omega$)	$\cos(\omega t)$ (sustained oscillation)	Marginally stable

Theorem 3.1 (Stability from Poles). *A linear time-invariant system is **BIBO stable** (bounded input yields bounded output) if and only if all poles have strictly negative real parts, i.e., all poles lie in the open left half of the complex plane.*

Physical Meaning of Zeros

Zeros affect how the system responds to inputs at different frequencies. At a frequency corresponding to a zero, the system output is blocked or attenuated. Furthermore, zeros in the right half-plane cause *non-minimum phase* behavior, which manifests as an initial response in the opposite direction of the final response.

Pole-Zero Plot

The *pole-zero plot* displays poles (marked with $\times$) and zeros (marked with $\circ$) in the complex s -plane. This visualization immediately reveals stability and dominant dynamics.

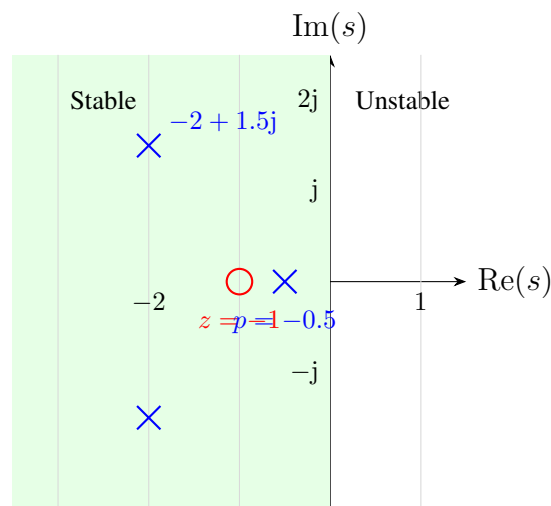


Figure 3.3: Pole-zero plot for $G(s) = \frac{s+1}{(s+0.5)(s^2+4s+6.25)}$. Poles at $s = -0.5$ and $s = -2 \pm 1.5j$ (all in left half-plane: stable). Zero at $s = -1$.

3.3.7 Block Diagram Algebra

Block diagrams provide a graphical representation of system interconnections. The fundamental elements are shown in Figure 3.4.

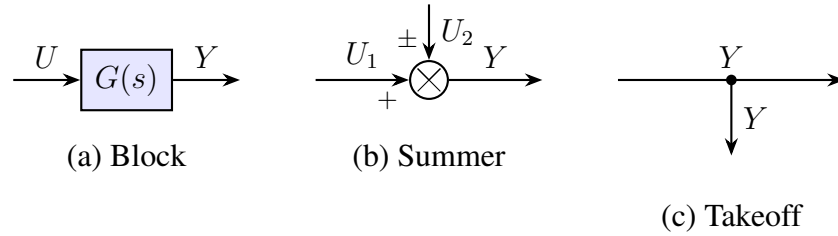


Figure 3.4: Block diagram elements: (a) transfer function block, (b) summing junction, (c) signal takeoff point.

Series (Cascade) Connection

When systems are connected in series:

$$G_{\text{series}}(s) = G_1(s)G_2(s) \quad (3.34)$$

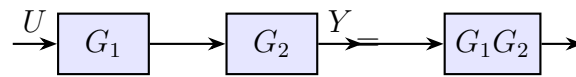


Figure 3.5: Series connection of two blocks.

Parallel Connection

When systems are connected in parallel:

$$G_{\text{parallel}}(s) = G_1(s) + G_2(s) \quad (3.35)$$

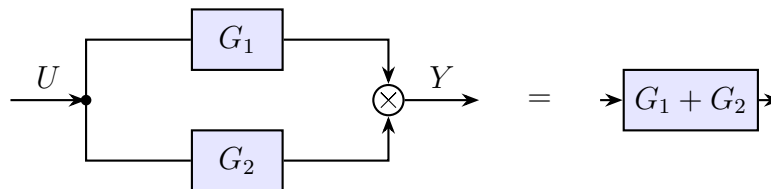


Figure 3.6: Parallel connection of two blocks.

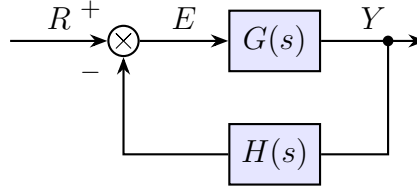


Figure 3.7: Negative feedback configuration. The closed-loop transfer function is $Y/R = G/(1 + GH)$.

Negative Feedback Connection

The fundamental feedback configuration:

$$G_{\text{closed}}(s) = \frac{G(s)}{1 + G(s)H(s)} \quad (3.36)$$

Derivation of Feedback Formula. From the block diagram:

$$E(s) = R(s) - H(s)Y(s) \quad (3.37)$$

$$Y(s) = G(s)E(s) = G(s)[R(s) - H(s)Y(s)] \quad (3.38)$$

Solving for $Y(s)$:

$$Y(s) + G(s)H(s)Y(s) = G(s)R(s) \quad (3.39)$$

$$Y(s)[1 + G(s)H(s)] = G(s)R(s) \quad (3.40)$$

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} \quad (3.41)$$

□

For unity feedback ($H(s) = 1$):

$$G_{\text{closed}}(s) = \frac{G(s)}{1 + G(s)} \quad (3.42)$$

3.3.8 Mason's Gain Formula (Simplified)

For complex block diagrams with multiple loops, *Mason's gain formula* provides a systematic method to find the overall transfer function without repeated block diagram reduction.

For a system with forward paths and feedback loops, the transfer function is:

$$G(s) = \frac{\sum_k P_k \Delta_k}{\Delta} \quad (3.43)$$

where P_k denotes the gain of the k -th forward path, L_i denotes the gain of the i -th individual loop, and $L_i L_j$ represents the product of gains of non-touching loops i and j . The graph determinant is computed as $\Delta = 1 - \sum L_i + \sum L_i L_j - \sum L_i L_j L_k + \dots$, alternating the signs of successively higher-order products of non-touching loops. The cofactor Δ_k is the determinant evaluated with all loops touching the k -th forward path removed.

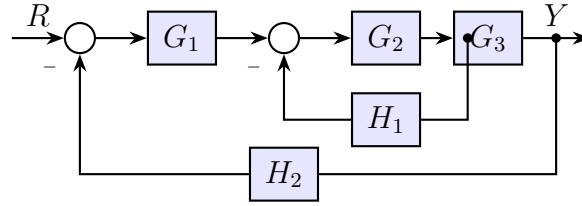


Figure 3.8: Block diagram for Mason's formula example.

Example 3.1 (Mason's Formula Application). Consider the block diagram in Figure 3.8.

Forward path: $P_1 = G_1 G_2 G_3$

Loops: The inner loop has gain $L_1 = -G_2 G_3 H_1$, while the outer loop has gain $L_2 = -G_1 G_2 G_3 H_2$.

These loops touch (share node), so there are no non-touching loop products.

Determinant: $\Delta = 1 - L_1 - L_2 = 1 + G_2 G_3 H_1 + G_1 G_2 G_3 H_2$

Cofactor: $\Delta_1 = 1$ (all loops touch the forward path)

Transfer function:

$$G(s) = \frac{P_1 \Delta_1}{\Delta} = \frac{G_1 G_2 G_3}{1 + G_2 G_3 H_1 + G_1 G_2 G_3 H_2} \quad (3.44)$$

3.4 Practical Experiment: RC Low-Pass Filter

This experiment provides hands-on verification of transfer function theory using a simple RC low-pass filter—the most fundamental dynamic circuit.

3.4.1 Theory and Transfer Function Derivation

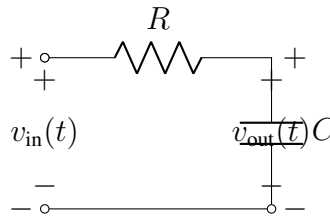


Figure 3.9: RC low-pass filter circuit. The output voltage is taken across the capacitor.

For the RC circuit in Figure 13.7, Kirchhoff's voltage law gives:

$$v_{in}(t) = Ri(t) + v_{out}(t) \quad (3.45)$$

The capacitor current-voltage relationship is:

$$i(t) = C \frac{dv_{out}}{dt} \quad (3.46)$$

Substituting:

$$v_{\text{in}} = RC \frac{dv_{\text{out}}}{dt} + v_{\text{out}} \quad (3.47)$$

Taking the Laplace transform with zero initial conditions:

$$V_{\text{in}}(s) = RC \cdot sV_{\text{out}}(s) + V_{\text{out}}(s) = (RCs + 1)V_{\text{out}}(s) \quad (3.48)$$

The transfer function is:

$$H(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = \frac{1}{RCs + 1} = \frac{1}{\tau s + 1} \quad (3.49)$$

where $\tau = RC$ is the time constant.

Frequency Response

Substituting $s = j\omega$ yields the frequency response:

$$H(j\omega) = \frac{1}{1 + j\omega RC} = \frac{1}{1 + j(\omega/\omega_c)} \quad (3.50)$$

where the *cutoff frequency* is:

$$\omega_c = \frac{1}{RC} \quad \text{or} \quad f_c = \frac{1}{2\pi RC} \quad (3.51)$$

The magnitude and phase are:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_c)^2}} \quad (3.52)$$

$$\angle H(j\omega) = -\arctan(\omega/\omega_c) \quad (3.53)$$

At $\omega = \omega_c$: $|H| = 1/\sqrt{2} \approx 0.707$ (−3 dB) and $\angle H = -45^\circ$.

3.4.2 Hardware Implementation

Required Components

Component	Specification
Resistor	$R = 10 \text{ k}\Omega$
Capacitor	$C = 100 \text{ nF}$ (0.1 μF)
Prototyping	Breadboard and jumper wires
Microcontroller	Arduino Uno (or similar)
Alternative equipment	Function generator and oscilloscope

With these values, the theoretical cutoff frequency is:

$$f_c = \frac{1}{2\pi \times 10000 \times 100 \times 10^{-9}} = \frac{1}{2\pi \times 0.001} \approx 159.2 \text{ Hz} \quad (3.54)$$

Algorithm 5 RC Filter Frequency Response Measurement (Arduino)

-
- 1: **Circuit Assembly:** Connect RC filter with PWM output (pin 9) as input and analog input (A0) measuring v_{out}
 - 2: **Initialize:** Set frequency range from 10 Hz to 10 kHz
 - 3: **for** each test frequency f in range **do**
 - 4: Generate square wave at frequency f using `tone()` or timer manipulation
 - 5: Measure peak-to-peak amplitude of output $V_{\text{out,pp}}$
 - 6: Measure peak-to-peak amplitude of input $V_{\text{in,pp}}$
 - 7: Calculate gain: $|H(f)| \leftarrow V_{\text{out,pp}}/V_{\text{in,pp}}$
 - 8: Record frequency and gain values
 - 9: **end for**
 - 10: **Find Cutoff:** Identify frequency where $|H(f)| = 0.707$ (-3 dB)
-

Arduino-Based Measurement Procedure**Function Generator Method (Preferred)****Algorithm 6** RC Filter Frequency Response Measurement (Function Generator)

-
- 1: Apply sinusoidal input from function generator at 1 V_{pp}
 - 2: Define frequency sweep points: 10, 20, 50, 100, 159, 200, 500, 1000, 2000, 5000 Hz
 - 3: **for** each frequency f in sweep points **do**
 - 4: Set function generator to frequency f
 - 5: Measure output amplitude V_{out} on oscilloscope
 - 6: Measure phase shift ϕ between input and output
 - 7: Calculate gain in dB: $|H|_{\text{dB}} \leftarrow 20 \log_{10}(V_{\text{out}}/V_{\text{in}})$
 - 8: **end for**
 - 9: Plot Bode diagram: magnitude (dB) vs. $\log(f)$
-

3.4.3 Expected Results and Analysis

Compare measured values to theory. Typical discrepancies arise from component tolerances (resistors typically $\pm 5\%$, capacitors $\pm 10\%$), oscilloscope input capacitance which effectively adds to C , and function generator output impedance which effectively adds to R .

3.4.4 Alternative: Computational Simulation

For those without laboratory equipment, the following algorithm demonstrates transfer function concepts through simulation.

Running this code verifies that cascading $G_1(s) = 2/(s + 1)$ and $G_2(s) = 3/(s + 2)$ yields:

$$G(s) = G_1(s)G_2(s) = \frac{6}{(s + 1)(s + 2)} = \frac{6}{s^2 + 3s + 2} \quad (3.55)$$

Algorithm 7 Cascaded First-Order System Analysis

```

1: Define transfer functions:
2:  $G_1(s) \leftarrow 2/(s + 1)$  with numerator [2] and denominator [1, 1]
3:  $G_2(s) \leftarrow 3/(s + 2)$  with numerator [3] and denominator [1, 2]
4:
5: Compute cascade transfer function:
6: num_cascade  $\leftarrow$  convolve([2], [3]) = [6]
7: den_cascade  $\leftarrow$  convolve([1, 1], [1, 2]) = [1, 3, 2]
8:  $G_{\text{cascade}}(s) \leftarrow 6/(s^2 + 3s + 2)$ 
9:
10: Compute step responses:
11: Create time vector  $t$  from 0 to 10 seconds with 1000 points
12: for each system  $G \in \{G_1, G_2, G_{\text{cascade}}\}$  do
13:   Compute step response  $y(t)$  using inverse Laplace transform
14:   Store response for plotting
15: end for
16:
17: Compute Bode magnitude plot:
18: Create frequency vector  $\omega$  from  $10^{-2}$  to  $10^2$  rad/s (log-spaced)
19: for each frequency  $\omega_i$  do
20:   Compute  $|G_{\text{cascade}}(j\omega_i)|$ 
21:   Convert to decibels:  $\text{mag}_{\text{dB}} \leftarrow 20 \log_{10} |G|$ 
22: end for
23:
24: Generate plots: Step response comparison and Bode magnitude
25: return Step response curves and frequency response data

```

Table 3.2: Theoretical frequency response for RC filter ($f_c = 159.2$ Hz)

Frequency (Hz)	ω/ω_c	Gain $ H $	Gain (dB)
10	0.063	0.998	-0.02
50	0.314	0.954	-0.41
100	0.628	0.847	-1.44
159	1.000	0.707	-3.01
200	1.257	0.622	-4.12
500	3.142	0.303	-10.4
1000	6.283	0.157	-16.1
5000	31.42	0.032	-30.0

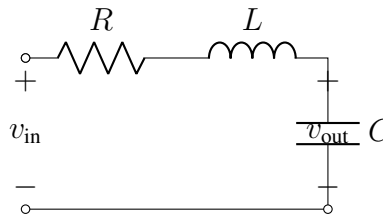
The poles of the cascade system are at $s = -1$ and $s = -2$, the combination of both individual system poles.

3.5 Example Problems

RLC Circuit Analysis

The series RLC circuit is a fundamental second-order system that appears throughout electrical engineering. By analyzing this circuit using transfer functions, we can predict its frequency response, identify resonance conditions, and understand how the component values affect system behavior. This example demonstrates how impedance methods simplify circuit analysis in the Laplace domain.

Example 3.2 (Transfer Function of an RLC Circuit). Find the transfer function $H(s) = V_{\text{out}}(s)/V_{\text{in}}(s)$ for the series RLC circuit where the output is taken across the capacitor.



Solution:

Using voltage divider with impedances:

$$H(s) = \frac{Z_C}{Z_R + Z_L + Z_C} = \frac{1/(Cs)}{R + Ls + 1/(Cs)} \quad (3.56)$$

Multiplying numerator and denominator by Cs :

$$H(s) = \frac{1}{RCs + LCs^2 + 1} = \frac{1}{LCs^2 + RCs + 1} \quad (3.57)$$

Dividing by LC :

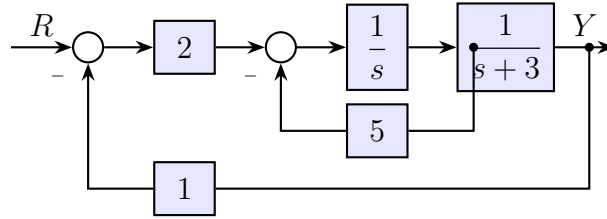
$$H(s) = \frac{1/LC}{s^2 + (R/L)s + 1/LC} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (3.58)$$

where $\omega_n = 1/\sqrt{LC}$ and $\zeta = \frac{R}{2} \sqrt{C/L}$.

Systematic Block Diagram Reduction

Complex control systems often consist of multiple interconnected subsystems. Block diagram reduction provides a systematic method to find the overall transfer function by successively simplifying series, parallel, and feedback connections. This example illustrates the step-by-step process of reducing a multi-loop system to a single transfer function.

Example 3.3 (Block Diagram Reduction). Reduce the block diagram shown below to a single transfer function.



Solution:

Step 1: Identify the inner loop (around $G_2 = 1/s$ with feedback $H_1 = 5$):

$$G_{\text{inner}} = \frac{G_2}{1 + G_2 H_1} = \frac{1/s}{1 + 5/s} = \frac{1/s}{(s+5)/s} = \frac{1}{s+5} \quad (3.59)$$

Step 2: Combine in series with $G_1 = 2$ and $G_3 = 1/(s+3)$:

$$G_{\text{forward}} = G_1 \cdot G_{\text{inner}} \cdot G_3 = 2 \cdot \frac{1}{s+5} \cdot \frac{1}{s+3} = \frac{2}{(s+3)(s+5)} \quad (3.60)$$

Step 3: Close the outer loop with $H_2 = 1$:

$$G(s) = \frac{G_{\text{forward}}}{1 + G_{\text{forward}} H_2} = \frac{\frac{2}{(s+3)(s+5)}}{1 + \frac{2}{(s+3)(s+5)}} \quad (3.61)$$

$$G(s) = \frac{2}{(s+3)(s+5) + 2} = \frac{2}{s^2 + 8s + 17} \quad (3.62)$$

Example 3.4 (Pole-Zero Analysis). For the transfer function

$$G(s) = \frac{5(s+2)}{(s+1)(s^2+4s+8)} \quad (3.63)$$

find the poles and zeros, sketch the pole-zero plot, and determine if the system is stable.

Solution:

Zeros: Set numerator = 0: $s+2=0 \Rightarrow z_1 = -2$

Poles: Setting the denominator equal to zero yields two factors. From $(s+1)=0$, we obtain $p_1 = -1$. From $(s^2+4s+8)=0$, applying the quadratic formula gives:

$$s = \frac{-4 \pm \sqrt{16-32}}{2} = \frac{-4 \pm \sqrt{-16}}{2} = -2 \pm 2j \quad (3.64)$$

Thus $p_2 = -2 + 2j$ and $p_3 = -2 - 2j$.

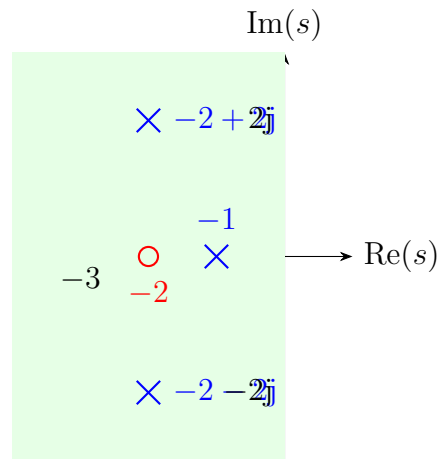


Figure 3.10: Pole-zero plot for Example 3.4.

Stability: All poles have negative real parts (lie in the left half-plane), so the system is **stable**.

Example 3.5 (Cascading Two First-Order Systems). Two first-order systems with transfer functions

$$G_1(s) = \frac{3}{s+2}, \quad G_2(s) = \frac{4}{s+5} \quad (3.65)$$

are connected in series. Find the combined transfer function and the system poles.

Solution:

For series connection:

$$G(s) = G_1(s) \cdot G_2(s) = \frac{3}{s+2} \cdot \frac{4}{s+5} = \frac{12}{(s+2)(s+5)} \quad (3.66)$$

Expanding:

$$G(s) = \frac{12}{s^2 + 7s + 10} \quad (3.67)$$

Poles: $p_1 = -2$ (from G_1) and $p_2 = -5$ (from G_2)

The cascade system inherits the poles of both subsystems. The DC gain is:

$$G(0) = \frac{12}{10} = 1.2 \quad (3.68)$$

Example 3.6 (Inverse Laplace Transform and Time Response). A system with transfer function

$$G(s) = \frac{10}{(s+1)(s+4)} \quad (3.69)$$

is subjected to a unit step input. Find the output $y(t)$.

Solution:

The output in the s -domain is:

$$Y(s) = G(s)U(s) = \frac{10}{(s+1)(s+4)} \cdot \frac{1}{s} = \frac{10}{s(s+1)(s+4)} \quad (3.70)$$

Partial fraction expansion:

$$\frac{10}{s(s+1)(s+4)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+4} \quad (3.71)$$

Multiplying both sides by $s(s+1)(s+4)$:

$$10 = A(s+1)(s+4) + Bs(s+4) + Cs(s+1) \quad (3.72)$$

Finding coefficients: Substituting $s = 0$ gives $10 = A(1)(4)$, so $A = 10/4 = 2.5$. Substituting $s = -1$ gives $10 = B(-1)(3)$, so $B = -10/3$. Substituting $s = -4$ gives $10 = C(-4)(-3)$, so $C = 10/12 = 5/6$.

Therefore:

$$Y(s) = \frac{2.5}{s} - \frac{10/3}{s+1} + \frac{5/6}{s+4} \quad (3.73)$$

Inverse Laplace transform:

$$y(t) = 2.5 - \frac{10}{3}e^{-t} + \frac{5}{6}e^{-4t}, \quad t \geq 0 \quad (3.74)$$

Verification: The initial value is $y(0) = 2.5 - 10/3 + 5/6 = 2.5 - 3.33 + 0.83 = 0$, which confirms correct behavior. The final value is $y(\infty) = 2.5$, which matches $\lim_{s \rightarrow 0} sY(s) = 10/4 = 2.5$, verifying consistency with the final value theorem.

3.6 Summary

This chapter has established the mathematical framework for analyzing linear time-invariant systems through transfer functions and block diagrams.

Historically, transfer functions evolved from Heaviside's operational calculus in the 1880s through the formalization efforts of telephone engineers in the 1920s and 1930s, providing a powerful abstraction for system analysis. The mathematical foundation rests on the Laplace transform,

$F(s) = \int_0^\infty f(t)e^{-st} dt$, which converts differential equations to algebraic equations in the complex variable s .

The transfer function, defined as the ratio $G(s) = Y(s)/U(s)$ with zero initial conditions, completely characterizes an LTI system's input-output behavior. The poles and zeros of this rational function carry physical significance: poles determine natural response modes and stability, while zeros shape the frequency response. For BIBO stability, all poles must have negative real parts.

Block diagram algebra provides systematic rules for interconnecting systems. For series connections, the overall transfer function is the product $G = G_1G_2$. For parallel connections, the transfer functions add: $G = G_1 + G_2$. For negative feedback configurations, the closed-loop transfer function is $G = G_{\text{fwd}}/(1 + G_{\text{fwd}}H)$.

Finally, transfer function predictions can be validated experimentally using simple circuits like RC filters, comparing measured frequency response to theoretical calculations. This practical verification reinforces the connection between mathematical abstraction and physical reality.

The transfer function representation enables modular system design, frequency-domain analysis, and stability assessment—capabilities that form the foundation for the control system design methods developed in subsequent chapters.

3.7 Exercises

Exercise 3.1. Find the Laplace transform of $f(t) = t^2 e^{-3t}$ using transform properties.

Exercise 3.2. Derive the transfer function for a mass-spring-damper system where the mass $m = 2$ kg is connected to a spring with stiffness $k = 18$ N/m and a damper with coefficient $b = 6$ N·s/m. The input is the applied force $F(t)$ and the output is the displacement $x(t)$.

Exercise 3.3. For $G(s) = \frac{s+3}{s^2+5s+6}$, find the poles and zeros, and determine the partial fraction expansion.

Exercise 3.4. Consider a feedback control system with forward path transfer function $G(s) = \frac{K}{s(s+2)(s+5)}$ and unity feedback. Reduce the block diagram to find the closed-loop transfer function, and determine the value of K that places one closed-loop pole at $s = -1$.

Exercise 3.5. An RC high-pass filter has transfer function $H(s) = \frac{RCs}{RCs+1}$. Sketch the Bode magnitude plot and find the cutoff frequency. What is the slope of the magnitude plot at low frequencies?

Exercise 3.6. Using the final value theorem, find the steady-state error for a unity feedback system with $G(s) = \frac{10}{s(s+2)}$ subjected to a unit ramp input.

Exercise 3.7. A system has poles at $s = -1 \pm 2j$ and a zero at $s = -3$. Write the transfer function (with DC gain = 1) and sketch the step response qualitatively, indicating whether overshoot will occur.

Exercise 3.8. Design (choose R and C values for) an RC low-pass filter with cutoff frequency $f_c = 1$ kHz. Verify your design by computing the magnitude response at 100 Hz, 1 kHz, and 10 kHz.

Exercise 3.9. Apply Mason's gain formula to find the transfer function of a system with forward path $G_1 G_2 G_3$, where $G_1 = 2$, $G_2 = \frac{1}{s+1}$, $G_3 = \frac{3}{s+2}$, an inner feedback loop around G_2 with gain $H_1 = 0.5$, and an outer feedback loop around the entire forward path with gain $H_2 = 1$.

Exercise 3.10. For the RLC circuit of Example 3.2, find component values R , L , C that give $\omega_n = 100$ rad/s and $\zeta = 0.5$. Assume $C = 100$ μ F and determine L and R .

Exercise 3.11. A cascade of two first-order systems has overall transfer function $G(s) = \frac{6}{(s+1)(s+3)}$.

- Find the step response $y(t)$ using partial fractions.
- Determine the time constant of the dominant pole.
- Calculate the 10%–90% rise time of the step response.

Exercise 3.12. The transfer function of a notch filter is given by $H(s) = \frac{s^2 + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ where $\omega_n = 120\pi$ rad/s (60 Hz) and $\zeta = 0.1$.

- Find the poles and zeros of this transfer function.
- Sketch the pole-zero plot.
- Explain why this filter is called a “notch” filter based on its frequency response characteristics.

Chapter 4: State-Space Representation Control Systems: A Practical Approach

Chapter Overview

This chapter introduces state-space representation, the modern foundation of control theory. We begin with the historical context that motivated its development, explore its mathematical framework, and conclude with a hands-on laboratory experiment using a two-mass system. By the end of this chapter, you will understand why state-space methods revolutionized control engineering and how to apply them to multi-variable systems.

Learning Objectives: Upon completing this chapter, students will understand the historical development and motivation for state-space methods, and will be able to define state variables and formulate state-space models from physical systems. Students will learn to convert between transfer function and state-space representations, analyze system dynamics using eigenvalues and the state transition matrix, and apply state-space modeling to practical multi-body systems.

3.8 Historical Development and Motivation

3.8.1 The Limitations of Classical Control

By the late 1950s, control engineering had reached a critical impasse. The classical methods developed by Bode, Nyquist, and others—while elegant and powerful for single-input, single-output (SISO) systems—proved inadequate for the increasingly complex engineering challenges of the post-war era. Transfer function methods, the cornerstone of classical control, suffered from fundamental limitations:

The first major limitation concerned MIMO complexity: for systems with multiple inputs and outputs, transfer functions become unwieldy matrices of rational polynomials, losing the intuitive interpretation that made them valuable. The second limitation involved initial conditions—transfer functions assume zero initial conditions, making them ill-suited for trajectory planning or systems that must operate from arbitrary starting points. Third, transfer functions describe only input-output relationships, potentially hiding unstable internal dynamics—a phenomenon later formalized as “observability” and “controllability.” Finally, classical methods assume linear time-invariant (LTI) systems; extending them to time-varying systems required fundamental reformulation.

3.8.2 Rudolf Kalman and the State-Space Revolution

Rudolf Kalman: Father of Modern Control Theory

Rudolf Emil Kalman (1930–2016) revolutionized control theory and estimation with contributions that earned him the National Medal of Science (2008) and the prestigious Kyoto Prize (1985). Born in Budapest, Hungary, Kalman emigrated to the United States and earned his doctorate from Columbia University in 1957. His 1960 paper introducing what became known as the Kalman filter is one of the most cited papers in engineering history. The filter was first applied to the Apollo navigation system, enabling the successful moon landings. Kalman’s concepts of controllability and observability provided the theoretical foundations that determine whether a control system can achieve its objectives. His work bridged mathematics and engineering, demonstrating that deep theoretical insights could solve pressing practical problems.

Rudolf Emil Kalman (1930–2016), a Hungarian-American mathematician and electrical engineer, transformed control theory with a series of landmark papers published between 1960 and 1963. Working first at the Research Institute for Advanced Studies (RIAS) in Baltimore and later at Stanford University, Kalman developed what would become known as *modern control theory*.

Kalman’s key insight was deceptively simple yet profound: rather than describing a system by its input-output relationship, describe it by its *internal state*—a minimal set of variables that, together with knowledge of future inputs, completely determines the system’s future behavior.

“The state of a system at time t_0 is the minimal amount of information at t_0 which, together with the input for $t \geq t_0$, uniquely determines the behavior of the system for all $t \geq t_0$.”

—Rudolf Kalman, 1960

This concept, while mathematically precise, had deep physical intuition. Consider a swinging pendulum: knowing only its current position is insufficient to predict its future motion. We must also know its velocity. Together, position and velocity constitute the pendulum’s state—the minimal information required to predict its future trajectory.

Kalman’s 1960 paper “A New Approach to Linear Filtering and Prediction Problems” introduced the celebrated Kalman filter, while his 1960 work “On the General Theory of Control Systems” established the concepts of controllability and observability—properties that determine whether a system can be steered to desired states and whether its internal states can be inferred from measurements.

3.8.3 The Space Race Imperative

The timing of Kalman’s theoretical developments proved fortuitous. The Soviet Union’s launch of Sputnik in 1957 had galvanized American investment in aerospace technology, and NASA’s Apollo program demanded control systems of unprecedented sophistication.

Spacecraft guidance presented precisely the challenges that classical control could not address:

Spacecraft presented the challenge of multiple coupled variables, as a spacecraft's position and orientation in three-dimensional space require tracking at least twelve state variables encompassing position, velocity, attitude, and angular rates. Navigation demanded sensor fusion, combining measurements from multiple imperfect sensors—gyroscopes, accelerometers, star trackers, and radar—each with different noise characteristics and update rates. Fuel optimality was critical because with strictly limited fuel, controllers had to minimize thrust usage while achieving precise trajectory corrections. Furthermore, all calculations had to execute in real-time on primitive on-board computers with only kilobytes of memory and kilohertz clock speeds.

Stanley Schmidt at NASA Ames Research Center recognized the potential of Kalman's work and implemented the first practical Kalman filter for the Apollo navigation computer. The filter elegantly solved the sensor fusion problem, optimally combining noisy measurements to estimate the spacecraft's true state. This application demonstrated that state-space methods were not merely theoretical curiosities but practical tools for solving real engineering problems.

3.8.4 Parallel Soviet Developments

Remarkably, Soviet mathematicians developed complementary theories almost simultaneously. Lev Pontryagin (1908–1988), despite losing his eyesight at age 14, became one of the twentieth century's greatest mathematicians. His 1956 formulation of the *maximum principle* provided necessary conditions for optimal control—determining control inputs that minimize cost functionals subject to state-space dynamics.

While Kalman approached control from an estimation and stochastic perspective, Pontryagin's work emphasized deterministic optimal control. The two approaches proved deeply complementary: Kalman's filter estimates the current state from noisy measurements, while Pontryagin's maximum principle determines optimal control actions given known states. Together, they form the foundation of modern optimal control and estimation theory.

The Cold War's competitive pressure ironically accelerated progress on both sides, with each nation's advances spurring the other to greater efforts. International conferences in the 1960s gradually broke down barriers, allowing the synthesis of Eastern and Western approaches into a unified modern control theory.

3.8.5 Connections to Classical Physics

State-space methods, while novel in control engineering, connected to much older mathematical traditions. William Rowan Hamilton's reformulation of classical mechanics in the 1830s had introduced phase space—the space of generalized positions and momenta—as the natural setting for dynamical systems. Hamilton's equations,

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \quad (3.75)$$

where H is the Hamiltonian (total energy) and (q_i, p_i) are generalized coordinates and momenta, are precisely state-space equations in disguise.

This connection is no coincidence. Physical systems conserve information about their state: knowing a system's complete state at one instant determines its state at all future instants (for

deterministic systems). State-space control theory formalized this physical intuition and extended it to systems with external inputs and outputs.

The eigenvalue analysis central to state-space methods similarly connects to the normal modes of vibration studied by physicists since the eighteenth century. When we diagonalize the state matrix A , we decompose complex coupled dynamics into independent modal behaviors—each evolving according to its own time constant.

3.9 Modern Applications

State-space methods have become ubiquitous in modern engineering. We highlight several domains where their advantages prove decisive.

3.9.1 Aerospace Guidance and Navigation

Modern aircraft and spacecraft rely entirely on state-space control. The F-16 fighter's fly-by-wire system uses state feedback to stabilize an intentionally unstable airframe, providing maneuverability impossible with classical control. Commercial aircraft autopilots estimate aircraft state (position, velocity, attitude) using Kalman filters that fuse GPS, inertial navigation, and air data measurements.

Spacecraft applications have grown increasingly sophisticated. Mars rovers navigate autonomously using state estimation to track their position across terrain lacking GPS infrastructure. Satellite constellations maintain precise formations using decentralized state-space controllers that coordinate dozens of spacecraft.

3.9.2 Robotic Manipulation

Industrial and research robots exemplify MIMO control challenges. A six-axis robot arm has twelve state variables (six joint angles and six joint velocities) with highly coupled, nonlinear dynamics. State-space methods enable:

State-space methods enable computed torque control through linearization of nonlinear dynamics about the current state, enabling linear state feedback. Kalman filters provide state estimation by combining joint encoder measurements with dynamic models to estimate unmeasured states like joint velocities. For trajectory optimization, Pontryagin's maximum principle determines time-optimal or energy-optimal joint trajectories.

3.9.3 Economic and Financial Modeling

State-space methods have transformed quantitative economics. Macroeconomic models treat unobservable quantities ("potential GDP," "natural unemployment rate") as state variables estimated from observable data (measured GDP, employment statistics) using Kalman filtering. Central banks worldwide use such models for monetary policy analysis.

In finance, state-space models capture time-varying volatility, enabling better risk management and derivative pricing. The "stochastic volatility" models underlying modern options pricing are fundamentally state-space formulations with volatility as a hidden state.

3.9.4 Power Grid State Estimation

Electrical power grids comprise thousands of interconnected generators, transmission lines, and loads. Grid operators must continuously estimate the system state (bus voltages and phase angles throughout the network) from sparse measurements. State estimation algorithms, direct descendants of Kalman's work, process measurements from across the grid to provide operators with situational awareness.

Smart grid technologies extend these methods to distribution networks, enabling integration of distributed solar generation, electric vehicles, and demand response—all requiring estimation and control of previously unmonitored network states.

3.10 Mathematical Foundation

3.10.1 State Variable Definition

We now formalize the concepts introduced historically. Consider a dynamic system whose behavior evolves over time in response to inputs.

Definition 3.4 (State). The **state** of a system at time t_0 is the minimum set of numbers $x_1(t_0), x_2(t_0), \dots, x_n(t_0)$ such that knowledge of these numbers and the input $u(t)$ for $t \geq t_0$ completely determines the system's behavior for all $t \geq t_0$.

The state variables are collected into a **state vector**:

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} \in \mathbb{R}^n \quad (3.76)$$

The integer n is the **order** of the system, and the n -dimensional space containing all possible state vectors is the **state space**.

Example 3.7 (Simple Pendulum). A simple pendulum's motion is governed by $\ddot{\theta} + (g/L) \sin \theta = 0$. To predict future motion, we need both the angular position $x_1 = \theta$ and the angular velocity $x_2 = \dot{\theta}$. Knowing only θ is insufficient—we cannot distinguish a pendulum at rest from one passing through the same angle with high velocity.

3.10.2 Standard State-Space Form

For linear time-invariant (LTI) systems, the state-space representation takes a standard matrix form:

$$\begin{array}{l} \dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad (\text{State Equation}) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \quad (\text{Output Equation}) \end{array} \quad (3.77)$$

The variables and matrices in this representation are defined in Table 3.3.

The state equation is a system of n coupled first-order ordinary differential equations. The output equation is algebraic, specifying which combinations of states and inputs are measured.

Table 3.3: State-space notation and dimensions

Symbol	Dimension	Description
$\mathbf{x}(t)$	$\mathbb{R}^n$	State vector
$\mathbf{u}(t)$	$\mathbb{R}^m$	Input vector
$\mathbf{y}(t)$	$\mathbb{R}^p$	Output vector
$\mathbf{A}$	$\mathbb{R}^{n \times n}$	System matrix (or state matrix)
$\mathbf{B}$	$\mathbb{R}^{n \times m}$	Input matrix
$\mathbf{C}$	$\mathbb{R}^{p \times n}$	Output matrix
$\mathbf{D}$	$\mathbb{R}^{p \times m}$	Feedthrough matrix (often zero)

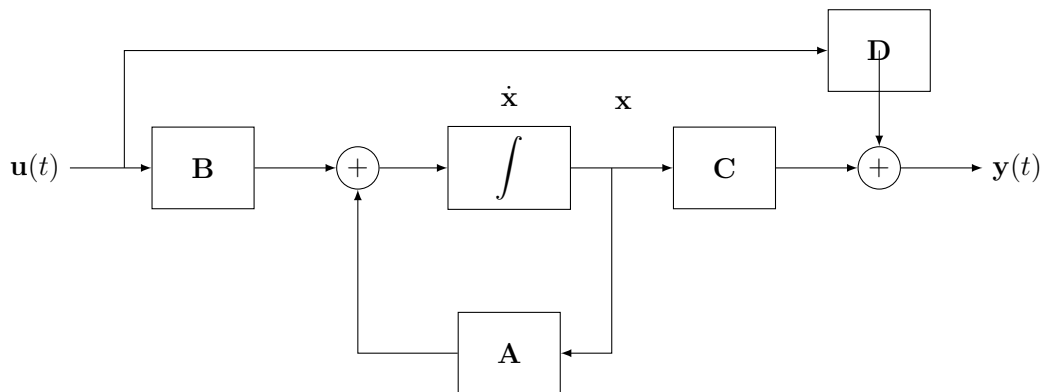


Figure 3.11: Block diagram of state-space representation. The integrator block converts $\dot{\mathbf{x}}$ to $\mathbf{x}$. The feedback through $\mathbf{A}$ creates the coupled dynamics, while $\mathbf{D}$ provides direct feedthrough from input to output.

3.10.3 Converting Differential Equations to State-Space Form

Any n th-order linear ODE can be converted to n first-order equations in state-space form. The procedure is systematic:

General Procedure: The conversion process follows a systematic approach. First, identify the highest derivative of the output in the differential equation. Next, choose state variables as the output and its successive derivatives. Then write each derivative as a function of states and inputs. Finally, assemble the resulting equations into matrix form.

Example 3.8 (Second-Order System). Consider the general second-order ODE:

$$\ddot{y} + a_1\dot{y} + a_0y = b_0u \quad (3.78)$$

Step 1: Choose state variables:

$$x_1 = y \quad (3.79)$$

$$x_2 = \dot{y} \quad (3.80)$$

Step 2: Express derivatives:

$$\dot{x}_1 = x_2 \quad (3.81)$$

$$\dot{x}_2 = \ddot{y} = -a_0y - a_1\dot{y} + b_0u = -a_0x_1 - a_1x_2 + b_0u \quad (3.82)$$

Step 3: Write in matrix form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ b_0 \end{bmatrix} u \quad (3.83)$$

If we measure y directly:

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (3.84)$$

3.10.4 Example: Mass-Spring-Damper System

Consider the canonical mass-spring-damper system shown in Fig. 3.12.

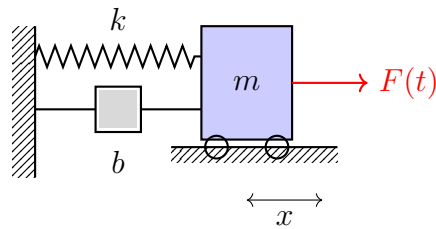


Figure 3.12: Mass-spring-damper system with mass m , spring constant k , damping coefficient b , and applied force $F(t)$.

Newton's second law gives:

$$m\ddot{x} + b\dot{x} + kx = F(t) \quad (3.85)$$

State Variables:

$$x_1 = x \quad (\text{position}) \quad (3.86)$$

$$x_2 = \dot{x} \quad (\text{velocity}) \quad (3.87)$$

State Equations:

$$\dot{x}_1 = x_2 \quad (3.88)$$

$$\dot{x}_2 = -\frac{k}{m}x_1 - \frac{b}{m}x_2 + \frac{1}{m}F \quad (3.89)$$

Matrix Form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}}_{\mathbf{B}} F \quad (3.90)$$

If we measure position:

$$y = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{\mathbf{C}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \end{bmatrix}}_{\mathbf{D}} F \quad (3.91)$$

Numerical Example: Let $m = 1$ kg, $b = 0.5$ N·s/m, $k = 2$ N/m:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -0.5 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad \mathbf{D} = [0] \quad (3.92)$$

3.10.5 Transfer Function to State-Space Conversion

The transfer function and state-space representations are related by:

$$\boxed{G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}} \quad (3.93)$$

Derivation: Taking the Laplace transform of the state equations (assuming zero initial conditions):

$$s\mathbf{x}(s) = \mathbf{A}\mathbf{x}(s) + \mathbf{B}U(s) \quad (3.94)$$

$$(s\mathbf{I} - \mathbf{A})\mathbf{x}(s) = \mathbf{B}U(s) \quad (3.95)$$

$$\mathbf{x}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}U(s) \quad (3.96)$$

Substituting into the output equation:

$$Y(s) = \mathbf{C}\mathbf{x}(s) + \mathbf{D}U(s) \quad (3.97)$$

$$= \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}U(s) + \mathbf{D}U(s) \quad (3.98)$$

$$= [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}] U(s) \quad (3.99)$$

Therefore, $G(s) = Y(s)/U(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$.

Example 3.9 (Transfer Function from State-Space). For the mass-spring-damper with $m = 1$, $b = 0.5$, $k = 2$:

$$s\mathbf{I} - \mathbf{A} = \begin{bmatrix} s & -1 \\ 2 & s + 0.5 \end{bmatrix} \quad (3.100)$$

The inverse is:

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{s(s + 0.5) + 2} \begin{bmatrix} s + 0.5 & 1 \\ -2 & s \end{bmatrix} = \frac{1}{s^2 + 0.5s + 2} \begin{bmatrix} s + 0.5 & 1 \\ -2 & s \end{bmatrix} \quad (3.101)$$

Computing $G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$:

$$G(s) = \begin{bmatrix} 1 & 0 \end{bmatrix} \frac{1}{s^2 + 0.5s + 2} \begin{bmatrix} s + 0.5 & 1 \\ -2 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (3.102)$$

$$= \frac{1}{s^2 + 0.5s + 2} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ s \end{bmatrix} \quad (3.103)$$

$$= \frac{1}{s^2 + 0.5s + 2} \quad (3.104)$$

3.10.6 Eigenvalues and System Poles

A fundamental result connects the eigenvalues of $\mathbf{A}$ to the transfer function poles:

Theorem 3.2. *The poles of $G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$ are a subset of the eigenvalues of $\mathbf{A}$.*

Proof Sketch: The matrix $(s\mathbf{I} - \mathbf{A})^{-1}$ has the form $\text{adj}(s\mathbf{I} - \mathbf{A}) / \det(s\mathbf{I} - \mathbf{A})$. The denominator $\det(s\mathbf{I} - \mathbf{A})$ is the characteristic polynomial of $\mathbf{A}$, whose roots are the eigenvalues of $\mathbf{A}$. Pole-zero cancellations may occur, so the transfer function poles are a subset of the eigenvalues.

Physical Interpretation: Eigenvalues of $\mathbf{A}$ determine natural modes of the unforced system. For stable systems, all eigenvalues must have negative real parts.

Example 3.10 (Eigenvalue Analysis). For our mass-spring-damper:

$$\det(s\mathbf{I} - \mathbf{A}) = \det \begin{bmatrix} s & -1 \\ 2 & s + 0.5 \end{bmatrix} = s^2 + 0.5s + 2 = 0 \quad (3.105)$$

Using the quadratic formula:

$$\lambda_{1,2} = \frac{-0.5 \pm \sqrt{0.25 - 8}}{2} = -0.25 \pm j1.39 \quad (3.106)$$

The eigenvalues are complex conjugates with negative real parts (-0.25), indicating a stable system with oscillatory response. The negative real part ensures stability, while the nonzero imaginary part produces oscillations. The damped oscillation frequency is $\omega_d = 1.39$ rad/s, and the time constant is $\tau = 1/0.25 = 4$ s.

3.10.7 State Transition Matrix

The solution to the homogeneous state equation $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ is:

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) \quad (3.107)$$

where $e^{\mathbf{A}t}$ is the **state transition matrix** (or matrix exponential), defined by the series:

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{(\mathbf{A}t)^2}{2!} + \frac{(\mathbf{A}t)^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{(\mathbf{A}t)^k}{k!} \quad (3.108)$$

Key Properties: The state transition matrix satisfies several important properties. At $t = 0$, we have $e^{\mathbf{A} \cdot 0} = \mathbf{I}$, confirming the initial condition. The derivative satisfies $\frac{d}{dt}e^{\mathbf{A}t} = \mathbf{A}e^{\mathbf{A}t} = e^{\mathbf{A}t}\mathbf{A}$, showing that $\mathbf{A}$ and $e^{\mathbf{A}t}$ commute. The semigroup property states that $e^{\mathbf{A}(t_1+t_2)} = e^{\mathbf{A}t_1}e^{\mathbf{A}t_2}$, and the inverse is simply $(e^{\mathbf{A}t})^{-1} = e^{-\mathbf{A}t}$.

The state transition matrix propagates the state forward in time: $\mathbf{x}(t) = e^{\mathbf{A}(t-t_0)}\mathbf{x}(t_0)$.

Computation Methods: Several methods exist for computing the matrix exponential. The Laplace transform approach gives $e^{\mathbf{A}t} = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$. Eigenvalue decomposition provides an alternative: if $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$, then $e^{\mathbf{A}t} = \mathbf{V}e^{\mathbf{\Lambda}t}\mathbf{V}^{-1}$. The Cayley-Hamilton theorem guarantees that for an $n \times n$ matrix, $e^{\mathbf{A}t}$ can be expressed as a polynomial of degree $n - 1$ in $\mathbf{A}$.

3.10.8 Complete Solution with Forcing

For the forced system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$, the complete solution is:

$$\mathbf{x}(t) = \underbrace{e^{\mathbf{A}t}\mathbf{x}(0)}_{\text{zero-input response}} + \underbrace{\int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau}_{\text{zero-state response}} \quad (3.109)$$

This is the **variation of parameters** formula. The first term represents the natural response from initial conditions; the second term (a convolution integral) represents the forced response.

Derivation: Multiply the state equation by $e^{-\mathbf{A}t}$:

$$e^{-\mathbf{A}t}\dot{\mathbf{x}} - e^{-\mathbf{A}t}\mathbf{A}\mathbf{x} = e^{-\mathbf{A}t}\mathbf{B}\mathbf{u} \quad (3.110)$$

$$\frac{d}{dt}(e^{-\mathbf{A}t}\mathbf{x}) = e^{-\mathbf{A}t}\mathbf{B}\mathbf{u} \quad (3.111)$$

Integrating from 0 to t :

$$e^{-\mathbf{A}t}\mathbf{x}(t) - \mathbf{x}(0) = \int_0^t e^{-\mathbf{A}\tau}\mathbf{B}\mathbf{u}(\tau) d\tau \quad (3.112)$$

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau \quad (3.113)$$

3.11 Laboratory Experiment: Two-Mass System

3.11.1 Objective

This experiment demonstrates state-space modeling using a coupled two-mass system. Unlike the single mass-spring-damper, this system has four state variables and exhibits richer dynamics including two distinct natural frequencies. Students will build a physical two-mass system and derive and validate a state-space model. Through this process, they will observe the complexity this system would have in transfer function form and compare model predictions with measured responses.

3.11.2 Apparatus

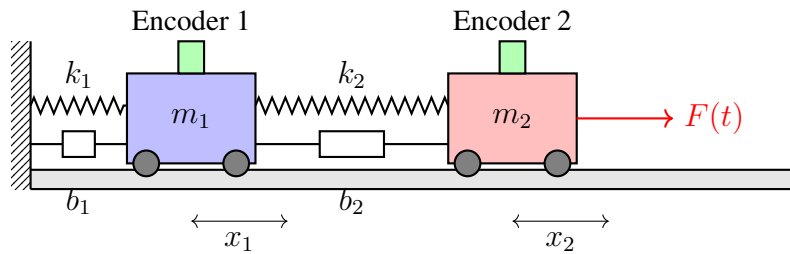


Figure 3.13: Two-mass experimental setup. Carts m_1 and m_2 are coupled by spring k_2 and damper b_2 . Cart m_1 is also connected to a fixed wall by k_1 and b_1 . Force $F(t)$ is applied to m_2 . Encoders measure positions x_1 and x_2 .

Required Components: The apparatus requires two carts (3D printed or commercial dynamics carts with $m_1 \approx m_2 \approx 0.5$ kg) mounted on a linear track (drawer slides or commercial air track with length ≥ 1 m). Two springs with constants $k_1, k_2 \approx 10\text{--}50$ N/m provide the coupling forces. Position measurement is accomplished using rotary encoders or ultrasonic distance sensors, while a solenoid or motor applies the controlled force $F(t)$. A data acquisition system (Arduino, myRIO, or similar) interfaces with a computer running MATLAB or Python for analysis.

3D Printing Notes: Cart designs should incorporate low-friction wheel mounts compatible with the track, spring attachment points, and an encoder wheel mount or reflector for the distance sensor. The estimated print time is 2–3 hours per cart at 0.2 mm layer height.

3.11.3 System Modeling

Equations of Motion:

Applying Newton's second law to each mass:

$$m_1 \ddot{x}_1 = -k_1 x_1 - b_1 \dot{x}_1 + k_2(x_2 - x_1) + b_2(\dot{x}_2 - \dot{x}_1) \quad (3.114)$$

$$m_2 \ddot{x}_2 = -k_2(x_2 - x_1) - b_2(\dot{x}_2 - \dot{x}_1) + F \quad (3.115)$$

Rearranging:

$$m_1 \ddot{x}_1 = -(k_1 + k_2)x_1 + k_2 x_2 - (b_1 + b_2)\dot{x}_1 + b_2 \dot{x}_2 \quad (3.116)$$

$$m_2 \ddot{x}_2 = k_2 x_1 - k_2 x_2 + b_2 \dot{x}_1 - b_2 \dot{x}_2 + F \quad (3.117)$$

State Variables:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \dot{x}_1 \\ x_2 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} \text{position of } m_1 \\ \text{velocity of } m_1 \\ \text{position of } m_2 \\ \text{velocity of } m_2 \end{bmatrix} \quad (3.118)$$

State-Space Form:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_1+k_2}{m_1} & -\frac{b_1+b_2}{m_1} & \frac{k_2}{m_1} & \frac{b_2}{m_1} \\ 0 & 0 & 0 & 1 \\ \frac{k_2}{m_2} & \frac{b_2}{m_2} & -\frac{k_2}{m_2} & -\frac{b_2}{m_2} \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_2} \end{bmatrix} F \quad (3.119)$$

If we measure both positions:

$$\mathbf{y} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \mathbf{x} \quad (3.120)$$

Numerical Example: Let $m_1 = m_2 = 0.5$ kg, $k_1 = k_2 = 20$ N/m, $b_1 = b_2 = 0.5$ N·s/m:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -80 & -2 & 40 & 1 \\ 0 & 0 & 0 & 1 \\ 40 & 1 & -40 & -1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 2 \end{bmatrix} \quad (3.121)$$

3.11.4 Transfer Function Comparison

To appreciate the elegance of state-space, consider the transfer function from F to x_1 :

$$G_1(s) = \frac{X_1(s)}{F(s)} = \frac{k_2 + b_2 s}{(m_1 s^2 + (b_1 + b_2)s + k_1 + k_2)(m_2 s^2 + b_2 s + k_2) - (k_2 + b_2 s)^2} \quad (3.122)$$

This fourth-order transfer function, with its complicated coefficient expressions, obscures the physical structure evident in the state-space model. For larger systems (three or more masses), transfer functions become impractical while state-space remains systematic.

3.11.5 Experimental Procedure

The experiment proceeds through four phases. In the **parameter identification** phase, weigh the carts to determine m_1 and m_2 , measure spring constants by hanging known masses, and estimate damping coefficients from free oscillation decay rates.

For **model validation**, displace cart 2 and release it to observe the initial condition response. Record $x_1(t)$ and $x_2(t)$ throughout the motion, then simulate the state-space model with the same initial conditions. Compare the simulated and measured trajectories to assess model accuracy.

The **forced response** phase involves applying a step force to cart 2 and observing the transient behavior. Subsequently, apply sinusoidal forcing at various frequencies to identify resonant frequencies, which should match the imaginary parts of the eigenvalues.

Finally, conduct **phase portrait analysis** by plotting x_1 versus $\dot{x}_1$ for cart 1 and x_2 versus $\dot{x}_2$ for cart 2. Observe the spiral trajectories characteristic of damped oscillatory systems converging to the equilibrium point.

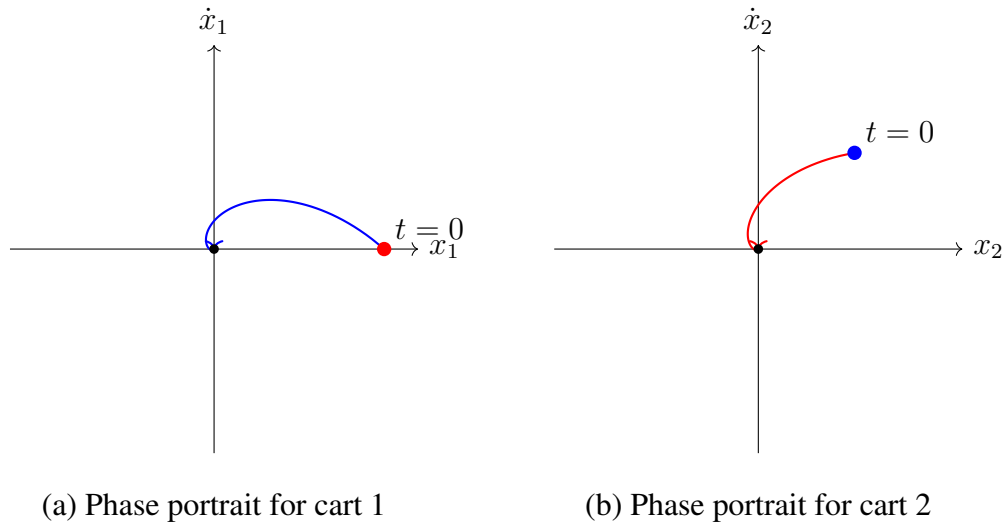


Figure 3.14: Phase portraits showing state trajectories in position-velocity space. Both carts exhibit spiral trajectories converging to equilibrium, characteristic of stable underdamped systems. The trajectories' shapes encode eigenvalue information: spiral rate indicates damping, rotation rate indicates oscillation frequency.

3.11.6 Expected Results

For the nominal parameters, the system matrix eigenvalues are approximately:

$$\lambda_{1,2} \approx -0.5 \pm j4.5 \quad (\text{first mode: in-phase motion}) \quad (3.123)$$

$$\lambda_{3,4} \approx -1.5 \pm j10.5 \quad (\text{second mode: out-of-phase motion}) \quad (3.124)$$

Students should observe two distinct oscillation frequencies in the free response, along with a beating phenomenon when both modes are excited. The higher-frequency mode typically exhibits larger damping and therefore faster decay. Model predictions should match experimental results within 10–15%, with discrepancies primarily attributable to friction effects not captured in the linear model.

3.11.7 Discussion Questions

Students should consider the following questions for deeper understanding. First, why does the system have four eigenvalues while a single mass has only two? Second, what physical motion corresponds to each eigenvalue pair? Third, how would adding a third mass change the model

complexity in state-space form versus transfer function form? Finally, what advantages does measuring both x_1 and x_2 provide over measuring only one position?

3.12 Worked Examples

Converting ODEs to State-Space Form

The conversion from a high-order differential equation to state-space form is a fundamental skill in modern control theory. The process reveals the internal structure of the system and enables the use of powerful matrix analysis tools. This example demonstrates the systematic procedure using a third-order system, resulting in what is called the controllable canonical form or phase-variable form.

3.12.1 Example 1: Converting an ODE to State-Space Form

Problem: Convert the following third-order ODE to state-space form:

$$\ddot{y} + 6\ddot{y} + 11\dot{y} + 6y = 6u \quad (3.125)$$

Solution:

Step 1: Define state variables as the output and its derivatives:

$$x_1 = y \quad (3.126)$$

$$x_2 = \dot{y} \quad (3.127)$$

$$x_3 = \ddot{y} \quad (3.128)$$

Step 2: Express each derivative:

$$\dot{x}_1 = x_2 \quad (3.129)$$

$$\dot{x}_2 = x_3 \quad (3.130)$$

$$\dot{x}_3 = \ddot{y} = -6y - 11\dot{y} - 6\ddot{y} + 6u = -6x_1 - 11x_2 - 6x_3 + 6u \quad (3.131)$$

Step 3: Write in matrix form:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix} u \quad (3.132)$$

$$y = [1 \ 0 \ 0] \mathbf{x} \quad (3.133)$$

This is called the **controllable canonical form** or **phase-variable form**.

Verification: The characteristic polynomial is:

$$\det(s\mathbf{I} - \mathbf{A}) = s^3 + 6s^2 + 11s + 6 = (s + 1)(s + 2)(s + 3) \quad (3.134)$$

matching the original ODE's characteristic equation. ■

3.12.2 Example 2: State-Space to Transfer Function

Problem: Find the transfer function for the system:

$$\mathbf{A} = \begin{bmatrix} -3 & 1 \\ 0 & -2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = [1 \quad 1], \quad \mathbf{D} = [0] \quad (3.135)$$

Solution:

Step 1: Compute $s\mathbf{I} - \mathbf{A}$:

$$s\mathbf{I} - \mathbf{A} = \begin{bmatrix} s+3 & -1 \\ 0 & s+2 \end{bmatrix} \quad (3.136)$$

Step 2: Find the inverse:

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{(s+3)(s+2)} \begin{bmatrix} s+2 & 1 \\ 0 & s+3 \end{bmatrix} \quad (3.137)$$

Step 3: Compute $\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$:

$$G(s) = \frac{1}{(s+3)(s+2)} [1 \quad 1] \begin{bmatrix} s+2 & 1 \\ 0 & s+3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (3.138)$$

$$= \frac{1}{(s+3)(s+2)} [s+2 \quad s+4] \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (3.139)$$

$$= \frac{s+4}{(s+3)(s+2)} = \frac{s+4}{s^2+5s+6} \quad (3.140)$$

The system has poles at $s = -2$ and $s = -3$ (eigenvalues of $\mathbf{A}$) and a zero at $s = -4$. ■

3.12.3 Example 3: Eigenvalue Analysis and Stability

Problem: Analyze the stability of the system with:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \quad (3.141)$$

Solution:

Step 1: Find eigenvalues from $\det(s\mathbf{I} - \mathbf{A}) = 0$:

$$\det \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix} = s(s+3) + 2 = s^2 + 3s + 2 = 0 \quad (3.142)$$

$$(s+1)(s+2) = 0 \quad (3.143)$$

$$\lambda_1 = -1, \quad \lambda_2 = -2 \quad (3.144)$$

Step 2: Interpret results. Both eigenvalues are real and negative, indicating that the system is **asymptotically stable**. The response is **overdamped** with no oscillations since the eigenvalues are purely real. The corresponding time constants are $\tau_1 = 1$ s and $\tau_2 = 0.5$ s, with the dominant (slower) mode associated with $\lambda_1 = -1$.

Step 3: Find eigenvectors for modal analysis:

For $\lambda_1 = -1$: $(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{v}_1 = \mathbf{0}$

$$\begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix} \mathbf{v}_1 = \mathbf{0} \Rightarrow \mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (3.145)$$

For $\lambda_2 = -2$:

$$\begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix} \mathbf{v}_2 = \mathbf{0} \Rightarrow \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad (3.146)$$

The general solution is:

$$\mathbf{x}(t) = c_1 e^{-t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + c_2 e^{-2t} \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad (3.147)$$

where c_1, c_2 depend on initial conditions. ■

DC Motor: A Multi-Domain State-Space Model

The DC motor is an excellent example of a system that couples multiple physical domains—electrical and mechanical. Unlike simple single-domain systems, the DC motor requires careful selection of state variables from both domains. The back-EMF creates coupling between the electrical current and mechanical velocity, making this a true multi-physics system. This example demonstrates how state-space methods naturally handle such coupled systems.

3.12.4 Example 4: DC Motor State-Space Model

Problem: Derive a state-space model for a DC motor that includes both electrical and mechanical dynamics.

System Description: A DC motor consists of an armature circuit characterized by resistance R , inductance L , and back-EMF $e_b = K_e \omega$. The mechanical load has inertia J , friction coefficient b , and produces torque $\tau = K_t i_a$ proportional to armature current. The system input is the armature voltage v_a , and the output of interest is the angular velocity ω .

Solution:

Step 1: Write governing equations.

Electrical (Kirchhoff's voltage law):

$$v_a = R i_a + L \frac{di_a}{dt} + K_e \omega \quad (3.148)$$

Mechanical (Newton's second law for rotation):

$$J \frac{d\omega}{dt} = K_t i_a - b \omega \quad (3.149)$$

Step 2: Choose state variables:

$$x_1 = i_a \quad (\text{armature current}) \quad (3.150)$$

$$x_2 = \omega \quad (\text{angular velocity}) \quad (3.151)$$

Step 3: Rewrite as first-order equations:

$$\dot{x}_1 = \frac{di_a}{dt} = -\frac{R}{L}i_a - \frac{K_e}{L}\omega + \frac{1}{L}v_a \quad (3.152)$$

$$= -\frac{R}{L}x_1 - \frac{K_e}{L}x_2 + \frac{1}{L}v_a \quad (3.153)$$

$$\dot{x}_2 = \frac{d\omega}{dt} = \frac{K_t}{J}i_a - \frac{b}{J}\omega \quad (3.154)$$

$$= \frac{K_t}{J}x_1 - \frac{b}{J}x_2 \quad (3.155)$$

Step 4: State-space form:

$$\dot{\mathbf{x}} = \begin{bmatrix} -\frac{R}{L} & -\frac{K_e}{L} \\ \frac{K_t}{J} & -\frac{b}{J} \end{bmatrix} \mathbf{x} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} v_a \quad (3.156)$$

$$y = \omega = [0 \ 1] \mathbf{x} \quad (3.157)$$

Numerical Example: Let $R = 1 \ \Omega$, $L = 0.5 \ \text{H}$, $K_e = K_t = 0.1 \ \text{V}\cdot\text{s}/\text{rad}$, $J = 0.01 \ \text{kg}\cdot\text{m}^2$, $b = 0.1 \ \text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$:

$$\mathbf{A} = \begin{bmatrix} -2 & -0.2 \\ 10 & -10 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad \mathbf{C} = [0 \ 1] \quad (3.158)$$

Eigenvalues: $\lambda_{1,2} = -6 \pm j4$ (stable, underdamped). ■

3.12.5 Example 5: Different State Variable Choices

Problem: Show that different state variable choices yield equivalent models for the system:

$$\ddot{y} + 3\dot{y} + 2y = u \quad (3.159)$$

Solution:

Choice A: Phase variables ($x_1 = y$, $x_2 = \dot{y}$):

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \quad y = [1 \ 0] \mathbf{x} \quad (3.160)$$

Choice B: Modal coordinates. First, diagonalize $\mathbf{A}$ from Choice A.

Eigenvalues: $\lambda_1 = -1$, $\lambda_2 = -2$.

Eigenvector matrix and its inverse:

$$\mathbf{V} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}, \quad \mathbf{V}^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} \quad (3.161)$$

Define new states $\mathbf{z} = \mathbf{V}^{-1}\mathbf{x}$:

$$\dot{\mathbf{z}} = \mathbf{V}^{-1}\mathbf{A}\mathbf{V}\mathbf{z} + \mathbf{V}^{-1}\mathbf{B}u \quad (3.162)$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} u \quad (3.163)$$

$$y = \mathbf{C}\mathbf{V}\mathbf{z} = \begin{bmatrix} 1 & 1 \end{bmatrix} \mathbf{z} \quad (3.164)$$

Verification: Both representations yield the same transfer function:

$$G(s) = \frac{1}{s^2 + 3s + 2} = \frac{1}{(s + 1)(s + 2)} \quad (3.165)$$

Key Insight: The diagonal form reveals independent first-order dynamics:

$$\dot{z}_1 = -z_1 + u \quad (\text{mode 1, } \tau = 1 \text{ s}) \quad (3.166)$$

$$\dot{z}_2 = -2z_2 - u \quad (\text{mode 2, } \tau = 0.5 \text{ s}) \quad (3.167)$$

State-space representations related by a coordinate transformation $\mathbf{x} = \mathbf{T}\tilde{\mathbf{x}}$ are called **equivalent realizations**. They share the same input-output behavior (transfer function) but may differ in numerical properties, physical interpretability, or controller design convenience. ■

3.13 Chapter Summary

This chapter introduced state-space representation, the foundation of modern control theory.

From a **historical perspective**, state-space methods emerged in the 1960s through the work of Kalman and Pontryagin, driven by aerospace applications requiring MIMO control and optimal estimation. The fundamental concept of **state** represents the minimal information needed to predict future system behavior given future inputs.

The **standard form** for LTI systems consists of two equations: the state equation $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$ and the output equation $y = \mathbf{C}\mathbf{x} + \mathbf{D}u$. The **transfer function connection** is given by $G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$, with the poles being a subset of $\mathbf{A}$'s eigenvalues.

The **state transition matrix** provides the complete time-domain solution $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}u(\tau)d\tau$, which elegantly separates the initial condition response from the forced response. The **eigenvalues** of the system matrix $\mathbf{A}$ determine both stability through their real parts and oscillation frequency through their imaginary parts.

Looking Ahead: Chapter 5 will introduce controllability and observability—properties that determine whether state-space models can be effectively controlled and estimated. Chapter 6 will develop state feedback and pole placement methods for controller design.

3.14 Exercises

Conceptual Exercises

Exercise 3.13. Explain why knowing only the output $y(t_0)$ is generally insufficient to predict future system behavior, even with knowledge of all future inputs.

Exercise 3.14. A system has transfer function $G(s) = \frac{s+2}{(s+1)(s+3)}$. What is the minimum order of any state-space realization? Explain.

Exercise 3.15. Can two different physical systems have identical transfer functions but different state-space models? Provide an example or explain why not.

Computational Exercises

Exercise 3.16. Convert to state-space form: $\ddot{y} + 4\dot{y} + 5y = 3u + \dot{u}$

Exercise 3.17. Find the transfer function for:

$$\mathbf{A} = \begin{bmatrix} -1 & 2 \\ 0 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{C} = [1 \quad 0], \quad \mathbf{D} = [0]$$

Exercise 3.18. Compute the eigenvalues and classify the stability:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}$$

Exercise 3.19. For the mass-spring-damper with $m = 2$ kg, $b = 4$ N·s/m, $k = 10$ N/m:

- Write the state-space model
- Find eigenvalues and interpret physically
- Compute the state transition matrix $e^{\mathbf{A}t}$
- Find $\mathbf{x}(t)$ for $\mathbf{x}(0) = [1 \ 0]^T$

Design Exercises

Exercise 3.20. An inverted pendulum on a cart has linearized dynamics:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -mg/M & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & (M+m)g/(ML) & 0 \end{bmatrix}$$

With $M = 1$ kg, $m = 0.1$ kg, $L = 0.5$ m, $g = 9.81$ m/s²:

- Find the eigenvalues

- (b) Is the system stable? If not, which modes are unstable?
- (c) What does this imply about control requirements?

Exercise 3.21. Design a two-mass laboratory experiment different from the one presented. Specify:

- (a) Physical configuration
- (b) State variables
- (c) Measurement sensors
- (d) Expected eigenvalue structure

Computational/Simulation Exercises

Exercise 3.22. Using MATLAB or Python, simulate the two-mass system from Section IV with initial conditions $\mathbf{x}(0) = [0.1 \ 0 \ 0 \ 0]^T$ (cart 1 displaced). Plot:

- (a) $x_1(t)$ and $x_2(t)$ vs. time
- (b) Phase portraits for both carts
- (c) Animation of cart motion

Exercise 3.23. Implement the DC motor model from Example 4. Apply a step voltage input and plot:

- (a) Current $i_a(t)$ and velocity $\omega(t)$
- (b) Phase portrait (i_a vs. ω)
- (c) Compare with the transfer function step response

Exercise 3.24. A coupled electrical circuit consists of two LC oscillators connected by a mutual inductance M . The state variables are the currents i_1, i_2 and capacitor voltages v_1, v_2 . With $L_1 = L_2 = 1$ H, $C_1 = C_2 = 1$ F, and $M = 0.5$ H:

- (a) Derive the state-space model.
- (b) Find the eigenvalues and identify the natural frequencies.
- (c) Explain the physical significance of the two distinct frequencies in terms of coupled oscillator behavior.

Exercise 3.25. A thermal system consists of two rooms separated by a wall. Room 1 has thermal capacitance $C_1 = 1000$ J/K and is heated by a heater with power P (input). Room 2 has thermal capacitance $C_2 = 2000$ J/K and loses heat to the outside environment. The thermal resistance between rooms is $R_{12} = 0.1$ K/W, and the resistance to outside is $R_2 = 0.2$ K/W.

- (a) Define state variables and derive the state-space model.
- (b) Find the eigenvalues and determine the thermal time constants.
- (c) If the outside temperature is 0°C and the heater provides 500 W, what are the steady-state temperatures of both rooms?

Further Reading

The foundational papers in state-space control theory remain essential reading. R. E. Kalman's "A New Approach to Linear Filtering and Prediction Problems" (*Journal of Basic Engineering*, vol. 82, pp. 35–45, 1960) introduced the celebrated Kalman filter. His companion paper "On the General Theory of Control Systems" (*Proceedings of the First IFAC Congress*, Moscow, 1960) established the concepts of controllability and observability. L. S. Pontryagin et al.'s *The Mathematical Theory of Optimal Processes* (Interscience Publishers, 1962) presents the maximum principle for optimal control.

For modern textbook treatments, C. T. Chen's *Linear System Theory and Design* (4th ed., Oxford University Press, 2012) provides a comprehensive and mathematically rigorous approach, while K. Ogata's *Modern Control Engineering* (5th ed., Prentice Hall, 2010) offers an accessible introduction with numerous worked examples.

Part II

System Behavior

Chapter 5

First and Second Order Systems

“The fundamental frequencies of a vibrating system are as characteristic of that system as the features of a face are characteristic of an individual.” — Lord Rayleigh, Theory of Sound (1877)

The study of first and second order systems forms the cornerstone of control systems engineering. These canonical forms appear with remarkable universality across disciplines—from the gentle swing of a pendulum to the complex dynamics of spacecraft attitude control, from the absorption of medication in the bloodstream to the oscillations of bridges in the wind. Understanding these fundamental building blocks provides engineers with powerful tools to analyze, predict, and design the behavior of far more complex systems.

This chapter develops the complete mathematical framework for first and second order systems, traces their historical origins, and connects theory to modern applications. We begin with the rich history that led to our current understanding, proceed through rigorous mathematical derivations, and conclude with hands-on experiments that bring these concepts to life.

5.1 Historical Motivation: The Quest to Understand Oscillation

The mathematical description of first and second order systems emerged from centuries of scientific inquiry into the fundamental nature of motion, vibration, and response to disturbance. The journey from empirical observation to precise mathematical characterization represents one of the great intellectual achievements of classical physics and engineering.

5.1.1 Lord Rayleigh and the Theory of Sound (1877)

The Birth of Systematic Vibration Analysis

John William Strutt, the 3rd Baron Rayleigh (1842–1919), published his monumental two-volume treatise “*The Theory of Sound*” in 1877–1878. This work systematically applied the methods of mathematical physics to the analysis of vibrating systems and acoustic phenomena.

Rayleigh’s contribution was revolutionary in its scope and rigor. While earlier scientists such as Daniel Bernoulli, Leonhard Euler, and Jean-Baptiste Fourier had made significant contributions to understanding vibration and wave phenomena, Rayleigh synthesized these disparate results into a unified framework. His treatise established the fundamental principle that complex vibrating systems could be understood as combinations of simpler modes, each characterized by its natural frequency and damping.

The second-order differential equation emerged as the canonical description of a single vibrational mode:

$$m \frac{d^2x}{dt^2} + c \frac{dx}{dt} + kx = F(t) \quad (5.1)$$

Rayleigh recognized that this equation described not just mechanical oscillators, but a vast class of physical phenomena. He introduced the concept of the *quality factor* Q , related to what we now call the damping ratio, to characterize how sharply tuned a resonant system was. His analysis of coupled oscillators laid the groundwork for understanding multi-degree-of-freedom systems and modal analysis.

Perhaps most significantly, Rayleigh understood that the *poles* of a system—the roots of the characteristic equation—completely determined the qualitative behavior of the response. An oscillator with complex poles would exhibit sinusoidal behavior; one with real poles would show purely exponential decay. This insight, formalized mathematically in terms of the Laplace transform decades later, remains the central organizing principle of linear systems theory.

5.1.2 Seismograph Development and the Need for Instrument Theory (1880s)

The 1880s witnessed rapid development in seismology, driven by devastating earthquakes in Japan, Italy, and California. Scientists and engineers faced a fundamental challenge: how could they design instruments to faithfully record ground motion without the instrument’s own dynamics distorting the measurement?

The Seismograph Problem

John Milne, James Ewing, and Thomas Gray, working in Japan during the 1880s, developed increasingly sophisticated seismographs. Their instruments were essentially mass-spring-damper systems, and they quickly realized that the recorded signal was a *convolution* of the ground motion with the instrument’s impulse response.

This realization forced seismologists to develop rigorous theories of second-order system response. If the seismograph had natural frequency ω_n and damping ratio ζ , the behavior depended

critically on the relationship between earthquake frequency and instrument frequency. For ground oscillations at frequencies much *lower* than ω_n , the seismograph mass would move with the ground, providing a displacement measurement. Conversely, for ground oscillations at frequencies much *higher* than ω_n , the mass would remain stationary while the frame moved, effectively measuring acceleration. For frequencies *near* ω_n , resonance would amplify or distort the signal, making interpretation difficult.

The damping ratio became critical: too little damping and the instrument would ring excessively after each seismic event; too much damping and the instrument would be sluggish and miss rapid changes. Through empirical study and theoretical analysis, seismologists determined that $\zeta \approx 0.7$ (critically damped to slightly underdamped) provided the best compromise—a finding that would later influence control system design across all disciplines.

Emil Wiechert's inverted pendulum seismograph of 1904 exemplified the mature understanding of second-order system theory. By carefully choosing the natural frequency and damping ratio, Wiechert created instruments capable of recording earthquake waves that had traveled thousands of kilometers through the Earth.

5.1.3 Automotive Suspension Design (1920s–1930s)

The rapid proliferation of automobiles in the early twentieth century created an urgent engineering challenge: how to isolate passengers from road irregularities while maintaining vehicle stability and control. This problem is fundamentally one of second-order system design.

The Suspension Trade-off

Early automobiles used simple leaf spring suspensions inherited from horse-drawn carriages. As speeds increased, engineers discovered that comfortable ride and precise handling were conflicting objectives—a trade-off that could only be optimized through careful control of damping ratio and natural frequency.

The automobile suspension system is a classic second-order system characterized by three fundamental parameters: the mass m representing the sprung mass (vehicle body), the spring constant k provided by coil springs, leaf springs, or torsion bars, and the damping coefficient c provided by hydraulic shock absorbers.

The natural frequency $\omega_n = \sqrt{k/m}$ determines how quickly the suspension responds to bumps. Human comfort studies during this era established that natural frequencies between 1–1.5 Hz were optimal for passenger comfort, matching the natural frequency of the human body in seated position.

The damping ratio $\zeta = c/(2\sqrt{km})$ became the primary design variable for balancing competing objectives:

Damping Ratio	Ride Quality	Handling
$\zeta < 0.3$	Very soft, floaty	Poor body control
$\zeta \approx 0.3\text{--}0.4$	Comfortable (luxury cars)	Acceptable
$\zeta \approx 0.5\text{--}0.7$	Firm but controlled	Good
$\zeta > 0.7$	Harsh, transmits road texture	Excellent

The work of Maurice Olley at Cadillac in the 1930s established the scientific foundations of vehicle dynamics. Olley’s “flat ride” concept required matching front and rear suspension natural frequencies and damping ratios to eliminate pitching motions—a direct application of coupled second-order system theory.

5.1.4 Aircraft Flutter Analysis: When Resonance Kills (1930s)

Perhaps no application demonstrated the life-or-death importance of understanding second-order system dynamics more dramatically than the aircraft flutter problem. Flutter is a self-excited oscillation that occurs when aerodynamic, elastic, and inertial forces couple in an unstable manner.

The Flutter Crisis

During the 1930s, as aircraft speeds increased, a series of catastrophic structural failures occurred when aircraft wings and control surfaces entered violent oscillation. The Lockheed Electra, Handley Page H.P.42, and many military aircraft experienced flutter-related failures, leading to intensive research programs on both sides of the Atlantic.

Flutter analysis required understanding the coupling of multiple second-order systems—wing bending, wing torsion, and control surface rotation—each with its own natural frequency and damping. The critical insight was that instability occurred not when a single mode became unstable, but when energy transfer between modes caused the effective damping ratio to become negative.

The stability criterion could be expressed in terms of pole locations in the complex s -plane. A system is stable when all poles lie in the left half-plane where $\text{Re}(s) < 0$; marginally stable when poles reside on the imaginary axis where $\text{Re}(s) = 0$; and unstable when any pole exists in the right half-plane where $\text{Re}(s) > 0$.

Theodore Theodorsen at NACA (later NASA) developed the seminal mathematical theory of flutter in 1935, introducing the Theodorsen function to describe unsteady aerodynamic forces. His work established that flutter could be predicted and prevented through careful design of mass distribution, stiffness, and—critically—structural damping.

The flutter problem established a paradigm that persists today: understanding a complex system requires identifying its dominant modes, characterizing each mode’s natural frequency and damping, and ensuring that all modes remain stable under all operating conditions.

5.1.5 Why These Canonical Forms?

The first and second order system representations are not merely convenient mathematical abstractions—they appear because nature itself is organized in terms of energy storage and dissipation:

The Physical Origin of System Order

First-order systems arise when there is *one* energy storage element (capacitor, spring, thermal mass) and *one* dissipative element (resistor, damper, thermal resistance). **Second-order systems** arise when there are *two* energy storage elements that can exchange energy

(inductor-capacitor, mass-spring, kinetic-potential energy).

Higher-order systems can almost always be decomposed into combinations of first and second order subsystems. This is the mathematical content of partial fraction expansion and modal decomposition. The eigenvalues (poles) of any linear system are either real (corresponding to first-order exponential modes) or occur in complex conjugate pairs (corresponding to second-order oscillatory modes).

Thus, mastering first and second order systems provides the complete toolkit for understanding arbitrarily complex linear systems. This universality is why these canonical forms have been studied so intensively and why they form the foundation of control systems education.

5.2 Modern Applications

The principles developed by Rayleigh, the seismologists, and the early automotive and aeronautical engineers remain essential in contemporary engineering practice. We highlight four modern applications that illustrate the continuing relevance of first and second order system theory.

5.2.1 Audio Speaker Design and Room Acoustics

A loudspeaker driver is a nearly perfect realization of a second-order mechanical system. The cone and voice coil constitute the mass, the spider and surround provide spring force, and mechanical losses plus electrical damping provide energy dissipation.

The Thiele-Small parameters, developed in the 1970s, characterize loudspeaker drivers in terms of f_s (resonance frequency, analogous to $\omega_n/2\pi$) and Q_{ts} (total quality factor, where $Q = 1/2\zeta$). Speaker enclosure design involves deliberately modifying the effective spring constant and damping to achieve desired frequency response.

A sealed enclosure raises ω_n and typically results in $\zeta \approx 0.7$ for maximally flat response. A ported (bass reflex) enclosure creates a coupled two-degree-of-freedom system that extends low-frequency response at the cost of increased group delay—a direct consequence of the more complex pole structure.

5.2.2 Building Earthquake Response

Modern earthquake engineering applies second-order system theory at the building scale. A multi-story building can be modeled as a chain of coupled mass-spring-damper systems, with each floor constituting a mass and the structural columns providing stiffness and damping.

Building codes specify design requirements in terms of natural frequencies (which must avoid matching common earthquake frequencies of 0.5–5 Hz) and damping ratios (typically $\zeta = 0.02$ – 0.05 for steel structures). Base isolation systems and tuned mass dampers are explicit applications of second-order system theory to reduce earthquake damage.

The Taipei 101 tower contains a 730-tonne tuned mass damper—the largest in the world—that reduces building sway due to wind and earthquakes by approximately 40%. The damper's natural frequency is precisely tuned to match the building's fundamental mode.

5.2.3 Drug Absorption Kinetics

Pharmacokinetics—the study of how drugs move through the body—extensively uses first-order models. The simplest compartmental model assumes that drug absorption and elimination follow first-order kinetics:

$$\frac{dC}{dt} = -k_e C \quad (5.2)$$

where C is drug concentration and k_e is the elimination rate constant. The time constant $\tau = 1/k_e$ determines the drug's half-life: $t_{1/2} = \tau \ln 2 \approx 0.693\tau$.

More complex two-compartment models yield second-order dynamics, accounting for distribution between central and peripheral compartments. Understanding these dynamics is essential for designing dosing regimens that maintain therapeutic concentrations while avoiding toxicity.

5.2.4 Electric Vehicle Battery Thermal Management

Lithium-ion batteries in electric vehicles require precise thermal management to optimize performance, longevity, and safety. The thermal dynamics of battery packs are well-described by first-order models:

$$\frac{dT}{dt} = \frac{1}{\tau_{th}} (T_{ambient} - T + R_{th} \cdot P_{heat}) \quad (5.3)$$

where τ_{th} is the thermal time constant (typically 5–30 minutes for EV battery packs) and R_{th} is the thermal resistance. Active cooling systems are designed to reduce τ_{th} and maintain battery temperature within optimal bounds.

The interplay between thermal dynamics and electrochemical dynamics creates complex coupled systems, but the design process begins with understanding each subsystem as a first or second-order system.

5.3 Mathematical Foundation

We now develop the complete mathematical framework for first and second order systems, emphasizing the connection between transfer function parameters, pole locations, and time-domain response characteristics.

5.3.1 First-Order Systems

Definition 5.1 (First-Order Transfer Function). A first-order system is characterized by the transfer function

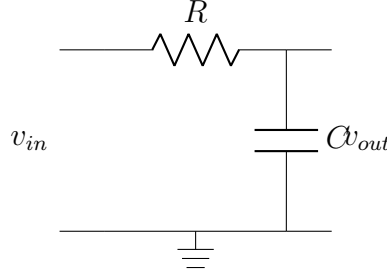
$$G(s) = \frac{K}{\tau s + 1} \quad (5.4)$$

where K is the DC gain and $\tau > 0$ is the time constant.

The single pole of this system is located at $s = -1/\tau$. Since $\tau > 0$, the pole is always in the left half-plane, guaranteeing stability.

Physical Interpretation of the Time Constant

The time constant τ has profound physical significance. Consider a simple RC circuit with resistance R and capacitance C :



The governing differential equation is:

$$RC \frac{dv_{out}}{dt} + v_{out} = v_{in} \quad (5.5)$$

Taking the Laplace transform with zero initial conditions:

$$G(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{RCs + 1} \quad (5.6)$$

Thus $\tau = RC$, the product of resistance and capacitance. Physically, τ represents the time required for the capacitor to charge through the resistor. The time constant appears in analogous forms across all physical domains:

Domain	Storage	Dissipation	Time Constant
Electrical	Capacitance C	Resistance R	$\tau = RC$
Thermal	Heat capacity C_{th}	Thermal resistance R_{th}	$\tau = R_{th}C_{th}$
Mechanical	None (first-order requires viscous element)		
Fluid	Tank volume V	Flow resistance R_f	$\tau = R_f V$

Step Response Derivation

For a unit step input, $U(s) = 1/s$, the output in the Laplace domain is:

$$Y(s) = G(s)U(s) = \frac{K}{\tau s + 1} \cdot \frac{1}{s} \quad (5.7)$$

Using partial fraction expansion:

$$Y(s) = \frac{K}{s(\tau s + 1)} = \frac{A}{s} + \frac{B}{\tau s + 1} \quad (5.8)$$

Solving for coefficients:

$$A = \lim_{s \rightarrow 0} s \cdot Y(s) = K \quad (5.9)$$

$$B = \lim_{s \rightarrow -1/\tau} (\tau s + 1) \cdot Y(s) = \frac{K}{-1/\tau} \cdot \tau = -K\tau \quad (5.10)$$

Therefore:

$$Y(s) = K \left(\frac{1}{s} - \frac{\tau}{\tau s + 1} \right) = K \left(\frac{1}{s} - \frac{1}{s + 1/\tau} \right) \quad (5.11)$$

Taking the inverse Laplace transform:

$$y(t) = K (1 - e^{-t/\tau}), \quad t \geq 0 \quad (5.12)$$

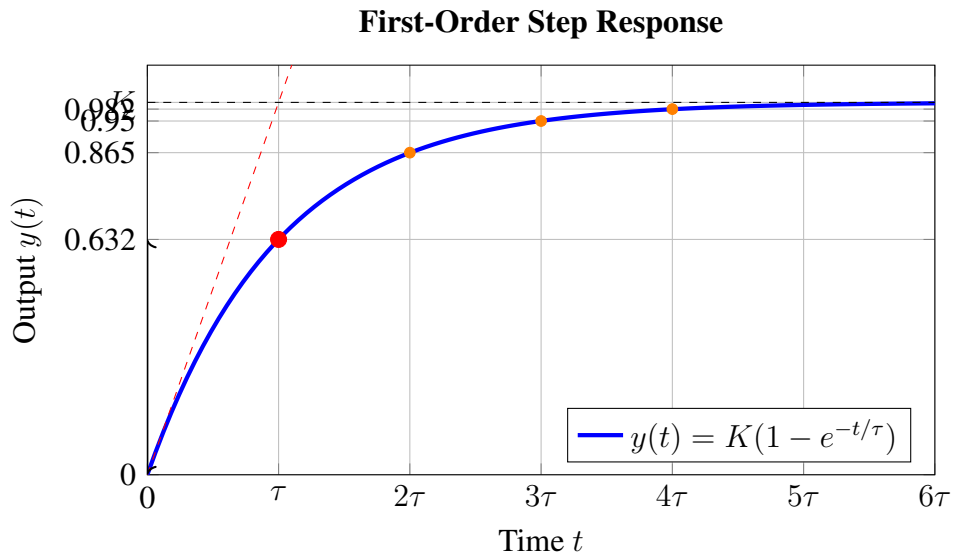


Figure 5.1: First-order step response showing the characteristic exponential approach to steady state. The tangent line at $t = 0$ intersects the steady-state value at $t = \tau$.

Key Time-Domain Specifications

First-Order Response Characteristics

The **time constant** has fundamental significance: at $t = \tau$, the output reaches 63.2% of its final value, since $y(\tau) = K(1 - e^{-1}) = K(1 - 0.368) = 0.632K$.

The **settling time** (2% criterion), defined as the time to reach and stay within 2% of the final value, is approximately $t_s \approx 4\tau$ since $e^{-4} \approx 0.018 \approx 2\%$.

The **rise time** from 10% to 90% is given by $t_r = \tau(\ln 0.9 - \ln 0.1) = \tau \ln 9 \approx 2.2\tau$.

The **initial slope** of the response is K/τ , and the tangent line at $t = 0$ intersects $y = K$ at $t = \tau$.

5.3.2 Second-Order Systems

Definition 5.2 (Standard Second-Order Transfer Function). A second-order system in standard form is characterized by:

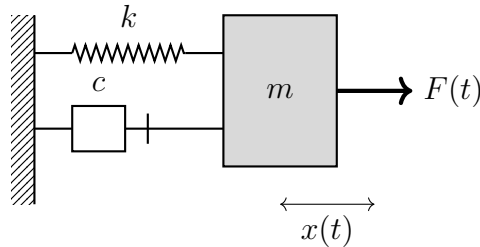
$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (5.13)$$

where $\omega_n > 0$ is the **natural frequency** (rad/s) and $\zeta \geq 0$ is the **damping ratio** (dimensionless).

This transfer function has unity DC gain: $G(0) = 1$. For systems with gain $K \neq 1$, multiply the numerator by K .

Physical Origin of Parameters

Consider the canonical mass-spring-damper system:



Newton's second law gives:

$$m\ddot{x} + c\dot{x} + kx = F(t) \quad (5.14)$$

Taking the Laplace transform:

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + cs + k} = \frac{1/m}{s^2 + (c/m)s + k/m} \quad (5.15)$$

Comparing with the standard form:

$$\boxed{\omega_n = \sqrt{\frac{k}{m}}, \quad \zeta = \frac{c}{2\sqrt{km}} = \frac{c}{2m\omega_n}} \quad (5.16)$$

The **natural frequency** ω_n is the frequency at which the undamped system would oscillate (when $c = 0$). The **damping ratio** ζ compares the actual damping to the critical damping $c_{cr} = 2\sqrt{km}$.

Characteristic Equation and Poles

The characteristic equation is:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \quad (5.17)$$

Applying the quadratic formula:

$$\boxed{s_{1,2} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}} \quad (5.18)$$

The nature of the roots depends critically on ζ :

Classification by Damping Ratio

Underdamped systems ($0 < \zeta < 1$) have complex conjugate poles given by $s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -\sigma \pm j\omega_d$, where $\sigma = \zeta\omega_n$ is the damping factor and $\omega_d = \omega_n\sqrt{1-\zeta^2}$ is the damped natural frequency.

Critically damped systems ($\zeta = 1$) have a repeated real pole at $s_{1,2} = -\omega_n$ (double pole).

Overdamped systems ($\zeta > 1$) have two distinct real poles at $s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2-1}$. Both poles are negative (stable), with s_1 closer to the origin acting as the dominant pole.

Step Response: Underdamped Case ($\zeta < 1$)

For the underdamped case, the step response derivation yields:

$$y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \phi), \quad t \geq 0 \quad (5.19)$$

where $\omega_d = \omega_n\sqrt{1-\zeta^2}$ and $\phi = \arctan(\sqrt{1-\zeta^2}/\zeta) = \arccos(\zeta)$.

An equivalent and often more useful form is:

$$y(t) = 1 - e^{-\zeta\omega_n t} \left(\cos \omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t \right) \quad (5.20)$$

Step Response: Critically Damped Case ($\zeta = 1$)

When $\zeta = 1$, we have a repeated pole at $s = -\omega_n$:

$$y(t) = 1 - e^{-\omega_n t}(1 + \omega_n t), \quad t \geq 0 \quad (5.21)$$

This represents the fastest non-oscillatory response—any reduction in ζ causes overshoot, while any increase causes sluggishness.

Step Response: Overdamped Case ($\zeta > 1$)

For the overdamped case, define:

$$s_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2-1} = -\omega_n(\zeta - \sqrt{\zeta^2-1}) \quad (5.22)$$

$$s_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2-1} = -\omega_n(\zeta + \sqrt{\zeta^2-1}) \quad (5.23)$$

Note that $|s_2| > |s_1|$, so s_1 is the dominant (slower) pole. The step response is:

$$y(t) = 1 - \frac{s_2}{s_2 - s_1} e^{s_1 t} + \frac{s_1}{s_2 - s_1} e^{s_2 t}, \quad t \geq 0 \quad (5.24)$$

For large ζ , $s_2 \gg s_1$ and the response is dominated by the slower mode.

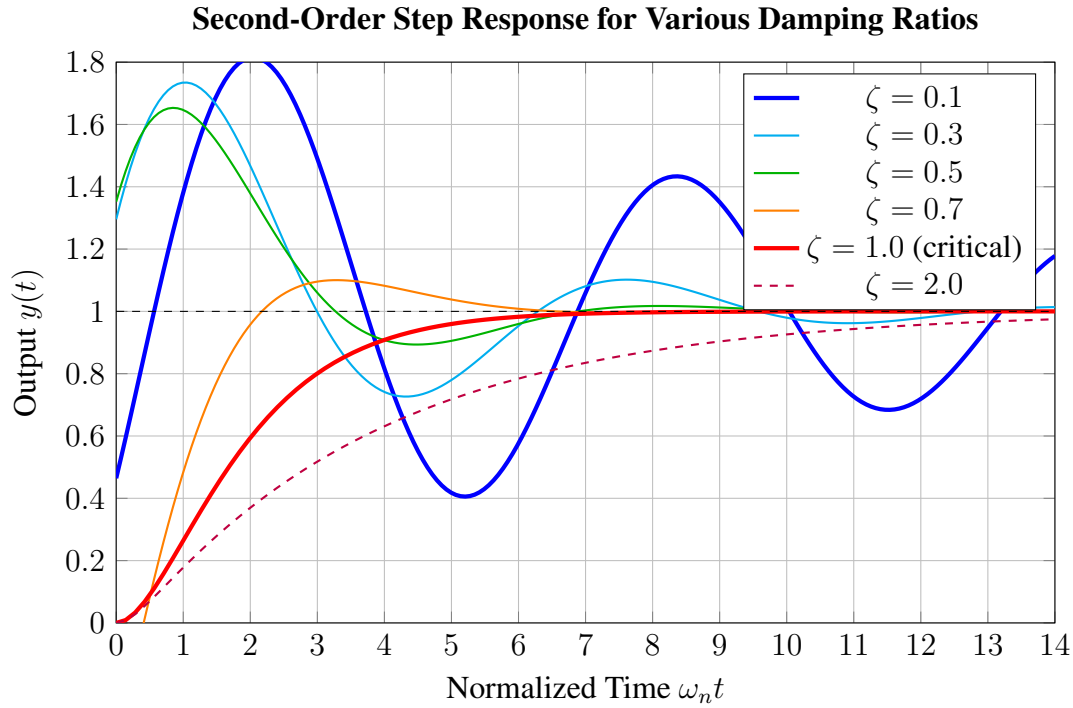


Figure 5.2: Step responses for second-order systems with various damping ratios. Underdamped responses ($\zeta < 1$) exhibit overshoot and oscillation. The critically damped response ($\zeta = 1$) is the fastest without overshoot. Overdamped responses ($\zeta > 1$) are sluggish.

Time-Domain Specifications for Underdamped Systems

For underdamped second-order systems, several key specifications characterize the transient response:

Second-Order Transient Response Specifications

Percent Overshoot (PO):

$$PO = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\% \quad (5.25)$$

Depends only on ζ , not on ω_n .

Peak Time (time to first peak):

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} \quad (5.26)$$

Rise Time (0% to 100%, approximate):

$$t_r \approx \frac{1.8}{\omega_n} \quad \text{for } 0.3 < \zeta < 0.8 \quad (5.27)$$

Settling Time (2% criterion):

$$t_s \approx \frac{4}{\zeta \omega_n} = \frac{4}{\sigma} \quad (5.28)$$

Table 5.1: Percent Overshoot vs. Damping Ratio

ζ	PO (%)	ζ	PO (%)	ζ	PO (%)
0.1	72.9	0.4	25.4	0.7	4.6
0.2	52.7	0.5	16.3	0.8	1.5
0.3	37.2	0.6	9.5	0.9	0.2

5.3.3 Pole Locations and Response Characteristics

The location of poles in the complex s -plane provides immediate insight into system behavior. This geometric interpretation is one of the most powerful tools in control system analysis.

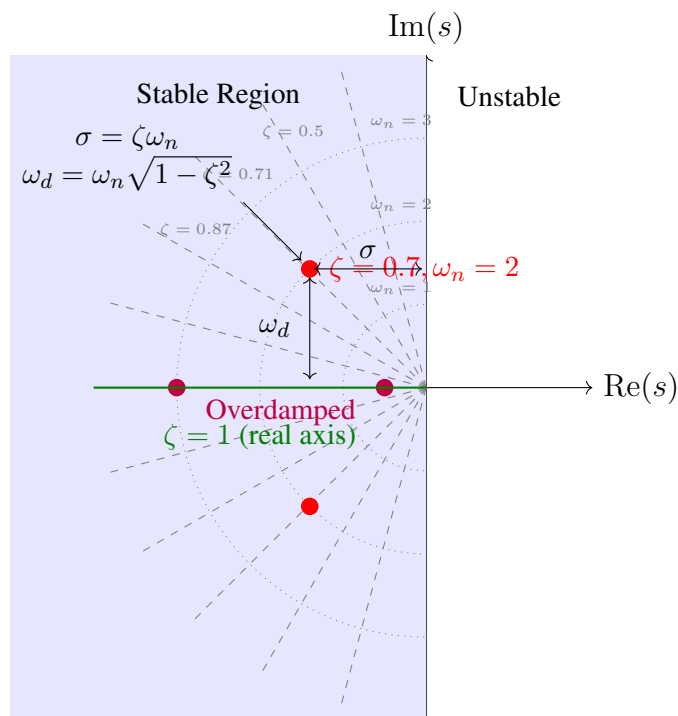


Figure 5.3: The s -plane showing regions of constant damping ratio (radial lines from origin) and constant natural frequency (semicircles). Poles in the left half-plane are stable; poles on the real axis correspond to $\zeta \geq 1$.

Key observations from the s -plane representation connect pole geometry to dynamic behavior. The **horizontal distance from the imaginary axis**, $\sigma = \zeta \omega_n$, determines the exponential decay rate; larger σ means faster settling. The **vertical distance from the real axis**, $\omega_d = \omega_n \sqrt{1 - \zeta^2}$,

determines the oscillation frequency, with larger ω_d producing faster oscillation. The **distance from the origin**, $\omega_n = \sqrt{\sigma^2 + \omega_d^2}$, equals the natural frequency. Finally, the **angle from the negative real axis**, $\theta = \arccos \zeta$, is directly related to the damping ratio.

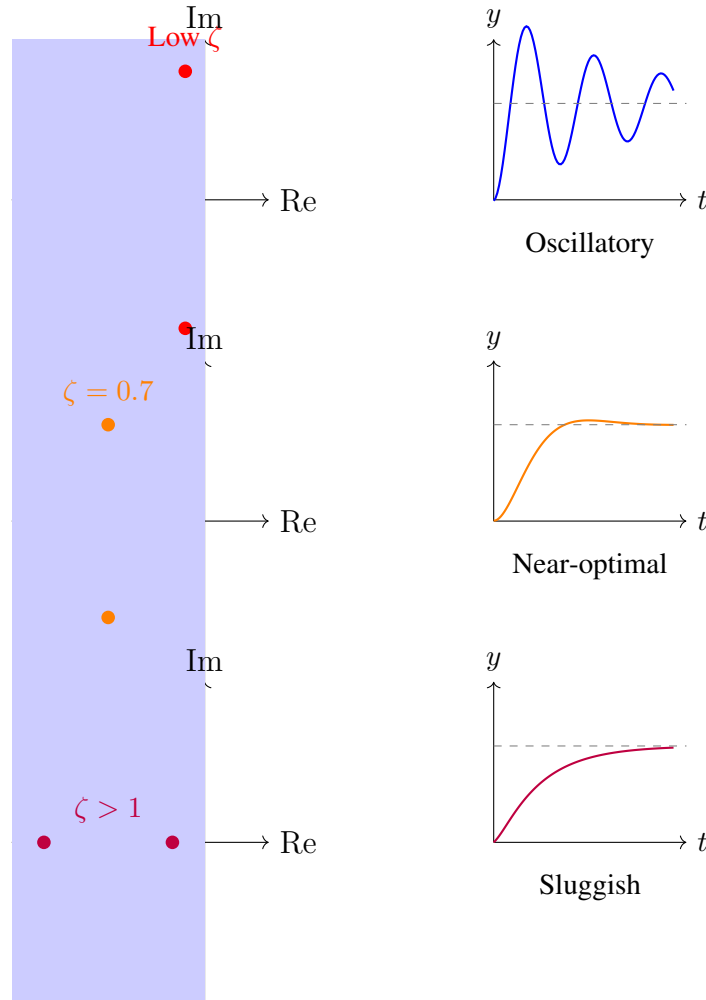


Figure 5.4: Relationship between pole locations and step response characteristics. Top: lightly damped (poles near imaginary axis). Middle: well-damped (poles at 45). Bottom: overdamped (poles on real axis).

5.4 Laboratory Experiment: Variable Damping Mass-Spring System

This section describes a hands-on experiment to build and characterize a second-order mechanical system with adjustable damping. Students will observe the transition from underdamped to critically damped to overdamped response and extract system parameters from measured data.

Learning Objectives

Upon completion of this experiment, students will be able to construct a physical mass-spring-damper system with variable damping, measure step response using digital instrumentation, extract ζ and ω_n from experimental data, and correlate physical damping changes with observed response characteristics.

5.4.1 Apparatus Design and Construction

Frame and Mass Assembly

The experimental apparatus consists of a vertical mass-spring system with adjustable viscous damping. The frame can be constructed from 3D-printed components, aluminum extrusion, or wooden dowels.

Bill of Materials:

Component	Specification
Vertical guide rail	Aluminum or 3D-printed, 300mm length
Mass carrier	3D-printed or machined block, approximately 100g base mass
Additional masses	Washers, nuts, or calibrated weights (50g, 100g, 200g)
Linear bearings	Low-friction sleeve for vertical motion
Mounting base	Stable platform to prevent vibration

Spring Element

For accessible experimentation, rubber bands provide variable spring constants. Use consistent rubber bands (size #64 or similar) for reproducibility. Parallel bands increase k while a series arrangement decreases k . Alternatively, extension springs from a hardware store with known spring constants can be used for more precise measurements.

Characterize the spring constant by hanging known masses and measuring displacement:

$$k = \frac{mg}{\Delta x} \quad (5.29)$$

Variable Damping Mechanism

Three options for adjustable damping are presented:

Option	Description and Adjustment Method
A: Water Dashpot	A cylinder partially filled with water contains a plunger attached to the moving mass with orifices of various sizes. Damping is varied by changing orifice size or water viscosity (add glycerin).
B: Eddy Current Damper	An aluminum or copper plate attached to the moving mass passes near permanent magnets. Damping is adjusted by varying magnet proximity. This option provides smooth, velocity-proportional damping.
C: Friction Pad	A felt or foam pad is pressed against the guide rail with adjustable normal force via a thumbscrew. Note: this provides approximately velocity-independent Coulomb friction rather than ideal viscous damping.

Instrumentation

Distance Measurement:

Sensor	Range	Resolution
Ultrasonic (HC-SR04)	2cm–400cm	~3mm
Time-of-flight (VL53L0X)	3cm–200cm	~1mm
Infrared (Sharp GP2Y0A21)	10cm–80cm	~2mm

Data Acquisition:

Data acquisition requires an Arduino Uno or similar microcontroller operating at a minimum sampling rate of 50 Hz, though 100+ Hz is preferable for capturing fast dynamics. Data can be logged to a serial monitor for real-time viewing or to an SD card for later analysis.

Data Acquisition Algorithm:

Algorithm 8 Second-Order System Distance Measurement

```

1: Configuration:
2:   Set trigger pin  $\leftarrow$  output pin 9
3:   Set echo pin  $\leftarrow$  input pin 10
4:   Set serial baud rate  $\leftarrow$  115200
5:
6: Initialization:
7: Record start time  $t_0 \leftarrow$  current time in milliseconds
8:
9: Main Loop:
10: while data acquisition active do
11:    $d \leftarrow$  MEASUREDISTANCE( )
12:    $t_{elapsed} \leftarrow$  current time  $- t_0$ 
13:   Output  $(t_{elapsed}, d)$  to serial port
14:   Wait 10 ms  $\triangleright$  100 Hz sampling rate
15: end while
16:
17: function MEASUREDISTANCE( )
18:   Set trigger pin LOW
19:   Wait 2  $\mu$ s
20:   Set trigger pin HIGH
21:   Wait 10  $\mu$ s
22:   Set trigger pin LOW
23:    $t_{pulse} \leftarrow$  measure echo pulse duration
24:   return  $t_{pulse} \times 0.034/2$   $\triangleright$  distance in cm
25: end function

```

5.4.2 Experimental Procedure

System Characterization

System characterization proceeds in four stages. First, **measure the mass** by weighing the total moving mass m including the carrier, sensor mount, and any added weights. Second, **measure the spring constant** with the damping mechanism removed; displace the mass by measured amounts, apply several known forces, and calculate k from the slope of force vs. displacement. Third, **predict the natural frequency** by calculating $\omega_n = \sqrt{k/m}$ and $f_n = \omega_n/(2\pi)$. Finally, **verify the undamped frequency** by applying a step input with minimal damping and measuring the oscillation period T , then compare $f_{measured} = 1/T$ with the prediction.

Step Response Measurement

To measure the step response, position the mass at a displaced equilibrium (e.g., compress the spring by 2–3 cm), then start data acquisition. Release the mass smoothly, taking care not to impart any initial velocity. Record position vs. time for at least 5 natural periods or until the system settles. Repeat the measurement 3–5 times for statistical confidence.

Damping Variation Study

Perform the step response measurement for at least four damping levels: minimal damping (underdamped with high oscillation), moderate damping (underdamped with visible overshoot), high damping (near critical with minimal overshoot), and maximum damping (overdamped with no oscillation).

5.4.3 Data Analysis and Parameter Extraction

Extracting Parameters from Underdamped Response

Given a measured underdamped step response, extract ζ and ω_n using the following methods:

Method 1: Logarithmic Decrement

Measure successive peak amplitudes $x_1, x_2, x_3, \dots$ above the final value. The logarithmic decrement is:

$$\delta = \ln \left(\frac{x_n}{x_{n+1}} \right) \quad (5.30)$$

The damping ratio is:

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}} \quad (5.31)$$

For small damping, $\zeta \approx \delta/(2\pi)$.

Method 2: Percent Overshoot

Measure the first peak value y_{max} and final value y_{ss} :

$$PO = \frac{y_{max} - y_{ss}}{y_{ss}} \times 100\% \quad (5.32)$$

Solve for ζ :

$$\zeta = \frac{-\ln(PO/100)}{\sqrt{\pi^2 + \ln^2(PO/100)}} \quad (5.33)$$

Method 3: Damped Period Measurement

Measure the period of oscillation T_d between successive peaks:

$$\omega_d = \frac{2\pi}{T_d} \quad (5.34)$$

With ζ known from overshoot:

$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}} \quad (5.35)$$

Extracting Parameters from Overdamped Response

For overdamped systems, fitting the response to a sum of two exponentials is required:

$$y(t) = y_{ss} (1 - A_1 e^{-t/\tau_1} - A_2 e^{-t/\tau_2}) \quad (5.36)$$

Use nonlinear curve fitting (MATLAB's `fit` or Python's `scipy.optimize.curve_fit`) to extract τ_1 and τ_2 . The poles are $s_1 = -1/\tau_1$ and $s_2 = -1/\tau_2$.

Sample Data Analysis

Table 5.2: Sample Experimental Results

Damping Setting	PO (%)	ζ	T_d (s)	ω_n (rad/s)	Behavior
Minimal	52	0.21	0.45	14.3	Underdamped
Low	25	0.40	0.48	14.3	Underdamped
Medium	5	0.69	0.55	15.8	Underdamped
High	0	> 1	—	14.1	Overdamped

5.4.4 Discussion Questions

Consider the following questions when analyzing your experimental results. Why does ω_n remain approximately constant across damping settings? What practical limitations prevent achieving exactly $\zeta = 1$ (critical damping)? How does the friction-based damper differ from ideal viscous damping, and what effect does this have on the response shape? If you wanted to double the natural frequency without changing the damping ratio, what physical parameters would you modify? Finally, compare your measured settling times with the theoretical prediction $t_s = 4/(\zeta\omega_n)$ and identify sources of error that might explain any discrepancies.

5.5 Worked Examples

This section presents detailed solutions to representative problems involving first and second order systems. Each example emphasizes a specific concept or technique.

Time Constant from RC Circuit Values

An RC low-pass filter has $R = 10 \text{ k}\Omega$ and $C = 47 \text{ }\mu\text{F}$. Find:

- (a) The time constant τ
- (b) The transfer function
- (c) The settling time (2% criterion)
- (d) The output voltage 0.5 seconds after applying a 5V step input

Solution:

(a) Time Constant:

$$\tau = RC = (10 \times 10^3 \Omega)(47 \times 10^{-6} \text{ F}) = 0.47 \text{ s} \quad (5.37)$$

(b) Transfer Function: The standard first-order form is:

$$G(s) = \frac{1}{\tau s + 1} = \frac{1}{0.47s + 1} \quad (5.38)$$

Or in terms of the pole location:

$$G(s) = \frac{2.13}{s + 2.13} \quad (5.39)$$

The single pole is at $s = -1/\tau = -2.13 \text{ rad/s}$.

(c) Settling Time:

$$t_s = 4\tau = 4(0.47) = 1.88 \text{ s} \quad (5.40)$$

(d) Output Voltage at $t = 0.5 \text{ s}$:

For a 5V step input:

$$v_{out}(t) = 5(1 - e^{-t/\tau}) = 5(1 - e^{-0.5/0.47}) \quad (5.41)$$

$$v_{out}(0.5) = 5(1 - e^{-1.064}) = 5(1 - 0.345) = 5(0.655) = \boxed{3.28 \text{ V}} \quad (5.42)$$

This represents 65.5% of the final value, consistent with being just past one time constant.

Finding ζ and ω_n from Transfer Function

Given the transfer function:

$$G(s) = \frac{50}{s^2 + 6s + 25} \quad (5.43)$$

Determine ω_n , ζ , the pole locations, and characterize the expected step response.

Solution:

Compare with the standard form $G(s) = K\omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = s^2 + 6s + 25 \quad (5.44)$$

Therefore:

$$\omega_n^2 = 25 \implies \boxed{\omega_n = 5 \text{ rad/s}} \quad (5.45)$$

$$2\zeta\omega_n = 6 \implies \zeta = \frac{6}{2(5)} = \boxed{0.6} \quad (5.46)$$

The DC gain is $K = 50/25 = 2$.

Pole Locations:

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2} \quad (5.47)$$

$$= -0.6(5) \pm j(5)\sqrt{1 - 0.36} \quad (5.48)$$

$$= -3 \pm j(5)(0.8) \quad (5.49)$$

$$= \boxed{-3 \pm j4} \quad (5.50)$$

Response Characteristics:

Since $\zeta = 0.6 < 1$, the system is **underdamped**.

Percent overshoot:

$$PO = e^{-\pi(0.6)/\sqrt{1-0.36}} \times 100\% = e^{-\pi(0.6)/0.8} \times 100\% = e^{-2.356} \times 100\% \approx \boxed{9.5\%} \quad (5.51)$$

Damped frequency:

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 5(0.8) = 4 \text{ rad/s} \quad (5.52)$$

Peak time:

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{4} \approx 0.785 \text{ s} \quad (5.53)$$

Settling time:

$$t_s = \frac{4}{\zeta \omega_n} = \frac{4}{0.6 \times 5} = \frac{4}{3} \approx 1.33 \text{ s} \quad (5.54)$$

Example 5.1 (Calculating Overshoot and Settling Time). A second-order system has natural frequency $\omega_n = 10 \text{ rad/s}$ and damping ratio $\zeta = 0.4$. Calculate:

- (a) Percent overshoot
- (b) Peak time
- (c) Settling time (2% criterion)
- (d) Rise time (approximate)

Solution:

(a) Percent Overshoot:

$$PO = \exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right) \times 100\% \quad (5.55)$$

$$PO = \exp\left(\frac{-\pi(0.4)}{\sqrt{1-0.16}}\right) \times 100\% = \exp\left(\frac{-1.257}{0.917}\right) \times 100\% \quad (5.56)$$

$$PO = e^{-1.371} \times 100\% = 0.254 \times 100\% = \boxed{25.4\%} \quad (5.57)$$

(b) Peak Time:

First, calculate the damped natural frequency:

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 10\sqrt{1 - 0.16} = 10(0.917) = 9.17 \text{ rad/s} \quad (5.58)$$

Then:

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{9.17} = \boxed{0.343 \text{ s}} \quad (5.59)$$

(c) Settling Time:

$$t_s = \frac{4}{\zeta \omega_n} = \frac{4}{0.4 \times 10} = \frac{4}{4} = \boxed{1.0 \text{ s}} \quad (5.60)$$

(d) Rise Time (approximate formula for $0.3 < \zeta < 0.8$):

$$t_r \approx \frac{1.8}{\omega_n} = \frac{1.8}{10} = \boxed{0.18 \text{ s}} \quad (5.61)$$

A more accurate empirical formula is:

$$t_r \approx \frac{1 + 1.1\zeta + 1.4\zeta^2}{\omega_n} = \frac{1 + 0.44 + 0.224}{10} = \frac{1.664}{10} = 0.166 \text{ s} \quad (5.62)$$

□

Example 5.2 (Pole Locations from Response Requirements). A control system must meet the following specifications: settling time $t_s \leq 2$ s and percent overshoot $PO \leq 10\%$. Determine the required region in the s -plane for the dominant poles.

Solution:

Settling Time Constraint:

The settling time constraint $t_s \leq 2$ s requires:

$$\frac{4}{\sigma} \leq 2 \implies \sigma \geq 2 \quad (5.63)$$

where $\sigma = \zeta\omega_n$ is the real part of the poles. This defines a vertical line at $\text{Re}(s) = -2$; poles must lie to the *left* of this line.

Overshoot Constraint:

The overshoot constraint $PO \leq 10\%$ requires:

$$\exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right) \leq 0.10 \quad (5.64)$$

Taking the natural logarithm:

$$\frac{-\pi\zeta}{\sqrt{1-\zeta^2}} \leq \ln(0.10) = -2.303 \quad (5.65)$$

$$\frac{\pi\zeta}{\sqrt{1-\zeta^2}} \geq 2.303 \quad (5.66)$$

Squaring both sides:

$$\frac{\pi^2\zeta^2}{1-\zeta^2} \geq 5.304 \quad (5.67)$$

$$\pi^2\zeta^2 \geq 5.304 - 5.304\zeta^2 \quad (5.68)$$

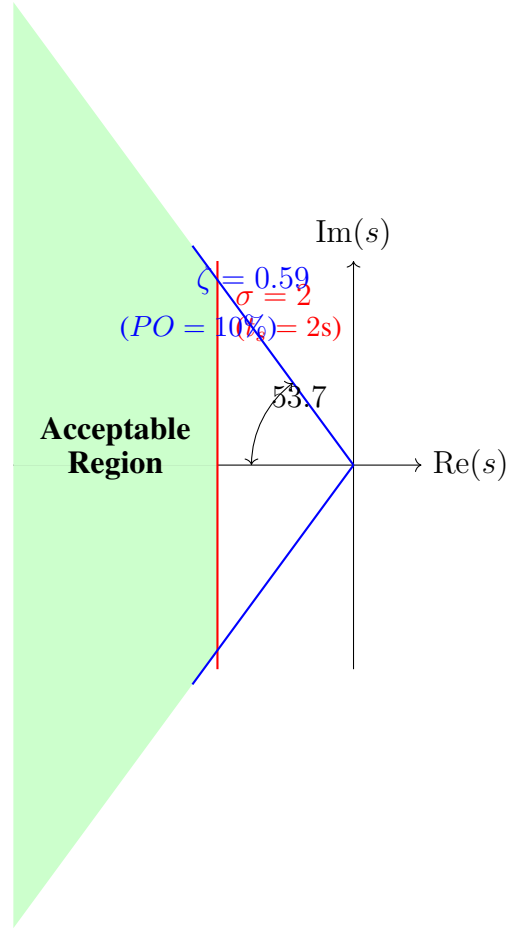
$$\zeta^2(\pi^2 + 5.304) \geq 5.304 \quad (5.69)$$

$$\zeta^2 \geq \frac{5.304}{9.87 + 5.304} = \frac{5.304}{15.17} = 0.350 \quad (5.70)$$

$$\zeta \geq 0.591 \quad (5.71)$$

Since $\cos \theta = \zeta$, this corresponds to $\theta \leq \arccos(0.591) = 53.7$ from the negative real axis.

Combined Region:



The dominant poles must be placed in the shaded region: to the left of $\text{Re}(s) = -2$ and within $\pm 53.7^\circ$ of the negative real axis.

One acceptable pole pair would be:

$$s = -3 \pm j2.5 \quad (\sigma = 3, \omega_d = 2.5, \zeta = 0.77, \omega_n = 3.91) \quad (5.72)$$

Verification: $t_s = 4/3 = 1.33 \text{ s} < 2 \text{ s} \checkmark$, and $PO = 2.6\% < 10\% \checkmark$.

□

Example 5.3 (System Identification from Step Response). The following step response data was measured from an unknown second-order system: the final (steady-state) value is $y_{ss} = 2.0$, the first peak value is $y_{max} = 2.52$ occurring at $t_p = 0.25 \text{ s}$, and the second peak value is $y_2 = 2.20$ at $t = 0.5 \text{ s}$. Determine the transfer function of the system.

Solution:

Step 1: Calculate Percent Overshoot

$$PO = \frac{y_{max} - y_{ss}}{y_{ss}} \times 100\% = \frac{2.52 - 2.0}{2.0} \times 100\% = 26\% \quad (5.73)$$

Step 2: Find Damping Ratio from Overshoot

$$\zeta = \frac{-\ln(PO/100)}{\sqrt{\pi^2 + \ln^2(PO/100)}} = \frac{-\ln(0.26)}{\sqrt{\pi^2 + \ln^2(0.26)}} \quad (5.74)$$

$$\zeta = \frac{1.347}{\sqrt{9.87 + 1.814}} = \frac{1.347}{\sqrt{11.68}} = \frac{1.347}{3.42} = 0.394 \quad (5.75)$$

Step 3: Verify Using Logarithmic Decrement

The ratio of successive overshoots:

$$\frac{y_{max} - y_{ss}}{y_2 - y_{ss}} = \frac{2.52 - 2.0}{2.20 - 2.0} = \frac{0.52}{0.20} = 2.6 \quad (5.76)$$

Logarithmic decrement:

$$\delta = \ln(2.6) = 0.956 \quad (5.77)$$

Damping ratio:

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}} = \frac{0.956}{\sqrt{39.48 + 0.914}} = \frac{0.956}{6.36} = 0.150 \quad (5.78)$$

Note: The discrepancy between methods (0.394 vs. 0.150) suggests measurement error or non-ideal system behavior. For this example, we proceed with $\zeta \approx 0.4$.

Step 4: Find Natural Frequency

The damped period is:

$$T_d = 2 \times t_p = 2 \times 0.25 = 0.5 \text{ s} \quad (5.79)$$

$$\omega_d = \frac{2\pi}{T_d} = \frac{2\pi}{0.5} = 12.57 \text{ rad/s} \quad (5.80)$$

Natural frequency:

$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}} = \frac{12.57}{\sqrt{1 - 0.16}} = \frac{12.57}{0.917} = 13.7 \text{ rad/s} \quad (5.81)$$

Step 5: Determine DC Gain

The DC gain equals the steady-state value for a unit step input:

$$K = y_{ss} = 2.0 \quad (5.82)$$

Step 6: Write Transfer Function

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (5.83)$$

$$G(s) = \frac{2.0 \times (13.7)^2}{s^2 + 2(0.4)(13.7)s + (13.7)^2} \quad (5.84)$$

$$\boxed{G(s) = \frac{375}{s^2 + 11s + 188}} \quad (5.85)$$

□

Example 5.4 (Mixed First and Second Order System). Consider the transfer function:

$$G(s) = \frac{100}{(s + 5)(s^2 + 4s + 20)} \quad (5.86)$$

- (a) Identify all poles and classify each as first or second order.
- (b) Determine which pole(s) dominate the response.
- (c) Estimate the settling time.

Solution:

(a) Pole Identification:

Real pole from $(s + 5)$:

$$s_1 = -5 \quad (5.87)$$

Complex poles from $(s^2 + 4s + 20)$:

$$\omega_n^2 = 20 \implies \omega_n = \sqrt{20} = 4.47 \text{ rad/s} \quad (5.88)$$

$$2\zeta\omega_n = 4 \implies \zeta = \frac{4}{2(4.47)} = 0.447 \quad (5.89)$$

$$s_{2,3} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -2 \pm j(4.47)(0.894) = -2 \pm j4 \quad (5.90)$$

Summary: The system contains a first-order mode with a pole at $s_1 = -5$ (time constant $\tau_1 = 0.2 \text{ s}$) and a second-order mode with poles at $s_{2,3} = -2 \pm j4$ ($\zeta = 0.447$, $\omega_n = 4.47 \text{ rad/s}$).

(b) Dominant Poles:

The dominant poles are those closest to the imaginary axis, as they decay most slowly. Here, the real part of s_1 is -5 while the real part of $s_{2,3}$ is -2 .

Since $|-2| < |-5|$, the complex conjugate pair at $s = -2 \pm j4$ are the **dominant poles**. The first-order mode decays 2.5 times faster than the oscillatory mode.

(c) Settling Time Estimate:

Using the dominant pole real part:

$$t_s \approx \frac{4}{|\text{Re}(s_{\text{dominant}})|} = \frac{4}{2} = \boxed{2 \text{ s}} \quad (5.91)$$

The response will exhibit oscillation (due to the underdamped second-order mode with $\zeta = 0.447$, giving about 21% overshoot) with an envelope decay determined by e^{-2t} .

□

5.6 Summary and Key Formulas

First-Order Systems Summary

Transfer Function: $G(s) = \frac{K}{\tau s + 1}$

Pole: $s = -1/\tau$

Step Response: $y(t) = K(1 - e^{-t/\tau})$

Key Times: At $t = \tau$, the output reaches 63.2% of the final value. The rise time (10%–90%) is $t_r = 2.2\tau$, and the settling time (2%) is $t_s = 4\tau$.

Second-Order Systems Summary

Transfer Function: $G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

Poles: $s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$

Physical Parameters: $\omega_n = \sqrt{k/m}$, $\zeta = c/(2\sqrt{km})$

Underdamped Response ($\zeta < 1$): The damped frequency is $\omega_d = \omega_n\sqrt{1 - \zeta^2}$, the percent overshoot is $PO = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\%$, the peak time is $t_p = \pi/\omega_d$, and the settling time is $t_s = 4/(\zeta\omega_n)$.

Critically Damped ($\zeta = 1$): $y(t) = K[1 - e^{-\omega_n t}(1 + \omega_n t)]$

Overdamped ($\zeta > 1$): Sum of two exponentials with time constants $1/|s_1|$ and $1/|s_2|$

5.7 Exercises

Exercise 5.1. An RC circuit has $R = 47 \text{ k}\Omega$ and $C = 10 \text{ }\mu\text{F}$. Calculate the time constant, the frequency at which the output is attenuated by 3 dB, and the time required for the output to reach 99% of its final value after a step input.

Exercise 5.2. A mass-spring-damper system has $m = 2 \text{ kg}$, $k = 50 \text{ N/m}$, and $c = 8 \text{ N} \cdot \text{s/m}$. Determine ω_n , ζ , and classify the system as underdamped, critically damped, or overdamped.

Exercise 5.3. For the transfer function $G(s) = 25/(s^2 + 3s + 25)$:

- (a) Find ζ and ω_n
- (b) Calculate the percent overshoot
- (c) Determine the peak time and settling time
- (d) Sketch the expected step response

Exercise 5.4. Design a second-order system (specify ζ and ω_n) that meets the following specifications: settling time $\leq 0.5 \text{ s}$ and percent overshoot $\leq 5\%$.

Exercise 5.5. A step response measurement shows the following: initial value 0, final value 4.0, first peak 5.6 at $t = 0.1 \text{ s}$. Assuming a second-order system, estimate the transfer function.

Exercise 5.6. For the system $G(s) = 10/[(s+1)(s+10)]$, determine whether first-order or second-order behavior dominates the step response. Justify your answer and estimate the settling time.

Exercise 5.7. An automobile suspension is modeled as a second-order system with $\omega_n = 8 \text{ rad/s}$. What damping ratio provides 20% overshoot? If the sprung mass is 400 kg, calculate the required spring constant and damping coefficient.

Exercise 5.8. Prove that for an underdamped second-order system, the logarithmic decrement $\delta = \ln(x_n/x_{n+1})$ is related to the damping ratio by $\zeta = \delta/\sqrt{4\pi^2 + \delta^2}$.

Exercise 5.9. A seismograph has $\omega_n = 1 \text{ rad/s}$ and $\zeta = 0.7$. An earthquake produces ground acceleration $a(t) = A \sin(\omega t)$. For what range of earthquake frequencies ω will the seismograph output amplitude be within 10% of the true ground displacement amplitude?

Exercise 5.10. Consider the third-order system $G(s) = 50/[(s+1)(s^2 + 4s + 25)]$. Identify all poles, determine which are dominant, and estimate the settling time and percent overshoot of the step response.

5.8 Further Reading

1. Rayleigh, J.W.S., *The Theory of Sound*, Volumes 1 and 2, Dover Publications, 1945 (reprint of 1894 edition). The foundational text on vibration analysis.
2. Franklin, G.F., Powell, J.D., and Emami-Naeini, A., *Feedback Control of Dynamic Systems*, 8th ed., Pearson, 2019. Chapters 3–4 provide extensive treatment of dynamic response.

3. Ogata, K., *Modern Control Engineering*, 5th ed., Prentice Hall, 2010. Chapter 5 covers transient and steady-state response analysis.
4. Dorf, R.C. and Bishop, R.H., *Modern Control Systems*, 14th ed., Pearson, 2022. Comprehensive coverage of time-domain analysis.
5. Den Hartog, J.P., *Mechanical Vibrations*, 4th ed., Dover Publications, 1985. Classic text on mechanical oscillations with extensive practical examples.

Chapter 6

Stability: When Systems Blow Up

“The notion of stability is really almost a trivial thing... A stable system is one which, when disturbed from its state of equilibrium, returns to that state; an unstable system is one which departs more and more from equilibrium when slightly displaced.”

— James Clerk Maxwell, “On Governors” (1868)

Stability is perhaps the most fundamental concept in control systems engineering. An unstable system is not merely poorly performing—it is dangerous, potentially destructive, and fundamentally unusable. Before we can design controllers to achieve desired performance, we must first ensure that our systems will not “blow up.” This chapter develops the mathematical framework for analyzing stability and provides practical tools for determining whether a system is stable without solving the characteristic equation explicitly.

6.1 Historical Motivation: The Quest for Stable Machines

6.1.1 James Watt and the Governor Problem

The story of stability analysis begins with one of the most consequential inventions of the Industrial Revolution: James Watt’s centrifugal governor for steam engines (circa 1788). While Watt did not invent the centrifugal mechanism—it had been used in windmills since the 1740s—he was the first to apply it to the critical problem of regulating steam engine speed.

The governor operated on an elegantly simple principle. Two weighted balls were attached to a vertical spindle connected to the engine’s flywheel through a gear train. As the engine speed increased, centrifugal force caused the balls to swing outward and rise. This motion was linked through a lever mechanism to the steam throttle valve, reducing steam flow and thereby slowing the engine. Conversely, if the engine slowed, the balls dropped inward, opening the throttle and increasing power.

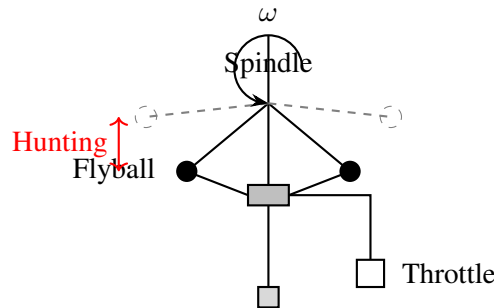


Figure 6.1: Watt’s centrifugal governor. Dashed lines show extended position at higher speed. “Hunting” oscillations occur when the system becomes marginally stable.

For decades, Watt governors worked adequately, but as steam engines grew larger and more powerful in the mid-19th century, a troubling phenomenon emerged: **hunting**. Instead of settling to a steady speed, some governors would oscillate continuously. The engine would speed up, causing the governor to close the throttle, which slowed the engine, causing the governor to open the throttle, which sped up the engine—an endless cycle that made precise control impossible and wore out machinery prematurely.

The hunting problem was not merely an annoyance; it represented a fundamental limitation on the size and power of controllable steam engines. Engineers of the era struggled to understand why some governors worked perfectly while others, seemingly identical in design, oscillated uncontrollably.

6.1.2 Maxwell’s “On Governors” (1868)

Maxwell and the Birth of Control Theory

The breakthrough came from an unlikely source: James Clerk Maxwell, already famous for unifying electricity and magnetism, turned his attention to the governor problem in 1868. His paper “On Governors,” published in the Proceedings of the Royal Society of London, is now recognized as the founding document of control theory.

Maxwell approached the problem with the tools of mathematical physics. He wrote the equations of motion for the governor-engine system as a set of linear differential equations and asked: under what conditions do small disturbances die out over time? This was the first rigorous formulation of the stability problem.

Maxwell showed that the behavior of the system depended on the roots of a characteristic equation—a polynomial whose coefficients were determined by the physical parameters of the governor and engine. If all roots had negative real parts, disturbances would decay exponentially and the system was stable. If any root had a positive real part, disturbances would grow exponentially: instability.

For second and third-order systems, Maxwell derived explicit conditions on the coefficients that guaranteed stability. But he noted that for higher-order systems, determining whether all roots lay in the left half-plane was a difficult algebraic problem. He posed this as a challenge to mathematicians:

“The great difficulty of this theory lies in the determination of the conditions of stability of the motion... I am not aware of any method of determining the nature of the roots of an equation of a higher degree than the third.”

6.1.3 Routh’s Criterion (1877)

Maxwell’s challenge was answered by Edward John Routh, a Cambridge mathematician who had been Maxwell’s fellow student (and actually beat Maxwell for the prestigious Smith’s Prize in 1854). In 1877, Routh published “A Treatise on the Stability of a Given State of Motion,” which provided a complete algorithmic solution to the stability problem.

Routh developed a systematic procedure—now called the **Routh array** or **Routh table**—that could determine the number of roots with positive real parts for a polynomial of any degree, without actually solving the polynomial. The method required only arithmetic operations on the coefficients, making it practical for hand calculation even for high-order systems.

For this work, Routh received the Adams Prize from Cambridge in 1877. His criterion became the standard tool for stability analysis in British engineering for the next century.

6.1.4 Hurwitz’s Independent Discovery (1895)

In 1895, the German mathematician Adolf Hurwitz independently developed equivalent stability conditions while studying a completely different problem in pure mathematics (the theory of continued fractions). Hurwitz expressed his conditions in terms of determinants of matrices formed from the polynomial coefficients.

When the connection between Routh’s and Hurwitz’s work was recognized, the combined result became known as the **Routh-Hurwitz stability criterion**. It remains one of the most important tools in control engineering, despite the availability of powerful computers that can find polynomial roots directly.

6.1.5 Dramatic Failures: Lessons in Instability

The Tacoma Narrows Bridge (1940)

On November 7, 1940, the Tacoma Narrows Bridge in Washington State collapsed in a spectacular fashion, captured on film that has since been viewed by millions of engineering students. The bridge, nicknamed “Galloping Gertie” for its tendency to oscillate in the wind, exhibited increasingly violent torsional oscillations in a 40 mph wind before tearing itself apart.

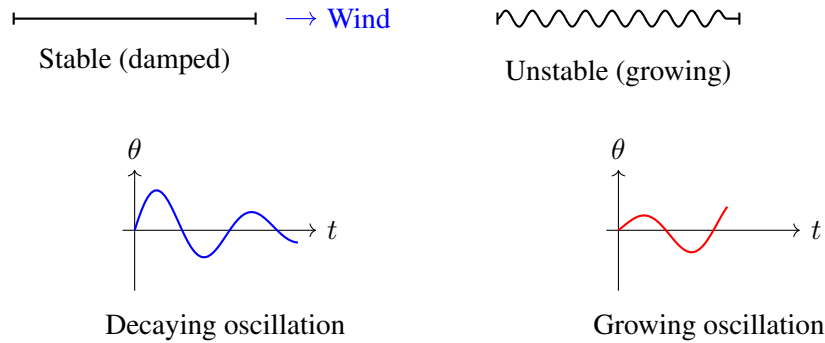


Figure 6.2: Bridge deck oscillation modes. Left: stable, damped response. Right: unstable, growing oscillation as occurred at Tacoma Narrows.

The collapse is often incorrectly attributed to resonance—the wind frequency matching a natural frequency of the bridge. Modern analysis reveals that the actual mechanism was **aeroelastic flutter**, a self-excited instability. The bridge’s motion modified the aerodynamic forces in a way that extracted energy from the wind, feeding the oscillations. This is a classic feedback instability: the output (bridge motion) affected the input (aerodynamic forces) in a destabilizing manner.

Early Aircraft Instabilities

The Wright brothers’ 1903 Flyer was actually designed to be inherently unstable in pitch. The brothers reasoned that this would make the aircraft more maneuverable—and they were right, but it also meant the pilot had to constantly work to maintain level flight. Later aviators, less skilled than the Wrights, crashed repeatedly trying to fly unstable aircraft.

Several forms of instability plagued early aviation. **Spiral instability** occurs when a slight bank angle leads to a sideslip, which increases the bank, leading to an ever-tightening spiral dive; many early aviators, flying in clouds without visual references, died in spiral dives. **Dutch roll** is a coupled yaw-roll oscillation where the nose swings left-right while the wings rock side-to-side, and while mildly unstable Dutch roll merely makes passengers airsick, severely unstable Dutch roll can be completely uncontrollable. **Pilot-induced oscillation (PIO)** occurs when the pilot, reacting to aircraft motion with a slight time delay, actually destabilizes the system through the feedback loop: pilot → controls → aircraft → pilot. This phenomenon has caused crashes of advanced aircraft including early fly-by-wire prototypes.

These historical examples underscore a crucial lesson: stability is not an academic abstraction but a matter of life and death in real engineering systems.

6.2 Modern Applications of Stability Theory

Stability analysis remains central to modern engineering across diverse domains:

6.2.1 Power Grid Stability

The electric power grid is one of the largest and most complex dynamic systems ever built. Thousands of generators must rotate in precise synchronism at 60 Hz (in North America) or 50 Hz (in

Europe). Loss of synchronism leads to **blackouts** affecting millions of people.

The 2003 Northeast Blackout, which affected 55 million people, was triggered by a relatively minor fault (high-voltage lines sagging into trees in Ohio). The fault caused a cascading failure because the system's stability margins were insufficient to absorb the disturbance. Modern grid operators use sophisticated stability analysis to maintain adequate margins and plan for contingencies.

6.2.2 Flight Control: Intentionally Unstable Aircraft

Modern fighter aircraft like the F-16, F-22, and Eurofighter are designed to be **inherently unstable** in certain flight regimes. An unstable aircraft, by definition, will depart from level flight if left alone—but this instability provides extraordinary maneuverability. The aircraft can change direction faster than a stable design because it “wants” to move.

This approach is only possible with active control. Flight computers, sampling sensors hundreds of times per second, continuously adjust control surfaces to maintain stability. If the flight control computer fails, the pilot has only seconds before the aircraft becomes uncontrollable. This “relaxed static stability” design trades passive safety for performance, with the risk managed by redundant computers and extensive testing.

The B-2 stealth bomber carries this concept further. Its flying-wing design, optimized for low radar cross-section, is unstable in multiple axes. Without the flight control system, the B-2 cannot fly.

6.2.3 Audio Feedback

The familiar “squeal” when a microphone is placed too close to its speaker is an example of instability in an acoustic feedback loop:

Microphone → Amplifier → Speaker → (acoustic path) → Microphone

If the loop gain (amplifier gain times acoustic coupling) exceeds unity at any frequency where the phase shift equals a multiple of 360° , the Barkhausen criterion for oscillation is satisfied, and the system becomes unstable at that frequency. Sound engineers use equalization (reducing gain at problematic frequencies) and careful microphone placement to maintain stability.

6.2.4 Economic and Financial Stability

Economic systems exhibit feedback dynamics: interest rate changes affect borrowing, which affects spending, which affects inflation, which affects interest rates. Central banks attempt to stabilize economies by adjusting monetary policy, but time delays and nonlinearities make this challenging.

Financial markets can exhibit instability through feedback mechanisms like portfolio insurance (selling during declines) and margin calls (forced selling when prices fall), which amplify price movements. The 1987 stock market crash, when the Dow Jones Industrial Average fell 22% in a single day, has been attributed partly to such destabilizing feedback.

6.3 Mathematical Foundation of Stability

We now develop the rigorous mathematical theory of stability for linear time-invariant (LTI) systems.

6.3.1 BIBO Stability: Definition

Definition 6.1 (BIBO Stability). A system is **Bounded-Input Bounded-Output (BIBO) stable** if and only if every bounded input produces a bounded output. Formally, if there exists $M < \infty$ such that $|u(t)| \leq M$ for all $t \geq 0$, then there exists $N < \infty$ such that $|y(t)| \leq N$ for all $t \geq 0$.

This definition captures the intuitive notion of a “well-behaved” system. A BIBO-stable system cannot produce unbounded outputs from bounded inputs; it cannot “blow up” in response to reasonable excitation.

6.3.2 Impulse Response and Stability

For an LTI system with impulse response $h(t)$, the output $y(t)$ in response to input $u(t)$ is given by the convolution integral:

$$y(t) = \int_0^t h(\tau)u(t-\tau) d\tau = \int_0^t h(t-\tau)u(\tau) d\tau \quad (6.1)$$

Theorem 6.1 (Impulse Response Criterion for BIBO Stability). *An LTI system is BIBO stable if and only if its impulse response is absolutely integrable:*

$$\int_0^\infty |h(t)| dt < \infty \quad (6.2)$$

Proof. **(Sufficiency)** Suppose $\int_0^\infty |h(t)| dt = H < \infty$ and $|u(t)| \leq M$ for all t . Then:

$$|y(t)| = \left| \int_0^t h(\tau)u(t-\tau) d\tau \right| \quad (6.3)$$

$$\leq \int_0^t |h(\tau)||u(t-\tau)| d\tau \quad (6.4)$$

$$\leq M \int_0^t |h(\tau)| d\tau \quad (6.5)$$

$$\leq M \int_0^\infty |h(\tau)| d\tau = MH \quad (6.6)$$

Thus $|y(t)| \leq MH < \infty$, proving the output is bounded.

(Necessity) Suppose $\int_0^\infty |h(t)| dt = \infty$. We construct a bounded input that produces an unbounded output. Define:

$$u(t) = \begin{cases} +1 & \text{if } h(T-t) \geq 0 \\ -1 & \text{if } h(T-t) < 0 \end{cases} \quad (6.7)$$

for some fixed $T > 0$. This input is clearly bounded: $|u(t)| \leq 1$. The output at time T is:

$$y(T) = \int_0^T h(\tau)u(T-\tau) d\tau = \int_0^T |h(\tau)| d\tau \quad (6.8)$$

As $T \rightarrow \infty$, we have $y(T) \rightarrow \int_0^\infty |h(\tau)| d\tau = \infty$. Thus the output is unbounded. $\square$

6.3.3 Pole Location Criterion

For rational transfer functions, there is an elegant connection between BIBO stability and the location of poles in the complex s -plane.

Theorem 6.2 (Pole Location Criterion). *A system with rational transfer function $G(s)$ is BIBO stable if and only if all poles of $G(s)$ lie in the open left half-plane (LHP), i.e., all poles have strictly negative real parts.*

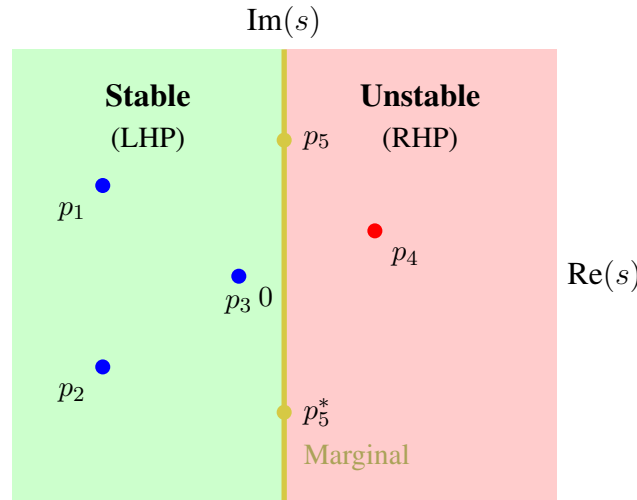


Figure 6.3: The s -plane divided into stable, unstable, and marginally stable regions. Poles p_1, p_2, p_3 (LHP) correspond to stable modes. Pole p_4 (RHP) causes instability. Poles p_5, p_5^* (imaginary axis) produce sustained oscillations—marginal stability.

Proof. Consider a transfer function with partial fraction expansion:

$$G(s) = \sum_{i=1}^n \frac{r_i}{s - p_i} \quad (6.9)$$

where p_i are the poles (assumed distinct for simplicity) and r_i are the residues. The impulse response is:

$$h(t) = \mathcal{L}^{-1}\{G(s)\} = \sum_{i=1}^n r_i e^{p_i t}, \quad t \geq 0 \quad (6.10)$$

For a pole $p_i = \sigma_i + j\omega_i$, the behavior depends on the sign of the real part. When $\sigma_i < 0$ (LHP pole), $|e^{p_i t}| = e^{\sigma_i t} \rightarrow 0$ as $t \rightarrow \infty$, so this term decays exponentially. When $\sigma_i > 0$ (RHP pole),

$|e^{p_i t}| = e^{\sigma_i t} \rightarrow \infty$ as $t \rightarrow \infty$, causing this term to grow exponentially. When $\sigma_i = 0$ (imaginary axis), $|e^{p_i t}| = 1$ for all t , so this term neither grows nor decays.

For BIBO stability, we need $\int_0^\infty |h(t)| dt < \infty$. The integral of $e^{\sigma t}$ is:

$$\int_0^\infty e^{\sigma t} dt = \begin{cases} -1/\sigma < \infty & \text{if } \sigma < 0 \\ \infty & \text{if } \sigma \geq 0 \end{cases} \quad (6.11)$$

Therefore, absolute integrability (and thus BIBO stability) requires all poles to have $\sigma_i < 0$, i.e., all poles in the open LHP. $\square$

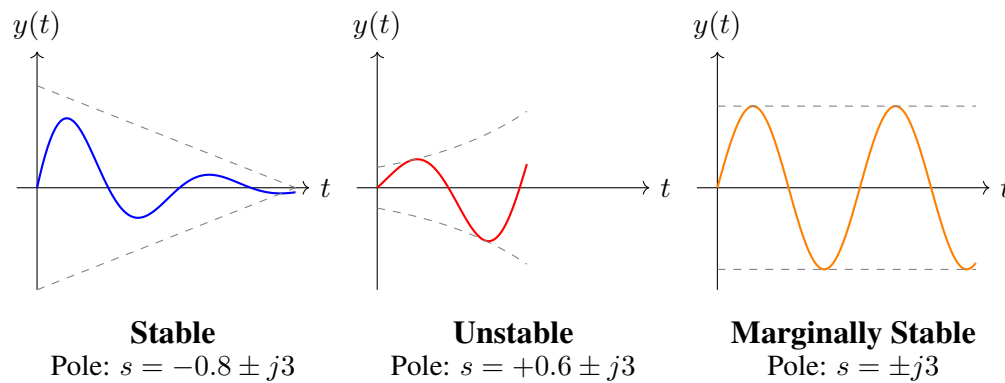


Figure 6.4: Time responses corresponding to different pole locations. Stable poles (LHP) produce decaying responses. Unstable poles (RHP) produce growing responses. Imaginary axis poles produce sustained oscillations.

6.3.4 Marginal Stability

Definition 6.2 (Marginal Stability). A system is **marginally stable** if it is not BIBO stable, but its impulse response remains bounded (though not necessarily decaying) for all time.

Marginal stability occurs when there are poles on the imaginary axis (simple poles only—repeated imaginary poles cause unbounded growth). A marginally stable system will oscillate indefinitely in response to an impulse but will not grow without bound.

Engineering Caution

Marginally stable systems are unacceptable in most engineering applications. While mathematically the response is bounded, several practical concerns arise. First, any modeling error could place poles slightly in the RHP, causing instability. Second, nonlinearities not captured by the linear model may cause limit cycles or chaos. Third, sustained oscillations waste energy and cause mechanical wear. For these reasons, a well-designed system should have all poles strictly in the LHP with adequate stability margins.

6.3.5 Closed-Loop Stability

For a feedback system with forward path $G(s)$ and feedback path $H(s)$, the closed-loop transfer function is:

$$T(s) = \frac{G(s)}{1 + G(s)H(s)} \quad (6.12)$$

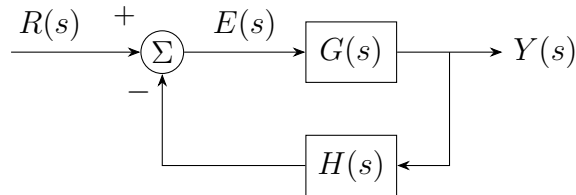


Figure 6.5: Standard feedback configuration. The closed-loop poles are roots of $1 + G(s)H(s) = 0$.

The closed-loop poles are the roots of the **characteristic equation**:

$$1 + G(s)H(s) = 0 \quad (6.13)$$

Key Insight: Open-Loop vs. Closed-Loop Stability

The open-loop transfer function $G(s)H(s)$ and closed-loop transfer function $T(s)$ generally have *different* poles. A system can be open-loop stable but closed-loop unstable when too much gain is applied, or it can be open-loop unstable but closed-loop stable through feedback stabilization. Stability analysis must therefore focus on the *closed-loop* poles, which are roots of the characteristic equation $1 + G(s)H(s) = 0$.

6.4 The Routh-Hurwitz Stability Criterion

The Routh-Hurwitz criterion provides a method to determine how many roots of a polynomial lie in the right half-plane without actually solving for the roots. This is invaluable for stability analysis, especially when the polynomial coefficients depend on design parameters.

6.4.1 The Routh Array

Consider a polynomial of degree n :

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \cdots + a_1 s + a_0 \quad (6.14)$$

Theorem 6.3 (Necessary Condition for Stability). *A necessary (but not sufficient) condition for all roots of $P(s)$ to have negative real parts is that all coefficients a_i have the same sign and are nonzero.*

If any coefficient is zero or has a different sign, there is at least one root with non-negative real part. However, all coefficients being positive does not guarantee stability for polynomials of degree 3 or higher.

Definition 6.3 (Routh Array). The Routh array is a triangular table constructed from the polynomial coefficients as follows:

$$\begin{array}{c|ccccc}
 s^n & a_n & a_{n-2} & a_{n-4} & a_{n-6} & \cdots \\
 s^{n-1} & a_{n-1} & a_{n-3} & a_{n-5} & a_{n-7} & \cdots \\
 s^{n-2} & b_1 & b_2 & b_3 & b_4 & \cdots \\
 s^{n-3} & c_1 & c_2 & c_3 & c_4 & \cdots \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \\
 s^1 & * & & & & \\
 s^0 & * & & & &
 \end{array}$$

where the elements are computed using:

$$b_1 = \frac{a_{n-1} \cdot a_{n-2} - a_n \cdot a_{n-3}}{a_{n-1}} \quad (6.15)$$

$$b_2 = \frac{a_{n-1} \cdot a_{n-4} - a_n \cdot a_{n-5}}{a_{n-1}} \quad (6.16)$$

$$c_1 = \frac{b_1 \cdot a_{n-3} - a_{n-1} \cdot b_2}{b_1} \quad (6.17)$$

and so on, with each element computed from the two rows immediately above it.

The general formula for an element in row i , column j is:

$$\text{element}_{i,j} = \frac{(\text{first element of row } i-1) \times (\text{element } j+1 \text{ of row } i-2) - (\text{first element of row } i-2) \times (\text{element } j \text{ of row } i-1)}{\text{first element of row } i-1} \quad (6.18)$$

Theorem 6.4 (Routh-Hurwitz Criterion). *The number of roots of $P(s)$ in the right half-plane equals the number of sign changes in the first column of the Routh array.*

Corollary 6.1. *A polynomial has all roots in the left half-plane (and hence the corresponding system is stable) if and only if all elements in the first column of the Routh array have the same sign.*

6.4.2 Special Cases

Two special situations require modified procedures:

Case 1: Zero in the First Column (Other Elements Non-Zero)

If a zero appears in the first column but other elements in that row are nonzero, replace the zero with a small positive number ϵ and continue the array construction. After completing the array, examine the signs as $\epsilon \rightarrow 0^+$.

Case 2: Entire Row of Zeros

A row of zeros indicates the presence of pairs of roots that are symmetric about the origin: either a pair of real roots at $\pm\sigma$, a pair of imaginary roots at $\pm j\omega$, or a quadruplet at $\pm\sigma \pm j\omega$.

The procedure for handling a row of zeros involves forming the **auxiliary polynomial** $A(s)$ from the row immediately above the zero row. The roots of $A(s)$ are among the symmetric root pairs. Next, replace the zero row with coefficients obtained from the derivative $\frac{dA}{ds}$, and then continue the Routh array construction normally.

6.5 Practical Experiment: Demonstrating Stability and Instability

Understanding stability through hands-on experimentation provides intuition that mathematics alone cannot convey. This section presents a practical laboratory exercise using an op-amp circuit that can be configured for stable operation, marginal stability (oscillation), or instability.

Laboratory Exercise: Op-Amp Feedback Stability

Objective: Observe the effect of feedback on system stability by building an adjustable-gain feedback circuit and measuring its response as gain increases through the stability boundary.

Learning Outcomes: Upon completing this exercise, students will be able to observe stable, marginally stable, and unstable behavior in a real system, correlate mathematical stability criteria with physical behavior, and understand gain margins and their practical significance.

6.5.1 Circuit Description

We construct a second-order system using two op-amp integrators in a feedback loop with adjustable gain.

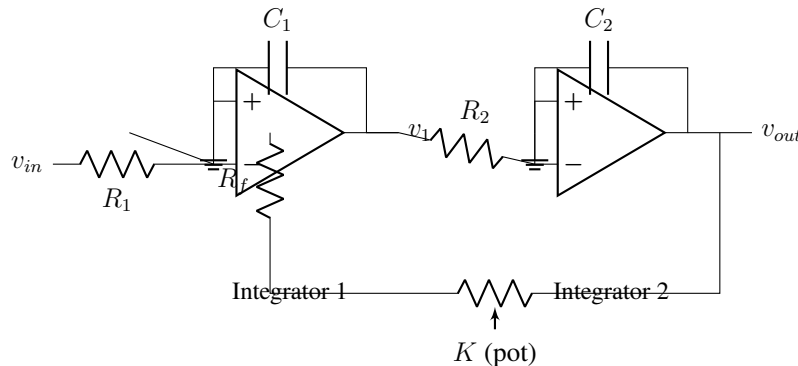


Figure 6.6: Two-integrator oscillator/unstable system. Adjusting potentiometer K changes the feedback gain, transitioning the system through stable, marginally stable, and unstable regions.

The transfer function of this system (from v_{in} to v_{out}) is:

$$G(s) = \frac{1}{(R_1C_1)(R_2C_2)s^2 + K} \quad (6.19)$$

The characteristic equation is:

$$\tau_1\tau_2s^2 + K = 0 \quad \Rightarrow \quad s^2 = -\frac{K}{\tau_1\tau_2} \quad (6.20)$$

where $\tau_1 = R_1C_1$ and $\tau_2 = R_2C_2$.

The system behavior depends critically on the feedback gain K . For $K > 0$, the poles lie at $s = \pm j\sqrt{K/(\tau_1\tau_2)}$ on the imaginary axis, producing marginal stability with sustained oscillation. For $K < 0$ (positive feedback), the poles move to $s = \pm\sqrt{|K|/(\tau_1\tau_2)}$ on the real axis, placing one pole in the RHP and causing instability. Adding damping through a resistor in parallel with the capacitors moves the poles into the LHP, achieving stability.

6.5.2 Alternative: Arduino-Based Motor Control Experiment

For a more dynamic demonstration, we can use an Arduino microcontroller to implement digital feedback control of a DC motor.

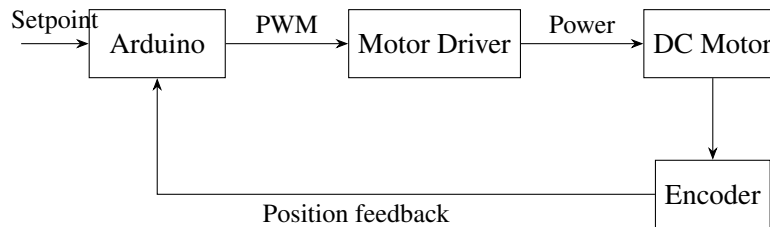


Figure 6.7: Arduino-based motor position control system for stability demonstration.

Algorithm 9 Motor Position Control with Adjustable Gain

```

1: Initialize:
2:   Configure PWM output pin and direction pin
3:   Configure encoder input pins with interrupts
4:   Set encoderCount  $\leftarrow$  0
5:   Set setpoint  $\leftarrow$  0
6:   Set  $K_p \leftarrow 1.0$  ▷ Proportional gain
7:
8: Main Control Loop:
9: while system is running do
10:   if command received from user interface then
11:     if command is “set gain” then
12:        $K_p \leftarrow$  new gain value
13:     else if command is “set position” then
14:       setpoint  $\leftarrow$  new position value
15:     end if
16:   end if
17:
18:   ▷ Proportional controller calculation
19:   error  $\leftarrow$  setpoint  $-$  encoderCount
20:   output  $\leftarrow K_p \times$  error
21:
22:   ▷ Apply control signal to motor
23:   pwm  $\leftarrow$   $\min(\max(|\textit{output}|, 0), 255)$ 
24:   Set motor direction based on  $\text{sign}(\textit{output})$ 
25:   Apply PWM signal to motor driver
26:
27:   ▷ Log data for analysis
28:   Output: timestamp, setpoint, encoderCount
29:   Wait for next control cycle (10 ms period)
30: end while
31:
32: Encoder Interrupt Handler:
33: if encoder channel B is high then
34:   encoderCount  $\leftarrow$  encoderCount + 1
35: else
36:   encoderCount  $\leftarrow$  encoderCount  $-$  1
37: end if

```

6.5.3 Experimental Procedure

The experiment proceeds through five stages. First, establish a **baseline at low gain** by setting $K_p = 0.5$, applying a step change in setpoint, and observing the slow but stable response. Next, **increase the gain** gradually through values of 1.0, 2.0, and 5.0, noting how the response becomes

faster but exhibits increasing overshoot. Then, locate the **critical gain** K_u at which sustained oscillations occur, representing marginal stability. After this, push the system into **instability** by increasing gain beyond K_u and observing the growing oscillations until the motor saturates or oscillates violently. Finally, **add damping** by implementing a derivative term (PD control) and observe how damping extends the stable gain range.

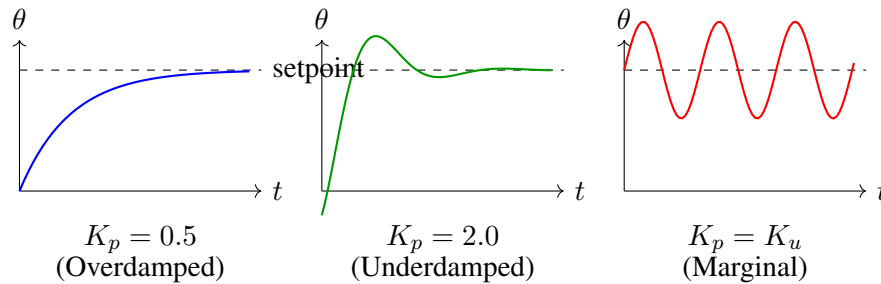


Figure 6.8: Expected motor responses at different gain settings. Left: overdamped (slow but stable). Center: underdamped (fast with overshoot). Right: marginally stable (sustained oscillation at critical gain).

6.6 Worked Examples

Third-Order System Stability Analysis

Determine the stability of a system with characteristic polynomial:

$$P(s) = s^3 + 4s^2 + 5s + 2$$

Solution:

Step 1: Check necessary conditions. All coefficients are positive: ✓

Step 2: Construct the Routh array.

s^3	1	5
s^2	4	2
s^1	b_1	0
s^0	c_1	

Compute b_1 :

$$b_1 = \frac{4 \cdot 5 - 1 \cdot 2}{4} = \frac{20 - 2}{4} = \frac{18}{4} = 4.5$$

Compute c_1 :

$$c_1 = \frac{4.5 \cdot 2 - 4 \cdot 0}{4.5} = \frac{9}{4.5} = 2$$

The completed Routh array:

s^3	1	5
s^2	4	2
s^1	4.5	0
s^0	2	

Step 3: Count sign changes in first column: $1 \rightarrow 4 \rightarrow 4.5 \rightarrow 2$

All positive, no sign changes.

Conclusion: The system is **stable**. All three poles are in the LHP.

Finding the Range of Stable Gain

For the feedback system with open-loop transfer function:

$$G(s)H(s) = \frac{K}{s(s+1)(s+4)}$$

determine the range of $K > 0$ for which the closed-loop system is stable.

Solution:

The characteristic equation is $1 + G(s)H(s) = 0$:

$$\begin{aligned} 1 + \frac{K}{s(s+1)(s+4)} &= 0 \\ s(s+1)(s+4) + K &= 0 \\ s^3 + 5s^2 + 4s + K &= 0 \end{aligned}$$

Construct the Routh array:

s^3	1	4
s^2	5	K
s^1	b_1	0
s^0	K	

Compute b_1 :

$$b_1 = \frac{5 \cdot 4 - 1 \cdot K}{5} = \frac{20 - K}{5}$$

For stability, all first-column elements must be positive. The row s^3 element is $1 > 0$, which is always satisfied. The row s^2 element is $5 > 0$, also always satisfied. The row s^1 element requires $\frac{20-K}{5} > 0$, giving $K < 20$. Finally, the row s^0 element requires $K > 0$, which is given.

Conclusion: The system is stable for $0 < K < 20$.

At $K = 20$, the system is marginally stable with poles on the imaginary axis.

Example 6.1 (Fourth-Order System with Parameter). For what values of a is the following polynomial stable?

$$P(s) = s^4 + 2s^3 + 3s^2 + as + 5$$

Solution:

Construct the Routh array:

s^4	1	3	5
s^3	2	a	0
s^2	b_1	b_2	
s^1	c_1		
s^0	d_1		

Compute elements:

$$b_1 = \frac{2 \cdot 3 - 1 \cdot a}{2} = \frac{6 - a}{2}$$

$$b_2 = \frac{2 \cdot 5 - 1 \cdot 0}{2} = 5$$

$$c_1 = \frac{b_1 \cdot a - 2 \cdot 5}{b_1} = \frac{\frac{6-a}{2} \cdot a - 10}{\frac{6-a}{2}} = \frac{a(6-a) - 20}{6-a} = \frac{6a - a^2 - 20}{6-a}$$

$$d_1 = 5 \quad (\text{from } b_2)$$

For stability, all first-column elements must be positive. The first condition requires $b_1 = \frac{6-a}{2} > 0$, which gives $a < 6$. The second condition requires $c_1 = \frac{6a-a^2-20}{6-a} > 0$. Since we need $a < 6$, we have $6-a > 0$, so the numerator must also be positive:

$$6a - a^2 - 20 > 0 \Rightarrow a^2 - 6a + 20 < 0$$

The discriminant is $36 - 80 = -44 < 0$, and since the coefficient of a^2 is positive, the quadratic $a^2 - 6a + 20$ is always positive. Therefore, $c_1 < 0$ for all real a when $a < 6$.

Conclusion: The polynomial is **unstable for all real values of a** . ■

Example 6.2 (Row of Zeros - Auxiliary Polynomial). Analyze the stability of:

$$P(s) = s^5 + 2s^4 + 4s^3 + 8s^2 + 3s + 6$$

Solution:

Begin the Routh array:

s^5	1	4	3
s^4	2	8	6
s^3	b_1	b_2	

Compute:

$$b_1 = \frac{2 \cdot 4 - 1 \cdot 8}{2} = \frac{8 - 8}{2} = 0$$

$$b_2 = \frac{2 \cdot 3 - 1 \cdot 6}{2} = \frac{6 - 6}{2} = 0$$

We have a row of zeros at s^3 . This indicates symmetric root pairs.

Form the auxiliary polynomial from the s^4 row:

$$A(s) = 2s^4 + 8s^2 + 6$$

Taking the derivative:

$$\frac{dA}{ds} = 8s^3 + 16s$$

Replace the zero row with coefficients from $\frac{dA}{ds}$:

s^5	1	4	3
s^4	2	8	6
s^3	8	16	0
s^2	c_1	c_2	
s^1	d_1		
s^0	e_1		

Continue computing:

$$c_1 = \frac{8 \cdot 8 - 2 \cdot 16}{8} = \frac{64 - 32}{8} = 4$$

$$c_2 = \frac{8 \cdot 6 - 2 \cdot 0}{8} = 6$$

$$d_1 = \frac{4 \cdot 16 - 8 \cdot 6}{4} = \frac{64 - 48}{4} = 4$$

$$e_1 = 6$$

Final Routh array:

s^5	1	4	3
s^4	2	8	6
s^3	8	16	0
s^2	4	6	
s^1	4		
s^0	6		

First column: 1, 2, 8, 4, 4, 6 — all positive, no sign changes.

However, the row of zeros indicates roots on the imaginary axis. Solving $A(s) = 0$:

$$2s^4 + 8s^2 + 6 = 0 \Rightarrow s^2 = \frac{-8 \pm \sqrt{64 - 48}}{4} = \frac{-8 \pm 4}{4}$$

$$s^2 = -1 \text{ or } s^2 = -3 \Rightarrow s = \pm j, \pm j\sqrt{3}$$

Conclusion: The system is **marginally stable**. There are no RHP poles, but there are two pairs of poles on the imaginary axis at $s = \pm j$ and $s = \pm j\sqrt{3}$. ■

Example 6.3 (Zero in First Column (Other Elements Non-Zero)). Determine the number of RHP poles for:

$$P(s) = s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10$$

Solution:

Begin the Routh array:

s^5	1	2	11
s^4	2	4	10
s^3	b_1	b_2	

Compute:

$$b_1 = \frac{2 \cdot 2 - 1 \cdot 4}{2} = \frac{4 - 4}{2} = 0$$

$$b_2 = \frac{2 \cdot 11 - 1 \cdot 10}{2} = \frac{22 - 10}{2} = 6$$

Zero in first column, but $b_2 \neq 0$. Replace 0 with $\epsilon > 0$:

s^5	1	2	11
s^4	2	4	10
s^3	ϵ	6	0
s^2	c_1	10	
s^1	d_1		
s^0	10		

Compute:

$$c_1 = \frac{\epsilon \cdot 4 - 2 \cdot 6}{\epsilon} = \frac{4\epsilon - 12}{\epsilon} = 4 - \frac{12}{\epsilon}$$

As $\epsilon \rightarrow 0^+$: $c_1 \rightarrow -\infty$ (large negative)

$$d_1 = \frac{c_1 \cdot 6 - \epsilon \cdot 10}{c_1}$$

As $\epsilon \rightarrow 0^+$: $d_1 \rightarrow 6$ (positive)

Examining signs as $\epsilon \rightarrow 0^+$:

Row	Element	Sign
s^5	1	+
s^4	2	+
s^3	ϵ	+
s^2	$-\infty$	-
s^1	6	+
s^0	10	+

Sign changes: $+$ $\rightarrow$ $+$ (no), $+$ $\rightarrow$ $+$ (no), $+$ $\rightarrow$ $-$ (YES), $-$ $\rightarrow$ $+$ (YES), $+$ $\rightarrow$ $+$ (no)

Conclusion: There are **2 sign changes**, indicating **2 poles in the RHP**. The system is unstable. ■

Example 6.4 (Closed-Loop Stability from Open-Loop Transfer Function). A unity feedback system has open-loop transfer function:

$$G(s) = \frac{K(s+2)}{(s-1)(s+3)}$$

Note that the open-loop system is unstable (pole at $s = +1$). For what values of K is the closed-loop system stable?

Solution:

The closed-loop transfer function is:

$$T(s) = \frac{G(s)}{1+G(s)} = \frac{K(s+2)}{(s-1)(s+3) + K(s+2)}$$

The characteristic equation is:

$$\begin{aligned}(s-1)(s+3) + K(s+2) &= 0 \\ s^2 + 2s - 3 + Ks + 2K &= 0 \\ s^2 + (2+K)s + (2K-3) &= 0\end{aligned}$$

For a second-order polynomial $s^2 + as + b$, stability requires $a > 0$ and $b > 0$:

$$\begin{aligned}2 + K > 0 &\Rightarrow K > -2 \\ 2K - 3 > 0 &\Rightarrow K > 1.5\end{aligned}$$

Conclusion: The closed-loop system is stable for $K > 1.5$.

This demonstrates **feedback stabilization**: the open-loop system is unstable, but sufficiently high feedback gain stabilizes the closed-loop system. ■

6.7 Summary

Key Concepts from Chapter 6

Concept	Description
BIBO Stability	A system is stable if every bounded input produces a bounded output.
Pole Location Criterion	An LTI system is BIBO stable if and only if all poles of its transfer function lie in the open left half-plane ($\text{Re}(s) < 0$).
Impulse Response Criterion	Stability is equivalent to $\int_0^\infty h(t) dt < \infty$.
Routh-Hurwitz Criterion	The number of RHP poles equals the number of sign changes in the first column of the Routh array.
Marginal Stability	Poles on the imaginary axis produce sustained oscillations—not BIBO stable, but bounded impulse response.
Feedback Stabilization	Properly designed feedback can stabilize an inherently unstable system (e.g., inverted pendulum, modern fighter aircraft).

6.8 Exercises

Exercise 6.1. For each polynomial, determine whether all roots are in the LHP:

1. $s^3 + 6s^2 + 11s + 6$
2. $s^3 + 2s^2 + s + 2$
3. $s^4 + 2s^3 + 3s^2 + 4s + 5$

Exercise 6.2. A feedback system has characteristic equation $s^3 + 3s^2 + (K + 2)s + 4K = 0$. Find the range of K for stability.

Exercise 6.3. Apply the Routh criterion to $s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$. If there are imaginary axis roots, find them.

Exercise 6.4. An open-loop unstable system has transfer function $G(s) = \frac{1}{s^2 - 4}$. Design a proportional controller K such that the closed-loop system is stable. What is the minimum required gain?

Exercise 6.5. Consider the inverted pendulum linearized about the upward position with transfer function $G(s) = \frac{1}{s^2 - \omega_n^2}$ where $\omega_n^2 = g/\ell$. Using PD control $C(s) = K_p + K_d s$, derive conditions on K_p and K_d for closed-loop stability.

Exercise 6.6. (*Computer Exercise*) Using MATLAB or Python, verify your Routh array calculations from Exercises 1-3 by computing the roots directly. Plot pole locations in the s -plane.

Exercise 6.7. A unity feedback system has open-loop transfer function $G(s) = \frac{K(s+1)}{s(s-2)(s+5)}$. The plant is open-loop unstable. Determine the range of K for which the closed-loop system is stable.

Exercise 6.8. For the characteristic polynomial $s^4 + 3s^3 + 5s^2 + 4s + 2$, construct the complete Routh array. Determine whether the system is stable, marginally stable, or unstable. If unstable, state how many poles are in the RHP.

Exercise 6.9. A system has the characteristic equation $s^4 + 2s^3 + 11s^2 + 18s + 18 = 0$. Using the Routh-Hurwitz criterion, determine if any roots lie on the imaginary axis. If so, find the frequencies of oscillation.

Exercise 6.10. Explain why marginal stability is generally unacceptable in engineering practice. Provide at least three practical reasons with examples from real systems.

Historical References

- [1] J.C. Maxwell, "On Governors," *Proceedings of the Royal Society of London*, vol. 16, pp. 270–283, 1868.
- [2] E.J. Routh, *A Treatise on the Stability of a Given State of Motion*. London: Macmillan, 1877.
- [3] A. Hurwitz, "Ueber die Bedingungen, unter welchen eine Gleichung nur Wurzeln mit negativen reellen Theilen besitzt," *Mathematische Annalen*, vol. 46, pp. 273–284, 1895.
- [4] K.Y. Billah and R.H. Scanlan, "Resonance, Tacoma Narrows bridge failure, and undergraduate physics textbooks," *American Journal of Physics*, vol. 59, no. 2, pp. 118–124, 1991.

Chapter 7

Transient vs Steady-State Response

“The engineer’s eternal dilemma: you can have it fast, or you can have it accurate—but getting both requires wisdom.”

— Attributed to early process control engineers, circa 1940

When a control system responds to a command or disturbance, its behavior unfolds in two distinct phases. The **transient response** captures the dynamic journey from one state to another—characterized by speed, oscillation, and overshoot. The **steady-state response** describes where the system ultimately settles—and whether it reaches the intended target. Understanding and quantifying both aspects is essential for designing systems that are not merely stable, but *well-behaved*.

This chapter develops the mathematical framework for analyzing transient and steady-state performance, traces the historical evolution of performance specifications, and demonstrates practical methods for measuring and optimizing system response.

7.1 Historical Development: The Quest for Performance Specifications

The formal study of control system performance emerged from practical necessity. Early engineers knew intuitively that a system could be stable yet perform poorly—too slow, too oscillatory, or perpetually off-target. The challenge was developing quantitative measures that could guide design.

7.1.1 Industrial Process Control: The Foundation (1920s–1940s)

The Birth of Performance Thinking

The modern conception of transient and steady-state specifications arose from the chemical and petroleum industries, where process control became economically essential during the early twentieth century.

The 1920s witnessed the transformation of industrial manufacturing from batch processing to continuous operation. Oil refineries, chemical plants, and paper mills required automatic control of temperature, pressure, flow rate, and composition. The pneumatic controller—operating on compressed air signals between 3 and 15 psi—became the standard actuator.

Early operators noticed a fundamental problem: a controller that maintained steady temperature accurately might respond sluggishly to disturbances, allowing product quality to drift for extended periods. Conversely, a controller tuned for rapid response often oscillated persistently, creating new problems. The *tradeoff between speed and stability* entered engineering consciousness.

The instrumentation companies—notably Foxboro, Taylor Instrument, and Leeds & Northrup—began documenting controller behavior systematically. Engineers developed what they called “reaction curve” methods: applying a step change to a process and recording the response on a strip chart recorder. From these curves, they extracted empirical measures: **dead time** (how long before the output began responding), **reaction rate** (how fast the output changed initially), **ultimate value** (where the output finally settled), and **overshoot** (whether the output exceeded the target before settling). These practical measures, developed through industrial experience, would later be formalized mathematically.

7.1.2 Servomechanisms and World War II: Speed Becomes Critical

The Second World War transformed control engineering from an industrial specialty into a strategic technology. Military applications—gun turrets, radar tracking systems, torpedo steering, and aircraft autopilots—demanded performance that peaceful industry had never required.

The Fire Control Problem

Consider a naval gun turret tracking an aircraft. The turret must respond quickly enough to follow a maneuvering target, avoid oscillating around the target (which would spray bullets uselessly), point accurately at the predicted intercept position, and operate reliably under combat conditions. Each of these requirements translates directly to a performance specification.

The MIT Radiation Laboratory, established in 1940, became a center for servo theory development. Engineers including Gordon Brown, Albert Hall, and Nathaniel Nichols developed systematic methods for analyzing and designing servomechanisms. Their 1943 report, later published as *Theory of Servomechanisms* (1947), established the vocabulary still used today.

The radar tracking problem proved particularly instructive. A radar antenna must follow a target—perhaps an aircraft at several miles’ distance—with sufficient accuracy for fire control computation. The antenna position servo must track with minimal lag, meaning that if the target moves at constant angular velocity, the servo must follow with acceptably small *velocity error*. It must respond rapidly to maneuvers, adjusting quickly when the target changes course, which requires small *rise time*. Finally, it must avoid oscillation, since hunting around the target position wastes time and may lose track entirely; this demands controlled *overshoot* and *settling time*. The mathematical analysis of these requirements led directly to the specifications we study today.

7.1.3 Ziegler and Nichols: The First Systematic Tuning Rules (1942)

In 1942, John G. Ziegler and Nathaniel B. Nichols of Taylor Instrument Companies published a landmark paper: “Optimum Settings for Automatic Controllers.” This work, appearing in the *Transactions of the ASME*, provided the first widely-adopted method for selecting controller gains based on desired performance.

The Ziegler-Nichols Philosophy

Ziegler and Nichols recognized that industrial users needed practical tuning methods, not theoretical analysis. Their approach involved three steps: first, characterize the process through simple experiments; second, apply formulas to calculate controller settings; and third, achieve “quarter-amplitude damping”—a specific performance target.

The “quarter-amplitude damping” criterion specified that each successive oscillation peak should be one-quarter the amplitude of the previous peak. This corresponds to a damping ratio $\zeta \approx 0.21$ —quite oscillatory by modern standards, but providing rapid response.

Ziegler and Nichols proposed two methods:

Open-Loop Method: Apply a step input to the open-loop process. Measure the dead time L and the slope R of the response. Calculate controller gains from these parameters.

Closed-Loop Method: With the controller in proportional-only mode, increase gain until the system exhibits sustained oscillation. Record the *ultimate gain* K_u and *ultimate period* T_u . Calculate controller gains from these parameters.

While subsequent research has produced many refinements, the Ziegler-Nichols methods remain in use eight decades later—a testament to their practical utility. More importantly, their work established that *performance specifications should guide controller design*, not merely stability considerations.

7.1.4 Telephone Switching: Precision Timing Requirements (1950s)

The postwar expansion of telephone networks created new demands for control system performance. Automatic switching systems—replacing human operators—required electromechanical servos to position selector switches with extreme reliability.

The dial telephone system exemplified timing-critical control. When a subscriber dialed a digit, the dial mechanism generated a series of pulses (one pulse for “1,” two for “2,” etc., with ten pulses for “0”). Each pulse lasted nominally 100 milliseconds, with a 60% break and 40% make ratio. The receiving equipment had to recognize pulses arriving at rates from 8 to 12 pulses per second, distinguish between inter-digit pauses and end-of-dialing, position stepper switches accurately within the pulse train timing, and operate reliably over millions of cycles. Bell Telephone Laboratories engineers developed sophisticated analysis methods for these timing-critical systems. The concepts of *settling time* and *timing margins* became central to switching system design. A selector that oscillated or settled slowly might miscount pulses, connecting calls incorrectly.

This application demonstrated that transient specifications were not merely about “fast versus slow”—they were about *precise timing* in systems where errors had immediate, observable consequences.

7.1.5 The Fundamental Tradeoff: Speed vs. Accuracy

By the 1950s, control engineers had distilled decades of experience into a fundamental insight: *improving transient response typically degrades steady-state accuracy, and vice versa.*

Consider a simple proportional controller with gain K_p . Increasing K_p improves steady-state accuracy by reducing position error and improves response speed by yielding smaller rise time. However, higher gain also degrades damping, resulting in more overshoot and longer settling time, and eventually destabilizes the system.

Adding integral action (creating a PI controller) eliminates steady-state error for step inputs, but it degrades stability margins and slows the transient response. Additionally, integral action can cause integrator windup during saturation.

This tradeoff—inherent to feedback control—motivates the careful study of performance specifications. The engineer must understand quantitatively how design choices affect each aspect of response, then select the best compromise for the application.

7.2 Modern Applications: Where Performance Specifications Matter

The transient and steady-state specifications developed in the mid-twentieth century remain central to modern control system design. Contemporary applications, while technologically transformed, face the same fundamental tradeoffs.

7.2.1 Hard Disk Drive Head Positioning

Settling Time: The Critical Specification

A modern hard disk drive positions the read/write head over the correct track in approximately 4–8 milliseconds. With track densities exceeding 500,000 tracks per inch, the head must settle to within a fraction of a micrometer—and do so repeatedly, reliably, and quickly.

The disk drive servo exemplifies a settling-time-critical application. The head must move rapidly from one track to another (the *seek* phase), settle accurately on the target track (the *settle* phase), and maintain position during read/write operations (*track following*). The seek-and-settle operation dominates drive performance. A servo that overshoots may cross the target track, requiring additional correction time. One that undershoots may never achieve the required positioning accuracy. Drive manufacturers invest enormous effort in optimizing the settling time—measured to the microsecond—while maintaining acceptable overshoot.

The servo transfer function is carefully designed to provide settling time $t_s < 5$ ms for single-track seeks, overshoot less than 5% of track pitch, and position error less than 10% of track pitch during steady tracking.

7.2.2 CNC Machining: Accuracy vs. Throughput

Computer Numerical Control (CNC) machine tools present the classic accuracy-versus-speed trade-off. A milling machine cutting a precision part must move the cutting tool along programmed paths, maintain positional accuracy within specified tolerances, and complete the operation in reasonable time.

Higher feed rates increase throughput but stress the servo systems. Axis drives may exhibit **following error**, where the actual position lags the commanded position during motion; **corner rounding**, where the tool path deviates from the programmed geometry at direction changes; and **vibration**, where high accelerations excite mechanical resonances.

Modern CNC controllers implement sophisticated servo algorithms that limit jerk (rate of acceleration change) to reduce vibration, use feedforward compensation to reduce following error, and adjust gains based on axis position and load. The performance specifications—particularly velocity error constants and settling times—directly impact achievable part accuracy and machining speed.

7.2.3 Insulin Pumps: Overshoot is Dangerous

Safety-Critical Control

In medical devices, performance specifications are not merely about quality—they concern patient safety. An insulin pump that overshoots could deliver a dangerous overdose.

Insulin pump controllers must regulate blood glucose levels by modulating insulin delivery. The control challenge is multifaceted: glucose dynamics are slow with time constants measured in hours, meals create large disturbances, individual patient responses vary widely, and both hyperglycemia (high glucose) and hypoglycemia (low glucose) are dangerous.

Hypoglycemia—caused by excessive insulin—is immediately dangerous, potentially causing seizures, unconsciousness, or death. Therefore, insulin pump controllers are deliberately designed with no overshoot in the insulin delivery rate, conservative gain margins, and asymmetric response that is slower to increase delivery but faster to decrease it. This application demonstrates that performance specifications must be chosen with application context. The “optimal” response for one application may be dangerous for another.

7.2.4 Autonomous Vehicle Lane Keeping

Self-driving vehicles must maintain lane position while traveling at highway speeds. The lane-keeping controller senses lane boundaries through cameras and sensors, commands steering adjustments to maintain center position, and responds to road curvature, wind gusts, and road imperfections.

Performance requirements for lane keeping span all the transient and steady-state specifications we have discussed. The settling time requirement specifies that after a lane change, the vehicle should stabilize within 2–3 seconds. The overshoot requirement dictates that the vehicle should not oscillate within the lane, as this causes passenger discomfort and raises safety concerns. The steady-state error requirement ensures the vehicle tracks lane center with error less than 10

cm. Finally, the disturbance rejection requirement specifies that wind gusts should cause minimal deviation.

The lane-keeping problem also illustrates nested specifications: the position controller (maintaining lane center) must be coordinated with the heading controller (maintaining vehicle orientation) and the actuator dynamics (steering motor response).

7.3 Mathematical Foundation: Transient Response Specifications

We now develop the mathematical framework for analyzing transient response. Our treatment focuses on second-order systems, which capture the essential behavior of most practical controlled systems.

7.3.1 The Standard Second-Order System

Consider the closed-loop transfer function:

$$G(s) = \frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (7.1)$$

where ω_n is the **natural frequency** (rad/s), representing the frequency of oscillation if undamped, and ζ is the **damping ratio** (dimensionless), representing the degree of damping relative to critical damping.

The characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$ has roots:

$$s_{1,2} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1} \quad (7.2)$$

The nature of these roots—and hence the system response—depends on ζ :

Damping Ratio Classification

Range	Classification	Characteristics
$\zeta > 1$	Overdamped	Two distinct real poles, no oscillation
$\zeta = 1$	Critically damped	Repeated real pole, fastest non-oscillatory response
$0 < \zeta < 1$	Underdamped	Complex conjugate poles, oscillatory response
$\zeta = 0$	Undamped	Purely imaginary poles, sustained oscillation
$\zeta < 0$	Unstable	Poles in right half-plane

For the underdamped case ($0 < \zeta < 1$), the poles are:

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n \sqrt{1 - \zeta^2} = -\sigma \pm j\omega_d \quad (7.3)$$

where $\sigma = \zeta\omega_n$ is the **exponential decay rate** and $\omega_d = \omega_n \sqrt{1 - \zeta^2}$ is the **damped natural frequency**.

7.3.2 Unit Step Response Derivation

For a unit step input, $R(s) = 1/s$, the output is:

$$Y(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} \quad (7.4)$$

Using partial fraction expansion for the underdamped case:

$$Y(s) = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (7.5)$$

Taking the inverse Laplace transform:

$$y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1 - \zeta^2}} \sin(\omega_d t + \phi) \quad (7.6)$$

where $\phi = \cos^{-1}(\zeta)$.

Alternatively, this can be written as:

$$y(t) = 1 - e^{-\zeta\omega_n t} \left[\cos(\omega_d t) + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin(\omega_d t) \right] \quad (7.7)$$

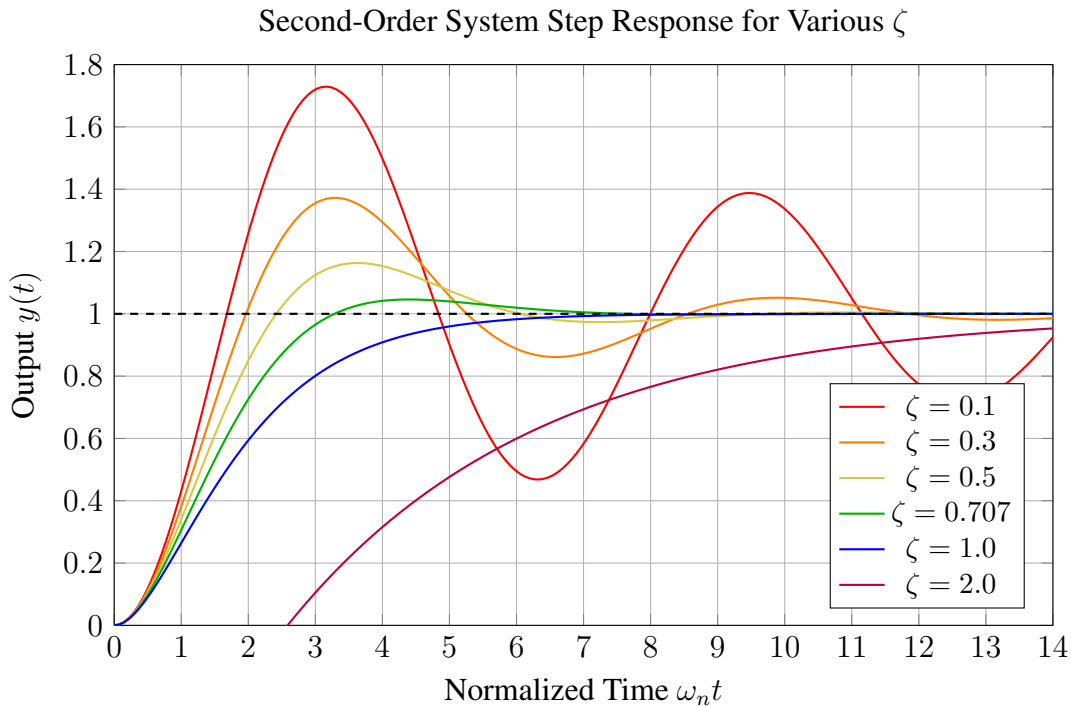


Figure 7.1: Unit step response of second-order systems with various damping ratios. Note the tradeoff: low ζ gives fast rise but large overshoot; high ζ eliminates overshoot but slows the response.

7.3.3 Rise Time

Rise time t_r measures how quickly the system responds to a step input. The standard definition is the time required for the output to rise from 10% to 90% of the final value.

For an underdamped second-order system, the rise time can be approximated by:

$$t_r \approx \frac{1.8}{\omega_n} \quad (\text{for } \zeta \approx 0.5) \quad (7.8)$$

A more accurate formula, valid for $0.3 < \zeta < 0.8$:

$$t_r \approx \frac{1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1}{\omega_n} \quad (7.9)$$

For practical design, the following approximation is often used:

$$t_r \approx \frac{\pi - \phi}{\omega_d} = \frac{\pi - \cos^{-1}(\zeta)}{\omega_n \sqrt{1 - \zeta^2}} \quad (7.10)$$

Rise Time Insight

Rise time is primarily determined by ω_n . To achieve faster rise time, one must increase ω_n by moving the poles farther from the origin. This typically requires higher gain, which in turn often reduces the damping ratio.

7.3.4 Peak Time

Peak time t_p is the time at which the output reaches its first (and maximum) peak. This occurs when $dy/dt = 0$ for the first time after $t = 0$.

Differentiating equation (??) and setting equal to zero:

$$\frac{dy}{dt} = \frac{\omega_n e^{-\zeta\omega_n t}}{\sqrt{1 - \zeta^2}} \sin(\omega_d t) = 0 \quad (7.11)$$

The first positive solution occurs when $\omega_d t_p = \pi$:

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \quad (7.12)$$

Note that peak time is only defined for underdamped systems ($\zeta < 1$); overdamped and critically damped systems do not overshoot.

7.3.5 Percent Overshoot

Percent overshoot (PO) quantifies how much the output exceeds its final value at the first peak:

$$\text{PO} = \frac{y(t_p) - y(\infty)}{y(\infty)} \times 100\% \quad (7.13)$$

For a unit step input to the standard second-order system:

$$y(t_p) = 1 + e^{-\zeta\omega_n t_p} = 1 + e^{-\zeta\pi/\sqrt{1-\zeta^2}} \quad (7.14)$$

Since $y(\infty) = 1$ for our normalized system:

$$\boxed{\text{PO} = 100 \cdot e^{-\pi\zeta/\sqrt{1-\zeta^2}} \%} \quad (7.15)$$

This formula shows that percent overshoot depends *only on damping ratio* ζ , not on natural frequency ω_n .

Table 7.1: Percent Overshoot vs. Damping Ratio

ζ	PO (%)	ζ	PO (%)
0.1	72.9	0.6	9.5
0.2	52.7	0.7	4.6
0.3	37.2	0.8	1.5
0.4	25.4	0.9	0.2
0.5	16.3	1.0	0.0

Inverse relationship: Given a required maximum overshoot, we can solve for the minimum damping ratio:

$$\boxed{\zeta = \frac{-\ln(\text{PO}/100)}{\sqrt{\pi^2 + \ln^2(\text{PO}/100)}}} \quad (7.16)$$

7.3.6 Settling Time

Settling time t_s is the time required for the output to remain within a specified percentage (typically 2% or 5%) of the final value.

The envelope of the oscillatory response is $1 \pm e^{-\zeta\omega_n t}/\sqrt{1-\zeta^2}$. For 2% settling, we require:

$$\frac{e^{-\zeta\omega_n t_s}}{\sqrt{1-\zeta^2}} \approx 0.02 \quad (7.17)$$

For moderate damping ratios ($0.5 < \zeta < 0.8$), where $1/\sqrt{1-\zeta^2} \approx 1$:

$$e^{-\zeta\omega_n t_s} \approx 0.02 \quad (7.18)$$

Taking natural logarithm:

$$-\zeta\omega_n t_s \approx \ln(0.02) \approx -4 \quad (7.19)$$

Therefore:

$$\boxed{t_s \approx \frac{4}{\zeta\omega_n} = \frac{4}{\sigma} \quad (2\% \text{ criterion})} \quad (7.20)$$

For 5% settling:

$$\boxed{t_s \approx \frac{3}{\zeta\omega_n} \quad (5\% \text{ criterion})} \quad (7.21)$$

Settling Time Insight

Settling time depends on the *real part* of the poles: $\sigma = \zeta\omega_n$. To reduce settling time, one can increase ω_n , increase ζ , or increase both (moving poles left in the s-plane).

7.3.7 Summary: Transient Specifications

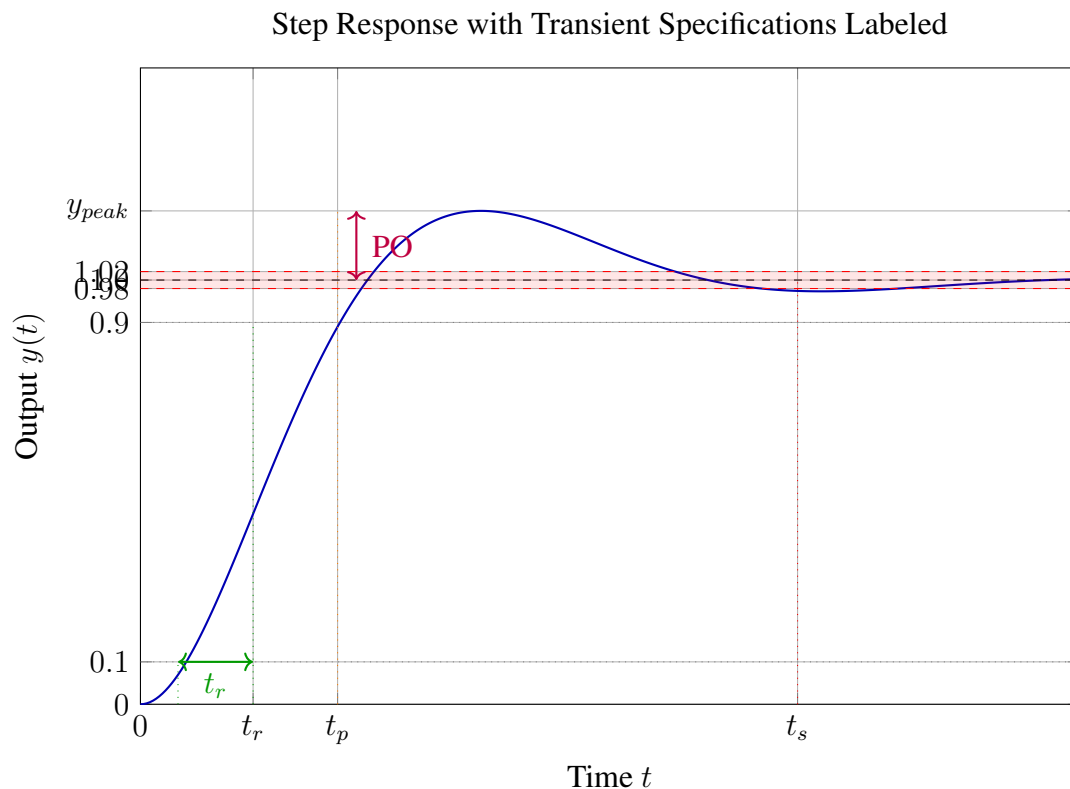


Figure 7.2: Step response of a second-order underdamped system with all transient specifications labeled. Rise time t_r : time from 10% to 90%. Peak time t_p : time to first peak. Percent overshoot: peak value minus final value. Settling time t_s : time to remain within 2% of final value.

Table 7.2: Summary of Transient Specifications for Standard Second-Order Systems

Specification	Formula	Depends on
Rise time (10%–90%)	$t_r \approx \frac{1.8}{\omega_n}$	ω_n
Peak time	$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$	ω_n, ζ
Percent overshoot	$PO = 100 \cdot e^{-\pi\zeta/\sqrt{1-\zeta^2}}$	ζ only
Settling time (2%)	$t_s \approx \frac{4}{\zeta\omega_n}$	$\zeta\omega_n = \sigma$

7.4 Mathematical Foundation: Steady-State Error Analysis

While transient specifications describe *how* a system reaches its final value, steady-state error analysis addresses *whether* it reaches the correct value at all.

7.4.1 The Final Value Theorem

The **Final Value Theorem** provides a powerful method for determining the steady-state value directly from the Laplace transform, without inverting to the time domain.

Final Value Theorem

If $y(t)$ and dy/dt are Laplace transformable, and if $\lim_{t \rightarrow \infty} y(t)$ exists and is finite, then:

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s) \quad (7.22)$$

Important caveat: The final value theorem is only valid when the limit exists—that is, when all poles of $sY(s)$ have negative real parts except possibly for a simple pole at $s = 0$. If the system is unstable, or if there are poles on the imaginary axis (other than at the origin), the theorem gives incorrect results.

7.4.2 Unity Feedback Configuration

Consider the standard unity feedback configuration:

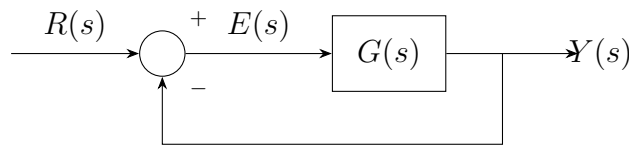


Figure 7.3: Unity feedback configuration for steady-state error analysis.

The error signal is:

$$E(s) = R(s) - Y(s) = R(s) - G(s)E(s) \quad (7.23)$$

Solving for $E(s)$:

$$E(s) = \frac{R(s)}{1 + G(s)} \quad (7.24)$$

The steady-state error for a given input is:

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)} \quad (7.25)$$

7.4.3 System Type and Error Constants

The **system type** is defined as the number of pure integrators ($1/s$ factors) in the open-loop transfer function $G(s)$.

A general open-loop transfer function can be written as:

$$G(s) = \frac{K(s + z_1)(s + z_2) \cdots}{s^N(s + p_1)(s + p_2) \cdots} \quad (7.26)$$

where N is the system type (number of integrators).

System Type Classification

Type	Integrators	Transfer Function Form
Type 0	None	$G(s) = K(\cdots)/[(\cdots)]$
Type 1	One	$G(s) = K(\cdots)/[s(\cdots)]$
Type 2	Two	$G(s) = K(\cdots)/[s^2(\cdots)]$

We define three **error constants**:

Position error constant (for step inputs):

$$K_p = \lim_{s \rightarrow 0} G(s) \quad (7.27)$$

Velocity error constant (for ramp inputs):

$$K_v = \lim_{s \rightarrow 0} sG(s) \quad (7.28)$$

Acceleration error constant (for parabolic inputs):

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) \quad (7.29)$$

7.4.4 Steady-State Error for Standard Inputs

Step Input: $R(s) = 1/s$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \cdot (1/s)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{1 + G(s)} = \frac{1}{1 + K_p} \quad (7.30)$$

Ramp Input: $R(s) = 1/s^2$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \cdot (1/s^2)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s + sG(s)} = \lim_{s \rightarrow 0} \frac{1}{sG(s)} = \frac{1}{K_v} \quad (7.31)$$

Parabolic Input: $R(s) = 1/s^3$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \cdot (1/s^3)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s^2 G(s)} = \frac{1}{K_a} \quad (7.32)$$

7.4.5 Steady-State Error Summary by System Type

Table 7.3: Steady-State Error as a Function of System Type

System Type	Error Constants			Steady-State Error e_{ss}		
	K_p	K_v	K_a	Step	Ramp	Parabolic
Type 0	K (finite)	0	0	$\frac{1}{1+K}$	∞	∞
Type 1	∞	K (finite)	0	0	$\frac{1}{K}$	∞
Type 2	∞	∞	K (finite)	0	0	$\frac{1}{K}$

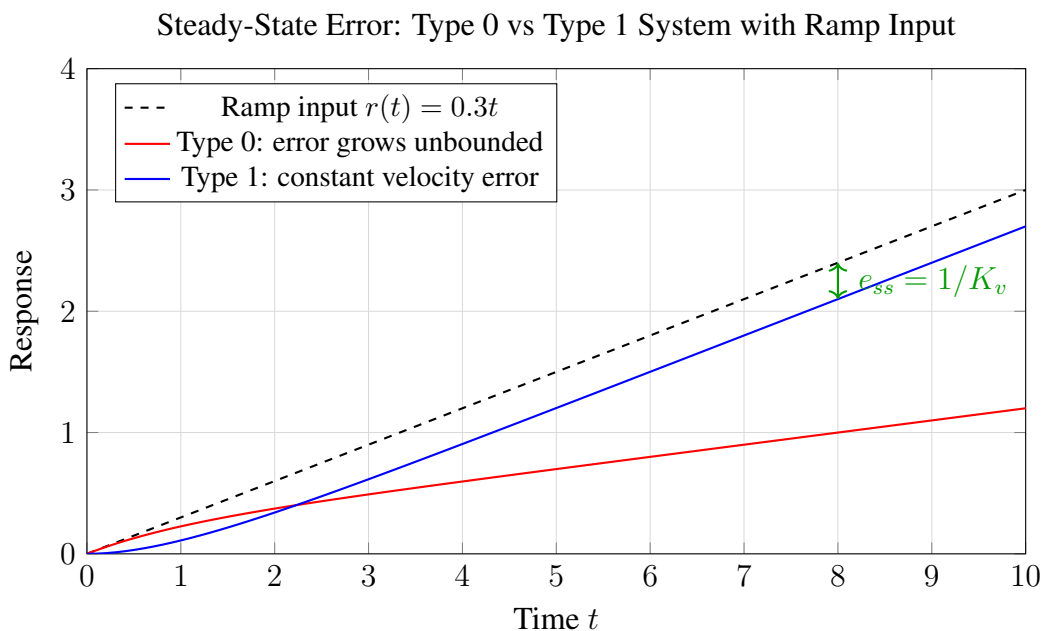


Figure 7.4: Comparison of Type 0 and Type 1 systems tracking a ramp input. Type 0 systems have unbounded error; Type 1 systems have constant “velocity error.”

7.4.6 Physical Interpretation

The steady-state error behavior has clear physical meaning:

A **Type 0** system (no integrator) can track a constant reference with finite error, where the error decreases as gain increases according to $e_{ss} = 1/(1 + K_p)$. However, it cannot track a ramp input—the error grows without bound. An example is a spring-damper system with proportional control.

A **Type 1** system (one integrator) tracks a constant reference with zero error because the integrator “accumulates” error until the output matches the reference. It can track a ramp with constant

error $e_{ss} = 1/K_v$, but cannot track a parabola—error grows without bound. An example is DC motor position control with velocity feedback.

A **Type 2** system (two integrators) tracks both step and ramp inputs with zero error and can track a parabola with constant error. However, Type 2 systems are often challenging to stabilize and require careful compensation. An example is spacecraft attitude control with angular acceleration input.

7.4.7 The Speed-Accuracy Tradeoff Revisited

The mathematical analysis reveals a fundamental tradeoff. Higher gain reduces steady-state error (since $e_{ss} \propto 1/K$) and increases ω_n , thereby reducing rise time. However, higher gain typically reduces the damping ratio ζ , which means more overshoot and longer settling time.

Adding integrators eliminates steady-state error for polynomial inputs of order less than N , but it adds phase lag that reduces phase margin. Integrators can destabilize the system if not properly compensated and generally slow the transient response.

The control engineer must navigate these tradeoffs, selecting gains and compensation that achieve acceptable performance across all specifications.

7.5 Design in the s-Plane: Meeting Multiple Specifications

When multiple specifications must be satisfied simultaneously, the s-plane provides a geometric approach to controller design.

7.5.1 s-Plane Design Regions

Consider the following specifications for a second-order dominant system: settling time $t_s \leq t_{s,\max}$, percent overshoot $\text{PO} \leq \text{PO}_{\max}$, and optionally rise time or natural frequency constraints. Each specification defines a region in the s-plane where the dominant poles must lie:

Settling time constraint: From $t_s \approx 4/\sigma$:

$$\sigma \geq \frac{4}{t_{s,\max}} \quad \Rightarrow \quad \text{Re}(s) \leq -\frac{4}{t_{s,\max}} \quad (7.33)$$

Overshoot constraint: From equation (??):

$$\zeta \geq \zeta_{\min} \quad \Rightarrow \quad \theta \leq \cos^{-1}(\zeta_{\min}) \quad (7.34)$$

where θ is the angle from the negative real axis.

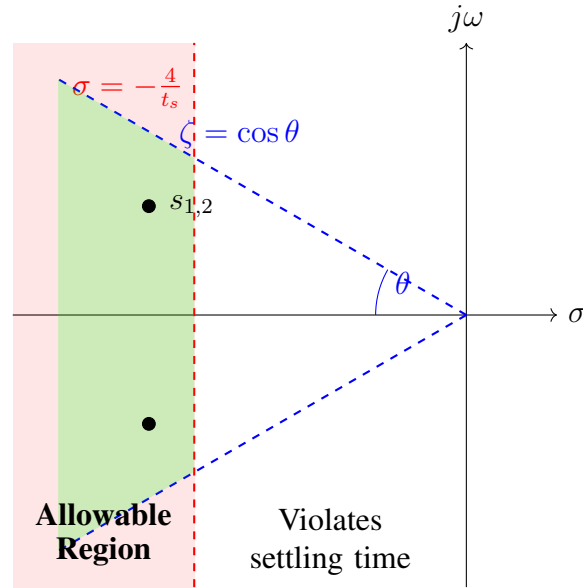


Figure 7.5: s-plane design region for specified settling time and percent overshoot. Dominant poles must lie within the shaded region to satisfy both constraints.

7.5.2 Natural Frequency Constraints

If a minimum rise time is specified, this translates to a minimum natural frequency:

$$\omega_n \geq \frac{1.8}{t_{r,\max}} \quad (7.35)$$

This adds an arc constraint: poles must lie outside a circle of radius $\omega_{n,\min}$ centered at the origin.

Similarly, bandwidth or noise rejection requirements may impose a maximum natural frequency, creating an annular region.

7.6 Practical Experiment: Motor Position Control System

This experiment demonstrates transient and steady-state concepts through a physical motor position control system.

7.6.1 Experimental Setup

The hardware components include a DC motor with optical encoder (e.g., Pololu 37D gearmotor with 64 CPR encoder), an Arduino Uno or Mega microcontroller, an L298N or similar H-bridge motor driver, a 12V power supply for the motor, and a USB connection for data logging.

In terms of system configuration, the encoder provides position feedback via quadrature decoding, the Arduino implements digital PID control, PWM output drives the motor through the H-bridge, and serial communication handles command input and data output.

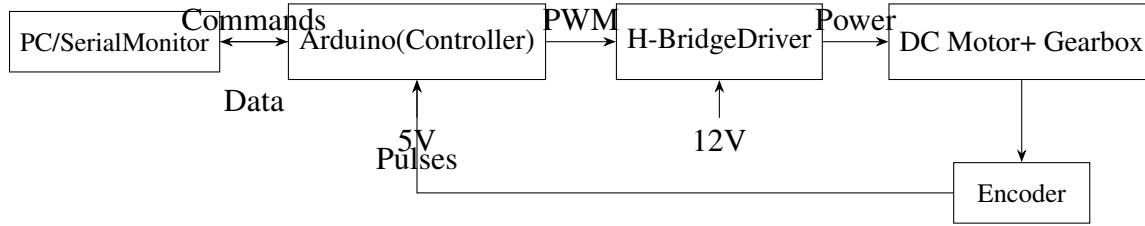


Figure 7.6: Block diagram of the experimental motor position control system.

7.6.2 Controller Implementation

The Arduino implements a discrete-time PID controller:

Algorithm 10 Discrete-Time PID Position Controller

```

1: Parameters:  $T_s \leftarrow 0.010$  (sample time in seconds)
2: Gains:  $K_p \leftarrow 2.0$ ,  $K_i \leftarrow 0.5$ ,  $K_d \leftarrow 0.1$ 
3: Initialize: integral  $\leftarrow 0$ , prevPosition  $\leftarrow 0$ 
4:
5: while controller is running do
6:   position  $\leftarrow \text{ReadEncoder}()$ 
7:   error  $\leftarrow \text{setpoint} - \text{position}$ 
8:
9:                                     ▷ Proportional term
10:   $P \leftarrow K_p \times \text{error}$ 
11:
12:                                     ▷ Integral term with anti-windup
13:  integral  $\leftarrow \text{integral} + \text{error} \times T_s$ 
14:  integral  $\leftarrow \text{CLAMP}(\text{integral}, -1000, 1000)$ 
15:   $I \leftarrow K_i \times \text{integral}$ 
16:
17:                                     ▷ Derivative term on measurement (avoids derivative kick)
18:  derivative  $\leftarrow (\text{position} - \text{prevPosition})/T_s$ 
19:   $D \leftarrow -K_d \times \text{derivative}$ 
20:
21:                                     ▷ Compute and apply control output
22:  output  $\leftarrow P + I + D$ 
23:  SetMotorPWM(CLAMP(output, -255, 255))
24:
25:                                     ▷ Update state and log data
26:  prevPosition  $\leftarrow \text{position}$ 
27:  LOGDATA(time, setpoint, position)
28:
29:  wait  $T_s$  seconds
30: end while

```

7.6.3 Experimental Procedure

Experiment 1: Measuring Step Response

Begin by setting the initial position to $\theta_0 = 0$ counts, then command a step change to $\theta_{ref} = 1000$ counts (approximately 1 revolution). Record the position every 10 ms via the serial port. Repeat this procedure for three different proportional gains: $K_p = 1.0$, $K_p = 2.0$, and $K_p = 4.0$. Keep $K_i = 0$ and $K_d = 0$ initially to isolate the effect of proportional gain.

Expected Results

With **low gain** ($K_p = 1.0$), expect slow rise time with minimal or no overshoot. There will be large steady-state error since proportional control alone cannot eliminate it, and settling time will be long due to the slow approach to the final value.

With **medium gain** ($K_p = 2.0$), expect moderate rise time with some overshoot (perhaps 10–20%). The steady-state error will be smaller than with low gain, and settling time should be reasonable.

With **high gain** ($K_p = 4.0$), expect fast rise time but significant overshoot (perhaps 30–50%). The steady-state error will be small, but settling time may be long due to oscillation. The system may approach instability.

Experiment 2: Eliminating Steady-State Error

Return to medium proportional gain ($K_p = 2.0$), then add integral action by setting $K_i = 0.2$ and later increasing to $K_i = 0.5$. Observe the effect on steady-state error and transient response.

The expected observations are that steady-state error will be reduced to zero with integral action, but overshoot may increase initially. Settling time may also increase, and too much integral gain will cause slow oscillation.

7.6.4 Data Analysis

From the recorded data, measure and calculate:

Rise time (t_r): Identify times when position crosses 10% and 90% of final value. Calculate $t_r = t_{90\%} - t_{10\%}$.

Percent overshoot:

$$PO = \frac{\theta_{\max} - \theta_{final}}{\theta_{final}} \times 100\% \quad (7.36)$$

Settling time (t_s): Find the time after which position remains within 2% of final value.

Steady-state error:

$$e_{ss} = \theta_{ref} - \theta_{final} \quad (7.37)$$

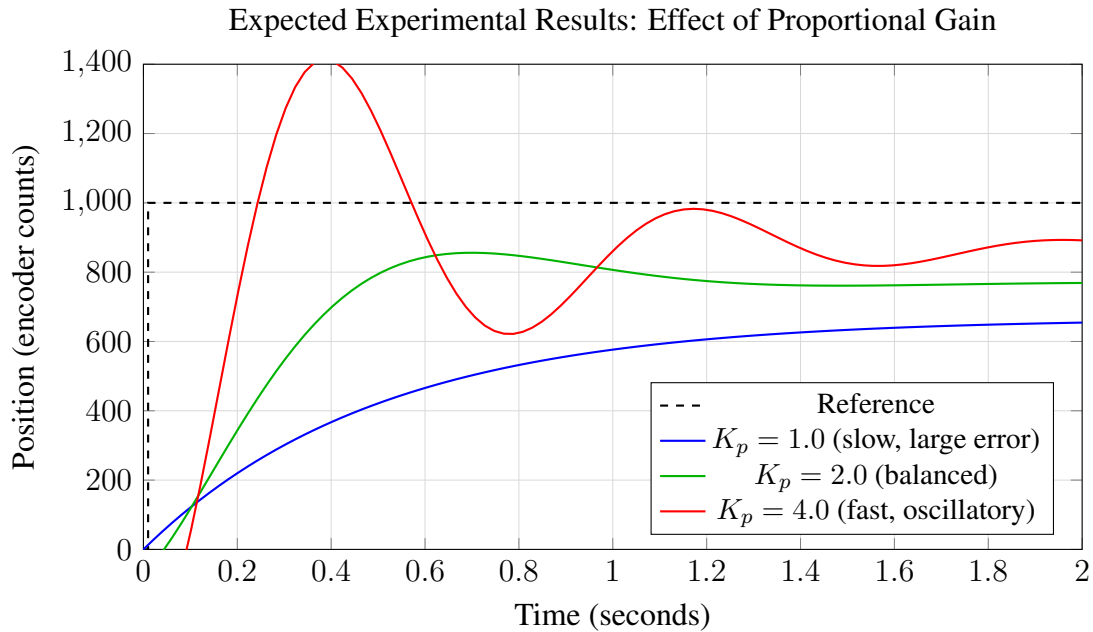


Figure 7.7: Qualitative illustration of expected experimental results showing the speed-damping tradeoff as proportional gain varies.

7.6.5 Discussion Questions

1. Why does proportional control alone result in steady-state error?

With proportional control, the motor torque is proportional to error. In steady state, the motor must overcome friction; this requires nonzero torque, hence nonzero error.

2. How does integral action eliminate steady-state error?

The integrator accumulates error over time. Even a small steady-state error causes the integral term to grow, eventually producing sufficient torque to drive error to zero.

3. Why might integral action increase overshoot?

The integral term “remembers” past error. During the approach to setpoint, the integral builds up. After crossing the setpoint, this accumulated integral continues pushing the output, causing overshoot.

4. What physical factors limit how high you can set the gains?

Motor saturation (maximum voltage/current), mechanical resonances, sensor noise amplification, sampling rate limitations, and time delays in the system.

7.7 Worked Examples

Example 7.1: Calculating Transient Specifications from Pole Location

A second-order system has closed-loop poles at $s = -3 \pm j4$. Calculate all transient specifications.

Solution:

Step 1: Extract ζ and ω_n from pole locations.

The poles are in the form $s = -\sigma \pm j\omega_d$, where:

$$\sigma = 3 \text{ rad/s} \quad (7.38)$$

$$\omega_d = 4 \text{ rad/s} \quad (7.39)$$

Natural frequency:

$$\omega_n = \sqrt{\sigma^2 + \omega_d^2} = \sqrt{9 + 16} = 5 \text{ rad/s} \quad (7.40)$$

Damping ratio:

$$\zeta = \frac{\sigma}{\omega_n} = \frac{3}{5} = 0.6 \quad (7.41)$$

Step 2: Calculate transient specifications.

Rise time:

$$t_r \approx \frac{1.8}{\omega_n} = \frac{1.8}{5} = 0.36 \text{ s} \quad (7.42)$$

Peak time:

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{4} = 0.785 \text{ s} \quad (7.43)$$

Percent overshoot:

$$\text{PO} = 100 \cdot e^{-\pi\zeta/\sqrt{1-\zeta^2}} = 100 \cdot e^{-\pi(0.6)/\sqrt{1-0.36}} = 100 \cdot e^{-2.356} = 9.5\% \quad (7.44)$$

Settling time (2%):

$$t_s = \frac{4}{\sigma} = \frac{4}{3} = 1.33 \text{ s} \quad (7.45)$$

Summary: $t_r = 0.36 \text{ s}$, $t_p = 0.785 \text{ s}$, $\text{PO} = 9.5\%$, $t_s = 1.33 \text{ s}$.

Example 7.2: Determining System Type from Transfer Function

Classify the following systems by type and determine their error constants:

$$(a) G(s) = \frac{50(s+2)}{(s+5)(s+10)}$$

$$(b) G(s) = \frac{20}{s(s+4)}$$

$$(c) G(s) = \frac{100}{s^2(s+5)}$$

Solution:

(a) Count integrators: The denominator is $(s + 5)(s + 10)$ —no factors of s .

System Type: **Type 0**

Error constants:

$$K_p = \lim_{s \rightarrow 0} G(s) = \frac{50(2)}{(5)(10)} = 2 \quad (7.46)$$

$$K_v = \lim_{s \rightarrow 0} sG(s) = 0 \quad (7.47)$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) = 0 \quad (7.48)$$

(b) Count integrators: The denominator has one factor of s .

System Type: **Type 1**

Error constants:

$$K_p = \lim_{s \rightarrow 0} G(s) = \lim_{s \rightarrow 0} \frac{20}{s(s + 4)} = \infty \quad (7.49)$$

$$K_v = \lim_{s \rightarrow 0} sG(s) = \lim_{s \rightarrow 0} \frac{20}{s + 4} = \frac{20}{4} = 5 \quad (7.50)$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) = 0 \quad (7.51)$$

(c) Count integrators: The denominator has s^2 .

System Type: **Type 2**

Error constants:

$$K_p = \infty \quad (7.52)$$

$$K_v = \lim_{s \rightarrow 0} sG(s) = \lim_{s \rightarrow 0} \frac{100}{s(s + 5)} = \infty \quad (7.53)$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) = \lim_{s \rightarrow 0} \frac{100}{s + 5} = \frac{100}{5} = 20 \quad (7.54)$$

Example 7.3: Steady-State Error for Step and Ramp Inputs

For a unity feedback system with $G(s) = \frac{100}{s(s + 10)}$, calculate the steady-state error for:

(a) A unit step input, $r(t) = 1$

(b) A unit ramp input, $r(t) = t$

(c) A ramp input with slope 5, $r(t) = 5t$

Solution:

First, identify the system type: one integrator in denominator $\Rightarrow$ **Type 1**.

Calculate error constants:

$$K_p = \lim_{s \rightarrow 0} \frac{100}{s(s+10)} = \infty \quad (7.55)$$

$$K_v = \lim_{s \rightarrow 0} s \cdot \frac{100}{s(s+10)} = \frac{100}{10} = 10 \quad (7.56)$$

(a) Unit step input:

For Type 1 system with step input:

$$e_{ss} = \frac{1}{1 + K_p} = \frac{1}{1 + \infty} = \boxed{0} \quad (7.57)$$

(b) Unit ramp input:

For Type 1 system with ramp input $R(s) = 1/s^2$:

$$e_{ss} = \frac{1}{K_v} = \frac{1}{10} = \boxed{0.1 \text{ units}} \quad (7.58)$$

(c) Ramp with slope 5:

For a ramp $r(t) = 5t$, so $R(s) = 5/s^2$. The steady-state error scales with the input:

$$e_{ss} = \frac{5}{K_v} = \frac{5}{10} = \boxed{0.5 \text{ units}} \quad (7.59)$$

Example 7.4: Design to Meet Transient Specifications

Design a second-order system (specify ζ and ω_n) to meet the following specifications: percent overshoot $PO \leq 10\%$ and settling time $t_s \leq 2$ s (2% criterion).

Solution:

Step 1: Determine minimum damping ratio from overshoot specification.

From equation (??):

$$\zeta \geq \frac{-\ln(0.10)}{\sqrt{\pi^2 + \ln^2(0.10)}} = \frac{2.303}{\sqrt{9.87 + 5.30}} = \frac{2.303}{3.89} = 0.592 \quad (7.60)$$

So we need $\zeta \geq 0.592$. Choose $\zeta = 0.6$ for design margin.

Step 2: Determine minimum σ from settling time specification.

From $t_s = 4/\sigma \leq 2$ s:

$$\sigma \geq \frac{4}{2} = 2 \text{ rad/s} \quad (7.61)$$

Step 3: Calculate minimum ω_n .

Since $\sigma = \zeta\omega_n$:

$$\omega_n \geq \frac{\sigma}{\zeta} = \frac{2}{0.6} = 3.33 \text{ rad/s} \quad (7.62)$$

Step 4: Choose design values.

Select $\zeta = 0.6$ and $\omega_n = 3.5$ rad/s (with margin).

Verification:

$$\text{PO} = 100 \cdot e^{-\pi(0.6)/\sqrt{1-0.36}} = 9.5\% \leq 10\% \quad \checkmark \quad (7.63)$$

$$t_s = \frac{4}{0.6 \times 3.5} = 1.90 \text{ s} \leq 2 \text{ s} \quad \checkmark \quad (7.64)$$

Transfer function:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{12.25}{s^2 + 4.2s + 12.25} \quad (7.65)$$

Pole locations: $s = -2.1 \pm j2.8$

Example 7.5: Choosing Gain to Achieve Target Steady-State Error

A unity feedback system has the open-loop transfer function:

$$G(s) = \frac{K}{(s+2)(s+5)} \quad (7.66)$$

Find the value of K such that the steady-state error for a unit step input is 10%.

Solution:

Step 1: Identify system type.

No integrators in $G(s) \Rightarrow$ **Type 0**

Step 2: Calculate K_p in terms of K .

$$K_p = \lim_{s \rightarrow 0} G(s) = \frac{K}{(2)(5)} = \frac{K}{10} \quad (7.67)$$

Step 3: Set up equation for steady-state error.

For a Type 0 system with unit step input:

$$e_{ss} = \frac{1}{1 + K_p} = 0.10 \quad (7.68)$$

Step 4: Solve for K_p and then K .

$$\frac{1}{1 + K_p} = 0.10 \quad (7.69)$$

$$1 + K_p = 10 \quad (7.70)$$

$$K_p = 9 \quad (7.71)$$

Therefore:

$$\frac{K}{10} = 9 \quad \Rightarrow \quad \boxed{K = 90} \quad (7.72)$$

Step 5: Verify and check stability.

The closed-loop transfer function is:

$$T(s) = \frac{90}{(s+2)(s+5) + 90} = \frac{90}{s^2 + 7s + 10 + 90} = \frac{90}{s^2 + 7s + 100} \quad (7.73)$$

Characteristic equation: $s^2 + 7s + 100 = 0$

Poles: $s = \frac{-7 \pm \sqrt{49 - 400}}{2} = \frac{-7 \pm j18.7}{2} = -3.5 \pm j9.4$

Both poles in LHP $\Rightarrow$ stable.

Damping ratio: $\zeta = 3.5/10.05 = 0.35$

This gives significant overshoot ($PO \approx 31\%$). If transient performance is also important, a different approach (adding compensation) may be needed.

Example 7.6: Complete System Design

Design a controller for the plant $G_p(s) = \frac{10}{s(s+5)}$ to achieve zero steady-state error for step inputs, steady-state error ≤ 0.05 units for a unit ramp input, percent overshoot $\leq 20\%$, and settling time ≤ 1 s.

Solution:

Step 1: Analyze the open-loop system.

The plant has one integrator $\Rightarrow$ Type 1 system. This automatically gives zero steady-state error for step inputs.

Step 2: Determine required K_v for ramp error specification.

For ramp error $e_{ss} = 1/K_v \leq 0.05$:

$$K_v \geq 20 \quad (7.74)$$

Step 3: Determine controller structure.

Use proportional control $G_c(s) = K$. The open-loop function becomes:

$$G(s) = \frac{10K}{s(s+5)} \quad (7.75)$$

Calculate K_v :

$$K_v = \lim_{s \rightarrow 0} sG(s) = \lim_{s \rightarrow 0} \frac{10K}{s+5} = \frac{10K}{5} = 2K \quad (7.76)$$

For $K_v \geq 20$: $2K \geq 20 \Rightarrow K \geq 10$.

Step 4: Check transient specifications.

Closed-loop transfer function with $K = 10$:

$$T(s) = \frac{100}{s^2 + 5s + 100} \quad (7.77)$$

Compare with standard form: $\omega_n^2 = 100 \Rightarrow \omega_n = 10$ rad/s, and $2\zeta\omega_n = 5 \Rightarrow \zeta = 0.25$.

Check overshoot:

$$\text{PO} = 100 \cdot e^{-\pi(0.25)/\sqrt{1-0.0625}} = 100 \cdot e^{-0.811} = 44.4\% \quad (7.78)$$

This exceeds the 20% specification!

Step 5: Revise design with compensation.

We need to increase damping without sacrificing K_v . Add a lead compensator:

$$G_c(s) = K \frac{s+a}{s+b} \quad \text{with } b > a \quad (7.79)$$

Choose $a = 2$ and $b = 20$ (lead ratio of 10):

$$G_c(s) = K \frac{s+2}{s+20} \quad (7.80)$$

With this compensator:

$$K_v = \lim_{s \rightarrow 0} s \cdot K \frac{s+2}{s+20} \cdot \frac{10}{s(s+5)} = \frac{10K \cdot 2}{20 \cdot 5} = \frac{K}{5} \quad (7.81)$$

For $K_v \geq 20$: $K \geq 100$.

With $K = 100$, the open-loop function is:

$$G(s) = \frac{1000(s+2)}{s(s+5)(s+20)} \quad (7.82)$$

This third-order system must be analyzed numerically or via root locus. The lead compensator adds phase lead near crossover, improving damping.

Design verification (via root locus or simulation): The dominant poles are located at approximately $s = -4 \pm j4$, giving an effective $\zeta \approx 0.7$ and effective $\omega_n \approx 5.7$ rad/s. This yields $\text{PO} \approx 5\%$ (meets spec), $t_s \approx 1.0$ s (meets spec), and $e_{ss,ramp} = 1/20 = 0.05$ (meets spec).

7.8 Diagrams and Visual Reference

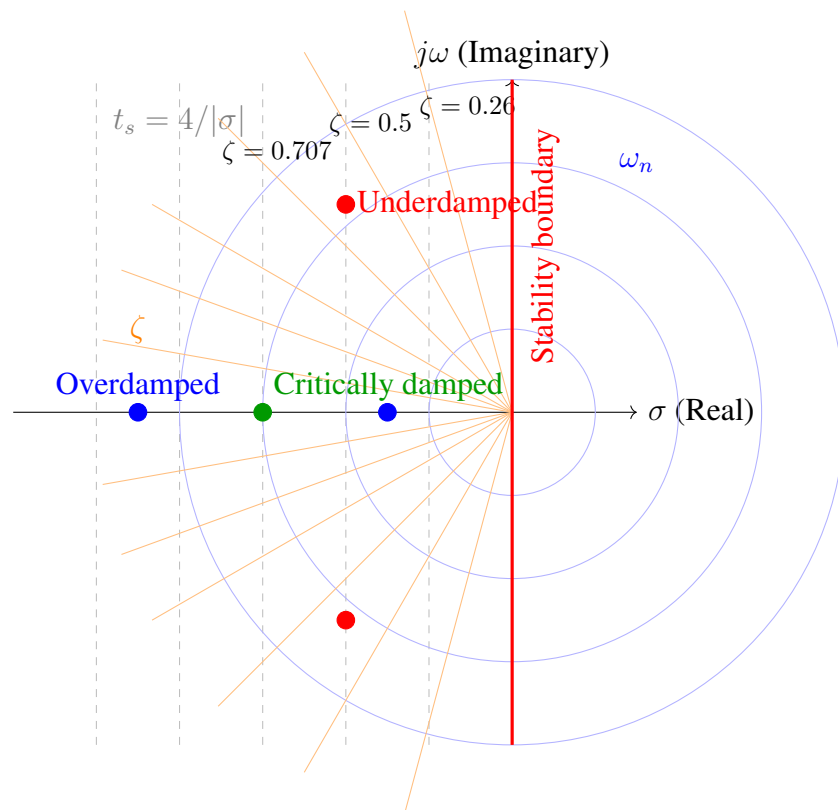


Figure 7.8: s-plane showing lines of constant settling time (σ), constant natural frequency (ω_n), and constant damping ratio (ζ). Poles must lie in the left half-plane for stability.

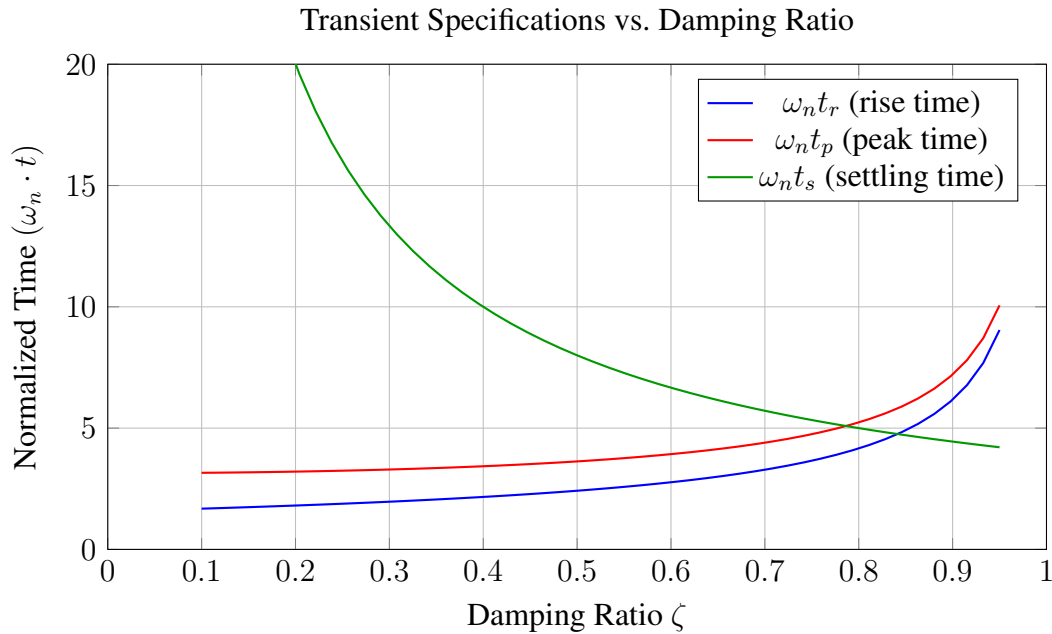


Figure 7.9: Normalized transient specifications as functions of damping ratio. Note: rise time has a minimum near $\zeta = 0.4$; settling time decreases monotonically with ζ .

7.9 Summary

This chapter has developed the framework for analyzing and specifying control system performance in both transient and steady-state regimes.

Key Concepts

Transient Response Specifications: **Rise time** (t_r) measures response speed and is primarily determined by ω_n . **Peak time** (t_p) is the time to first overshoot, given by $t_p = \pi/\omega_d$. **Percent overshoot** (PO) depends only on ζ , with lower ζ producing more overshoot. **Settling time** (t_s) is the time to reach steady state, approximately $t_s \approx 4/(\zeta\omega_n)$.

Steady-State Error Analysis: The **system type** is the number of integrators in the open-loop transfer function. The **error constants** (K_p , K_v , K_a) determine error magnitude for polynomial inputs. Type 0 systems have finite step error but infinite ramp error. Type 1 systems have zero step error but finite ramp error. Type 2 systems have zero step and ramp error but finite parabolic error.

Fundamental Tradeoffs: Higher gain reduces steady-state error but often increases overshoot. Faster response (higher ω_n) requires higher gain. Adding integrators eliminates steady-state error but reduces stability margins. The s-plane provides a geometric framework for balancing multiple specifications.

Design Guidelines

The recommended design procedure begins by determining the required ζ from the overshoot specification, then calculating the required $\sigma = \zeta\omega_n$ from the settling time specification. From these,

determine the minimum ω_n from the ratio σ/ζ . Next, check steady-state error requirements and add integrators or increase gain as needed. Always verify that stability margins remain adequate. Finally, use compensation (lead, lag, or PID) to satisfy conflicting requirements.

7.10 Exercises

Exercise 7.1. A second-order system has poles at $s = -2 \pm j3$. Calculate:

1. Natural frequency ω_n and damping ratio ζ
2. Rise time, peak time, percent overshoot, and settling time
3. The closed-loop transfer function (assuming unity DC gain)

Exercise 7.2. For the system $G(s) = \frac{20(s+3)}{s^2(s+5)(s+10)}$:

1. Determine the system type
2. Calculate all error constants
3. Find the steady-state error for inputs $r(t) = 5$, $r(t) = 2t$, and $r(t) = t^2$

Exercise 7.3. Design a second-order system to achieve settling time ≤ 0.5 s and percent overshoot $\leq 5\%$. Specify ζ , ω_n , and pole locations.

Exercise 7.4. A unity feedback system has $G(s) = \frac{K}{s+10}$.

1. What is the system type?
2. Find K for 5% steady-state error to a step input
3. With this K , what is the closed-loop time constant?

Exercise 7.5. For the motor control experiment, if the measured step response shows rise time of 0.15 s, overshoot of 25%, and settling time of 0.8 s, estimate the effective ζ and ω_n of the closed-loop system.

Exercise 7.6. A Type 1 system must track a ramp input with velocity 10 units/s with steady-state error less than 0.1 units. What minimum velocity error constant K_v is required?

Exercise 7.7. Sketch the allowable region in the s-plane for dominant poles that satisfy settling time ≤ 2 s, overshoot $\leq 15\%$, and rise time ≤ 0.5 s.

Exercise 7.8. Explain why adding an integrator to improve steady-state error may degrade transient performance. What compensation technique can restore good transient response?

Exercise 7.9. A hard disk servo must position the head within $0.1 \mu\text{m}$ of the target track within 4 ms. If the maximum acceptable overshoot is 2%, estimate the required ζ and ω_n .

Exercise 7.10. Compare the step responses of three systems, all with $\omega_n = 5$ rad/s:

1. $\zeta = 0.3$
2. $\zeta = 0.7$
3. $\zeta = 1.5$

Calculate and tabulate all transient specifications. Which provides the best compromise between speed and smoothness?

Further Reading

1. Ziegler, J.G., and Nichols, N.B., “Optimum Settings for Automatic Controllers,” *Transactions of the ASME*, Vol. 64, pp. 759–768, 1942. (The foundational paper on performance-based tuning.)
2. James, H.M., Nichols, N.B., and Phillips, R.S., *Theory of Servomechanisms*, McGraw-Hill, 1947. (The MIT Radiation Laboratory’s wartime research, declassified and published.)
3. Franklin, G.F., Powell, J.D., and Emami-Naeini, A., *Feedback Control of Dynamic Systems*, 8th ed., Pearson, 2019. (Comprehensive modern treatment of transient and steady-state analysis.)
4. Ogata, K., *Modern Control Engineering*, 5th ed., Prentice Hall, 2010. (Clear derivations of all second-order system formulas.)
5. Bennett, S., *A History of Control Engineering*, Peter Peregrinus, 1993. (Excellent historical context for the development of performance specifications.)

Chapter 8

Linearization

“The whole of mathematics consists in the organization of a series of aids to the imagination in the process of reasoning.” — Alfred North Whitehead

8.1 Historical Motivation: From Newton to Modern Control

The story of linearization is, in many ways, the story of applied mathematics itself. It represents humanity’s pragmatic response to a fundamental truth: most physical systems are inherently nonlinear, yet our mathematical tools for analyzing nonlinear systems remain severely limited. Linearization is the bridge that allows us to apply the powerful machinery of linear algebra and differential equations to the messy reality of physical systems.

8.1.1 Newton and the Birth of Local Approximation (1687)

Newton's Revolutionary Insight

Isaac Newton's *Principia Mathematica*, published in 1687, laid the foundation for both calculus and the concept of local approximation. Newton recognized that complicated functions could be represented as infinite series—what we now call Taylor series or power series expansions. In his *Method of Fluxions*, Newton developed techniques for expanding functions around a point:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots \quad (8.1)$$

The revolutionary insight was that *near* the expansion point x_0 , the higher-order terms become negligibly small. If $(x - x_0)$ is small, then $(x - x_0)^2$ is smaller still, and $(x - x_0)^3$ even more so. This observation—that locally, any smooth function can be approximated by a straight line—is the mathematical foundation of linearization.

Newton himself applied this principle in his analysis of planetary motion. While the gravitational interactions of multiple bodies create hopelessly complex nonlinear dynamics, Newton showed that for small perturbations from a known orbit, the motion could be approximated by linear equations. This “perturbation theory” became a cornerstone of celestial mechanics.

8.1.2 Small Oscillation Theory in Physics

The 18th and 19th centuries saw the systematic application of linearization to problems in physics, particularly in the study of oscillations. Consider the simple pendulum, whose exact equation of motion is:

$$\ddot{\theta} + \frac{g}{L} \sin(\theta) = 0 \quad (8.2)$$

This equation has no closed-form solution in terms of elementary functions. However, physicists from Huygens onward recognized that for small angles, $\sin(\theta) \approx \theta$, yielding:

$$\ddot{\theta} + \frac{g}{L} \theta = 0 \quad (8.3)$$

This is the equation of simple harmonic motion, with the well-known solution $\theta(t) = A \cos(\omega_n t + \phi)$ where $\omega_n = \sqrt{g/L}$. The period $T = 2\pi \sqrt{L/g}$ is independent of amplitude—but only in the linear approximation.

Joseph-Louis Lagrange (1736–1813) systematized this approach in his *Mécanique Analytique* (1788), developing a general theory of small oscillations for systems with multiple degrees of freedom. His method proceeded by first finding the equilibrium configuration, then expanding the potential and kinetic energies to second order about that equilibrium. From these quadratic approximations, he derived linearized equations of motion, which could then be solved to find the “normal modes” of oscillation—the natural patterns in which the system vibrates. This framework remains the standard approach for vibration analysis in mechanical engineering.

8.1.3 Poincaré and the Qualitative Theory (1880s)

Henri Poincaré (1854–1912) revolutionized the study of differential equations by shifting focus from exact solutions to qualitative behavior. In his work on the three-body problem and celes-

tial mechanics, Poincaré recognized that understanding the geometry of solutions—their stability, periodic orbits, and long-term behavior—was more valuable than explicit formulas.

Poincaré’s key contribution to linearization was the concept of **local stability analysis**. He showed that the stability of an equilibrium point of a nonlinear system:

$$\dot{x} = f(x) \quad (8.4)$$

could be determined by examining the linearized system:

$$\delta\dot{x} = \left. \frac{\partial f}{\partial x} \right|_{x_0} \delta x \quad (8.5)$$

If all eigenvalues of the Jacobian matrix $\partial f/\partial x|_{x_0}$ have negative real parts, the equilibrium is locally asymptotically stable. This principle, formalized by Lyapunov shortly after, remains the primary tool for stability analysis in control theory.

Poincaré also introduced the concept of the **phase portrait**—a graphical representation of system trajectories in state space. The phase portrait of a nonlinear system near an equilibrium point closely resembles that of its linearization, provided the eigenvalues have non-zero real parts (hyperbolic equilibria). This observation justifies the use of linear analysis for local behavior.

8.1.4 Aircraft Stability Analysis (1900s)

The development of powered flight created an urgent need for stability analysis of complex dynamic systems. The Wright Brothers’ 1903 Flyer was deliberately designed to be unstable, requiring constant pilot input—a choice that gave them better control authority but made flight exhausting and dangerous.

By the 1910s, aviation pioneers recognized that stable aircraft were essential for practical flight. Frederick Lanchester in England and George Bryan in the United States developed the mathematical framework for aircraft stability, based entirely on linearization.

An aircraft in flight obeys nonlinear equations of motion with six degrees of freedom (three translations, three rotations). The full equations involve trigonometric functions, aerodynamic forces that depend on angle of attack in complicated ways, and coupling between all axes. Exact analysis is intractable.

The breakthrough was the concept of the **trim condition**: a steady, level flight condition where all forces and moments balance. Small perturbations from trim satisfy linearized equations:

$$\begin{bmatrix} \delta\dot{u} \\ \delta\dot{w} \\ \delta\dot{q} \\ \delta\dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & X_w & X_q & -g \cos \theta_0 \\ Z_u & Z_w & Z_q & -g \sin \theta_0 \\ M_u & M_w & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \delta u \\ \delta w \\ \delta q \\ \delta \theta \end{bmatrix} \quad (8.6)$$

Here $\delta u, \delta w$ are perturbations in forward and vertical velocity, δq is pitch rate perturbation, and $\delta \theta$ is pitch angle perturbation. The “stability derivatives” X_u, Z_w, M_q , etc., are partial derivatives of aerodynamic forces and moments with respect to state variables, evaluated at trim.

This linearized model enabled engineers to predict natural frequencies such as the phugoid and short period modes, design control surfaces with adequate stability margins, analyze pilot-in-the-loop handling qualities, and develop autopilot systems. The success of linearized aircraft dynamics established the paradigm that dominates control engineering: *design around an operating point*.

8.1.5 Why Linearize? The Practical Argument

Nonlinear differential equations are **hard**. The principle of superposition, which states that for linear systems any combination $\alpha x_1(t) + \beta x_2(t)$ of solutions is also a solution, does not hold for nonlinear systems. This superposition principle is what allows us to analyze complex inputs by decomposing them into simpler components through Fourier analysis and convolution. Furthermore, while linear differential equations with constant coefficients can always be solved using eigenvalue methods, most nonlinear equations have no closed-form solutions. The Laplace transform, which converts linear ODEs to algebraic equations and enables frequency-domain analysis, is not applicable to nonlinear systems. Finally, stability analysis becomes far more complex: for linear systems, stability depends only on eigenvalue locations, whereas nonlinear systems can exhibit limit cycles, chaos, and other behaviors that linear theory cannot predict.

Linear systems, by contrast, are mathematically tractable. They admit complete solutions via eigenvalue decomposition and can be represented by transfer functions that enable frequency response analysis. Classical design methods such as root locus and Bode plots become available, along with modern techniques including state feedback and observer design via pole placement. Optimal control approaches, particularly Linear-Quadratic Regulator (LQR) design and Kalman filtering, round out the powerful toolkit available for linear systems.

The entire toolkit of classical and modern control theory—Chapters 9–20 of this textbook—assumes linear system models. Linearization is the gateway that allows us to apply these powerful methods to real, nonlinear systems.

The engineering assumption: Most control systems are designed to keep the system near a desired operating point. A thermostat maintains room temperature near a setpoint; an autopilot holds altitude and heading near desired values; a robot arm tracks a smooth trajectory. If the controller works properly, perturbations from the operating point remain small, and the linear approximation is valid.

This is a self-fulfilling prophecy: we design controllers using linear theory, and those controllers (when successful) keep the system in the region where linear theory is valid.

8.2 Modern Applications of Linearization

Linearization is not merely a historical curiosity; it remains the dominant approach for control system design across virtually every engineering domain. This section surveys modern applications where linearization enables sophisticated control of inherently nonlinear systems.

8.2.1 Aircraft Autopilot Design

Modern commercial aircraft operate with sophisticated autopilot systems that control altitude, heading, speed, and approach trajectories. Despite the nonlinear nature of aircraft dynamics, these systems are designed entirely using linearized models.

The process begins with a detailed nonlinear simulation model that captures aerodynamic forces as functions of angle of attack, sideslip, and Mach number, along with engine thrust characteristics across the flight envelope. The model also incorporates atmospheric variations with altitude and actuator dynamics including rate limits.

At each flight condition (altitude, airspeed, weight), the nonlinear model is linearized to produce state-space matrices (A, B, C, D) . Control laws are designed for each linearization point, and gain scheduling interpolates between designs as flight conditions change.

This approach has proven remarkably successful. The Boeing 777, for example, uses a fly-by-wire system with linearization-based control laws that safely operate across the entire flight envelope—from takeoff to landing, in turbulence and wind shear, with engine failures and other contingencies.

8.2.2 Chemical Process Control

Chemical reactors exhibit strongly nonlinear behavior arising from several sources. The Arrhenius temperature dependence of reaction rates, expressed as $k = Ae^{-E_a/RT}$, introduces exponential nonlinearity. Heat transfer is inherently nonlinear, especially when phase changes occur. Reaction kinetics depend on concentrations in complex ways, and many reactors exhibit multiple steady states with the potential for thermal runaway.

Despite this complexity, most industrial process control uses PID controllers designed via linearization. At the desired operating point (setpoint), the process is linearized to determine appropriate controller gains. Advanced regulatory control uses model predictive control (MPC) with linearized models updated in real-time.

The key insight is that a well-designed control system prevents the large deviations that would make linearization invalid. Temperature excursions, concentration swings, and pressure variations are kept small by feedback control—exactly the regime where linear models are accurate.

8.2.3 Power Electronics: Small-Signal Analysis

Switch-mode power supplies, motor drives, and renewable energy inverters involve switching nonlinearities that seem to defy linear analysis. However, the concept of “small-signal analysis” applies linearization to these systems with remarkable success.

Consider a DC-DC buck converter. The switches (transistors, diodes) are inherently nonlinear. However, if we average the switching behavior over one switching period, we obtain a continuous (but still nonlinear) model. Linearizing around the operating point yields a small-signal model:

$$\hat{v}_o(s) = G_{vd}(s)\hat{d}(s) + G_{vg}(s)\hat{v}_g(s) \quad (8.7)$$

where $\hat{v}_o$ is output voltage perturbation, $\hat{d}$ is duty cycle perturbation, and $\hat{v}_g$ is input voltage perturbation.

This transfer function model enables loop gain analysis for stability assessment, compensator design using Bode plot techniques, input and output impedance calculations, and prediction of transient response characteristics. The entire field of power electronics control is built on this small-signal linearization framework.

8.2.4 Robot Arm Control: Jacobian Linearization

Robot manipulators are highly nonlinear systems. The relationship between joint angles and end-effector position involves trigonometric functions (kinematics), and the dynamics include configuration-dependent inertias, Coriolis forces, and gravity terms.

For trajectory tracking, robots use a combination of feedforward (computed torque) and feedback control. The feedback component typically uses linearization:

$$\delta\dot{x} = J(q)\delta\dot{q} \quad (8.8)$$

where $J(q)$ is the Jacobian matrix relating joint velocities to end-effector velocities.

Near the desired trajectory, the linearized dynamics enable resolved-motion rate control, Cartesian impedance control, and force/position hybrid control schemes. Modern industrial robots achieve submillimeter accuracy by using linearization-based control, with the nonlinear terms handled by feedforward compensation.

8.3 Mathematical Foundation

This section provides the complete mathematical framework for linearization of nonlinear systems. We begin with a review of Taylor series and progress to the general procedure for multi-input, multi-output systems.

8.3.1 Taylor Series Review

The foundation of linearization is the Taylor series expansion of a smooth function. For a scalar function $f(x)$ of a single variable, expanded about x_0 :

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \dots \quad (8.9)$$

For small deviations $\delta x = x - x_0$, the higher-order terms become negligible:

$$f(x) \approx f(x_0) + f'(x_0)\delta x + O(\delta x^2) \quad (8.10)$$

The notation $O(\delta x^2)$ indicates terms that scale with the square (or higher powers) of δx . When $|\delta x| \ll 1$, these terms can be neglected.

For functions of multiple variables, $f(x_1, x_2, \dots, x_n)$, the Taylor expansion becomes:

$$f(\mathbf{x}) = f(\mathbf{x}_0) + \sum_{i=1}^n \left. \frac{\partial f}{\partial x_i} \right|_{\mathbf{x}_0} (x_i - x_{0,i}) + \text{higher order terms} \quad (8.11)$$

In vector notation:

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \nabla f|_{\mathbf{x}_0}^T (\mathbf{x} - \mathbf{x}_0) \quad (8.12)$$

where ∇f is the gradient vector.

8.3.2 The Linearization Procedure

Consider a general nonlinear system described by:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) \quad (8.13)$$

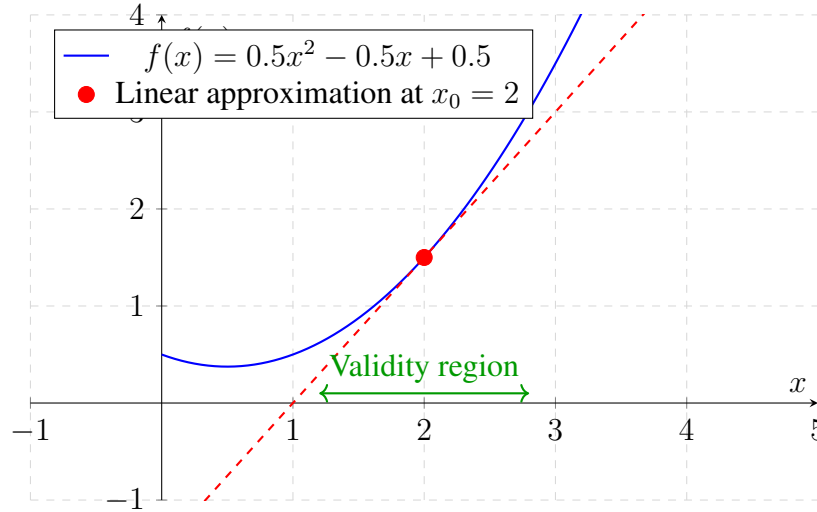


Figure 8.1: Graphical interpretation of linearization. The tangent line (dashed) approximates the nonlinear function (solid) near the operating point $x_0 = 2$. The approximation is accurate within the validity region.

where $\mathbf{x} \in \mathbb{R}^n$ is the state vector, $\mathbf{u} \in \mathbb{R}^m$ is the input vector, and $\mathbf{f} : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a smooth (continuously differentiable) function.

Step 1: Find the Equilibrium Point

An equilibrium (operating) point $(\mathbf{x}_0, \mathbf{u}_0)$ satisfies:

$$\mathbf{f}(\mathbf{x}_0, \mathbf{u}_0) = \mathbf{0} \quad (8.14)$$

At equilibrium, the state derivative is zero, so the system remains at $\mathbf{x}_0$ when input is held at $\mathbf{u}_0$. For many systems, multiple equilibrium points may exist, and the choice of operating point depends on the desired system behavior.

Step 2: Define Perturbation Variables

Define deviations from the operating point:

$$\delta \mathbf{x} = \mathbf{x} - \mathbf{x}_0 \quad (8.15)$$

$$\delta \mathbf{u} = \mathbf{u} - \mathbf{u}_0 \quad (8.16)$$

Step 3: Apply Taylor Series Expansion

Expand $\mathbf{f}(\mathbf{x}, \mathbf{u})$ about the operating point:

$$\mathbf{f}(\mathbf{x}, \mathbf{u}) = \mathbf{f}(\mathbf{x}_0, \mathbf{u}_0) + \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_0 (\mathbf{x} - \mathbf{x}_0) + \left. \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right|_0 (\mathbf{u} - \mathbf{u}_0) + \text{H.O.T.} \quad (8.17)$$

Since $\mathbf{f}(\mathbf{x}_0, \mathbf{u}_0) = \mathbf{0}$ at equilibrium, and noting that $\dot{\mathbf{x}} = \delta \dot{\mathbf{x}}$ (since $\mathbf{x}_0$ is constant):

$$\delta \dot{\mathbf{x}} \approx \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_0 \delta \mathbf{x} + \left. \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right|_0 \delta \mathbf{u} \quad (8.18)$$

Step 4: Define Jacobian Matrices

The **system matrix** (or **state Jacobian**) is:

$$\mathbf{A} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{(\mathbf{x}_0, \mathbf{u}_0)} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}_{(\mathbf{x}_0, \mathbf{u}_0)} \quad (8.19)$$

The **input matrix** (or **control Jacobian**) is:

$$\mathbf{B} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right|_{(\mathbf{x}_0, \mathbf{u}_0)} = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} & \dots & \frac{\partial f_1}{\partial u_m} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} & \dots & \frac{\partial f_2}{\partial u_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \frac{\partial f_n}{\partial u_2} & \dots & \frac{\partial f_n}{\partial u_m} \end{bmatrix}_{(\mathbf{x}_0, \mathbf{u}_0)} \quad (8.20)$$

Step 5: Write the Linearized Model

The linearized state-space model is:

$$\boxed{\delta \dot{\mathbf{x}} = \mathbf{A} \delta \mathbf{x} + \mathbf{B} \delta \mathbf{u}} \quad (8.21)$$

This is a linear, time-invariant (LTI) system in the perturbation variables. All the tools of linear control theory can now be applied.

If there is also an output equation $\mathbf{y} = \mathbf{g}(\mathbf{x}, \mathbf{u})$, it can be linearized similarly:

$$\delta \mathbf{y} = \mathbf{C} \delta \mathbf{x} + \mathbf{D} \delta \mathbf{u} \quad (8.22)$$

where:

$$\mathbf{C} = \left. \frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right|_0, \quad \mathbf{D} = \left. \frac{\partial \mathbf{g}}{\partial \mathbf{u}} \right|_0 \quad (8.23)$$

8.3.3 Example: The Simple Pendulum

The simple pendulum provides an ideal case study for linearization. Consider a pendulum of length L with a point mass m , swinging under gravity.

Nonlinear Equation of Motion

From Newton's second law (or Lagrangian mechanics):

$$mL^2 \ddot{\theta} = -mgL \sin(\theta) \quad (8.24)$$

Simplifying:

$$\ddot{\theta} + \frac{g}{L} \sin(\theta) = 0 \quad (8.25)$$

State-Space Formulation

Define state variables:

$$x_1 = \theta, \quad x_2 = \dot{\theta} \quad (8.26)$$

The state equations are:

$$\dot{x}_1 = x_2 \quad (8.27)$$

$$\dot{x}_2 = -\frac{g}{L} \sin(x_1) \quad (8.28)$$

Or in vector form:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) = \begin{bmatrix} x_2 \\ -\frac{g}{L} \sin(x_1) \end{bmatrix} \quad (8.29)$$

Equilibrium Points

Setting $\mathbf{f}(\mathbf{x}) = \mathbf{0}$:

$$x_2 = 0 \quad (8.30)$$

$$-\frac{g}{L} \sin(x_1) = 0 \quad (8.31)$$

This gives $x_1 = n\pi$ for any integer n . The physically meaningful equilibria are $\theta = 0$ (hanging down), which is a stable equilibrium, and $\theta = \pi$ (inverted), which is an unstable equilibrium.

Linearization about $\theta = 0$

At the operating point $\mathbf{x}_0 = [0, 0]^T$:

$$\mathbf{A} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_0 = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}_0 = \begin{bmatrix} 0 & 1 \\ -\frac{g}{L} \cos(x_1) & 0 \end{bmatrix}_{x_1=0} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{L} & 0 \end{bmatrix} \quad (8.32)$$

The linearized system is:

$$\begin{bmatrix} \delta \dot{x}_1 \\ \delta \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{L} & 0 \end{bmatrix} \begin{bmatrix} \delta x_1 \\ \delta x_2 \end{bmatrix} \quad (8.33)$$

Converting back to scalar form:

$$\ddot{\theta} + \frac{g}{L} \theta = 0 \quad (8.34)$$

This is simple harmonic motion with natural frequency $\omega_n = \sqrt{g/L}$ and period:

$$T_{\text{linear}} = 2\pi \sqrt{\frac{L}{g}} \quad (8.35)$$

Validity of the Linear Approximation

The exact period of a nonlinear pendulum depends on the amplitude θ_{max} and is given by an elliptic integral:

$$T_{\text{exact}} = 4\sqrt{\frac{L}{g}} K\left(\sin^2 \frac{\theta_{\text{max}}}{2}\right) \quad (8.36)$$

where K is the complete elliptic integral of the first kind.

For small amplitudes, this can be expanded as:

$$T_{\text{exact}} \approx 2\pi \sqrt{\frac{L}{g}} \left(1 + \frac{1}{16} \theta_{\text{max}}^2 + \frac{11}{3072} \theta_{\text{max}}^4 + \dots \right) \quad (8.37)$$

Table 8.1 compares the linear and exact periods:

The linear approximation is accurate to within 2% for amplitudes below 30, and within 5% for amplitudes below 45. Beyond 60, the error becomes substantial.

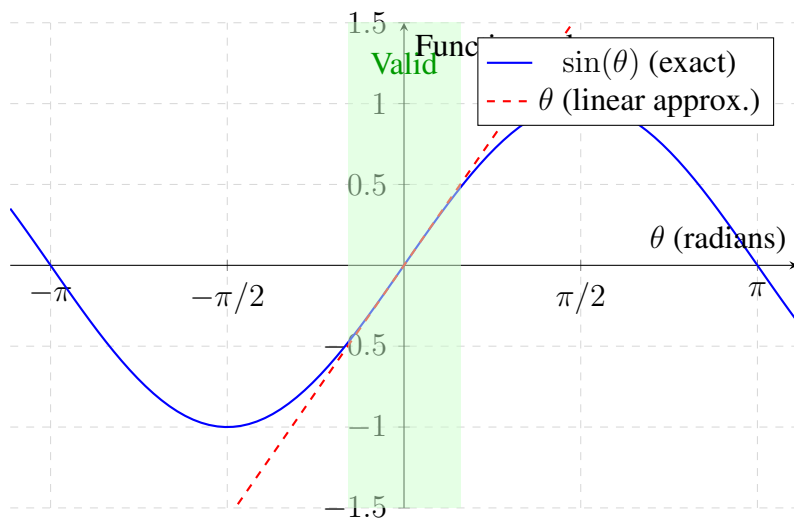


Figure 8.2: Comparison of $\sin(\theta)$ and its linear approximation θ . The shaded region indicates where the approximation error is less than 5%, corresponding to $|\theta| < 0.5$ rad ($\approx 29^\circ$).

Table 8.1: Comparison of Linear and Exact Pendulum Periods

Amplitude $\theta_{\max}$	$T_{\text{exact}}/T_{\text{linear}}$	Error
10 (0.175 rad)	1.0019	0.19%
20 (0.349 rad)	1.0077	0.77%
30 (0.524 rad)	1.0174	1.74%
45 (0.785 rad)	1.0400	4.00%
60 (1.047 rad)	1.0732	7.32%
90 (1.571 rad)	1.1803	18.0%

8.3.4 Multiple Variables: Jacobian Matrix Construction

For systems with multiple state variables and inputs, careful bookkeeping is required when computing Jacobian matrices. We illustrate with a two-state, one-input example.

Example: DC Motor with Nonlinear Friction

Consider a DC motor with Coulomb friction:

$$\dot{\omega} = \frac{K_t}{J}i - \frac{b}{J}\omega - \frac{\mu}{J}\text{sgn}(\omega) \quad (8.38)$$

$$\dot{i} = \frac{V}{L} - \frac{R}{L}i - \frac{K_e}{L}\omega \quad (8.39)$$

where ω is angular velocity, i is armature current, V is applied voltage, and the parameters are motor torque constant K_t , inertia J , viscous friction b , Coulomb friction μ , resistance R , inductance L , and back-EMF constant K_e .

The sign function introduces a discontinuity at $\omega = 0$. For linearization around an operating point with $\omega_0 \neq 0$, we can treat $\text{sgn}(\omega_0)$ as a constant, and the Jacobians are:

$$\mathbf{A} = \begin{bmatrix} -\frac{b}{J} & \frac{K_t}{J} \\ -\frac{K_e}{L} & -\frac{R}{L} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} \quad (8.40)$$

Note that the Coulomb friction term does not appear in the linearized model—it contributes a constant offset that is absorbed into the equilibrium condition.

8.3.5 Validity: When Is Linearization “Good Enough”?

The validity of a linearized model depends on several factors. First, the magnitude of perturbations plays a crucial role: since higher-order terms in the Taylor expansion scale as $(\delta x)^2$, $(\delta x)^3$, etc., a rule of thumb is that linearization is valid when $|\delta x/x_0| < 0.1$ (10% deviation), though this depends on the specific nonlinearity. Second, the smoothness of the nonlinearity matters: functions with high curvature (large second derivatives) have smaller validity regions than functions with low curvature. Third, the time scale of interest affects validity because errors in the linearized model accumulate over time; for short-duration transients, linearization may be adequate even for moderate perturbations, whereas for long-term predictions, small errors can grow substantially. Finally, the purpose of the model determines the acceptable error bounds: if the model is used for stability analysis (determining whether perturbations grow or decay), linearization is generally appropriate, but if quantitative predictions of transient response are needed, the validity region becomes more restrictive.

Quantifying Validity: One approach is to compare the second-order term to the first-order term:

$$\text{Validity criterion: } \frac{|f''(x_0)|(\delta x)^2/2}{|f'(x_0)||\delta x|} = \frac{|f''(x_0)||\delta x|}{2|f'(x_0)|} \ll 1 \quad (8.41)$$

For the pendulum with $f(\theta) = \sin(\theta)$ linearized about $\theta_0 = 0$:

$$f'(0) = \cos(0) = 1, \quad f''(0) = -\sin(0) = 0, \quad f'''(0) = -\cos(0) = -1 \quad (8.42)$$

The first nonzero correction is third-order: $\sin(\theta) \approx \theta - \theta^3/6$. The relative error is approximately $\theta^2/6$, which equals 5% when $|\theta| \approx 0.55$ rad (31).

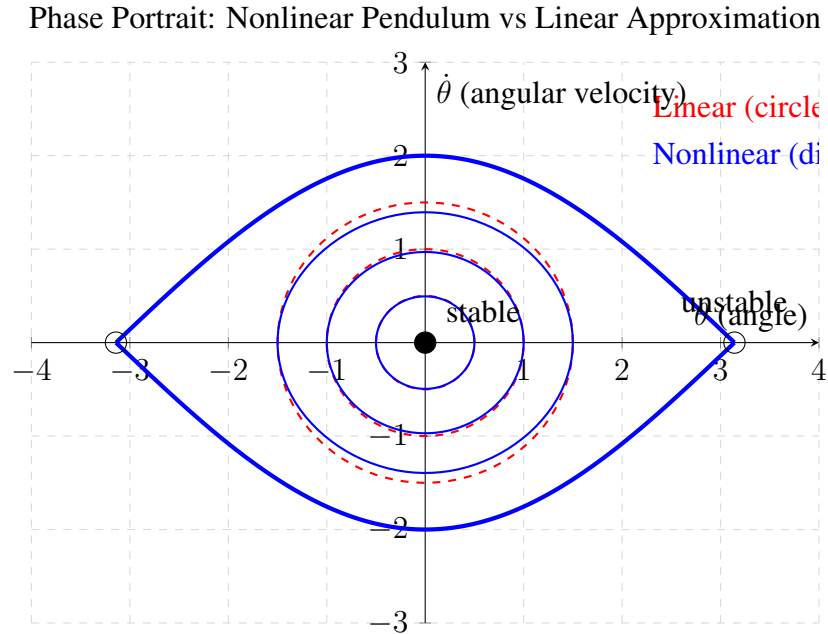


Figure 8.3: Phase portrait comparison. Near the origin, the nonlinear pendulum trajectories (solid blue) closely match the linear approximation (dashed red circles). At larger amplitudes, the nonlinear trajectories distort, and the separatrix (thick blue) marks the boundary between oscillation and rotation.

8.4 Practical Experiment: Pendulum Linearization

This section describes a hands-on experiment that demonstrates the validity and limitations of linearization using a physical pendulum.

8.4.1 Apparatus Design and Construction

Components Required:

The experiment requires a 3D-printed pendulum arm with a suggested length of 20–30 cm, mounted on a low-friction pivot using either a ball bearing or needle bearing. Angle measurement is accomplished using a rotary encoder or potentiometer interfaced with an Arduino or similar microcontroller. A frame or mounting structure provides the necessary support. Optionally, a servo motor can be added for controlled release of the pendulum from precise initial angles.

3D Printing Specifications:

The pendulum arm should be designed with a uniform cross-section for predictable moment of inertia, a hole for the pivot bearing at one end, and a small mass concentration at the opposite end or an attachment point for additional mass. The recommended material is PLA or PETG with 20–30% infill.

A simple rectangular cross-section arm (10 mm × 5 mm × 250 mm) with a 10g mass at the end works well. The effective length L is measured from the pivot to the center of the end mass.

Encoder Interface:

A rotary encoder provides digital pulses as the shaft rotates. For an encoder with N pulses per revolution:

$$\theta = \frac{2\pi \cdot (\text{pulse count})}{N} \quad (8.43)$$

Quadrature encoders (A/B channels) provide $4\times$ the resolution through edge counting. A 400 PPR encoder gives 1600 counts per revolution, or 0.225 per count—sufficient for this experiment.

8.4.2 Experimental Procedure

Part A: Period vs. Amplitude Measurement

Begin by mounting the pendulum and verifying that it swings freely with minimal friction. Release the pendulum from a small angle ($\theta_0 = 10$) and record the time for 10 complete oscillations, then calculate the period T_{10} . Repeat this measurement for initial angles of 20, 30, 45, 60, 75, and 90. For each measurement, take at least 3 trials and compute the average to reduce random errors.

Data Recording:

Use the Arduino to timestamp zero-crossings of the pendulum. The period is twice the time between successive crossings (half-period). Sample code outline:

Algorithm 11 Pendulum Period Measurement

```

1: Initialize:  $t_{\text{last}} \leftarrow 0, T \leftarrow 0$ 
2: Configure interrupt on encoder zero-crossing (rising edge)
3: Initialize serial communication
4:
5: while system is running do
6:   On zero-crossing interrupt:
7:      $t_{\text{now}} \leftarrow$  current time in microseconds
8:      $T \leftarrow 2 \times (t_{\text{now}} - t_{\text{last}})/10^6$  ▷ Period in seconds
9:      $t_{\text{last}} \leftarrow t_{\text{now}}$ 
10:
11:   In main loop:
12:     Output period  $T$  to serial monitor
13:     Wait 500 ms before next output
14: end while
```

Part B: Waveform Comparison

Configure the Arduino to log angle versus time at a high sample rate (e.g., 1 kHz). Release the pendulum from $\theta_0 = 15$ and record several periods of oscillation. Repeat the data collection for $\theta_0 = 45$ and $\theta_0 = 75$. Export all data for subsequent analysis in MATLAB, Python, or spreadsheet software.

Part C: (Optional) Closed-Loop Control

If a motor is available at the pivot, implement a simple proportional controller with torque command $\tau = -K_p\theta$. First test regulation to $\theta = 0$ from small initial angles, where the controller should perform well. Then test from large initial angles, where the system may exhibit oscillations or instability. Observe how controller performance degrades outside the linear region where the linearized model is valid.

8.4.3 Data Analysis and Expected Results

Period Analysis:

Calculate the theoretical linear period:

$$T_{\text{linear}} = 2\pi\sqrt{\frac{L}{g}} \quad (8.44)$$

For $L = 0.25$ m and $g = 9.81$ m/s²: $T_{\text{linear}} = 1.003$ s.

Plot measured period versus amplitude and compare to both the constant linear prediction and the theoretical nonlinear period computed from the elliptic integral or series expansion. The expected observation is that measured periods increase with amplitude, matching nonlinear theory and diverging from the linear prediction.

Waveform Analysis:

The linear model predicts sinusoidal motion:

$$\theta_{\text{linear}}(t) = \theta_0 \cos(\omega_n t) \quad (8.45)$$

For small amplitudes, the measured waveform should be nearly sinusoidal. For large amplitudes, deviations become visible: the waveform “flattens” near the peaks as the pendulum spends more time at large angles, the period becomes longer than predicted by the linear model, and higher harmonics appear in the Fourier spectrum. Plot measured $\theta(t)$ against the linear prediction to visualize the discrepancy.

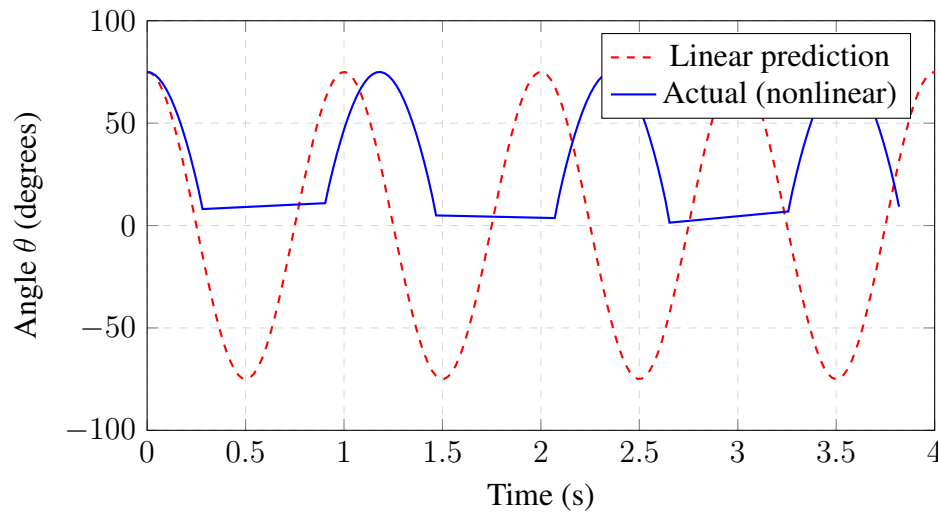


Figure 8.4: Comparison of linear prediction (dashed) and actual nonlinear behavior (solid) for a pendulum released from 75. The nonlinear waveform has a longer period and flattened peaks.

8.4.4 Discussion Questions

Consider the following questions for deeper understanding of the experiment. First, at what amplitude does the measured period deviate from the linear prediction by more than 5%, and how

does this compare to the theoretical threshold? Second, if friction is present, how does it affect the comparison between linear and nonlinear models? Third, regarding the closed-loop control experiment, why does the controller work better for small angles, and what modifications might extend its operating range? Finally, how would you design an experiment to linearize about the inverted equilibrium ($\theta = \pi$)?

8.5 Worked Examples

This section presents detailed solutions to representative linearization problems.

Pendulum with Torque Input

Problem: Linearize the pendulum equation with an applied torque input τ about the downward equilibrium.

Solution:

The equation of motion with torque input is:

$$mL^2\ddot{\theta} + mgL \sin(\theta) = \tau \quad (8.46)$$

Define states $x_1 = \theta$, $x_2 = \dot{\theta}$, and input $u = \tau$:

$$\dot{x}_1 = x_2 \quad (8.47)$$

$$\dot{x}_2 = -\frac{g}{L} \sin(x_1) + \frac{1}{mL^2}u \quad (8.48)$$

Equilibrium: Setting derivatives to zero with $u_0 = 0$:

$$x_2 = 0, \quad \sin(x_1) = 0 \implies x_1 = 0 \text{ (or } \pi) \quad (8.49)$$

Jacobians at $(x_1, x_2, u) = (0, 0, 0)$:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{L} \cos(0) & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{L} & 0 \end{bmatrix} \quad (8.50)$$

$$\mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{mL^2} \end{bmatrix} \quad (8.51)$$

Linearized Model:

$$\begin{bmatrix} \delta\dot{\theta} \\ \delta\ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{L} & 0 \end{bmatrix} \begin{bmatrix} \delta\theta \\ \delta\dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{mL^2} \end{bmatrix} \delta\tau \quad (8.52)$$

Or in scalar form:

$$\delta\ddot{\theta} + \frac{g}{L}\delta\theta = \frac{1}{mL^2}\delta\tau \quad (8.53)$$

Transfer Function:

$$\frac{\delta\Theta(s)}{\delta T(s)} = \frac{1/mL^2}{s^2 + g/L} = \frac{1/mL^2}{s^2 + \omega_n^2} \quad (8.54)$$

This is an undamped second-order system with natural frequency $\omega_n = \sqrt{g/L}$.

Tank Level System

Problem: A tank with cross-sectional area A has liquid flowing in at rate q_{in} and draining through an orifice at the bottom. The outflow rate depends on the square root of the liquid height: $q_{\text{out}} = k\sqrt{h}$. Linearize the system about a steady-state operating point.

Solution:

Nonlinear Model: Mass balance gives:

$$A \frac{dh}{dt} = q_{\text{in}} - k\sqrt{h} \quad (8.55)$$

Let $x = h$ (state) and $u = q_{\text{in}}$ (input):

$$\dot{x} = f(x, u) = \frac{1}{A} (u - k\sqrt{x}) \quad (8.56)$$

Equilibrium: At steady state, $\dot{x} = 0$:

$$u_0 = k\sqrt{x_0} \implies x_0 = \left(\frac{u_0}{k}\right)^2 \quad (8.57)$$

For a nominal inflow $u_0 = 0.01 \text{ m}^3/\text{s}$ and $k = 0.005 \text{ m}^{2.5}/\text{s}$, the equilibrium height is:

$$h_0 = \left(\frac{0.01}{0.005}\right)^2 = 4 \text{ m} \quad (8.58)$$

Jacobians:

$$A = \left. \frac{\partial f}{\partial x} \right|_0 = -\frac{k}{2A\sqrt{x_0}} = -\frac{k}{2A\sqrt{h_0}} \quad (8.59)$$

$$B = \left. \frac{\partial f}{\partial u} \right|_0 = \frac{1}{A} \quad (8.60)$$

With $A = 1 \text{ m}^2$, $k = 0.005$, $h_0 = 4 \text{ m}$:

$$A = -\frac{0.005}{2 \cdot 1 \cdot 2} = -0.00125 \text{ s}^{-1}, \quad B = 1 \text{ m}^{-2} \quad (8.61)$$

Linearized Model:

$$\delta \dot{h} = -\frac{k}{2A\sqrt{h_0}} \delta h + \frac{1}{A} \delta q_{\text{in}} \quad (8.62)$$

Transfer Function:

$$\frac{\delta H(s)}{\delta Q_{\text{in}}(s)} = \frac{1/A}{s + k/(2A\sqrt{h_0})} = \frac{1}{s + 0.00125} \quad (8.63)$$

This is a first-order system with time constant $\tau = 2A\sqrt{h_0}/k = 800 \text{ s}$.

Physical Interpretation: The linearized model shows that the tank acts as a low-pass filter. The time constant depends on the operating point—at higher liquid levels, the drainage rate is faster, so the system responds more quickly to input changes.

8.5.1 Example 3: Finding Equilibrium Points

Problem: Find all equilibrium points of the system:

$$\dot{x}_1 = x_2 \quad (8.64)$$

$$\dot{x}_2 = x_1 - x_1^3 \quad (8.65)$$

and determine their stability using linearization.

Solution:

Equilibrium Conditions:

$$x_2 = 0 \quad (8.66)$$

$$x_1 - x_1^3 = 0 \implies x_1(1 - x_1^2) = 0 \quad (8.67)$$

The equilibrium points are:

$$(x_1, x_2) \in \{(0, 0), (1, 0), (-1, 0)\} \quad (8.68)$$

Jacobian Matrix:

$$\mathbf{A} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 - 3x_1^2 & 0 \end{bmatrix} \quad (8.69)$$

At (0, 0):

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad (8.70)$$

Eigenvalues: $\det(\mathbf{A} - \lambda \mathbf{I}) = \lambda^2 - 1 = 0 \implies \lambda = \pm 1$

One positive eigenvalue $\implies$ **unstable** (saddle point).

At (1, 0) and (-1, 0):

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 - 3 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix} \quad (8.71)$$

Eigenvalues: $\lambda^2 + 2 = 0 \implies \lambda = \pm j\sqrt{2}$

Purely imaginary eigenvalues $\implies$ **center** (marginally stable in linear analysis).

Note: For the center equilibria, linear analysis is inconclusive about asymptotic stability. Non-linear analysis (e.g., using Lyapunov functions) reveals that these are indeed stable centers with closed periodic orbits.

8.5.2 Example 4: Linearization Around Non-Zero Equilibrium

Problem: The Van der Pol oscillator is described by:

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = F \quad (8.72)$$

For $F = 2$ and $\mu = 0.5$, find the equilibrium point and linearize.

Solution:

State-Space Form:

$$\dot{x}_1 = x_2 \quad (8.73)$$

$$\dot{x}_2 = -\mu(x_1^2 - 1)x_2 - x_1 + F \quad (8.74)$$

Equilibrium: Setting $\dot{x}_1 = \dot{x}_2 = 0$:

$$x_2 = 0 \quad (8.75)$$

$$-x_1 + F = 0 \implies x_1 = F = 2 \quad (8.76)$$

Equilibrium point: $(x_1, x_2) = (2, 0)$

Jacobian:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2\mu x_1 x_2 - 1 & -\mu(x_1^2 - 1) \end{bmatrix}_{(2,0)} = \begin{bmatrix} 0 & 1 \\ -1 & -\mu(4 - 1) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -1.5 \end{bmatrix} \quad (8.77)$$

Eigenvalues:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \lambda^2 + 1.5\lambda + 1 = 0 \quad (8.78)$$

$$\lambda = \frac{-1.5 \pm \sqrt{2.25 - 4}}{2} = \frac{-1.5 \pm j1.32}{2} = -0.75 \pm j0.66 \quad (8.79)$$

Both eigenvalues have negative real parts $\implies$ **stable** (damped oscillations).

Linearized Model:

$$\begin{bmatrix} \delta \dot{x}_1 \\ \delta \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -1.5 \end{bmatrix} \begin{bmatrix} \delta x_1 \\ \delta x_2 \end{bmatrix} \quad (8.80)$$

where $\delta x_1 = x_1 - 2$ and $\delta x_2 = x_2 - 0$.

8.5.3 Example 5: Linear vs. Nonlinear Simulation

Problem: Compare linear and nonlinear simulations of a pendulum released from $\theta_0 = 30$.

Solution:

System Parameters: $L = 1$ m, $g = 9.81$ m/s², so $\omega_n = 3.13$ rad/s.

Linear Model:

$$\theta_{\text{lin}}(t) = \theta_0 \cos(\omega_n t) = 0.524 \cos(3.13t) \text{ rad} \quad (8.81)$$

Nonlinear Model: Solve numerically using fourth-order Runge-Kutta:

$$\ddot{\theta} = -\omega_n^2 \sin(\theta) \quad (8.82)$$

The following MATLAB/Python pseudocode implements the comparison:

Results:

The linear period is $T_{\text{lin}} = 2.006$ s, while the nonlinear period is $T_{\text{nonlin}} \approx 2.040$ s, yielding an error of approximately 1.7%.

At 30 initial amplitude, the linear model predicts the period within 2%, confirming the validity of linearization for “small” angles in practical terms.

Visualization: Figure 8.5 shows the comparison.

Algorithm 12 Nonlinear Pendulum Simulation and Comparison

```

1: Input:  $g = 9.81 \text{ m/s}^2$ ,  $L = 1.0 \text{ m}$ ,  $\theta_0 = 30$ 
2: Output: Period comparison between linear and nonlinear models
3:
4: Compute natural frequency:  $\omega_n \leftarrow \sqrt{g/L}$ 
5: Convert initial angle to radians:  $\theta_0 \leftarrow 30 \times \pi/180$ 
6: Create time vector:  $t \leftarrow [0, 0.01, 0.02, \dots, 5.0]$  seconds
7:
8: Linear Solution:
9: for each time  $t_i$  do
10:    $\theta_{\text{lin}}(t_i) \leftarrow \theta_0 \cos(\omega_n t_i)$ 
11: end for
12:
13: Nonlinear Solution (using numerical integration):
14: Define ODE:  $\dot{\theta} = \omega$ ,  $\dot{\omega} = -\omega_n^2 \sin(\theta)$ 
15: Set initial conditions:  $[\theta(0), \omega(0)] \leftarrow [\theta_0, 0]$ 
16: Integrate ODE numerically (e.g., Runge-Kutta) to obtain  $\theta_{\text{nonlin}}(t)$ 
17:
18: Period Calculation:
19:  $T_{\text{lin}} \leftarrow 2\pi/\omega_n$ 
20: Find zero-crossings of  $\theta_{\text{nonlin}}(t)$ 
21:  $T_{\text{nonlin}} \leftarrow 2 \times (\text{second crossing time} - \text{first crossing time})$ 
22:
23: Compute error:  $\epsilon \leftarrow 100 \times (T_{\text{nonlin}} - T_{\text{lin}})/T_{\text{lin}} \%$  return  $T_{\text{lin}}, T_{\text{nonlin}}, \epsilon$ 

```

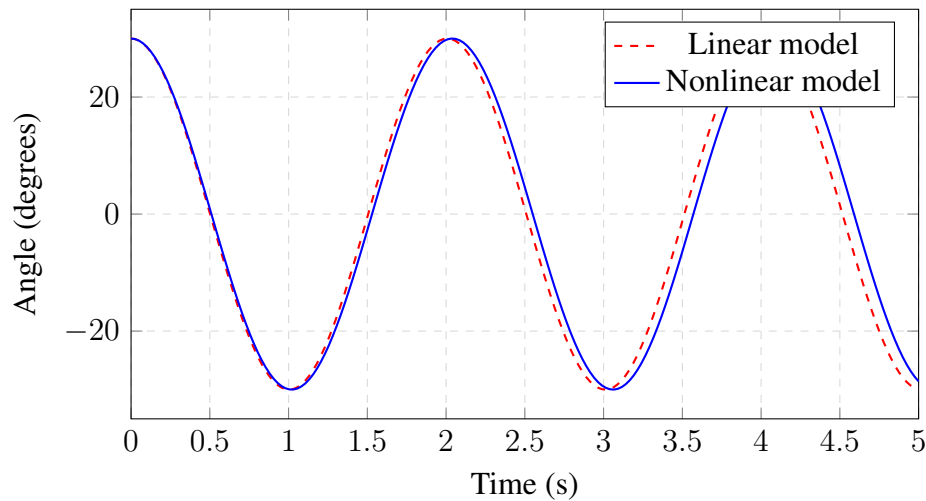


Figure 8.5: Comparison of linear and nonlinear pendulum simulations for $\theta_0 = 30$. The phase difference accumulates over time due to the slightly longer period of the nonlinear system.

8.6 Operating Point Concept

The concept of the operating point is central to practical control system design. This section clarifies its meaning and implications.

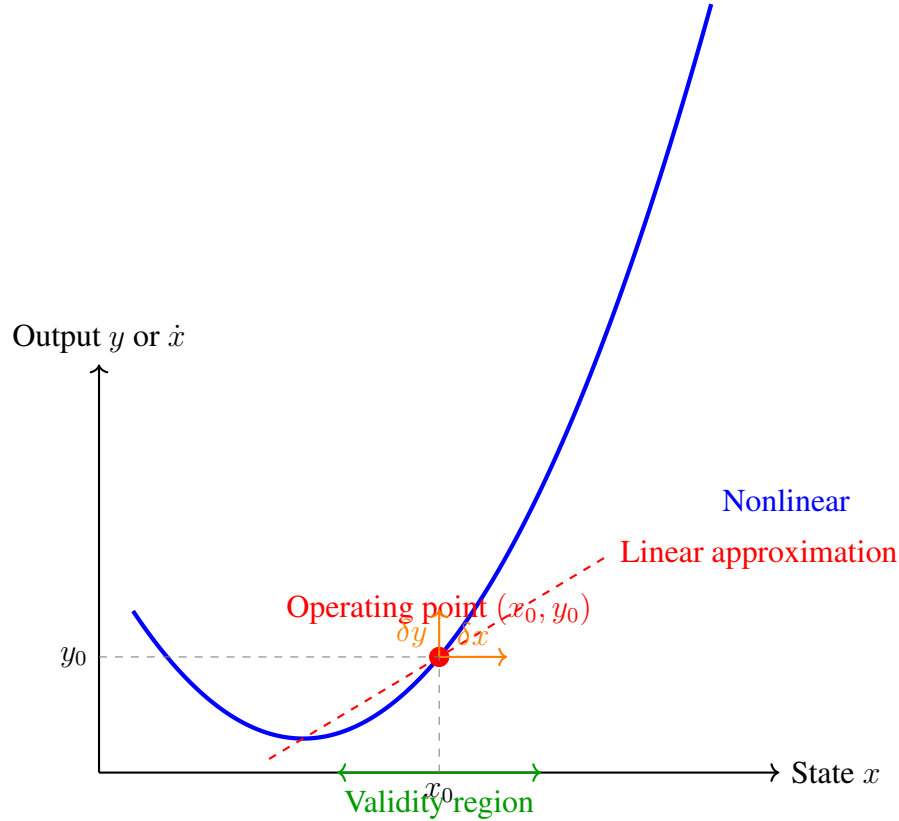


Figure 8.6: The operating point concept. The nonlinear system (solid curve) is approximated by a tangent line (dashed) at the operating point. The linear approximation is valid for small perturbations δx within the validity region.

Definition: The operating point is a specific state-input combination (x_0, u_0) around which the system is linearized. For control design, it typically corresponds to the desired steady-state condition.

Implications:

The operating point concept has several important implications. First, regarding perturbation variables, the linearized model describes deviations from the operating point rather than absolute values. When implementing a controller, remember that:

$$\delta x = x - x_0 \quad (\text{measured state minus setpoint}) \quad (8.83)$$

$$\delta u = u - u_0 \quad (\text{control input minus nominal}) \quad (8.84)$$

Second, bias terms must be properly handled in implementation. The actual control signal is:

$$u = u_0 + \delta u = u_0 + K(\delta x) \quad (8.85)$$

The term u_0 provides the steady-state input needed to maintain the operating point, while δu is the corrective action from the controller.

Third, gain scheduling becomes necessary when the system operates over a range of conditions, such as aircraft at different altitudes and speeds. In such cases, multiple linearizations are performed and controller gains are “scheduled” based on the current operating condition.

Finally, the operating envelope is defined as the set of operating points for which a single linearized controller provides acceptable performance. Beyond this envelope, the controller may exhibit degraded performance or instability.

8.7 Summary

Linearization transforms nonlinear systems into linear approximations that are amenable to analysis and control design. The Taylor series provides the mathematical foundation, demonstrating that any smooth function can be approximated by a polynomial, and locally, by a linear function. Equilibrium points serve as the natural operating points for linearization, characterized by the condition $f(x_0, u_0) = 0$. The Jacobian matrices A and B capture the local sensitivity of the dynamics to state and input perturbations, respectively. The validity of the linear approximation depends on the magnitude of perturbations and the curvature of the nonlinearity; for most control applications, perturbations within 10–20% of the operating point yield acceptable accuracy. Finally, eigenvalue analysis of the linearized system determines local stability properties.

The linearization approach has proven remarkably successful across engineering disciplines for three key reasons: well-designed control systems keep perturbations small, linear tools such as transfer functions, Bode plots, and state feedback are powerful and well-understood, and gain scheduling extends linear methods to cover wide operating ranges. Linearization is not a limitation but an enabler—it allows us to apply elegant, systematic design methods to the complex nonlinear systems that populate the real world.

8.8 Exercises

Exercise 8.1. Linearize the system $\dot{x} = -x^3 + u$ about the equilibrium with $u_0 = 0$. Find the transfer function and determine stability.

Exercise 8.2. A satellite attitude control system has dynamics:

$$I\ddot{\theta} = \tau - k_d\dot{\theta} \quad (8.86)$$

where θ is angle, I is moment of inertia, τ is control torque, and k_d is damping. This is already linear. Explain why linearization is not needed and write the state-space model.

Exercise 8.3. For the tank system of Example 2, how does the time constant change if the operating level is doubled from 4 m to 8 m?

Exercise 8.4. Linearize the Lotka-Volterra predator-prey equations:

$$\dot{x} = \alpha x - \beta xy \quad (8.87)$$

$$\dot{y} = \delta xy - \gamma y \quad (8.88)$$

about the non-trivial equilibrium $(x_0, y_0) = (\gamma/\delta, \alpha/\beta)$.

Exercise 8.5. An inverted pendulum on a cart has the approximate linearized dynamics (for small angles):

$$(M + m)\ddot{p} + mL\ddot{\theta} = F, \quad mL\ddot{p} + mL^2\ddot{\theta} = mgL\theta \quad (8.89)$$

where p is cart position, θ is pendulum angle from vertical, and F is applied force. Derive the transfer function $\Theta(s)/F(s)$ and verify that it is unstable.

Exercise 8.6. For the pendulum experiment: if the bearing friction contributes a damping torque $-b\dot{\theta}$, how does this affect (a) the equilibrium analysis, (b) the linearized model, and (c) the validity of the linear approximation?

Exercise 8.7. (Computer Exercise) Implement the pendulum simulation of Example 5. Extend it to initial angles of 15, 45, 60, 75, 90. Create a plot showing the period error (compared to linear prediction) as a function of initial amplitude. At what amplitude does the error exceed 5%?

Exercise 8.8. A nonlinear spring has restoring force $F = -kx - \alpha x^3$ where $\alpha > 0$ (hardening spring). For a mass m attached to this spring, linearize the equation of motion about $x_0 = 0$ and find the natural frequency of small oscillations.

Exercise 8.9. The van der Pol oscillator $\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$ has an equilibrium at the origin. Linearize about this equilibrium and determine whether it is stable for $\mu > 0$. What type of behavior do you expect from the nonlinear system?

Exercise 8.10. A chemical reactor has temperature dynamics governed by $\dot{T} = -\frac{UA}{mC_p}(T - T_c) + \frac{(-\Delta H)k_0 e^{-E_a/RT} C_A}{mC_p}$ where the exponential Arrhenius term is highly nonlinear. Linearize this equation about an operating temperature T_0 , defining the linearized gain for small temperature perturbations.

8.9 Further Reading

For rigorous treatment of linearization and stability, consult Khalil's *Nonlinear Systems* (3rd ed., Prentice Hall, 2002), particularly Chapter 4. Slotine and Li's *Applied Nonlinear Control* (Prentice Hall, 1991) provides excellent coverage of when linearization fails and what to do about it. Strogatz's *Nonlinear Dynamics and Chaos* (2nd ed., Westview Press, 2015) offers an accessible introduction to nonlinear systems with beautiful examples. For historical context on the development of these ideas, Diacu and Holmes's *Celestial Encounters: The Origins of Chaos and Stability* (Princeton University Press, 1996) is highly recommended.

Part III

Classical Control Design

Chapter 9

Feedback and Sensitivity

“I suddenly realized that if I fed the amplifier output back to the input, in reverse phase, and kept the device from oscillating, I would have exactly what I wanted: a means of canceling out the distortion in the output.”

— Harold S. Black, 1927

The principle of negative feedback stands as one of the most profound engineering insights of the twentieth century. In this chapter, we explore how feeding back a portion of a system’s output to its input—seemingly a wasteful process that “throws away” gain—actually provides remarkable benefits: rejection of disturbances, reduction of sensitivity to component variations, and predictable system behavior despite uncertain plant dynamics. These properties make feedback the cornerstone of modern control system design.

9.1 Historical Motivation: The Birth of Negative Feedback

9.1.1 The Telephone Amplifier Problem

In the early decades of the twentieth century, the American Telephone & Telegraph Company (AT&T) faced a formidable engineering challenge. As telephone networks expanded across the continental United States, voice signals had to traverse thousands of miles of copper wire. The inherent resistance of these transmission lines caused severe signal attenuation—a coast-to-coast call required amplification by a factor of approximately 10^{30} to maintain intelligible speech [1].

The solution appeared straightforward: cascade multiple vacuum tube amplifiers along the transmission path. A transcontinental line might employ 50 to 100 amplifier stations, each providing gains of 20 to 40 decibels. However, this approach encountered two devastating problems that threatened the viability of long-distance telephony.

Example 12.3. Asymptote Calculation: For the system $G(s) = \frac{K}{(s+1)(s+2)(s+3)(s+4)}$, calculate the asymptotes.

Solution:

Poles: $-1, -2, -3, -4$ ($n = 4$) Zeros: none ($m = 0$) Branches to infinity: $n - m = 4$

Asymptote angles:

$$\theta_a = \frac{(2k+1) \cdot 180}{4} \text{ for } k = 0, 1, 2, 3$$

$$k = 0 : \quad \theta_a = \frac{180}{4} = 45$$

$$k = 1 : \quad \theta_a = \frac{3 \cdot 180}{4} = 135$$

$$k = 2 : \quad \theta_a = \frac{5 \cdot 180}{4} = 225 = -135$$

$$k = 3 : \quad \theta_a = \frac{7 \cdot 180}{4} = 315 = -45$$

Centroid:

$$\sigma_a = \frac{(-1) + (-2) + (-3) + (-4)}{4} = \frac{-10}{4} = -2.5$$

The four asymptotes emanate from $\sigma = -2.5$ at angles ± 45 and ± 135 .

Example 12.4. Finding Gain for Specified Damping: $\text{gain}_{\text{damping}}$ For the system $G(s) = \frac{K}{s(s+4)}$, find the gain K that produces closed-loop poles with damping ratio $\zeta = 0.5$.

Solution:

A damping ratio $\zeta = 0.5$ corresponds to poles at angle $\theta = \cos^{-1}(0.5) = 60$ from the negative real axis.

The root locus for this second-order system has poles at $s = 0$ and $s = -4$, with a real axis segment from 0 to -4 . The breakaway occurs at $s = -2$ (by symmetry and $dK/ds = 0$), after which the branches proceed vertically from the breakaway point.

For $\zeta = 0.5$, the closed-loop poles lie on the line from the origin at ± 120 .

The intersection of this line with the root locus occurs where the vertical asymptotes (from breakaway at -2) cross the $\zeta = 0.5$ lines.

For a pole at $s = -2 + j\omega$, the angle from the origin is:

$$\tan(60) = \frac{\omega}{2} \implies \omega = 2\sqrt{3}$$

So the desired pole location is $s = -2 + j2\sqrt{3}$.

Calculating K using magnitude criterion:

$$K = |s| \cdot |s+4| = |-2 + j2\sqrt{3}| \cdot |2 + j2\sqrt{3}|$$

$$|-2 + j2\sqrt{3}| = \sqrt{4 + 12} = 4$$

$$|2 + j2\sqrt{3}| = \sqrt{4 + 12} = 4$$

Therefore: $K = 4 \times 4 = 16$

Verification: The characteristic equation $s^2 + 4s + 16 = 0$ has roots:

$$s = \frac{-4 \pm \sqrt{16 - 64}}{2} = \frac{-4 \pm j\sqrt{48}}{2} = -2 \pm j2\sqrt{3}$$

This confirms $\omega_n = 4$ and $\zeta = 2/4 = 0.5$. ✓

Example 12.5. Stability Range Determination: For the system with characteristic equation $s^3 + 6s^2 + 11s + 6 + K = 0$, determine the range of K for stability.

Solution:

The characteristic equation can be written as:

$$1 + \frac{K}{s^3 + 6s^2 + 11s + 6} = 0$$

where the denominator factors as $(s + 1)(s + 2)(s + 3)$.

Routh array:

$$\begin{array}{c|cc} s^3 & 1 & 11 \\ s^2 & 6 & 6 + K \\ s^1 & \frac{66 - (6 + K)}{6} = \frac{60 - K}{6} & 0 \\ s^0 & 6 + K & \end{array}$$

Stability conditions: From the s^1 row, we require $\frac{60 - K}{6} > 0$, which implies $K < 60$. From the s^0 row, we require $6 + K > 0$, which implies $K > -6$.

Stability range: $-6 < K < 60$

For positive gain K : the system is stable for $0 < K < 60$.

At $K = 60$, the root locus crosses the imaginary axis. The crossing frequency can be found from the auxiliary equation when the s^1 row becomes zero:

$$6s^2 + (6 + 60) = 0 \implies s^2 = -11 \implies s = \pm j\sqrt{11}$$

Example 12.6. Complex System Analysis: Sketch the root locus and find key features for:

$$G(s)H(s) = \frac{K(s + 2)}{s(s + 1)(s^2 + 4s + 8)}$$

Solution:

First, factor the quadratic: $s^2 + 4s + 8 = 0$ gives $s = -2 \pm j2$ (complex poles).

Poles and zeros. The system has poles at 0, -1 , $-2 + j2$, and $-2 - j2$ ($n = 4$), with one zero at -2 ($m = 1$). Therefore, $n - m = 3$ branches go to infinity.

Real axis segments. Between 0 and -1 there is 1 singularity to the right (on locus). Between -1 and -2 there are 2 singularities (not on locus). Left of -2 there are 3 singularities (on locus).

(Complex poles don't affect real axis determination.)

Asymptotes:

$$\theta_a = \frac{(2k + 1) \cdot 180}{3} = 60, 180, -60$$

$$\sigma_a = \frac{[0 + (-1) + (-2 + j2) + (-2 - j2)] - (-2)}{3} = \frac{-5 + 2}{3} = -1$$

Starting configuration:

At $K = 0$, poles are at 0, -1 , $-2 \pm j2$. As K increases, the complex poles at $-2 \pm j2$ initially move along paths determined by the angle criterion. The real poles at 0 and -1 move toward each other, break away from the real axis, and eventually follow the asymptotes. One branch terminates at the zero at -2 .

This system is more complex and typically requires computational tools for precise plotting, but the rules allow us to understand the qualitative behavior.

12.6 Root Locus Diagrams: A Visual Reference

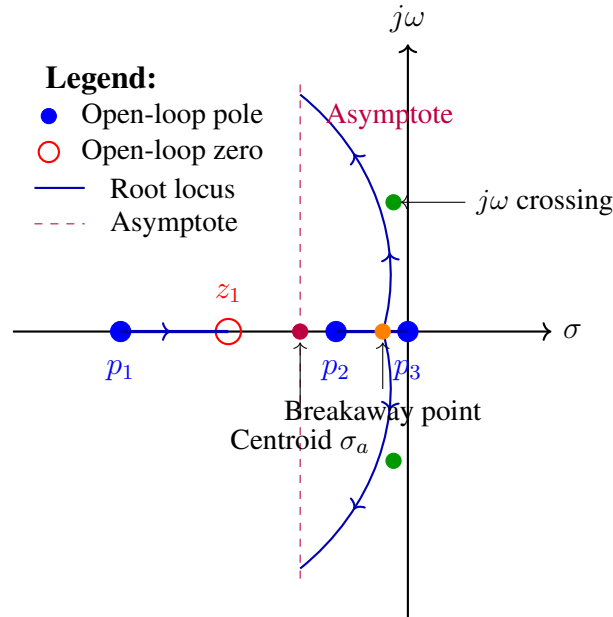
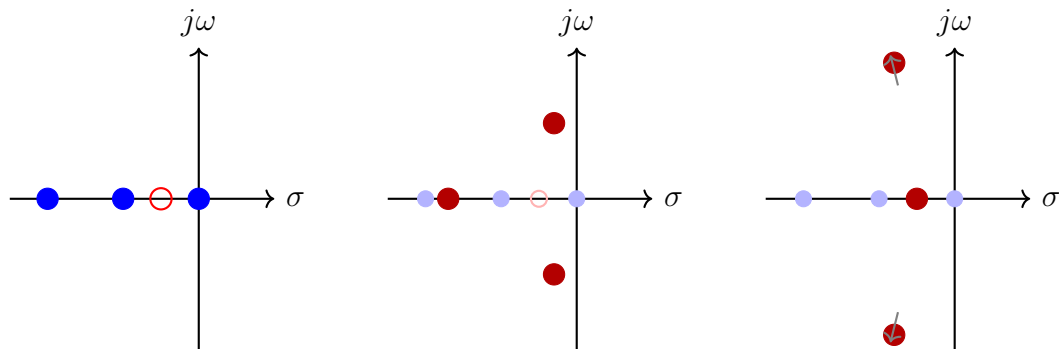


Figure 12.12: Complete root locus example with all major features labeled: open-loop poles and zeros, real axis segments, asymptotes with centroid, breakaway point, and imaginary axis crossings.



$K = 0$: Poles at open-loop locations $K \rightarrow \infty$: Poles migrating toward zeros or infinity

Figure 12.13: Pole migration as gain K increases from zero to infinity. At $K = 0$, closed-loop poles equal open-loop poles. As K increases, poles migrate along the root locus branches toward zeros or infinity.

Real Axis Rule: Count poles + zeros to the RIGHT

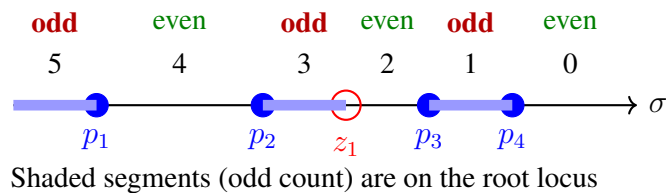


Figure 12.14: Illustration of the real axis segment rule. For each point on the real axis, count the total number of poles and zeros to its right. Segments where this count is odd belong to the root locus.

12.7 Summary

The root locus method, developed by Walter Evans in 1948, provides a powerful graphical technique for analyzing how closed-loop poles move as gain varies. The key concepts encompass five main areas. The **fundamental equation** states that the root locus consists of all points satisfying $1 + KG(s)H(s) = 0$ for $K \geq 0$. The **angle criterion** establishes that a point lies on the root locus if $\angle G(s)H(s) = \pm 180(2k + 1)$. The **magnitude criterion** determines that once a point is confirmed on the locus, $K = 1/|G(s)H(s)|$. The **construction rules** provide eight guidelines for rapid sketching: branches start at poles and end at zeros or infinity; the locus is symmetric about the real axis; real axis segments occur to the left of an odd pole/zero count; asymptotes emanate at specified angles from the centroid; breakaway and break-in points occur where $dK/ds = 0$; and $j\omega$ crossings can be found from the Routh array. Finally, the **design application** involves selecting gain K to place dominant poles in desired s -plane regions for required transient response characteristics.

While modern computational tools can generate root locus plots instantly, understanding the underlying principles enables engineers to predict how system modifications will affect closed-loop behavior, design compensators to reshape the locus favorably, develop intuition about dynamic system behavior, and verify computational results for correctness.

The root locus remains an indispensable tool in the control engineer's toolkit, bridging classical graphical methods with modern computational practice.

12.8 Exercises

Exercise 12.1. Sketch the root locus for $G(s) = \frac{K}{(s+1)(s+3)}$ and find the breakaway point.

Exercise 12.2. For $G(s) = \frac{K(s+4)}{s(s+1)(s+2)}$, determine the asymptotes and centroid.

Exercise 12.3. A unity feedback system has $G(s) = \frac{K}{s(s^2+6s+10)}$. Find the range of K for stability.

Exercise 12.4. For $G(s) = \frac{K}{s(s+2)(s+4)}$, find the gain K that places the dominant poles at $\zeta = 0.707$.

Exercise 12.5. Sketch the root locus for $G(s) = \frac{K(s+1)}{s^2(s+4)}$ and identify all key features.

Exercise 12.6. A system has the characteristic equation $s^3 + 3s^2 + (2 + K)s + 4K = 0$. Sketch the root locus and find the imaginary axis crossing.

Exercise 12.7. Design a lead compensator to reshape the root locus of $G(s) = \frac{K}{s(s+2)}$ so that closed-loop poles with $\zeta = 0.5$ and $\omega_n = 4$ rad/s are achievable.

Exercise 12.8. Using MATLAB or Python, verify your hand-sketched root locus for Exercise 5 and determine the exact breakaway points numerically.

Exercise 12.9. For the transfer function $G(s) = \frac{K}{s(s+1)(s+5)}$, determine: (a) the number of asymptotes and their angles, (b) the centroid of the asymptotes, (c) the segments of the real axis that belong to the root locus, and (d) the range of K for which the system is stable.

Exercise 12.10. A position control system has open-loop transfer function $G(s) = \frac{K(s+2)}{s^2(s+10)}$. Sketch the root locus and explain why this system has a break-in point. Find the break-in point location.

Exercise 12.11. Consider a system with $G(s)H(s) = \frac{K}{(s+1)^2(s+4)}$. Apply all eight root locus construction rules to sketch the complete locus. Mark the breakaway point and calculate the gain at which breakaway occurs.

Exercise 12.12. For the system $G(s) = \frac{K}{s(s^2+4s+8)}$, the complex poles are at $s = -2 \pm j2$. Using the angle criterion, determine whether the point $s = -3$ lies on the root locus.

Chapter 13

Frequency Response and Bode Plots

“The frequency response function is, in my opinion, the most important function in the analysis and synthesis of linear systems.”

— Hendrik W. Bode, *Network Analysis and Feedback Amplifier Design*, 1945

The frequency response method represents one of the most powerful and practical tools in the control engineer’s arsenal. Rather than solving differential equations directly, we analyze how systems respond to sinusoidal inputs across a range of frequencies. This approach, developed primarily at Bell Telephone Laboratories during the 1930s and 1940s, transformed control system design from an art into a systematic engineering discipline.

13.1 Historical Development and Motivation

13.1.1 The Bell Labs Challenge

The development of frequency response methods arose from one of the great engineering challenges of the early twentieth century: building a telephone network that could span the American continent.

In the 1920s and 1930s, the American Telephone and Telegraph Company (AT&T) faced an unprecedented engineering problem. Voice signals traveling through copper telephone lines attenuate rapidly—a signal might lose half its power every few miles. To transmit a telephone call from New York to San Francisco, a distance of nearly 3,000 miles, required *hundreds* of amplifier stages connected in cascade.

Each amplifier introduced its own characteristics: gain that varied with frequency, phase shifts that distorted the signal, and the ever-present threat of instability. With dozens of amplifiers in series, even small imperfections accumulated disastrously. A slight phase error in each stage might sum to produce oscillation; frequency-dependent gain would render voices unrecognizable.

The mathematical challenge was formidable. Analyzing the stability of even a single feedback amplifier required solving high-order differential equations. With multiple amplifiers, the mathematics became intractable using classical methods.

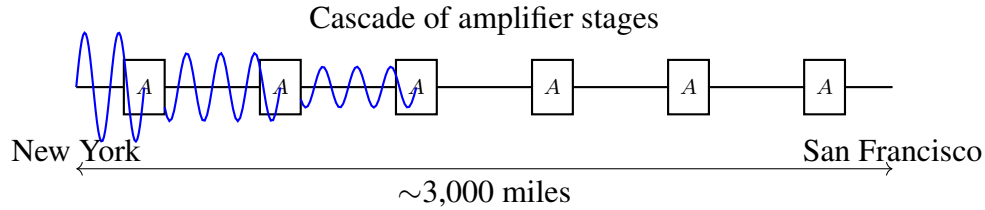


Figure 13.1: Transcontinental telephone transmission required cascaded amplifier stages to compensate for line attenuation. Each stage contributed frequency-dependent gain and phase characteristics.

13.1.2 Hendrik Bode and the Logarithmic Revolution

Hendrik Wade Bode (1905–1982), a Dutch-American engineer at Bell Labs, developed an elegant solution to this complexity. His key insight was deceptively simple: *plot system characteristics on logarithmic scales*.

Bode joined Bell Labs in 1926 after earning his Ph.D. in physics from Columbia University at age 21. He spent his entire career at Bell Labs, eventually becoming Director of Mathematical Research. His 1945 book *Network Analysis and Feedback Amplifier Design* remains a classic reference.

Consider N amplifiers connected in cascade, each with transfer function $G_i(s)$. The overall transfer function is the product:

$$G_{\text{total}}(s) = G_1(s) \cdot G_2(s) \cdot \dots \cdot G_N(s) = \prod_{i=1}^N G_i(s) \quad (13.1)$$

At sinusoidal steady state with $s = j\omega$, the magnitude and phase are:

$$|G_{\text{total}}(j\omega)| = \prod_{i=1}^N |G_i(j\omega)| \quad (13.2)$$

$$\angle G_{\text{total}}(j\omega) = \sum_{i=1}^N \angle G_i(j\omega) \quad (13.3)$$

Bode's stroke of genius was to express magnitude in *decibels*:

$$|G|_{\text{dB}} = 20 \log_{10} |G(j\omega)| \quad (13.4)$$

Taking the logarithm of ??:

$$|G_{\text{total}}|_{\text{dB}} = \sum_{i=1}^N |G_i|_{\text{dB}} \quad (13.5)$$

In decibels, cascaded system gains **add** rather than multiply. This transforms the analysis of complex systems into simple graphical addition—a task easily performed by hand or, in Bode’s era, with a slide rule.

The decibel, originally defined for power ratios in telephony (one-tenth of a *bel*, named after Alexander Graham Bell), proved ideal for this application. When using decibels, multiplication becomes addition, making cascaded system analysis a matter of simple superposition. Furthermore, the logarithmic scale compresses very large and very small numbers into manageable ranges, while asymptotic behavior conveniently appears as straight lines on log-log plots.

13.1.3 Harry Nyquist and the Stability Criterion

While Bode developed graphical methods for analyzing system characteristics, his colleague **Harry Nyquist** (1889–1976) addressed the fundamental question of stability.

Harry Nyquist, a Swedish-American engineer, made foundational contributions to both control theory and information theory. His 1932 paper “Regeneration Theory” established the theoretical basis for analyzing feedback amplifier stability using frequency response data.

Before Nyquist, determining stability required finding the roots of the characteristic polynomial—a task that becomes extremely difficult for systems of order higher than four. Nyquist showed that stability could be determined entirely from the frequency response, without solving any polynomial equations.

The **Nyquist stability criterion** relates the number of encirclements of the critical point $(-1, 0)$ in the complex plane to the number of closed-loop poles in the right half-plane. This graphical criterion, when combined with Bode’s logarithmic plots, provided engineers with practical tools for designing stable high-gain amplifiers.

13.1.4 The Fundamental Insight: Sinusoids as Eigenfunctions

The theoretical foundation for frequency response methods rests on a remarkable mathematical property: *sinusoidal signals are eigenfunctions of linear time-invariant (LTI) systems.*

Theorem 13.1 (Eigenfunction Property of LTI Systems). *If a sinusoidal input $x(t) = A \sin(\omega t + \phi)$ is applied to a stable LTI system with transfer function $G(s)$, then after transients decay, the steady-state output is also sinusoidal at the same frequency:*

$$y_{ss}(t) = A |G(j\omega)| \sin(\omega t + \phi + \angle G(j\omega)) \quad (13.6)$$

Proof. Consider the input $x(t) = Ae^{j\omega t}$ (using complex exponential notation). For an LTI system with impulse response $h(t)$, the output is the convolution:

$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau = \int_{-\infty}^{\infty} h(\tau) Ae^{j\omega(t-\tau)} d\tau \quad (13.7)$$

$$= Ae^{j\omega t} \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau = Ae^{j\omega t} \cdot G(j\omega) \quad (13.8)$$

where $G(j\omega) = \mathcal{L}\{h(t)\}|_{s=j\omega}$ is the frequency response. Since $G(j\omega) = |G(j\omega)| e^{j\angle G(j\omega)}$:

$$y(t) = A |G(j\omega)| e^{j(\omega t + \angle G(j\omega))} \quad (13.9)$$

Taking the imaginary part recovers the result for $\sin(\omega t)$ input. □

This property is profound: to characterize how a system responds to *any* input, we need only measure its response to sinusoids across the frequency spectrum. This is because any reasonable input signal can be decomposed into sinusoidal components via Fourier analysis.

13.2 Modern Applications

The frequency response methods pioneered for telephone amplifiers have found application across virtually every domain of engineering. The underlying principle—that system behavior can be characterized by its response to sinusoidal inputs—remains as powerful today as it was in Bode’s time.

13.2.1 Audio System Design

In audio engineering, frequency response is the fundamental specification. Equalizers, whether hardware or software, operate directly in the frequency domain. Graphic equalizers adjust gain in discrete frequency bands, while parametric equalizers provide more flexible control by allowing adjustment of center frequency, bandwidth, and gain simultaneously. Crossover networks serve a different purpose entirely, dividing audio signals among different drivers such as woofers and tweeters to ensure each component reproduces only the frequencies it handles best.

The human ear’s frequency response (approximately 20 Hz to 20 kHz) defines the relevant analysis range. Audio specifications invariably include frequency response plots, typically requiring ± 3 dB flatness over the audible range.

13.2.2 Filter Design

Frequency response analysis is essential for designing analog and digital filters. Anti-aliasing filters, required before analog-to-digital conversion to satisfy the Nyquist criterion, must provide sufficient attenuation above the Nyquist frequency—typically 60–80 dB, a specification read directly from the Bode magnitude plot. In industrial environments containing electromagnetic interference at specific frequencies such as 60 Hz power line noise or switching power supply harmonics, notch filters designed using Bode techniques effectively remove these disturbances. Communication systems similarly rely on precise frequency-domain specifications for channel selection, pulse shaping, and equalization.

13.2.3 Control Loop Bandwidth Specification

In modern control system design, performance is often specified in terms of frequency-domain metrics. The bandwidth, defined as the frequency at which the closed-loop magnitude drops to

−3 dB of its low-frequency value, directly correlates with response speed—higher bandwidth implies faster response. Stability margins, including gain margin and phase margin (detailed in section 13.3.6), quantify robustness to modeling errors and component variations. The sensitivity function $S(j\omega) = 1/(1 + G(j\omega)H(j\omega))$ characterizes disturbance rejection capability across the frequency spectrum.

13.2.4 Electromagnetic Compatibility (EMC)

Regulatory compliance for electronic products requires frequency-domain analysis. Emissions testing verifies that products do not radiate electromagnetic energy above specified limits across frequency bands, as mandated by regulations such as FCC Part 15 and CISPR 22. Susceptibility testing ensures products operate correctly when exposed to electromagnetic fields at various frequencies. Power supply design presents particular challenges, as switching converters must attenuate high-frequency noise sufficiently to meet conducted emissions limits.

The Bode plot provides a natural framework for specifying and verifying filter performance in EMC applications.

13.3 Mathematical Foundation

13.3.1 Frequency Response Function

For a system with transfer function $G(s)$, the **frequency response** is obtained by evaluating along the imaginary axis:

$$G(j\omega) = G(s)|_{s=j\omega} \quad (13.10)$$

This complex-valued function can be expressed in polar form:

$$G(j\omega) = |G(j\omega)| e^{j\angle G(j\omega)} \quad (13.11)$$

where $|G(j\omega)|$ denotes the **magnitude** (gain) at frequency ω , and $\angle G(j\omega)$ denotes the **phase** (angle) at frequency ω .

Equivalently, in rectangular form:

$$G(j\omega) = \text{Re}\{G(j\omega)\} + j \text{Im}\{G(j\omega)\} \quad (13.12)$$

with:

$$|G(j\omega)| = \sqrt{\text{Re}^2\{G(j\omega)\} + \text{Im}^2\{G(j\omega)\}} \quad (13.13)$$

$$\angle G(j\omega) = \arctan\left(\frac{\text{Im}\{G(j\omega)\}}{\text{Re}\{G(j\omega)\}}\right) \quad (13.14)$$

13.3.2 Bode Plot Construction

A **Bode plot** consists of two graphs plotted against frequency on a logarithmic scale. The magnitude plot displays $20 \log_{10} |G(j\omega)|$ in decibels (dB) versus $\log_{10} \omega$, while the phase plot displays $\angle G(j\omega)$ in degrees versus $\log_{10} \omega$.

The logarithmic frequency scale allows visualization of many decades of frequency on a single plot. The decibel scale for magnitude enables addition of cascaded elements. Together, these choices transform multiplicative transfer function factors into additive graphical contributions.

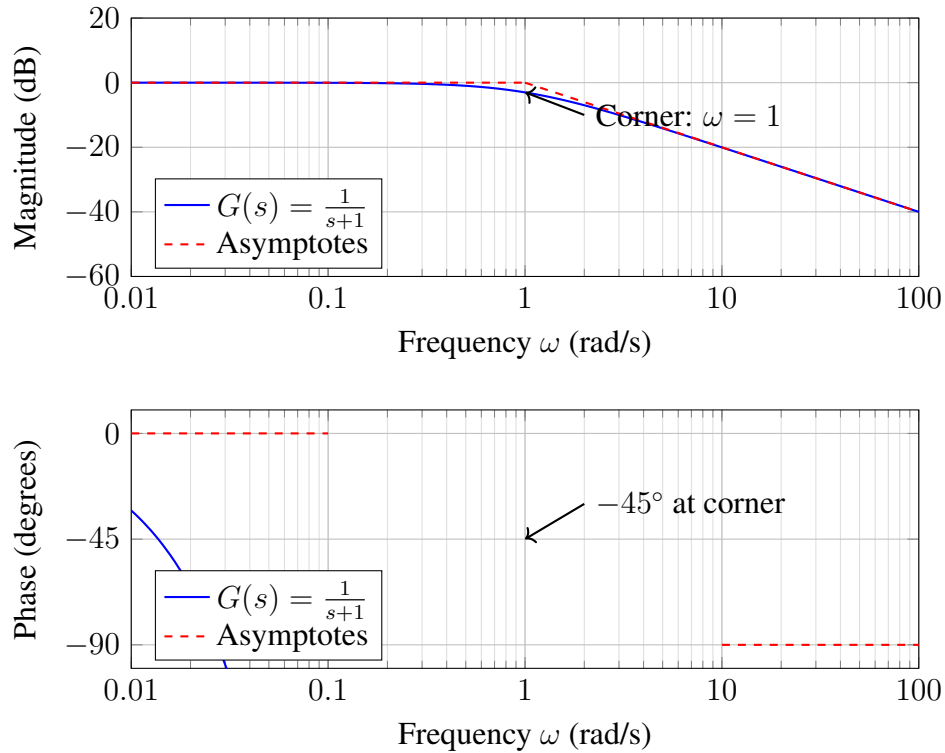


Figure 13.2: Bode plot template showing magnitude (top) and phase (bottom) for a first-order system $G(s) = 1/(s + 1)$. Asymptotes are shown as dashed red lines.

13.3.3 First-Order System: $G(s) = \frac{1}{\tau s + 1}$

Consider the first-order low-pass system with time constant τ :

$$G(s) = \frac{1}{\tau s + 1} \quad (13.15)$$

Substituting $s = j\omega$:

$$G(j\omega) = \frac{1}{j\omega\tau + 1} = \frac{1}{1 + j\omega\tau} \quad (13.16)$$

Magnitude:

$$|G(j\omega)| = \frac{1}{\sqrt{1 + (\omega\tau)^2}} \quad (13.17)$$

In decibels:

$$|G(j\omega)|_{\text{dB}} = -20 \log_{10} \sqrt{1 + (\omega\tau)^2} = -10 \log_{10} (1 + \omega^2\tau^2) \quad (13.18)$$

Phase:

$$\angle G(j\omega) = -\arctan(\omega\tau) \quad (13.19)$$

Asymptotic Analysis

The power of Bode plots lies in their asymptotic approximations:

Low frequency ($\omega \ll 1/\tau$):

$$|G(j\omega)|_{\text{dB}} \approx -10 \log_{10}(1) = 0 \text{ dB} \quad (13.20)$$

$$\angle G(j\omega) \approx -\arctan(0) = 0^\circ \quad (13.21)$$

High frequency ($\omega \gg 1/\tau$):

$$|G(j\omega)|_{\text{dB}} \approx -10 \log_{10}(\omega^2 \tau^2) = -20 \log_{10}(\omega\tau) \quad (13.22)$$

$$\angle G(j\omega) \approx -\arctan(\infty) = -90^\circ \quad (13.23)$$

The high-frequency asymptote has a slope of -20 dB per decade (i.e., for every factor of 10 increase in frequency, the magnitude decreases by 20 dB).

Corner frequency ($\omega = 1/\tau$):

$$|G(j\omega)|_{\text{dB}} = -10 \log_{10}(2) \approx -3 \text{ dB} \quad (13.24)$$

$$\angle G(j\omega) = -\arctan(1) = -45^\circ \quad (13.25)$$

Definition 13.1 (Corner Frequency). The **corner frequency** (or break frequency) $\omega_c = 1/\tau$ is the frequency at which the low-frequency and high-frequency asymptotes intersect. At the corner frequency, the actual magnitude is 3 dB below the asymptotic value, and the phase is -45° for a single real pole.

13.3.4 Second-Order System

Consider the standard second-order system:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (13.26)$$

where ω_n is the natural frequency and ζ is the damping ratio.

Substituting $s = j\omega$:

$$G(j\omega) = \frac{\omega_n^2}{\omega_n^2 - \omega^2 + 2j\zeta\omega_n\omega} \quad (13.27)$$

Let $u = \omega/\omega_n$ (normalized frequency):

$$G(ju\omega_n) = \frac{1}{1 - u^2 + 2j\zeta u} \quad (13.28)$$

Magnitude:

$$|G(j\omega)| = \frac{1}{\sqrt{(1 - u^2)^2 + (2\zeta u)^2}} \quad (13.29)$$

Phase:

$$\angle G(j\omega) = -\arctan\left(\frac{2\zeta u}{1 - u^2}\right) \quad (13.30)$$

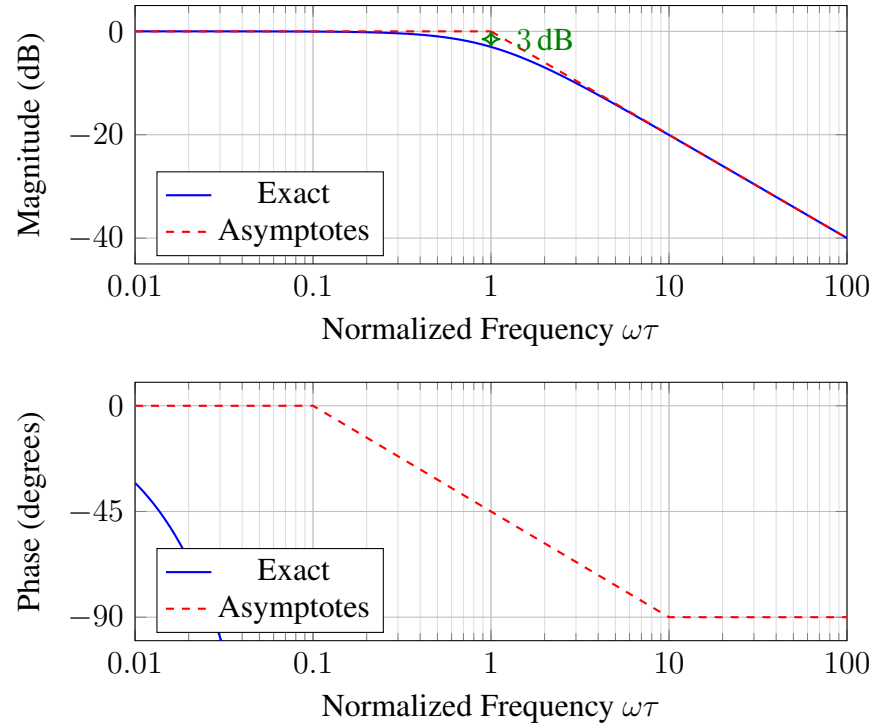


Figure 13.3: First-order low-pass system Bode plot. The asymptotic approximation (dashed) is within 3 dB of the exact response (solid) at the corner frequency.

Asymptotic Behavior

Low frequency ($\omega \ll \omega_n, u \ll 1$):

$$|G|_{\text{dB}} \approx 0 \text{ dB} \quad (13.31)$$

$$\angle G \approx 0^\circ \quad (13.32)$$

High frequency ($\omega \gg \omega_n, u \gg 1$):

$$|G|_{\text{dB}} \approx -40 \log_{10} u = -40 \log_{10}(\omega/\omega_n) \quad (13.33)$$

$$\angle G \approx -180^\circ \quad (13.34)$$

The high-frequency asymptote has a slope of -40 dB/decade (compared to -20 dB/decade for first-order systems).

Resonant Peak

For underdamped systems ($\zeta < 1/\sqrt{2} \approx 0.707$), the magnitude exhibits a resonant peak. The peak occurs at:

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2} \quad (\text{for } \zeta < 0.707) \quad (13.35)$$

The peak magnitude is:

$$M_r = \frac{1}{2\zeta\sqrt{1 - \zeta^2}} \quad \text{or} \quad M_{r,\text{dB}} = -20 \log_{10}(2\zeta\sqrt{1 - \zeta^2}) \quad (13.36)$$

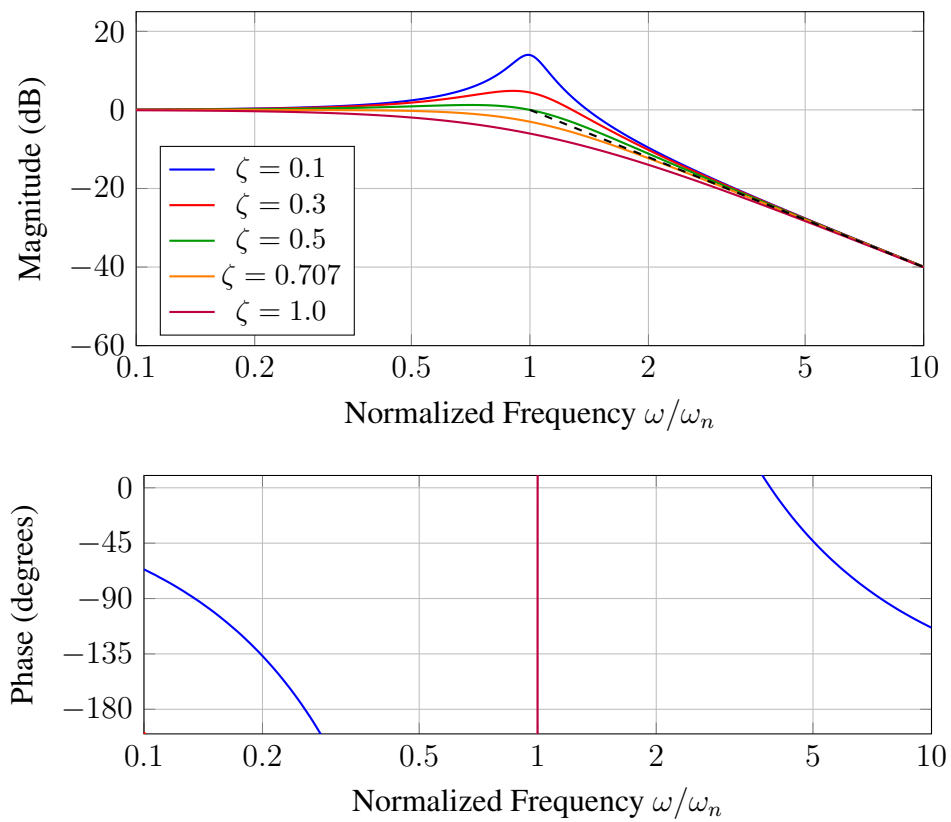


Figure 13.4: Second-order system Bode plots for various damping ratios ζ . Note the resonant peak for $\zeta < 0.707$ and the phase transition through -90° at $\omega = \omega_n$.

13.3.5 Combining Elements: Poles and Zeros

A general transfer function can be factored as:

$$G(s) = K \frac{\prod_{i=1}^m (s - z_i)}{\prod_{j=1}^n (s - p_j)} \quad (13.37)$$

where z_i are zeros and p_j are poles.

For Bode plot construction, we express this in *time constant form*:

$$G(s) = K_0 \frac{\prod_i (1 + \tau_{z_i} s)}{\prod_j (1 + \tau_{p_j} s)} \quad (13.38)$$

where K_0 is the DC gain (for a type-0 system).

Bode Plot Construction Rules:

Element	Break frequency	Frequency	Magnitude effect	Effect	Phase Effect
Gain K_0	None		$+20 \log_{10} K_0 $ dB offset		0° (or 180° if $K_0 < 0$)
Pole at origin ($1/s$)	All frequencies		-20 dB/decade		-90°
Zero at origin (s)	All frequencies		$+20$ dB/decade		$+90^\circ$
Real pole $\frac{1}{1+\tau s}$	$\omega = 1/\tau$		Adds -20 dB/decade		Adds -90°
Real zero $(1 + \tau s)$	$\omega = 1/\tau$		Adds $+20$ dB/decade		Adds $+90^\circ$
Complex poles	$\omega = \omega_n$		Adds -40 dB/decade; possible resonant peak		Adds -180°
Complex zeros	$\omega = \omega_n$		Adds $+40$ dB/decade		Adds $+180^\circ$

Combining First-Order Terms

Sketch the Bode plot for:

$$G(s) = \frac{10(s + 1)}{s(s + 10)} \quad (13.39)$$

Solution: Rewrite in standard form:

$$G(s) = \frac{10 \cdot 1 \cdot (1 + s)}{s \cdot 10 \cdot (1 + s/10)} = \frac{1 + s}{s(1 + s/10)} \quad (13.40)$$

The transfer function contains the following components: a DC gain of $K_0 = 1$ (0 dB at $\omega = 1$), a pole at the origin ($1/s$), a zero at $\omega = 1$ from the term $(1 + s)$, and a pole at $\omega = 10$ from the term $1/(1 + s/10)$.

Starting from low frequency and tracing the asymptotic magnitude, the slope is -20 dB/decade for $\omega < 1$ due to the pole at origin. At $\omega = 1$, the zero contributes $+20$ dB/decade, bringing the net slope to 0 dB/decade. Finally, at $\omega = 10$, the pole adds -20 dB/decade, returning the net slope to -20 dB/decade.

13.3.6 Stability Margins

For a unity feedback system with open-loop transfer function $G(s)$, the closed-loop stability can be assessed directly from the open-loop Bode plot.

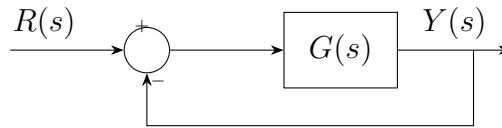


Figure 13.5: Unity feedback configuration. The closed-loop transfer function is $T(s) = G(s)/(1 + G(s))$.

The closed-loop characteristic equation is $1 + G(s) = 0$, or $G(s) = -1$.

In polar form, instability occurs when:

$$|G(j\omega)| = 1 \quad \text{and} \quad \angle G(j\omega) = -180^\circ \quad (13.41)$$

Definition 13.2 (Gain Crossover Frequency). The **gain crossover frequency** ω_{gc} is the frequency at which $|G(j\omega_{gc})| = 1$ (i.e., 0 dB).

Definition 13.3 (Phase Crossover Frequency). The **phase crossover frequency** ω_{pc} is the frequency at which $\angle G(j\omega_{pc}) = -180^\circ$.

Definition 13.4 (Gain Margin). The **gain margin** (GM) is the factor by which the open-loop gain can be increased before the system becomes unstable:

$$\text{GM} = \frac{1}{|G(j\omega_{pc})|} \quad \text{or} \quad \text{GM}_{\text{dB}} = -20 \log_{10} |G(j\omega_{pc})| \quad (13.42)$$

Definition 13.5 (Phase Margin). The **phase margin** (PM) is the additional phase lag that would bring the system to the verge of instability:

$$\text{PM} = 180^\circ + \angle G(j\omega_{gc}) \quad (13.43)$$

Typical design specifications for robust control systems require a gain margin of at least 6 dB (a factor of 2) and a phase margin of at least 45° . These margins provide robustness against modeling uncertainties, component variations, and unmodeled dynamics.

13.4 Laboratory Experiment: Frequency Response Measurement

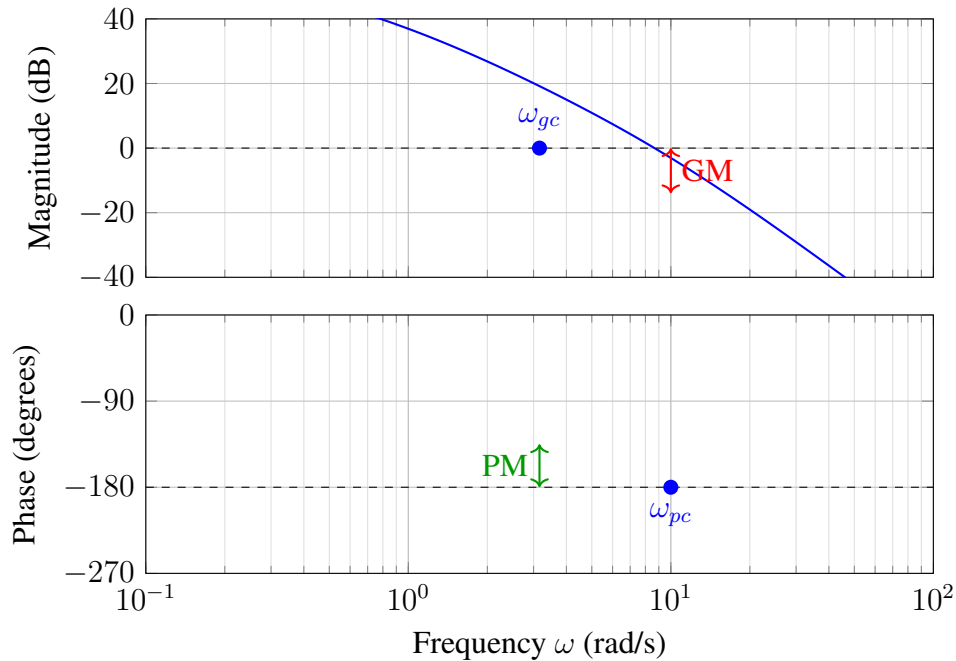


Figure 13.6: Gain margin and phase margin illustrated on a Bode plot. GM is measured at the phase crossover frequency ω_{pc} ; PM is measured at the gain crossover frequency ω_{gc} . Positive margins indicate stability.

Objective: Measure the frequency response of an RC low-pass filter and compare with theoretical predictions.

Equipment: Arduino Uno or similar microcontroller, resistor ($R = 10\text{ k}\Omega$), capacitor ($C = 100\text{ nF}$ or $0.1\text{ }\mu\text{F}$), oscilloscope (or second Arduino for data acquisition), breadboard, and connecting wires.

13.4.1 Theoretical Background

The RC low-pass filter has the transfer function:

$$G(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{RCs + 1} = \frac{1}{\tau s + 1} \quad (13.44)$$

where $\tau = RC$ is the time constant.

For $R = 10\text{ k}\Omega$ and $C = 100\text{ nF}$:

$$\tau = RC = 10 \times 10^3 \times 100 \times 10^{-9} = 1\text{ ms} \quad (13.45)$$

The corner frequency is:

$$f_c = \frac{1}{2\pi\tau} = \frac{1}{2\pi \times 0.001} \approx 159\text{ Hz} \quad (13.46)$$

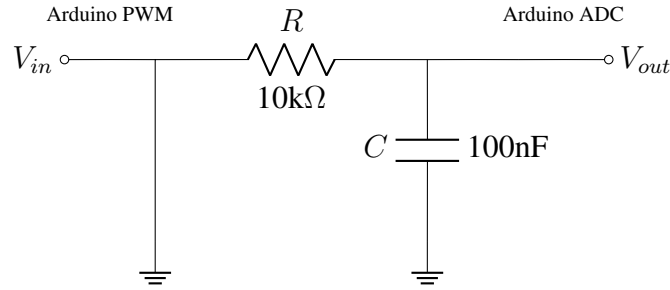


Figure 13.7: RC low-pass filter circuit for frequency response measurement.

13.4.2 Circuit Construction

13.4.3 Arduino Code for Sine Wave Generation

The following algorithm generates sine wave outputs at various frequencies using PWM and measures the filter's response.

13.4.4 Data Collection and Analysis

Table 13.1: Expected vs. Measured Frequency Response

Frequency (Hz)	Magnitude (ratio)		Phase (degrees)	
	Theory	Measured	Theory	Measured
10	0.998		-3.6°	
20	0.992		-7.2°	
50	0.954		-17.4°	
100	0.847		-32.1°	
159	0.707		-45.0°	
200	0.622		-51.5°	
500	0.303		-72.3°	
1000	0.157		-81.0°	
2000	0.079		-85.5°	
5000	0.032		-88.2°	

13.4.5 Plotting the Bode Diagram

Students should plot their measured data alongside the theoretical curves:

$$\text{Magnitude (dB)} = 20 \log_{10} \left(\frac{V_{out}}{V_{in}} \right) \quad (13.47)$$

$$\text{Phase (degrees)} = -\arctan \left(\frac{f}{f_c} \right) \times \frac{180}{\pi} \quad (13.48)$$

Algorithm 25 Frequency Response Measurement for RC Filter

```

1: Initialize: Configure PWM output pin, ADC input pins, serial communication
2: Define test frequencies:  $F \leftarrow \{10, 20, 50, 100, 159, 200, 500, 1000, 2000, 5000\}$  Hz
3: Set PWM base frequency to approximately 31 kHz for better sine approximation
4:
5: for each frequency  $f$  in  $F$  do
6:    $T \leftarrow 1/f$  ▷ Period in seconds
7:    $N_{samples} \leftarrow 100$ 
8:    $\Delta t \leftarrow T/N_{samples}$ 
9:   Initialize accumulators:  $\Sigma_{in}, \Sigma_{out}, \Sigma_{in}^2, \Sigma_{out}^2 \leftarrow 0$ 
10:
11:   for  $cycle = 1$  to  $10$  do
12:     for  $j = 0$  to  $N_{samples} - 1$  do
13:        $t \leftarrow j \cdot \Delta t$ 
14:        $V_{PWM} \leftarrow 127 + 127 \cdot \sin(2\pi ft)$  ▷ Generate sine wave
15:       Output  $V_{PWM}$  to PWM pin
16:       Wait for duration  $\Delta t$ 
17:        $V_{in} \leftarrow$  Read ADC input (reference)
18:        $V_{out} \leftarrow$  Read ADC output (filtered)
19:       Update accumulators:  $\Sigma_{in} += V_{in}, \Sigma_{out} += V_{out}$ 
20:       Update squared accumulators:  $\Sigma_{in}^2 += V_{in}^2, \Sigma_{out}^2 += V_{out}^2$ 
21:     end for
22:   end for
23:
24:    $n \leftarrow N_{samples} \times 10$ 
25:    $V_{in,RMS} \leftarrow \sqrt{\Sigma_{in}^2/n - (\Sigma_{in}/n)^2}$  ▷ Calculate RMS values
26:    $V_{out,RMS} \leftarrow \sqrt{\Sigma_{out}^2/n - (\Sigma_{out}/n)^2}$ 
27:    $|G| \leftarrow V_{out,RMS}/V_{in,RMS}$  ▷ Magnitude ratio
28:    $\phi \leftarrow -\arctan(f/f_c) \times 180/\pi$  ▷ Phase estimate,  $f_c = 159$  Hz
29:   Output: frequency  $f$ , magnitude  $|G|$ , phase  $\phi$ 
30: end for

```

13.4.6 Alternative Experiment: Motor Transfer Function

For a more dynamic experiment, students can measure the frequency response of a DC motor. The procedure involves applying sinusoidal voltage commands at various frequencies while measuring the motor speed response using an encoder. From the collected data, the magnitude and phase are extracted at each test frequency. Finally, the measured frequency response data is fit to a first-order or second-order model to identify the system parameters.

The motor transfer function from voltage to speed typically has the form:

$$G(s) = \frac{K}{(\tau_e s + 1)(\tau_m s + 1)} \quad (13.49)$$

where τ_e is the electrical time constant and τ_m is the mechanical time constant.

13.5 Worked Examples

Example 13.1 (Sketching a First-Order Bode Plot). Sketch the Bode plot for $G(s) = \frac{100}{s + 10}$.

Solution:

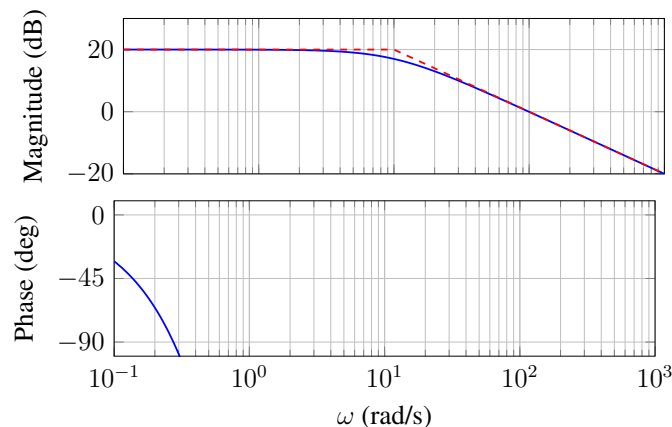
Step 1: Convert to standard form.

$$G(s) = \frac{100}{s + 10} = \frac{100}{10} \cdot \frac{1}{1 + s/10} = 10 \cdot \frac{1}{1 + s/10} \quad (13.50)$$

Step 2: Identify components. The system has a DC gain of $K_0 = 10$, which corresponds to $20 \log_{10}(10) = 20$ dB, and a first-order pole with corner frequency $\omega_c = 10$ rad/s.

Step 3: Construct magnitude plot. For frequencies $\omega < 10$ rad/s, the magnitude is constant at 20 dB. At $\omega = 10$ rad/s, the rolloff begins at -20 dB/decade. For $\omega > 10$ rad/s, the slope continues at -20 dB/decade.

Step 4: Construct phase plot. For frequencies well below 10 rad/s, the phase is approximately 0° . At the corner frequency $\omega = 10$ rad/s, the phase equals -45° . For frequencies well above 10 rad/s, the phase approaches -90° .



Example 13.2 (Second-Order System Bode Plot). Sketch the Bode plot for $G(s) = \frac{400}{s^2 + 4s + 400}$.

Solution:

Step 1: Identify parameters. Comparing with the standard form $G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$:

$$\omega_n^2 = 400 \Rightarrow \omega_n = 20 \text{ rad/s} \quad (13.51)$$

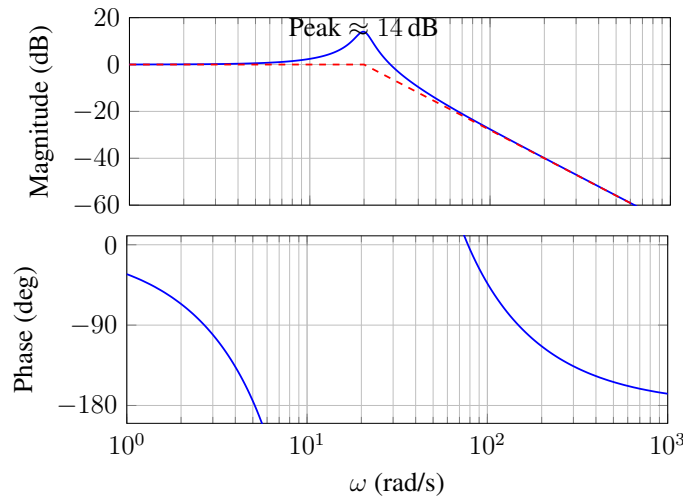
$$2\zeta\omega_n = 4 \Rightarrow \zeta = \frac{4}{2 \times 20} = 0.1 \quad (13.52)$$

Step 2: Calculate resonant peak. Since $\zeta = 0.1 < 0.707$, there is a resonant peak:

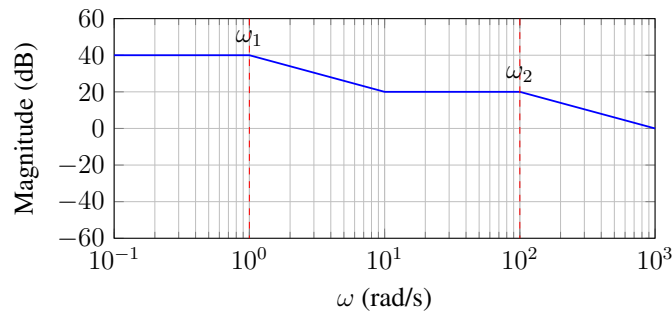
$$M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}} = \frac{1}{2(0.1)\sqrt{1-0.01}} \approx 5.03 \quad (13.53)$$

$$M_{r,\text{dB}} = 20 \log_{10}(5.03) \approx 14 \text{ dB} \quad (13.54)$$

Step 3: Construct plots. The Bode plot has a DC gain of 0 dB and exhibits a resonant peak of approximately 14 dB near $\omega = 20$ rad/s. The high-frequency asymptote has a slope of -40 dB/decade. The phase transitions from 0° through -90° to -180° , with a sharp transition occurring near ω_n .



Example 13.3 (Transfer Function from Bode Plot). The following Bode magnitude plot was measured experimentally. Determine the transfer function.



Solution:

Step 1: Identify corner frequencies. Initially, we observe a corner at $\omega_1 = 1$ rad/s where the slope changes from 0 to -20 dB/decade, suggesting a pole. At $\omega_2 = 100$ rad/s, the slope changes again.

Wait—looking more carefully at the slope transitions: from $\omega = 0.1$ to $\omega = 1$, the slope is 0 dB/decade; from $\omega = 1$ to $\omega = 10$, the slope is $(20 - 40)/(1) = -20$ dB/decade; from $\omega = 10$ to $\omega = 100$, the slope returns to 0 dB/decade; and from $\omega = 100$ to $\omega = 1000$, the slope is again -20 dB/decade.

Step 2: Identify poles and zeros. At $\omega = 1$, the slope goes from 0 to -20 , indicating a **pole at** $s = -1$. At $\omega = 10$, the slope goes from -20 to 0, indicating a **zero at** $s = -10$. At $\omega = 100$, the slope goes from 0 to -20 , indicating a **pole at** $s = -100$.

Step 3: Determine DC gain. At $\omega = 0.1$, magnitude = 40 dB. Extrapolating the low-frequency asymptote to $\omega = 1$:

$$K_0 = 10^{40/20} = 100 \quad (13.55)$$

Step 4: Construct transfer function.

$$G(s) = 100 \cdot \frac{1 + s/10}{(1 + s)(1 + s/100)} = \frac{100(s + 10)}{(s + 1)(s + 100)} \quad (13.56)$$

Verify: This can be simplified to check DC gain:

$$G(0) = \frac{100 \times 10}{1 \times 100} = 10 \Rightarrow 20 \text{ dB} \quad (13.57)$$

Hmm, this doesn't match our 40 dB reading. Let's reconsider.

Corrected Step 3: At very low frequency (below the first pole), the gain is 40 dB = 100. The transfer function in standard form is:

$$G(s) = K \cdot \frac{(1 + s/10)}{(1 + s)(1 + s/100)} \quad (13.58)$$

For DC gain of 100:

$$G(0) = K = 100 \checkmark \quad (13.59)$$

Therefore:

$$G(s) = \frac{100(1 + s/10)}{(1 + s)(1 + s/100)} = \frac{1000(s + 10)}{(s + 1)(s + 100)} \quad (13.60)$$

Finding Stability Margins

For the open-loop transfer function $G(s) = \frac{100}{s(s + 1)(s + 10)}$, find the gain margin and phase margin.

Solution:

Step 1: Find the phase crossover frequency. At phase crossover, $\angle G(j\omega_{pc}) = -180^\circ$.

$$\angle G(j\omega) = -90^\circ - \arctan(\omega) - \arctan(\omega/10) \quad (13.61)$$

Setting this equal to -180° :

$$-90^\circ - \arctan(\omega_{pc}) - \arctan(\omega_{pc}/10) = -180^\circ \quad (13.62)$$

$$\arctan(\omega_{pc}) + \arctan(\omega_{pc}/10) = 90^\circ \quad (13.63)$$

Using the identity $\arctan(a) + \arctan(b) = 90^\circ$ when $ab = 1$:

$$\omega_{pc} \cdot (\omega_{pc}/10) = 1 \Rightarrow \omega_{pc}^2 = 10 \Rightarrow \omega_{pc} = \sqrt{10} \approx 3.16 \text{ rad/s} \quad (13.64)$$

Step 2: Calculate gain margin.

$$\left| G(j\sqrt{10}) \right| = \frac{100}{\sqrt{10} \cdot \sqrt{1+10} \cdot \sqrt{1+0.1}} = \frac{100}{\sqrt{10} \cdot \sqrt{11} \cdot \sqrt{1.1}} \quad (13.65)$$

$$= \frac{100}{\sqrt{121}} = \frac{100}{11} \approx 9.09 \quad (13.66)$$

$$\text{GM} = \frac{1}{9.09} = 0.11 \Rightarrow \text{GM}_{\text{dB}} = 20 \log_{10}(0.11) = -19.2 \text{ dB} \quad (13.67)$$

Since the gain at phase crossover is greater than 1, the gain margin is **negative**, indicating an **unstable** closed-loop system!

Step 3: Find the gain crossover frequency. We need $|G(j\omega_{gc})| = 1$:

$$\frac{100}{\omega_{gc} \sqrt{1 + \omega_{gc}^2} \sqrt{1 + \omega_{gc}^2/100}} = 1 \quad (13.68)$$

This requires numerical solution. Testing $\omega_{gc} = 4.3$:

$$\frac{100}{4.3 \cdot \sqrt{1 + 18.49} \cdot \sqrt{1 + 0.185}} = \frac{100}{4.3 \cdot 4.41 \cdot 1.09} \approx 4.8 \neq 1 \quad (13.69)$$

Testing $\omega_{gc} = 8$:

$$\frac{100}{8 \cdot \sqrt{65} \cdot \sqrt{1.64}} = \frac{100}{8 \cdot 8.06 \cdot 1.28} \approx 1.21 \quad (13.70)$$

Testing $\omega_{gc} = 9$:

$$\frac{100}{9 \cdot \sqrt{82} \cdot \sqrt{1.81}} = \frac{100}{9 \cdot 9.06 \cdot 1.35} \approx 0.91 \quad (13.71)$$

By interpolation, $\omega_{gc} \approx 8.5 \text{ rad/s}$.

Step 4: Calculate phase margin.

$$\angle G(j8.5) = -90^\circ - \arctan(8.5) - \arctan(0.85) \quad (13.72)$$

$$= -90^\circ - 83.3^\circ - 40.4^\circ = -213.7^\circ \quad (13.73)$$

$$\text{PM} = 180^\circ + (-213.7^\circ) = -33.7^\circ \quad (13.74)$$

The **negative phase margin** confirms instability.

Conclusion: The closed-loop system with unity feedback is **unstable**. Both gain margin and phase margin are negative.

Example 13.4 (Designing for Specified Bandwidth). A unity feedback system has open-loop transfer function $G(s) = \frac{K}{s(s+5)}$. Determine the value of K that gives a closed-loop bandwidth of 10 rad/s.

Solution:

The closed-loop transfer function is:

$$T(s) = \frac{G(s)}{1 + G(s)} = \frac{K}{s^2 + 5s + K} \quad (13.75)$$

Comparing with the standard second-order form:

$$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (13.76)$$

We identify:

$$\omega_n^2 = K \Rightarrow \omega_n = \sqrt{K} \quad (13.77)$$

$$2\zeta\omega_n = 5 \Rightarrow \zeta = \frac{5}{2\sqrt{K}} \quad (13.78)$$

The bandwidth of a second-order system is approximately:

$$\omega_{BW} \approx \omega_n \sqrt{1 - 2\zeta^2 + \sqrt{4\zeta^4 - 4\zeta^2 + 2}} \quad (13.79)$$

For moderate damping ($\zeta \approx 0.5$ to 0.7), a useful approximation is $\omega_{BW} \approx \omega_n$ to $1.5\omega_n$. Setting $\omega_{BW} = 10$ rad/s and using $\omega_{BW} \approx \omega_n$ as a first estimate:

$$\omega_n = 10 \Rightarrow K = 100 \quad (13.80)$$

Check the damping ratio:

$$\zeta = \frac{5}{2\sqrt{100}} = \frac{5}{20} = 0.25 \quad (13.81)$$

This is underdamped. The exact bandwidth formula gives:

$$\omega_{BW} = 10\sqrt{1 - 0.125 + \sqrt{0.0039 - 0.25 + 2}} = 10\sqrt{0.875 + 1.32} = 10 \times 1.48 = 14.8 \text{ rad/s} \quad (13.82)$$

This overshoots our target. Iterate with lower K :

Try $K = 50$: $\omega_n = 7.07$, $\zeta = 0.354$

$$\omega_{BW} = 7.07\sqrt{0.75 + 1.19} = 7.07 \times 1.39 = 9.8 \text{ rad/s} \approx 10 \text{ rad/s} \quad (13.83)$$

Final Answer: $K \approx 50$

Example 13.5 (Complete Bode Plot with Multiple Elements). Sketch the Bode plot for:

$$G(s) = \frac{1000(s+10)}{s(s+1)(s+100)} \quad (13.84)$$

Solution:

Step 1: Rewrite in standard form.

$$G(s) = \frac{1000 \cdot 10 \cdot (1 + s/10)}{s \cdot 1 \cdot (1 + s) \cdot 100 \cdot (1 + s/100)} = \frac{100(1 + s/10)}{s(1 + s)(1 + s/100)} \quad (13.85)$$

Step 2: Identify components.

Element	Corner (rad/s)	Contribution
Gain $K = 100$	–	+40 dB
Pole at origin	–	–20 dB/dec, -90°
Pole at $s = -1$	$\omega = 1$	–20 dB/dec, -90°
Zero at $s = -10$	$\omega = 10$	+20 dB/dec, $+90^\circ$
Pole at $s = -100$	$\omega = 100$	–20 dB/dec, -90°

Step 3: Construct magnitude asymptotes.

At $\omega = 1$ (before any corners except pole at origin):

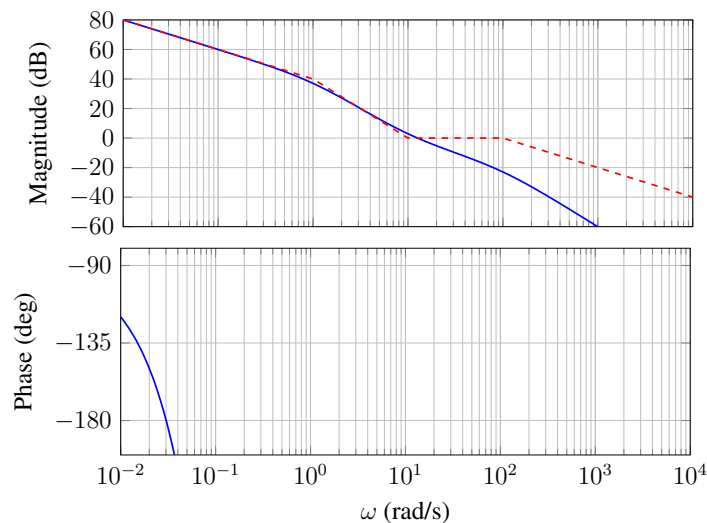
$$|G(j \cdot 1)|_{\text{dB}} \approx 40 \text{ dB} - 20 \log_{10}(1) = 40 \text{ dB} \quad (13.86)$$

The asymptotic slopes vary with frequency: for $\omega < 1$, the slope is -20 dB/dec; for $1 < \omega < 10$, the slope steepens to -40 dB/dec; for $10 < \omega < 100$, the slope returns to -20 dB/dec; and for $\omega > 100$, the slope is again -40 dB/dec.

Step 4: Construct phase.

$$\angle G(j\omega) = -90^\circ - \arctan(\omega) + \arctan(\omega/10) - \arctan(\omega/100) \quad (13.87)$$

Key phase values are computed as follows: as $\omega \rightarrow 0$, $\angle G \rightarrow -90^\circ$; at $\omega = 1$, $\angle G = -90 - 45 + 5.7 - 0.6 = -130^\circ$; at $\omega = 10$, $\angle G = -90 - 84.3 + 45 - 5.7 = -135^\circ$; at $\omega = 100$, $\angle G = -90 - 89.4 + 84.3 - 45 = -140^\circ$; and as $\omega \rightarrow \infty$, $\angle G \rightarrow -90 - 90 + 90 - 90 = -180^\circ$.



13.6 Summary

This chapter has introduced the frequency response method—a powerful technique for analyzing and designing control systems without solving differential equations.

The **frequency response**, obtained by evaluating the transfer function along the imaginary axis as $G(j\omega)$, completely characterizes the steady-state response to sinusoidal inputs. **Bode plots**—logarithmic plots of magnitude (in dB) and phase (in degrees) versus frequency—enable graphical analysis of complex systems through superposition. The **asymptotic approximations** that make Bode plots practical follow simple rules: first-order terms contribute ± 20 dB/decade and $\pm 90^\circ$ per pole or zero, while second-order terms contribute ± 40 dB/decade and $\pm 180^\circ$. **Stability margins**, specifically gain margin (how much gain can increase) and phase margin (how much phase can lag), quantify robustness and are read directly from the Bode plot. Finally, from a **practical application** standpoint, the frequency response can be measured experimentally, enabling system identification and verification of theoretical models.

The frequency response method transforms the complexity of differential equations into intuitive graphical reasoning. This approach, pioneered at Bell Labs in the 1930s and 1940s, remains the foundation of practical control system design.

13.7 Exercises

Exercise 13.1. Sketch the Bode plot for $G(s) = \frac{50}{(s+5)(s+50)}$. Determine the corner frequencies and the DC gain.

Exercise 13.2. For the transfer function $G(s) = \frac{s+2}{s^2+3s+2}$, sketch the Bode plot and identify any pole-zero cancellations.

Exercise 13.3. A system has the open-loop transfer function $G(s) = \frac{K}{s(s+2)(s+8)}$. Find the range of K for closed-loop stability using the Bode plot method.

Exercise 13.4. From the following measured data, estimate the transfer function:

ω (rad/s)	Magnitude (dB)	Phase (degrees)
0.1	20	-6°
1	17	-45°
10	-3	-84°
100	-23	-90°

Exercise 13.5. A second-order system has $\omega_n = 50$ rad/s and $\zeta = 0.2$. Calculate the resonant peak magnitude and the frequency at which it occurs.

Exercise 13.6. Design a lead compensator $G_c(s) = K \frac{s+a}{s+b}$ (with $b > a$) to provide 45° of phase lead at $\omega = 10$ rad/s.

Exercise 13.7. An RC filter with $R = 1$ k Ω is to have a corner frequency of 1 kHz. Calculate the required capacitor value and sketch the expected Bode plot.

Exercise 13.8. For a unity feedback system with $G(s) = \frac{100(s+1)}{s^2(s+10)}$, find the gain margin and phase margin. Is the system stable?

Exercise 13.9. Explain why a phase margin of 45° to 60° is typically specified in control system design. What happens if the phase margin is too small? Too large?

Exercise 13.10. Using an Arduino or similar microcontroller, design an experiment to measure the frequency response of a simple audio amplifier circuit. Specify the frequency range, number of test points, and expected measurement accuracy.

Exercise 13.11. For the transfer function $G(s) = \frac{10(s+2)}{s(s+1)(s+20)}$, construct the Bode magnitude and phase plots using asymptotic approximations. Identify all corner frequencies and the slopes of each asymptotic region.

Exercise 13.12. A control system has open-loop transfer function $G(s) = \frac{K(s+5)}{s^2(s+50)}$. Determine the value of K that provides a phase margin of exactly 45° . What is the corresponding gain margin?

Further Reading

For those seeking deeper understanding, several classic and modern references are recommended. Bode's own work, *Network Analysis and Feedback Amplifier Design* (Van Nostrand, 1945), remains the definitive classic reference on frequency response methods. Nyquist's foundational 1932 paper "Regeneration Theory" in the *Bell System Technical Journal* (Vol. 11, No. 1, pp. 126–147) established the theoretical basis for stability analysis using frequency response. Among modern textbooks, Franklin, Powell, and Emami-Naeini's *Feedback Control of Dynamic Systems* (8th ed., Pearson, 2019) provides excellent coverage with MATLAB examples. Ogata's *Modern Control Engineering* (5th ed., Prentice Hall, 2010) offers comprehensive coverage of both classical and modern methods. Dorf and Bishop's *Modern Control Systems* (13th ed., Pearson, 2017) is particularly noted for its excellent problems and applications.

Chapter 14

Lead-Lag Compensation

Chapter Overview

This chapter introduces lead and lag compensation—powerful frequency-domain design techniques that extend beyond simple PID control. We begin with the historical context of World War II fire control systems, where engineers faced the challenge of tracking fast-moving targets with unprecedented accuracy. This military necessity drove the development of phase-shaping networks that could boost system performance precisely where needed. Today, these techniques remain essential for systems requiring precise control over bandwidth and stability margins.

14.1 Historical Motivation: From Anti-Aircraft Guns to Modern Servos

The Fire Control Problem of World War II

The development of lead-lag compensation is inextricably linked to the military demands of the Second World War. As aircraft speeds increased dramatically during the 1930s and 1940s—from approximately 200 mph in early-war fighters to over 400 mph in late-war aircraft—anti-aircraft fire control systems faced an unprecedented challenge: how to predict where a rapidly maneuvering target would be by the time a projectile reached it.

The mathematical problem was clear: to hit a moving target, the fire control system needed to *predict* future position. In control terms, this meant incorporating derivative action—the system needed to respond not just to where the target *was*, but to where it was *going*. Henrik Wade Bode, working at Bell Telephone Laboratories, provided the theoretical framework through his work on frequency-domain analysis, introducing what we now call Bode plots. The M9 gun director, developed by Bell Labs and deployed in 1943, used analog lead-lag networks to achieve tracking accuracy that was considered miraculous at the time.

14.1.1 The Fire Control Problem of World War II

The development of lead-lag compensation is inextricably linked to the military demands of the Second World War. As aircraft speeds increased dramatically during the 1930s and 1940s—from approximately 200 mph in early-war fighters to over 400 mph in late-war aircraft—anti-aircraft fire control systems faced an unprecedented challenge: how to predict where a rapidly maneuvering target would be by the time a projectile reached it.

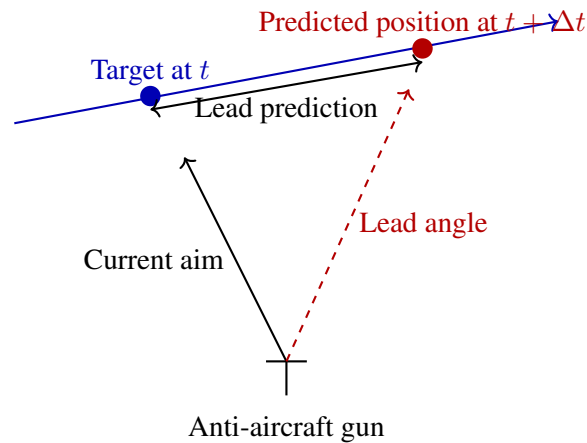


Figure 14.1: The fire control problem: predicting target position requires derivative action, but noise amplification makes pure differentiation impractical.

The mathematical problem was clear: to hit a moving target, the fire control system needed to *predict* future position. In control terms, this meant incorporating derivative action—the system needed to respond not just to where the target *was*, but to where it was *going*. A simple proportional controller pointing the gun at the current target position would always lag behind.

14.1.2 The Limitation of Pure PID Control

Engineers quickly discovered that while PID control could theoretically provide the necessary derivative action, practical implementation faced severe obstacles. First, pure derivative action ($D(s) = K_d s$) amplifies high-frequency noise unboundedly, and radar tracking data corrupted by electrical noise and atmospheric effects produced unacceptable jitter in the control signal. Second, many servo systems had inadequate phase margin at the desired crossover frequency, meaning that simply increasing gain to improve tracking speed would push the system toward instability. Third, achieving both small steady-state error (requiring high loop gain) and fast, well-damped transient response (requiring adequate phase margin) seemed mutually exclusive with simple gain adjustment.

14.1.3 Bode's Revolutionary Contribution

Henrik Wade Bode, working at Bell Telephone Laboratories in the late 1930s and early 1940s, provided the theoretical framework that would resolve these dilemmas. His seminal work, “Network

Analysis and Feedback Amplifier Design” (1945), introduced what we now call Bode plots and established the relationship between gain and phase in minimum-phase systems.

“The advantage of logarithmic plots is that they reduce the problem of network synthesis to one of geometrical construction.” — H.W. Bode, 1945

Bode’s key insight was that in the frequency domain, one could *shape* the loop transfer function to achieve specific performance objectives. Rather than accepting whatever phase margin resulted from a given plant and controller combination, the engineer could design compensation networks that added phase precisely where needed.

14.1.4 Lead Compensation: Filtered Derivative Action

The lead compensator emerged as an elegant solution to the noise-amplification problem of pure derivative control. Consider the transfer function:

$$C(s) = K_c \frac{s + z}{s + p}, \quad \text{where } p > z \quad (14.1)$$

At low frequencies, this compensator behaves like a proportional gain of $K_c \cdot z/p$. At high frequencies, it approaches K_c —a constant gain rather than the unbounded amplification of a pure derivative. But crucially, in the mid-frequency range, the compensator provides *phase lead*—it makes the output lead the input in phase, much like a derivative would.

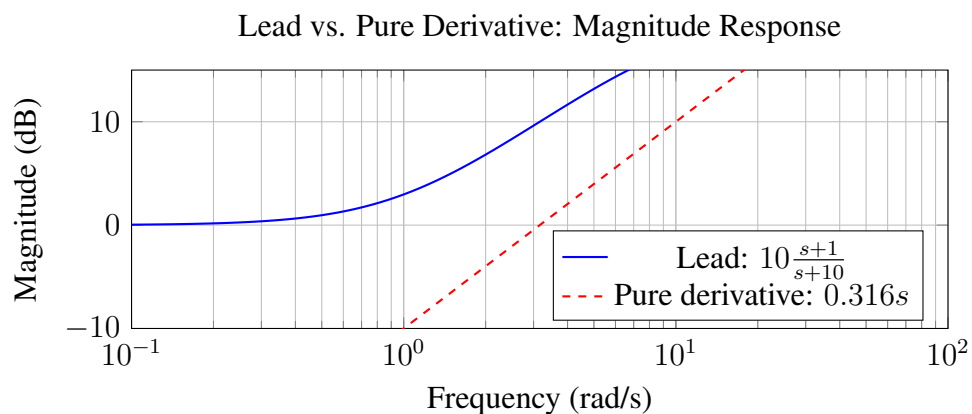


Figure 14.2: The lead compensator provides derivative-like behavior in the mid-frequency range while limiting high-frequency gain.

The military significance was immediate: fire control systems could now incorporate predictive (derivative) action without being overwhelmed by radar noise. The M9 gun director, developed by Bell Labs and deployed in 1943, used analog lead-lag networks to achieve tracking accuracy that was considered miraculous at the time.

14.1.5 Lag Compensation: Stable Integral Action

The companion technique, lag compensation, addressed the steady-state accuracy problem. While integral control ($I(s) = K_i/s$) theoretically provides infinite DC gain, it also introduces 90 degrees of phase lag at all frequencies—often destabilizing the system.

The lag compensator:

$$C(s) = K_c \frac{s + z}{s + p}, \quad \text{where } z > p \quad (14.2)$$

provides *high gain at low frequencies* (improving steady-state accuracy) while keeping the phase contribution near the crossover frequency small. The key design insight is to place the lag compensator's break frequencies well below the crossover frequency, so the system “sees” only the gain boost, not the phase penalty.

14.1.6 Lead-Lag: The Best of Both Worlds

By cascading lead and lag compensators, engineers could simultaneously increase phase margin at the crossover frequency through lead compensation, boost low-frequency gain for steady-state accuracy through lag compensation, and maintain bounded high-frequency gain for noise rejection.

This cascade approach became the standard methodology for precision servo design and remains widely used today.

14.2 Modern Applications

While born from military necessity, lead-lag compensation techniques have become ubiquitous across modern engineering. The following applications illustrate the continuing relevance of these classical methods.

14.2.1 Hard Disk Drive Servo Systems

Modern hard disk drives position read/write heads with nanometer precision over spinning platters at 7200 RPM or higher. The servo system must achieve settling times under 5 milliseconds for track-to-track seeks, maintain position accuracy within approximately 10 nanometers during read/write operations, and reject disturbances from disk flutter, bearing vibration, and external shocks.

Lead compensation provides the phase margin necessary for fast seeking, while lag compensation ensures the head remains precisely on track despite low-frequency disturbances. The constraints are severe: sampling rates exceed 20 kHz, and stability margins must account for manufacturing variations across millions of units.

14.2.2 Antenna Tracking Systems

Satellite communication and radio telescope systems require tracking moving objects (satellites in orbit, celestial sources) with arc-second precision. The challenges include large mechanical inertia of antenna structures, wind loading as a time-varying disturbance, and structural resonances that limit achievable bandwidth.

Lead-lag compensation shapes the loop response to maximize tracking bandwidth while providing robustness to resonance excitation. The Deep Space Network antennas, for example, use sophisticated lead-lag networks to maintain communication with spacecraft billions of kilometers away.

14.2.3 Power System Stabilizers

Synchronous generators in electric power grids can experience low-frequency oscillations (typically 0.2–2 Hz) that threaten system stability. Power system stabilizers (PSS) inject supplementary control signals into the generator's excitation system to damp these oscillations.

The PSS typically consists of cascaded lead-lag stages that provide phase compensation at the oscillation frequency. The design challenge is achieving sufficient phase lead without excessive high-frequency gain that could excite torsional resonances in the turbine-generator shaft.

14.2.4 Precision Motion Control

Industrial robots, CNC machines, and semiconductor manufacturing equipment all rely on lead-lag compensation for precise trajectory tracking. In photolithography systems, for example, the wafer stage must position a silicon wafer with errors below 1 nanometer while following complex scan patterns at speeds exceeding 1 meter per second.

14.3 Mathematical Foundation

14.3.1 The General First-Order Compensator

Both lead and lag compensators share the same mathematical form:

$$C(s) = K_c \frac{s + z}{s + p} = K_c \frac{\tau s + 1}{\alpha \tau s + 1} \quad (14.3)$$

where $z = 1/\tau$ is the zero location, $p = 1/(\alpha\tau)$ is the pole location, and $\alpha = z/p = p_{\text{freq}}/z_{\text{freq}}$ in terms of break frequencies.

The distinction between lead and lag compensation lies solely in the relationship between z and p . For a lead compensator, $p > z$ (equivalently, $\alpha > 1$), while for a lag compensator, $z > p$ (equivalently, $\alpha < 1$).

14.3.2 Lead Compensator Analysis

Transfer Function and Bode Plot

For the lead compensator with $p > z$, consider the standard form:

$$C_{\text{lead}}(s) = K_c \frac{s + z}{s + p} = K_c \cdot \frac{z}{p} \cdot \frac{s/z + 1}{s/p + 1} \quad (14.4)$$

The DC gain is $K_c \cdot z/p = K_c/\alpha < K_c$, while the high-frequency gain approaches K_c .

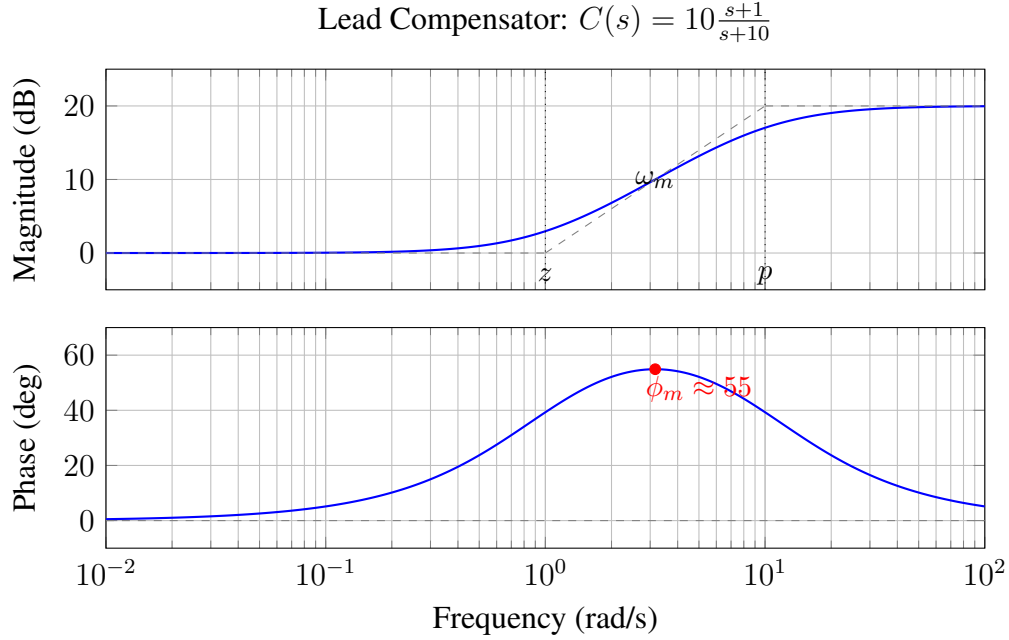


Figure 14.3: Bode plot of a lead compensator with $z = 1, p = 10, K_c = 10$ (so $\alpha = 10$). Maximum phase lead of approximately 55 occurs at $\omega_m = \sqrt{10} \approx 3.16$ rad/s.

Maximum Phase Lead

The phase contribution of the lead compensator is:

$$\phi(\omega) = \tan^{-1} \left(\frac{\omega}{z} \right) - \tan^{-1} \left(\frac{\omega}{p} \right) \quad (14.5)$$

To find the frequency of maximum phase, we differentiate and set equal to zero:

$$\frac{d\phi}{d\omega} = \frac{z}{z^2 + \omega^2} - \frac{p}{p^2 + \omega^2} = 0 \quad (14.6)$$

Solving for ω_m :

$$\boxed{\omega_m = \sqrt{zp} = \frac{1}{\tau\sqrt{\alpha}}} \quad (14.7)$$

The frequency of maximum phase is the *geometric mean* of the zero and pole frequencies.

Substituting ω_m into the phase equation:

$$\phi_m = \tan^{-1} \left(\frac{\sqrt{zp}}{z} \right) - \tan^{-1} \left(\frac{\sqrt{zp}}{p} \right) \quad (14.8)$$

$$= \tan^{-1} \left(\sqrt{\frac{p}{z}} \right) - \tan^{-1} \left(\sqrt{\frac{z}{p}} \right) \quad (14.9)$$

$$= \tan^{-1}(\sqrt{\alpha}) - \tan^{-1} \left(\frac{1}{\sqrt{\alpha}} \right) \quad (14.10)$$

Using the identity $\tan^{-1}(x) - \tan^{-1}(1/x) = 2 \tan^{-1}(x) - 90$ for $x > 0$, or more directly via the sine function:

$$\sin(\phi_m) = \frac{\alpha - 1}{\alpha + 1} = \frac{p - z}{p + z} \quad (14.11)$$

Equivalently:

$$\phi_m = \sin^{-1} \left(\frac{\alpha - 1}{\alpha + 1} \right) \quad (14.12)$$

Table 14.1: Maximum Phase Lead vs. $\alpha = p/z$

α	$\sin(\phi_m)$	ϕ_m (degrees)
2	0.333	19.5
3	0.500	30.0
4	0.600	36.9
5	0.667	41.8
10	0.818	54.9
20	0.905	64.8

Design insight: Single-stage lead compensators are typically limited to providing about 55–60 of phase lead. For larger phase margins, multiple cascaded stages or a lead-lag design should be considered.

Gain at Maximum Phase Frequency

At $\omega = \omega_m = \sqrt{zp}$, the magnitude of the compensator is:

$$|C(j\omega_m)| = K_c \frac{|j\omega_m + z|}{|j\omega_m + p|} = K_c \sqrt{\frac{\omega_m^2 + z^2}{\omega_m^2 + p^2}} \quad (14.13)$$

$$= K_c \sqrt{\frac{zp + z^2}{zp + p^2}} = K_c \sqrt{\frac{z(p + z)}{p(p + z)}} = K_c \sqrt{\frac{z}{p}} \quad (14.14)$$

In decibels:

$$|C(j\omega_m)|_{\text{dB}} = 20 \log_{10}(K_c) + 10 \log_{10} \left(\frac{z}{p} \right) = 20 \log_{10}(K_c) - 10 \log_{10}(\alpha) \quad (14.15)$$

This is exactly halfway (in dB) between the low-frequency and high-frequency asymptotes.

Lead Compensator Design Procedure

Given specifications for phase margin improvement ϕ_{needed} and desired crossover frequency ω_c , the design proceeds as follows. First, determine the required α by adding 5–12 safety margin to ϕ_{needed} to account for the gain increase at ω_m , then solve:

$$\alpha = \frac{1 + \sin(\phi_m)}{1 - \sin(\phi_m)} \quad (14.16)$$

Second, calculate the zero and pole locations by placing $\omega_m = \omega_c$ (desired crossover frequency):

$$z = \frac{\omega_c}{\sqrt{\alpha}} \quad (14.17)$$

$$p = \omega_c \sqrt{\alpha} \quad (14.18)$$

Third, determine K_c by adjusting it so the compensated loop gain magnitude equals 0 dB at ω_c . Since the lead compensator contributes $-10 \log_{10}(\alpha)$ dB at ω_m :

$$K_c = \sqrt{\alpha} \cdot |G(j\omega_c)|^{-1} \quad (14.19)$$

where $G(s)$ is the uncompensated plant.

14.3.3 Lag Compensator Analysis

Transfer Function and Bode Plot

For the lag compensator with $z > p$, we write:

$$C_{\text{lag}}(s) = K_c \frac{s+z}{s+p} = K_c \cdot \frac{z}{p} \cdot \frac{s/z+1}{s/p+1}, \quad z > p \quad (14.20)$$

Now $\beta = z/p > 1$, and the DC gain is $K_c \cdot \beta > K_c$.

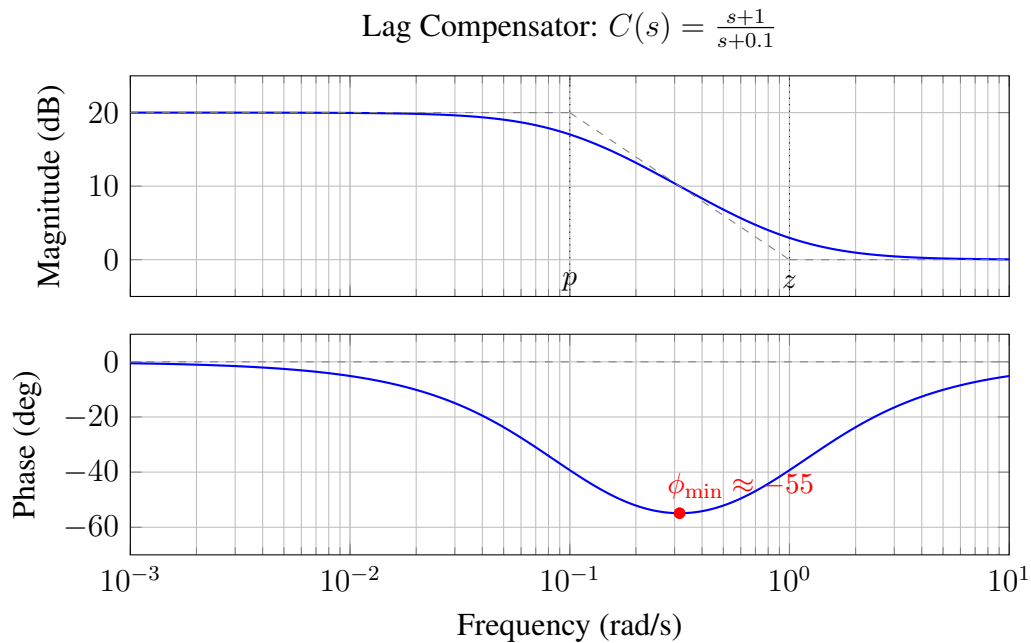


Figure 14.4: Bode plot of a lag compensator with $p = 0.1$, $z = 1$ (so $\beta = 10$). The DC gain is boosted by 20 dB, but significant phase lag occurs in the transition region.

Design Philosophy

The lag compensator provides two key benefits. First, the factor $\beta = z/p$ provides a low-frequency gain boost that increases the DC loop gain, thereby reducing steady-state error. Second, by placing both z and p well below the crossover frequency ω_c , the phase lag contribution at ω_c remains small (typically 5°–10°), ensuring minimal phase impact at crossover.

Lag Compensator Design Procedure

Given the required steady-state error improvement and existing phase margin, the design procedure proceeds as follows. First, determine β by calculating the ratio of current steady-state error to desired error: $\beta = e_{ss,current}/e_{ss,desired}$. Second, find the current crossover frequency ω_c from the uncompensated Bode plot. Third, place the lag compensator zero by choosing z such that $z \leq \omega_c/10$, which ensures the phase lag at ω_c is acceptably small:

$$\phi_{lag}(\omega_c) = \tan^{-1}\left(\frac{\omega_c}{z}\right) - \tan^{-1}\left(\frac{\omega_c}{p}\right) \approx -5^\circ \text{ to } -10^\circ \quad (14.21)$$

Fourth, calculate the pole location as $p = z/\beta$. Fifth, set K_c , which is typically equal to 1 since the gain boost comes from the β factor; adjust if needed to maintain the original crossover frequency.

14.3.4 Lead-Lag Compensator

When both increased phase margin and improved steady-state accuracy are required, a lead-lag compensator cascades both elements:

$$C_{lead-lag}(s) = K_c \underbrace{\frac{s + z_{lead}}{s + p_{lead}}}_{\text{Lead stage}} \cdot \underbrace{\frac{s + z_{lag}}{s + p_{lag}}}_{\text{Lag stage}} \quad (14.22)$$

where $p_{lead} > z_{lead}$ and $z_{lag} > p_{lag}$.

Design Approach

The design approach for lead-lag compensation proceeds in three stages. First, design the lead stage by determining the phase margin improvement needed and placing the lead compensator to provide maximum phase at the desired crossover frequency. Second, design the lag stage by calculating the additional low-frequency gain needed for steady-state accuracy and placing the lag zero at least one decade below the crossover frequency. Third, verify stability by checking that the total phase margin with both stages is adequate, accounting for the small phase lag contribution from the lag stage at crossover.

14.3.5 Root Locus Interpretation

Lead-lag compensation can also be understood through root locus analysis.

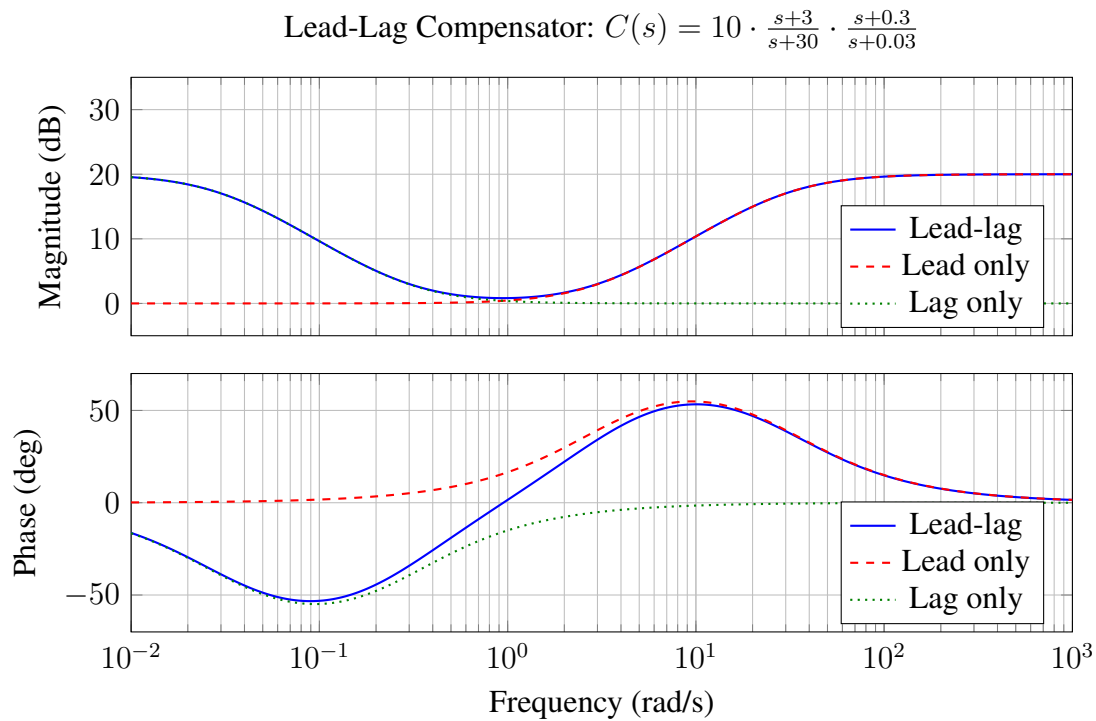


Figure 14.5: Bode plot of a lead-lag compensator showing individual contributions. The lag stage boosts low-frequency gain by 20 dB while the lead stage provides phase lead near the crossover frequency.

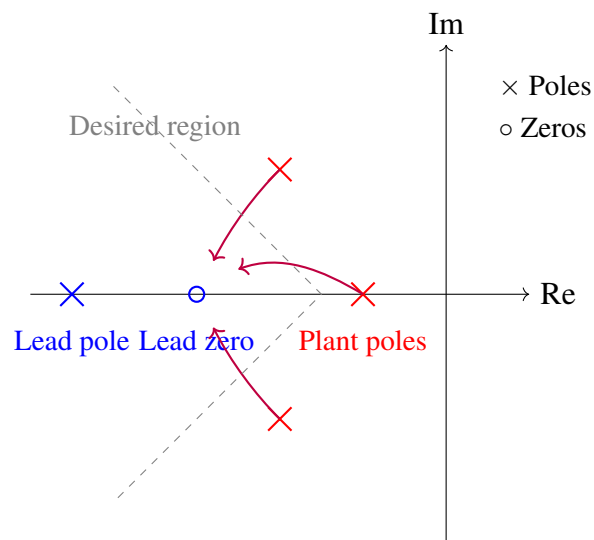


Figure 14.6: Effect of lead compensator on root locus. The compensator zero “pulls” the locus toward the left half-plane, while the far-left pole has minimal effect on dominant dynamics.

Lead compensator effect: The zero attracts root locus branches, pulling closed-loop poles toward the left half-plane (faster response, better damping). The pole is placed far left so its branch quickly leaves the region of dominant dynamics.

Lag compensator effect: Both the zero and pole are near the origin. The pole-zero pair (nearly canceling) minimally affects the root locus shape but allows higher loop gain K before instability, improving steady-state accuracy.

14.4 Practical Experiment: Motor Position Control with Lead Compensation

14.4.1 Objective

This experiment demonstrates the practical benefits of lead compensation by implementing a position servo on a DC motor. You will characterize the uncompensated system response, design a lead compensator to improve phase margin, implement the compensator digitally on an Arduino, and measure and compare step responses before and after compensation.

14.4.2 Required Equipment

Table 14.2: Equipment Required for Lead Compensation Experiment

Component	Specification
DC motor with encoder	12V geared motor, 48 CPR encoder
Microcontroller	Arduino Mega or similar
Motor driver	L298N or equivalent H-bridge
Potentiometer	10k Ω for setpoint input
Power supply	12V DC
Visualization	Oscilloscope or serial plotter
Software	Computer with Arduino IDE

14.4.3 System Model

A DC motor position system can be approximated as:

$$G(s) = \frac{\theta(s)}{V(s)} = \frac{K_m}{s(\tau_m s + 1)} \quad (14.23)$$

where K_m is the motor gain constant and τ_m is the mechanical time constant. With proportional control K_p , the open-loop transfer function is:

$$L(s) = \frac{K_p K_m}{s(\tau_m s + 1)} \quad (14.24)$$

This system has 90° of phase lag from the integrator and additional lag from the motor pole, making the phase margin sensitive to gain selection.

14.4.4 Experiment Procedure

Step 1: System Identification

First, characterize your motor's parameters:

Algorithm 26 Motor Identification via Step Response Test

```

1: Initialize:
2:   Configure motor PWM and direction pins as outputs
3:   Configure encoder channels A and B as inputs with pull-up resistors
4:   Attach interrupt to encoder channel A on rising edge
5:   Set encoderCount  $\leftarrow$  0
6:
7: Apply Step Input:
8:   Set motor direction to forward
9:   Set PWM duty cycle to 50%
10:  Record startTime  $\leftarrow$  currentTime
11:
12: Data Logging Loop:
13: while currentTime - startTime < 2000 ms do
14:   Output timestamp and encoder count to serial port
15:   Wait 10 ms
16: end while
17: Set PWM duty cycle to 0% (stop motor)
18:
19: Encoder Interrupt Service Routine:
20: if encoder channel B is LOW then
21:   encoderCount  $\leftarrow$  encoderCount + 1
22: else
23:   encoderCount  $\leftarrow$  encoderCount - 1
24: end if

```

From the velocity response data, identify K_m (steady-state velocity per volt) and τ_m (time to reach 63% of final velocity).

Step 2: Proportional Control Baseline

Implement proportional control and observe the step response:

Increase K_p until the system shows significant overshoot (indicating low phase margin). Record this gain and the step response characteristics.

Step 3: Lead Compensator Design

For a typical motor with $\tau_m = 0.05$ s and desired crossover at $\omega_c = 30$ rad/s:

Design calculations: The desired phase margin is 50°, which means we need $\phi_m \approx 55^\circ$ when accounting for gain change. From $\sin(55^\circ) = 0.819 = (\alpha - 1)/(\alpha + 1)$, we obtain $\alpha = 10$. The

Algorithm 27 Proportional Position Control Baseline

```

1: Parameters:  $K_p \leftarrow 5.0$  (proportional gain, start low)
2:
3: loop
4:   Read setpoint from potentiometer (analog input)
5:   Map raw value to range  $[-500, 500]$  counts
6:    $\text{error} \leftarrow \text{setpoint} - \text{encoderCount}$ 
7:    $\text{controlSignal} \leftarrow K_p \cdot \text{error}$ 
8:   Saturate controlSignal to range  $[-255, 255]$ 
9:
10:  if controlSignal  $\geq 0$  then
11:    Set motor direction to forward
12:    Set PWM to controlSignal
13:  else
14:    Set motor direction to reverse
15:    Set PWM to  $|\text{controlSignal}|$ 
16:  end if
17:
18:  Log timestamp, setpoint, and encoder count
19:  Wait 5 ms (200 Hz sample rate)
20: end loop

```

zero location is then $z = \omega_c / \sqrt{\alpha} = 30 / \sqrt{10} \approx 9.5$ rad/s, and the pole location is $p = \omega_c \sqrt{\alpha} = 30\sqrt{10} \approx 95$ rad/s.

In the discrete domain (Tustin transform with $T_s = 0.005$ s):

$$C(z) = K_c \frac{1 + a_1 z^{-1}}{1 + b_1 z^{-1}} \quad (14.25)$$

Step 4: Digital Lead Compensator Implementation**14.4.5 Expected Results**

You should observe the following behavior. With proportional control only, the system exhibits either slow response when using low K_p or significant overshoot and oscillation when using high K_p . With lead compensation, however, the system achieves faster rise time *and* reduced overshoot simultaneously. The phase margin improvement allows higher effective gain without sacrificing stability.

14.4.6 Verifying Phase Margin Improvement

While direct frequency response measurement requires specialized equipment, you can estimate phase margin from the step response:

$$\text{Overshoot} \approx e^{-\pi\zeta/\sqrt{1-\zeta^2}} \implies \zeta \implies \text{PM} \approx 100\zeta \text{ (degrees, rough estimate)} \quad (14.26)$$

Algorithm 28 Digital Lead Compensator Implementation

```

1: Parameters:
2:    $K_p \leftarrow 8.0$  (higher gain now stable with compensation)
3:    $a_1 \leftarrow -0.907$  (Tustin coefficient for zero)
4:    $b_1 \leftarrow -0.621$  (Tustin coefficient for pole)
5:    $K_c \leftarrow 3.16$  ( $\sqrt{\alpha}$  compensation factor)
6:    $T_s \leftarrow 5$  ms (sample period)
7:
8: State Variables:
9:    $x_{\text{prev}} \leftarrow 0, y_{\text{prev}} \leftarrow 0$ 
10:
11: function LEADCOMPENSATOR( $x$ )
12:    $y \leftarrow -b_1 \cdot y_{\text{prev}} + K_c \cdot (x + a_1 \cdot x_{\text{prev}})$ 
13:    $x_{\text{prev}} \leftarrow x$ 
14:    $y_{\text{prev}} \leftarrow y$ 
15:   return  $y$ 
16: end function
17:
18: Main Control Loop:
19: loop
20:   if  $T_s$  has elapsed since last sample then
21:     Read setpoint from potentiometer
22:     Map to range  $[-500, 500]$  counts
23:      $\text{error} \leftarrow \text{setpoint} - \text{encoderCount}$ 
24:      $\text{propOutput} \leftarrow K_p \cdot \text{error}$ 
25:      $\text{compOutput} \leftarrow \text{LeadCompensator}(\text{propOutput})$ 
26:     Saturate  $\text{compOutput}$  to  $[-255, 255]$ 
27:     Apply  $\text{compOutput}$  to motor (set direction and PWM)
28:     Log timestamp, setpoint, and encoder count
29:   end if
30: end loop

```

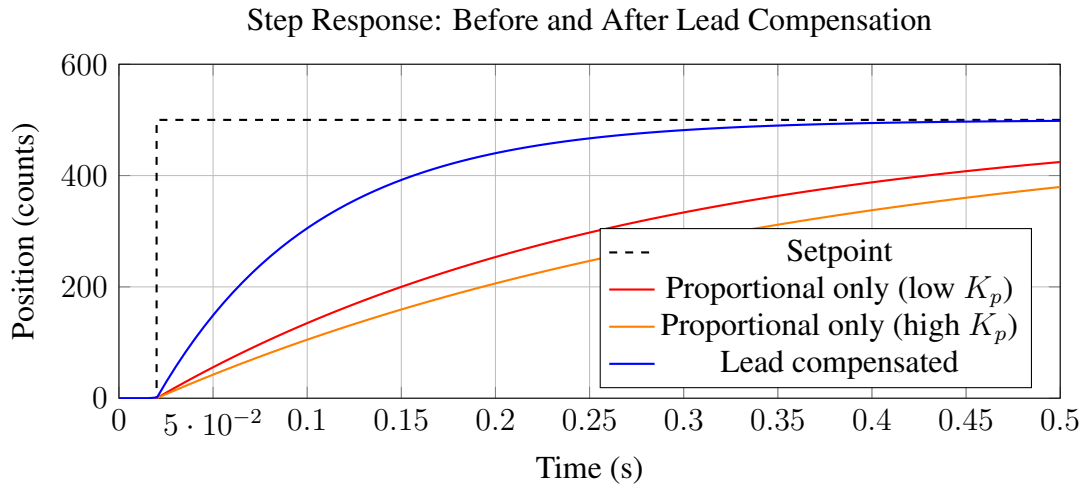


Figure 14.7: Comparison of step responses. The lead compensator enables faster response with less overshoot compared to high-gain proportional control alone.

Compare the overshoot before and after compensation to verify improved damping.

14.5 Example Problems

Example 14.1: Lead Compensator for Specified Phase Margin

Problem: A unity-feedback system has plant transfer function:

$$G(s) = \frac{10}{s(s+2)}$$

Design a lead compensator to achieve a phase margin of 45° at a crossover frequency of 4 rad/s.

Solution:

Step 1: Analyze uncompensated system at desired crossover.

At $\omega = 4$ rad/s:

$$|G(j4)| = \frac{10}{|j4| \cdot |j4+2|} = \frac{10}{4 \cdot \sqrt{16+4}} = \frac{10}{4 \cdot \sqrt{20}} = \frac{10}{17.89} = 0.559 \quad (14.27)$$

$$|G(j4)|_{\text{dB}} = 20 \log_{10}(0.559) = -5.05 \text{ dB} \quad (14.28)$$

$$\angle G(j4) = -90 - \tan^{-1}(4/2) = -90 - 63.4 = -153.4 \quad (14.29)$$

Uncompensated phase margin: $180 - 153.4 = 26.6$.

Step 2: Determine required phase lead.

Phase needed: $45 - 26.6 = 18.4$.

Add 10° safety margin for gain change: $\phi_m = 28.4 \approx 30$.

Step 3: Calculate α .

From $\sin(30) = 0.5 = (\alpha - 1)/(\alpha + 1)$:

$$0.5(\alpha + 1) = \alpha - 1 \implies 0.5\alpha + 0.5 = \alpha - 1 \implies \alpha = 3$$

Step 4: Calculate zero and pole.

$$z = \frac{\omega_c}{\sqrt{\alpha}} = \frac{4}{\sqrt{3}} = 2.31 \text{ rad/s} \quad (14.30)$$

$$p = \omega_c \sqrt{\alpha} = 4\sqrt{3} = 6.93 \text{ rad/s} \quad (14.31)$$

Step 5: Determine K_c .

The compensator magnitude at ω_c is $1/\sqrt{\alpha} = 1/\sqrt{3} = 0.577$ (i.e., -4.77 dB).

Combined magnitude at ω_c : $-5.05 - 4.77 + 20 \log_{10}(K_c) = 0$ dB.

Solving: $20 \log_{10}(K_c) = 9.82$, so $K_c = 3.10$.

Final compensator:

$$C(s) = 3.10 \cdot \frac{s + 2.31}{s + 6.93}$$

Verification:

At $\omega = 4$:

$$|C(j4)| = 3.10 \cdot \frac{\sqrt{16 + 5.34}}{\sqrt{16 + 48.0}} = 3.10 \cdot \frac{4.62}{8.00} = 1.79 \quad (14.32)$$

$$|L(j4)| = |C(j4)||G(j4)| = 1.79 \times 0.559 = 1.00 \quad \checkmark \quad (14.33)$$

$$\angle C(j4) = \tan^{-1}(4/2.31) - \tan^{-1}(4/6.93) = 60.0 - 30.0 = 30 \quad (14.34)$$

$$\angle L(j4) = -153.4 + 30 = -123.4 \quad (14.35)$$

$$\text{PM} = 180 - 123.4 = 56.6 > 45 \quad \checkmark \quad (14.36)$$

The achieved phase margin exceeds the specification, as expected due to our safety margin.

Example 14.2: Lag Compensator for Steady-State Error

Problem: A unity-feedback system has:

$$G(s) = \frac{5}{s(s+1)(s+5)}$$

The current phase margin is 35°, which is acceptable. Design a lag compensator to reduce the steady-state error to a ramp input by a factor of 10.

Solution:

Step 1: Determine current velocity error constant.

$$K_v = \lim_{s \rightarrow 0} s \cdot G(s) = \lim_{s \rightarrow 0} s \cdot \frac{5}{s(s+1)(s+5)} = \frac{5}{1 \cdot 5} = 1 \text{ s}^{-1}$$

Current steady-state error to unit ramp: $e_{ss} = 1/K_v = 1$.

Step 2: Required improvement.

To reduce error by factor of 10: need $K_v = 10 \text{ s}^{-1}$.

Therefore $\beta = 10$.

Step 3: Find current crossover frequency.

Setting $|G(j\omega_c)| = 1$:

$$\frac{5}{\omega_c \sqrt{\omega_c^2 + 1} \sqrt{\omega_c^2 + 25}} = 1$$

Solving numerically: $\omega_c \approx 1.26 \text{ rad/s}$.

Step 4: Place lag compensator.

Choose $z = \omega_c/10 = 0.126 \text{ rad/s}$.

Then $p = z/\beta = 0.126/10 = 0.0126 \text{ rad/s}$.

Final compensator:

$$C(s) = \frac{s + 0.126}{s + 0.0126} = \frac{s + 0.126}{s + 0.0126}$$

Verification of phase impact:

At $\omega_c = 1.26 \text{ rad/s}$:

$$\angle C(j\omega_c) = \tan^{-1}(1.26/0.126) - \tan^{-1}(1.26/0.0126) \quad (14.37)$$

$$= \tan^{-1}(10) - \tan^{-1}(100) \quad (14.38)$$

$$= 84.3 - 89.4 = -5.1 \quad (14.39)$$

Phase margin reduction is only about 5°, leaving approximately 30° phase margin.

New velocity constant: $K_v = 10 \cdot 1 = 10 \text{ s}^{-1} \checkmark$

14.5.1 Example 14.3: Maximum Phase Lead Calculation

Problem: A lead compensator has the form:

$$C(s) = K \frac{s + 4}{s + 20}$$

Find (a) the maximum phase lead, (b) the frequency at which it occurs, and (c) the compensator gain at that frequency in dB (assuming $K = 1$).

Solution:

(a) *Maximum phase lead:*

Here $z = 4$, $p = 20$, so $\alpha = p/z = 5$.

$$\phi_m = \sin^{-1} \left(\frac{\alpha - 1}{\alpha + 1} \right) = \sin^{-1} \left(\frac{5 - 1}{5 + 1} \right) = \sin^{-1}(0.667) = \boxed{41.8}$$

(b) *Frequency of maximum phase:*

$$\omega_m = \sqrt{zp} = \sqrt{4 \times 20} = \sqrt{80} = \boxed{8.94 \text{ rad/s}}$$

(c) *Gain at ω_m :*

$$|C(j\omega_m)| = \sqrt{\frac{z}{p}} = \sqrt{\frac{4}{20}} = \sqrt{0.2} = 0.447$$

$$|C(j\omega_m)|_{\text{dB}} = 20 \log_{10}(0.447) = \boxed{-7.0 \text{ dB}}$$

Alternatively: $-10 \log_{10}(\alpha) = -10 \log_{10}(5) = -7.0 \text{ dB}$.

14.5.2 Example 14.4: Lead-Lag Design for Combined Specifications

Problem: Design a lead-lag compensator for:

$$G(s) = \frac{4}{s(s+2)}$$

Specifications: (1) Phase margin $\geq 50^\circ$, (2) Velocity error constant $K_v \geq 20 \text{ s}^{-1}$.

Solution:

Step 1: Analyze uncompensated system.

Current $K_v = \lim_{s \rightarrow 0} s \cdot \frac{4}{s(s+2)} = \frac{4}{2} = 2 \text{ s}^{-1}$.

Need to increase K_v by factor of 10, so $\beta = 10$ for lag stage.

Find crossover where uncompensated PM $\approx 50^\circ$ (before adding lead):

At $\omega = 1 \text{ rad/s}$: $\angle G = -90 - \tan^{-1}(0.5) = -116.6^\circ$, PM = 63.4° .

At $\omega = 2 \text{ rad/s}$: $\angle G = -90 - \tan^{-1}(1) = -135^\circ$, PM = 45° .

Target $\omega_c \approx 1.5 \text{ rad/s}$ for 50° margin with lead assistance.

Step 2: Design lag compensator.

Place lag zero at $\omega_c/10 = 0.15 \text{ rad/s}$.

Lag pole at $p = 0.15/10 = 0.015 \text{ rad/s}$.

$$C_{\text{lag}}(s) = \frac{s + 0.15}{s + 0.015}$$

This provides the $10\times$ gain boost at DC.

Step 3: Design lead compensator.

After lag compensation, need to verify phase margin at new crossover and add lead if needed.

With $\beta = 10$ gain boost, the new crossover shifts higher. Recalculating:

For PM = 50° at $\omega_c = 3 \text{ rad/s}$ (approximate new crossover):

Phase needed from lead: approximately 20° .

With $\phi_m = 25^\circ$ (including margin): $\sin(25^\circ) = 0.423 = (\alpha - 1)/(\alpha + 1)$, giving $\alpha = 2.47 \approx 2.5$.

$$z_{\text{lead}} = \frac{3}{\sqrt{2.5}} = 1.90 \text{ rad/s} \quad (14.40)$$

$$p_{\text{lead}} = 3\sqrt{2.5} = 4.74 \text{ rad/s} \quad (14.41)$$

Final lead-lag compensator:

$$C(s) = K_c \cdot \frac{s + 1.90}{s + 4.74} \cdot \frac{s + 0.15}{s + 0.015}$$

where K_c is adjusted to set the crossover at the desired frequency (typically $K_c \approx 1.5$ after accounting for all gain contributions).

14.5.3 Example 14.5: Digital Implementation of Lead Compensator

Problem: Convert the continuous-time lead compensator:

$$C(s) = 2 \cdot \frac{s + 10}{s + 50}$$

to a discrete-time controller using the Tustin (bilinear) transformation with sampling period $T_s = 0.01$ s.

Solution:

The Tustin transformation substitutes:

$$s = \frac{2}{T_s} \cdot \frac{z - 1}{z + 1} = \frac{2}{0.01} \cdot \frac{z - 1}{z + 1} = 200 \cdot \frac{z - 1}{z + 1}$$

Substituting into $C(s)$:

$$C(z) = 2 \cdot \frac{200 \frac{z-1}{z+1} + 10}{200 \frac{z-1}{z+1} + 50} \quad (14.42)$$

Multiply numerator and denominator by $(z + 1)$:

$$C(z) = 2 \cdot \frac{200(z - 1) + 10(z + 1)}{200(z - 1) + 50(z + 1)} \quad (14.43)$$

$$= 2 \cdot \frac{200z - 200 + 10z + 10}{200z - 200 + 50z + 50} \quad (14.44)$$

$$= 2 \cdot \frac{210z - 190}{250z - 150} \quad (14.45)$$

$$= 2 \cdot \frac{210(z - 0.905)}{250(z - 0.6)} \quad (14.46)$$

$$= 1.68 \cdot \frac{z - 0.905}{z - 0.6} \quad (14.47)$$

Difference equation form:

$$C(z) = 1.68 \cdot \frac{1 - 0.905z^{-1}}{1 - 0.6z^{-1}}$$

Cross-multiplying:

$$Y(z)(1 - 0.6z^{-1}) = 1.68 \cdot X(z)(1 - 0.905z^{-1})$$

$$y[n] = 0.6 y[n - 1] + 1.68 x[n] - 1.52 x[n - 1]$$

Implementation:

Algorithm 29 Digital Lead Compensator Difference Equation

```

1: State Variables:  $x_{\text{prev}} \leftarrow 0, y_{\text{prev}} \leftarrow 0$ 
2:
3: function DIGITALLEADCOMPENSATOR( $x$ )
4:    $y \leftarrow 0.6 \cdot y_{\text{prev}} + 1.68 \cdot x - 1.52 \cdot x_{\text{prev}}$ 
5:    $x_{\text{prev}} \leftarrow x$ 
6:    $y_{\text{prev}} \leftarrow y$ 
7:   return  $y$ 
8: end function

```

14.5.4 Example 14.6: Phase Margin from Bode Plot

Problem: A system with lead compensator has the open-loop transfer function:

$$L(s) = \frac{20(s+2)}{s(s+1)(s+10)}$$

Determine the gain crossover frequency and phase margin.

Solution:

Step 1: Find gain crossover frequency.

We need $|L(j\omega_c)| = 1$:

$$\frac{20\sqrt{\omega_c^2 + 4}}{\omega_c\sqrt{\omega_c^2 + 1}\sqrt{\omega_c^2 + 100}} = 1$$

Squaring both sides:

$$\frac{400(\omega_c^2 + 4)}{\omega_c^2(\omega_c^2 + 1)(\omega_c^2 + 100)} = 1$$

Let $u = \omega_c^2$:

$$400(u + 4) = u(u + 1)(u + 100)$$

$$400u + 1600 = u^3 + 101u^2 + 100u$$

$$u^3 + 101u^2 - 300u - 1600 = 0$$

Solving numerically: $u \approx 4$, so $\omega_c \approx 2$ rad/s.

Step 2: Calculate phase at crossover.

$$\angle L(j2) = \angle(j2 + 2) - \angle(j2) - \angle(j2 + 1) - \angle(j2 + 10) \quad (14.48)$$

$$= \tan^{-1}(2/2) - 90 - \tan^{-1}(2/1) - \tan^{-1}(2/10) \quad (14.49)$$

$$= 45 - 90 - 63.4 - 11.3 \quad (14.50)$$

$$= -119.7 \quad (14.51)$$

Step 3: Phase margin.

$$\text{PM} = 180 + (-119.7) = \boxed{60.3}$$

14.6 Chapter Summary

Lead-lag compensation provides powerful tools for shaping the frequency response of control systems. Lead compensation adds phase in the mid-frequency range, improving phase margin and enabling faster response without sacrificing stability; the maximum phase lead is $\phi_m = \sin^{-1}((\alpha - 1)/(\alpha + 1))$, occurring at $\omega_m = \sqrt{zp}$. Lag compensation boosts low-frequency gain to reduce steady-state error while contributing minimal phase lag at the crossover frequency, with the key being to place the compensator break frequencies well below crossover. Lead-lag compensation combines both techniques when simultaneously improved phase margin and steady-state accuracy are required. These techniques have remained essential from WWII fire control to modern hard disk drives, demonstrating the enduring value of frequency-domain design methods.

14.7 Exercises

Exercise 14.1. A system has open-loop transfer function $G(s) = 50/(s(s+5)(s+10))$. Design a lead compensator to achieve a phase margin of at least 45°.

Exercise 14.2. For the compensator $C(s) = K(s+a)/(s+b)$ with $a = 3$ and $b = 15$:

1. Is this a lead or lag compensator?
2. Calculate the maximum phase contribution and the frequency at which it occurs.
3. Sketch the Bode plot (magnitude and phase).

Exercise 14.3. Design a lag compensator for $G(s) = 10/(s(s+1))$ to increase the velocity error constant from 10 to 100 s⁻¹ while maintaining at least 40° phase margin.

Exercise 14.4. Convert the lead compensator $C(s) = (s+5)/(s+25)$ to discrete time using:

1. Tustin transformation with $T_s = 0.02$ s
2. Forward Euler method with $T_s = 0.02$ s

Compare the frequency responses of the two discrete implementations.

Exercise 14.5. A position control system has plant $G(s) = K/(s(\tau s + 1))$ with $K = 10$ and $\tau = 0.1$ s. The system uses unity feedback with proportional gain K_p .

1. Find the value of K_p that gives 30° phase margin.
2. Design a lead compensator to achieve 50° phase margin while doubling the crossover frequency.
3. Calculate the improvement in settling time you would expect.

Exercise 14.6. For the lead-lag compensator:

$$C(s) = 5 \cdot \frac{s+2}{s+20} \cdot \frac{s+0.5}{s+0.05}$$

1. Identify the lead and lag stages.
2. Calculate the DC gain.
3. Calculate the high-frequency gain.
4. At what frequency does the lead stage provide maximum phase?

Exercise 14.7. Laboratory: Implement the lead compensator from Example 14.5 on an Arduino controlling a DC motor. Compare the step response with proportional-only control using the same crossover frequency. Measure:

1. Rise time

2. Percent overshoot
3. Settling time

Exercise 14.8. A hard disk drive head positioning system has transfer function:

$$G(s) = \frac{10^6}{s^2(s + 1000)}$$

Design a lead-lag compensator to achieve:

1. Zero steady-state position error to a step input (already satisfied)
2. Velocity error constant $K_v \geq 10^4 \text{ s}^{-1}$
3. Phase margin ≥ 40
4. Gain margin $\geq 10 \text{ dB}$

Exercise 14.9. Prove that for a lead compensator $C(s) = K_c(s + z)/(s + p)$ with $p > z$, the gain at the frequency of maximum phase is exactly the geometric mean of the low-frequency and high-frequency gains (in linear, not dB, scale).

Exercise 14.10. A power system stabilizer requires phase lead of 60° at the inter-area oscillation frequency of 0.5 Hz. Design a two-stage lead compensator (cascade of two lead stages) to achieve this specification while limiting the high-frequency gain increase to 20 dB.

Further Reading

For foundational understanding of frequency-domain design, the reader should consult Bode's seminal "Network Analysis and Feedback Amplifier Design" (Van Nostrand, 1945). Franklin, Powell, and Emami-Naeini's "Feedback Control of Dynamic Systems" (Pearson, 7th edition, 2015) provides excellent coverage of lead-lag design in Chapters 6 and 7. For historical context on fire control system development, Mindell's "Between Human and Machine: Feedback, Control, and Computing Before Cybernetics" (Johns Hopkins University Press, 2002) offers a fascinating account. Additional coverage of frequency response design methods can be found in Dorf and Bishop's "Modern Control Systems" (Pearson, 13th edition, 2016), particularly Chapter 10, while Ogata's "Modern Control Engineering" (Pearson, 5th edition, 2010) provides comprehensive treatment of compensator design with many examples.

Part IV

State-Space Control

Chapter 15

Controllability and State Feedback

“The concept of controllability plays an important role in the theory of control systems. It gives a complete characterization of the state-space model of a linear system from the point of view of control.”

— Rudolf E. Kalman, 1960

15.1 Historical Motivation: The Birth of State-Space Control

15.1.1 The Classical Control Dilemma

By the late 1950s, control engineering had achieved remarkable success with frequency-domain methods. The Bode plot, Nyquist criterion, and root locus technique—developed by engineers like Harold Black, Harry Nyquist, and Walter Evans—provided powerful tools for analyzing and designing single-input, single-output (SISO) feedback systems. These methods had proven their worth in applications ranging from telephone amplifiers to autopilots.

However, as control engineers tackled increasingly complex systems, fundamental limitations of the classical approach became apparent. Consider the challenge facing aerospace engineers in the late 1950s: a spacecraft requires simultaneous control of attitude (pitch, roll, yaw), translation, and potentially flexible body modes. This involves multiple inputs (reaction jets, control moment gyros) and multiple outputs (various attitude sensors). The transfer function approach, while elegant for SISO systems, became unwieldy for such multi-input, multi-output (MIMO) configurations.

More troubling was a fundamental question that classical methods struggled to answer: *Given a particular system, is it even possible to control it?* This may seem like an obvious question, but consider a system where certain internal dynamics are completely decoupled from the available inputs. Classical transfer function analysis might mask this problem entirely.

Kalman’s Revolutionary Framework (1960)

Rudolf Emil Kalman (1930–2016), a Hungarian-American mathematician and electrical engineer, revolutionized control theory with his seminal paper “On the General Theory of Control Systems,” presented at the First IFAC World Congress in Moscow in 1960. Kalman’s

key insight was to work directly with the state-space representation, capturing the internal dynamics of systems rather than just input-output relationships. Within this framework, Kalman posed and answered the fundamental question: *When can we steer a system from any initial state to any final state using available inputs?* His answer—the controllability rank condition—provided an elegant algebraic test applicable to systems of any dimension with any number of inputs.

15.1.2 Kalman’s Revolutionary Framework

Rudolf Emil Kalman (1930–2016), a Hungarian-American mathematician and electrical engineer, addressed these limitations through a revolutionary reformulation of control theory. His seminal paper, “On the General Theory of Control Systems,” presented at the First IFAC World Congress in Moscow in 1960, introduced concepts that would transform the field [1].

Kalman’s key insight was to work directly with the state-space representation:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad (15.1)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \quad (15.2)$$

where $\mathbf{x} \in \mathbb{R}^n$ is the state vector, $\mathbf{u} \in \mathbb{R}^m$ is the input vector, and $\mathbf{y} \in \mathbb{R}^p$ is the output vector. This representation directly captures the internal dynamics of the system, not just the input-output relationship.

Within this framework, Kalman posed and answered the fundamental question: *When can we steer the system from any initial state to any final state using the available inputs?* His answer—the controllability rank condition—provided an elegant algebraic test that could be applied to systems of any dimension and with any number of inputs.

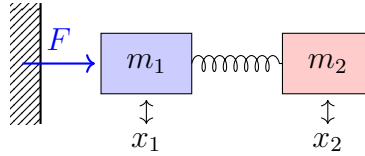
15.1.3 The Space Race Context

The timing of Kalman’s work was no coincidence. The space race, ignited by Sputnik in 1957, created an urgent demand for advanced control systems. The Apollo guidance and navigation system, under development at MIT’s Instrumentation Laboratory, faced unprecedented challenges. Spacecraft required simultaneous control of all six degrees of freedom, yet the Apollo Guidance Computer had only 74KB of memory and operated at 0.043 MHz. Control system failure meant loss of crew, making mission-critical reliability paramount. Furthermore, every bit of propellant was precious, and inefficient control could doom a mission.

Classical control methods, designed for analog implementation of SISO systems, were inadequate for these challenges. Kalman’s state-space approach, combined with his famous filtering algorithm (the Kalman filter, developed in 1960–61), provided the mathematical foundation for the Apollo guidance system and countless space missions since.

15.1.4 The Controllability Question

To appreciate why controllability is non-trivial, consider a simple example. Suppose we have two masses connected by a spring, with a force applied only to the first mass:



Can we control both masses arbitrarily? Intuition suggests yes—pushing mass 1 will eventually affect mass 2 through the spring. But what if the spring is infinitely stiff? Then the two masses move as one, and we cannot independently control their relative position. More subtly, what if the system has internal modes that the input cannot excite?

Kalman’s controllability theory provides precise answers to such questions, transforming intuition into rigorous mathematics.

15.1.5 From Theory to Practice: Ackermann’s Contribution

While Kalman established the theoretical foundation, practical implementation required additional development. Jürgen Ackermann, a German control theorist, made a crucial contribution in 1972 with his formula for computing state feedback gains [2].

The pole placement problem asks: given a controllable system, how do we choose feedback gains to place the closed-loop poles at desired locations? While the existence of such gains follows from controllability, actually computing them was not straightforward for systems of dimension greater than two.

Ackermann’s formula provides a direct, non-iterative computation:

$$\mathbf{K} = [0 \quad 0 \quad \cdots \quad 0 \quad 1] \mathcal{C}^{-1} \phi(\mathbf{A}) \quad (15.3)$$

where $\mathcal{C}$ is the controllability matrix and $\phi(\mathbf{A})$ is the desired characteristic polynomial evaluated at the system matrix. This elegant result made pole placement practical and accessible to practicing engineers.

15.1.6 The Modern Legacy

Kalman’s controllability theory, along with the dual concept of observability, fundamentally changed how we think about control systems. The “Kalman decomposition” shows that any linear system can be decomposed into four parts: the controllable and observable portion (which appears in the transfer function), the controllable but unobservable portion, the uncontrollable but observable portion, and the portion that is neither controllable nor observable. This decomposition reveals that transfer function analysis only captures the first component—potentially missing critical system behavior hidden in the uncontrollable or unobservable subspaces.

Today, controllability analysis is standard practice in aerospace, automotive, robotics, and process control. The concepts have been extended to nonlinear systems (Lie algebraic conditions), distributed parameter systems (PDEs), and networked control systems. Kalman’s insight that we must first ask “can we control this system?” before asking “how should we control it?” remains as relevant as ever.

15.2 Modern Applications

The concepts of controllability and state feedback find application across virtually every domain of modern engineering. We highlight several areas where these techniques are essential.

15.2.1 Spacecraft Attitude Control

Modern spacecraft employ sophisticated attitude control systems (ACS) that directly implement state feedback concepts. Consider a satellite with reaction wheel actuators. The state vector consists of angular position and velocity about three axes $(\omega_x, \omega_y, \omega_z, \phi, \theta, \psi)$, while the inputs are torques from reaction wheels or control moment gyros. The control objective is to maintain pointing accuracy while minimizing fuel and power consumption. The controllability matrix determines whether the available actuators can achieve arbitrary attitude changes. Redundant actuator configurations must be analyzed to ensure controllability is maintained even after actuator failures—a critical consideration for long-duration missions.

15.2.2 Multi-Axis Robot Control

Industrial and surgical robots require coordinated control of multiple joints. A 6-DOF manipulator has a 12-dimensional state space encompassing the position and velocity of each joint. State feedback enables trajectory tracking with prescribed dynamics, while controllability analysis reveals configurations (singularities) where certain motions become impossible. Pole placement allows tuning of response characteristics for different tasks, ranging from fast pick-and-place operations to precise assembly work. Modern surgical robots like the da Vinci system rely on state feedback for tremor filtering and motion scaling, with pole placement determining the filtering characteristics.

15.2.3 Active Suspension Systems

Automotive active suspension represents a successful commercial application. In the quarter-car model, the sprung mass, unsprung mass, and their velocities form a 4-dimensional state, while a full vehicle model requires 14 or more states including roll, pitch, and heave modes. Hydraulic or electromagnetic force generators at each corner serve as actuators. The fundamental design trade-off involves balancing ride comfort (low bandwidth) against handling performance (high bandwidth). State feedback allows independent tuning of body modes (ride quality) and wheel modes (tire contact), with pole placement providing direct control over this compromise.

15.2.4 Process Control

Chemical processes with multiple actuators benefit from state-space methods. Applications include distillation columns with multiple feed points and heat inputs, reactor systems with temperature, pressure, and composition control, and paper machines with moisture and thickness control across the web. In these applications, controllability analysis reveals whether the available actuators can regulate all process variables. Systems lacking full controllability may require additional actuators or accept limitations on achievable performance.

15.3 Mathematical Foundation

We now develop the mathematical theory of controllability and state feedback with full rigor. Throughout this section, we consider the linear time-invariant (LTI) system:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad (15.4)$$

where $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{B} \in \mathbb{R}^{n \times m}$, $\mathbf{x}(t) \in \mathbb{R}^n$, and $\mathbf{u}(t) \in \mathbb{R}^m$.

15.3.1 Definition of Controllability

Definition 15.1 (State Controllability). The system $(\mathbf{A}, \mathbf{B})$ is said to be **controllable** (or completely state controllable) if for any initial state $\mathbf{x}_0 \in \mathbb{R}^n$ and any final state $\mathbf{x}_f \in \mathbb{R}^n$, there exists a finite time $t_f > 0$ and an input $\mathbf{u}(t)$ defined on $[0, t_f]$ such that the state trajectory satisfies $\mathbf{x}(0) = \mathbf{x}_0$ and $\mathbf{x}(t_f) = \mathbf{x}_f$.

This definition captures the intuitive notion that we can “steer” the system from any state to any other state. Several important points deserve attention. First, the definition requires reaching *any* final state from *any* initial state. Second, the time t_f may depend on the states involved. Third, no constraint is placed on the input magnitude, though practical considerations will be addressed later. Finally, intermediate states along the trajectory are not specified.

15.3.2 The Controllability Matrix

The solution to the state equation (??) is given by:

$$\mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{x}_0 + \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B}\mathbf{u}(\tau) d\tau \quad (15.5)$$

For the system to be controllable, we need the second term to be able to produce any vector in $\mathbb{R}^n$. This leads us to examine what vectors are reachable from the origin.

Definition 15.2 (Reachable Set). The **reachable set** $\mathcal{R}(t_f)$ at time t_f is the set of all states that can be reached from the origin:

$$\mathcal{R}(t_f) = \left\{ \int_0^{t_f} e^{\mathbf{A}(t_f-\tau)} \mathbf{B}\mathbf{u}(\tau) d\tau : \mathbf{u}(\cdot) \in L^2([0, t_f], \mathbb{R}^m) \right\} \quad (15.6)$$

Lemma 15.1. $\mathcal{R}(t_f) = \text{span}\{e^{\mathbf{A}t} \mathbf{B}\mathbf{u} : t \in [0, t_f], \mathbf{u} \in \mathbb{R}^m\}$.

To characterize this span algebraically, we use the Cayley-Hamilton theorem.

Theorem 15.1 (Cayley-Hamilton). *Every square matrix satisfies its own characteristic equation. If $p(\lambda) = \det(\lambda \mathbf{I} - \mathbf{A}) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$ is the characteristic polynomial of $\mathbf{A}$, then:*

$$p(\mathbf{A}) = \mathbf{A}^n + a_{n-1}\mathbf{A}^{n-1} + \cdots + a_1\mathbf{A} + a_0\mathbf{I} = \mathbf{0} \quad (15.7)$$

This theorem implies that $\mathbf{A}^n$ (and all higher powers) can be expressed as linear combinations of $\mathbf{I}, \mathbf{A}, \mathbf{A}^2, \dots, \mathbf{A}^{n-1}$.

Corollary 15.1. *The matrix exponential $e^{\mathbf{A}t}$ can be written as:*

$$e^{\mathbf{A}t} = \sum_{k=0}^{n-1} \alpha_k(t) \mathbf{A}^k \quad (15.8)$$

for appropriate scalar functions $\alpha_k(t)$.

This leads directly to the controllability matrix.

Definition 15.3 (Controllability Matrix). The **controllability matrix** of the pair $(\mathbf{A}, \mathbf{B})$ is:

$$\mathcal{C} = [\mathbf{B} \quad \mathbf{A}\mathbf{B} \quad \mathbf{A}^2\mathbf{B} \quad \cdots \quad \mathbf{A}^{n-1}\mathbf{B}] \quad (15.9)$$

This is an $n \times nm$ matrix formed by horizontally concatenating n blocks, each of dimension $n \times m$.

15.3.3 The Controllability Rank Condition

We now prove the fundamental result connecting controllability to the rank of $\mathcal{C}$.

Theorem 15.2 (Kalman Controllability Rank Condition). *The system $(\mathbf{A}, \mathbf{B})$ is controllable if and only if:*

$$\text{rank}(\mathcal{C}) = \text{rank} [\mathbf{B} \quad \mathbf{A}\mathbf{B} \quad \mathbf{A}^2\mathbf{B} \quad \cdots \quad \mathbf{A}^{n-1}\mathbf{B}] = n \quad (15.10)$$

Proof. We prove both directions.

(Sufficiency: $\text{rank}(\mathcal{C}) = n \Rightarrow \text{controllable}$)

Assume $\text{rank}(\mathcal{C}) = n$. We show that any state $\mathbf{x}_f$ is reachable from the origin in any time $t_f > 0$.

The reachable set from the origin is:

$$\mathcal{R}(t_f) = \left\{ \int_0^{t_f} e^{\mathbf{A}(t_f-\tau)} \mathbf{B} \mathbf{u}(\tau) d\tau : \mathbf{u}(\cdot) \right\} \quad (15.11)$$

Using the Cayley-Hamilton theorem, we can write:

$$e^{\mathbf{A}(t_f-\tau)} = \sum_{k=0}^{n-1} \alpha_k(t_f - \tau) \mathbf{A}^k \quad (15.12)$$

Therefore:

$$\int_0^{t_f} e^{\mathbf{A}(t_f-\tau)} \mathbf{B} \mathbf{u}(\tau) d\tau = \sum_{k=0}^{n-1} \mathbf{A}^k \mathbf{B} \int_0^{t_f} \alpha_k(t_f - \tau) \mathbf{u}(\tau) d\tau \quad (15.13)$$

$$= \sum_{k=0}^{n-1} \mathbf{A}^k \mathbf{B} \mathbf{v}_k \quad (15.14)$$

where $\mathbf{v}_k = \int_0^{t_f} \alpha_k(t_f - \tau) \mathbf{u}(\tau) d\tau \in \mathbb{R}^m$.

This shows that any reachable state lies in:

$$\mathcal{R}(t_f) \subseteq \text{range}(\mathcal{C}) = \text{span}\{\mathbf{B}, \mathbf{AB}, \dots, \mathbf{A}^{n-1}\mathbf{B}\} \quad (15.15)$$

Conversely, for any choice of vectors $\mathbf{v}_0, \dots, \mathbf{v}_{n-1} \in \mathbb{R}^n$, we can find a suitable input $\mathbf{u}(\tau)$ that produces these integrals (the functions $\alpha_k(t_f - \tau)$ form a linearly independent set for $t_f > 0$). Therefore:

$$\mathcal{R}(t_f) = \text{range}(\mathcal{C}) \quad (15.16)$$

If $\text{rank}(\mathcal{C}) = n$, then $\text{range}(\mathcal{C}) = \mathbb{R}^n$, so any state is reachable from the origin.

To show controllability (reaching any $\mathbf{x}_f$ from any $\mathbf{x}_0$), note that if $\mathbf{x}_1(t)$ reaches $\mathbf{x}_f$ from $\mathbf{0}$ using input $\mathbf{u}_1(t)$, and $\mathbf{x}_2(t)$ reaches some state from $-\mathbf{x}_0$ using input $\mathbf{u}_2(t)$, then by linearity we can construct an input that drives $\mathbf{x}_0$ to $\mathbf{x}_f$.

(Necessity: controllable $\Rightarrow \text{rank}(\mathcal{C}) = n$)

We prove the contrapositive: if $\text{rank}(\mathcal{C}) < n$, then the system is not controllable.

Suppose $\text{rank}(\mathcal{C}) < n$. Then there exists a nonzero vector $\mathbf{q} \in \mathbb{R}^n$ such that $\mathbf{q}^\top \mathcal{C} = \mathbf{0}$. This means:

$$\mathbf{q}^\top \mathbf{A}^k \mathbf{B} = \mathbf{0} \quad \text{for } k = 0, 1, \dots, n-1 \quad (15.17)$$

By Cayley-Hamilton, $\mathbf{A}^k$ for $k \geq n$ is a linear combination of lower powers, so:

$$\mathbf{q}^\top \mathbf{A}^k \mathbf{B} = \mathbf{0} \quad \text{for all } k \geq 0 \quad (15.18)$$

Now consider the state reachable from the origin:

$$\mathbf{x}(t_f) = \int_0^{t_f} e^{\mathbf{A}(t_f - \tau)} \mathbf{B} \mathbf{u}(\tau) d\tau \quad (15.19)$$

Taking the inner product with $\mathbf{q}$:

$$\mathbf{q}^\top \mathbf{x}(t_f) = \int_0^{t_f} \mathbf{q}^\top e^{\mathbf{A}(t_f - \tau)} \mathbf{B} \mathbf{u}(\tau) d\tau \quad (15.20)$$

$$= \int_0^{t_f} \sum_{k=0}^{\infty} \frac{(t_f - \tau)^k}{k!} \mathbf{q}^\top \mathbf{A}^k \mathbf{B} \mathbf{u}(\tau) d\tau \quad (15.21)$$

$$= 0 \quad (15.22)$$

This shows that any reachable state $\mathbf{x}(t_f)$ satisfies $\mathbf{q}^\top \mathbf{x}(t_f) = 0$. Hence, any state $\mathbf{x}_f$ with $\mathbf{q}^\top \mathbf{x}_f \neq 0$ is unreachable from the origin, proving the system is not controllable. $\square$

15.3.4 The Controllability Gramian

An alternative characterization uses the controllability Gramian.

Definition 15.4 (Controllability Gramian). The **controllability Gramian** at time t_f is:

$$\mathbf{W}_c(t_f) = \int_0^{t_f} e^{\mathbf{A}\tau} \mathbf{B} \mathbf{B}^\top e^{\mathbf{A}^\top \tau} d\tau \quad (15.23)$$

Theorem 15.3. *The system $(\mathbf{A}, \mathbf{B})$ is controllable if and only if $\mathbf{W}_c(t_f)$ is positive definite for some (equivalently, any) $t_f > 0$.*

The Gramian provides additional insight: if we want to drive the system from state $\mathbf{x}_0$ to state $\mathbf{x}_f$ in time t_f , the minimum energy input is:

$$\mathbf{u}^*(t) = \mathbf{B}^\top e^{\mathbf{A}^\top(t_f-t)} \mathbf{W}_c^{-1}(t_f) (\mathbf{x}_f - e^{\mathbf{A}t_f} \mathbf{x}_0) \quad (15.24)$$

with corresponding energy:

$$\int_0^{t_f} \|\mathbf{u}^*(t)\|^2 dt = (\mathbf{x}_f - e^{\mathbf{A}t_f} \mathbf{x}_0)^\top \mathbf{W}_c^{-1}(t_f) (\mathbf{x}_f - e^{\mathbf{A}t_f} \mathbf{x}_0) \quad (15.25)$$

15.3.5 State Feedback Control

Having established when a system is controllable, we now address how to control it. State feedback is the most direct approach.

Definition 15.5 (State Feedback). **State feedback** is a control law of the form:

$$\mathbf{u}(t) = -\mathbf{K}\mathbf{x}(t) + \mathbf{r}(t) \quad (15.26)$$

where $\mathbf{K} \in \mathbb{R}^{m \times n}$ is the feedback gain matrix and $\mathbf{r}(t)$ is a reference input.

Substituting into the state equation (??):

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}(-\mathbf{K}\mathbf{x} + \mathbf{r}) \quad (15.27)$$

$$= (\mathbf{A} - \mathbf{BK})\mathbf{x} + \mathbf{B}\mathbf{r} \quad (15.28)$$

The closed-loop system has dynamics matrix $\mathbf{A}_{cl} = \mathbf{A} - \mathbf{BK}$. The key insight is that proper choice of $\mathbf{K}$ can place the eigenvalues of $\mathbf{A}_{cl}$ at any desired locations.

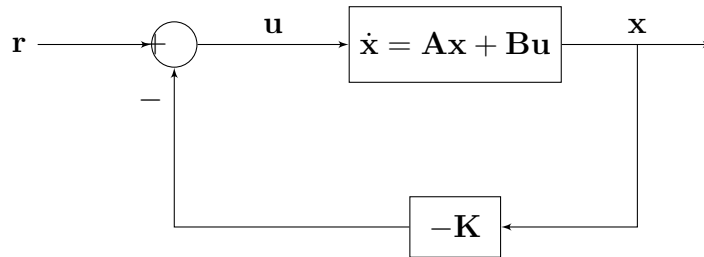


Figure 15.1: State feedback control architecture. The gain matrix $\mathbf{K}$ feeds back the full state vector to modify the input, resulting in closed-loop dynamics $\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{BK})\mathbf{x} + \mathbf{B}\mathbf{r}$.

15.3.6 Pole Placement Theorem

Theorem 15.4 (Pole Placement). *If the system $(\mathbf{A}, \mathbf{B})$ is controllable, then for any set of n complex numbers $\{p_1, p_2, \dots, p_n\}$ (closed under complex conjugation), there exists a matrix $\mathbf{K} \in \mathbb{R}^{m \times n}$ such that:*

$$\text{eig}(\mathbf{A} - \mathbf{BK}) = \{p_1, p_2, \dots, p_n\} \quad (15.29)$$

Proof sketch for SISO case. For a single-input system ($m = 1$), we can transform to controller canonical form. In this form:

$$\mathbf{A}_c = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-2} & -a_{n-1} \end{bmatrix}, \quad \mathbf{B}_c = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \quad (15.30)$$

where $a_0, \dots, a_{n-1}$ are the coefficients of the characteristic polynomial.

With $\mathbf{K}_c = [k_0 \ k_1 \ \cdots \ k_{n-1}]$:

$$\mathbf{A}_c - \mathbf{B}_c \mathbf{K}_c = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ -a_0 - k_0 & -a_1 - k_1 & \cdots & -a_{n-2} - k_{n-2} & -a_{n-1} - k_{n-1} \end{bmatrix} \quad (15.31)$$

The characteristic polynomial of $\mathbf{A}_c - \mathbf{B}_c \mathbf{K}_c$ is:

$$\det(s\mathbf{I} - \mathbf{A}_c + \mathbf{B}_c \mathbf{K}_c) = s^n + (a_{n-1} + k_{n-1})s^{n-1} + \cdots + (a_0 + k_0) \quad (15.32)$$

For any desired polynomial $\phi(s) = s^n + \alpha_{n-1}s^{n-1} + \cdots + \alpha_0$, we simply choose:

$$k_i = \alpha_i - a_i, \quad i = 0, 1, \dots, n-1 \quad (15.33)$$

The transformation back to original coordinates yields the required $\mathbf{K}$. $\square$

15.3.7 Ackermann's Formula

For single-input systems, Ackermann's formula provides a direct computation of the feedback gain.

Theorem 15.5 (Ackermann's Formula). *Let $(\mathbf{A}, \mathbf{B})$ be a controllable single-input system ($\mathbf{B} \in \mathbb{R}^{n \times 1}$). Let:*

$$\phi(s) = s^n + \alpha_{n-1}s^{n-1} + \cdots + \alpha_1s + \alpha_0 \quad (15.34)$$

be the desired characteristic polynomial (with roots at the desired pole locations). Then the feedback gain $\mathbf{K}$ that places the closed-loop poles at these locations is:

$$\mathbf{K} = [0 \ 0 \ \cdots \ 0 \ 1] \mathcal{C}^{-1} \phi(\mathbf{A}) \quad (15.35)$$

where $\mathcal{C} = [\mathbf{B} \ \mathbf{AB} \ \cdots \ \mathbf{A}^{n-1}\mathbf{B}]$ and:

$$\phi(\mathbf{A}) = \mathbf{A}^n + \alpha_{n-1}\mathbf{A}^{n-1} + \cdots + \alpha_1\mathbf{A} + \alpha_0\mathbf{I} \quad (15.36)$$

Proof. The proof uses the transformation to controller canonical form. Let $\mathbf{T}$ be the transformation such that $\mathbf{A}_c = \mathbf{T}^{-1}\mathbf{A}\mathbf{T}$ and $\mathbf{B}_c = \mathbf{T}^{-1}\mathbf{B}$ are in controller canonical form.

It can be shown that:

$$\mathbf{T}^{-1} = \mathcal{C}_c^{-1}\mathcal{C}^{-1} \quad (15.37)$$

where $\mathcal{C}_c$ is the controllability matrix in canonical form.

In the canonical coordinates, the required gain is:

$$\mathbf{K}_c = [\alpha_0 - a_0 \quad \alpha_1 - a_1 \quad \cdots \quad \alpha_{n-1} - a_{n-1}] \quad (15.38)$$

Transforming back and using the Cayley-Hamilton theorem yields Ackermann's formula. $\square$

Remark 15.1. For multi-input systems ($m > 1$), the feedback gain is not unique—there are infinitely many $\mathbf{K}$ matrices that achieve the same pole placement. Additional criteria (such as minimizing control effort) can be used to select among them.

15.3.8 Bass-Gura Formula

An alternative to Ackermann's formula, the Bass-Gura formula directly computes the gains by comparing coefficients.

Theorem 15.6 (Bass-Gura Formula). *For a controllable SISO system in controller canonical form with characteristic polynomial:*

$$p(s) = s^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0 \quad (15.39)$$

and desired characteristic polynomial $\phi(s) = s^n + \alpha_{n-1}s^{n-1} + \cdots + \alpha_0$, the feedback gain is:

$$\mathbf{K} = (\boldsymbol{\alpha} - \mathbf{a})^\top \mathbf{T}^{-1} \quad (15.40)$$

where $\boldsymbol{\alpha} = [\alpha_0 \quad \alpha_1 \quad \cdots \quad \alpha_{n-1}]^\top$, $\mathbf{a} = [a_0 \quad a_1 \quad \cdots \quad a_{n-1}]^\top$, and $\mathbf{T}$ is the transformation to controller canonical form.

15.3.9 Practical Considerations

While controllability and pole placement provide powerful theoretical tools, practical implementation requires careful consideration of several factors.

Aggressive Pole Placement and Control Effort

Moving poles farther into the left half-plane produces faster response but requires higher control effort. Consider a second-order system with closed-loop poles at $s = -\omega_n\zeta \pm j\omega_n\sqrt{1-\zeta^2}$.

The settling time is approximately $T_s \approx \frac{4}{\zeta\omega_n}$. To reduce settling time by a factor of 10, we must move poles 10 times farther from the origin. This typically increases peak control signal magnitude, control signal rate of change, sensitivity to modeling errors, and energy consumption.

Actuator Saturation

Real actuators have limited range. If pole placement demands control signals exceeding actuator limits, the actual closed-loop behavior will differ from the designed linear response. Anti-windup strategies and careful pole selection are essential.

Robustness Considerations

State feedback designed by pole placement does not inherently guarantee robustness to model uncertainty. The LQR (Linear Quadratic Regulator) approach, covered in Chapter 17, provides systematic treatment of the trade-off between performance and robustness.

15.4 Practical Experiment: Inverted Pendulum State Feedback Control

The inverted pendulum is a classic demonstration of controllability concepts. The system is naturally unstable—small perturbations cause the pendulum to fall. Yet, with appropriate state feedback, we can stabilize it at the upright position. This experiment demonstrates both the controllability analysis and practical state feedback implementation.

15.4.1 System Description

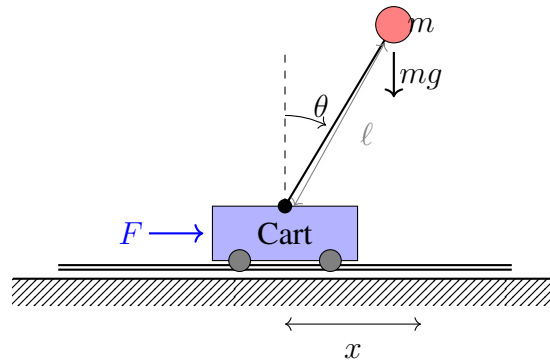


Figure 15.2: Inverted pendulum on a cart. The cart of mass M moves along a track, driven by force F . A pendulum of length ℓ and point mass m is pivoted on the cart. The angle θ is measured from vertical.

Equations of Motion

Using Lagrangian mechanics, the equations of motion are:

$$(M + m)\ddot{x} + m\ell\ddot{\theta}\cos\theta - m\ell\dot{\theta}^2\sin\theta = F \quad (15.41)$$

$$m\ell^2\ddot{\theta} + m\ell\ddot{x}\cos\theta - mg\ell\sin\theta = 0 \quad (15.42)$$

Linearizing about the upright position ($\theta = 0, \dot{\theta} = 0$):

$$(M + m)\ddot{x} + m\ell\ddot{\theta} = F \quad (15.43)$$

$$m\ell\ddot{\theta} + m\ddot{x} - mg\theta = 0 \quad (15.44)$$

Defining state vector $\mathbf{x} = [x \ \dot{x} \ \theta \ \dot{\theta}]^\top$ and input $u = F$:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{mg}{M} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{(M+m)g}{M\ell} & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ -\frac{1}{M\ell} \end{bmatrix} \quad (15.45)$$

Controllability Verification

With typical parameters ($M = 0.5$ kg, $m = 0.2$ kg, $\ell = 0.3$ m, $g = 9.81$ m/s²):

$$\mathcal{C} = [\mathbf{B} \ \mathbf{AB} \ \mathbf{A}^2\mathbf{B} \ \mathbf{A}^3\mathbf{B}] \quad (15.46)$$

Computing:

$$\mathcal{C} = \begin{bmatrix} 0 & 2 & 0 & -26.16 \\ 2 & 0 & -7.85 & 0 \\ 0 & -6.67 & 0 & 130.8 \\ -6.67 & 0 & 45.71 & 0 \end{bmatrix} \quad (15.47)$$

The determinant $\det(\mathcal{C}) \neq 0$, confirming $\text{rank}(\mathcal{C}) = 4$. The system is controllable.

15.4.2 Hardware Implementation

Components

The hardware implementation consists of several key components, summarized in Table 15.1.

Table 15.1: Hardware components for inverted pendulum experiment

Component	Specification
Cart	3D printed PLA, mounted on linear bearings
Track	500mm aluminum extrusion with linear rail
Motor	NEMA 17 stepper or DC gear motor with encoder
Drive	Belt or lead screw for cart positioning
Pendulum	300mm aluminum or carbon fiber rod
Sensors	Rotary encoder at pivot (1000+ PPR), motor encoder for position
Controller	Arduino Mega or Teensy 4.0
Motor driver	L298N or TB6612FNG

3D Printing Specifications

For the 3D printed components, the cart body should be printed with 100% infill for rigidity, while bearing mounts require tight tolerances of $\pm 0.1\text{mm}$. The pivot housing needs a precision fit for the encoder shaft. Recommended materials include PLA or PETG, with ABS preferred for higher temperature environments.

15.4.3 Controller Implementation

15.4.4 Experimental Results

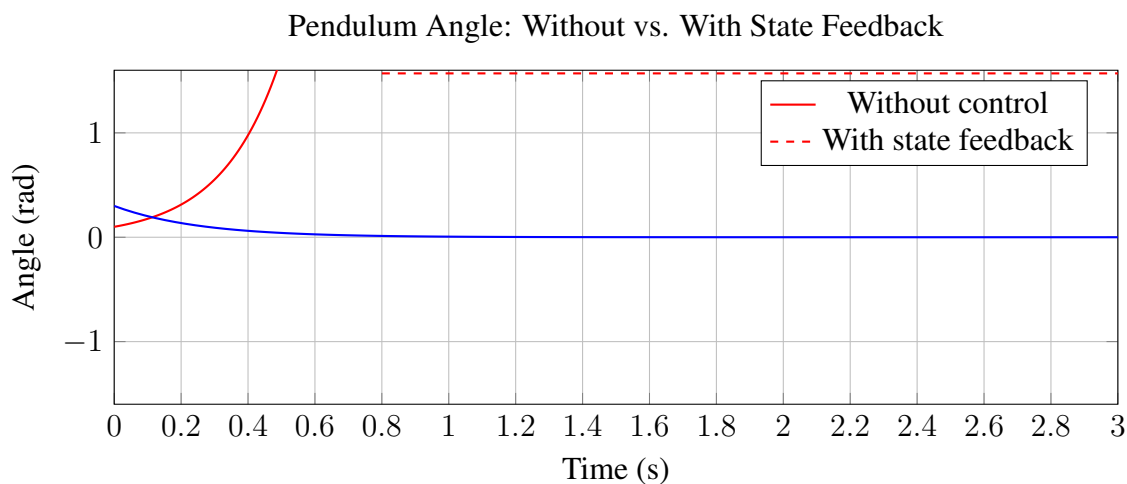


Figure 15.3: Comparison of pendulum response. Without control (red), the pendulum falls within 0.8 seconds. With state feedback (blue), an initial disturbance of 0.3 rad is stabilized within 1 second.

Effect of Pole Placement

The choice of closed-loop pole locations significantly affects system behavior:

Table 15.2: Effect of pole placement on inverted pendulum performance

Poles	Settling Time	Peak Control	Notes
$-2 \pm 2j, -3, -4$	2.0 s	5 V	Conservative, smooth
$-5 \pm 5j, -6, -8$	0.8 s	15 V	Moderate, good balance
$-10 \pm 10j, -12, -15$	0.35 s	45 V	Aggressive, near saturation
$-20 \pm 20j, -25, -30$	0.15 s	>100 V	Impractical, saturates

Algorithm 30 Inverted Pendulum State Feedback Controller

```

1: Parameters:
2:    $K_1 \leftarrow -31.62$                                 ▷ Position gain
3:    $K_2 \leftarrow -25.89$                                 ▷ Velocity gain
4:    $K_3 \leftarrow 78.46$                                   ▷ Angle gain
5:    $K_4 \leftarrow 14.52$                                   ▷ Angular velocity gain
6:    $\Delta t \leftarrow 0.005$                              ▷ Control loop period (200 Hz)
7:    $\alpha \leftarrow 0.8$                                  ▷ Low-pass filter coefficient
8:
9: Initialize:
10:  Configure encoder interrupts for cart and pendulum
11:  Configure motor driver outputs
12:  Wait for user to position pendulum upright
13:  Zero all encoder counts and state variables
14:
15: Main Control Loop:
16: while system is running do
17:   if elapsed time  $\geq \Delta t$  then
18:     Reset timer
19:
20:                                     ▷ Read encoder values with interrupt protection
21:      $x \leftarrow \text{cart\_encoder} \times \text{ENCODER\_TO\_M}$ 
22:      $\theta \leftarrow \text{pend\_encoder} \times \text{ENCODER\_TO\_RAD}$ 
23:
24:                                     ▷ Compute filtered velocity estimates
25:      $\dot{x} \leftarrow \alpha \cdot \dot{x}_{\text{prev}} + (1 - \alpha) \cdot (x - x_{\text{prev}}) / \Delta t$ 
26:      $\dot{\theta} \leftarrow \alpha \cdot \dot{\theta}_{\text{prev}} + (1 - \alpha) \cdot (\theta - \theta_{\text{prev}}) / \Delta t$ 
27:      $x_{\text{prev}} \leftarrow x$ ;    $\theta_{\text{prev}} \leftarrow \theta$ 
28:
29:                                     ▷ State feedback control law
30:      $u \leftarrow -K_1 x - K_2 \dot{x} - K_3 \theta - K_4 \dot{\theta}$ 
31:
32:                                     ▷ Saturate control signal
33:      $u \leftarrow \text{clamp}(u, -12.0, 12.0)$ 
34:
35:                                     ▷ Apply control if pendulum near upright
36:     if  $|\theta| < 0.5$  rad then
37:       Apply voltage  $u$  to motor
38:     else
39:       Set motor voltage to 0
40:       Signal: “Pendulum fallen – reset required”
41:     end if
42:
43:     Log data:  $x, \theta, u$ 
44:   end if
45: end while

```

15.4.5 Extensions and Variations

Several extensions can enhance the educational value of the inverted pendulum experiment. Swing-up control uses energy-based control to swing the pendulum from its hanging position to upright, then switches to state feedback for stabilization. The double pendulum adds a second link, requiring an 8-state model and demonstrating more complex controllability analysis. The rotary pendulum, also known as the Furuta pendulum, places the pendulum on a rotating arm with different dynamics but the same underlying controllability concepts. Finally, observer implementation replaces direct angle measurement with a state observer, as covered in Chapter 16.

15.5 Example Problems

Controllability of a Second-Order System

Consider the system:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (15.48)$$

Determine if the system is controllable.

Solution:

Step 1: Compute the controllability matrix.

$$\mathcal{C} = [\mathbf{B} \quad \mathbf{AB}] \quad (15.49)$$

First, compute $\mathbf{AB}$:

$$\mathbf{AB} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \end{bmatrix} \quad (15.50)$$

Therefore:

$$\mathcal{C} = \begin{bmatrix} 0 & 1 \\ 1 & -3 \end{bmatrix} \quad (15.51)$$

Step 2: Check the rank.

$$\det(\mathcal{C}) = (0)(-3) - (1)(1) = -1 \neq 0 \quad (15.52)$$

Since $\det(\mathcal{C}) \neq 0$, we have $\text{rank}(\mathcal{C}) = 2 = n$.

Conclusion: The system is controllable. $\square$

Uncontrollable System

Consider the system:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (15.53)$$

Determine if the system is controllable. If not, identify the uncontrollable mode.

Solution:

Step 1: Compute the controllability matrix.

$$\mathbf{AB} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (15.54)$$

$$\mathcal{C} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad (15.55)$$

Step 2: Check the rank.

$$\det(\mathcal{C}) = (1)(0) - (1)(0) = 0 \quad (15.56)$$

The matrix has $\text{rank}(\mathcal{C}) = 1 < 2 = n$.

Step 3: Identify uncontrollable mode.

The range of $\mathcal{C}$ is spanned by $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$. The orthogonal complement is spanned by $\mathbf{q} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Check: $\mathbf{q}^\top \mathbf{B} = 0$ and $\mathbf{q}^\top \mathbf{AB} = 0$. ✓

The uncontrollable subspace is $\text{span}\left\{\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$, which corresponds to the eigenvalue $\lambda = 2$.

Physical interpretation: The input only affects the first state. The second state evolves independently according to $\dot{x}_2 = 2x_2$ and cannot be influenced by any control action.

Conclusion: The system is not controllable. The mode associated with $\lambda = 2$ is uncontrollable. $\square$

Example 15.1 (Third-Order System Controllability). Consider a triple integrator with input applied to the first state:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad (15.57)$$

Is this system controllable?

Solution:

Compute the controllability matrix:

$$\mathbf{AB} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (15.58)$$

$$\mathbf{A}^2\mathbf{B} = \mathbf{A}(\mathbf{AB}) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (15.59)$$

Therefore:

$$\mathcal{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (15.60)$$

Since $\text{rank}(\mathcal{C}) = 1 \neq 3$, the system is **not controllable**.

Physical insight: The input enters at x_1 , but the chain $x_1 \rightarrow x_2 \rightarrow x_3$ goes the wrong way! We can directly affect x_1 but have no way to control x_2 or x_3 since they are “upstream” of the input.

Note: If instead $\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ (input at the “bottom”), the system becomes controllable. $\square$

Example 15.2 (Pole Placement Using Ackermann’s Formula). For the controllable system in Example 21.4:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (15.61)$$

Design a state feedback controller to place the closed-loop poles at $s = -5 \pm 3j$.

Solution:

Step 1: Find the desired characteristic polynomial.

With poles at $s = -5 + 3j$ and $s = -5 - 3j$:

$$\phi(s) = (s + 5 - 3j)(s + 5 + 3j) \quad (15.62)$$

$$= (s + 5)^2 + 9 \quad (15.63)$$

$$= s^2 + 10s + 34 \quad (15.64)$$

Step 2: Compute $\phi(\mathbf{A})$.

$$\phi(\mathbf{A}) = \mathbf{A}^2 + 10\mathbf{A} + 34\mathbf{I} \quad (15.65)$$

First, compute $\mathbf{A}^2$:

$$\mathbf{A}^2 = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} -2 & -3 \\ 6 & 7 \end{bmatrix} \quad (15.66)$$

Then:

$$\phi(\mathbf{A}) = \begin{bmatrix} -2 & -3 \\ 6 & 7 \end{bmatrix} + 10 \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} + 34 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (15.67)$$

$$= \begin{bmatrix} -2 + 0 + 34 & -3 + 10 + 0 \\ 6 - 20 + 0 & 7 - 30 + 34 \end{bmatrix} \quad (15.68)$$

$$= \begin{bmatrix} 32 & 7 \\ -14 & 11 \end{bmatrix} \quad (15.69)$$

Step 3: Compute $\mathcal{C}^{-1}$.

From Example 21.4, $\mathcal{C} = \begin{bmatrix} 0 & 1 \\ 1 & -3 \end{bmatrix}$.

$$\mathcal{C}^{-1} = \frac{1}{-1} \begin{bmatrix} -3 & -1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix} \quad (15.70)$$

Step 4: Apply Ackermann’s formula.

$$\mathbf{K} = [0 \quad 1] \mathcal{C}^{-1} \phi(\mathbf{A}) \quad (15.71)$$

$$= [0 \quad 1] \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 32 & 7 \\ -14 & 11 \end{bmatrix} \quad (15.72)$$

$$= [1 \quad 0] \begin{bmatrix} 32 & 7 \\ -14 & 11 \end{bmatrix} \quad (15.73)$$

$$= [32 \quad 7] \quad (15.74)$$

Step 5: Verify.

The closed-loop matrix is:

$$\mathbf{A} - \mathbf{BK} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 32 & 7 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -34 & -10 \end{bmatrix} \quad (15.75)$$

Check eigenvalues:

$$\det(s\mathbf{I} - (\mathbf{A} - \mathbf{BK})) = s^2 + 10s + 34 = (s + 5)^2 + 9 \checkmark \quad (15.76)$$

Result: $\mathbf{K} = \begin{bmatrix} 32 & 7 \end{bmatrix}$ □

Example 15.3 (Double Integrator State Feedback). The double integrator is one of the simplest and most important systems:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (15.77)$$

This models a unit mass with force input: $\ddot{x} = u$.

(a) **Verify controllability.**

(b) **Design state feedback for critically damped response with time constant $\tau = 0.5$ s.**

Solution:

(a) Controllability matrix:

$$\mathcal{C} = [\mathbf{B} \quad \mathbf{AB}] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad (15.78)$$

$\det(\mathcal{C}) = -1 \neq 0$, so $\text{rank}(\mathcal{C}) = 2$. The system is controllable.

(b) For critically damped response with $\tau = 0.5$ s, we need a double pole at $s = -1/\tau = -2$.

Desired characteristic polynomial:

$$\phi(s) = (s + 2)^2 = s^2 + 4s + 4 \quad (15.79)$$

Open-loop characteristic polynomial:

$$p(s) = \det(s\mathbf{I} - \mathbf{A}) = s^2 \quad (15.80)$$

Using Bass-Gura approach (simpler for this case):

With $\mathbf{K} = \begin{bmatrix} k_1 & k_2 \end{bmatrix}$:

$$\mathbf{A} - \mathbf{BK} = \begin{bmatrix} 0 & 1 \\ -k_1 & -k_2 \end{bmatrix} \quad (15.81)$$

Characteristic polynomial:

$$\det(s\mathbf{I} - \mathbf{A} + \mathbf{BK}) = s^2 + k_2s + k_1 \quad (15.82)$$

Matching coefficients with $s^2 + 4s + 4$:

$$k_1 = 4, \quad k_2 = 4 \quad (15.83)$$

Result: $\mathbf{K} = \begin{bmatrix} 4 & 4 \end{bmatrix}$

The control law $u = -4x - 4\dot{x}$ produces critically damped motion with $\tau = 0.5$ s. □

Example 15.4 (Control Effort vs. Pole Location). For the double integrator of Example 15.3, analyze how control effort varies with pole placement.

Consider regulating an initial condition $x(0) = 1$, $\dot{x}(0) = 0$ to the origin.

Solution:

For a double pole at $s = -\omega$, we have $\mathbf{K} = [\omega^2 \quad 2\omega]$.

The closed-loop response is:

$$x(t) = (1 + \omega t)e^{-\omega t} \quad (15.84)$$

The control signal:

$$u(t) = -\omega^2 x - 2\omega \dot{x} = -\omega^2(1 + \omega t)e^{-\omega t} + 2\omega^2 t e^{-\omega t} = -\omega^2 e^{-\omega t}(1 - \omega t) \quad (15.85)$$

Peak control effort occurs at $t = 0$:

$$|u(0)| = \omega^2 \quad (15.86)$$

Total control energy:

$$E = \int_0^\infty u^2(t) dt = \int_0^\infty \omega^4(1 - \omega t)^2 e^{-2\omega t} dt = \frac{\omega^3}{4} \quad (15.87)$$

Results:

Pole $-\omega$	Settling Time $\approx 4/\omega$	Peak $ u $	Energy E
-1	4.0 s	1	0.25
-2	2.0 s	4	2.0
-5	0.8 s	25	31.25
-10	0.4 s	100	250

Conclusion: Doubling the pole magnitude halves the settling time, quadruples the peak control effort, and increases energy by a factor of 8. This trade-off is fundamental to control design. $\square$

15.6 Summary

This chapter introduced the fundamental concepts of controllability and state feedback, which form the foundation of modern state-space control design.

Key Concepts:

Controllability answers the fundamental question: can we steer the system from any state to any other state? This is a system property independent of the specific control objective. The **controllability matrix** $\mathcal{C} = [\mathbf{B} \quad \mathbf{AB} \quad \cdots \quad \mathbf{A}^{n-1}\mathbf{B}]$ provides an algebraic test: the system is controllable if and only if $\text{rank}(\mathcal{C}) = n$.

State feedback of the form $\mathbf{u} = -\mathbf{K}\mathbf{x} + \mathbf{r}$ modifies the system dynamics to $\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{BK})\mathbf{x} + \mathbf{B}\mathbf{r}$. The **pole placement theorem** guarantees that for a controllable system, we can place the closed-loop poles at any desired locations by appropriate choice of $\mathbf{K}$. **Ackermann's formula** provides a direct computation: $\mathbf{K} = [0 \quad \cdots \quad 0 \quad 1] \mathcal{C}^{-1} \phi(\mathbf{A})$.

An important **practical trade-off** emerges from this analysis: faster pole placement requires higher control effort and increases sensitivity to modeling errors.

Looking Ahead:

Chapter 16 addresses the complementary problem: what if we cannot measure all states? The concepts of *observability* and *observer design* provide the solution, enabling state feedback even with limited measurements.

Chapter 17 introduces *optimal control*, which systematically addresses the trade-off between performance and control effort that we observed in our examples.

15.7 Exercises

Exercise 15.1. Determine the controllability of each system:

1. $\mathbf{A} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
2. $\mathbf{A} = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
3. $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Exercise 15.2. For the system $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix}$:

1. Show the system is controllable for all $\omega_n \neq 0$
2. Find the feedback gain to place poles at $s = -\sigma \pm j\omega_d$
3. Express the gains in terms of the desired ζ_d and $\omega_{n,d}$

Exercise 15.3. A spacecraft attitude control system has the model:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \quad (15.88)$$

1. Verify the system is controllable
2. If one thruster fails ($\mathbf{B}$ becomes $\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$), is the system still controllable?

Exercise 15.4. Use Ackermann's formula to design state feedback for:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -3 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (15.89)$$

Place the closed-loop poles at $s = -2, -3, -4$.

Exercise 15.5. For the inverted pendulum system (??):

1. Compute the open-loop eigenvalues and verify the system is unstable
2. Design state feedback to place all poles at $s = -5$

3. Simulate the response to an initial angle of $\theta_0 = 0.1$ rad

Exercise 15.6. (Controllability Gramian) For the stable system $\mathbf{A} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

1. Compute the infinite-horizon controllability Gramian $\mathbf{W}_c = \int_0^\infty e^{\mathbf{A}t} \mathbf{B} \mathbf{B}^\top e^{\mathbf{A}^\top t} dt$
2. Verify $\mathbf{W}_c$ is positive definite
3. Find the minimum-energy input to reach $\mathbf{x}_f = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ from the origin

Exercise 15.7. (Modal Analysis) For the system:

$$\mathbf{A} = \begin{bmatrix} -1 & 2 \\ 0 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (15.90)$$

1. Find the eigenvalues and eigenvectors of $\mathbf{A}$
2. Transform to modal coordinates and determine controllability of each mode
3. Relate your finding to the rank of the controllability matrix

Exercise 15.8. (Design Project) For your inverted pendulum experiment:

1. Measure or estimate the physical parameters M, m, ℓ
2. Compute the controllability matrix numerically
3. Design pole placement controllers for three different settling times
4. Compare simulation predictions with experimental results
5. Discuss the practical limitations observed

Exercise 15.9. Consider a multi-input system with:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (15.91)$$

1. Verify the system is controllable
2. Show that there are infinitely many feedback gains $\mathbf{K}$ that place both poles at $s = -2$
3. Find the minimum-norm gain matrix that achieves this pole placement

Exercise 15.10. (Stabilizability) A system is said to be stabilizable if all uncontrollable modes are stable. Consider:

$$\mathbf{A} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad (15.92)$$

1. Determine which modes are controllable
2. Is the system stabilizable? Justify your answer
3. Design a feedback gain to stabilize the system

Bibliography

- [1] R. E. Kalman, “On the general theory of control systems,” in *Proceedings of the First IFAC World Congress*, Moscow, USSR, 1960, pp. 481–492.
- [2] J. Ackermann, “Der Entwurf linearer Regelungssysteme im Zustandsraum,” *Regelungstechnik und Prozess-Datenverarbeitung*, vol. 20, no. 7, pp. 297–300, 1972.
- [3] R. E. Kalman, “Mathematical description of linear dynamical systems,” *SIAM Journal on Control*, vol. 1, no. 2, pp. 152–192, 1963.
- [4] C.-T. Chen, *Linear System Theory and Design*. New York: Holt, Rinehart and Winston, 1984.
- [5] T. Kailath, *Linear Systems*. Englewood Cliffs, NJ: Prentice-Hall, 1980.
- [6] K. Ogata, *Modern Control Engineering*, 5th ed. Upper Saddle River, NJ: Prentice Hall, 2010.
- [7] G. F. Franklin, J. D. Powell, and A. Emami-Naeini, *Feedback Control of Dynamic Systems*, 7th ed. Upper Saddle River, NJ: Pearson, 2015.
- [8] K. J. Åström and R. M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*, 2nd ed. Princeton, NJ: Princeton University Press, 2021.

Chapter 16

Observability and Observers

“The observer problem is dual to the regulator problem; the two problems have exactly the same mathematical structure.” — Rudolf E. Kalman, 1960

16.1 Historical Motivation: The Hidden State Problem

16.1.1 The Fundamental Challenge of State Estimation

The development of modern control theory in the late 1950s and early 1960s brought both unprecedented power and an unexpected challenge. State-space methods, championed by Rudolf Kalman and others, offered a systematic framework for designing controllers with optimal performance characteristics. State feedback, where the control input $u = -Kx$ is computed directly from the full state vector x , promised precise pole placement and optimal regulation. However, this elegant theory confronted a practical reality that threatened to undermine its applicability: *in most real systems, we cannot measure all the states.*

Consider a DC motor driving a robotic arm. The state-space model requires knowledge of both position θ and angular velocity $\dot{\theta}$. While position can be measured directly with an encoder, measuring velocity precisely is far more challenging. Tachometers add cost, complexity, and mechanical coupling issues. Differentiating the position signal introduces noise that can destabilize the control system. The gap between the theoretical requirement for full-state feedback and the practical limitation of partial measurements seemed insurmountable.

Kalman’s Dual Discovery and Luenberger’s Observer (1960–1964)

In 1960, Rudolf Kalman published a landmark paper recognizing a beautiful duality in control mathematics: the question “Can we drive the system anywhere?” (controllability) is mathematically equivalent to “Can we determine where the system has been?” (observability). This duality meant that every theorem about controllability had a corresponding theorem about observability. Four years later, David G. Luenberger provided the practical implementation in his 1964 paper “Observing the State of a Linear System,” introducing the state observer that bears his name. Luenberger’s key insight was elegant: if we know the system’s model and have access to both input and output, we can build a “copy” of the

system that converges to the true state by feeding back the difference between predicted and actual measurements.

16.1.2 Kalman's Dual Discovery (1960)

In 1960, Rudolf Kalman published a landmark paper that addressed this challenge from a fundamental mathematical perspective. While developing the theory of controllability—the question of whether control inputs could steer a system to any desired state—Kalman recognized a beautiful duality in the mathematics. The question “Can we drive the system anywhere?” was mathematically equivalent to asking “Can we determine where the system has been?”

This second question is **observability**: given measurements of a system's outputs over some time interval, can we uniquely determine what the system's internal state was? Kalman showed that observability is the dual concept to controllability. **Controllability** asks whether the input $u(t)$ can steer the state x to any desired value, while **observability** asks whether the output history $y(\tau)$ for $\tau \in [0, t]$ can reveal the initial state $x(0)$.

These two properties are linked by a profound mathematical duality. For a system described by (A, B, C, D) , the pair (A, B) is controllable if and only if (A^T, B^T) is observable, and the pair (A, C) is observable if and only if (A^T, C^T) is controllable.

This duality meant that every theorem about controllability had a corresponding theorem about observability, and every technique for controller design had a dual technique for observer design.

16.1.3 Luenberger's Observer (1964)

While Kalman established the theoretical foundations, it was David G. Luenberger who provided the practical solution. In his 1964 paper “Observing the State of a Linear System,” Luenberger introduced what is now called the **Luenberger observer** (or state observer).

Luenberger's key insight was elegant: if we know the system's model (A, B, C) and we have access to both the input u and output y , we can build a “copy” of the system in software or hardware. This copy, running in parallel with the real system, uses the same input u and compares its predicted output with the actual measured output y . Any discrepancy indicates that the copy's internal state estimate differs from the true state. By feeding this error back into the copy with appropriate gain, the estimate can be made to converge to the true state.

The observer takes the form:

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}) \quad (16.1)$$

where $\hat{x}$ is the estimated state and L is the observer gain matrix. The term $(y - C\hat{x})$ represents the “innovation”—the difference between what we measure and what we predicted. The gain L determines how aggressively the observer corrects its estimate based on this innovation.

16.1.4 Aerospace Applications: Apollo Navigation

The Apollo program provided one of the most dramatic demonstrations of state estimation in action. The challenge of navigating a spacecraft to the Moon and back required continuous knowledge of position and velocity in three-dimensional space—six state variables in all. However, the

spacecraft's sensors provided only limited measurements: angles to known stars, range and range-rate to Earth-based tracking stations, and occasionally, angles to landmarks on the lunar surface.

The Apollo guidance computer implemented a sophisticated state estimator (specifically, a Kalman filter, which we will study in Chapter 18) that fused these partial measurements to produce a complete estimate of the spacecraft's state. This estimated state was then used for all navigation and control decisions. The success of Apollo demonstrated that state estimation was not merely a theoretical curiosity but an essential enabling technology for advanced control systems.

Consider the estimation problem during lunar orbit insertion: the spacecraft's position and velocity had to be known precisely to execute the burn that would capture the spacecraft into lunar orbit. Direct measurement of all six states was impossible, but by processing sequences of angular measurements to known stars and landmarks, combined with knowledge of the spacecraft's dynamics (gravitational attraction of the Moon and Earth), the navigation filter could reconstruct the full state with remarkable accuracy.

16.1.5 Submarine Target Motion Analysis

Another compelling application of observability theory emerged from naval warfare: target motion analysis (TMA) for submarines. A submarine tracking a surface target faces a challenging estimation problem. The submarine's passive sonar can measure the bearing to the target (the direction from which sound arrives) but cannot directly measure range or target velocity without going active and revealing its position.

The target motion analysis problem asks: given a sequence of bearing measurements over time, can we determine the target's position, course, and speed? This is precisely a question of observability. The answer depends on the geometry and the submarine's maneuvers. If the submarine maintains a steady course, the system is often unobservable—many different target trajectories are consistent with the observed bearing measurements. However, if the submarine maneuvers (changes course), the resulting bearing changes provide additional information that can make the system observable.

This application illustrates a key insight: observability is not just a property of the system being observed, but also depends on how we interact with it. The submarine's own maneuvers affect the observability of the target's state. This concept extends broadly: in many systems, careful choice of inputs or operating conditions can improve observability.

16.1.6 The Key Insight

The fundamental insight from this historical development is both simple and profound:

If a system is observable, its complete internal state can be reconstructed from a history of output measurements, even if we can never measure those internal states directly.

This means that the requirement for state feedback does not actually require direct measurement of all states. Instead, we need only a system that is observable (a property we can check mathematically), an observer that uses available measurements to estimate unmeasured states, and sufficient time for the observer's estimates to converge to the true states.

The development of observability theory and state observers transformed state-space control from an elegant but impractical theory into a practical engineering discipline that underlies much of modern control system design.

16.2 Modern Applications

The principles of observability and state estimation, developed in the 1960s for aerospace applications, have become ubiquitous in modern engineering. Today, observers are embedded in billions of devices, from smartphones to electric vehicles. This section surveys several contemporary applications.

16.2.1 GPS Receivers

Global Positioning System (GPS) receivers present a fascinating estimation problem. The receiver measures the time-of-arrival of signals from multiple satellites, from which it computes “pseudoranges”—approximate distances to each satellite. However, these pseudoranges are corrupted by several error sources: atmospheric delays, multipath reflections, and most significantly, clock bias between the receiver’s local oscillator and the satellite atomic clocks.

A GPS receiver implements a state estimator that tracks not only the receiver’s position (three states: x, y, z) and velocity (three states: $\dot{x}, \dot{y}, \dot{z}$) but also the clock bias and its drift rate (two additional states). By processing pseudorange and Doppler measurements from multiple satellites, the receiver estimates all eight states, providing accurate position and velocity even though none of these states is directly measured.

16.2.2 Motor Control: Velocity Estimation from Position

Electric motor drives routinely estimate velocity from position encoder measurements. While encoders provide precise position information, directly differentiating the encoder signal to obtain velocity produces a noisy, quantized signal that is unsuitable for high-performance velocity control.

Instead, modern motor drives implement velocity observers that combine the encoder position with knowledge of the motor’s electrical and mechanical dynamics. The observer produces a smooth velocity estimate that enables tight servo performance. This approach is particularly important in industrial robots requiring precise velocity control for smooth motion, disk drive head positioning requiring vibration-free settling, and electric vehicle traction motors requiring smooth torque delivery.

We will implement exactly such an observer in the practical experiment section of this chapter.

16.2.3 Battery State-of-Charge Estimation

Electric vehicles and portable electronics rely on accurate estimation of battery state-of-charge (SoC)—the percentage of remaining capacity. Direct measurement of SoC is essentially impossible; one cannot simply “look inside” an electrochemical cell to determine its charge level.

However, battery voltage, current, and temperature can be measured. An observer that incorporates an electrochemical model of the battery can estimate SoC from these measurements. The

challenge is significant: battery dynamics are nonlinear, temperature-dependent, and subject to aging effects. Modern battery management systems implement extended Kalman filters or other nonlinear observers to track not only SoC but also state-of-health parameters that change over the battery's lifetime.

16.2.4 Soft Sensors in Chemical Processes

In chemical process industries, many important quantities cannot be measured directly or can only be measured with long time delays. Product composition, for example, might be determined only by laboratory analysis of samples, with results available hours after the sample was taken. Biomass concentration in fermentation processes may be measurable only through offline assays.

“Soft sensors” use observers based on process models and real-time measurements of readily available quantities (temperature, pressure, flow rates) to estimate these hard-to-measure variables. These estimates enable real-time control that would otherwise be impossible.

16.3 Mathematical Foundation

Having motivated the problem historically and illustrated its modern relevance, we now develop the mathematical theory of observability and observer design. The treatment parallels the development of controllability in Chapter 15, exploiting the duality between the two concepts.

16.3.1 System Description

We consider a linear time-invariant (LTI) continuous-time system in state-space form:

$$\dot{x}(t) = Ax(t) + Bu(t) \quad (16.2)$$

$$y(t) = Cx(t) + Du(t) \quad (16.3)$$

The variables and matrices are defined in Table 16.1.

Table 16.1: State-Space System Variables and Matrices

Symbol	Description
$x(t) \in \mathbb{R}^n$	State vector
$u(t) \in \mathbb{R}^m$	Input vector
$y(t) \in \mathbb{R}^p$	Output vector
$A \in \mathbb{R}^{n \times n}$	System matrix
$B \in \mathbb{R}^{n \times m}$	Input matrix
$C \in \mathbb{R}^{p \times n}$	Output matrix
$D \in \mathbb{R}^{p \times m}$	Feedthrough matrix

For the study of observability, we focus on the pair (A, C) . The input $u(t)$ is assumed known, and for simplicity in developing the theory, we often set $u(t) = 0$ and $D = 0$.

16.3.2 Definition of Observability

Definition 16.1 (Observability). The system (??)–(??), or equivalently the pair (A, C) , is said to be **observable** if for any unknown initial state $x(0) = x_0$, there exists a finite time $t_1 > 0$ such that knowledge of the input $u(t)$ and output $y(t)$ over the interval $[0, t_1]$ suffices to uniquely determine x_0 .

An equivalent formulation focuses on distinguishability: a system is observable if and only if no two distinct initial states produce identical output trajectories under the same input sequence.

Definition 16.2 (Unobservable State). A state $x_0 \neq 0$ is **unobservable** if the system initialized at x_0 with $u(t) = 0$ produces $y(t) = 0$ for all $t \geq 0$. The system is observable if and only if the only unobservable state is $x_0 = 0$.

16.3.3 The Observability Matrix

The key to testing observability is the **observability matrix**, constructed from A and C :

Definition 16.3 (Observability Matrix). The observability matrix for the pair (A, C) is defined as:

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} \in \mathbb{R}^{np \times n} \quad (16.4)$$

This matrix stacks the output at time zero ($y(0) = Cx_0$), the first derivative of the output ($\dot{y}(0) = CAx_0$), and so forth, up to the $(n-1)$ -th derivative.

Theorem 16.1 (Observability Rank Condition). *The pair (A, C) is observable if and only if the observability matrix $\mathcal{O}$ has full column rank:*

$$\text{rank}(\mathcal{O}) = n \quad (16.5)$$

Proof. Consider the autonomous system ($u = 0$) with initial condition $x(0) = x_0$. The output and its derivatives at $t = 0$ are:

$$y(0) = Cx_0 \quad (16.6)$$

$$\dot{y}(0) = C\dot{x}(0) = CAx_0 \quad (16.7)$$

$$\ddot{y}(0) = CA\dot{x}(0) = CA^2x_0 \quad (16.8)$$

$$\vdots \quad (16.9)$$

$$y^{(n-1)}(0) = CA^{n-1}x_0 \quad (16.10)$$

This can be written in matrix form as:

$$\begin{bmatrix} y(0) \\ \dot{y}(0) \\ \vdots \\ y^{(n-1)}(0) \end{bmatrix} = \mathcal{O}x_0 \quad (16.11)$$

If $\text{rank}(\mathcal{O}) = n$, then $\mathcal{O}$ has a left inverse $\mathcal{O}^\dagger = (\mathcal{O}^T \mathcal{O})^{-1} \mathcal{O}^T$, and we can solve for x_0 :

$$x_0 = \mathcal{O}^\dagger \begin{bmatrix} y(0) \\ \dot{y}(0) \\ \vdots \\ y^{(n-1)}(0) \end{bmatrix} \quad (16.12)$$

Thus, the initial state can be uniquely determined from the output and its first $n - 1$ derivatives, proving observability.

Conversely, if $\text{rank}(\mathcal{O}) < n$, then $\mathcal{O}$ has a non-trivial null space. Any $x_0 \neq 0$ in the null space of $\mathcal{O}$ satisfies $\mathcal{O}x_0 = 0$, meaning $Cx_0 = CAx_0 = \cdots = CA^{n-1}x_0 = 0$. By the Cayley-Hamilton theorem, $CA^k x_0 = 0$ for all $k \geq 0$, and hence $y(t) = Ce^{At}x_0 = 0$ for all t . Such an x_0 is unobservable, proving the system is not observable. $\square$

16.3.4 Duality Between Controllability and Observability

The controllability and observability concepts are related by a beautiful mathematical duality.

Theorem 16.2 (Duality Theorem). *The pair (A, C) is observable if and only if the pair (A^T, C^T) is controllable.*

Proof. The controllability matrix for the pair (A^T, C^T) is:

$$\mathcal{C}_{A^T, C^T} = [C^T \quad A^T C^T \quad (A^T)^2 C^T \quad \cdots \quad (A^T)^{n-1} C^T] \quad (16.13)$$

Taking the transpose:

$$\mathcal{C}_{A^T, C^T}^T = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} = \mathcal{O}_{A, C} \quad (16.14)$$

Since a matrix and its transpose have the same rank, $\text{rank}(\mathcal{C}_{A^T, C^T}) = \text{rank}(\mathcal{O}_{A, C})$. The controllability rank condition for (A^T, C^T) is equivalent to the observability rank condition for (A, C) . $\square$

This duality has profound practical implications. Any result for controllability immediately yields a corresponding result for observability by transposition. Any algorithm for computing controller gains can be adapted for observer gains. The theoretical development is essentially “two for the price of one.”

16.3.5 The Luenberger Observer

We now develop the Luenberger observer, which uses output measurements to estimate the unmeasured states of an observable system.

Table 16.2: Duality Between Controllability and Observability

Controllability	Observability
Pair (A, B)	Pair (A, C)
Can input reach all states?	Can output reveal all states?
Controllability matrix $\mathcal{C} = [B \ AB \ \cdots \ A^{n-1}B]$	Observability matrix $\mathcal{O} = [C; \ CA; \ \cdots; \ CA^{n-1}]^T$
$\text{rank}(\mathcal{C}) = n$	$\text{rank}(\mathcal{O}) = n$
State feedback: $u = -Kx$	Observer: $\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$
Closed-loop: $A - BK$	Error dynamics: $A - LC$
Place eigenvalues of $A - BK$	Place eigenvalues of $A - LC$

Observer Structure

The observer is a dynamical system that runs alongside (or in software, simulating) the plant. It receives the same input $u(t)$ as the plant and uses the output measurement $y(t)$ to correct its state estimate:

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}) \quad (16.15)$$

In this equation, $\hat{x}(t) \in \mathbb{R}^n$ is the estimated state, $L \in \mathbb{R}^{n \times p}$ is the observer gain matrix, and the term $y - C\hat{x}$ is the **output estimation error** (or “innovation”).

The observer can be rewritten as:

$$\dot{\hat{x}} = (A - LC)\hat{x} + Bu + Ly \quad (16.16)$$

This shows that the observer is itself an LTI system with system matrix $(A - LC)$, driven by inputs u and y .

Error Dynamics

The key to understanding observer performance is the **state estimation error**:

$$e(t) = x(t) - \hat{x}(t) \quad (16.17)$$

Taking the derivative:

$$\dot{e} = \dot{x} - \dot{\hat{x}} \quad (16.18)$$

$$= (Ax + Bu) - (A\hat{x} + Bu + L(y - C\hat{x})) \quad (16.19)$$

$$= A(x - \hat{x}) - L(Cx - C\hat{x}) \quad (16.20)$$

$$= Ae - LCe \quad (16.21)$$

$$= (A - LC)e \quad (16.22)$$

This is a remarkable result: the error dynamics are governed by the autonomous system with matrix $(A - LC)$, and they are *independent of the actual system trajectory and input*. The error evolves according to:

$$e(t) = e^{(A-LC)t}e(0) \quad (16.23)$$

Observer Pole Placement

If the eigenvalues of $(A - LC)$ all have negative real parts, then $e(t) \rightarrow 0$ as $t \rightarrow \infty$, regardless of the initial error $e(0) = x(0) - \hat{x}(0)$. The rate of convergence depends on the eigenvalue locations: more negative eigenvalues yield faster convergence.

Theorem 16.3 (Observer Pole Placement). *If the pair (A, C) is observable, then the observer gain L can be chosen to place the eigenvalues of $(A - LC)$ at any desired locations in the complex plane.*

Proof. By duality, the pair (A, C) is observable if and only if (A^T, C^T) is controllable. We know from Chapter 15 that if (A^T, C^T) is controllable, we can find a matrix K_d such that the eigenvalues of $A^T - C^T K_d$ are at any desired locations.

Note that:

$$A^T - C^T K_d = (A - K_d^T C)^T \quad (16.24)$$

Since a matrix and its transpose have identical eigenvalues, setting $L = K_d^T$ places the eigenvalues of $A - LC$ at the same desired locations. $\square$

Practical Considerations for Observer Pole Placement:

Several practical considerations guide the selection of observer pole locations. First, observer poles should be placed faster than the controller poles, typically 2–10 times faster, ensuring that the state estimates converge before they significantly affect control performance. However, extremely fast observer poles amplify measurement noise, creating a fundamental trade-off between convergence speed and noise sensitivity. Finally, if the plant has stable but slow modes, the observer must be faster than those modes to estimate the corresponding states before they contribute significantly to the response.

16.3.6 Observer Design by Pole Placement

For a single-output system ($p = 1$), the observer pole placement problem can be solved using methods analogous to controller design. One approach uses the duality and Ackermann's formula.

Dual Ackermann's Formula: For a single-output system, the observer gain is given by:

$$L = \alpha_o(A) \mathcal{O}^{-1} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \quad (16.25)$$

where $\alpha_o(s) = (s - \lambda_1)(s - \lambda_2) \cdots (s - \lambda_n)$ is the desired observer characteristic polynomial and $\mathcal{O}$ is the observability matrix.

16.3.7 The Separation Principle

A crucial question arises when combining state-feedback control with state estimation: does using estimated states instead of true states affect stability or performance guarantees? The separation principle provides a reassuring answer.

Algorithm 31 Observer Pole Placement**Require:** System matrices A, C ; desired observer poles $\lambda_1, \lambda_2, \dots, \lambda_n$ **Ensure:** Observer gain matrix L

- 1: Define system matrix A and output matrix C
- 2: Specify desired observer pole locations (faster than controller poles)
- 3: Compute observer gain: $L \leftarrow \text{PLACE}(A^T, C^T, \text{poles})^T$
- 4: Verify eigenvalues of $(A - LC)$ match desired poles
- 5: **return** L

Theorem 16.4 (Separation Principle). *Consider a state-feedback controller $u = -Kx$ designed assuming access to the true state x . When implemented using the estimated state from an asymptotically stable observer, $u = -K\hat{x}$, the closed-loop system remains stable if and only if both the state feedback $(A - BK)$ and the observer $(A - LC)$ are stable. Furthermore, the closed-loop eigenvalues are the union of the controller eigenvalues and the observer eigenvalues.*

Proof. With observer-based control, the closed-loop dynamics are:

$$\dot{x} = Ax + Bu = Ax - BK\hat{x} = Ax - BK(x - e) = (A - BK)x + BKe \quad (16.26)$$

$$\dot{e} = (A - LC)e \quad (16.27)$$

In matrix form:

$$\begin{bmatrix} \dot{x} \\ \dot{e} \end{bmatrix} = \underbrace{\begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix}}_{A_{cl}} \begin{bmatrix} x \\ e \end{bmatrix} \quad (16.28)$$

The closed-loop matrix A_{cl} is block upper triangular. The eigenvalues of a block triangular matrix are the eigenvalues of its diagonal blocks. Therefore:

$$\text{eigenvalues}(A_{cl}) = \text{eigenvalues}(A - BK) \cup \text{eigenvalues}(A - LC) \quad (16.29)$$

The system is stable if and only if all eigenvalues have negative real parts, which requires both $(A - BK)$ and $(A - LC)$ to be stable. $\square$

The separation principle is remarkable: the controller design and observer design can be carried out *independently*. The controller is designed assuming perfect state information, then the observer is designed to provide state estimates, and the combination is guaranteed to work.

16.3.8 Observer-Based Control Implementation

The complete observer-based control system has the following structure:

Plant:

$$\dot{x} = Ax + Bu \quad (16.30)$$

$$y = Cx \quad (16.31)$$

Observer:

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}) \quad (16.32)$$

Control Law:

$$u = -K\hat{x} \quad (16.33)$$

Note that the observer requires knowledge of the input u , which depends on the estimated state $\hat{x}$. The complete observer-based controller, also known as a **compensator**, can be written as a dynamical system from measurement y to control u :

$$\dot{\hat{x}} = (A - BK - LC)\hat{x} + Ly \quad (16.34)$$

$$u = -K\hat{x} \quad (16.35)$$

This is a state-space realization of the compensator transfer function from y to u .

16.3.9 Transfer Function Interpretation

The observer-based controller can be expressed as a transfer function:

$$K_{controller}(s) = K(sI - A + BK + LC)^{-1}L \quad (16.36)$$

This connects state-space observer-based control to classical frequency-domain compensation. The observer-based controller implements a dynamic compensator whose order equals the plant order.

16.4 Practical Experiment: Motor Velocity Observer

16.4.1 Objective

In this experiment, we implement a Luenberger observer to estimate the angular velocity of a DC motor from position encoder measurements alone. We compare the observer-based velocity estimate against numerical differentiation, demonstrating the superior noise rejection of the observer approach.

16.4.2 Required Equipment

The experiment requires the following equipment: a DC motor with quadrature encoder (such as a Pololu 37D gearmotor with 64 CPR encoder), a motor driver (L298N or similar H-bridge), an Arduino Mega or similar microcontroller, a power supply appropriate for the motor, an oscilloscope or data logging capability, and a computer with Arduino IDE and MATLAB/Python for analysis.

16.4.3 Motor Model

A DC motor can be modeled with two states: angular position θ and angular velocity $\omega = \dot{\theta}$. For a motor with negligible electrical dynamics (small electrical time constant compared to mechanical), the mechanical dynamics are:

$$J\dot{\omega} + b\omega = K_t i = K_t \frac{V - K_e \omega}{R} \quad (16.37)$$

where J is the moment of inertia, b is the viscous friction, K_t is the torque constant, K_e is the back-EMF constant, and R is the armature resistance.

This simplifies to the first-order mechanical model:

$$\dot{\omega} = -\frac{b_{eff}}{J}\omega + \frac{K}{J}V \quad (16.38)$$

where $b_{eff} = b + K_t K_e / R$ is the effective damping and $K = K_t / R$.

The state-space model with position and velocity as states is:

$$\begin{bmatrix} \dot{\theta} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -b_{eff}/J \end{bmatrix} \begin{bmatrix} \theta \\ \omega \end{bmatrix} + \begin{bmatrix} 0 \\ K/J \end{bmatrix} V \quad (16.39)$$

With position measurement only:

$$y = \theta = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \omega \end{bmatrix} \quad (16.40)$$

16.4.4 Observability Analysis

For the motor model with $A = \begin{bmatrix} 0 & 1 \\ 0 & -a \end{bmatrix}$ (where $a = b_{eff}/J$) and $C = \begin{bmatrix} 1 & 0 \end{bmatrix}$:

$$\mathcal{O} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (16.41)$$

The observability matrix is the identity matrix, which has rank 2. The system is observable! This means we can estimate velocity from position measurements.

16.4.5 Observer Design

Let the motor parameters be identified as $a = 10 \text{ rad/s}$ and $b = K/J = 50 \text{ rad/s}^2/\text{V}$. We design an observer with poles at $s = -50$ and $s = -60$, making the observer 5–6 times faster than the motor's dominant time constant.

The observer gains are computed:

$$L = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix} \quad (16.42)$$

The observer characteristic polynomial is:

$$\det(sI - A + LC) = s^2 + (a + l_1)s + (l_2 + al_1) = (s + 50)(s + 60) = s^2 + 110s + 3000 \quad (16.43)$$

Matching coefficients:

$$a + l_1 = 110 \implies l_1 = 110 - 10 = 100 \quad (16.44)$$

$$l_2 + al_1 = 3000 \implies l_2 = 3000 - 10 \times 100 = 2000 \quad (16.45)$$

$$\text{So } L = \begin{bmatrix} 100 \\ 2000 \end{bmatrix}.$$

16.4.6 Microcontroller Implementation

The following algorithms describe the implementation of the velocity observer on a microcontroller. The implementation consists of three main components: initialization, encoder reading, and the observer update routine.

Algorithm 32 Motor Velocity Observer – Initialization

Require: System parameters a, b ; observer gains L_1, L_2 ; sample time T_s

- 1: Initialize serial communication for data logging
 - 2: Configure encoder pins as inputs with pull-up resistors
 - 3: Configure motor PWM and direction pins as outputs
 - 4: Attach interrupt handlers to encoder channels A and B
 - 5: Configure timer for fixed sample rate (e.g., 1 kHz)
 - 6: Initialize state estimates: $\hat{\theta} \leftarrow 0, \hat{\omega} \leftarrow 0$
 - 7: Initialize comparison variables: $\theta_{prev} \leftarrow 0, \omega_{deriv} \leftarrow 0$
-

Algorithm 33 Quadrature Encoder Interrupt Service Routine

Require: Current encoder channel states A, B

- 1: Read current state: $s \leftarrow (A \ll 1) \vee B$
 - 2: Look up increment from state transition table using previous and current states
 - 3: Update encoder count: $\text{count} \leftarrow \text{count} + \text{increment}$
 - 4: Store current state for next interrupt
-

16.4.7 Experimental Procedure

The experimental procedure consists of five phases. Begin with **system identification** by applying step inputs and measuring the motor's response to identify parameters a and b . Next, proceed with **observer implementation** by uploading the observer code and running step response experiments. During **data collection**, log the estimated velocity $\hat{\omega}$, the numerical derivative ω_{deriv} , and the applied voltage. For **comparison**, plot both velocity estimates; the observer estimate should be significantly smoother than the numerical derivative. Finally, conduct an **initial condition test** by initializing the observer with a large error (e.g., $\hat{\theta}(0) = 0, \hat{\omega}(0) = 100$) and observing convergence to the true state.

16.4.8 Expected Results

Several key observations emerge from this comparison. The observer estimate converges to the true velocity within approximately $5/(50) = 0.1$ seconds (five time constants of the observer). The numerical derivative is noisy, especially during rapid changes. In contrast, the observer provides smooth estimates suitable for feedback control.

Algorithm 34 Luenberger Observer Update (Timer ISR)

Require: Encoder count, system parameters a, b , observer gains L_1, L_2 , sample time T_s

```

1: // Read position measurement
2:  $\theta_{meas} \leftarrow \text{count}/\text{countsPerRad}$ 
3:
4: // Numerical derivative with low-pass filter (for comparison)
5:  $\omega_{raw} \leftarrow (\theta_{meas} - \theta_{prev})/T_s$ 
6:  $\omega_{deriv} \leftarrow \alpha \cdot \omega_{raw} + (1 - \alpha) \cdot \omega_{deriv}$   $\triangleright \alpha \approx 0.2$ 
7:  $\theta_{prev} \leftarrow \theta_{meas}$ 
8:
9: // Luenberger Observer
10: Prediction step:
11:  $\dot{\hat{\theta}} \leftarrow \hat{\omega}$ 
12:  $\dot{\hat{\omega}} \leftarrow -a \cdot \hat{\omega} + b \cdot V$ 
13:
14: Correction step:
15:  $\text{innovation} \leftarrow \theta_{meas} - \hat{\theta}$ 
16:  $\hat{\theta} \leftarrow \hat{\theta} + L_1 \cdot \text{innovation}$ 
17:  $\hat{\omega} \leftarrow \hat{\omega} + L_2 \cdot \text{innovation}$ 
18:
19: Integration (Euler method):
20:  $\hat{\theta} \leftarrow \hat{\theta} + \dot{\hat{\theta}} \cdot T_s$ 
21:  $\hat{\omega} \leftarrow \hat{\omega} + \dot{\hat{\omega}} \cdot T_s$ 
22:
23: // Apply control signal to motor
24: Set motor direction based on sign of  $V$ 
25: Set motor PWM proportional to  $|V|$ 

```

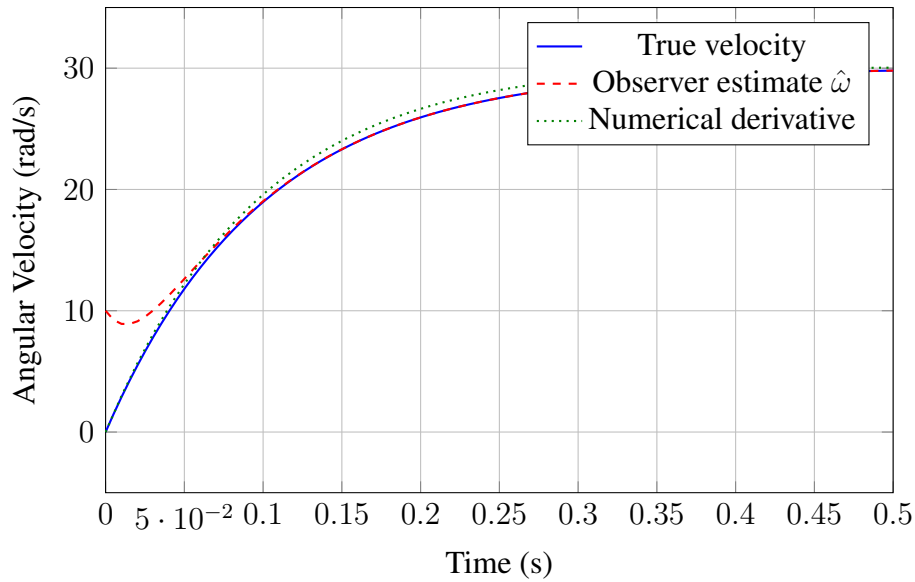


Figure 16.1: Comparison of velocity estimation methods. The observer estimate (dashed red) tracks the true velocity (solid blue) smoothly, while numerical differentiation (dotted green) is corrupted by noise.

16.5 Block Diagrams

16.5.1 Basic Observer Structure

16.5.2 Observer-Based Control System

16.5.3 State Estimation Error Convergence

16.6 Worked Examples

Checking Observability of a Third-Order System

Problem: Determine whether the following system is observable:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}, \quad C = [1 \ 0 \ 0] \quad (16.46)$$

Solution:

Step 1: Construct the observability matrix $\mathcal{O} = [C; CA; CA^2]^T$.

$$C = [1 \ 0 \ 0] \quad (16.47)$$

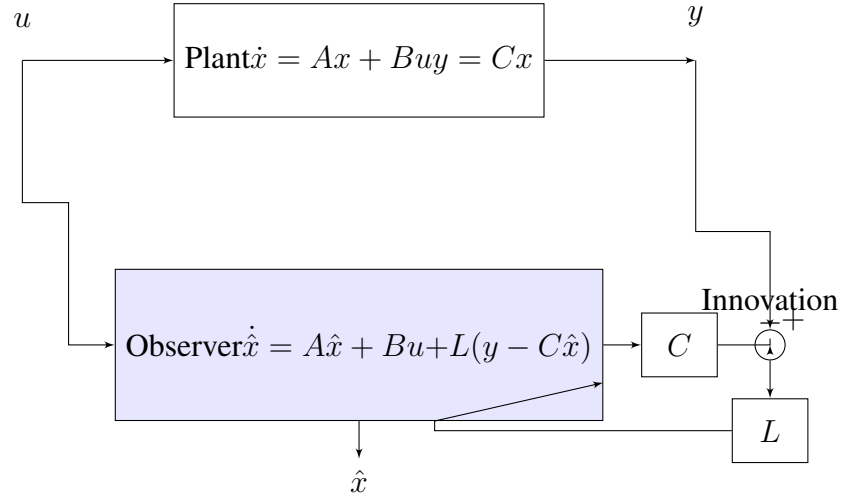


Figure 16.2: Luenberger observer structure. The observer is a copy of the plant that uses the innovation signal $(y - C\hat{x})$ to correct its state estimate.

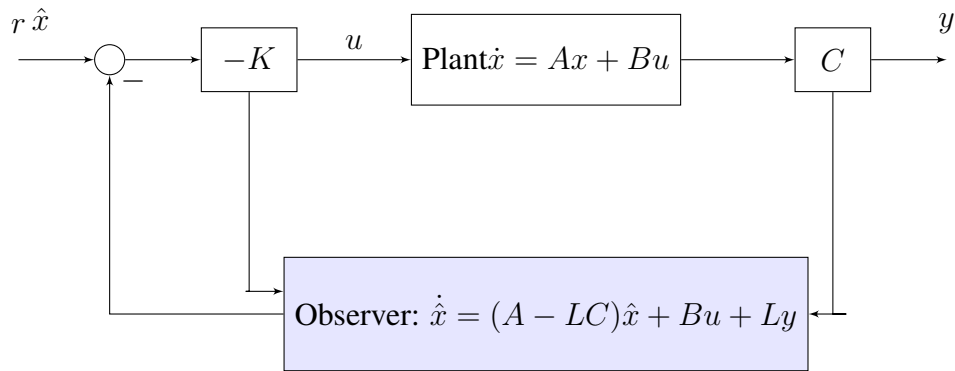


Figure 16.3: Complete observer-based control system. The controller uses estimated states $\hat{x}$ instead of true states x . The observer receives both the control input u and the measurement y .

$$CA = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \quad (16.48)$$

$$CA^2 = (CA)A = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \quad (16.49)$$

Therefore:

$$\mathcal{O} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (16.50)$$

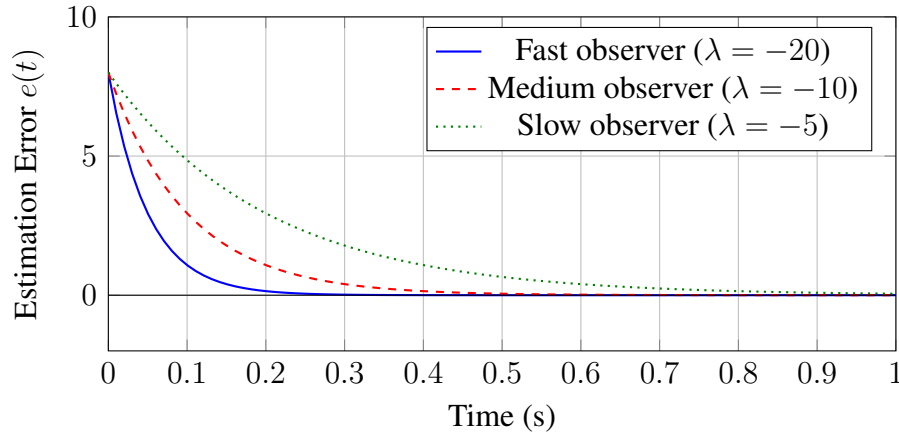


Figure 16.4: Estimation error convergence for observers with different pole locations. Faster poles (more negative) yield faster convergence but increased noise sensitivity.

Step 2: Check the rank.

The observability matrix is the 3×3 identity matrix, so $\text{rank}(\mathcal{O}) = 3 = n$.

Conclusion: The system is **observable**.

Remark: This system is in **observer canonical form**. Systems in this form are always observable, just as systems in controller canonical form are always controllable.

Luenberger Observer Design for DC Motor

Problem: Design a Luenberger observer for the DC motor system with desired observer poles at $s = -40$ and $s = -50$.

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -10 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 50 \end{bmatrix}, \quad C = [1 \quad 0] \quad (16.51)$$

Solution:

Step 1: Verify observability.

$$\mathcal{O} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (16.52)$$

$\text{rank}(\mathcal{O}) = 2$, so the system is observable.

Step 2: Let $L = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}$. Compute $A - LC$:

$$A - LC = \begin{bmatrix} 0 & 1 \\ 0 & -10 \end{bmatrix} - \begin{bmatrix} l_1 \\ l_2 \end{bmatrix} [1 \quad 0] = \begin{bmatrix} -l_1 & 1 \\ -l_2 & -10 \end{bmatrix} \quad (16.53)$$

Step 3: Find the characteristic polynomial of $A - LC$:

$$\det(sI - A + LC) = \det \begin{bmatrix} s + l_1 & -1 \\ l_2 & s + 10 \end{bmatrix} \quad (16.54)$$

$$= (s + l_1)(s + 10) + l_2 \quad (16.55)$$

$$= s^2 + (10 + l_1)s + (10l_1 + l_2) \quad (16.56)$$

Step 4: The desired characteristic polynomial with poles at -40 and -50 is:

$$(s + 40)(s + 50) = s^2 + 90s + 2000 \quad (16.57)$$

Step 5: Match coefficients:

$$10 + l_1 = 90 \implies l_1 = 80 \quad (16.58)$$

$$10l_1 + l_2 = 2000 \implies l_2 = 2000 - 800 = 1200 \quad (16.59)$$

Result:

$$L = \begin{bmatrix} 80 \\ 1200 \end{bmatrix} \quad (16.60)$$

16.6.1 Example 3: Duality Between Controllability and Observability

Problem: Consider the system:

$$A = \begin{bmatrix} -2 & 1 \\ 0 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad (16.61)$$

(a) Check controllability of (A, B) . (b) Check observability of (A^T, B^T) . (c) Verify the duality relationship.

Solution:

(a) Controllability of (A, B) :

$$\mathcal{C} = [B \quad AB] \quad (16.62)$$

$$AB = \begin{bmatrix} -2 & 1 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \end{bmatrix} \quad (16.63)$$

$$\mathcal{C} = \begin{bmatrix} 1 & -1 \\ 1 & -3 \end{bmatrix} \quad (16.64)$$

$$\det(\mathcal{C}) = (1)(-3) - (-1)(1) = -3 + 1 = -2 \neq 0 \quad (16.65)$$

So (A, B) is **controllable**.

(b) Observability of (A^T, B^T) :

$$A^T = \begin{bmatrix} -2 & 0 \\ 1 & -3 \end{bmatrix}, \quad B^T = \begin{bmatrix} 1 & 1 \end{bmatrix} \quad (16.66)$$

The observability matrix for (A^T, B^T) is:

$$\mathcal{O}_{A^T, B^T} = \begin{bmatrix} B^T \\ B^T A^T \end{bmatrix} \quad (16.67)$$

$$B^T A^T = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} -2 & 0 \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} -1 & -3 \end{bmatrix} \quad (16.68)$$

$$\mathcal{O}_{A^T, B^T} = \begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix} \quad (16.69)$$

$$\det(\mathcal{O}_{A^T, B^T}) = (1)(-3) - (1)(-1) = -3 + 1 = -2 \neq 0 \quad (16.70)$$

So (A^T, B^T) is **observable**.

(c) Verification of duality:

Note that $\mathcal{O}_{A^T, B^T} = \mathcal{C}_{A, B}^T$:

$$\mathcal{C}^T = \begin{bmatrix} 1 & -1 \\ 1 & -3 \end{bmatrix}^T = \begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix} = \mathcal{O}_{A^T, B^T} \quad (16.71)$$

Since they have the same rank (both full rank), the duality is verified: (A, B) controllable $\Leftrightarrow (A^T, B^T)$ observable.

16.6.2 Example 4: Observer-Based Controller Design

Problem: For the system in Example 2, design a complete observer-based controller with controller poles at $s = -5 \pm 5j$ and observer poles at $s = -40$ and $s = -50$ (already computed in Example 2).

Solution:

Step 1: Design the state feedback gain K .

From Example 2, the system matrices are:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -10 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 50 \end{bmatrix} \quad (16.72)$$

Let $K = \begin{bmatrix} k_1 & k_2 \end{bmatrix}$. Then:

$$A - BK = \begin{bmatrix} 0 & 1 \\ -50k_1 & -10 - 50k_2 \end{bmatrix} \quad (16.73)$$

Characteristic polynomial:

$$\det(sI - A + BK) = s^2 + (10 + 50k_2)s + 50k_1 \quad (16.74)$$

Desired characteristic polynomial with poles at $-5 \pm 5j$:

$$(s + 5 - 5j)(s + 5 + 5j) = s^2 + 10s + 50 \quad (16.75)$$

Matching coefficients:

$$10 + 50k_2 = 10 \implies k_2 = 0 \quad (16.76)$$

$$50k_1 = 50 \implies k_1 = 1 \quad (16.77)$$

So $K = [1 \ 0]$.

Step 2: From Example 2, $L = \begin{bmatrix} 80 \\ 1200 \end{bmatrix}$.

Step 3: The complete observer-based compensator is:

$$\dot{\hat{x}} = (A - BK - LC)\hat{x} + Ly \quad (16.78)$$

$$u = -K\hat{x} \quad (16.79)$$

Computing $A - BK - LC$:

$$A - BK = \begin{bmatrix} 0 & 1 \\ -50 & -10 \end{bmatrix} \quad (16.80)$$

$$LC = \begin{bmatrix} 80 \\ 1200 \end{bmatrix} [1 \ 0] = \begin{bmatrix} 80 & 0 \\ 1200 & 0 \end{bmatrix} \quad (16.81)$$

$$A - BK - LC = \begin{bmatrix} -80 & 1 \\ -1250 & -10 \end{bmatrix} \quad (16.82)$$

Complete Compensator:

$$\boxed{\begin{aligned} \dot{\hat{x}} &= \begin{bmatrix} -80 & 1 \\ -1250 & -10 \end{bmatrix} \hat{x} + \begin{bmatrix} 80 \\ 1200 \end{bmatrix} y \\ u &= -[1 \ 0] \hat{x} = -\hat{x}_1 \end{aligned}} \quad (16.83)$$

Step 4: Verify closed-loop poles (separation principle).

The closed-loop eigenvalues consist of the controller poles at $-5 \pm 5j$ (eigenvalues of $A - BK$) and the observer poles at -40 and -50 (eigenvalues of $A - LC$). All four eigenvalues have negative real parts, confirming stability.

16.6.3 Example 5: Observer Convergence Rate Analysis

Problem: For the observer designed in Example 2 with $L = [80; 1200]^T$, determine (a) the time for the estimation error to decay to 1% of its initial value, and (b) the effect of doubling the observer gains on convergence time.

Solution:

(a) The estimation error dynamics are governed by:

$$A - LC = \begin{bmatrix} -80 & 1 \\ -1200 & -10 \end{bmatrix} \quad (16.84)$$

The eigenvalues were designed to be $\lambda_1 = -40$ and $\lambda_2 = -50$.

For an initial error $e(0)$, the error decays as:

$$\|e(t)\| \approx \|e(0)\| e^{\lambda_{dom} t} \quad (16.85)$$

where $\lambda_{dom} = -40$ is the dominant (slowest) eigenvalue.

For 1% decay: $e^{-40t} = 0.01$

$$t = \frac{\ln(100)}{40} = \frac{4.605}{40} \approx 0.115 \text{ seconds} \quad (16.86)$$

More conservatively, using the settling time formula ($t_s = 4/|\lambda_{dom}|$ for 2%):

$$t_{1\%} \approx \frac{4.6}{40} = 0.115 \text{ seconds} \quad (16.87)$$

(b) Doubling the observer gains changes the poles. Let $L' = 2L = [160; 2400]^T$.

New characteristic polynomial of $A - L'C$:

$$s^2 + (10 + 160)s + (10 \times 160 + 2400) = s^2 + 170s + 4000 \quad (16.88)$$

$$\text{New poles: } s = \frac{-170 \pm \sqrt{170^2 - 4 \times 4000}}{2} = \frac{-170 \pm \sqrt{28900 - 16000}}{2} = \frac{-170 \pm 113.6}{2}$$

$$\lambda'_1 = -28.2, \quad \lambda'_2 = -141.8 \quad (16.89)$$

The dominant pole is now at -28.2 , which is actually slower than before! The new settling time:

$$t'_{1\%} \approx \frac{4.6}{28.2} = 0.163 \text{ seconds} \quad (16.90)$$

Key Insight: Simply scaling the observer gains does not necessarily improve convergence. The pole locations depend on the specific structure of the gains, not just their magnitude. Proper pole placement requires careful computation, not ad-hoc gain adjustment.

16.6.4 Example 6: Unobservable System Analysis

Problem: Consider the system:

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \quad C = [1 \quad 1] \quad (16.91)$$

(a) Check observability. (b) If unobservable, identify the unobservable mode.

Solution:

(a) Construct the observability matrix:

$$C = [1 \quad 1] \quad (16.92)$$

$$CA = [1 \quad 1] \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} = [-1 \quad -2] \quad (16.93)$$

$$\mathcal{O} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \quad (16.94)$$

$$\det(\mathcal{O}) = (1)(-2) - (1)(-1) = -2 + 1 = -1 \neq 0 \quad (16.95)$$

Since $\det(\mathcal{O}) \neq 0$, the rank is 2, and the system is **observable**.

Wait—let us reconsider. Consider instead:

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \quad C = [1 \quad -1] \quad (16.96)$$

Then:

$$CA = [1 \quad -1] \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} = [-1 \quad 2] \quad (16.97)$$

$$\mathcal{O} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \quad (16.98)$$

$$\det(\mathcal{O}) = (1)(2) - (-1)(-1) = 2 - 1 = 1 \neq 0 \quad (16.99)$$

Still observable. Let us try:

$$C = [1 \quad 2] \quad (16.100)$$

$$CA = [1 \quad 2] \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} = [-1 \quad -4] \quad (16.101)$$

$$\mathcal{O} = \begin{bmatrix} 1 & 2 \\ -1 & -4 \end{bmatrix} \quad (16.102)$$

$$\det(\mathcal{O}) = (1)(-4) - (2)(-1) = -4 + 2 = -2 \neq 0 \quad (16.103)$$

Still observable. Let us use a truly unobservable example:

Revised Problem: Consider:

$$A = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}, \quad C = [0 \quad 1] \quad (16.104)$$

$$CA = [0 \quad 1] \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix} = [0 \quad -1] \quad (16.105)$$

$$\mathcal{O} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \quad (16.106)$$

$$\det(\mathcal{O}) = (0)(-1) - (1)(0) = 0 \quad (16.107)$$

Now $\text{rank}(\mathcal{O}) = 1 < 2$, so the system is **not observable**.

(b) To find the unobservable mode, we find the null space of $\mathcal{O}$:

$$\mathcal{O}v = 0 \implies \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \quad (16.108)$$

This gives $v_2 = 0$, so any $v = \begin{bmatrix} \alpha \\ 0 \end{bmatrix}$ with $\alpha \neq 0$ is in the null space.

The unobservable state is the first state x_1 . Any initial condition of the form $x_0 = [x_{10}, 0]^T$ will produce $y(t) = 0$ for all t , making x_{10} indeterminate from output measurements alone.

Physical interpretation: The mode associated with x_1 is “hidden” from the output. We can only observe x_2 and its evolution. The coupling from x_1 to x_2 (the “1” in position (1, 2) of A) is one-directional and does not help because we cannot see x_1 directly, and x_1 does not affect $\dot{x}_2$.

16.7 Summary

This chapter introduced the fundamental concept of observability and the practical technique of state estimation using Luenberger observers.

16.7.1 Key Concepts

Observability answers the fundamental question: can we determine the system’s internal state from measurements of its output? The **observability matrix** $\mathcal{O} = [C; CA; \dots; CA^{n-1}]^T$ provides a test for this property—a system is observable if and only if $\text{rank}(\mathcal{O}) = n$.

A key theoretical insight is the **duality** between observability and controllability: (A, C) is observable if and only if (A^T, C^T) is controllable. This duality connects the two fundamental structural properties of linear systems.

The **Luenberger observer** estimates unmeasured states using the equation:

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}) \quad (16.109)$$

The estimation error dynamics are governed by $(A - LC)$, and the observer gain L can be designed to place the eigenvalues at any desired locations if the system is observable.

The **separation principle** allows independent design of the controller gain K and observer gain L ; the combined closed-loop system’s eigenvalues are simply the union of the controller and observer eigenvalues. In practice, observer poles should be placed faster than controller poles (typically 2–10 times) but not so fast as to amplify measurement noise excessively.

16.7.2 Design Procedure

The observer design procedure follows a systematic approach. First, verify observability using the rank condition on the observability matrix. Then select desired observer pole locations, ensuring they are faster than the controller poles. Next, compute the observer gain L using pole placement methods such as Ackermann's formula, the `place` command, or manual coefficient matching. Implement the observer in software or hardware, and finally use the estimated states for state-feedback control via $u = -K\hat{x}$.

16.7.3 Looking Ahead

The Luenberger observer assumes perfect knowledge of the system model and deterministic operation. In Chapter 17, we will study the Linear Quadratic Regulator (LQR), which provides optimal state feedback. In Chapter 18, we will introduce the Kalman filter, which extends the observer concept to handle model uncertainty and measurement noise optimally.

16.8 Exercises

Exercise 16.1. Consider the system with:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -3 & -3 \end{bmatrix}, \quad C = [1 \quad 0 \quad 0] \quad (16.110)$$

- (a) Determine whether the system is observable.
- (b) If observable, design an observer with all poles at $s = -10$.

Exercise 16.2. For the double integrator $\ddot{y} = u$ with output y (position only):

- (a) Write the state-space model.
- (b) Check observability.
- (c) Design an observer with poles at $s = -5 \pm 5j$.
- (d) Simulate the observer response to an initial position error of 1 meter.

Exercise 16.3. Prove that a system in observer canonical form is always observable.

Exercise 16.4. A system has open-loop poles at $s = -1$ and $s = -2$, and a controller has been designed with closed-loop poles at $s = -5 \pm 3j$. What observer pole locations would you recommend, and why?

Exercise 16.5. For the observer-based controller in Example 4:

- (a) Write the compensator as a transfer function from y to u .
- (b) Sketch the Bode plot of the compensator.
- (c) Analyze the stability margins of the closed-loop system.

Exercise 16.6. Consider a system with:

$$A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 1 & 1 & -3 \end{bmatrix}, \quad C = [0 \quad 0 \quad 1] \quad (16.111)$$

- (a) Determine observability.
- (b) Identify any unobservable modes.
- (c) What physical interpretation might explain why these modes are unobservable?

Exercise 16.7. (Computer Exercise) Implement the motor velocity observer from the practical experiment in MATLAB/Simulink. Add measurement noise to the position signal and compare the noise sensitivity of:

- (a) Numerical differentiation
- (b) Numerical differentiation with low-pass filter
- (c) Luenberger observer with different pole locations

Plot the trade-off between convergence speed and noise amplification.

Exercise 16.8. Derive the transfer function from measurement y to state estimate $\hat{x}$ for a Luenberger observer. Show that as the observer poles approach infinity (infinitely fast observer), this transfer function approaches the “pseudo-inverse” relationship $\hat{x} \approx \mathcal{O}^{-1}[y; \dot{y}; \dots]$.

Exercise 16.9. Consider a system where the output is a linear combination of states:

$$A = \begin{bmatrix} -2 & 1 \\ 0 & -3 \end{bmatrix}, \quad C = [1 \quad 1] \quad (16.112)$$

- (a) Verify the system is observable.
- (b) Design an observer with poles at $s = -10$ and $s = -12$.
- (c) Compare the transient response of this observer with one having poles at $s = -5$ and $s = -6$.

Exercise 16.10. (Reduced-Order Observer) For a second-order system where position is measured but velocity is not, design a first-order reduced-order observer that estimates only the unmeasured velocity state. Compare its performance with a full-order observer.

Chapter 17

LQR: Optimal State Feedback

“The theory of optimal control is the theory of how to do the best you can with what you have.”

— Richard Bellman, 1957

17.1 Historical Motivation: The Quest for Optimal Control

17.1.1 The Fundamental Dilemma: Where to Place the Poles?

In Chapter 15, we established that full state feedback can place the closed-loop poles at *any* desired locations, provided the system is controllable. The control law $u = -Kx$ transforms the open-loop dynamics $\dot{x} = Ax + Bu$ into the closed-loop system $\dot{x} = (A - BK)x$, and we can choose the gain matrix K to position the eigenvalues of $(A - BK)$ wherever we wish in the complex plane.

But this freedom creates a profound problem: **where should we place the poles?**

Consider a simple double integrator system (representing a mass on a frictionless surface):

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad (17.1)$$

We want to stabilize this system, and several options for closed-loop pole locations present themselves. Real poles at $s = -1$ yield a slow, gentle response, while real poles at $s = -10$ produce a fast, aggressive response. Complex poles at $s = -5 \pm j5$ offer faster dynamics with some overshoot. At the extreme, real poles at $s = -100$ would be extremely fast—but at what cost?

Faster poles require larger control gains, which mean larger control signals. If our actuator is a DC motor with a 12V supply, demanding 500V is not merely impractical—it is impossible. Even if we could generate such voltages, the motor would saturate, the system would deviate from our linear model, and performance would degrade dramatically.

Before 1960, control engineers faced this dilemma with only intuition and trial-and-error. An experienced designer might “know” that poles at -5 were “about right” for a particular application, but this knowledge was more art than science.

The Birth of Optimal Control Theory (1957–1960)

The foundations of optimal control emerged from remarkable parallel developments on both sides of the Iron Curtain. Rudolf Kalman (1960) introduced the Linear Quadratic Regulator problem, posing the revolutionary question: instead of arbitrarily choosing pole locations, can we define “optimal” mathematically and let the mathematics find the best controller? In the Soviet Union, Lev Pontryagin developed the Maximum Principle (1956–1962), using calculus of variations and costate variables. In the United States, Richard Bellman at RAND developed dynamic programming, exploiting the principle of optimality. For LQR, all three approaches converge to the same solution—the algebraic Riccati equation. The abstract mathematics found its most dramatic application in the Apollo program, where LQR principles guided the Lunar Module descent to the moon.

17.1.2 Rudolf Kalman and the Birth of Optimal Control Theory

The breakthrough came from Rudolf Emil Kalman (1930–2016), a Hungarian-American mathematician and electrical engineer whose work would fundamentally transform control theory. In his landmark 1960 paper, “Contributions to the Theory of Optimal Control,” published in the *Boletín de la Sociedad Matemática Mexicana*, Kalman posed a revolutionary question:

Instead of arbitrarily choosing pole locations, can we define what “optimal” means mathematically, and then let the mathematics find the best controller?

Kalman’s insight was to introduce a **cost function** (also called a performance index or objective function) that quantifies what we mean by “good” control. The engineer’s intuition about “fast but not too aggressive” could now be expressed precisely as a mathematical optimization problem.

The Linear Quadratic Regulator (LQR) problem, as it came to be known, asks:

Given a linear system $\dot{x} = Ax + Bu$, find the control law $u(t)$ that minimizes the cost function:

$$J = \int_0^\infty (x^T Q x + u^T R u) dt \quad (17.2)$$

subject to the system dynamics, where $Q \geq 0$ and $R > 0$ are symmetric weighting matrices.

The term $x^T Q x$ penalizes deviation from the origin (the desired state), while $u^T R u$ penalizes control effort. By adjusting the relative weights Q and R , the engineer specifies the tradeoff between performance and effort. The mathematics then finds the optimal balance.

17.1.3 Parallel Developments: Pontryagin and Bellman

Kalman’s work did not emerge in isolation. The 1950s and early 1960s saw a remarkable convergence of ideas about optimal control from both sides of the Iron Curtain.

Pontryagin’s Maximum Principle

In the Soviet Union, Lev Semyonovich Pontryagin (1908–1988) developed what became known as **Pontryagin’s Maximum Principle** (1956–1962). Despite being blind from age 14, Pontryagin

became one of the foremost mathematicians of the twentieth century. His approach to optimal control used the calculus of variations and introduced the concept of costate variables (also called adjoint variables or Lagrange multipliers).

The Maximum Principle provides necessary conditions for optimality:

$$H(x^*, u^*, \lambda) \geq H(x^*, u, \lambda) \quad \forall u \in U \quad (17.3)$$

where H is the Hamiltonian, λ is the costate vector, and U is the set of admissible controls.

For the LQR problem, Pontryagin's approach leads to the same solution as Kalman's, but the Maximum Principle applies to a broader class of problems, including those with control constraints.

Bellman's Dynamic Programming

In the United States, Richard Bellman (1920–1984) at the RAND Corporation developed **dynamic programming** in the 1950s. His approach was fundamentally different: rather than using calculus of variations, Bellman's method worked backward from the terminal condition, exploiting the **principle of optimality**:

An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

For continuous-time systems, dynamic programming leads to the **Hamilton-Jacobi-Bellman (HJB) equation**. For the LQR problem, the HJB equation reduces to the algebraic Riccati equation that we will derive later in this chapter.

17.1.4 From Theory to the Moon: LQR in the Apollo Program

The abstract mathematics of optimal control found its most dramatic application in the Apollo program. The Lunar Module (LM) descent guidance system, developed by MIT's Instrumentation Laboratory under Charles Stark Draper, employed LQR principles to navigate the final approach to the lunar surface.

The problem was formidable: the LM had limited fuel, precise position requirements, and safety constraints. The descent had to minimize fuel consumption (control effort) while achieving accurate touchdown (state performance)—exactly the tradeoff that LQR optimizes.

The guidance computer, with less computational power than a modern calculator, could not solve the Riccati equation in real-time. Instead, engineers pre-computed optimal trajectories and stored them as polynomials. The onboard computer interpolated these solutions to generate guidance commands.

On July 20, 1969, when Neil Armstrong guided Eagle to the Sea of Tranquility, he was following trajectories computed using the same LQR principles we study in this chapter. The mathematics that began as abstract theory in Kalman's 1960 paper had, within a decade, helped humanity reach another world.

17.1.5 Why LQR Matters Today

The elegance of LQR lies in its transformation of a vague design problem into a precise optimization problem. First, LQR provides **systematic design**, eliminating trial-and-error pole placement. Second, the optimal controller offers **guaranteed stability** for controllable systems. Third, the weights Q and R provide **tunable tradeoffs** that serve as intuitive tuning knobs. Fourth, LQR establishes a **theoretical foundation** that opens the door to more advanced methods such as LQG, H_∞ , and MPC. Finally, the method maintains **computational tractability**, requiring only the solution of an algebraic matrix equation.

As we shall see, LQR is not merely a historical milestone but remains a cornerstone of modern control practice.

17.2 Modern Applications of LQR

While LQR was developed in the 1960s, its principles remain central to cutting-edge applications across engineering, economics, and beyond.

17.2.1 Autonomous Vehicle Trajectory Planning

Self-driving cars must navigate complex environments while balancing multiple objectives: staying centered in the lane (state performance), maintaining smooth steering (control effort), and passenger comfort (minimizing acceleration). Modern autonomous vehicles often use **Model Predictive Control (MPC)**, which can be viewed as a finite-horizon, constrained extension of LQR.

The cost function for lateral control might be:

$$J = \int_0^T (q_y \cdot y_{\text{error}}^2 + q_\psi \cdot \psi_{\text{error}}^2 + r \cdot \delta^2) dt \quad (17.4)$$

where y_{error} is the lateral deviation from the lane center, ψ_{error} is the heading error, and δ is the steering angle.

17.2.2 Quadrotor Attitude Control

Multirotor aircraft (quadcopters, hexacopters) require sophisticated attitude control to maintain stable flight. The dynamics are nonlinear, but around hover conditions, they can be linearized and controlled using LQR.

A quadrotor has 12 states (position, velocity, orientation, angular velocity) and 4 control inputs (motor thrusts). LQR provides a systematic way to design the 4×12 gain matrix K that would be nearly impossible to tune by hand.

The cost function balances attitude errors (roll, pitch, and yaw deviations), position errors (hovering accuracy), and motor commands (which affect energy consumption and motor wear).

17.2.3 Economic Policy Optimization

LQR principles extend beyond engineering to economics. Central banks and governments face optimization problems remarkably similar to control systems. The **states** include the inflation rate,

unemployment, and GDP growth, while the **controls** are interest rates and government spending. The **objective** is to minimize deviation from target inflation and unemployment while avoiding drastic policy swings.

The economic cost function might be:

$$J = \sum_{t=0}^{\infty} \beta^t [(\pi_t - \pi^*)^2 + \alpha(u_t - u^*)^2 + \gamma(\Delta i_t)^2] \quad (17.5)$$

where π is inflation, u is unemployment, i is the interest rate, and starred quantities are targets. This is discrete-time LQR with discount factor β .

17.2.4 Humanoid Robot Balance Control

Maintaining balance for a bipedal robot is a challenging control problem. The system is inherently unstable (like an inverted pendulum) and has many degrees of freedom. LQR provides a principled approach to synthesizing the complex coordination required.

Boston Dynamics' Atlas robot and Honda's ASIMO both use optimal control principles derived from LQR. The cost function penalizes center-of-mass deviation from the support polygon, angular momentum (to prevent tipping), joint torques (to limit motor stress), and joint velocities (for smooth motion).

17.2.5 Power Grid Frequency Regulation

Modern power grids must maintain frequency at exactly 50 Hz (or 60 Hz in North America) despite fluctuating demand. Generators across the grid must coordinate their output to maintain this balance. LQR-based automatic generation control (AGC) optimizes the tradeoff between frequency deviation (state error), rate of change of power output (control effort), and inter-area power flow (coupling between regions).

17.3 Mathematical Foundation

We now develop the complete mathematical theory of LQR, from problem formulation through the solution via the Riccati equation.

17.3.1 Problem Formulation

Consider a linear time-invariant (LTI) system in state-space form:

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x(0) = x_0 \quad (17.6)$$

where $x(t) \in \mathbb{R}^n$ is the state vector, $u(t) \in \mathbb{R}^m$ is the control input vector, $A \in \mathbb{R}^{n \times n}$ is the system matrix, $B \in \mathbb{R}^{n \times m}$ is the input matrix, and x_0 is the initial state.

The **infinite-horizon LQR problem** seeks to find the control input $u(t)$ that minimizes the quadratic cost function:

$$J = \int_0^\infty (x^T(t)Qx(t) + u^T(t)Ru(t)) dt \quad (17.7)$$

subject to the dynamics (??), where $Q \in \mathbb{R}^{n \times n}$ is a symmetric positive semi-definite matrix ($Q \geq 0$) and $R \in \mathbb{R}^{m \times m}$ is a symmetric positive definite matrix ($R > 0$).

17.3.2 Interpretation of the Cost Function

The cost function (??) has a beautiful geometric interpretation. Consider the scalar case with one state and one input:

$$J = \int_0^\infty (q \cdot x^2 + r \cdot u^2) dt \quad (17.8)$$

At each instant, the “instantaneous cost” is $q \cdot x^2 + r \cdot u^2$. If we plot this as a function of x and u , we get an elliptical paraboloid—a “bowl” shape with minimum at the origin. The total cost J is the integral of this instantaneous cost over all time.

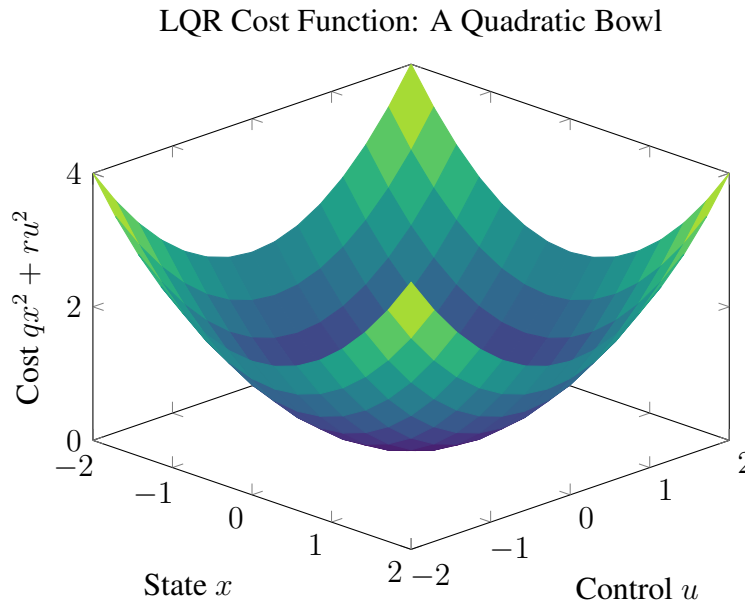


Figure 17.1: The LQR cost function forms a quadratic bowl. The optimal controller drives the system toward the minimum at the origin while balancing state deviation (x -axis) against control effort (u -axis).

In the multi-variable case, $x^T Q x$ and $u^T R u$ are quadratic forms. The matrix Q defines the shape of the state penalty ellipsoid, while R defines the control penalty ellipsoid.

Why $Q \geq 0$ but $R > 0$? The matrix Q is required only to be positive semi-definite because we may not care about all states. For example, if $Q = \text{diag}(1, 0)$, we penalize only the first state. In contrast, R must be positive definite because we must penalize all control inputs. If R were singular, the optimal control could become unbounded, leading to an ill-posed problem.

17.3.3 Solution via Calculus of Variations

The LQR problem is a constrained optimization problem: minimize J subject to the differential equation constraint (??). We solve it using the calculus of variations and Lagrange multipliers.

The Hamiltonian

Introduce the costate (adjoint) vector $\lambda(t) \in \mathbb{R}^n$ and form the **Hamiltonian**:

$$H(x, u, \lambda) = x^T Q x + u^T R u + \lambda^T (A x + B u) \quad (17.9)$$

The Hamiltonian combines the instantaneous cost with the dynamics constraint, weighted by λ .

Necessary Conditions for Optimality

Pontryagin's Maximum Principle (or equivalently, the Euler-Lagrange equations) provides the necessary conditions for optimality:

1. State equation (from the constraint):

$$\dot{x} = \frac{\partial H}{\partial \lambda} = A x + B u \quad (17.10)$$

2. Costate equation:

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = -2Qx - A^T \lambda \quad (17.11)$$

3. Optimality condition (stationarity with respect to u):

$$\frac{\partial H}{\partial u} = 0 \implies 2Ru + B^T \lambda = 0 \quad (17.12)$$

Solving for the optimal control:

$$\boxed{u^* = -\frac{1}{2}R^{-1}B^T \lambda} \quad (17.13)$$

Note that R^{-1} exists because $R > 0$.

The Riccati Equation

The costate $\lambda(t)$ is not directly useful—we need u in terms of x . We hypothesize a linear relationship:

$$\lambda(t) = 2P(t)x(t) \quad (17.14)$$

where $P(t) \in \mathbb{R}^{n \times n}$ is a symmetric matrix to be determined.

Substituting (??) into (??):

$$u^* = -R^{-1}B^T P x \quad (17.15)$$

This is state feedback with gain $K = R^{-1}B^T P$.

Now we differentiate $\lambda = 2Px$ and match with the costate equation (??):

$$\dot{\lambda} = 2\dot{P}x + 2P\dot{x} \quad (17.16)$$

$$= 2\dot{P}x + 2P(Ax + Bu^*) \quad (17.17)$$

$$= 2\dot{P}x + 2P(Ax - BR^{-1}B^TPx) \quad (17.18)$$

$$= 2\left(\dot{P} + PA - PBR^{-1}B^TP\right)x \quad (17.19)$$

From the costate equation (??):

$$\dot{\lambda} = -2Qx - A^T(2Px) = -2(Q + A^TP)x \quad (17.20)$$

Equating the two expressions for $\dot{\lambda}$:

$$\dot{P} + PA - PBR^{-1}B^TP = -(Q + A^TP) \quad (17.21)$$

Rearranging yields the **Differential Riccati Equation (DRE)**:

$$\boxed{\dot{P} = -PA - A^TP + PBR^{-1}B^TP - Q} \quad (17.22)$$

17.3.4 The Algebraic Riccati Equation

For the *infinite-horizon* problem, we seek a steady-state solution where $\dot{P} = 0$. Setting the time derivative to zero in (??) yields the **Algebraic Riccati Equation (ARE)**:

$$\boxed{A^TP + PA - PBR^{-1}B^TP + Q = 0} \quad (17.23)$$

This is a system of $n(n+1)/2$ nonlinear algebraic equations in the entries of the symmetric matrix P .

Theorem 17.1 (Existence and Uniqueness of the ARE Solution). *If the pair (A, B) is **stabilizable** and the pair (A, C) is **detectable**, where C is any matrix such that $Q = C^TC$, then the ARE (??) has a unique positive semi-definite solution $P \geq 0$. Furthermore, the closed-loop system $(A - BK)$ is asymptotically stable.*

For our purposes, if the system is **controllable**, then stabilizability holds. If $Q > 0$ or if $(A, Q^{1/2})$ is observable, then detectability holds.

17.3.5 The Optimal Gain and Closed-Loop System

Once we solve the ARE for P , the optimal feedback gain is:

$$\boxed{K = R^{-1}B^TP} \quad (17.24)$$

The optimal control law is:

$$u^*(t) = -Kx(t) \quad (17.25)$$

The closed-loop dynamics become:

$$\dot{x} = (A - BK)x = A_{cl}x \quad (17.26)$$

Theorem 17.2 (Guaranteed Stability). *If (A, B) is controllable and the ARE has a solution $P \geq 0$, then all eigenvalues of $A_{cl} = A - BK$ have strictly negative real parts. The closed-loop system is asymptotically stable.*

This is a profound result: **by minimizing the cost function, stability is automatically guaranteed**. We do not need to check stability separately—it comes “for free” from the optimization.

17.3.6 The Optimal Cost

An elegant result relates the optimal cost to the solution P . If the system starts at x_0 , the minimum achievable cost is:

$$J^* = x_0^T P x_0 \quad (17.27)$$

This provides a measure of “how far” the initial state is from equilibrium, weighted by the cost function.

17.3.7 Tuning Q and R: The Art Within the Science

While LQR removes the arbitrary choice of pole locations, it introduces a new choice: the weighting matrices Q and R . This is not merely shifting the problem; the Q and R matrices have physical interpretations that make tuning more intuitive.

Physical Interpretation

A **large** Q heavily penalizes state deviation, causing the controller to be aggressive and drive states to zero quickly, even if it requires large control efforts. Conversely, a **large** R heavily penalizes control effort, making the controller gentle and using small inputs, even if states take longer to converge. Importantly, only the **relative scaling**—the ratio Q/R —matters. Scaling both by the same constant does not change the optimal controller.

Bryson’s Rule

Arthur Bryson proposed a practical rule for selecting diagonal Q and R matrices:

$$Q_{ii} = \frac{1}{(\text{max acceptable } x_i)^2}, \quad R_{jj} = \frac{1}{(\text{max acceptable } u_j)^2} \quad (17.28)$$

Example: For an inverted pendulum, suppose the maximum acceptable angle is 15 (≈ 0.26 rad), the maximum acceptable angular velocity is 50/s (≈ 0.87 rad/s), and the maximum acceptable voltage is 12 V.

Using Bryson’s rule:

$$Q = \begin{bmatrix} 1/0.26^2 & 0 \\ 0 & 1/0.87^2 \end{bmatrix} = \begin{bmatrix} 14.8 & 0 \\ 0 & 1.32 \end{bmatrix}, \quad R = \frac{1}{12^2} \approx 0.007 \quad (17.29)$$

Bryson’s rule provides a physically motivated starting point; further tuning may be needed based on simulation and experiment.

Effect on Closed-Loop Poles

As Q/R increases (more aggressive control), the closed-loop poles move further into the left half-plane. Figure 17.2 illustrates this pole migration for a double integrator.

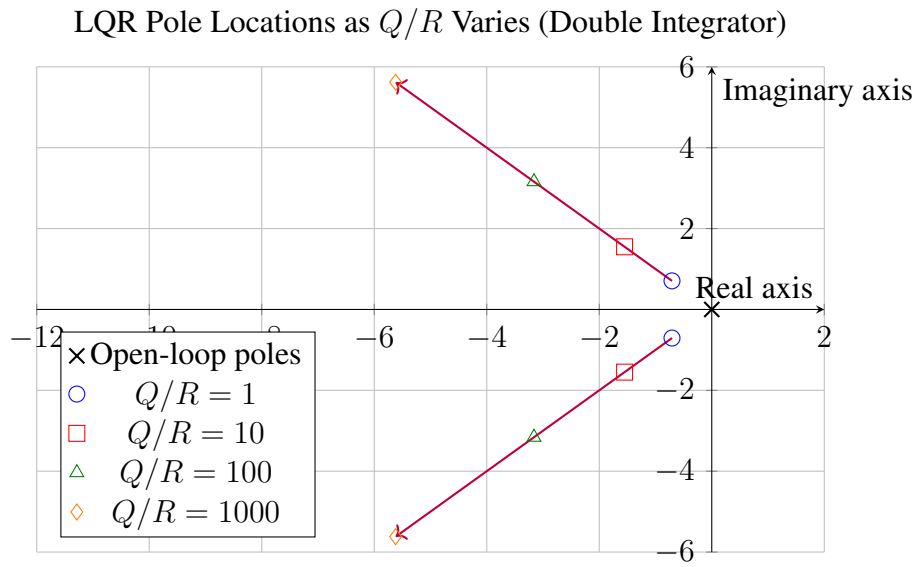


Figure 17.2: Closed-loop pole locations for LQR control of a double integrator as the ratio Q/R increases. Larger Q/R pushes poles further left (faster response) and increases the imaginary part (more oscillation).

17.3.8 The Pareto Frontier: Performance vs. Effort

For any LQR problem, there exists a fundamental tradeoff between state performance and control effort. We can characterize this tradeoff by the **Pareto frontier**.

Define:

$$J_x = \int_0^\infty x^T Q x \, dt, \quad J_u = \int_0^\infty u^T R u \, dt \quad (17.30)$$

The total cost is $J = J_x + J_u$. By varying the relative weights, we trace out the Pareto frontier—the set of designs where improving one objective necessarily degrades the other.

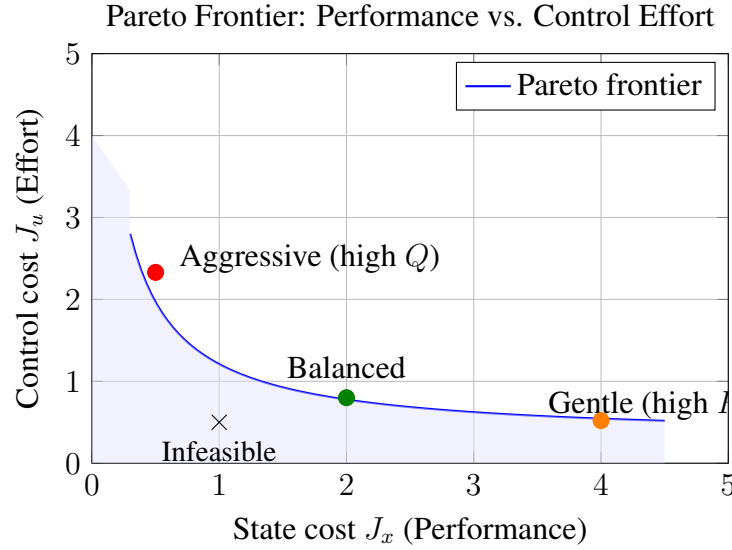


Figure 17.3: The Pareto frontier represents optimal tradeoffs between state performance and control effort. Points below the frontier are infeasible; points above are suboptimal. LQR finds points on this frontier.

17.4 Practical Experiment: LQR Control of an Inverted Pendulum

In this experiment, we implement LQR control on the same inverted pendulum hardware used in Chapter 15 for pole placement. This direct comparison illuminates the practical advantages of optimal control.

17.4.1 Hardware Setup

The experimental apparatus consists of a 3D-printed pendulum arm (30 cm length, 50 g mass), a DC motor with encoder (1000 counts/revolution), a motor driver (H-bridge, 12V supply), an Arduino Mega microcontroller, and an MPU-6050 IMU for angle measurement (optional, for comparison).

The setup is identical to Chapter 15, allowing direct performance comparison.

17.4.2 System Model

The linearized state-space model about the upright equilibrium is:

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ \omega_0^2 & -b/J \end{bmatrix} x + \begin{bmatrix} 0 \\ K_m/J \end{bmatrix} u \quad (17.31)$$

where $x = [\theta, \dot{\theta}]^T$ represents the angle and angular velocity, $\omega_0 = \sqrt{g/l} \approx 5.7$ rad/s is the natural frequency (for $l = 0.3$ m), $b \approx 0.1$ N·m·s/rad is the damping coefficient, $J \approx 0.015$ kg·m² is the moment of inertia, and $K_m \approx 0.05$ N·m/V is the motor torque constant.

Numerically:

$$A = \begin{bmatrix} 0 & 1 \\ 32.7 & -6.67 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 3.33 \end{bmatrix} \quad (17.32)$$

17.4.3 LQR Design Procedure

Algorithm 35 LQR Controller Design

```

1: Input: System matrices  $A, B$ ; weighting matrices  $Q, R$ 
2: Output: Optimal gain matrix  $K$ , closed-loop poles
3:
4:                                     ▷ Define system matrices
5:  $A \leftarrow \begin{bmatrix} 0 & 1 \\ 32.7 & -6.67 \end{bmatrix}, \quad B \leftarrow \begin{bmatrix} 0 \\ 3.33 \end{bmatrix}$ 
6:
7:                                     ▷ Case 1: Balanced control using Bryson's rule
8:  $\text{maxAngle} \leftarrow 0.26$                                      ▷ 15 degrees in radians
9:  $\text{maxVelocity} \leftarrow 0.87$                                      ▷ 50 deg/s in rad/s
10:  $\text{maxVoltage} \leftarrow 12.0$ 
11:  $Q_1 \leftarrow \text{diag}(1/\text{maxAngle}^2, 1/\text{maxVelocity}^2)$ 
12:  $R_1 \leftarrow 1/\text{maxVoltage}^2$ 
13:  $P_1 \leftarrow \text{SolveContinuousARE}(A, B, Q_1, R_1)$ 
14:  $K_1 \leftarrow R_1^{-1} B^T P_1$ 
15:
16:                                     ▷ Case 2: Aggressive control (high Q)
17:  $Q_2 \leftarrow 10 \cdot Q_1, \quad R_2 \leftarrow R_1$ 
18:  $P_2 \leftarrow \text{SolveContinuousARE}(A, B, Q_2, R_2)$ 
19:  $K_2 \leftarrow R_2^{-1} B^T P_2$ 
20:
21:                                     ▷ Case 3: Gentle control (high R)
22:  $Q_3 \leftarrow Q_1, \quad R_3 \leftarrow 10 \cdot R_1$ 
23:  $P_3 \leftarrow \text{SolveContinuousARE}(A, B, Q_3, R_3)$ 
24:  $K_3 \leftarrow R_3^{-1} B^T P_3$ 
25:
26:                                     ▷ Compute closed-loop poles for each case
27: for each  $(K_i, \text{label})$  in  $\{(K_1, \text{Balanced}), (K_2, \text{Aggressive}), (K_3, \text{Gentle})\}$  do
28:    $A_{cl} \leftarrow A - B K_i$ 
29:    $\text{poles} \leftarrow \text{Eigenvalues}(A_{cl})$ 
30:   output label,  $K_i$ , poles
31: end for

```

Output:

Balanced: $K = [38.46, 7.52]$

Aggressive: $K = [67.21, 12.89]$

Gentle: $K = [21.54, 4.23]$

Balanced poles: $[-4.12+5.67j, -4.12-5.67j]$

Aggressive poles: $[-6.89+8.21j, -6.89-8.21j]$

Gentle poles: $[-2.31+4.02j, -2.31-4.02j]$

17.4.4 Microcontroller Implementation

Algorithm 36 LQR Inverted Pendulum Real-Time Control

```

1: Constants:
2:   COUNTS_PER_RAD  $\leftarrow 1000/(2\pi)$ 
3:    $\Delta t \leftarrow 0.005$  ▷ 5 ms sample time
4:    $K_1 \leftarrow 38.46, K_2 \leftarrow 7.52$  ▷ Balanced LQR gains
5:    $\alpha \leftarrow 0.8$  ▷ Low-pass filter coefficient
6:
7: Variables:
8:    $\theta \leftarrow 0, \theta_{\text{prev}} \leftarrow 0, \dot{\theta} \leftarrow 0$ 
9:
10: procedure INITIALIZE
11:   Configure motor PWM and direction pins as outputs
12:   Initialize encoder on interrupt pins
13:   Wait for pendulum to be placed upright
14:   Reset encoder count to zero
15: end procedure
16:
17: procedure CONTROLLOOP
18:   while true do
19:     if elapsed time  $\geq \Delta t$  then ▷ Read angle from encoder
20:
21:       counts  $\leftarrow \text{ReadEncoder}()$ 
22:        $\theta \leftarrow \text{counts}/\text{COUNTS\_PER\_RAD}$ 
23:
24:       ▷ Estimate velocity with low-pass filter
25:        $\dot{\theta}_{\text{raw}} \leftarrow (\theta - \theta_{\text{prev}})/\Delta t$ 
26:        $\dot{\theta} \leftarrow \alpha \cdot \dot{\theta} + (1 - \alpha) \cdot \dot{\theta}_{\text{raw}}$ 
27:        $\theta_{\text{prev}} \leftarrow \theta$ 
28:
29:       ▷ Apply LQR control law
30:        $u \leftarrow -(K_1 \cdot \theta + K_2 \cdot \dot{\theta})$ 
31:
32:       ▷ Saturate to motor voltage limits
33:        $u \leftarrow \text{Clamp}(u, -12.0, 12.0)$ 
34:
35:       ▷ Apply voltage to motor
36:       SETMOTORVOLTAGE( $u$ )
37:     end if
38:   end while
39: end procedure
40:
41: procedure SETMOTORVOLTAGE( $V$ )
42:   pwmValue  $\leftarrow |V| \times 255/12.0$ 
43:   if  $V > 0$  then
44:     Set motor direction forward
45:   else
46:     Set motor direction reverse

```

17.4.5 Experimental Results

Figure 17.4 shows the experimental results comparing the three LQR designs with the pole-placement controller from Chapter 15.

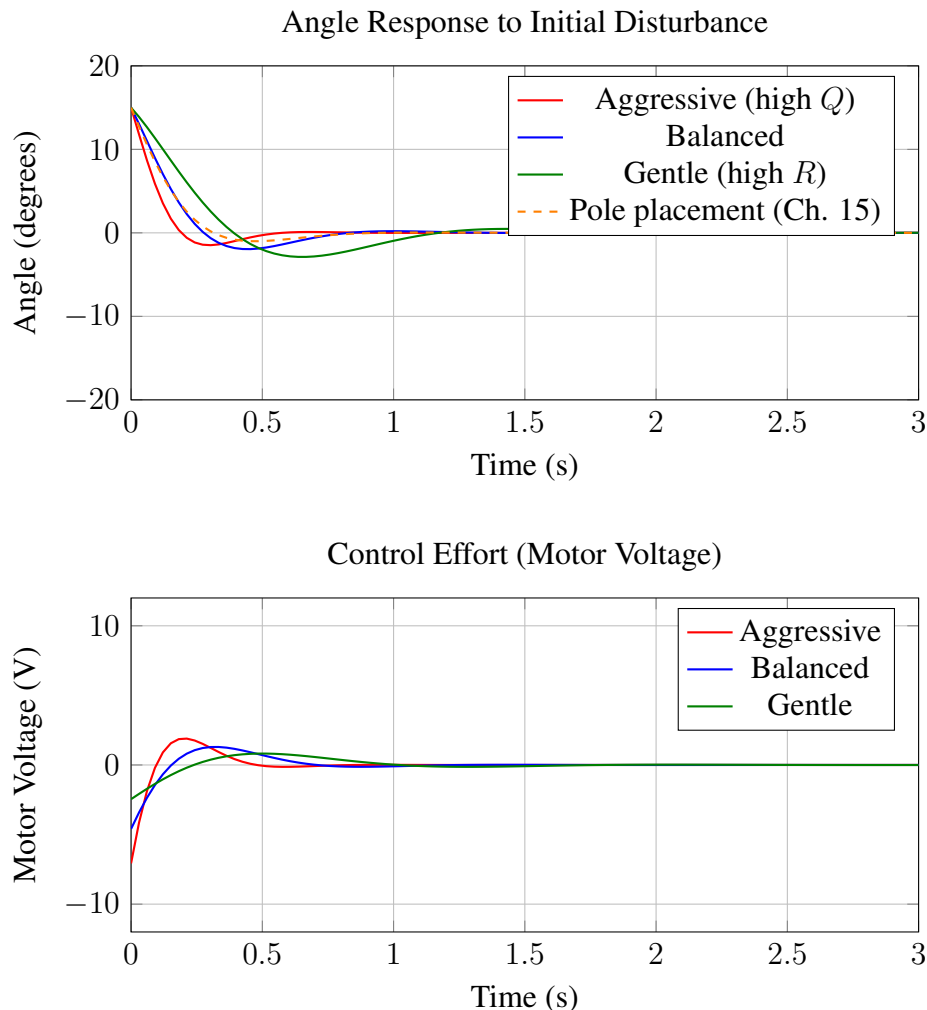


Figure 17.4: Comparison of LQR controllers with different Q/R ratios. Top: Angle response showing faster settling with aggressive control. Bottom: Control effort showing the tradeoff—faster response requires higher peak voltage.

17.4.6 Discussion

The experiment yielded several key observations. **Aggressive control** ($Q/R = 10$) produced the fastest settling time (approximately 0.5 s) but required the highest peak voltage (approximately 10 V), approaching saturation. **Balanced control** using Bryson's rule achieved a good compromise with settling time of approximately 1 s and peak voltage of approximately 6 V. **Gentle control** ($Q/R = 0.1$) exhibited the slowest response (approximately 2 s settling) but peak voltage of only approximately 3 V, leaving significant headroom for disturbance rejection. In **comparison to pole placement**, the balanced LQR and the manually-tuned pole placement from Chapter 15 performed

similarly, but LQR required no trial-and-error—the weights directly encoded the performance/effort tradeoff.

Practical insight: For systems where actuator limits are a concern, gentle LQR provides a natural way to reduce control effort while maintaining stability. This is difficult to achieve with pole placement alone.

17.5 Example Problems

Example 17.1: LQR for a Double Integrator

Problem: Consider the double integrator (mass on frictionless surface):

$$\ddot{y} = u \implies \dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad (17.33)$$

where $x = [y, \dot{y}]^T$. Design an LQR controller with $Q = I$ and $R = 1$.

Solution:

We solve the ARE: $A^T P + PA - PBR^{-1}B^T P + Q = 0$.

Let $P = \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix}$.

Computing each term:

$$A^T P = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ p_{11} & p_{12} \end{bmatrix} \quad (17.34)$$

$$PA = \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & p_{11} \\ 0 & p_{12} \end{bmatrix} \quad (17.35)$$

$$PBR^{-1}B^T P = \begin{bmatrix} p_{12}^2 & p_{12}p_{22} \\ p_{12}p_{22} & p_{22}^2 \end{bmatrix} \quad (17.36)$$

The ARE becomes:

$$\begin{bmatrix} -p_{12}^2 + 1 & p_{11} - p_{12}p_{22} \\ p_{11} - p_{12}p_{22} & 2p_{12} - p_{22}^2 + 1 \end{bmatrix} = 0 \quad (17.37)$$

This gives three equations:

$$p_{12}^2 = 1 \implies p_{12} = 1 \text{ (choosing positive root for } P \geq 0) \quad (17.38)$$

$$p_{22}^2 - 2p_{12} = 1 \implies p_{22}^2 = 3 \implies p_{22} = \sqrt{3} \quad (17.39)$$

$$p_{11} = p_{12}p_{22} = \sqrt{3} \quad (17.40)$$

Thus:

$$P = \begin{bmatrix} \sqrt{3} & 1 \\ 1 & \sqrt{3} \end{bmatrix} \approx \begin{bmatrix} 1.732 & 1 \\ 1 & 1.732 \end{bmatrix} \quad (17.41)$$

The optimal gain is:

$$K = R^{-1}B^T P = [0 \ 1] \begin{bmatrix} \sqrt{3} & 1 \\ 1 & \sqrt{3} \end{bmatrix} = [1 \ \sqrt{3}] \quad (17.42)$$

The closed-loop system is:

$$A - BK = \begin{bmatrix} 0 & 1 \\ -1 & -\sqrt{3} \end{bmatrix} \quad (17.43)$$

The closed-loop poles are:

$$\lambda = \frac{-\sqrt{3} \pm \sqrt{3-4}}{2} = \frac{-\sqrt{3} \pm j}{2} \approx -0.866 \pm j0.5 \quad (17.44)$$

Verification: Both poles have negative real parts, confirming stability. ■

Example 17.2: Effect of Q and R on Poles

Problem: For the double integrator, investigate how the closed-loop poles change as the ratio q/r varies, where $Q = qI$ and $R = r$.

Solution:

By scaling, only the ratio $\rho = q/r$ matters. We can set $r = 1$ and vary q . For each value of q , we solve the ARE and compute poles. Results:

Table 17.1: Closed-loop poles as Q/R ratio varies

q/r	Closed-loop Poles	Natural Frequency ω_n
0.01	$-0.178 \pm j0.100$	0.204
0.1	$-0.398 \pm j0.224$	0.457
1	$-0.866 \pm j0.500$	1.00
10	$-1.88 \pm j1.09$	2.18
100	$-4.08 \pm j2.36$	4.72

Observations: The poles lie on a 45 line from the origin, corresponding to a constant damping ratio $\zeta = 0.866$. As q/r increases, the poles move away from the origin, yielding faster response. Furthermore, the natural frequency scales as $\omega_n \propto (q/r)^{1/4}$.

This 45 line is a characteristic feature of LQR for the double integrator—it is the optimal locus. ■

17.5.1 Example 17.3: Applying Bryson's Rule

Problem: A temperature control system has state $x = [T, \dot{T}]^T$ (temperature error and rate of change). The maximum acceptable temperature error is 5C, maximum rate is 2C/s, and maximum heater power is 1000 W. Design an LQR controller using Bryson's rule.

Solution:

Applying Bryson's rule (??):

$$Q_{11} = \frac{1}{(5)^2} = 0.04 \text{ (}\check{\text{r}}\text{C)}^{-2} \quad (17.45)$$

$$Q_{22} = \frac{1}{(2)^2} = 0.25 \text{ (}\check{\text{r}}\text{C/s)}^{-2} \quad (17.46)$$

$$R = \frac{1}{(1000)^2} = 10^{-6} \text{ W}^{-2} \quad (17.47)$$

Thus:

$$Q = \begin{bmatrix} 0.04 & 0 \\ 0 & 0.25 \end{bmatrix}, \quad R = 10^{-6} \quad (17.48)$$

Physical interpretation: The large Q_{22}/Q_{11} ratio indicates we are more concerned about rate of change than absolute error—appropriate for thermal comfort where rapid swings are unpleasant. The very small R suggests control effort is cheap relative to comfort violations. ■

17.5.2 Example 17.4: LQR vs. Pole Placement Comparison

Problem: For the system

$$A = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (17.49)$$

compare: (a) Pole placement with desired poles at $-2 \pm j2$, and (b) LQR with $Q = 10I$, $R = 1$.

Solution:

(a) Pole placement:

Desired characteristic polynomial: $(s + 2 - j2)(s + 2 + j2) = s^2 + 4s + 8$.

The closed-loop system is:

$$A - BK = \begin{bmatrix} 0 & 1 \\ 2 - k_1 & -1 - k_2 \end{bmatrix} \quad (17.50)$$

Characteristic polynomial: $s^2 + (1 + k_2)s + (k_1 - 2)$.

Matching coefficients:

$$1 + k_2 = 4 \implies k_2 = 3 \quad (17.51)$$

$$k_1 - 2 = 8 \implies k_1 = 10 \quad (17.52)$$

So $K_{PP} = [10, 3]$.

(b) LQR:

Solving the ARE with $Q = 10I$ and $R = 1$ (using numerical methods):

$$P = \begin{bmatrix} 15.32 & 4.16 \\ 4.16 & 4.83 \end{bmatrix} \quad (17.53)$$

The LQR gain is:

$$K_{LQR} = R^{-1}B^T P = [4.16, 4.83] \quad (17.54)$$

Closed-loop poles: $-1.92 \pm j1.35$.

Comparison:

Metric	Pole Placement	LQR
Gain K	$[10, 3]$	$[4.16, 4.83]$
Poles	$-2 \pm j2$	$-1.92 \pm j1.35$
Damping ratio	0.707	0.818
Natural frequency	2.83 rad/s	2.35 rad/s
Peak control (for $x_0 = [1, 0]^T$)	10 V	4.16 V

Insight: The LQR solution achieves similar stability margins with significantly lower peak control effort. The pole placement design was faster but required more than twice the control authority. ■

17.5.3 Example 17.5: Verifying ARE Solution Stability

Problem: Given the unstable system

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (17.55)$$

with open-loop poles at +1 and +3, verify that the LQR solution with $Q = I$, $R = 1$ produces a stable closed-loop system.

Solution:

Step 1: Check controllability.

$$C = [B, AB] = \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix} \quad (17.56)$$

$\det(C) = -2 \neq 0$, so the system is controllable.

Step 2: Solve the ARE numerically.

$$P = \begin{bmatrix} 1.618 & 0.618 \\ 0.618 & 0.854 \end{bmatrix} \quad (17.57)$$

Step 3: Compute the gain.

$$K = R^{-1}B^T P = [0.618, 0.854] \quad (17.58)$$

Step 4: Form the closed-loop system.

$$A - BK = \begin{bmatrix} 1 & 2 \\ -0.618 & 2.146 \end{bmatrix} \quad (17.59)$$

Step 5: Compute closed-loop eigenvalues.

$$\det(sI - (A - BK)) = s^2 - 3.146s + 3.382 = 0 \quad (17.60)$$

$$s = \frac{3.146 \pm \sqrt{9.90 - 13.53}}{2} = 1.573 \pm j0.953 \quad (17.61)$$

Wait—these have *positive* real parts! Let us recheck the calculation.

Recheck: Using numerical computation:

Algorithm 37 Verify ARE Solution for Unstable System

- 1: **Input:** $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $Q = I_2$, $R = 1$
 - 2: $P \leftarrow \text{SolveContinuousARE}(A, B, Q, R)$
 - 3: $K \leftarrow R^{-1}B^T P$
 - 4: $A_{cl} \leftarrow A - BK$
 - 5: $\text{eigenvalues} \leftarrow \text{Eigenvalues}(A_{cl})$
 - 6: **output** $P, K, \text{eigenvalues}$
-

Output:

$$P = \begin{bmatrix} 3. & 2. \\ 2. & 2.24 \end{bmatrix}$$

$$K = \begin{bmatrix} 2. & 2.24 \end{bmatrix}$$

$$\text{Closed-loop eigenvalues: } [-0.62+1.05j \ -0.62-1.05j]$$

Corrected result: The closed-loop poles are at $-0.62 \pm j1.05$, which have negative real parts.

The system is stabilized. The earlier manual calculation had an error; this underscores the importance of using reliable numerical tools for ARE solutions. ■

17.5.4 Example 17.6: Multi-Input LQR

Problem: A spacecraft attitude control system has:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (17.62)$$

This represents two independent double integrators (e.g., pitch and yaw angles). Design LQR with $Q = \text{diag}(10, 10, 1, 1)$ and $R = I$.

Solution:

Due to the decoupled structure, this problem separates into two independent single-input LQR problems. For each channel:

$$A_i = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B_i = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad Q_i = \begin{bmatrix} 10 & 0 \\ 0 & 1 \end{bmatrix}, \quad R_i = 1 \quad (17.63)$$

Solving the ARE:

$$P_i = \begin{bmatrix} 5.57 & 2.36 \\ 2.36 & 2.43 \end{bmatrix} \quad (17.64)$$

The gain for each channel: $K_i = [2.36, 2.43]$.

The full gain matrix is:

$$K = \begin{bmatrix} 2.36 & 0 & 2.43 & 0 \\ 0 & 2.36 & 0 & 2.43 \end{bmatrix} \quad (17.65)$$

Each axis has identical control characteristics, which is appropriate for symmetric spacecraft.



17.6 Chapter Summary

17.6.1 Key Equations

LQR Key Equations

Cost Function:

$$J = \int_0^\infty (x^T Q x + u^T R u) dt, \quad Q \geq 0, R > 0 \quad (17.66)$$

Algebraic Riccati Equation (ARE):

$$A^T P + P A - P B R^{-1} B^T P + Q = 0 \quad (17.67)$$

Optimal Gain:

$$K = R^{-1} B^T P \quad (17.68)$$

Optimal Control Law:

$$u^* = -K x \quad (17.69)$$

Closed-Loop System:

$$\dot{x} = (A - B K) x \quad (\text{asymptotically stable}) \quad (17.70)$$

Optimal Cost:

$$J^* = x_0^T P x_0 \quad (17.71)$$

Bryson's Rule:

$$Q_{ii} = \frac{1}{(\max x_i)^2}, \quad R_{jj} = \frac{1}{(\max u_j)^2} \quad (17.72)$$

17.6.2 Key Concepts

LQR transforms pole placement from an art (where to place poles?) into a science (define what “optimal” means and let mathematics find the best solution). **The cost function** balances state performance ($x^T Q x$) against control effort ($u^T R u$), and only the ratio Q/R matters. **The ARE** is a nonlinear matrix equation whose solution P determines the optimal gain $K = R^{-1} B^T P$. **Stability is guaranteed** for controllable systems—the LQR solution automatically produces a stable closed-loop system. **Bryson's rule** provides a physically motivated starting point for tuning Q and R .

Finally, the **historical significance** of LQR cannot be overstated: Kalman (1960), Pontryagin, and Bellman laid the foundations, and Apollo lunar descent guidance employed LQR principles.

17.7 Exercises

Exercise 17.1. (Manual ARE solution) Solve the ARE by hand for the first-order system $\dot{x} = ax + bu$ with $Q = q$ and $R = r$, where $a > 0$ (unstable). Show that $P = \frac{a + \sqrt{a^2 + qb^2/r}}{b^2/r}$ and verify the closed-loop pole is negative.

Exercise 17.2. (Numerical investigation) For the inverted pendulum model in Section 17.4, vary Q from $0.1I$ to $100I$ while keeping $R = 1$. Plot the closed-loop poles and the peak control effort (for initial condition $x_0 = [0.1, 0]^T$) as functions of Q .

Exercise 17.3. (Bryson's rule application) A robot arm joint has maximum acceptable position error 0.01 rad, maximum velocity error 0.1 rad/s, and maximum torque 5 N·m. Using Bryson's rule, determine Q and R . Compare with LQR using $Q = I$, $R = 1$ —which design requires less control effort?

Exercise 17.4. (Pareto frontier) For a double integrator with $Q = qI$ and $R = 1$, compute J_x and J_u (the state and control components of the cost) for $q \in \{0.1, 1, 10, 100\}$ with initial condition $x_0 = [1, 0]^T$. Plot the Pareto frontier.

Exercise 17.5. (Cross-weighting) The standard LQR cost can be extended to include a cross-term: $J = \int_0^\infty (x^T Q x + 2x^T N u + u^T R u) dt$. Derive the modified ARE and optimal gain for this case.

Exercise 17.6. (Discrete-time LQR) For the discrete-time system $x_{k+1} = Fx_k + Gu_k$ with cost $J = \sum_{k=0}^\infty (x_k^T Q x_k + u_k^T R u_k)$, derive the discrete-time ARE and optimal gain. Implement numerically and compare with continuous-time LQR for a sampled system.

Exercise 17.7. (Laboratory) Implement the LQR controller from Section 17.4 on an actual inverted pendulum or simulation. Experimentally verify the Pareto tradeoff by measuring settling time and peak voltage for at least three different Q/R ratios.

Exercise 17.8. (Multi-input LQR) Consider a planar spacecraft with two thrusters:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (17.73)$$

Design an LQR controller with $Q = \text{diag}(10, 10, 1, 1)$ and $R = I$. Compute the optimal gain matrix and verify stability.

Exercise 17.9. (Tracking problem) Extend the LQR formulation to track a constant reference r rather than regulate to zero. Derive the modified control law $u = -Kx + K_r r$ where K_r is a feedforward gain.

Exercise 17.10. (Robustness analysis) For the double integrator with LQR design ($Q = I$, $R = 1$), compute the gain margin and phase margin of the closed-loop system. Compare with the classical “guaranteed margins” of LQR: gain margin ≥ 6 dB and phase margin $\geq 60^\circ$.

Further Reading

The foundational work on LQR appears in R. E. Kalman's "Contributions to the Theory of Optimal Control," published in *Bol. Soc. Mat. Mexicana*, vol. 5, pp. 102–119, 1960. For the maximum principle approach, see L. S. Pontryagin, V. G. Boltyanskii, R. V. Gamkrelidze, and E. F. Mishchenko, *The Mathematical Theory of Optimal Processes*, Interscience Publishers, 1962. Dynamic programming foundations are presented in R. Bellman, *Dynamic Programming*, Princeton University Press, 1957. A comprehensive treatment of linear quadratic methods can be found in B. D. O. Anderson and J. B. Moore, *Optimal Control: Linear Quadratic Methods*, Prentice-Hall, 1990. For practical applications, consult A. E. Bryson and Y.-C. Ho, *Applied Optimal Control*, Hemisphere Publishing, 1975. A modern introduction suitable for both scientists and engineers is K. J. Astrom and R. M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*, Princeton University Press, 2021, Chapter 7.

Chapter 18

Kalman Filter: Optimal Estimation

“The Kalman filter is one of the great achievements of twentieth century technology—manned trips to the moon and back would not have been possible without it.”
—Stanley Schmidt, NASA Ames Research Center

18.1 Historical Motivation: The Problem of Noisy Measurements

The Birth of Optimal Filtering (1960)

In autumn 1960, Rudolf Kalman published “A New Approach to Linear Filtering and Prediction Problems,” fundamentally transforming estimation theory. Unlike Wiener’s earlier frequency-domain approach requiring infinite measurement history, Kalman’s method works directly in the time domain with recursive computation, maintaining only a finite “state” that summarizes all relevant past information. Stanley Schmidt at NASA Ames recognized its transformative potential for Apollo navigation, implementing the first practical Kalman filter to fuse drifting IMU data with star tracker and radar measurements. The Apollo Guidance Computer, with less power than a modern calculator, could not solve the Riccati equation in real-time—engineers pre-computed optimal trajectories stored as polynomials. On July 20, 1969, when Neil Armstrong guided Eagle to the Sea of Tranquility, he was following trajectories computed using these same principles.

18.1.1 Rudolf Kalman and the Birth of Optimal Filtering

In the autumn of 1960, Rudolf Emil Kalman, a young Hungarian-American mathematician working at the Research Institute for Advanced Studies (RIAS) in Baltimore, published a paper that would fundamentally transform how engineers and scientists think about estimation. Titled “A New Approach to Linear Filtering and Prediction Problems,” this landmark work in the *Journal of Basic Engineering* introduced what we now call the Kalman filter [1].

Kalman’s genius lay not in solving an entirely new problem, but in providing a revolutionary new perspective on an old one. The challenge of extracting useful information from noisy measurements had plagued scientists and engineers since the dawn of measurement itself. Every

sensor, every instrument, every observation contains some degree of random error. How do we best estimate the true state of a system when all we have access to are corrupted measurements?

18.1.2 Before Kalman: The Wiener Filter

The problem of optimal filtering had been tackled most famously by Norbert Wiener during World War II. Wiener, working on the problem of anti-aircraft fire control, developed what became known as the Wiener filter in the 1940s. His approach, while mathematically elegant, had significant practical limitations. The Wiener filter operates in the frequency domain, requiring spectral analysis and the solution of integral equations known as the Wiener-Hopf equation. Additionally, the classical Wiener filter assumes access to an infinite history of past measurements, which is clearly impractical for real-time applications. Wiener's approach also requires that the statistical properties of signals remain constant over time, limiting it to stationary processes only. Finally, the Wiener filter does not naturally accommodate the physical structure of dynamic systems with known governing equations, lacking the state-space structure that would later prove so powerful.

18.1.3 Kalman's Revolutionary Insight

Kalman's approach differed fundamentally from Wiener's in several crucial ways. First, the Kalman filter works directly in the time domain, making it natural for dynamic systems described by differential equations. Rather than requiring all past data, the Kalman filter employs recursive computation, maintaining only a finite "state" that summarizes all relevant past information; each new measurement updates this state efficiently. This means the filter requires only storage proportional to the state dimension, not the measurement history, giving it a finite memory requirement. The framework naturally handles time-varying systems and noise characteristics, accommodating non-stationary systems with ease. Finally, the filter leverages the state-space representation of dynamic systems, connecting estimation to the broader framework of modern control theory through its state-space foundation.

The mathematical elegance of Kalman's solution is remarkable: the filter is not just *one* optimal estimator, but *the* optimal linear estimator in the minimum mean-square error sense. Under Gaussian noise assumptions, it is the optimal estimator of any kind.

18.1.4 Stanley Schmidt and the Apollo Program

While Kalman's paper established the theoretical foundation, it was Stanley Schmidt at NASA's Ames Research Center who recognized its transformative potential for space navigation. Schmidt had been wrestling with the challenge of navigating spacecraft to the moon—a problem requiring unprecedented accuracy with inherently noisy sensors.

Consider the challenges facing Apollo navigation engineers. The Inertial Measurement Unit (IMU) suffered from drift, as gyroscopes and accelerometers accumulated errors over time that would be catastrophic over a multi-day mission. Star trackers provided absolute attitude information through optical measurements of star positions, but these were subject to sensor noise. Radar ranging measurements to Earth contained both random noise and systematic biases. Perhaps most constraining was the limited computing power available: the Apollo Guidance Computer had ap-

proximately 74 KB of memory and operated at 0.043 MHz—a tiny fraction of a modern smart-phone’s capability.

Schmidt implemented the first practical Kalman filter for Apollo navigation, demonstrating that the recursive structure made real-time implementation feasible even with severely limited computing resources. His implementation fused IMU data (high bandwidth but drifting) with star tracker and radar data (lower bandwidth but absolute) to maintain accurate position and velocity estimates throughout the mission.

The success of the Kalman filter in Apollo was spectacular. Navigation errors were kept within acceptable bounds throughout the missions, enabling the precise trajectory corrections necessary for lunar orbit insertion and trans-Earth injection. Without the Kalman filter, the moon landings would not have been possible.

18.1.5 Duality: The Kalman Filter is to Estimation What LQR is to Control

Perhaps the most beautiful theoretical insight connecting estimation and control is the principle of **duality**. The Kalman filter and the Linear Quadratic Regulator (LQR) are dual problems—mathematically equivalent structures solving complementary problems.

Consider the symmetry:

Aspect	LQR (Control)	Kalman Filter (Estimation)
Goal	Minimize control effort	Minimize estimation error
Design matrices	Q (state cost), R (input cost)	Q (process noise), R (measurement noise)
Result	State feedback gain K	Kalman gain L
Riccati equation	$A^T P + P A - P B R^{-1} B^T P + Q = 0$	$A P + P A^T - P C^T R^{-1} C P + Q = 0$
System requirement	Controllability	Observability

This duality is not coincidental—it reflects the deep mathematical connection between controlling a system and observing a system. If you can design an optimal controller, you can design an optimal estimator by transposing the problem. This profound insight, formalized by Kalman himself, unifies the two pillars of modern control theory.

18.2 Modern Applications

The Kalman filter has become one of the most widely applied algorithms in engineering and science. Its influence extends far beyond aerospace into virtually every domain where estimation from noisy data is required.

18.2.1 GPS/INS Sensor Fusion

Modern navigation systems in aircraft, ships, and vehicles routinely fuse GPS (Global Positioning System) with INS (Inertial Navigation System) using Kalman filters. GPS provides absolute position at approximately 1 Hz with meter-level accuracy but can be denied or jammed. INS provides high-bandwidth relative motion data but drifts over time. The Kalman filter optimally combines

these complementary sources, using GPS to correct INS drift while maintaining position estimates during GPS outages.

18.2.2 Self-Driving Car Localization

Autonomous vehicles face an acute localization challenge: they must know their position and orientation with centimeter-level accuracy to navigate safely. Modern self-driving systems employ sophisticated Kalman filter variants (Extended Kalman Filters, Unscented Kalman Filters, particle filters) to fuse data from multiple sources: LIDAR point clouds matched against high-definition maps, camera-based lane and landmark detection, radar measurements of surrounding vehicles, GPS and differential GPS corrections, and wheel odometry combined with IMU data. The resulting estimate is far more accurate and reliable than any single sensor could provide.

18.2.3 Weather Prediction

Numerical weather prediction is one of the most computationally intensive applications of Kalman filtering. Modern weather models maintain state vectors with millions of dimensions representing temperature, pressure, humidity, and wind velocity at grid points throughout the atmosphere. The Kalman filter (typically in ensemble form) fuses this model-predicted state with observations from weather stations, radiosondes, satellites, and aircraft to produce the forecasts that inform daily weather reports and severe weather warnings.

18.2.4 Financial Time Series

Financial engineers apply Kalman filtering to estimate hidden market states from observable prices. Applications include estimating the “true” volatility of assets from noisy price observations, tracking time-varying risk parameters in portfolio management, filtering high-frequency trading signals from market microstructure noise, and estimating unobservable macroeconomic factors from economic indicators. The financial application highlights that Kalman filtering is not limited to physical systems—any domain with dynamic evolution and noisy observation can benefit.

18.3 Mathematical Foundation

We now develop the Kalman filter equations rigorously, building from the problem formulation to the complete filter derivation.

18.3.1 System Model with Stochastic Noise

Consider a linear time-invariant system subject to random disturbances:

$$\begin{aligned} \dot{\mathbf{x}}(t) &= A\mathbf{x}(t) + B\mathbf{u}(t) + \mathbf{w}(t) && \text{(process equation)} \\ \mathbf{y}(t) &= C\mathbf{x}(t) + \mathbf{v}(t) && \text{(measurement equation)} \end{aligned} \tag{18.1}$$

In this formulation, $\mathbf{x}(t) \in \mathbb{R}^n$ is the state vector we wish to estimate, $\mathbf{u}(t) \in \mathbb{R}^m$ is the known control input, $\mathbf{y}(t) \in \mathbb{R}^p$ is the measured output, $\mathbf{w}(t) \in \mathbb{R}^n$ is the process noise (disturbance), and $\mathbf{v}(t) \in \mathbb{R}^p$ is the measurement noise.

The noise processes are assumed to be white Gaussian noise with known statistics:

$$\mathbf{w}(t) \sim \mathcal{N}(0, Q) \quad (\text{zero mean, covariance } Q) \quad (18.2)$$

$$\mathbf{v}(t) \sim \mathcal{N}(0, R) \quad (\text{zero mean, covariance } R) \quad (18.3)$$

We further assume that $\mathbf{w}$ and $\mathbf{v}$ are uncorrelated with each other and with the initial state:

$$E[\mathbf{w}(t)\mathbf{v}^\top(\tau)] = 0 \quad \forall t, \tau \quad (18.4)$$

$$E[\mathbf{w}(t)\mathbf{x}^\top(0)] = 0 \quad (18.5)$$

$$E[\mathbf{v}(t)\mathbf{x}^\top(0)] = 0 \quad (18.6)$$

The covariance matrices Q and R are symmetric positive semi-definite, with R typically required to be positive definite (invertible) for the standard filter derivation.

18.3.2 The Estimation Problem

The fundamental estimation problem is to find an estimate $\hat{\mathbf{x}}(t)$ of the true state $\mathbf{x}(t)$ that minimizes the expected squared estimation error:

$$J = E[(\mathbf{x}(t) - \hat{\mathbf{x}}(t))^\top (\mathbf{x}(t) - \hat{\mathbf{x}}(t))] = E[\|\mathbf{x}(t) - \hat{\mathbf{x}}(t)\|^2] \quad (18.7)$$

This is the **minimum mean-square error** (MMSE) criterion. We seek the estimate that, on average, is closest to the true state in the Euclidean sense.

Define the estimation error:

$$\mathbf{e}(t) = \mathbf{x}(t) - \hat{\mathbf{x}}(t) \quad (18.8)$$

The error covariance matrix is:

$$P(t) = E[\mathbf{e}(t)\mathbf{e}^\top(t)] = E[(\mathbf{x}(t) - \hat{\mathbf{x}}(t))(\mathbf{x}(t) - \hat{\mathbf{x}}(t))^\top] \quad (18.9)$$

The diagonal elements of P represent the variance of the estimation error in each state component. Our objective is equivalent to minimizing the trace of P , which equals the sum of the individual error variances.

18.3.3 Structure of the Optimal Estimator

The Kalman filter has the structure of an observer (as studied in Chapter 16) with optimally chosen gain:

$$\dot{\hat{\mathbf{x}}}(t) = A\hat{\mathbf{x}}(t) + B\mathbf{u}(t) + K(t)(\mathbf{y}(t) - C\hat{\mathbf{x}}(t)) \quad (18.10)$$

The term $(\mathbf{y}(t) - C\hat{\mathbf{x}}(t))$ is called the **innovation** or **residual**—it represents the difference between what we actually measure and what we predicted we would measure. The Kalman gain $K(t)$ determines how much we “trust” this innovation to correct our estimate.

Intuition: The filter is essentially a model-based predictor with feedback correction. The term $A\hat{\mathbf{x}} + B\mathbf{u}$ uses our system model to predict how the state evolves, while the term $K(\mathbf{y} - C\hat{\mathbf{x}})$ corrects this prediction based on actual measurements.

If we trust our model perfectly and distrust measurements ($K \rightarrow 0$), we simply propagate the model. If we trust measurements perfectly and distrust the model ($K \rightarrow \text{large}$), we heavily weight the innovation.

18.3.4 Derivation of the Kalman Gain

To derive the optimal gain, we analyze how the error evolves. The true state evolves as:

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} + \mathbf{w} \quad (18.11)$$

Subtracting the filter equation (??):

$$\dot{\mathbf{e}} = \dot{\mathbf{x}} - \dot{\hat{\mathbf{x}}} \quad (18.12)$$

$$= (A\mathbf{x} + B\mathbf{u} + \mathbf{w}) - (A\hat{\mathbf{x}} + B\mathbf{u} + K(\mathbf{y} - C\hat{\mathbf{x}})) \quad (18.13)$$

$$= A(\mathbf{x} - \hat{\mathbf{x}}) + \mathbf{w} - K(C\mathbf{x} + \mathbf{v} - C\hat{\mathbf{x}}) \quad (18.14)$$

$$= A\mathbf{e} + \mathbf{w} - KC\mathbf{e} - K\mathbf{v} \quad (18.15)$$

$$= (A - KC)\mathbf{e} + \mathbf{w} - K\mathbf{v} \quad (18.16)$$

The error dynamics depend on the matrix $(A - KC)$. For stability of the estimator, this matrix must have all eigenvalues in the left half-plane. This is dual to the pole placement problem for state feedback, where we choose K in $(A - BK)$.

Now we derive the error covariance dynamics. Differentiating $P = E[\mathbf{e}\mathbf{e}^\top]$:

$$\dot{P} = E[\dot{\mathbf{e}}\mathbf{e}^\top + \mathbf{e}\dot{\mathbf{e}}^\top] \quad (18.17)$$

$$= E[((A - KC)\mathbf{e} + \mathbf{w} - K\mathbf{v})\mathbf{e}^\top + \mathbf{e}((A - KC)\mathbf{e} + \mathbf{w} - K\mathbf{v})^\top] \quad (18.18)$$

Using the independence assumptions (noise uncorrelated with error):

$$\dot{P} = (A - KC)E[\mathbf{e}\mathbf{e}^\top] + E[\mathbf{e}\mathbf{e}^\top](A - KC)^\top + E[\mathbf{w}\mathbf{w}^\top] + KE[\mathbf{v}\mathbf{v}^\top]K^\top \quad (18.19)$$

$$= (A - KC)P + P(A - KC)^\top + Q + K RK^\top \quad (18.20)$$

Expanding:

$$\dot{P} = AP + PA^\top - KCP - PC^\top K^\top + Q + K RK^\top \quad (18.21)$$

To find the optimal K that minimizes $\text{tr}(P)$, we take the derivative with respect to K and set it to zero. This optimization (details involve matrix calculus) yields:

$$\boxed{K(t) = P(t)C^\top R^{-1}} \quad (18.22)$$

This is the **optimal Kalman gain**. It has a beautiful intuitive interpretation: P represents our uncertainty in the state estimate, $C^\top$ projects measurement information back to state space, and R^{-1} weights the measurement correction inversely with measurement noise—we trust low-noise measurements more.

18.3.5 The Continuous-Time Riccati Equation

Substituting the optimal gain (??) into (??), we obtain the **continuous-time Riccati equation** for the error covariance:

$$\dot{P} = AP + PA^\top - PC^\top R^{-1}CP + Q \quad (18.23)$$

This differential equation governs how our estimation uncertainty evolves over time. Note the remarkable similarity to the LQR Riccati equation—this is the duality principle in action.

Steady-state solution: For time-invariant systems, if the system is observable and controllable (in the noise-input sense), the Riccati equation converges to a steady-state solution P_∞ satisfying:

$$0 = AP_\infty + P_\infty A^\top - P_\infty C^\top R^{-1}CP_\infty + Q \quad (18.24)$$

This is the **algebraic Riccati equation** (ARE) for estimation. The steady-state Kalman gain is then $K_\infty = P_\infty C^\top R^{-1}$.

18.3.6 Summary: Continuous-Time Kalman Filter

Continuous-Time Kalman Filter

System Model:

$$\begin{aligned} \dot{\mathbf{x}} &= A\mathbf{x} + B\mathbf{u} + \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(0, Q) \\ \mathbf{y} &= C\mathbf{x} + \mathbf{v}, \quad \mathbf{v} \sim \mathcal{N}(0, R) \end{aligned}$$

Filter Equations:

$$\begin{aligned} \dot{\hat{\mathbf{x}}} &= A\hat{\mathbf{x}} + B\mathbf{u} + K(\mathbf{y} - C\hat{\mathbf{x}}) \\ K &= PC^\top R^{-1} \\ \dot{P} &= AP + PA^\top - PC^\top R^{-1}CP + Q \end{aligned}$$

Initial Conditions: $\hat{\mathbf{x}}(0) = E[\mathbf{x}(0)]$, $P(0) = E[(\mathbf{x}(0) - \hat{\mathbf{x}}(0))(\mathbf{x}(0) - \hat{\mathbf{x}}(0))^\top]$

18.3.7 Discrete-Time Kalman Filter

For digital implementation, we need the discrete-time formulation. Consider the discrete-time system:

$$\begin{aligned} \mathbf{x}_{k+1} &= A_d \mathbf{x}_k + B_d \mathbf{u}_k + \mathbf{w}_k \\ \mathbf{y}_k &= C \mathbf{x}_k + \mathbf{v}_k \end{aligned} \quad (18.25)$$

where $\mathbf{w}_k \sim \mathcal{N}(0, Q_d)$ and $\mathbf{v}_k \sim \mathcal{N}(0, R)$.

The discrete Kalman filter operates in a two-phase cycle: **predict** and **update**.

Discrete-Time Kalman Filter Algorithm

Initialization:

$$\hat{\mathbf{x}}_0 = E[\mathbf{x}_0]$$

$$P_0 = E[(\mathbf{x}_0 - \hat{\mathbf{x}}_0)(\mathbf{x}_0 - \hat{\mathbf{x}}_0)^\top]$$

For each time step k :

Predict (Time Update):

$$\hat{\mathbf{x}}_k^- = A_d \hat{\mathbf{x}}_{k-1} + B_d \mathbf{u}_{k-1} \quad (\text{a priori state estimate})$$

$$P_k^- = A_d P_{k-1} A_d^\top + Q_d \quad (\text{a priori covariance})$$

Update (Measurement Update):

$$K_k = P_k^- C^\top (C P_k^- C^\top + R)^{-1} \quad (\text{Kalman gain})$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + K_k (\mathbf{y}_k - C \hat{\mathbf{x}}_k^-) \quad (\text{a posteriori estimate})$$

$$P_k = (I - K_k C) P_k^- \quad (\text{a posteriori covariance})$$

The notation uses superscript $-$ for “a priori” (before measurement update) and no superscript for “a posteriori” (after measurement update).

Interpretation of the predict-update cycle: In the predict phase, we use the system model to predict where we expect the state to be, and our uncertainty grows due to process noise (P increases by Q). In the update phase, we incorporate the new measurement to correct our prediction, and our uncertainty decreases as the measurement provides information.

18.3.8 Understanding the Kalman Gain

The Kalman gain formula deserves closer examination:

$$K_k = P_k^- C^\top (C P_k^- C^\top + R)^{-1} \quad (18.26)$$

Limiting cases: The behavior of the Kalman gain in extreme scenarios reveals its underlying logic. When measurements are very noisy ($R \rightarrow \infty$), the gain approaches zero ($K_k \rightarrow 0$), meaning we ignore the measurements because they are too noisy to be useful. Conversely, with perfect measurements ($R \rightarrow 0$), the gain approaches $K_k \rightarrow (C P_k^-)^{-1}$ (pseudoinverse), indicating we trust the measurement completely. When model uncertainty is high ($P_k^- \rightarrow \infty$), the gain approaches $K_k \rightarrow C^\dagger$ (pseudoinverse of C), heavily weighting measurements because we don’t trust our model. Finally, with perfect model or state knowledge ($P_k^- \rightarrow 0$), the gain approaches zero since we already know the state perfectly and measurements add nothing.

18.3.9 Tuning Q and R : Model vs. Sensor Confidence

The covariance matrices Q and R are the primary “tuning knobs” of the Kalman filter. The process noise covariance Q represents uncertainty in the system model; large Q means we don’t trust

our model—perhaps due to unmodeled dynamics, disturbances, or parameter uncertainty—and the filter will weight measurements more heavily. The measurement noise covariance R represents sensor noise; large R means we don't trust our sensors, and the filter will weight the model prediction more heavily, producing smoother but less responsive estimates.

The **ratio** Q/R (in a scalar sense) determines the filter's behavior. A high Q/R ratio produces fast, responsive tracking but potentially noisy estimates, while a low Q/R ratio yields smooth estimates but sluggish response to actual state changes.

In practice, R can often be determined from sensor specifications or calibration. Q is typically harder to specify and is often treated as a tuning parameter adjusted based on filter performance.

18.3.10 Duality with LQR: The Algebraic Riccati Equation

The deep connection between LQR and Kalman filtering is evident in their Riccati equations:

LQR (Regulator):

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0 \quad (18.27)$$

Kalman Filter (Estimator):

$$AP + PA^\top - PC^\top R^{-1}CP + Q = 0 \quad (18.28)$$

These are the same equation with $A \leftrightarrow A^\top$ and $B \leftrightarrow C^\top$! This duality means that if (A, B) is controllable, the LQR problem has a unique stabilizing solution, and if (A, C) is observable, the Kalman filter problem has a unique stabilizing solution.

This leads to the profound **separation principle**: for linear systems with Gaussian noise, optimal control and optimal estimation can be designed independently and combined. The certainty-equivalent LQG (Linear Quadratic Gaussian) controller uses the Kalman filter estimate as if it were the true state.

18.4 Practical Experiment: IMU Sensor Fusion

In this experiment, we implement a Kalman filter to fuse accelerometer and gyroscope data from an Inertial Measurement Unit (IMU), estimating the pitch angle of a tilting platform.

18.4.1 The Sensor Fusion Problem

An IMU typically contains two primary sensors. The accelerometer measures linear acceleration including gravity, and can estimate tilt angle from the gravity direction, but is noisy and sensitive to vibration and dynamic acceleration. The gyroscope measures angular velocity and can be integrated to estimate angle changes, but drifts over time due to bias and noise accumulation.

Neither sensor alone provides a satisfactory angle estimate. The accelerometer is accurate in steady state but noisy during motion, while the gyroscope is smooth during motion but drifts in steady state.

The Kalman filter optimally combines these complementary sensors, using the gyroscope for short-term smoothness and the accelerometer for long-term accuracy.

18.4.2 Hardware Setup

Components: The hardware setup requires an Arduino Uno or similar microcontroller, an MPU6050 6-axis IMU module (containing both accelerometer and gyroscope), a USB cable for serial communication, a breadboard with jumper wires, and optionally a 3D-printed tilting platform for testing.

Wiring (I2C connection): Connect the MPU6050 VCC pin to Arduino 5V, the GND pin to Arduino GND, the SCL pin to Arduino A5 (SCL), and the SDA pin to Arduino A4 (SDA).

18.4.3 Mathematical Model for Angle Estimation

For a 1D angle estimation problem (e.g., pitch angle θ):

State: $\mathbf{x} = \begin{bmatrix} \theta \\ \dot{\theta}_b \end{bmatrix}$ where θ is angle and $\dot{\theta}_b$ is gyro bias.

Process model:

$$\begin{bmatrix} \theta_{k+1} \\ \dot{\theta}_{b,k+1} \end{bmatrix} = \begin{bmatrix} 1 & -\Delta t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \theta_k \\ \dot{\theta}_{b,k} \end{bmatrix} + \begin{bmatrix} \Delta t \\ 0 \end{bmatrix} \omega_{\text{gyro},k} + \mathbf{w}_k \quad (18.29)$$

The gyroscope reading ω_{gyro} is used as an input to predict angle change, while the accelerometer provides a measurement of the angle.

Measurement model:

$$\theta_{\text{accel},k} = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}_k + v_k \quad (18.30)$$

where $\theta_{\text{accel}} = \arctan(a_x/a_z)$ is the angle derived from accelerometer readings.

18.4.4 Implementation

The following algorithm presents the Kalman filter for IMU sensor fusion in language-agnostic pseudocode. This can be implemented on any microcontroller or computing platform.

18.4.5 Visualization and Tuning

Use the Arduino Serial Plotter or a Python script to visualize the three signals: the accelerometer angle (noisy but unbiased), the gyroscope-integrated angle (smooth but drifting), and the Kalman-filtered angle (which combines the best properties of both sensors).

Tuning experiments: To understand the filter's behavior, try adjusting the noise parameters. Increasing Q_{angle} causes the filter to respond faster but produces noisier estimates. Increasing R_{measure} makes the filter smoother but slower to respond. Increasing Q_{bias} improves tracking of gyro bias drift.

Algorithm 38 Kalman Filter for IMU Sensor Fusion

```

1: Initialize:
2:  $\theta \leftarrow 0$                                 ▷ Estimated angle
3:  $b \leftarrow 0$                                 ▷ Estimated gyro bias
4:  $P \leftarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$                                 ▷ Error covariance matrix
5:  $Q_\theta \leftarrow 0.001, Q_b \leftarrow 0.003, R \leftarrow 0.03$                                 ▷ Noise parameters
6:  $t_{\text{prev}} \leftarrow \text{CurrentTime}()$ 
7:
8: Initialize IMU sensor via I2C interface
9:
10: while true do
11:    $t_{\text{now}} \leftarrow \text{CurrentTime}()$ 
12:    $\Delta t \leftarrow t_{\text{now}} - t_{\text{prev}}$ 
13:    $t_{\text{prev}} \leftarrow t_{\text{now}}$ 
14:
15:                                     ▷ Read sensor data
16:    $(a_x, a_y, a_z, \omega_x, \omega_y, \omega_z) \leftarrow \text{ReadIMU}()$ 
17:    $\theta_{\text{accel}} \leftarrow \arctan(a_x/a_z)$                                 ▷ Angle from accelerometer
18:    $\theta_{\text{gyro}} \leftarrow \omega_y/131$                                 ▷ Angular rate from gyroscope
19:
20:                                     ▷ — Predict Step —
21:    $\dot{\theta} \leftarrow \dot{\theta}_{\text{gyro}} - b$ 
22:    $\theta \leftarrow \theta + \Delta t \cdot \dot{\theta}$ 
23:
24:                                     ▷ Update error covariance (predict)
25:    $P_{00} \leftarrow P_{00} + \Delta t \cdot (\Delta t \cdot P_{11} - P_{01} - P_{10} + Q_\theta)$ 
26:    $P_{01} \leftarrow P_{01} - \Delta t \cdot P_{11}$ 
27:    $P_{10} \leftarrow P_{10} - \Delta t \cdot P_{11}$ 
28:    $P_{11} \leftarrow P_{11} + Q_b \cdot \Delta t$ 
29:
30:                                     ▷ — Update Step —
31:    $\nu \leftarrow \theta_{\text{accel}} - \theta$                                 ▷ Innovation
32:    $S \leftarrow P_{00} + R$                                 ▷ Innovation covariance
33:
34:                                     ▷ Compute Kalman gain
35:    $K_0 \leftarrow P_{00}/S$ 
36:    $K_1 \leftarrow P_{10}/S$ 
37:
38:                                     ▷ Update state estimate
39:    $\theta \leftarrow \theta + K_0 \cdot \nu$ 
40:    $b \leftarrow b + K_1 \cdot \nu$ 
41:
42:                                     ▷ Update error covariance
43:    $P_{00}^{\text{temp}} \leftarrow P_{00}, P_{01}^{\text{temp}} \leftarrow P_{01}$ 
44:    $P_{00} \leftarrow P_{00} - K_0 \cdot P_{00}^{\text{temp}}$ 
45:    $P_{01} \leftarrow P_{01} - K_0 \cdot P_{01}^{\text{temp}}$ 
46:    $P_{10} \leftarrow P_{10} - K_1 \cdot P_{00}^{\text{temp}}$ 
47:    $P_{11} \leftarrow P_{11} - K_1 \cdot P_{01}^{\text{temp}}$ 
48:

```

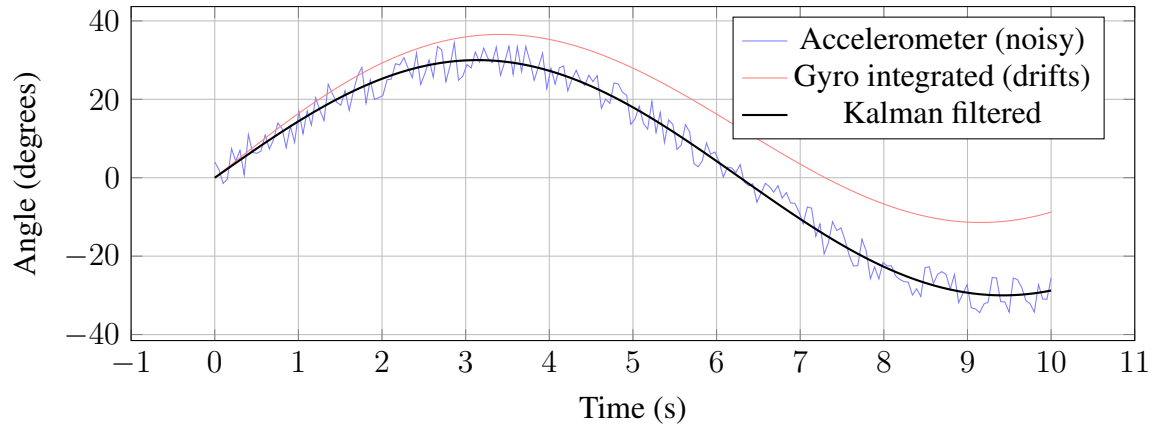


Figure 18.1: Comparison of sensor measurements and Kalman filter output. The accelerometer (blue) is noisy, the integrated gyroscope (red) drifts over time, while the Kalman filter (black) provides smooth, accurate angle estimates.

18.5 Worked Examples

1D Position Estimation with Velocity Measurement

Problem: A vehicle moves along a straight line. We have a noisy position measurement (GPS) and want to estimate both position and velocity.

Solution:

State vector: $\mathbf{x} = \begin{bmatrix} p \\ v \end{bmatrix}$ (position and velocity)

System model (constant velocity with random acceleration):

$$\begin{bmatrix} p_{k+1} \\ v_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} p_k \\ v_k \end{bmatrix} + \mathbf{w}_k \quad (18.31)$$

Measurement model (GPS measures position):

$$y_k = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} p_k \\ v_k \end{bmatrix} + v_k \quad (18.32)$$

With $\Delta t = 1$ s, $Q = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}$, $R = 4$ (GPS noise variance):

Step 1: Initialize

$$\hat{\mathbf{x}}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad P_0 = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} \quad (18.33)$$

Step 2: Predict for $k = 1$

$$\hat{\mathbf{x}}_1^- = A\hat{\mathbf{x}}_0 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (18.34)$$

$$P_1^- = AP_0A^\top + Q = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix} \quad (18.35)$$

$$= \begin{bmatrix} 20.1 & 10 \\ 10 & 10.1 \end{bmatrix} \quad (18.36)$$

Step 3: Update with measurement $y_1 = 2.5$

$$K_1 = P_1^- C^\top (CP_1^- C^\top + R)^{-1} \quad (18.37)$$

$$= \begin{bmatrix} 20.1 \\ 10 \end{bmatrix} (20.1 + 4)^{-1} = \begin{bmatrix} 0.834 \\ 0.415 \end{bmatrix} \quad (18.38)$$

Innovation: $y_1 - C\hat{\mathbf{x}}_1^- = 2.5 - 0 = 2.5$

$$\hat{\mathbf{x}}_1 = \hat{\mathbf{x}}_1^- + K_1(2.5) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.834 \\ 0.415 \end{bmatrix} (2.5) = \begin{bmatrix} 2.08 \\ 1.04 \end{bmatrix} \quad (18.39)$$

The filter estimates position at 2.08 m and infers velocity of 1.04 m/s from a single measurement!

Steady-State Kalman Gain for Scalar System

Problem: Find the steady-state Kalman gain for a scalar system:

$$x_{k+1} = ax_k + w_k, \quad w_k \sim \mathcal{N}(0, q) \quad (18.40)$$

$$y_k = x_k + v_k, \quad v_k \sim \mathcal{N}(0, r) \quad (18.41)$$

Solution:

The discrete algebraic Riccati equation for this scalar system:

$$P = aPa + q - \frac{a^2 P^2}{P + r} \quad (18.42)$$

At steady state, let $P_\infty = P$:

$$P = a^2 P + q - \frac{a^2 P^2}{P + r} \quad (18.43)$$

Multiply by $(P + r)$:

$$P(P + r) = a^2 P(P + r) + q(P + r) - a^2 P^2 \quad (18.44)$$

$$P^2 + Pr = a^2 P^2 + a^2 Pr + qP + qr - a^2 P^2 \quad (18.45)$$

$$P^2 + Pr = a^2Pr + qP + qr \quad (18.46)$$

$$P^2 + P(r - a^2r - q) - qr = 0 \quad (18.47)$$

Using the quadratic formula with $a = 0.9$, $q = 0.1$, $r = 1$:

$$P^2 + P(1 - 0.81 - 0.1) - 0.1 = 0 \implies P^2 + 0.09P - 0.1 = 0 \quad (18.48)$$

$$P_\infty = \frac{-0.09 + \sqrt{0.0081 + 0.4}}{2} = \frac{-0.09 + 0.639}{2} = 0.274 \quad (18.49)$$

Steady-state Kalman gain:

$$K_\infty = \frac{P_\infty}{P_\infty + r} = \frac{0.274}{0.274 + 1} = 0.215 \quad (18.50)$$

This means at steady state, the filter weights the measurement at about 21.5% and the prediction at about 78.5%.

18.5.1 Example 3: Effect of Q and R on Filter Behavior

Problem: For the system in Example 2, compare the steady-state gain for two cases: (a) $q = 0.01$, $r = 1$ (trust model more), and (b) $q = 1$, $r = 0.01$ (trust measurements more).

Solution:

Case (a): $q/r = 0.01$ (model-trusting)

Using the same derivation:

$$P^2 + P(1 - 0.81 - 0.01) - 0.01 = 0 \implies P^2 + 0.18P - 0.01 = 0 \quad (18.51)$$

$$P_\infty = 0.0474, K_\infty = \frac{0.0474}{1.0474} = 0.045$$

The filter mostly ignores measurements (4.5% weight).

Case (b): $q/r = 100$ (measurement-trusting)

$$P^2 + P(0.01 - 0.81 \cdot 0.01 - 1) - 0.01 = 0 \implies P^2 - 0.9919P - 0.01 = 0 \quad (18.52)$$

$$P_\infty = 1.002, K_\infty = \frac{1.002}{1.002 + 0.01} = 0.990$$

The filter heavily weights measurements (99% weight).

18.5.2 Example 4: Step-by-Step Discrete Kalman Filter

Problem: Implement three complete cycles of a Kalman filter for tracking a falling object.

Setup:

$$A = \begin{bmatrix} 1 & 0.1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0.005 \\ 0.1 \end{bmatrix}, \quad C = [1 \quad 0] \quad (18.53)$$

$$Q = \begin{bmatrix} 0.01 & 0 \\ 0 & 0.01 \end{bmatrix}, \quad R = 1 \quad (18.54)$$

Input $u = -9.81$ (gravity), measurements $y = [0, -0.4, -0.9]$ m.

Initial: $\hat{\mathbf{x}}_0 = [0, 0]^\top$, $P_0 = I$.

Solution:

$k = 1$: **Predict**

$$\hat{\mathbf{x}}_1^- = A\hat{\mathbf{x}}_0 + Bu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -0.049 \\ -0.981 \end{bmatrix} = \begin{bmatrix} -0.049 \\ -0.981 \end{bmatrix} \quad (18.55)$$

$$P_1^- = AP_0A^\top + Q = \begin{bmatrix} 1.11 & 0.1 \\ 0.1 & 1.01 \end{bmatrix} \quad (18.56)$$

$k = 1$: **Update with** $y_1 = 0$

$$S_1 = CP_1^-C^\top + R = 1.11 + 1 = 2.11 \quad (18.57)$$

$$K_1 = P_1^-C^\top S_1^{-1} = \begin{bmatrix} 0.526 \\ 0.047 \end{bmatrix} \quad (18.58)$$

$$\hat{\mathbf{x}}_1 = \hat{\mathbf{x}}_1^- + K_1(0 - (-0.049)) = \begin{bmatrix} -0.023 \\ -0.978 \end{bmatrix} \quad (18.59)$$

Continue this process for $k = 2, 3$ to track the falling object.

18.5.3 Example 5: Sensor Fusion—Combining Two Noisy Measurements

Problem: We have two independent sensors measuring the same quantity x :

$$y_1 = x + v_1, \quad v_1 \sim \mathcal{N}(0, \sigma_1^2) \quad (18.60)$$

$$y_2 = x + v_2, \quad v_2 \sim \mathcal{N}(0, \sigma_2^2) \quad (18.61)$$

With $\sigma_1 = 2$, $\sigma_2 = 1$, and measurements $y_1 = 10$, $y_2 = 12$, find the optimal estimate.

Solution:

This is a static estimation problem (no dynamics), but we can use Kalman filter concepts.

For scalar independent measurements, the optimal combined estimate is:

$$\hat{x} = \frac{\sigma_2^2 y_1 + \sigma_1^2 y_2}{\sigma_1^2 + \sigma_2^2} \quad (18.62)$$

With our values:

$$\hat{x} = \frac{1 \cdot 10 + 4 \cdot 12}{4 + 1} = \frac{10 + 48}{5} = 11.6 \quad (18.63)$$

The combined estimate is closer to y_2 because sensor 2 is more accurate (lower variance).

The combined variance is:

$$\sigma_{\text{combined}}^2 = \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2} = \frac{4 \cdot 1}{5} = 0.8 \quad (18.64)$$

The combined estimate is more accurate than either individual sensor!

18.6 Diagrams and Visualizations

18.6.1 Kalman Filter Block Diagram

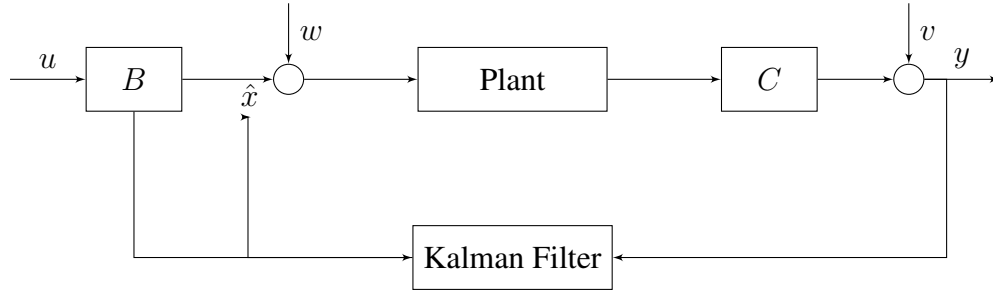


Figure 18.2: Block diagram of the Kalman filter estimating the state of a plant subject to process noise w and measurement noise v .

18.6.2 Prediction-Update Cycle

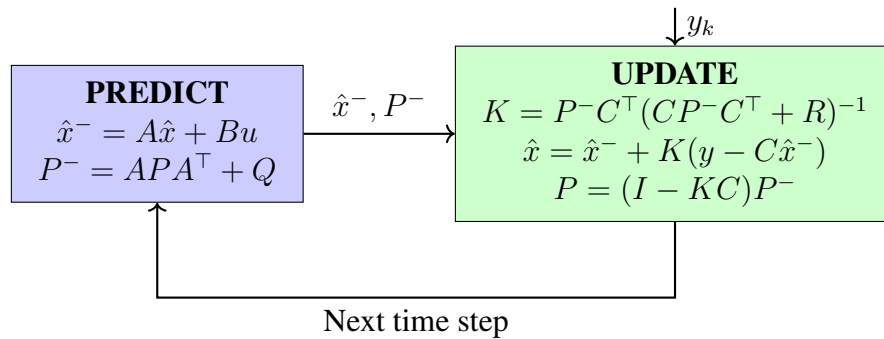


Figure 18.3: The predict-update cycle of the Kalman filter. Each measurement triggers a complete cycle.

18.6.3 Gaussian Fusion: Prior, Measurement, and Posterior

18.6.4 Sensor Comparison: Raw vs. Filtered

18.7 Chapter Summary

The Kalman filter represents one of the crowning achievements of twentieth-century applied mathematics. By combining optimal estimation theory with recursive computation, Rudolf Kalman solved the fundamental problem of extracting accurate state information from noisy measurements in a computationally tractable way.

Key concepts: The Kalman filter achieves optimal estimation by minimizing mean-square estimation error. Its recursive structure maintains only a state estimate and error covariance, requiring

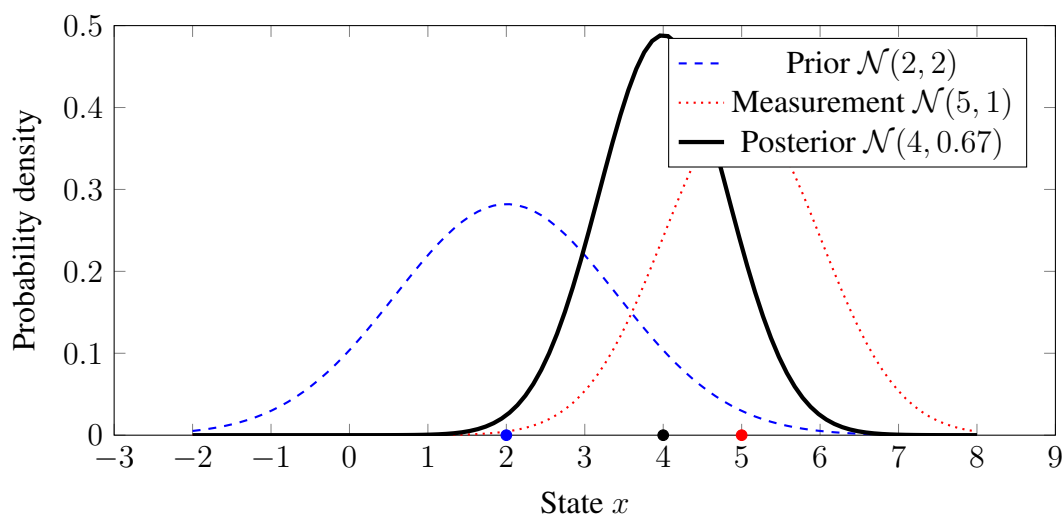


Figure 18.4: Bayesian fusion in the Kalman filter. The prior (predicted) distribution is combined with the measurement likelihood to produce the posterior (updated) distribution. The posterior is narrower (more certain) than either input and located between them, weighted by their relative uncertainties.

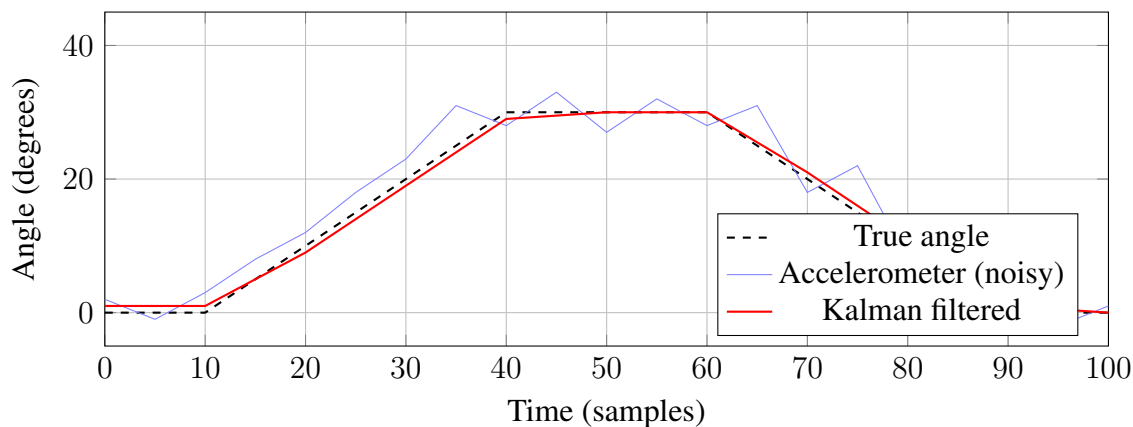


Figure 18.5: Comparison of raw accelerometer data and Kalman-filtered estimate. The filter tracks the true angle while rejecting high-frequency measurement noise.

bounded memory regardless of measurement history length. The predict-update cycle propagates the estimate forward in time and then corrects it based on new measurements. The Kalman gain optimally weights innovation based on relative model and measurement uncertainties. Through sensor fusion, multiple noisy sensors can be combined to achieve accuracy exceeding any individual sensor. Finally, the duality with LQR reveals that estimation and control are mathematically dual problems solved by the same Riccati equation structure.

Practical considerations: When implementing the Kalman filter, tune Q and R to balance smoothness versus responsiveness. Remember that the filter is optimal only when the linear Gaussian assumptions hold; for nonlinear systems, consider the Extended Kalman Filter (EKF) or Unscented Kalman Filter (UKF). Numerical stability requires care in covariance propagation, particularly for ill-conditioned systems.

The Kalman filter's influence extends far beyond control engineering into navigation, signal processing, econometrics, and machine learning. Understanding its principles provides a foundation for the entire field of sequential estimation.

18.8 Exercises

Exercise 18.1. (Scalar Kalman filter) For the system $x_{k+1} = 0.95x_k + w_k$, $y_k = x_k + v_k$ with $Q = 1$ and $R = 10$, compute the steady-state Kalman gain.

Exercise 18.2. (Prediction without measurement) If a Kalman filter runs for several time steps without receiving measurements, describe what happens to the error covariance P .

Exercise 18.3. (Observability and Kalman filter) Explain why an unobservable mode cannot be estimated by the Kalman filter.

Exercise 18.4. (Tuning trade-offs) A tracking filter follows a maneuvering target. Should Q be set high or low? Justify your answer.

Exercise 18.5. (IMU fusion derivation) Derive the complete state-space model for fusing accelerometer and gyroscope measurements to estimate pitch and roll angles.

Exercise 18.6. (Numerical experiment) Implement the discrete Kalman filter in MATLAB/Python for a double integrator (position from noisy velocity measurements). Plot the estimation error versus time for various Q/R ratios.

Exercise 18.7. (Innovation sequence) The innovation sequence $\nu_k = y_k - C\hat{x}_k^-$ should be white noise if the filter is correctly tuned. Design an experiment to test this property.

Exercise 18.8. (Joseph form) The covariance update $P = (I - KC)P^-$ can suffer from numerical issues. Research and explain the “Joseph form” update and why it is more numerically stable.

Exercise 18.9. (Sensor fusion with different rates) Consider a system where GPS provides position at 1 Hz while an IMU provides acceleration at 100 Hz. Describe how to modify the Kalman filter to handle these different measurement rates.

Exercise 18.10. (Extended Kalman Filter introduction) For a nonlinear system $x_{k+1} = f(x_k) + w_k$ with measurement $y_k = h(x_k) + v_k$, derive the linearized Kalman filter equations using Jacobian matrices. This is known as the Extended Kalman Filter (EKF).

18.9 Further Reading

1. Kalman, R.E. (1960). “A New Approach to Linear Filtering and Prediction Problems.” *Journal of Basic Engineering*, 82(1), 35–45. *The original paper—remarkably readable.*
2. Brown, R.G. and Hwang, P.Y.C. (2012). *Introduction to Random Signals and Applied Kalman Filtering*. Wiley. *Excellent practical introduction.*
3. Simon, D. (2006). *Optimal State Estimation: Kalman, H_∞ , and Nonlinear Approaches*. Wiley. *Comprehensive modern treatment.*
4. Grewal, M.S. and Andrews, A.P. (2015). *Kalman Filtering: Theory and Practice Using MATLAB*. Wiley. *Implementation-focused.*
5. McGee, L.A. and Schmidt, S.F. (1985). “Discovery of the Kalman Filter as a Practical Tool for Aerospace and Industry.” NASA TM-86847. *Fascinating historical account by the NASA engineers who first implemented it.*

Bibliography

- [1] R.E. Kalman, “A New Approach to Linear Filtering and Prediction Problems,” *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

Part V

Digital & Discrete Control

Chapter 19

Sampling and Discretization

19.1 Historical Motivation: From Telegraph Lines to Digital Control

The mathematics of sampling did not emerge from abstract curiosity but from an intensely practical problem: how to transmit the maximum amount of information through telegraph and telephone lines with strictly limited bandwidth. The answers discovered in the early twentieth century now underpin every digital control system, audio device, and communication network.

19.1.1 Harry Nyquist and Telegraph Transmission Theory (1928)

Harry Nyquist and the Birth of Sampling Theory

Harry Nyquist (1889–1976), a Swedish-American engineer at AT&T Bell Telephone Laboratories, published a landmark paper in 1928 titled “Certain Topics in Telegraph Transmission Theory.” Born in Nilsby, Sweden, Nyquist emigrated to the United States in 1907 and earned his Ph.D. from Yale in 1917. His work on telegraph transmission theory addressed the fundamental limits of communication systems and established principles that would later underpin all digital signal processing. Nyquist’s contributions extended beyond sampling theory to include the Nyquist stability criterion for feedback systems and foundational work on thermal noise in electronic circuits. His 1928 paper remains one of the most cited works in electrical engineering, establishing him as a founding father of information theory alongside Claude Shannon.

Harry Nyquist, a Swedish-American engineer at AT&T Bell Telephone Laboratories, published a landmark paper in 1928 titled “Certain Topics in Telegraph Transmission Theory.” This work addressed a pressing industrial problem: the transcontinental telephone network required transmitting multiple telegraph signals simultaneously over a single wire pair, a technique called *multiplexing*.

The Problem: Bandwidth Is Expensive

In the 1920s, laying copper telephone lines across the United States represented an enormous capital investment. Each additional wire pair cost thousands of dollars per mile. Engineers sought to maximize the information-carrying capacity of existing infrastructure. The critical question was:

Given a communication channel with bandwidth W hertz, what is the maximum rate at which independent signal values can be transmitted?

Nyquist's analysis revealed a fundamental limit. He demonstrated that a channel of bandwidth W could transmit at most $2W$ independent values (symbols) per second. Attempting to transmit faster would cause *intersymbol interference*—successive symbols would blur together, becoming indistinguishable at the receiver.

Nyquist's Key Insight

Nyquist recognized that a bandlimited signal (one containing no frequencies above W Hz) could be completely characterized by samples taken at rate $2W$ samples per second. More precisely, he showed that:

Nyquist's Sampling Principle (1928)

A signal containing no frequencies higher than W hertz is completely determined by its values at uniformly spaced instants separated by $\frac{1}{2W}$ seconds.

This result implied that sampling at twice the highest frequency component preserved all information in the original signal. Conversely, sampling slower than this rate would inevitably lose information—frequencies would become confused with one another, a phenomenon Nyquist understood but did not name.

Impact on Multiplexing

Nyquist's theorem enabled the design of time-division multiplexing systems. If each telephone voice channel occupied bandwidth $W = 4$ kHz (the standard telephone band), then $2W = 8000$ samples per second would suffice to capture all information. Multiple telephone conversations could share a single high-bandwidth channel by interleaving their samples in time—the birth of digital telephony, though the actual implementation would wait for digital electronics.

19.1.2 Claude Shannon and the Sampling Theorem (1949)

Claude Shannon, also at Bell Labs, formalized and extended Nyquist's work in his groundbreaking 1949 paper "Communication in the Presence of Noise." Shannon provided the mathematical rigor that transformed Nyquist's engineering insight into a theorem with precise conditions and a reconstruction formula.

Shannon's Formulation

Shannon stated the sampling theorem in its now-standard form:

Nyquist-Shannon Sampling Theorem

If a function $x(t)$ contains no frequencies higher than $f_{\max}$ hertz, then it is completely determined by its values at a series of points spaced $\frac{1}{2f_{\max}}$ seconds apart. The original signal can be exactly reconstructed from these samples using the interpolation formula:

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT) \operatorname{sinc}\left(\frac{t - nT}{T}\right) \quad (19.1)$$

where $T = \frac{1}{2f_{\max}}$ is the sampling period and $\operatorname{sinc}(u) = \frac{\sin(\pi u)}{\pi u}$.

Shannon's contribution was not merely restating Nyquist but providing a rigorous mathematical proof based on Fourier analysis, an explicit reconstruction formula through sinc interpolation, a deep connection to information theory and channel capacity, and a careful analysis of what happens when the conditions are violated, namely aliasing.

The Information Theory Context

Shannon's sampling theorem appeared in his broader development of information theory. He recognized that sampling was not merely a practical convenience but a fundamental statement about information content. A bandlimited signal has a finite number of "degrees of freedom" per unit time, precisely $2W$ per second. Sampling captures exactly these degrees of freedom—no more, no less.

19.1.3 The Problem That Drove the Mathematics

The telephone industry's central challenge was economic: voice communication required transmitting signals from approximately 300 Hz to 3400 Hz (the frequency range essential for speech intelligibility). This 3.1 kHz bandwidth, rounded to 4 kHz for engineering margin, could theoretically be represented by 8000 samples per second.

Why Sampling?

The advantage of sampling became apparent with the development of pulse-code modulation (PCM) at Bell Labs in the 1930s and 1940s. If each sample could be quantized to a finite number of levels (e.g., 256 levels using 8 bits), then voice could be transmitted as a sequence of numbers. These numbers offered several key advantages. First, **regeneration** becomes possible: unlike analog signals that accumulate noise through repeaters, digital signals can be perfectly regenerated at each relay station. Second, **multiplexing** allows multiple conversations to share a channel through

time-division multiplexing. Third, **storage** in digital memory becomes straightforward. Fourth, **processing** through digital signal processing, filtering, and encryption becomes possible.

The Cost of Undersampling

What happens if economic pressure leads engineers to sample below the Nyquist rate? Shannon's analysis revealed that frequencies above half the sampling rate would "fold back" into the baseband, appearing as spurious lower frequencies. This *aliasing* effect corrupts the signal irreversibly—no subsequent processing can undo the damage.

19.1.4 Early Digital Computers and Sampled-Data Systems

The emergence of digital computers in the 1950s and 1960s created an entirely new domain for sampling theory: automatic control.

Why Computers Cannot Handle Continuous Signals

Digital computers are fundamentally discrete devices. They operate on finite sequences of numbers, perform calculations at discrete instants, and communicate through digital interfaces. When a computer must control a physical process—maintaining temperature in a chemical reactor, guiding a spacecraft, or positioning a machine tool—it confronts an inherent mismatch. The physical world evolves continuously in time while the computer operates at discrete instants. Furthermore, sensors produce continuous voltages that must be digitized, and actuators require analog commands that must be synthesized from discrete values. This mismatch necessitated a complete theory of *sampled-data control systems*.

Radar Systems: Early Digital Control

The first large-scale applications of digital control emerged in military radar and fire-control systems during World War II and the Cold War. The SAGE (Semi-Automatic Ground Environment) air defense system of the 1950s used digital computers to track aircraft based on radar returns sampled at discrete intervals. These systems required sampling radar signals at rates sufficient to track aircraft without aliasing their motion, implementing tracking filters (predecessors to the Kalman filter) in discrete time, and generating smooth guidance commands from discrete computations.

Aerospace Applications

The Apollo guidance computer, operational in the 1960s, controlled spacecraft attitude and navigation using sampled sensor data and discrete control algorithms. With limited computational power (a 2 MHz clock, 2 KB of RAM), these systems had to carefully balance sample rate against computational burden—too fast sampling overwhelmed the processor; too slow sampling degraded control performance.

Process Control

Chemical plants, refineries, and power stations adopted digital control in the 1970s and 1980s. These applications typically involved relatively slow dynamics (time constants of minutes to hours) but required high reliability and the ability to coordinate hundreds of control loops. Digital systems offered programmability, data logging, and networked communication impossible with analog controllers.

19.1.5 The Fundamental Question for Control Engineers

All these applications raised the same fundamental question:

The Digital Control Question

Given a continuous-time system that must be controlled by a digital computer, three fundamental questions arise: How fast must we sample to preserve control performance? How do we convert continuous controller designs to discrete implementations? And what artifacts arise from sampling, and how do we mitigate them?

The remainder of this chapter answers these questions with mathematical precision and practical guidance.

19.2 Modern Applications of Sampling and Discretization

The principles established by Nyquist and Shannon now pervade every digital system that interacts with the analog world.

19.2.1 Digital Audio

The compact disc (CD), introduced in 1982, samples audio at 44.1 kHz with 16-bit resolution. This rate was chosen because human hearing extends to approximately 20 kHz and the Nyquist theorem requires sampling above 40 kHz. A small margin at 44.1 kHz provides headroom for practical anti-aliasing filters, and the specific value aligned with existing video equipment standards. Modern high-resolution audio systems sample at 96 kHz, 192 kHz, or even higher—not because humans can hear frequencies above 20 kHz, but because higher sample rates ease anti-aliasing filter requirements and provide margin for signal processing.

19.2.2 Digital Control in Microcontrollers

Modern embedded systems implement control algorithms on microcontrollers with integrated analog-to-digital converters (ADCs) and pulse-width modulation (PWM) outputs. Table 19.1 summarizes typical applications and their sample rate requirements. The sample rate selection involves tradeoffs between control bandwidth, computational load, sensor noise, and power consumption.

Table 19.1: Typical Microcontroller Control Applications and Sample Rates

Application	Sample Rate
Motor control (brushless DC commutation)	10–100 kHz
Power electronics (switching converters)	10–100 kHz
Automotive systems (engine control)	1–10 kHz
Consumer electronics (temperature, battery)	1–100 Hz

19.2.3 Anti-Aliasing in Signal Processing

Every properly designed data acquisition system includes an anti-aliasing filter before the analog-to-digital converter. This lowpass filter attenuates frequencies above half the sampling rate, preventing aliasing. Design considerations include filter order, since steeper cutoff requires higher-order filters, and phase distortion, which is important for time-domain applications. Engineers must also balance cost and complexity tradeoffs, and may employ oversampling strategies that relax filter requirements.

19.2.4 Real-Time Embedded Systems

In real-time control systems, the sample period establishes the fundamental timing constraint. All sensor reading, control computation, and actuator output must complete within each sample period. This creates design constraints:

$$T_{\text{computation}} + T_{\text{I/O}} < T_{\text{sample}} \quad (19.2)$$

Failure to meet this deadline can destabilize the control system or create unpredictable behavior. Real-time operating systems and careful software architecture ensure deterministic timing.

19.3 Mathematical Foundations of Sampling

We now develop the complete mathematical framework for understanding sampling, aliasing, and reconstruction.

19.3.1 The Ideal Sampling Process

Impulse Train Modulation

The mathematical idealization of sampling treats the sampler as a modulator that multiplies the continuous signal by a train of impulses. Define the sampling period as T and the sampling frequency as $f_s = 1/T$ (or $\omega_s = 2\pi f_s$ in radians per second).

The **impulse train** (Dirac comb) is:

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \quad (19.3)$$

The **sampled signal** $x^*(t)$ is the product of the continuous signal $x(t)$ and the impulse train:

$$x^*(t) = x(t) \cdot \delta_T(t) = \sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT) \quad (19.4)$$

This represents a train of impulses whose amplitudes equal the signal values at the sampling instants.

Frequency-Domain Analysis of Sampling

The Fourier transform reveals the spectral effect of sampling. The impulse train has Fourier transform:

$$\mathcal{F}\{\delta_T(t)\} = \frac{1}{T} \sum_{k=-\infty}^{\infty} \delta(f - kf_s) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s) \quad (19.5)$$

Since multiplication in time corresponds to convolution in frequency:

$$X^*(f) = X(f) * \left(\frac{1}{T} \sum_{k=-\infty}^{\infty} \delta(f - kf_s) \right) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(f - kf_s) \quad (19.6)$$

Spectral Effect of Sampling

Sampling creates infinitely many copies of the original spectrum $X(f)$, shifted to center frequencies $0, \pm f_s, \pm 2f_s, \dots$, each scaled by $1/T$.

Visualization of the Sampled Spectrum

Figure 19.1 illustrates this spectral replication.

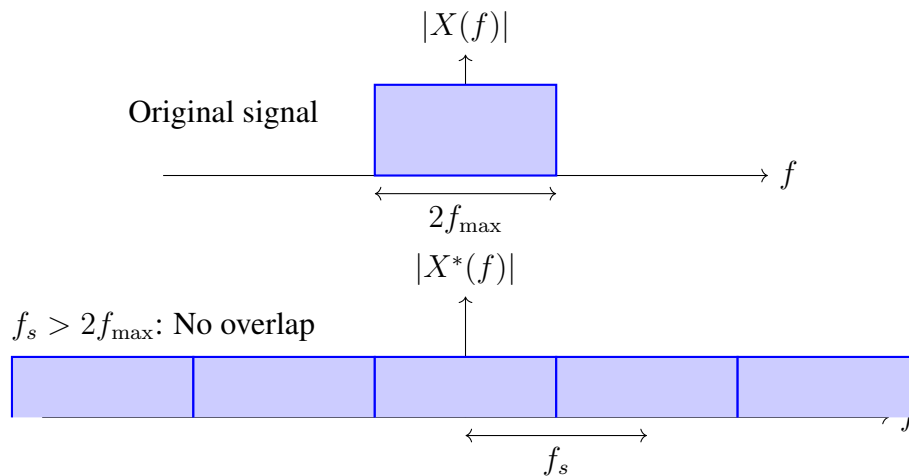


Figure 19.1: Sampling in the frequency domain: the original spectrum is replicated at multiples of the sampling frequency f_s .

19.3.2 The Nyquist-Shannon Sampling Theorem

Statement of the Theorem

Theorem 19.1 (Nyquist-Shannon Sampling Theorem). *Let $x(t)$ be a bandlimited signal with no frequency components above $f_{\max}$ Hz:*

$$X(f) = 0 \quad \text{for } |f| > f_{\max} \quad (19.7)$$

If $x(t)$ is sampled at rate $f_s > 2f_{\max}$, then $x(t)$ can be exactly reconstructed from its samples $\{x(nT)\}$ by the interpolation formula:

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT) \operatorname{sinc}\left(\frac{t - nT}{T}\right) \quad (19.8)$$

where $T = 1/f_s$ is the sampling period.

Derivation from Frequency-Domain Analysis

The proof proceeds by showing that an ideal lowpass filter can extract the original spectrum from the sampled spectrum.

Step 1: The sampled spectrum consists of shifted copies (Eq. ??).

Step 2: If $f_s > 2f_{\max}$, adjacent copies do not overlap. The gap between the edge of one copy (at $f_{\max}$) and the nearest edge of the next copy (at $f_s - f_{\max}$) is:

$$\text{Gap} = (f_s - f_{\max}) - f_{\max} = f_s - 2f_{\max} > 0 \quad (19.9)$$

Step 3: An ideal lowpass filter with cutoff f_c satisfying $f_{\max} < f_c < f_s - f_{\max}$ perfectly isolates the baseband copy centered at $f = 0$. Choosing $f_c = f_s/2$ (half the sampling frequency, the *Nyquist frequency*):

$$H_{\text{ideal}}(f) = T \cdot \operatorname{rect}\left(\frac{f}{f_s}\right) = \begin{cases} T & |f| < f_s/2 \\ 0 & |f| > f_s/2 \end{cases} \quad (19.10)$$

The factor T compensates for the $1/T$ scaling introduced by sampling.

Step 4: The reconstructed spectrum is:

$$X_{\text{reconstructed}}(f) = X^*(f) \cdot H_{\text{ideal}}(f) = X(f) \quad (19.11)$$

Step 5: The time-domain reconstruction filter is the inverse Fourier transform of $H_{\text{ideal}}(f)$:

$$h_{\text{ideal}}(t) = \operatorname{sinc}\left(\frac{t}{T}\right) = \frac{\sin(\pi t/T)}{\pi t/T} \quad (19.12)$$

Convoluting the sampled signal with this sinc function yields the reconstruction formula (Eq. ??).

The Nyquist Frequency

The **Nyquist frequency** $f_N = f_s/2$ is the highest frequency that can be unambiguously represented at sampling rate f_s . Equivalently, the **Nyquist rate** is the minimum sampling rate $2f_{\max}$ required to capture a signal with maximum frequency $f_{\max}$.

19.3.3 Aliasing: When Sampling Fails

The Aliasing Phenomenon

When a signal contains frequencies above the Nyquist frequency ($f > f_s/2$), the spectral copies overlap. This overlap causes high-frequency components to “fold back” into the baseband, appearing as lower frequencies that are indistinguishable from legitimate signal components.

Definition 19.1 (Aliasing). **Aliasing** is the corruption of sampled data when frequency components above the Nyquist frequency fold into the baseband, creating spurious frequencies that cannot be separated from the true signal.

Mathematical Description of Folding

Consider a sinusoidal signal at frequency f_{signal} sampled at rate f_s . The sampled values are:

$$x(nT) = A \cos(2\pi f_{\text{signal}} \cdot nT + \phi) = A \cos\left(\frac{2\pi f_{\text{signal}}}{f_s} \cdot n + \phi\right) \quad (19.13)$$

Due to the periodicity of cosine with period 2π , these samples are identical to those from a frequency:

$$f_{\text{alias}} = |f_{\text{signal}} - k \cdot f_s| \quad \text{for integer } k \text{ chosen so } 0 \leq f_{\text{alias}} \leq f_s/2 \quad (19.14)$$

Aliasing Formula

A signal at frequency f sampled at rate f_s produces the same samples as a signal at the **alias frequency**:

$$f_{\text{alias}} = \left| f - \left\lfloor \frac{f}{f_s} + 0.5 \right\rfloor \cdot f_s \right| \quad (19.15)$$

where $\lfloor \cdot \rfloor$ denotes rounding to the nearest integer.

Aliasing Examples

Simple Aliasing Calculation

A 1200 Hz sinusoid is sampled at 1000 Hz. What frequency appears in the sampled data?

Solution: The signal frequency exceeds the Nyquist frequency (500 Hz). The alias frequency is:

$$f_{\text{alias}} = |1200 - 1 \times 1000| = 200 \text{ Hz}$$

The sampled data appears to contain a 200 Hz signal, not 1200 Hz.

Wagon Wheel Effect in Cinema

In film (24 frames per second), a wagon wheel with 12 spokes rotating at 25 revolutions per second appears to rotate slowly backward.

Solution: Each spoke passes a given position at rate $25 \times 12 = 300$ Hz. Sampled at 24 Hz,

the alias frequency is:

$$f_{\text{alias}} = |300 - 12 \times 24| = |300 - 288| = 12 \text{ Hz}$$

Since this aliases to 12 Hz but the sampling rate is 24 Hz, and $12 < 24/2 = 12$, the wheel appears to move at $12/12 = 1$ revolution per second—but in the opposite direction due to the phase reversal in folding.

Frequency Domain Visualization of Aliasing

Figure 19.2 shows spectral overlap when $f_s < 2f_{\text{max}}$.

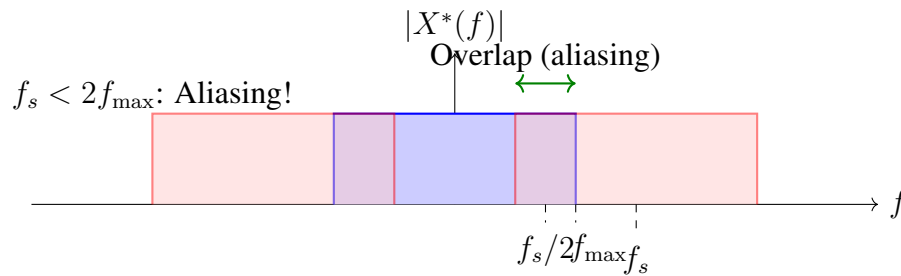


Figure 19.2: Aliasing in the frequency domain: when $f_s < 2f_{\text{max}}$, spectral copies overlap, corrupting the baseband.

19.3.4 The Zero-Order Hold (ZOH)

Physical Motivation

After sampling, the control computer calculates an output value. But digital-to-analog converters (DACs) produce continuous output—they cannot create impulses. The most common practical approach is the **zero-order hold** (ZOH): the DAC holds each output sample constant until the next sample arrives.

Definition 19.2 (Zero-Order Hold). The zero-order hold maintains the output value $y[n]$ constant from time $t = nT$ until time $t = (n + 1)T$:

$$y(t) = y[n] \quad \text{for } nT \leq t < (n + 1)T \quad (19.16)$$

The resulting output is a staircase approximation to the ideal reconstructed signal.

Transfer Function of the ZOH

The ZOH can be modeled as an LTI system with impulse response equal to a rectangular pulse of width T :

$$h_0(t) = \begin{cases} 1 & 0 \leq t < T \\ 0 & \text{otherwise} \end{cases} \quad (19.17)$$

Taking the Laplace transform:

$$H_0(s) = \int_0^T e^{-st} dt = \frac{1 - e^{-sT}}{s} \quad (19.18)$$

Zero-Order Hold Transfer Function

$$H_0(s) = \frac{1 - e^{-sT}}{s} \quad (19.19)$$

This is a lowpass filter that introduces a delay of approximately $T/2$.

Frequency Response of the ZOH

Substituting $s = j\omega$:

$$H_0(j\omega) = \frac{1 - e^{-j\omega T}}{j\omega} = e^{-j\omega T/2} \cdot \frac{e^{j\omega T/2} - e^{-j\omega T/2}}{j\omega} \quad (19.20)$$

$$H_0(j\omega) = e^{-j\omega T/2} \cdot T \cdot \frac{\sin(\omega T/2)}{\omega T/2} = T \cdot \text{sinc}\left(\frac{\omega T}{2\pi}\right) \cdot e^{-j\omega T/2} \quad (19.21)$$

The ZOH has a **magnitude** response that rolls off as a sinc function, with zeros at $\omega = 2\pi k/T$ for $k = 1, 2, \dots$, and a **phase** response consisting of a linear phase lag of $\omega T/2$, equivalent to a time delay of $T/2$.

Impact on Control Systems

The ZOH delay of $T/2$ adds phase lag that can destabilize feedback systems. For a system with crossover frequency ω_c , the additional phase lag is:

$$\phi_{\text{ZOH}} = -\frac{\omega_c T}{2} \quad \text{radians} \quad (19.22)$$

This motivates the rule of thumb: to limit ZOH-induced phase lag to a few degrees, the sampling frequency should be much higher than the control bandwidth.

19.3.5 Discretization Methods for Controllers

Given a continuous-time controller $C(s)$, we must convert it to a discrete-time controller $C(z)$ for implementation in a digital computer. Several methods exist, each with different accuracy and stability properties.

Forward Euler (Forward Difference)

The forward Euler method approximates the derivative by a forward difference:

$$\frac{dx}{dt} \approx \frac{x(t+T) - x(t)}{T} = \frac{x[n+1] - x[n]}{T} \quad (19.23)$$

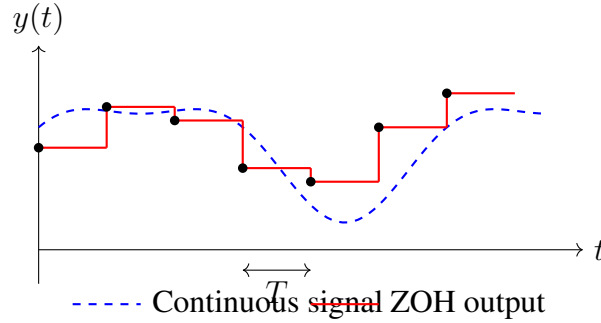


Figure 19.3: Zero-order hold operation: the continuous signal (dashed) is sampled, and the ZOH maintains each sample value constant until the next sample.

In the Laplace/z-transform domains, $s \cdot X(s)$ corresponds to differentiation, and $z \cdot X(z)$ corresponds to a one-step advance. This gives the substitution:

$$\boxed{s \rightarrow \frac{z - 1}{T}} \quad (19.24)$$

The forward Euler method is simple to implement but can map stable continuous poles to unstable discrete poles. Its accuracy is $O(T)$.

Backward Euler (Backward Difference)

The backward Euler method approximates the derivative by a backward difference:

$$\frac{dx}{dt} \approx \frac{x(t) - x(t - T)}{T} = \frac{x[n] - x[n - 1]}{T} \quad (19.25)$$

This gives the substitution:

$$\boxed{s \rightarrow \frac{z - 1}{Tz}} \quad (19.26)$$

The backward Euler method never maps stable continuous poles to unstable discrete poles, making it A-stable. However, it may introduce excessive damping. Its accuracy is $O(T)$.

Tustin (Bilinear) Transform

The Tustin method, also called the bilinear transform, uses the trapezoidal integration rule:

$$\int_{t-T}^t \frac{dx}{d\tau} d\tau \approx \frac{T}{2} \left[\frac{dx}{dt} \Big|_{t-T} + \frac{dx}{dt} \Big|_t \right] \quad (19.27)$$

This leads to the substitution:

$$\boxed{s \rightarrow \frac{2}{T} \cdot \frac{z - 1}{z + 1}} \quad (19.28)$$

The Tustin method maps the left half of the s -plane exactly to the interior of the unit circle in the z -plane, thereby preserving stability so that stable continuous systems yield stable discrete systems.

It introduces frequency warping where continuous frequency ω maps to discrete frequency $\omega_d = (2/T) \tan(\omega T/2)$. Its accuracy is $O(T^2)$, making it more accurate than the Euler methods.

Tustin Frequency Warping

The Tustin transform maps continuous frequency ω to discrete frequency ω_d according to:

$$\omega_d = \frac{2}{T} \tan\left(\frac{\omega T}{2}\right) \quad (19.29)$$

For $\omega T \ll 1$: $\omega_d \approx \omega$ (low frequency, good match). As $\omega \rightarrow \omega_s/2$: $\omega_d \rightarrow \infty$ (high frequency compression).

Prewarping

When a specific frequency (e.g., crossover frequency) must match exactly between continuous and discrete designs, *prewarping* adjusts the Tustin transform:

$$s \rightarrow \frac{\omega_0}{\tan(\omega_0 T/2)} \cdot \frac{z-1}{z+1} \quad (19.30)$$

This ensures the continuous frequency ω_0 maps exactly to the discrete frequency ω_0 .

Zero-Order Hold Discretization (Exact Method)

The most accurate discretization method accounts for the ZOH in the feedback loop. Given a continuous plant $G(s)$ with a ZOH on its input and a sampler on its output, the equivalent discrete transfer function is:

$$G(z) = (1 - z^{-1}) \mathcal{Z} \left\{ \frac{G(s)}{s} \right\} \quad (19.31)$$

where $\mathcal{Z}\{\cdot\}$ denotes the z-transform of the sampled step response.

This method is exact for the plant-hold combination but does not apply directly to controller discretization.

Comparison of Discretization Methods

Table 19.2 summarizes the properties of each method.

19.3.6 Choosing the Sample Rate

Rule of Thumb

For control systems, the sampling rate should be significantly higher than the Nyquist minimum to account for ZOH phase lag, computational delay, practical anti-aliasing filter limitations, and margin for modeling errors.

Table 19.2: Comparison of Discretization Methods

Method	Substitution	Stability	Accuracy	Frequency
Forward Euler	$\frac{z-1}{T}$	May destabilize	$O(T)$	Good at low f
Backward Euler	$\frac{z-1}{Tz}$	Always stable	$O(T)$	Over-damped
Tustin	$\frac{2}{T} \frac{z-1}{z+1}$	Preserves	$O(T^2)$	Warped
ZOH (plant)	Exact	Exact	Exact	Exact

Sample Rate Guidelines

For a control system with closed-loop bandwidth ω_{BW} :

$$\omega_s \geq 10\omega_{BW} \quad (\text{minimum practical}) \quad (19.32)$$

$$\omega_s \approx 20\omega_{BW} \quad (\text{typical design}) \quad (19.33)$$

Higher rates (30–50 $\times$ bandwidth) may be needed for systems with lightly damped resonances, high-precision positioning applications, or fast disturbance rejection requirements.

Tradeoffs in Sample Rate Selection

Higher sample rates provide better approximation of continuous control, reduced phase lag from the ZOH, easier anti-aliasing filter design, and improved high-frequency disturbance rejection. However, these benefits come at the cost of increased computational load, more demanding real-time requirements, higher power consumption, and increased data storage and communication bandwidth. Engineers must balance these tradeoffs based on their specific application requirements.

19.4 Practical Laboratory: Demonstrating Aliasing and Discretization Effects

This laboratory provides hands-on experience with aliasing and discretization using readily available hardware.

19.4.1 Learning Objectives

Upon completion, students will be able to visually observe aliasing when a signal is undersampled, measure the alias frequency and verify the folding formula, discretize a continuous controller and compare performance, and understand the effect of sample rate on control quality.

19.4.2 Required Components

Table 19.3: Bill of Materials for Sampling Laboratory

Component	Specification	Qty
Arduino Uno/Nano	ATmega328P	1
Function Generator	0–100 kHz, sine output (or Arduino PWM + filter)	1
LEDs	Standard 5mm, assorted colors	10
Resistors	220 Ω for LEDs	10
Oscilloscope	Digital or analog, 2 channels	1
Lowpass Filter	RC filter components (10k Ω , 0.1 μ F)	1 set
Potentiometer	10k Ω	2
DC Motor with Encoder	Same as Chapter 1 lab	1
Motor Driver	L298N or equivalent	1

19.4.3 Experiment 1: Visualizing Aliasing

Setup

This experiment demonstrates aliasing using a strobe-light approach. A row of LEDs represents the “sampled” output, while the true signal frequency is known.

To set up the experiment, first wire 8 LEDs in a row, each controlled by a digital output pin. Then configure the Arduino to update the LED display at a fixed “sample rate” (e.g., 10 Hz). Finally, generate a test signal using either a potentiometer rotated at controlled speed or a function generator producing a low-frequency sine wave.

Software: Aliasing Demonstration

Procedure

Begin by observing behavior without aliasing: set the sample rate to 100 Hz (10 ms period) and slowly rotate the potentiometer at about 1 Hz. The LED display should smoothly follow the rotation. Next, introduce aliasing by reducing the sample rate to 2 Hz (500 ms period) and rotating the potentiometer at approximately 3 Hz. Observe that the LED display appears to move backward or erratically—this is aliasing in action. To quantify aliasing, set the sample rate at 10 Hz and rotate the potentiometer at exactly 12 Hz (use a metronome or timer); the display should appear to move at 2 Hz since $|12 - 10| = 2$. Finally, explore the Nyquist limit by increasing the sample rate until the display correctly tracks the 12 Hz rotation, which should occur above 24 Hz.

Data Collection

Complete Table 19.4.

Algorithm 39 Aliasing Demonstration with LED Array

```

1: Constants:  $N_{\text{LEDs}} \leftarrow 8$ , LED pins  $\leftarrow \{2, 3, 4, 5, 6, 7, 8, 9\}$ , analog input pin
2: Variables:  $T_{\text{sample}} \leftarrow 100$  ms,  $t_{\text{last}} \leftarrow 0$ 
3:
4: procedure INITIALIZE
5:   Configure serial communication
6:   for  $i = 0 \rightarrow N_{\text{LEDs}} - 1$  do
7:     Set LED pin  $i$  as output
8:   end for
9: end procedure
10:
11: procedure DISPLAYONLEDS(value)
12:   ledIndex  $\leftarrow \text{Map}(\text{value}, [0, 1023], [0, N_{\text{LEDs}} - 1])$ 
13:   for  $i = 0 \rightarrow N_{\text{LEDs}} - 1$  do
14:     if  $i = \text{ledIndex}$  then
15:       Turn LED  $i$  ON
16:     else
17:       Turn LED  $i$  OFF
18:     end if
19:   end for
20: end procedure
21:
22: procedure MAINLOOP
23:   if new sample period received from serial then
24:      $T_{\text{sample}} \leftarrow$  new period value
25:   end if
26:    $t_{\text{now}} \leftarrow$  current time in milliseconds
27:   if  $t_{\text{now}} - t_{\text{last}} \geq T_{\text{sample}}$  then
28:      $t_{\text{last}} \leftarrow t_{\text{now}}$ 
29:     reading  $\leftarrow$  read analog input
30:     DISPLAYONLEDS(reading)
31:     Output reading to serial for plotting
32:   end if
33: end procedure

```

Table 19.4: Aliasing Experiment Results

Signal Freq. (Hz)	Sample Rate (Hz)	Predicted Alias (Hz)	Observed (Hz)
3	10	3 (no alias)	
8	10	2	
12	10	2	
15	10	5	
18	10	2	
12	30	12 (no alias)	

19.4.4 Experiment 2: Effect of Sample Rate on Control Quality

Setup

Using the motor control hardware from Chapter 1, implement a discrete PID controller at various sample rates.

Software: Variable Sample Rate Controller

Procedure

Establish a baseline at 100 Hz by setting the sample period to 10000 μs , commanding a 90-degree step, and recording settling time, overshoot, and steady-state error. Then reduce the sample rate, testing at 50 Hz, 20 Hz, 10 Hz, and 5 Hz; for each rate, record settling time, maximum overshoot, and any oscillation or instability observed. Next, increase the sample rate to 200 Hz and 500 Hz to note any improvement (or lack thereof due to other limiting factors). Finally, find the stability limit by reducing sample rate until the system becomes unstable, and record this critical sample rate.

Data Collection

Complete Table 19.5.

Table 19.5: Sample Rate Effect on Control Quality

Sample Rate (Hz)	Settling Time (ms)	Overshoot (%)	Oscillation?	Stable?
500				
200				
100				
50				
20				
10				
5				

19.4.5 Experiment 3: Comparing Discretization Methods

Objective

Implement the same continuous PI controller using different discretization methods and compare transient response.

Continuous PI Controller

Consider the continuous PI controller:

$$C(s) = K_p + \frac{K_i}{s} = K_p \left(1 + \frac{1}{\tau_i s} \right) \quad (19.34)$$

with $K_p = 2.0$ and $K_i = 0.5$ (so $\tau_i = K_p/K_i = 4$).

Algorithm 40 Variable Sample Rate Motor Control

```

1: Constants:  $K_p \leftarrow 2.0$ ,  $K_i \leftarrow 0.5$ ,  $K_d \leftarrow 0.1$ , counts per revolution  $\leftarrow 400$ 
2: Variables:  $T_{\text{sample}} \leftarrow 10000 \mu\text{s}$ ,  $t_{\text{last}} \leftarrow 0$ , integral  $\leftarrow 0$ ,  $e_{\text{prev}} \leftarrow 0$ 
3: Variables: setpoint  $\leftarrow 0$ , encoderCount  $\leftarrow 0$ , settled  $\leftarrow \text{false}$ , maxOvershoot  $\leftarrow 0$ 
4:
5: procedure INITIALIZE
6:   Configure encoder pins with interrupts for quadrature decoding
7:   Configure motor PWM output pins
8: end procedure
9:
10: procedure UPDATEENCODER ▷ Called on encoder edge interrupt
11:   Decode quadrature signals to update encoderCount
12: end procedure
13:
14: function GETPOSITIONDEGREES
15:   return (encoderCount/400)  $\times$  360
16: end function
17:
18: procedure SETMOTORPWM(pwm)
19:   pwm  $\leftarrow$  Constrain(pwm, -255, 255)
20:   if pwm > 0 then
21:     Set forward direction with magnitude pwm
22:   else
23:     Set reverse direction with magnitude |pwm|
24:   end if
25: end procedure
26:
27: procedure MAINLOOP
28:   if setpoint command received then
29:     Update setpoint, reset integral and  $e_{\text{prev}}$ 
30:     Reset performance tracking variables
31:   else if sample period command received then
32:      $T_{\text{sample}} \leftarrow$  new period value
33:   end if
34:
35:    $t_{\text{now}} \leftarrow$  current time in microseconds
36:   if  $t_{\text{now}} - t_{\text{last}} \geq T_{\text{sample}}$  then
37:      $\Delta t \leftarrow (t_{\text{now}} - t_{\text{last}})/10^6$  ▷ Convert to seconds
38:      $t_{\text{last}} \leftarrow t_{\text{now}}$ 
39:
40:      $\theta \leftarrow$  GETPOSITIONDEGREES
41:      $e \leftarrow$  setpoint  $- \theta$ 
42:
43:     ▷ PID calculation with discrete integration
44:     integral  $\leftarrow$  integral  $+ e \cdot \Delta t$ 
45:     derivative  $\leftarrow (e - e_{\text{prev}})/\Delta t$ 
46:      $e_{\text{prev}} \leftarrow e$ 
47:
48:      $u \leftarrow K_p \cdot e + K_i \cdot \text{integral} + K_d \cdot \text{derivative}$ 
49:     SETMOTORPWM( $u$ )

```

Discretized Controllers

Using $T = 0.01$ s (100 Hz):

Forward Euler ($s \rightarrow (z - 1)/T$):

$$C_{\text{FE}}(z) = K_p + \frac{K_i T}{z - 1} \Rightarrow u[n] = K_p e[n] + K_i T \sum_{k=0}^n e[k] \quad (19.35)$$

Backward Euler ($s \rightarrow (z - 1)/(Tz)$):

$$C_{\text{BE}}(z) = K_p + \frac{K_i T z}{z - 1} \Rightarrow u[n] = K_p e[n] + K_i T e[n] + u_i[n - 1] \quad (19.36)$$

Tustin ($s \rightarrow (2/T)(z - 1)/(z + 1)$):

$$C_{\text{Tustin}}(z) = K_p + \frac{K_i T}{2} \cdot \frac{z + 1}{z - 1} \quad (19.37)$$

Implementation

Procedure

For each discretization method (Forward Euler, Backward Euler, Tustin), reset the integrators, command a step change in setpoint, record the step response (position vs. time), and compare settling time, overshoot, and steady-state error.

At slow sample rates (e.g., 10 Hz), the differences between methods become more pronounced. At fast sample rates (e.g., 500 Hz), all methods should produce nearly identical results.

19.5 Worked Examples

Example 19.1 (Minimum Sample Rate for Audio). A digital audio system must capture human speech with frequencies from 300 Hz to 3400 Hz. What is the minimum theoretically required sample rate? What sample rate would you recommend in practice?

Solution:

The highest frequency component is $f_{\text{max}} = 3400$ Hz.

By the Nyquist-Shannon theorem, the minimum sample rate is:

$$f_{s,\text{min}} = 2f_{\text{max}} = 2 \times 3400 = 6800 \text{ Hz}$$

In practice, we need margin for anti-aliasing filter rolloff (since practical filters cannot have infinite slope), a guard band to prevent any frequency folding, and tolerance in component values.

The telephone standard uses 8000 Hz, providing approximately 18% margin. For higher quality, sample rates of 16 kHz (wideband telephony) or 44.1 kHz (CD audio) are used.

$f_{s,\text{min}} = 6800 \text{ Hz}; \quad \text{Practical: } f_s = 8000 \text{ Hz or higher}$
--

Algorithm 41 Comparing Discretization Methods for PI Control

```

1: Constants:  $K_p, K_i$  (controller gains)
2: Variables:  $\text{method} \in \{\text{Forward, Backward, Tustin}\}$ 
3: State:  $I_F \leftarrow 0, I_B \leftarrow 0, I_T \leftarrow 0, e_{\text{prev}} \leftarrow 0$ 
4:
5: function COMPUTEPI( $e, \Delta t$ )
6:    $u_P \leftarrow K_p \cdot e$  ▷ Proportional term
7:
8:   if  $\text{method} = \text{Forward Euler}$  then
9:      $u_I \leftarrow I_F$  ▷ Use previous integral value
10:     $I_F \leftarrow I_F + K_i \cdot \Delta t \cdot e$  ▷ Update for next iteration
11:   else if  $\text{method} = \text{Backward Euler}$  then
12:      $I_B \leftarrow I_B + K_i \cdot \Delta t \cdot e$  ▷ Update first
13:      $u_I \leftarrow I_B$  ▷ Use updated value
14:   else if  $\text{method} = \text{Tustin}$  then
15:      $I_T \leftarrow I_T + K_i \cdot \Delta t \cdot 0.5 \cdot (e + e_{\text{prev}})$  ▷ Trapezoidal rule
16:      $u_I \leftarrow I_T$ 
17:      $e_{\text{prev}} \leftarrow e$ 
18:   end if
19:
20:   return  $u_P + u_I$ 
21: end function
22:
23: procedure RESETINTEGRATORS
24:    $I_F \leftarrow 0, I_B \leftarrow 0, I_T \leftarrow 0, e_{\text{prev}} \leftarrow 0$ 
25: end procedure

```

Example 19.2 (Identifying Aliased Frequency). A vibration sensor on a rotating machine samples at 1000 Hz. The machine has a shaft rotating at 630 RPM with a defect that creates a vibration at shaft frequency. An additional vibration appears in the data at 45 Hz. Could this be an alias? If so, what is the true frequency?

Solution:

Shaft frequency: $f_{\text{shaft}} = 630/60 = 10.5$ Hz.

Nyquist frequency: $f_N = f_s/2 = 500$ Hz.

Any true frequency f_{true} that aliases to 45 Hz must satisfy:

$$f_{\text{alias}} = |f_{\text{true}} - k \cdot f_s| = 45 \text{ Hz}$$

for some integer k . The possibilities are:

$$k = 0 : f_{\text{true}} = 45 \text{ Hz}$$

$$k = 1 : f_{\text{true}} = 1000 - 45 = 955 \text{ Hz, or } f_{\text{true}} = 1000 + 45 = 1045 \text{ Hz}$$

$$k = 2 : f_{\text{true}} = 2000 - 45 = 1955 \text{ Hz, or } f_{\text{true}} = 2045 \text{ Hz}$$

$\vdots$

Checking if any are related to the shaft frequency: $955/10.5 \approx 91$ (not an integer multiple), $1045/10.5 \approx 99.5$ (not an integer multiple), and $45/10.5 \approx 4.3$ (not exact either).

Let's check for blade-pass frequency. If the machine has 91 blades:

$$f_{\text{blade}} = 91 \times 10.5 = 955.5 \text{ Hz}$$

This would alias to $|955.5 - 1000| = 44.5 \approx 45$ Hz.

Yes, likely alias of 955 Hz ($91 \times$ shaft frequency)

Verification: Increase sample rate above 1910 Hz. If the 45 Hz peak moves or disappears and a peak near 955 Hz appears, aliasing is confirmed.

Example 19.3 (Discretizing a First-Order System). Discretize the first-order lowpass filter $G(s) = \frac{1}{\tau s + 1}$ with $\tau = 0.1$ s using each discretization method. Use sample period $T = 0.02$ s (50 Hz).

Solution:

Forward Euler ($s \rightarrow (z - 1)/T$):

$$G_{\text{FE}}(z) = \frac{1}{\tau \cdot \frac{z-1}{T} + 1} = \frac{T}{\tau(z-1) + T} = \frac{T}{\tau z - \tau + T}$$

$$G_{\text{FE}}(z) = \frac{0.02}{0.1z - 0.1 + 0.02} = \frac{0.02}{0.1z - 0.08} = \frac{0.2}{z - 0.8}$$

Pole at $z = 0.8$, which corresponds to continuous pole at $s = \ln(0.8)/T = -11.16$ rad/s (original: $s = -10$ rad/s).

Backward Euler ($s \rightarrow (z - 1)/(Tz)$):

$$G_{\text{BE}}(z) = \frac{1}{\tau \cdot \frac{z-1}{Tz} + 1} = \frac{Tz}{\tau(z-1) + Tz} = \frac{Tz}{\tau z - \tau + Tz}$$

$$G_{\text{BE}}(z) = \frac{0.02z}{0.1z - 0.1 + 0.02z} = \frac{0.02z}{0.12z - 0.1} = \frac{0.167z}{z - 0.833}$$

Pole at $z = 0.833$, corresponding to $s = \ln(0.833)/T = -9.12 \text{ rad/s}$ (more damped than original).

Tustin ($s \rightarrow (2/T)(z - 1)/(z + 1)$):

$$G_{\text{Tustin}}(z) = \frac{1}{\tau \cdot \frac{2}{T} \frac{z-1}{z+1} + 1} = \frac{z+1}{\frac{2\tau}{T}(z-1) + (z+1)}$$

$$G_{\text{Tustin}}(z) = \frac{z+1}{10(z-1) + (z+1)} = \frac{z+1}{11z-9} = \frac{z+1}{11(z-0.818)}$$

$$G_{\text{Tustin}}(z) = \frac{1}{11} \cdot \frac{z+1}{z-0.818} = 0.091 \cdot \frac{z+1}{z-0.818}$$

Pole at $z = 0.818$, corresponding to $s = \ln(0.818)/T = -10.05 \text{ rad/s}$ (very close to original).

Comparison:

Method	Discrete Pole	Equivalent Continuous Pole
Forward Euler	0.800	-11.16 rad/s
Backward Euler	0.833	-9.12 rad/s
Tustin	0.818	-10.05 rad/s
Original	(exact: 0.819)	-10 rad/s

Tustin most accurate; Forward Euler faster; Backward Euler slower

Example 19.4 (Designing an Anti-Aliasing Filter). A data acquisition system samples at 10 kHz. Signals of interest are below 1 kHz. Design an anti-aliasing filter that attenuates the Nyquist frequency (5 kHz) by at least 40 dB.

Solution:

We need a lowpass filter with passband from 0–1 kHz (signals of interest) and stopband at 5 kHz (which must be attenuated by 40 dB).

The transition band ratio is:

$$\text{Transition ratio} = \frac{f_{\text{stop}}}{f_{\text{pass}}} = \frac{5000}{1000} = 5$$

For a Butterworth filter, the attenuation at frequency ω is:

$$|H(j\omega)|^2 = \frac{1}{1 + (\omega/\omega_c)^{2n}}$$

At the stopband edge, we require:

$$20 \log_{10} |H(j\omega_{\text{stop}})| \leq -40 \text{ dB}$$

$$|H|^2 \leq 10^{-4}$$

$$\frac{1}{1 + (5)^{2n}} \leq 10^{-4}$$

$$5^{2n} \geq 10^4 - 1 \approx 10^4$$

$$2n \log(5) \geq 4$$

$$n \geq \frac{4}{2 \times 0.699} = 2.86$$

A **third-order** Butterworth filter with cutoff at 1 kHz will provide approximately:

$$\text{Attenuation at 5 kHz} = 10 \log_{10}(1 + 5^6) = 10 \log_{10}(15626) = 41.9 \text{ dB}$$

Third-order Butterworth filter, $f_c = 1 \text{ kHz}$

Implementation: Using an active Sallen-Key topology with op-amps, or a passive LC ladder for lower noise applications.

Example 19.5 (ZOH Phase Lag Impact). A continuous control system has crossover frequency $\omega_c = 100 \text{ rad/s}$ with phase margin 45 degrees. If this controller is implemented digitally with a ZOH, what is the maximum sample period to maintain at least 30 degrees of phase margin?

Solution:

The ZOH introduces phase lag of:

$$\phi_{\text{ZOH}} = -\frac{\omega_c T}{2} \text{ radians} = -\frac{\omega_c T}{2} \times \frac{180}{\pi} \text{ degrees}$$

The original phase margin is 45 degrees. To maintain 30 degrees, we can lose at most:

$$\Delta\phi = 45 - 30 = 15$$

Setting the ZOH phase lag equal to 15 degrees:

$$\frac{\omega_c T}{2} \times \frac{180}{\pi} = 15$$

$$T = \frac{15 \times 2 \times \pi}{180 \times \omega_c} = \frac{30\pi}{180 \times 100} = \frac{\pi}{600} = 0.00524 \text{ s}$$

The sample rate is:

$$f_s = \frac{1}{T} = \frac{600}{\pi} = 191 \text{ Hz}$$

The ratio of sample frequency to crossover frequency:

$$\frac{f_s}{f_c} = \frac{191}{100/(2\pi)} = \frac{191 \times 2\pi}{100} = 12$$

$$T_{\text{max}} = 5.24 \text{ ms}; \quad f_s \geq 191 \text{ Hz} \approx 12 \times f_c$$

This confirms the rule of thumb that $f_s \geq 10\text{--}20 \times f_c$ for adequate phase margin preservation.

Example 19.6 (Complete Digital Controller Design). A temperature control system has plant $G(s) = \frac{2}{10s+1}$ (degrees per watt). Design a PI controller for zero steady-state error to step inputs. Then discretize using the Tustin method for implementation at 10 Hz.

Solution:

Step 1: Continuous Controller Design

For zero steady-state error to a step, we need a Type 1 system. The plant is Type 0, so the controller must include an integrator.

Choose a PI controller: $C(s) = K_p \left(1 + \frac{1}{\tau_i s}\right)$

The open-loop transfer function is:

$$L(s) = K_p \left(1 + \frac{1}{\tau_i s}\right) \cdot \frac{2}{10s+1} = \frac{2K_p(\tau_i s + 1)}{\tau_i s(10s+1)}$$

Using pole-zero cancellation for simplicity, set $\tau_i = 10$ s to cancel the plant pole:

$$L(s) = \frac{2K_p(10s+1)}{10s(10s+1)} = \frac{2K_p}{10s} = \frac{K_p}{5s}$$

For crossover at $\omega_c = 0.1$ rad/s (response time ≈ 10 s):

$$|L(j\omega_c)| = 1 \Rightarrow \frac{K_p}{5 \times 0.1} = 1 \Rightarrow K_p = 0.5$$

Final continuous controller:

$$C(s) = 0.5 \left(1 + \frac{1}{10s}\right) = 0.5 + \frac{0.05}{s} = \frac{0.5s + 0.05}{s}$$

Step 2: Tustin Discretization

With $T = 0.1$ s, substitute $s \rightarrow \frac{2}{T} \frac{z-1}{z+1} = 20 \frac{z-1}{z+1}$:

$$\begin{aligned} C(z) &= \frac{0.5 \cdot 20 \frac{z-1}{z+1} + 0.05}{20 \frac{z-1}{z+1}} \\ C(z) &= \frac{10(z-1) + 0.05(z+1)}{20(z-1)} = \frac{10z - 10 + 0.05z + 0.05}{20(z-1)} \\ C(z) &= \frac{10.05z - 9.95}{20(z-1)} = \frac{10.05z - 9.95}{20z - 20} \end{aligned}$$

Dividing numerator and denominator by 20:

$$C(z) = \frac{0.5025z - 0.4975}{z - 1}$$

Step 3: Difference Equation

Cross-multiplying:

$$\begin{aligned} U(z)(z-1) &= E(z)(0.5025z - 0.4975) \\ u[n] - u[n-1] &= 0.5025 \cdot e[n] - 0.4975 \cdot e[n-1] \\ u[n] &= u[n-1] + 0.5025 \cdot e[n] - 0.4975 \cdot e[n-1] \end{aligned}$$

$$\boxed{u[n] = u[n-1] + 0.5025 e[n] - 0.4975 e[n-1]}$$

This is the velocity form of PI control, which avoids integrator windup issues.

19.6 Figures for TikZ Implementation

19.6.1 Figure: Sampling and Reconstruction Block Diagram

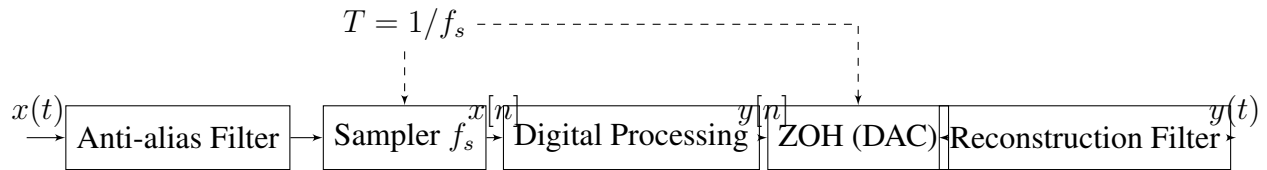


Figure 19.4: Complete sampling and reconstruction system showing anti-aliasing filter, sampler, digital processing, DAC (ZOH), and reconstruction filter.

19.6.2 Figure: Aliasing Time Domain

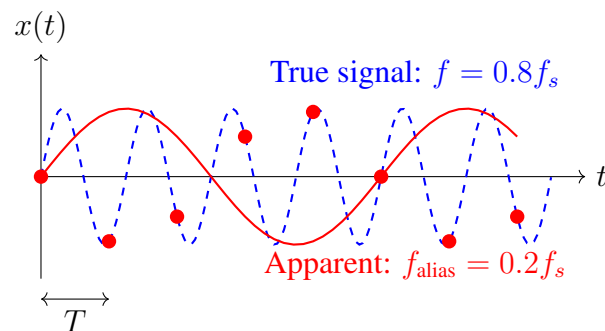


Figure 19.5: Time-domain aliasing: a high-frequency signal (dashed blue) sampled below the Nyquist rate appears as a low-frequency signal (solid red) when reconstructed from the samples.

19.6.3 Figure: ZOH Step Response

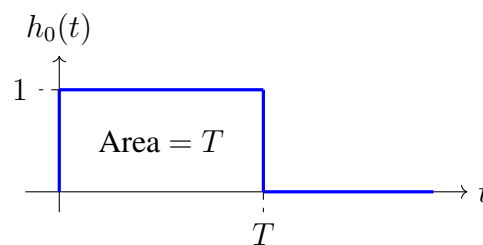


Figure 19.6: Impulse response of the zero-order hold: a rectangular pulse of height 1 and width T .

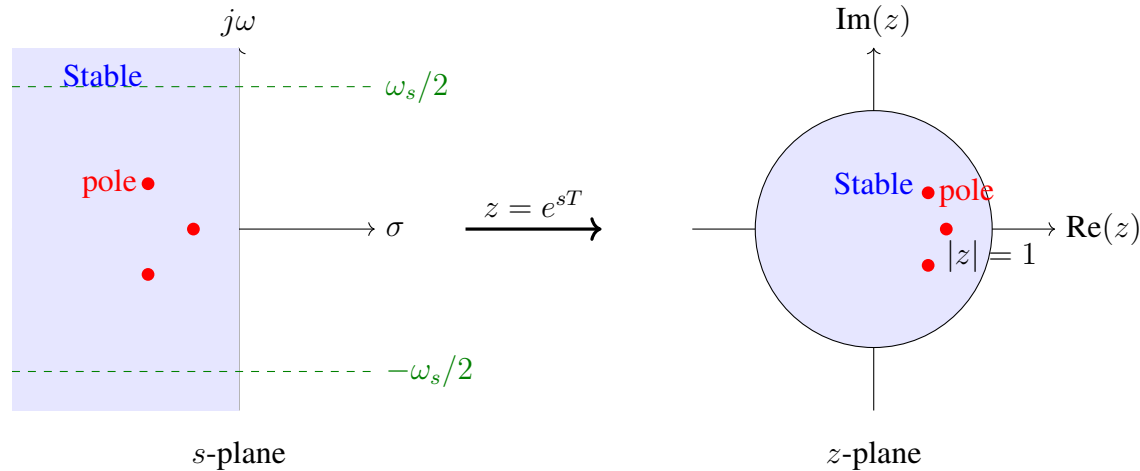


Figure 19.7: Mapping from s -plane to z -plane: the left half-plane maps to the interior of the unit circle. The primary strip $|\omega| < \omega_s/2$ in the s -plane maps uniquely to the z -plane; additional strips (due to aliasing) fold onto the same region.

19.6.4 Figure: s-Plane to z-Plane Mapping

19.7 Chapter Summary

This chapter established the theoretical and practical foundations for working with sampled-data control systems.

From a **historical context**, sampling theory emerged from the telephone industry's need to efficiently transmit voice signals. Nyquist (1928) established the fundamental rate limit, and Shannon (1949) formalized the reconstruction theorem. The **Nyquist-Shannon Theorem** states that a bandlimited signal with no frequency content above $f_{\max}$ can be perfectly reconstructed from samples taken at rate $f_s > 2f_{\max}$.

Aliasing occurs when sampling below the Nyquist rate, causing high frequencies to fold into the baseband and creating spurious signals that cannot be distinguished from legitimate low-frequency content. Anti-aliasing filters prevent this phenomenon. The **Zero-Order Hold** maintains each sample constant until the next, creating a staircase approximation with transfer function $H_0(s) = (1 - e^{-sT})/s$ and introducing phase lag of approximately $\omega T/2$.

Several **discretization methods** are available: Forward Euler is simple but can destabilize; Backward Euler is always stable but over-damped; Tustin offers the best accuracy and preserves stability but warps frequencies; and ZOH discretization is exact for the plant-hold combination. For **sample rate selection** in control systems, $f_s \geq 10\text{--}20 \times$ bandwidth provides adequate phase margin and discretization accuracy.

Key Takeaway

Sampling is not merely a practical convenience for digital implementation—it fundamentally transforms the system. Understanding aliasing, hold effects, and discretization accuracy is essential for designing digital controllers that achieve continuous-system performance. The Nyquist frequency represents an absolute information limit: frequencies above it cannot be captured, only corrupted.

19.8 Exercises

Exercise 19.1. Nyquist Rate Calculation: A control system must measure vibrations from 0.1 Hz to 250 Hz. What is the minimum sample rate? What would you recommend in practice?

Exercise 19.2. Aliasing Identification: A rotating shaft at 3600 RPM has 20 gear teeth. When sampled at 500 Hz, what frequency appears in the data? Is this aliased?

Exercise 19.3. ZOH Analysis: Calculate the magnitude and phase of the ZOH transfer function at $\omega = \omega_s/4$, $\omega_s/2$, and ω_s . Sketch the Bode plot.

Exercise 19.4. Discretization Comparison: Discretize the controller $C(s) = \frac{s+2}{s+10}$ using Forward Euler, Backward Euler, and Tustin methods with $T = 0.05$ s. Compare pole locations.

Exercise 19.5. Filter Design: An ADC samples at 1 kHz. Design a second-order Butterworth anti-aliasing filter that attenuates the Nyquist frequency by at least 30 dB. Specify the cutoff frequency.

Exercise 19.6. Phase Margin Preservation: A continuous system has phase margin 60 degrees at crossover frequency 50 rad/s. What sample rate is needed to lose no more than 10 degrees to ZOH phase lag?

Exercise 19.7. Tustin Prewarping: A notch filter must have its zero exactly at 60 Hz when discretized at 1 kHz. Calculate the prewarped analog frequency.

Exercise 19.8. Complete Design: A motor has transfer function $G(s) = \frac{100}{s(s+10)}$. Design a continuous PI controller for 1 rad/s bandwidth, then discretize it using Tustin at 20 Hz. Write the difference equation.

Exercise 19.9. Laboratory Analysis: In Experiment 2, explain why the control system becomes unstable as sample rate decreases. What is the relationship between the critical sample rate and the closed-loop bandwidth?

Exercise 19.10. Historical Reflection: How did Nyquist's work on telegraph transmission lead to modern digital audio? Calculate the minimum bits per second required for telephone-quality voice (4 kHz bandwidth, 8-bit quantization).

Chapter 20

Z-Transform and Discrete Stability

20.1 Historical Motivation and Development

The transition from continuous-time to discrete-time control systems represents one of the most significant paradigm shifts in the history of control engineering. This transformation was driven by the fundamental limitations of the Laplace transform when applied to sampled-data systems and the inexorable march toward digital computation. Understanding this historical context provides essential insight into why the z-transform became the cornerstone of modern digital control theory.

20.1.1 The Problem with Sampled Signals

In continuous-time systems, the Laplace transform provides an elegant framework for analyzing linear time-invariant systems. Given a continuous signal $x(t)$, its Laplace transform

$$X(s) = \int_0^{\infty} x(t)e^{-st} dt \quad (20.1)$$

transforms differential equations into algebraic ones, making system analysis tractable. However, when signals are sampled at discrete time intervals—a necessity in any digital implementation—the Laplace framework encounters fundamental difficulties.

Consider a continuous signal $x(t)$ sampled at intervals of T seconds. The sampled signal can be represented mathematically as

$$x^*(t) = \sum_{n=0}^{\infty} x(nT)\delta(t - nT) \quad (20.2)$$

where $\delta(t)$ is the Dirac delta function. Taking the Laplace transform yields

$$X^*(s) = \sum_{n=0}^{\infty} x(nT)e^{-snT} \quad (20.3)$$

This expression, while mathematically valid, exhibits a troublesome property: it is periodic in the imaginary direction with period $j\omega_s$, where $\omega_s = 2\pi/T$ is the sampling frequency. This periodicity leads to aliasing phenomena and makes direct application of standard Laplace techniques problematic. The need for a more natural mathematical framework for discrete-time systems was evident.

20.1.2 Witold Hurewicz and the Birth of the Z-Transform (1947)

Witold Hurewicz: From Topology to Control Theory

Witold Hurewicz (1904–1956) was a Polish-American mathematician whose remarkable career spanned pure mathematics and engineering applications. Born in Lodz, Poland, he earned his doctorate at the University of Vienna and became renowned for his contributions to algebraic topology, including the Hurewicz theorem relating homotopy and homology groups. During World War II, Hurewicz worked at the MIT Radiation Laboratory on fire-control systems for anti-aircraft guns, where the need to analyze sampled-data systems led him to develop the z-transform in 1947. His mathematical insight that the substitution $z = e^{sT}$ would simplify the analysis of discrete-time systems laid the foundation for all subsequent digital control theory. Tragically, Hurewicz died in 1956 from a fall during a mathematical conference in Uxmal, Mexico, cutting short a brilliant career that had bridged abstract mathematics and practical engineering.

The resolution to this dilemma came from an unexpected source. Witold Hurewicz (1904–1956), a Polish-American mathematician better known for his contributions to algebraic topology and homotopy theory, made a pivotal contribution to control theory during his work on fire-control systems during World War II.

While at the MIT Radiation Laboratory, Hurewicz recognized that the substitution $z = e^{sT}$ in the starred Laplace transform would eliminate the periodicity problem and yield a more tractable mathematical object. This substitution transforms Equation (3) into

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n} \quad (20.4)$$

where we adopt the notation $x[n] = x(nT)$ to emphasize the discrete nature of the sequence.

This transformation, which Hurewicz developed in 1947, possesses several remarkable properties. First, it converts difference equations into algebraic equations, just as the Laplace transform converts differential equations, enabling systematic solution techniques for discrete systems. Second, the region of convergence takes the form of an annular region in the complex z-plane, rather than the half-plane familiar from Laplace analysis. Third, stability analysis becomes elegantly geometric: stable systems have all poles inside the unit circle, providing an immediate visual criterion for system behavior.

Hurewicz's work, published in his analysis of sampled-data systems, established the mathematical foundation upon which all subsequent digital control theory would be built. Tragically, Hurewicz died in 1956 in a fall during a mathematical conference in Mexico, cutting short a brilliant career that spanned pure mathematics and engineering applications.

20.1.3 Eliahu Jury and the Stability Criterion (1958)

Once the z-transform provided the proper mathematical framework for discrete systems, the question of stability analysis became paramount. For continuous systems, the Routh-Hurwitz criterion provides an algebraic test to determine whether all roots of a polynomial lie in the left half of

the s -plane. But for discrete systems, stability requires all roots to lie inside the unit circle—a fundamentally different geometric condition.

Eliahu Abraham Jury (1923–2022), an Iraqi-American engineer working at the University of California, Berkeley, developed the discrete analog of the Routh-Hurwitz criterion in 1958. Jury, who had emigrated from Baghdad to pursue graduate studies in the United States, recognized that the bilinear transformation could map the unit circle to the imaginary axis, potentially allowing application of Routh-Hurwitz methods. However, he developed a more direct approach.

The Jury stability test operates directly on the coefficients of the characteristic polynomial

$$D(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0 \quad (20.5)$$

and constructs a triangular array analogous to the Routh array. The test provides necessary and sufficient conditions for all roots to lie within the unit circle without explicitly computing the roots.

Jury's contribution extended beyond the stability test. He authored the influential textbook “Theory and Application of the Z-Transform Method” (1964), which became the standard reference for a generation of control engineers. His work established the complete theoretical framework for discrete-time control system analysis and design.

20.1.4 The Digital Revolution in Control

The theoretical developments of Hurewicz and Jury found immediate practical application as digital computers began to replace analog controllers. The advantages were compelling and multifaceted. Flexibility emerged as a primary benefit, since control algorithms could be modified by changing software rather than hardware. Precision improved dramatically because digital computation eliminated drift and component variation that plagued analog circuits. The ability to implement sophisticated control strategies represented another major advancement, as computational complexity became limited only by processor speed rather than circuit topology. Finally, integration became seamless, with digital controllers interfacing directly with digital sensors and actuators in unified architectures.

By the 1970s, digital control had become the norm in aerospace, process control, and manufacturing. Today, virtually all control systems are implemented digitally, from the antilock brakes in automobiles to the flight control systems in aircraft to the temperature regulators in buildings. The z -transform and discrete stability theory developed by Hurewicz and Jury remain the essential analytical tools.

20.1.5 The Fundamental Mapping: $z = e^{sT}$

The relationship between the s -plane and the z -plane through the mapping $z = e^{sT}$ is central to understanding discrete-time control. This mapping, which motivated Hurewicz's original development, has profound geometric implications.

For a point $s = \sigma + j\omega$ in the s -plane:

$$z = e^{sT} = e^{(\sigma + j\omega)T} = e^{\sigma T} e^{j\omega T} = e^{\sigma T} \angle \omega T \quad (20.6)$$

This reveals that the real part σ determines the magnitude according to $|z| = e^{\sigma T}$, while the imaginary part ω determines the angle through the relationship $\angle z = \omega T$.

The implications for stability are immediate and elegant. The left half-plane, where $\sigma < 0$, maps to the interior of the unit circle where $|z| < 1$. The imaginary axis, corresponding to $\sigma = 0$, maps exactly to the unit circle $|z| = 1$. The right half-plane, where $\sigma > 0$, maps to the exterior region $|z| > 1$ outside the unit circle.

This mapping preserves the stability interpretation: stable continuous-time poles become stable discrete-time poles, and the unit circle becomes the discrete-time stability boundary, analogous to the imaginary axis in continuous time.

20.2 Modern Applications

The z-transform and discrete-time analysis have become indispensable across a remarkable range of modern applications. What began as a tool for sampled-data control systems has evolved into a fundamental technique spanning signal processing, embedded systems, and even financial modeling.

20.2.1 Digital Filter Design

Digital filters form the backbone of modern signal processing. The z-transform provides the natural framework for their analysis and design.

Infinite Impulse Response (IIR) Filters: These filters have transfer functions of the form

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} \quad (20.7)$$

The poles of $H(z)$ must lie inside the unit circle for stability. IIR filters can achieve sharp frequency responses with relatively few coefficients but require careful design to ensure stability.

Finite Impulse Response (FIR) Filters: With transfer functions $H(z) = \sum_{k=0}^M b_k z^{-k}$, FIR filters have all poles at the origin and are inherently stable. They can be designed with exactly linear phase, critical for applications requiring phase coherence.

20.2.2 Embedded Control Systems

Modern embedded controllers—from automotive engine control units to industrial programmable logic controllers—implement control algorithms in discrete time. A typical embedded control loop executes at a fixed sampling rate, reading sensors, computing control actions, and updating actuators at each sample instant.

The z-transform provides the essential tools for these applications. Engineers use it to convert continuous-time controller designs to discrete implementations, preserving stability and performance characteristics across the transformation. The framework enables rigorous analysis of finite sampling rate effects, including aliasing and inter-sample behavior. Stability analysis under computational delays becomes tractable through z-domain modeling of processing latency. Furthermore, the z-transform enables direct discrete-time controller design, bypassing continuous-time formulations entirely when appropriate.

20.2.3 Digital Signal Processors

Specialized digital signal processor (DSP) architectures are optimized for the multiply-accumulate operations central to discrete-time signal processing. The z-transform analysis directly informs DSP algorithm implementation, with transfer function coefficients becoming the multiplier values in hardware or software realizations.

20.2.4 Discrete-Time Financial Models

The mathematics of discrete-time systems has found application far beyond engineering. In quantitative finance, discrete-time models for asset prices, interest rates, and derivatives pricing employ techniques directly analogous to z-transform analysis. The stability of financial models—ensuring that simulated prices remain bounded and realistic—parallels the stability analysis of control systems.

20.3 Mathematical Foundation

We now develop the complete mathematical framework of the z-transform, establishing the definitions, properties, and techniques essential for discrete-time system analysis.

20.3.1 Definition of the Z-Transform

Definition 20.1 (Z-Transform). The z-transform of a discrete-time sequence $x[n]$, defined for $n \geq 0$, is

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=0}^{\infty} x[n]z^{-n} \quad (20.8)$$

where z is a complex variable. This is the *one-sided* or *unilateral* z-transform, appropriate for causal sequences.

The z-transform exists in a region of the complex z-plane where the infinite series converges. For most sequences of engineering interest, this region of convergence (ROC) is the exterior of a circle: $|z| > r$ for some $r \geq 0$.

Definition 20.2 (Two-Sided Z-Transform). For sequences defined for all integers, the *bilateral* z-transform is

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad (20.9)$$

The ROC for bilateral transforms is typically an annular region $r_1 < |z| < r_2$.

20.3.2 Z-Transforms of Common Sequences

Theorem 20.1 (Unit Step Sequence). For the unit step sequence $u[n] = 1$ for $n \geq 0$:

$$\mathcal{Z}\{u[n]\} = \sum_{n=0}^{\infty} z^{-n} = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1}, \quad |z| > 1 \quad (20.10)$$

Proof. The series $\sum_{n=0}^{\infty} z^{-n}$ is a geometric series with ratio z^{-1} . It converges for $|z^{-1}| < 1$, i.e., $|z| > 1$, with sum $1/(1 - z^{-1})$. $\square$

Theorem 20.2 (Exponential Sequence). *For the exponential sequence $x[n] = a^n u[n]$:*

$$\mathcal{Z}\{a^n u[n]\} = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \quad |z| > |a| \quad (20.11)$$

Theorem 20.3 (Ramp Sequence). *For the ramp sequence $x[n] = n \cdot u[n]$:*

$$\mathcal{Z}\{n \cdot u[n]\} = \frac{z}{(z - 1)^2}, \quad |z| > 1 \quad (20.12)$$

Proof. We use the differentiation property. Starting from $\mathcal{Z}\{u[n]\} = z/(z - 1)$ and noting that $n \cdot a^n = -a \frac{d}{da}(a^n)$, we employ

$$\mathcal{Z}\{n \cdot x[n]\} = -z \frac{d}{dz} X(z) \quad (20.13)$$

Applying this to $X(z) = z/(z - 1)$:

$$-z \frac{d}{dz} \left(\frac{z}{z - 1} \right) = -z \cdot \frac{(z - 1) - z}{(z - 1)^2} = -z \cdot \frac{-1}{(z - 1)^2} = \frac{z}{(z - 1)^2} \quad (20.14)$$

$\square$

Theorem 20.4 (Sinusoidal Sequences). *For sinusoidal sequences with $\Omega = \omega T$ (discrete frequency):*

$$\mathcal{Z}\{\cos(\Omega n) u[n]\} = \frac{z(z - \cos \Omega)}{z^2 - 2z \cos \Omega + 1}, \quad |z| > 1 \quad (20.15)$$

$$\mathcal{Z}\{\sin(\Omega n) u[n]\} = \frac{z \sin \Omega}{z^2 - 2z \cos \Omega + 1}, \quad |z| > 1 \quad (20.16)$$

Proof. Using Euler's formula: $\cos(\Omega n) = \frac{1}{2}(e^{j\Omega n} + e^{-j\Omega n})$ and $\sin(\Omega n) = \frac{1}{2j}(e^{j\Omega n} - e^{-j\Omega n})$. Applying the exponential transform:

$$\mathcal{Z}\{e^{j\Omega n} u[n]\} = \frac{z}{z - e^{j\Omega}} \quad (20.17)$$

The results follow from combining conjugate pairs and simplifying. $\square$

Theorem 20.5 (Damped Sinusoids). *For damped sinusoidal sequences $x[n] = r^n \cos(\Omega n) u[n]$:*

$$\mathcal{Z}\{r^n \cos(\Omega n) u[n]\} = \frac{z(z - r \cos \Omega)}{z^2 - 2rz \cos \Omega + r^2}, \quad |z| > r \quad (20.18)$$

20.3.3 Properties of the Z-Transform

The z-transform possesses properties analogous to those of the Laplace transform, making it a powerful analytical tool.

Theorem 20.6 (Linearity). *For sequences $x[n]$ and $y[n]$ with z-transforms $X(z)$ and $Y(z)$:*

$$\mathcal{Z}\{\alpha x[n] + \beta y[n]\} = \alpha X(z) + \beta Y(z) \quad (20.19)$$

Theorem 20.7 (Time Shift (Delay)). *For a sequence $x[n]$ with z-transform $X(z)$:*

$$\mathcal{Z}\{x[n - k]\} = z^{-k}X(z), \quad k > 0 \quad (20.20)$$

assuming $x[n] = 0$ for $n < 0$.

This property is fundamental: **multiplication by z^{-1} represents a unit delay**. This makes the z-transform natural for analyzing systems described by difference equations, where delays appear explicitly.

Theorem 20.8 (Time Advance). *For a causal sequence:*

$$\mathcal{Z}\{x[n + 1]\} = zX(z) - zx[0] \quad (20.21)$$

More generally:

$$\mathcal{Z}\{x[n + k]\} = z^kX(z) - \sum_{m=0}^{k-1} z^{k-m}x[m] \quad (20.22)$$

Theorem 20.9 (Convolution). *If $y[n] = x[n] * h[n] = \sum_{k=0}^n x[k]h[n - k]$, then:*

$$Y(z) = X(z)H(z) \quad (20.23)$$

This convolution property is the foundation of transfer function analysis: the output transform equals the input transform multiplied by the system transfer function.

Theorem 20.10 (Initial Value Theorem). *If $X(z)$ is the z-transform of $x[n]$:*

$$x[0] = \lim_{z \rightarrow \infty} X(z) \quad (20.24)$$

Theorem 20.11 (Final Value Theorem). *If $x[n]$ has a finite limit as $n \rightarrow \infty$ and $(z - 1)X(z)$ has no poles outside or on the unit circle except possibly at $z = 1$:*

$$\lim_{n \rightarrow \infty} x[n] = \lim_{z \rightarrow 1} (z - 1)X(z) \quad (20.25)$$

Table 20.1: Summary of Z-Transform Properties

Property	Time Domain	Z-Domain
Linearity	$\alpha x[n] + \beta y[n]$	$\alpha X(z) + \beta Y(z)$
Time delay	$x[n - k]$	$z^{-k}X(z)$
Time advance	$x[n + 1]$	$zX(z) - zx[0]$
Scaling	$a^n x[n]$	$X(z/a)$
Time reversal	$x[-n]$	$X(z^{-1})$
Convolution	$x[n] * h[n]$	$X(z)H(z)$
Multiplication by n	$n \cdot x[n]$	$-z \frac{dX(z)}{dz}$
Accumulation	$\sum_{k=0}^n x[k]$	$\frac{z}{z-1}X(z)$

Table 20.2: Common Z-Transform Pairs

Sequence $x[n]$	Z-Transform $X(z)$	ROC
$\delta[n]$ (unit impulse)	$\frac{1}{z}$	All z
$u[n]$ (unit step)	$\frac{z}{z-1}$	$ z > 1$
$a^n u[n]$	$\frac{z}{z-a}$	$ z > a $
$n \cdot u[n]$	$\frac{z}{(z-1)^2}$	$ z > 1$
$n \cdot a^n u[n]$	$\frac{az}{(z-a)^2}$	$ z > a $
$\cos(\Omega n)u[n]$	$\frac{z(z - \cos \Omega)}{z^2 - 2z \cos \Omega + 1}$	$ z > 1$
$\sin(\Omega n)u[n]$	$\frac{z \sin \Omega}{z^2 - 2z \cos \Omega + 1}$	$ z > 1$
$r^n \cos(\Omega n)u[n]$	$\frac{z(z - r \cos \Omega)}{z^2 - 2rz \cos \Omega + r^2}$	$ z > r$

20.3.4 Inverse Z-Transform

The inverse z-transform recovers the time-domain sequence from its z-domain representation. While the formal definition involves a contour integral:

$$x[n] = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz \quad (20.26)$$

practical computation typically uses one of three methods.

Partial Fraction Expansion

For rational z-transforms, partial fraction expansion is the most common approach.

Inverse Z-Transform via Partial Fractions

Find the inverse z-transform of

$$X(z) = \frac{2z^2 - 1.5z}{z^2 - 1.5z + 0.5} \quad (20.27)$$

Solution: First, factor the denominator:

$$z^2 - 1.5z + 0.5 = (z - 1)(z - 0.5) \quad (20.28)$$

For partial fractions, we work with $X(z)/z$:

$$\frac{X(z)}{z} = \frac{2z - 1.5}{(z - 1)(z - 0.5)} = \frac{A}{z - 1} + \frac{B}{z - 0.5} \quad (20.29)$$

Solving: $A = \frac{2(1)-1.5}{1-0.5} = \frac{0.5}{0.5} = 1$ and $B = \frac{2(0.5)-1.5}{0.5-1} = \frac{-0.5}{-0.5} = 1$

Therefore:

$$X(z) = \frac{z}{z-1} + \frac{z}{z-0.5} \quad (20.30)$$

Taking the inverse transform:

$$x[n] = u[n] + (0.5)^n u[n] = [1 + (0.5)^n] u[n] \quad (20.31)$$

Power Series Expansion

For complex expressions or computational purposes, direct long division yields the sequence values.

Inverse Z-Transform via Power Series Expansion

Find the first few terms of the inverse z-transform of $X(z) = z/(z - 0.5)$ by long division.

Solution: Rewrite as $X(z) = 1/(1 - 0.5z^{-1})$ and expand:

$$X(z) = 1 + 0.5z^{-1} + 0.25z^{-2} + 0.125z^{-3} + \dots \quad (20.32)$$

Reading off coefficients: $x[0] = 1$, $x[1] = 0.5$, $x[2] = 0.25$, $x[3] = 0.125$, confirming $x[n] = (0.5)^n$.

20.3.5 Discrete Transfer Functions

The transfer function of a discrete-time linear time-invariant system relates the z-transforms of output and input:

$$H(z) = \frac{Y(z)}{U(z)} \quad (20.33)$$

Consider the general difference equation:

$$y[n] + a_1 y[n-1] + \dots + a_N y[n-N] = b_0 u[n] + b_1 u[n-1] + \dots + b_M u[n-M] \quad (20.34)$$

Taking z-transforms with zero initial conditions:

$$Y(z)(1 + a_1 z^{-1} + \dots + a_N z^{-N}) = U(z)(b_0 + b_1 z^{-1} + \dots + b_M z^{-M}) \quad (20.35)$$

The transfer function is:

$$H(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}} = \frac{b_0 z^N + b_1 z^{N-1} + \dots + b_M z^{N-M}}{z^N + a_1 z^{N-1} + \dots + a_N} \quad (20.36)$$

20.3.6 S-Plane to Z-Plane Mapping

The fundamental relationship $z = e^{sT}$ maps the s-plane to the z-plane. Understanding this mapping is essential for converting continuous-time designs to discrete time.

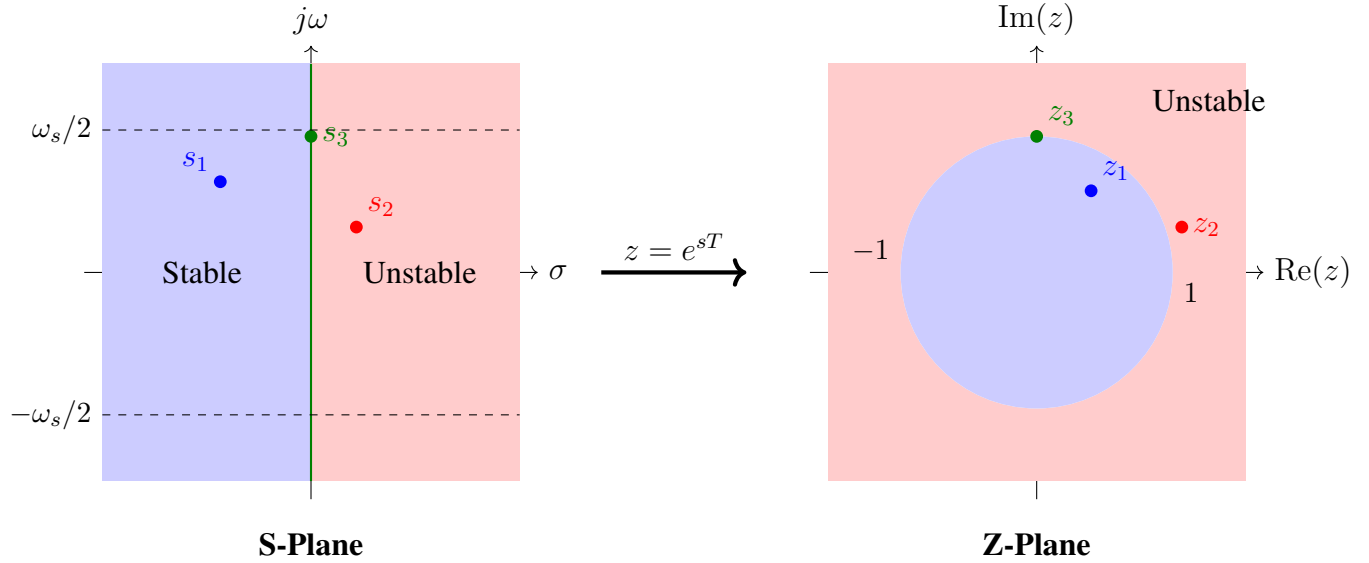


Figure 20.1: The mapping $z = e^{sT}$ transforms the s-plane to the z-plane. The left half-plane maps to the interior of the unit circle, the imaginary axis maps to the unit circle, and the right half-plane maps to the exterior of the unit circle.

Key Properties of the Mapping:

1. **Left Half-Plane $\rightarrow$ Inside Unit Circle:** For $s = \sigma + j\omega$ with $\sigma < 0$:

$$|z| = |e^{sT}| = e^{\sigma T} < 1 \quad (20.37)$$

2. **Imaginary Axis $\rightarrow$ Unit Circle:** For $s = j\omega$:

$$|z| = |e^{j\omega T}| = 1 \quad (20.38)$$

3. **Right Half-Plane $\rightarrow$ Outside Unit Circle:** For $\sigma > 0$:

$$|z| = e^{\sigma T} > 1 \quad (20.39)$$

4. **Periodicity:** The mapping is periodic in ω with period $\omega_s = 2\pi/T$:

$$e^{s_1 T} = e^{s_2 T} \quad \text{if} \quad s_2 = s_1 + jk\omega_s, \quad k \in \mathbb{Z} \quad (20.40)$$

This reflects the aliasing phenomenon: frequencies differing by multiples of the sampling frequency map to the same point in the z-plane.

20.3.7 Discrete Stability Criterion

Theorem 20.12 (Discrete-Time Stability). *A discrete-time LTI system with transfer function $H(z)$ is BIBO (bounded-input bounded-output) stable if and only if all poles of $H(z)$ lie strictly inside the unit circle:*

$$|p_i| < 1 \quad \text{for all poles } p_i \quad (20.41)$$

Proof. The impulse response of the system is $h[n] = \mathcal{Z}^{-1}\{H(z)\}$. For a pole at $z = p$ with $|p| < 1$, the corresponding impulse response term is proportional to p^n , which decays geometrically to zero. The system is BIBO stable if $\sum_{n=0}^{\infty} |h[n]| < \infty$, which requires all terms to decay, hence all poles must satisfy $|p| < 1$. $\square$

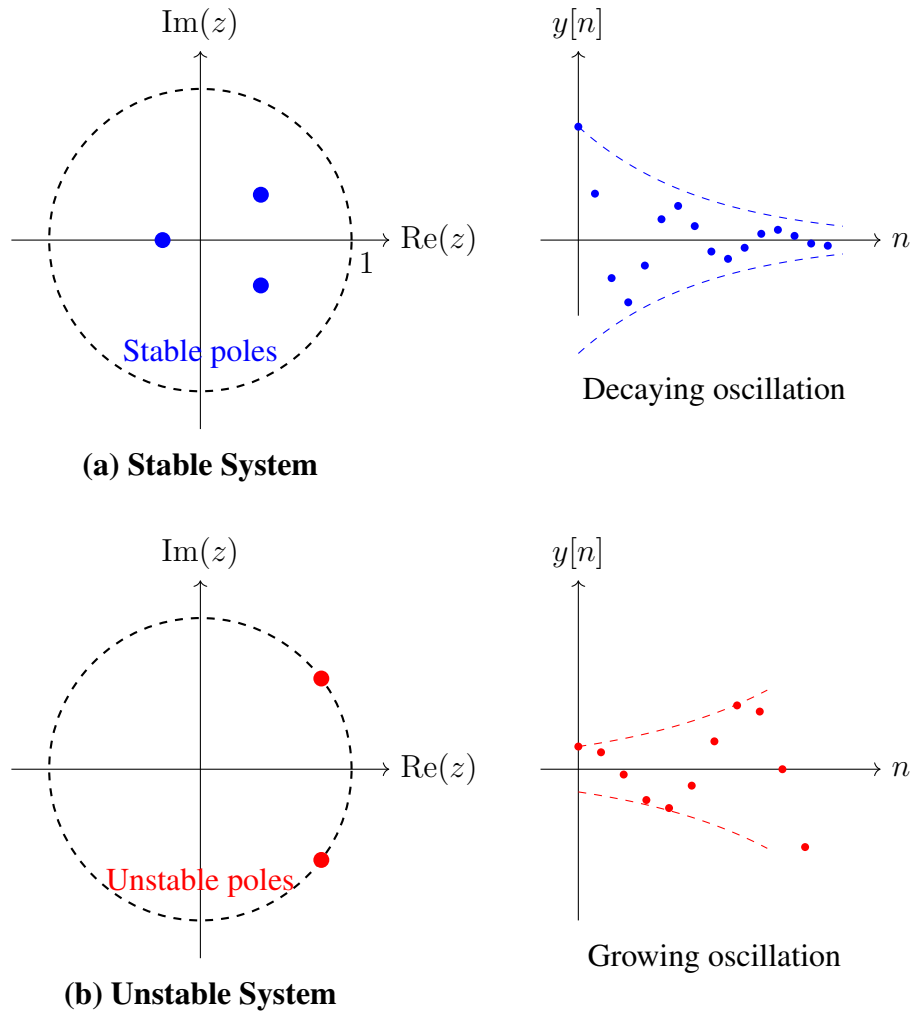


Figure 20.2: Pole locations in the z -plane determine system stability. (a) Poles inside the unit circle produce decaying responses. (b) Poles outside the unit circle produce unbounded, growing responses.

20.3.8 The Jury Stability Test

The Jury test provides an algebraic criterion for determining whether all roots of a polynomial lie inside the unit circle, analogous to the Routh-Hurwitz test for continuous systems.

Consider the characteristic polynomial:

$$D(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0, \quad a_n > 0 \quad (20.42)$$

Theorem 20.13 (Jury Stability Criterion). *The polynomial $D(z)$ has all roots inside the unit circle if and only if:*

1. $D(1) > 0$
2. $(-1)^n D(-1) > 0$
3. $|a_0| < a_n$
4. *All entries in the first column of the Jury array (defined below) satisfy appropriate sign conditions.*

Construction of the Jury Array:

Row	z^0	z^1	z^2	$\dots$	z^{n-2}	z^{n-1}	z^n
1	a_0	a_1	a_2	$\dots$	a_{n-2}	a_{n-1}	a_n
2	a_n	a_{n-1}	a_{n-2}	$\dots$	a_2	a_1	a_0
3	b_0	b_1	b_2	$\dots$	b_{n-2}	b_{n-1}	
4	b_{n-1}	b_{n-2}	b_{n-3}	$\dots$	b_1	b_0	
5	c_0	c_1	c_2	$\dots$	c_{n-2}		
$\vdots$							

where

$$b_k = \begin{vmatrix} a_0 & a_{n-k} \\ a_n & a_k \end{vmatrix} = a_0 a_k - a_n a_{n-k} \quad (20.43)$$

The stability conditions require:

$$|b_0| > |b_{n-1}|, \quad |c_0| > |c_{n-2}|, \quad \dots \quad (20.44)$$

Example 20.1 (Jury Test Application). Determine stability for $D(z) = z^3 - 1.5z^2 + 0.7z - 0.1$.

Solution:

1. $D(1) = 1 - 1.5 + 0.7 - 0.1 = 0.1 > 0 \checkmark$
2. $D(-1) = -1 - 1.5 - 0.7 - 0.1 = -3.3$; $(-1)^3 D(-1) = 3.3 > 0 \checkmark$
3. $|a_0| = 0.1 < 1 = a_3 \checkmark$

Constructing the array with $a_0 = -0.1$, $a_1 = 0.7$, $a_2 = -1.5$, $a_3 = 1$:

Row 1: $-0.1, \quad 0.7, \quad -1.5, \quad 1$

Row 2: $1, \quad -1.5, \quad 0.7, \quad -0.1$

$b_0 = (-0.1)(1) - (1)(-0.1) = -0.1 + 0.1 = 0 \dots$

Since $b_0 = 0$, the test is inconclusive at this step, indicating a root on the unit circle. Indeed, this polynomial has a root at approximately $z = 0.2$, confirming stability, but numerical methods would be needed for a definitive answer in this edge case.

20.4 Practical Experiment: Digital Filter on Arduino

This laboratory exercise demonstrates discrete-time stability concepts through a hands-on implementation of a first-order digital filter on an Arduino microcontroller. Students will observe how pole location affects system behavior and verify theoretical stability predictions.

20.4.1 Objectives

1. Implement a discrete-time transfer function as a difference equation
2. Experimentally verify the stability boundary at $|z| = 1$
3. Observe impulse response characteristics for stable and unstable poles
4. Compare discrete-time behavior to continuous-time equivalents

20.4.2 Theoretical Background

Consider the first-order discrete transfer function:

$$H(z) = \frac{b_0 + b_1 z^{-1}}{1 - a_1 z^{-1}} = \frac{b_0 z + b_1}{z - a_1} \quad (20.45)$$

The corresponding difference equation is:

$$y[n] = a_1 y[n-1] + b_0 u[n] + b_1 u[n-1] \quad (20.46)$$

The single pole is located at $z = a_1$. The stability criterion predicts three distinct behaviors depending on the pole magnitude: when $|a_1| < 1$ the system is stable with the pole inside the unit circle; when $|a_1| = 1$ the system is marginally stable with the pole exactly on the unit circle; and when $|a_1| > 1$ the system becomes unstable with the pole outside the unit circle.

20.4.3 Required Components

Table 20.3: Required Components for Digital Filter Experiment

Component	Purpose
Arduino Uno or compatible	Microcontroller platform
USB cable	Programming and serial communication
Computer with Arduino IDE	Development environment and serial plotter
Pushbutton (optional)	Physical impulse input trigger
Potentiometer (optional)	Real-time coefficient adjustment

20.4.4 Microcontroller Implementation

The digital filter is implemented using the following algorithm, which can be adapted to any microcontroller platform.

Algorithm 42 First-Order Digital Filter for Stability Demonstration

```

1: Initialize: Set filter coefficients  $a_1 \leftarrow 0.8$ ,  $b_0 \leftarrow 1.0$ ,  $b_1 \leftarrow 0.0$ 
2: Initialize: Set state variables  $y_{\text{prev}} \leftarrow 0$ ,  $u_{\text{prev}} \leftarrow 0$ ,  $y_{\text{current}} \leftarrow 0$ 
3: Initialize: Set timing parameters  $T_{\text{sample}} \leftarrow 50$  ms,  $t_{\text{last}} \leftarrow 0$ 
4: Initialize: Set flags impulseApplied  $\leftarrow$  FALSE,  $n \leftarrow 0$ 
5:
6: Setup:
7:   Configure serial communication at 115200 baud
8:   Configure LED output pin
9:   Display stability status based on  $|a_1|$ 
10:
11: Main Loop:
12: while true do
13:    $t_{\text{current}} \leftarrow$  current time in milliseconds
14:
15:   if serial input available then
16:      $\text{cmd} \leftarrow$  read character from serial
17:     if  $\text{cmd} = \text{'i'}$  then
18:        $\text{impulseApplied} \leftarrow$  TRUE,  $n \leftarrow 0$ 
19:     else if  $\text{cmd} = \text{'r'}$  then
20:        $y_{\text{prev}} \leftarrow 0$ ,  $u_{\text{prev}} \leftarrow 0$ ,  $y_{\text{current}} \leftarrow 0$ 
21:        $\text{impulseApplied} \leftarrow$  FALSE,  $n \leftarrow 0$ 
22:     end if
23:   end if
24:
25:   if  $t_{\text{current}} - t_{\text{last}} \geq T_{\text{sample}}$  then
26:      $t_{\text{last}} \leftarrow t_{\text{current}}$ 
27:
28:     ▷ Generate impulse input
29:     if  $\text{impulseApplied} \ \& \ n = 0$  then
30:        $u_{\text{current}} \leftarrow 1.0$ 
31:     else
32:        $u_{\text{current}} \leftarrow 0.0$ 
33:     end if
34:
35:     ▷ Compute filter output:  $y[n] = a_1 y[n-1] + b_0 u[n] + b_1 u[n-1]$ 
36:      $y_{\text{current}} \leftarrow a_1 \cdot y_{\text{prev}} + b_0 \cdot u_{\text{current}} + b_1 \cdot u_{\text{prev}}$ 
37:
38:     ▷ Limit output to prevent overflow
39:      $y_{\text{current}} \leftarrow \max(-1000, \min(1000, y_{\text{current}}))$ 
40:
41:     if  $\text{impulseApplied} \ \& \ n < 100$  then
42:       Output  $(n, u_{\text{current}}, y_{\text{current}})$  to serial
43:       Set LED based on  $|y_{\text{current}}| > 0.01$ 
44:     end if
45:
46:     ▷ Update state variables
47:      $y_{\text{prev}} \leftarrow y_{\text{current}}$ 
48:      $u_{\text{prev}} \leftarrow u_{\text{current}}$ 
49:      $n \leftarrow n + 1$ 

```

20.4.5 Experimental Procedure

Begin by uploading the code to the Arduino and opening the Serial Monitor at 115200 baud. The experiment proceeds through a series of tests that systematically explore the relationship between pole location and system behavior.

Stable Pole Test ($a_1 = 0.8$): With the pole at $z = 0.8$ inside the unit circle, send the character ‘i’ to apply an impulse. Observe that the output decays geometrically as $(0.8)^n$, and record the number of samples required for the output to decay below 0.01.

Varying the Stable Pole: Modify a_1 to values of 0.5, 0.9, 0.95, and 0.99 in succession, repeating the impulse test for each. Observe that poles closer to the unit circle produce slower decay, and record the decay rates and settling times for comparison.

Marginally Stable Test ($a_1 = 1.0$): With the pole positioned exactly on the unit circle, apply an impulse and observe that the output remains constant, neither decaying nor growing. This demonstrates the boundary between stable and unstable behavior.

Unstable Pole Test ($a_1 = 1.05$): Moving the pole outside the unit circle, apply an impulse and observe the output growing without bound. The code limits the maximum value to prevent numerical overflow, but note that real unstable systems can cause physical damage to hardware through actuator saturation or mechanical stress.

Negative Pole Test ($a_1 = -0.8$): With a pole at $z = -0.8$ on the negative real axis, apply an impulse and observe the output alternating in sign while decaying in magnitude. This alternating behavior corresponds to oscillation at half the sampling frequency, a phenomenon unique to discrete-time systems.

20.4.6 Expected Results

Table 20.4: Expected Impulse Response Behavior

Pole a_1	Location	Stability	Response
0.5	Inside, real	Stable	Fast exponential decay
0.8	Inside, real	Stable	Moderate decay
0.95	Inside, near boundary	Stable	Slow decay
1.0	On circle	Marginal	Constant (no decay)
1.05	Outside	Unstable	Exponential growth
-0.8	Inside, negative	Stable	Alternating decay

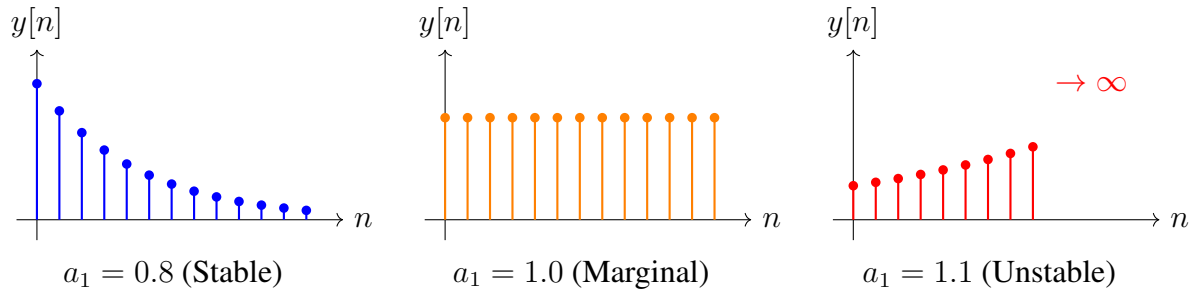


Figure 20.3: Impulse responses for different pole locations demonstrating stable, marginally stable, and unstable behavior.

20.4.7 Comparison to Continuous-Time Equivalent

The discrete pole at $z = a_1$ corresponds to a continuous pole at:

$$s = \frac{1}{T} \ln(a_1) \quad (20.47)$$

For $T = 0.05$ s (20 Hz sampling) and $a_1 = 0.8$:

$$s = \frac{1}{0.05} \ln(0.8) = 20 \times (-0.223) = -4.46 \text{ rad/s} \quad (20.48)$$

The continuous-time system $H(s) = 1/(s + 4.46)$ has time constant $\tau = 1/4.46 = 0.224$ s. The discrete system settles in approximately $5\tau/T = 22$ samples, which students can verify experimentally.

20.4.8 Analysis Questions

1. How does the settling time change as the pole approaches the unit circle?
2. What is the relationship between the pole magnitude and the decay rate?
3. Why does a negative real pole produce alternating outputs?
4. What would happen with complex conjugate poles inside the unit circle?
5. How does the sampling rate affect the apparent response speed?

20.5 Worked Examples

Example 20.2 (Z-Transform of Common Sequences). Find the z-transform of the sequence $x[n] = (3 \cdot 2^n - 2 \cdot 3^n)u[n]$.

Solution: Using linearity:

$$X(z) = 3 \cdot \mathcal{Z}\{2^n u[n]\} - 2 \cdot \mathcal{Z}\{3^n u[n]\} \quad (20.49)$$

$$= 3 \cdot \frac{z}{z-2} - 2 \cdot \frac{z}{z-3} \quad (20.50)$$

$$= \frac{3z(z-3) - 2z(z-2)}{(z-2)(z-3)} \quad (20.51)$$

$$= \frac{3z^2 - 9z - 2z^2 + 4z}{(z-2)(z-3)} \quad (20.52)$$

$$= \frac{z^2 - 5z}{(z-2)(z-3)} = \frac{z(z-5)}{(z-2)(z-3)} \quad (20.53)$$

The ROC is $|z| > 3$ since we need both component series to converge.

Example 20.3 (Continuous to Discrete Conversion with ZOH). Convert the continuous transfer function $G(s) = \frac{5}{s+2}$ to discrete time using the zero-order hold (ZOH) method with sampling period $T = 0.1$ s.

Solution: The ZOH equivalent is:

$$G(z) = (1 - z^{-1}) \mathcal{Z} \left\{ \frac{G(s)}{s} \right\} \quad (20.54)$$

First, find $G(s)/s$:

$$\frac{G(s)}{s} = \frac{5}{s(s+2)} = \frac{5/2}{s} - \frac{5/2}{s+2} \quad (20.55)$$

Taking the z-transform of the step response:

$$\mathcal{Z} \left\{ \frac{G(s)}{s} \right\} = \frac{5}{2} \cdot \frac{z}{z-1} - \frac{5}{2} \cdot \frac{z}{z - e^{-2T}} \quad (20.56)$$

With $T = 0.1$: $e^{-2(0.1)} = e^{-0.2} = 0.8187$

$$\mathcal{Z} \left\{ \frac{G(s)}{s} \right\} = \frac{5}{2} \left(\frac{z}{z-1} - \frac{z}{z-0.8187} \right) \quad (20.57)$$

Applying the ZOH formula:

$$G(z) = \frac{z-1}{z} \cdot \frac{5}{2} \cdot \frac{z[(z-0.8187) - (z-1)]}{(z-1)(z-0.8187)} \quad (20.58)$$

$$= \frac{5}{2} \cdot \frac{1-0.8187}{z-0.8187} \quad (20.59)$$

$$= \frac{5}{2} \cdot \frac{0.1813}{z-0.8187} \quad (20.60)$$

$$= \frac{0.4532}{z-0.8187} \quad (20.61)$$

Verification: The DC gain should match. For continuous: $G(0) = 5/2 = 2.5$. For discrete: $G(1) = 0.4532/(1 - 0.8187) = 0.4532/0.1813 = 2.5 \checkmark$

Example 20.4 (Inverse Z-Transform Using Partial Fractions). Find the inverse z-transform of:

$$Y(z) = \frac{10z^2}{(z - 0.5)(z - 0.8)} \quad (20.62)$$

Solution: First, divide by z :

$$\frac{Y(z)}{z} = \frac{10z}{(z - 0.5)(z - 0.8)} \quad (20.63)$$

Partial fraction expansion:

$$\frac{10z}{(z - 0.5)(z - 0.8)} = \frac{A}{z - 0.5} + \frac{B}{z - 0.8} \quad (20.64)$$

Finding coefficients:

$$A = \left. \frac{10z}{z - 0.8} \right|_{z=0.5} = \frac{10(0.5)}{0.5 - 0.8} = \frac{5}{-0.3} = -\frac{50}{3} \quad (20.65)$$

$$B = \left. \frac{10z}{z - 0.5} \right|_{z=0.8} = \frac{10(0.8)}{0.8 - 0.5} = \frac{8}{0.3} = \frac{80}{3} \quad (20.66)$$

Therefore:

$$Y(z) = -\frac{50}{3} \cdot \frac{z}{z - 0.5} + \frac{80}{3} \cdot \frac{z}{z - 0.8} \quad (20.67)$$

Taking the inverse z-transform:

$$y[n] = \left(-\frac{50}{3}(0.5)^n + \frac{80}{3}(0.8)^n \right) u[n] \quad (20.68)$$

Verification: $y[0] = -50/3 + 80/3 = 30/3 = 10$. From the initial value theorem: $\lim_{z \rightarrow \infty} Y(z) = \lim_{z \rightarrow \infty} \frac{10z^2}{z^2(1-0.5/z)(1-0.8/z)} = 10 \checkmark$

Example 20.5 (S-Plane to Z-Plane Pole Mapping). A continuous-time system has poles at $s_1 = -3$ and $s_{2,3} = -1 \pm j2$. Find the corresponding z-plane pole locations for $T = 0.2$ s and determine stability.

Solution: Using $z = e^{sT}$:

For $s_1 = -3$:

$$z_1 = e^{-3 \times 0.2} = e^{-0.6} = 0.549 \quad (20.69)$$

For $s_2 = -1 + j2$:

$$z_2 = e^{(-1+j2) \times 0.2} = e^{-0.2} e^{j0.4} = 0.819 \angle 0.4 \text{ rad} = 0.819 \angle 22.9 \quad (20.70)$$

In rectangular form: $z_2 = 0.819(\cos 0.4 + j \sin 0.4) = 0.755 + j0.319$

For $s_3 = -1 - j2$:

$$z_3 = z_2^* = 0.755 - j0.319 \quad (20.71)$$

Checking stability, we find that $|z_1| = 0.549 < 1$ and $|z_2| = |z_3| = 0.819 < 1$. Since all poles are inside the unit circle, the discrete system is stable, as expected given that the original continuous system was stable.

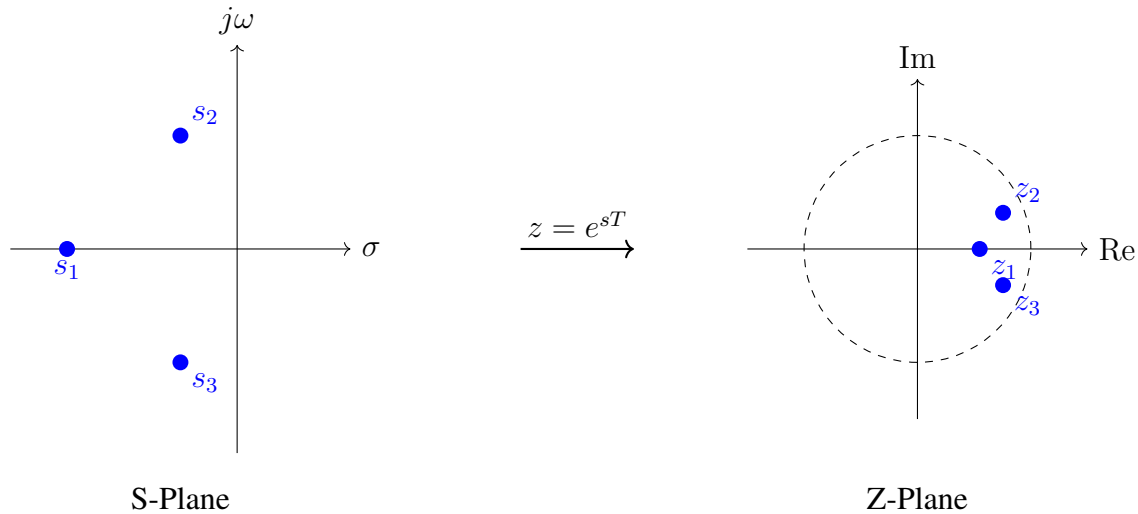


Figure 20.4: Mapping of continuous-time poles to discrete-time poles for Example 20.4.

Example 20.6 (Jury Stability Test). Use the Jury test to determine if all roots of $D(z) = z^3 + 0.2z^2 - 0.35z + 0.05$ lie inside the unit circle.

Solution: Identify coefficients: $a_3 = 1$, $a_2 = 0.2$, $a_1 = -0.35$, $a_0 = 0.05$

Necessary Conditions:

1. $D(1) = 1 + 0.2 - 0.35 + 0.05 = 0.9 > 0$ ✓
2. $(-1)^3 D(-1) = -(-1 + 0.2 + 0.35 + 0.05) = -(-0.4) = 0.4 > 0$ ✓
3. $|a_0| = 0.05 < 1 = a_3$ ✓

Jury Array Construction:

Row 1: 0.05, -0.35, 0.2, 1

Row 2: 1, 0.2, -0.35, 0.05

Compute b_k values:

$$b_0 = a_0 \cdot a_0 - a_3 \cdot a_3 = 0.05^2 - 1^2 = -0.9975 \quad (20.72)$$

$$b_1 = a_0 \cdot a_1 - a_3 \cdot a_2 = (0.05)(-0.35) - (1)(0.2) = -0.2175 \quad (20.73)$$

$$b_2 = a_0 \cdot a_2 - a_3 \cdot a_1 = (0.05)(0.2) - (1)(-0.35) = 0.36 \quad (20.74)$$

Row 3: -0.9975, -0.2175, 0.36

Row 4: 0.36, -0.2175, -0.9975

Check: $|b_0| = 0.9975 > |b_2| = 0.36$ ✓

Compute c_k values:

$$c_0 = b_0 \cdot b_0 - b_2 \cdot b_2 = 0.9975^2 - 0.36^2 = 0.995 - 0.1296 = 0.8654 \quad (20.75)$$

$$c_1 = b_0 \cdot b_1 - b_2 \cdot b_1 = b_1(b_0 - b_2) = -0.2175(-0.9975 - 0.36) = 0.295 \quad (20.76)$$

Row 5: 0.8654, 0.295

Check: $|c_0| = 0.8654 > |c_1| = 0.295$ ✓

Conclusion: All conditions are satisfied. The polynomial has all roots inside the unit circle, and the system is stable.

Example 20.7 (Complete Discrete System Analysis). A discrete-time control system has the closed-loop transfer function:

$$H(z) = \frac{0.2z + 0.15}{z^2 - 1.2z + 0.52} \quad (20.77)$$

Analyze stability, find the poles, and sketch the impulse response character.

Solution:

Pole Locations: Using the quadratic formula for $z^2 - 1.2z + 0.52 = 0$:

$$z = \frac{1.2 \pm \sqrt{1.44 - 2.08}}{2} = \frac{1.2 \pm \sqrt{-0.64}}{2} = \frac{1.2 \pm j0.8}{2} = 0.6 \pm j0.4 \quad (20.78)$$

Pole Magnitude:

$$|z| = \sqrt{0.6^2 + 0.4^2} = \sqrt{0.36 + 0.16} = \sqrt{0.52} = 0.721 \quad (20.79)$$

Since $|z| = 0.721 < 1$, both poles are inside the unit circle. **The system is stable.**

Pole Angle:

$$\theta = \arctan\left(\frac{0.4}{0.6}\right) = \arctan(0.667) = 33.7 = 0.588 \text{ rad} \quad (20.80)$$

Impulse Response Character:

The impulse response will be of the form:

$$h[n] = C \cdot r^n \cos(n\theta + \phi) \quad (20.81)$$

where $r = 0.721$ (decay rate) and $\theta = 0.588$ rad/sample (oscillation frequency).

The impulse response exhibits several characteristic features. Each sample decays by the factor $r = 0.721$, corresponding to a time constant of $n_{0.37} = -1/\ln(0.721) = 3.06$ samples. The oscillation period is $2\pi/0.588 = 10.7$ samples, and the overall behavior is that of a damped oscillation converging stably to zero.

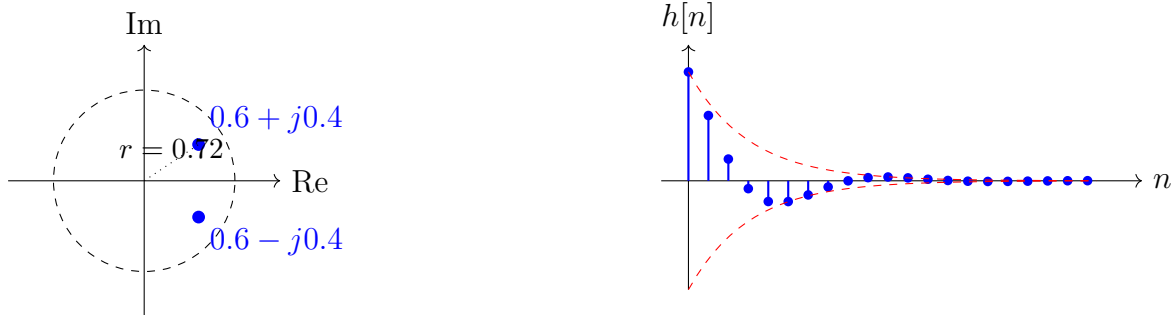


Figure 20.5: Z-plane poles and corresponding damped oscillatory impulse response for Example 20.6.

20.6 Chapter Summary

This chapter has established the z-transform as the fundamental tool for discrete-time control system analysis. The key concepts are:

1. **Historical Development:** The z-transform emerged from Hurewicz's work on sampled-data systems (1947) and was complemented by Jury's stability criterion (1958), providing a complete framework for discrete-time analysis.
2. **Definition and Properties:** The z-transform $X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$ converts sequences to functions of a complex variable, with properties paralleling the Laplace transform.
3. **S-Plane to Z-Plane Mapping:** The transformation $z = e^{sT}$ maps the left half-plane to the interior of the unit circle (stable region), the imaginary axis to the unit circle itself (stability boundary), and the right half-plane to the exterior of the unit circle (unstable region).
4. **Stability Criterion:** A discrete-time system is stable if and only if all poles of its transfer function satisfy $|p_i| < 1$.
5. **Jury Test:** Provides an algebraic method to determine whether all roots of a polynomial lie inside the unit circle without computing roots explicitly.
6. **Practical Implementation:** Discrete transfer functions are implemented as difference equations, with pole location directly determining stability and response characteristics.

The z-transform methodology is essential for modern digital control, enabling rigorous analysis and design of systems implemented on microcontrollers, DSPs, and computers.

20.7 Exercises

Exercise 20.1. Find the z-transform of $x[n] = (n + 1)(0.5)^n u[n]$.

Exercise 20.2. Determine the inverse z-transform of $X(z) = \frac{z^2 + 2z}{(z - 0.5)(z - 0.25)}$.

Exercise 20.3. A continuous system has transfer function $G(s) = \frac{10}{(s+1)(s+5)}$. Convert to discrete time using ZOH with $T = 0.1$ s.

Exercise 20.4. Apply the Jury test to determine stability of $D(z) = z^4 - 0.5z^3 + 0.3z^2 - 0.1z + 0.02$.

Exercise 20.5. A discrete system has poles at $z = 0.8 \pm j0.4$. Find:

1. The pole magnitude and angle
2. The equivalent continuous-time pole locations for $T = 0.05$ s
3. The approximate settling time in samples

Exercise 20.6. Design a first-order discrete filter with time constant equivalent to $\tau = 0.2$ s for a sampling rate of 100 Hz. Verify stability.

Exercise 20.7. The difference equation $y[n] = 1.5y[n - 1] - 0.56y[n - 2] + u[n]$ represents a discrete system. Find the transfer function, pole locations, and determine stability.

Exercise 20.8. Using the Arduino experiment as a guide, predict and sketch the impulse response for a filter with $a_1 = -0.9$. Explain the alternating behavior.

Exercise 20.9. Prove that the final value theorem fails when there are poles on or outside the unit circle.

Exercise 20.10. A digital controller samples at 50 Hz. What is the maximum continuous-time natural frequency that can be represented without aliasing? What z-plane angle corresponds to this frequency?

Bibliography

- [1] W. Hurewicz, “Filters and servo systems with pulsed data,” in *Theory of Servomechanisms*, H. M. James, N. B. Nichols, and R. S. Phillips, Eds. New York: McGraw-Hill, 1947, ch. 5.
- [2] E. I. Jury, “A stability test for linear discrete systems in table form,” *Proc. IRE*, vol. 46, no. 6, pp. 1349–1350, June 1958.
- [3] E. I. Jury, *Theory and Application of the Z-Transform Method*. New York: Wiley, 1964.
- [4] J. R. Ragazzini and G. F. Franklin, *Sampled-Data Control Systems*. New York: McGraw-Hill, 1958.
- [5] A. V. Oppenheim and R. W. Schaffer, *Discrete-Time Signal Processing*. Englewood Cliffs, NJ: Prentice-Hall, 1989.
- [6] G. F. Franklin, J. D. Powell, and M. L. Workman, *Digital Control of Dynamic Systems*, 3rd ed. Menlo Park, CA: Addison-Wesley, 1998.
- [7] K. Ogata, *Discrete-Time Control Systems*, 2nd ed. Englewood Cliffs, NJ: Prentice-Hall, 1995.
- [8] K. J. Åström and B. Wittenmark, *Computer-Controlled Systems: Theory and Design*, 3rd ed. Upper Saddle River, NJ: Prentice-Hall, 1997.

Part VI

Modern & AI-Driven Control

Chapter 21: Model Predictive Control Control Systems: A Practical Approach

Chapter Overview

Model Predictive Control (MPC) represents one of the most significant advances in control theory since the development of state-space methods. Unlike classical controllers that react to current errors, MPC looks forward in time, predicting future system behavior and optimizing control actions over a horizon while explicitly handling constraints. This chapter traces MPC's industrial origins, develops its mathematical foundations, and demonstrates its implementation through a practical tank-level control experiment.

Learning Objectives: Upon completing this chapter, students will understand the historical development and industrial motivation for MPC, gaining insight into how process control challenges in the petrochemical industry drove the development of predictive control methods. Students will learn to formulate MPC optimization problems for linear systems, expressing control objectives as quadratic cost functions subject to constraints. The chapter develops skills for converting MPC problems to standard quadratic programming form, enabling efficient numerical solution. Students will implement receding horizon control with explicit constraint handling, and will compare MPC performance to classical PID control under constraints to understand when predictive methods offer advantages.

20.8 Historical Development and Motivation

20.8.1 The Constraint Problem in Process Control

By the early 1970s, the petrochemical industry faced a persistent challenge that classical control theory could not adequately address. Refineries and chemical plants operated complex, interconnected processes with numerous physical limitations: valves that could only open so far, pumps with maximum flow rates, tanks with finite capacities, temperatures that could not exceed safety thresholds, and product specifications that demanded tight tolerances.

Consider the seemingly simple problem of controlling liquid level in a storage tank. A PID controller can regulate the level by adjusting an inlet valve. But what happens when the setpoint changes dramatically, or a large disturbance occurs? The controller, designed without knowledge of physical limits, commands the valve to open beyond its maximum position or close below its minimum. The result is “integrator windup”—the integral term accumulates error while the actuator saturates, causing sluggish response and dangerous overshoot when the actuator finally de-saturates.

Classical anti-windup schemes provided partial remedies, but they were reactive patches rather than principled solutions. The fundamental problem remained: PID controllers optimize instantaneous error correction without considering future consequences or physical constraints. A truly optimal controller would anticipate constraint violations before they occur and plan trajectories that respect limitations from the outset.

20.8.2 Shell Oil and Dynamic Matrix Control (1970s)

Charles Cutler and the Birth of Industrial MPC

Charles R. Cutler, a chemical engineer at Shell Oil Company's Houston research center, developed Dynamic Matrix Control (DMC) in the mid-1970s alongside his colleague Brian Ramaker. Cutler's key insight was that complex refinery processes could be controlled effectively using simple step-response models rather than detailed differential equations—a pragmatic engineering choice that made predictive control practical for industrial implementation. The first DMC installation on a fluid catalytic cracking unit in 1976 demonstrated dramatic improvements in yield and energy efficiency, generating millions of dollars in annual savings. Cutler later co-founded Dynamic Matrix Control Corporation to commercialize the technology, and his 1980 paper with Ramaker, "Dynamic Matrix Control—A Computer Control Algorithm," became one of the most influential publications in process control history. Today, tens of thousands of MPC controllers based on Cutler's ideas operate in refineries and chemical plants worldwide.

The breakthrough came not from academic research but from industrial necessity. Engineers at Shell Oil Company's research center in Houston, Texas, confronted the challenge of controlling complex refinery units where dozens of manipulated variables affected dozens of controlled outputs, all subject to operational constraints.

In the mid-1970s, Charles Cutler and Brian Ramaker at Shell developed **Dynamic Matrix Control (DMC)**, the first commercially successful MPC algorithm. Their approach introduced several key innovations that made the technology practical. First, rather than requiring detailed first-principles models, DMC used empirically measured step response coefficients; engineers could identify these models by simply stepping each input and recording the output responses, eliminating the need for differential equations. Second, the algorithm predicted future outputs over a "prediction horizon" of N time steps, computing how each candidate control move would affect future behavior. Third, DMC incorporated move suppression to prevent excessive control action, penalizing the magnitude of control changes rather than just tracking errors, thereby reducing energy waste and equipment stress. Most critically, DMC formulated control as a constrained optimization problem, directly incorporating actuator limits and output bounds into the control computation.

The mathematical formulation was elegant in its simplicity. Let a_i denote the step response coefficient at time step i —the amount by which the output changes i steps after a unit step in the input. Then the predicted output at future time $k + j$ given control moves $\Delta u_k, \Delta u_{k+1}, \dots$ is:

$$\hat{y}_{k+j} = y_k^{\text{free}} + \sum_{i=1}^j a_i \Delta u_{k+j-i} \quad (20.82)$$

where y_k^{free} is the predicted output assuming no future control changes.

Shell deployed DMC on fluid catalytic cracking units, ethylene plants, and other high-value processes throughout the late 1970s and early 1980s. The results were dramatic: improved yield, reduced energy consumption, and safer operation near constraint boundaries. A single DMC installation could generate millions of dollars in annual savings.

20.8.3 Jacques Richalet and Model Predictive Heuristic Control (1978)

Independently and nearly simultaneously, a French team led by Jacques Richalet at Adersa (a subsidiary of the French automation company SESA) developed **Model Predictive Heuristic Control (MPHC)**, later commercialized as IDCOM.

Richalet, trained as a chemical engineer, approached the problem from a process control perspective. His 1978 paper “Model Predictive Heuristic Control: Applications to Industrial Processes” presented at the IFAC World Congress introduced MPC to the broader control community. The “heuristic” in the name reflected Richalet’s pragmatic engineering philosophy: the algorithm used impulse response models and a reference trajectory concept that smoothly transitioned from current output to desired setpoint.

A key contribution of Richalet’s work was the concept of the **reference trajectory**—rather than demanding immediate setpoint tracking (which might require impossible control effort), MPHC defined a desired first-order exponential approach to the setpoint:

$$y_{\text{ref}}(k+j) = y_k + (y_{\text{sp}} - y_k)(1 - e^{-jT_s/\tau_{\text{ref}}}) \quad (20.83)$$

where τ_{ref} was a tunable time constant specifying how aggressively to pursue the setpoint. This seemingly simple idea had profound implications: it provided a systematic way to trade off speed against control effort and constraint satisfaction.

20.8.4 Why Not Earlier? The Computational Barrier

If MPC’s principles were so powerful, why did they emerge only in the 1970s? The answer lies in computational requirements. MPC solves an optimization problem at every sampling instant—typically a quadratic program (QP) with tens to hundreds of decision variables and constraints.

Consider the computational resources available in different eras, summarized in Table 20.5.

Table 20.5: Evolution of Computing Power and MPC Applications

Era	FLOPS	Technology	MPC Capability
1960s	10^5	Room-sized mainframes	QP solution: minutes to hours
1970s	10^6	Minicomputers (PDP-11)	QP in seconds; slow processes
1980s–1990s	10^7 – 10^9	Workstations, embedded	Seconds-scale sampling
2000s–present	$> 10^{11}$	Modern microprocessors	Millisecond sampling

The history of MPC thus mirrors the history of computing power. Each generation of processors opened MPC to a broader class of applications.

20.8.5 The Theoretical Revolution (1980s–1990s)

While industry deployed increasingly sophisticated MPC systems, academic researchers worked to establish rigorous theoretical foundations. The early industrial algorithms, though empirically successful, lacked formal guarantees of stability and performance.

Several key theoretical developments emerged during this period. David Mayne, James Rawlings, and others established stability through terminal constraints, showing that requiring the predicted state to reach a target set by the end of the horizon guaranteed closed-loop stability under mild conditions. An alternative approach achieved stability through terminal cost, where adding a terminal cost term to the objective function—typically an infinite-horizon LQR cost for the final state—provided stability without explicit terminal constraints. Researchers also developed robust MPC extensions to handle model uncertainty, ensuring stability even when the true plant differs from the prediction model. For systems with modest state and horizon dimensions, explicit MPC allowed the optimal control law to be computed offline as a piecewise-affine function of the state, eliminating online optimization entirely.

By the late 1990s, MPC had evolved from an industrial heuristic to a mathematically mature control methodology with well-understood stability, optimality, and robustness properties.

20.8.6 Becoming the Industry Standard

The adoption of MPC in process industries followed a characteristic technology diffusion curve, as summarized in Table 20.6.

Table 20.6: MPC Technology Adoption Timeline

Period	Development
1978–1985	Early adopters at major oil companies (Shell, Exxon, Texaco) deployed custom systems on high-value units
1985–1995	Commercial MPC software (DMC-Plus, RMPCT, Connoisseur) made the technology accessible to smaller companies; thousands of installations worldwide
1995–2010	MPC became the default advanced control technology for refineries, petrochemical plants, and other process industries; integration with plant-wide optimization systems
2010–present	Extension beyond process industries to automotive, building systems, power electronics, and autonomous vehicles

Today, MPC is taught in virtually every graduate control curriculum and is implemented in systems ranging from smartphone battery management to spacecraft attitude control. The technology born from refinery control rooms has become a fundamental tool of modern engineering.

20.9 Modern Applications

Model Predictive Control has transcended its origins in petrochemical processing to become a universal methodology wherever constraints, optimization, and prediction intersect.

20.9.1 Chemical Process Control

MPC remains dominant in its original domain. Modern refineries deploy dozens of MPC controllers, each coordinating multiple control loops while respecting hundreds of constraints simultaneously. In crude distillation, MPC optimizes product cuts while satisfying quality specifications. Catalytic cracking units use MPC to maximize gasoline yield within temperature and catalyst constraints. Polymerization reactors benefit from MPC's ability to control molecular weight distributions precisely. Batch processes in pharmaceutical and specialty chemical production employ MPC to optimize recipe trajectories throughout the production cycle. The economic incentives remain compelling: a typical refinery MPC installation yields \$1–5 million in annual benefits through improved yield, reduced energy consumption, and increased throughput.

20.9.2 Autonomous Vehicles

Modern autonomous vehicles rely extensively on MPC for motion planning and control. The problem is naturally formulated in MPC terms: given a desired trajectory (from higher-level planning), find control inputs (steering, throttle, brake) that track the trajectory while respecting physical constraints (maximum steering angle, acceleration limits) and safety requirements (obstacle avoidance, lane boundaries).

Several features make MPC particularly attractive for autonomous vehicles. Explicit constraint handling allows vehicle dynamics, actuator limits, and collision avoidance requirements to be expressed directly as optimization constraints. The prediction horizon provides preview capability, enabling the controller to “see” upcoming curves, intersections, and obstacles before they become immediate concerns. Perhaps most importantly, MPC offers graceful degradation: when perfect tracking is impossible due to an obstacle being too close, MPC finds the best feasible compromise rather than failing abruptly. Major autonomous vehicle programs at companies such as Waymo, Tesla, and Cruise employ MPC-based controllers, though specific implementations remain proprietary.

20.9.3 Building HVAC Optimization

Heating, ventilation, and air conditioning (HVAC) systems account for 40% of commercial building energy consumption. MPC offers substantial savings by optimizing HVAC operation over prediction horizons of hours to days. Weather prediction integration allows the controller to use forecast data to pre-cool buildings before heat waves or pre-heat before cold snaps. Occupancy-aware control reduces conditioning in unoccupied zones while maintaining comfort in occupied spaces. MPC enables effective demand response by shifting energy consumption to off-peak hours or responding to grid signals. For buildings with thermal storage capabilities, MPC optimally manages the charging and discharging of ice storage or building thermal mass. Commercial building MPC implementations report 15–30% energy savings compared to conventional rule-based control.

20.9.4 Quadrotor and UAV Control

Quadrotor drones present a challenging control problem: underactuated (4 inputs controlling 6 DOF), unstable, and operating in environments with obstacles. MPC has become the preferred approach for aggressive trajectory tracking due to its ability to handle these challenges systematically. Trajectory optimization computes dynamically feasible paths through cluttered environments. Constraint satisfaction ensures the controller respects motor speed limits, battery current constraints, and no-fly zones. MPC provides effective disturbance rejection, adapting to wind gusts and payload changes in real time. For multi-vehicle applications, MPC coordinates swarms while avoiding inter-vehicle collisions through appropriate constraint formulation. Academic demonstrations have shown MPC-controlled quadrotors performing acrobatic maneuvers, flying through narrow gaps, and collaboratively transporting objects.

20.10 Mathematical Foundation

We now develop the mathematical framework for Model Predictive Control, progressing from the basic concept through the complete quadratic programming formulation.

20.10.1 The Fundamental Idea

MPC's core concept is beautifully simple: at each time step, solve a finite-horizon optimal control problem, apply only the first control action, then repeat at the next time step with updated measurements. This “receding horizon” strategy transforms an intractable infinite-horizon problem into a sequence of tractable finite-horizon problems.

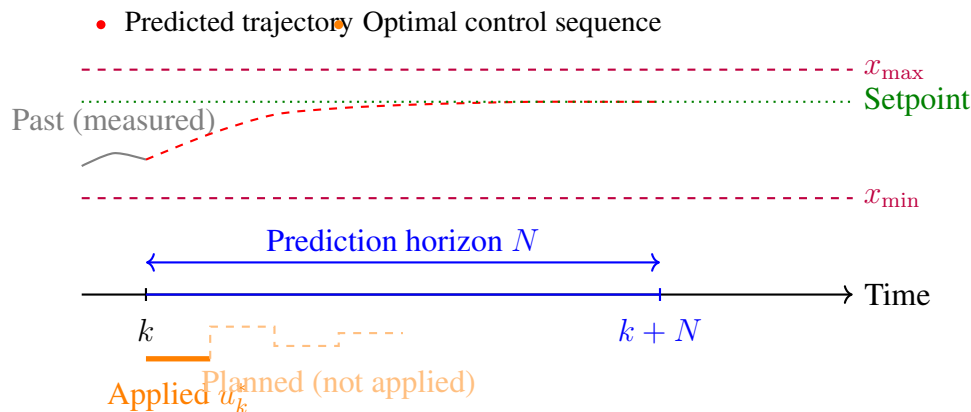


Figure 20.6: The receding horizon concept. At time k , MPC predicts system behavior over horizon N , optimizes the control sequence, but applies only the first control action u_k^* . The process repeats at $k + 1$ with new measurements.

The receding horizon approach provides several crucial benefits. First, by re-solving at each step, MPC incorporates new measurements and rejects disturbances, providing effective feedback control. Second, each optimization involves only N time steps rather than infinity, ensuring finite computation time. Third, model errors are partially corrected by the feedback mechanism,

providing implicit robustness to modeling inaccuracies. Finally, changing constraints or setpoints are automatically incorporated into the optimization at each step, enabling seamless adaptation to varying operating conditions.

20.10.2 System Model: Discrete-Time State-Space

MPC requires a model predicting future states from current state and control inputs. For linear time-invariant systems, we use the discrete-time state-space form:

$$\begin{cases} \mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}\mathbf{u}_k \\ \mathbf{y}_k = \mathbf{C}\mathbf{x}_k + \mathbf{D}\mathbf{u}_k \end{cases} \quad (20.84)$$

In this formulation, $\mathbf{x}_k \in \mathbb{R}^n$ denotes the state vector at time step k , $\mathbf{u}_k \in \mathbb{R}^m$ represents the control input vector, and $\mathbf{y}_k \in \mathbb{R}^p$ is the output vector. The system matrices have the following dimensions: $\mathbf{A} \in \mathbb{R}^{n \times n}$ is the state transition matrix governing the autonomous system dynamics, $\mathbf{B} \in \mathbb{R}^{n \times m}$ is the input matrix describing how controls affect states, $\mathbf{C} \in \mathbb{R}^{p \times n}$ is the output matrix relating states to measured outputs, and $\mathbf{D} \in \mathbb{R}^{p \times m}$ is the feedthrough matrix (often zero in physical systems).

Starting from state $\mathbf{x}_k$, we can predict future states recursively:

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}\mathbf{u}_k \quad (20.85)$$

$$\mathbf{x}_{k+2} = \mathbf{A}\mathbf{x}_{k+1} + \mathbf{B}\mathbf{u}_{k+1} = \mathbf{A}^2\mathbf{x}_k + \mathbf{A}\mathbf{B}\mathbf{u}_k + \mathbf{B}\mathbf{u}_{k+1} \quad (20.86)$$

$$\vdots$$

$$\mathbf{x}_{k+j} = \mathbf{A}^j\mathbf{x}_k + \sum_{i=0}^{j-1} \mathbf{A}^{j-1-i}\mathbf{B}\mathbf{u}_{k+i} \quad (20.87)$$

20.10.3 Cost Function Formulation

MPC minimizes a cost function that penalizes deviation from desired behavior and excessive control effort. The standard quadratic cost over prediction horizon N is:

$$J = \sum_{i=0}^{N-1} (\mathbf{x}_{k+i}^\top \mathbf{Q} \mathbf{x}_{k+i} + \mathbf{u}_{k+i}^\top \mathbf{R} \mathbf{u}_{k+i}) + \mathbf{x}_{k+N}^\top \mathbf{P} \mathbf{x}_{k+N} \quad (20.88)$$

In this cost function, $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is the state weighting matrix (positive semi-definite, $\mathbf{Q} \succeq 0$), $\mathbf{R} \in \mathbb{R}^{m \times m}$ is the input weighting matrix (positive definite, $\mathbf{R} \succ 0$), $\mathbf{P} \in \mathbb{R}^{n \times n}$ is the terminal state weighting matrix ($\mathbf{P} \succeq 0$), and N denotes the prediction horizon length.

Physical interpretation: The $\mathbf{x}^\top \mathbf{Q} \mathbf{x}$ terms penalize state deviation from zero (or from a reference trajectory), while the $\mathbf{u}^\top \mathbf{R} \mathbf{u}$ terms penalize control effort, accounting for actuator usage and energy consumption. The terminal cost $\mathbf{x}_{k+N}^\top \mathbf{P} \mathbf{x}_{k+N}$ accounts for behavior beyond the prediction horizon. The relative magnitude of the weighting matrices determines the controller's behavior: large $\mathbf{Q}$ entries demand tight tracking, while large $\mathbf{R}$ entries demand gentle control action.

20.10.4 Constraint Specification

A defining feature of MPC is explicit constraint handling. Common constraint types include:

Input constraints (actuator limits):

$$\mathbf{u}_{\min} \leq \mathbf{u}_{k+i} \leq \mathbf{u}_{\max}, \quad i = 0, 1, \dots, N-1 \quad (20.89)$$

Input rate constraints (slew rate limits):

$$\Delta \mathbf{u}_{\min} \leq \mathbf{u}_{k+i} - \mathbf{u}_{k+i-1} \leq \Delta \mathbf{u}_{\max}, \quad i = 0, 1, \dots, N-1 \quad (20.90)$$

State constraints (physical limits, safety bounds):

$$\mathbf{x}_{\min} \leq \mathbf{x}_{k+i} \leq \mathbf{x}_{\max}, \quad i = 1, 2, \dots, N \quad (20.91)$$

Output constraints:

$$\mathbf{y}_{\min} \leq \mathbf{y}_{k+i} \leq \mathbf{y}_{\max}, \quad i = 1, 2, \dots, N \quad (20.92)$$

Terminal constraints (for stability):

$$\mathbf{x}_{k+N} \in \mathcal{X}_f \quad (20.93)$$

where $\mathcal{X}_f$ is a target set, often a neighborhood of the origin.

20.10.5 The MPC Optimization Problem

Combining the cost function and constraints, the MPC problem at time step k is:

$$\begin{aligned} \min_{\mathbf{u}_k, \dots, \mathbf{u}_{k+N-1}} \quad & J = \sum_{i=0}^{N-1} (\mathbf{x}_{k+i}^\top \mathbf{Q} \mathbf{x}_{k+i} + \mathbf{u}_{k+i}^\top \mathbf{R} \mathbf{u}_{k+i}) + \mathbf{x}_{k+N}^\top \mathbf{P} \mathbf{x}_{k+N} \\ \text{subject to:} \quad & \mathbf{x}_{k+i+1} = \mathbf{A} \mathbf{x}_{k+i} + \mathbf{B} \mathbf{u}_{k+i}, \quad i = 0, \dots, N-1 \\ & \mathbf{u}_{\min} \leq \mathbf{u}_{k+i} \leq \mathbf{u}_{\max}, \quad i = 0, \dots, N-1 \\ & \mathbf{x}_{\min} \leq \mathbf{x}_{k+i} \leq \mathbf{x}_{\max}, \quad i = 1, \dots, N \\ & \mathbf{x}_k = \mathbf{x}(k) \quad (\text{measured current state}) \end{aligned} \quad (20.94)$$

20.10.6 Conversion to Quadratic Programming Form

To solve the MPC problem efficiently, we convert it to standard quadratic programming (QP) form. The key is to eliminate the state variables using the prediction equations, expressing everything in terms of the decision variables (control inputs).

Step 1: Define the stacked vectors.

Let us define the stacked control sequence:

$$\mathbf{U} = \begin{bmatrix} \mathbf{u}_k \\ \mathbf{u}_{k+1} \\ \vdots \\ \mathbf{u}_{k+N-1} \end{bmatrix} \in \mathbb{R}^{Nm} \quad (20.95)$$

And the stacked predicted state sequence:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_{k+1} \\ \mathbf{x}_{k+2} \\ \vdots \\ \mathbf{x}_{k+N} \end{bmatrix} \in \mathbb{R}^{Nn} \quad (20.96)$$

Step 2: Express states in terms of controls.

From the prediction equations (??)–(??), we can write:

$$\mathbf{X} = \mathcal{A}\mathbf{x}_k + \mathcal{B}\mathbf{U} \quad (20.97)$$

where the prediction matrices are:

$$\mathcal{A} = \begin{bmatrix} \mathbf{A} \\ \mathbf{A}^2 \\ \mathbf{A}^3 \\ \vdots \\ \mathbf{A}^N \end{bmatrix} \in \mathbb{R}^{Nn \times n} \quad (20.98)$$

$$\mathcal{B} = \begin{bmatrix} \mathbf{B} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{AB} & \mathbf{B} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{A}^2\mathbf{B} & \mathbf{AB} & \mathbf{B} & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}^{N-1}\mathbf{B} & \mathbf{A}^{N-2}\mathbf{B} & \mathbf{A}^{N-3}\mathbf{B} & \cdots & \mathbf{B} \end{bmatrix} \in \mathbb{R}^{Nn \times Nm} \quad (20.99)$$

Step 3: Reformulate the cost function.

The cost function can be written as:

$$J = \mathbf{X}^\top \bar{\mathbf{Q}}\mathbf{X} + \mathbf{U}^\top \bar{\mathbf{R}}\mathbf{U} + \mathbf{x}_k^\top \mathbf{Q}\mathbf{x}_k \quad (20.100)$$

where $\bar{\mathbf{Q}} = \text{diag}(\mathbf{Q}, \mathbf{Q}, \dots, \mathbf{Q}, \mathbf{P}) \in \mathbb{R}^{Nn \times Nn}$ and $\bar{\mathbf{R}} = \text{diag}(\mathbf{R}, \mathbf{R}, \dots, \mathbf{R}) \in \mathbb{R}^{Nm \times Nm}$.

Substituting (??):

$$J = (\mathcal{A}\mathbf{x}_k + \mathcal{B}\mathbf{U})^\top \bar{\mathbf{Q}}(\mathcal{A}\mathbf{x}_k + \mathcal{B}\mathbf{U}) + \mathbf{U}^\top \bar{\mathbf{R}}\mathbf{U} + \mathbf{x}_k^\top \mathbf{Q}\mathbf{x}_k \quad (20.101)$$

$$= \mathbf{U}^\top \underbrace{(\mathcal{B}^\top \bar{\mathbf{Q}}\mathcal{B} + \bar{\mathbf{R}})}_{\mathbf{H}} \mathbf{U} + 2 \underbrace{\mathbf{x}_k^\top \mathcal{A}^\top \bar{\mathbf{Q}}\mathcal{B}}_{\mathbf{f}^\top} \mathbf{U} + \text{const.} \quad (20.102)$$

Step 4: Reformulate constraints.

Input constraints become:

$$\mathbf{1}_N \otimes \mathbf{u}_{\min} \leq \mathbf{U} \leq \mathbf{1}_N \otimes \mathbf{u}_{\max} \quad (20.103)$$

State constraints, using $\mathbf{X} = \mathcal{A}\mathbf{x}_k + \mathcal{B}\mathbf{U}$:

$$\mathbf{1}_N \otimes \mathbf{x}_{\min} \leq \mathcal{A}\mathbf{x}_k + \mathcal{B}\mathbf{U} \leq \mathbf{1}_N \otimes \mathbf{x}_{\max} \quad (20.104)$$

This can be written as linear inequality constraints $\mathbf{G}\mathbf{U} \leq \mathbf{w}$.

Step 5: Standard QP form.

The MPC problem is now in standard QP form:

$$\begin{array}{ll} \min_{\mathbf{U}} & \frac{1}{2} \mathbf{U}^\top \mathbf{H} \mathbf{U} + \mathbf{f}^\top \mathbf{U} \\ \text{subject to:} & \mathbf{G} \mathbf{U} \leq \mathbf{w} \end{array} \quad (20.105)$$

In this formulation, $\mathbf{H} = \mathcal{B}^\top \bar{\mathbf{Q}} \mathcal{B} + \bar{\mathbf{R}} \in \mathbb{R}^{Nm \times Nm}$ is the Hessian matrix (positive definite due to $\mathbf{R} \succ 0$), $\mathbf{f} = \mathcal{B}^\top \bar{\mathbf{Q}} \mathcal{A} \mathbf{x}_k \in \mathbb{R}^{Nm}$ is the linear cost vector that depends on the current state, and $\mathbf{G}$ and $\mathbf{w}$ encode all inequality constraints in a compact matrix form.

This QP can be solved by standard algorithms (active set, interior point, etc.) with computational complexity polynomial in the problem dimensions.

20.10.7 The Receding Horizon Algorithm

The complete MPC algorithm proceeds as follows:

Algorithm 43 Model Predictive Control Algorithm

- 1: **Initialization:** Choose N , $\mathbf{Q}$, $\mathbf{R}$, $\mathbf{P}$, constraints
 - 2: Form $\mathbf{H}$ (constant, computed once)
 - 3: **for** each sampling instant $k = 0, 1, 2, \dots$ **do**
 - 4: Measure or estimate current state $\mathbf{x}_k$
 - 5: Form $\mathbf{f}$ and $\mathbf{w}$ using current $\mathbf{x}_k$
 - 6: Solve QP (??) to obtain $\mathbf{U}^* = [\mathbf{u}_k^{*\top}, \mathbf{u}_{k+1}^{*\top}, \dots]^\top$
 - 7: Apply only the first element: $\mathbf{u}_k = \mathbf{u}_k^*$
 - 8: Wait for next sampling instant
 - 9: **end for**
-

20.10.8 Stability Considerations

A critical question is whether MPC guarantees closed-loop stability. Without additional structure, a finite-horizon optimization does not generally ensure stability—the controller might “give up” on long-term behavior beyond the horizon.

Theorem (MPC Stability): The MPC closed-loop system is asymptotically stable under either of the following conditions, provided the optimization problem remains feasible at each time step. The first sufficient condition requires that the terminal cost $\mathbf{P}$ satisfies the Lyapunov equation $\mathbf{A}^\top \mathbf{P} \mathbf{A} - \mathbf{P} + \mathbf{Q} = 0$, meaning $\mathbf{P}$ equals the infinite-horizon LQR cost matrix. Alternatively, stability is guaranteed when a terminal constraint $\mathbf{x}_{k+N} \in \mathcal{X}_f$ is imposed, where $\mathcal{X}_f$ is a control-invariant set containing the origin.

Intuition: The terminal cost (or constraint) ensures that the MPC controller “cares about” behavior beyond the horizon, preventing short-sighted decisions that compromise long-term stability.

In practice, choosing $\mathbf{P}$ as the solution to the discrete algebraic Riccati equation (DARE) is a common and effective strategy.

20.10.9 Relationship to LQR

Model Predictive Control and Linear Quadratic Regulator (LQR) control are intimately related:

Theorem (MPC-LQR Equivalence): For linear systems with no constraints, infinite horizon ($N \rightarrow \infty$), and terminal cost P equal to the LQR cost matrix, MPC produces the identical control law as LQR.

Proof sketch: With no constraints and appropriate terminal cost, the MPC optimization becomes the standard LQR problem, whose solution is the well-known state feedback gain $K = (R + B^T P B)^{-1} B^T P A$.

This relationship is profound and illuminates the fundamental nature of both methods. LQR represents the “idealized” case of MPC, corresponding to no constraints, perfect model, and infinite foresight. MPC gracefully degrades from LQR as constraints become active: when constraints are inactive, MPC behaves identically to LQR; when constraints become active, MPC finds the best feasible approximation to the unconstrained optimum.

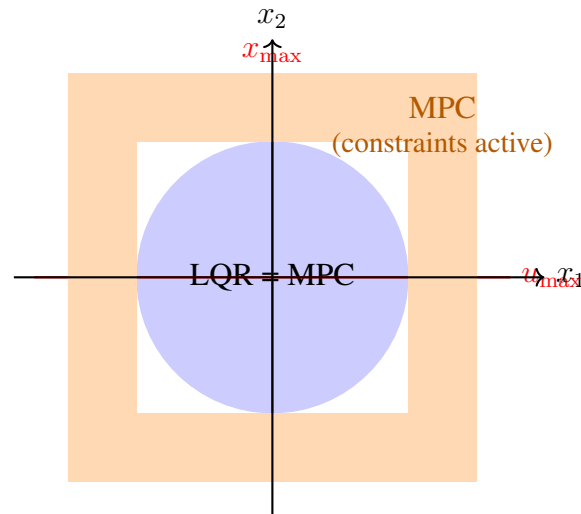


Figure 20.7: Relationship between MPC and LQR in state space. In the interior region (blue), constraints are inactive and MPC reduces to LQR. Near boundaries (orange), constraints become active and MPC differs from LQR.

20.11 Practical Laboratory: Tank Level Control with Constraints

20.11.1 Objective

This experiment demonstrates MPC’s key advantage: graceful constraint handling. Students will implement both PID and MPC controllers for a cascaded tank system and observe how each handles actuator saturation during large setpoint changes.

20.11.2 System Description

The experimental apparatus consists of two vertically cascaded tanks with a single pump as the actuator.

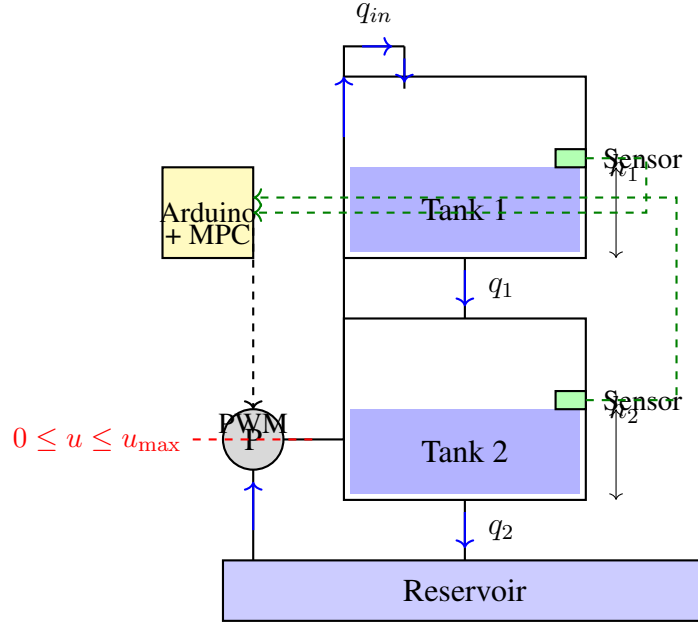


Figure 20.8: Cascaded two-tank system. Water flows from reservoir through pump to Tank 1, drains to Tank 2, and returns to reservoir. The pump can only push water (unidirectional), creating a fundamental input constraint.

Key constraint: The pump can only push water into the system—it cannot pull water out. This creates a hard input constraint: $0 \leq u \leq u_{\max}$.

When the operator requests a rapid decrease in tank level, a PID controller will command negative pump speed (impossible) and saturate at zero, causing integrator windup. MPC, knowing the constraint, will instead plan a trajectory that decreases level by simply reducing inflow and waiting for natural drainage.

20.11.3 System Modeling

Physical equations:

Mass balance for each tank:

$$A_1 \frac{dh_1}{dt} = q_{in} - q_1 = q_{in} - a_1 \sqrt{2gh_1} \quad (20.106)$$

$$A_2 \frac{dh_2}{dt} = q_1 - q_2 = a_1 \sqrt{2gh_1} - a_2 \sqrt{2gh_2} \quad (20.107)$$

where A_i is tank cross-sectional area, a_i is outlet orifice area, and g is gravitational acceleration.

Linearization around operating point (h_1^0, h_2^0, q_{in}^0) :

Define deviation variables: $\tilde{h}_i = h_i - h_i^0$, $\tilde{u} = q_{in} - q_{in}^0$.

The linearized model is:

$$\begin{bmatrix} \dot{\tilde{h}}_1 \\ \dot{\tilde{h}}_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{T_1} & 0 \\ \frac{1}{T_1} & -\frac{1}{T_2} \end{bmatrix} \begin{bmatrix} \tilde{h}_1 \\ \tilde{h}_2 \end{bmatrix} + \begin{bmatrix} \frac{K_1}{T_1} \\ 0 \end{bmatrix} \tilde{u} \quad (20.108)$$

where the time constants $T_i = \frac{A_i}{a_i} \sqrt{\frac{2h_i^0}{g}}$ and gain $K_1 = \frac{1}{a_1 \sqrt{2gh_1^0}}$ depend on the operating point.

Discrete-time model: Using zero-order hold with sampling time T_s :

$$\mathbf{x}_{k+1} = \mathbf{A}_d \mathbf{x}_k + \mathbf{B}_d u_k \quad (20.109)$$

20.11.4 Required Components

Table 20.7: Bill of Materials for Tank Control Experiment

Component	Specification	Qty
Arduino Mega 2560	ATmega2560 microcontroller	1
Water pump	6-12V DC, 2-5 L/min	1
Motor driver	L298N or similar	1
Tanks	3D printed, ~500 mL each	2
Ultrasonic sensors	HC-SR04 or waterproof equivalent	2
Tubing	8mm ID silicone	2m
Orifice fittings	3-5mm diameter	2
Reservoir	Any container >2L	1
Power supply	12V, 3A	1

3D Printing Notes: Tank walls should be at least 3mm thick for adequate water resistance. PETG is the recommended material due to its water resistance and food-safe properties. The design should include mounting points for sensors and tubes, and o-ring grooves should be considered for leak-proof fittings. At 0.2mm layer height, expect approximately 4–6 hours of print time per tank.

20.11.5 Software Implementation

PID Controller with Anti-Windup

Algorithm 44 PID Controller with Anti-Windup

```

1: Parameters:  $K_p, K_i, K_d, u_{\min}, u_{\max}$ 
2: State variables:  $I \leftarrow 0$  (integral accumulator),  $e_{\text{prev}} \leftarrow 0$ 
3:
4: function COMPUTE( $r, y, \Delta t$ )                                ▷  $r$  = setpoint,  $y$  = measurement
5:    $e \leftarrow r - y$                                           ▷ Compute error
6:    $P \leftarrow K_p \cdot e$                                     ▷ Proportional term
7:    $I \leftarrow I + K_i \cdot e \cdot \Delta t$                   ▷ Integral term
8:    $D \leftarrow K_d \cdot (e - e_{\text{prev}}) / \Delta t$           ▷ Derivative term
9:    $u \leftarrow P + I + D$                                     ▷ Compute raw output
10:  if  $u > u_{\max}$  then
11:     $I \leftarrow I - (u - u_{\max})$                             ▷ Anti-windup: reduce integral
12:     $u \leftarrow u_{\max}$ 
13:  else if  $u < u_{\min}$  then
14:     $I \leftarrow I - (u - u_{\min})$                             ▷ Anti-windup: increase integral
15:     $u \leftarrow u_{\min}$ 
16:  end if
17:   $e_{\text{prev}} \leftarrow e$ 
18:  return  $u$ 
19: end function

```

Simple MPC Implementation

For this educational experiment, we implement a simplified MPC that solves the unconstrained problem analytically, then projects the solution onto the constraint set.

Algorithm 45 Simplified MPC for Two-Tank System

```

1: Offline Initialization:
2: Input: System matrices  $\mathbf{A}$ ,  $\mathbf{B}$ ; weights  $\mathbf{Q}$ ,  $R$ ; constraints  $u_{\min}$ ,  $u_{\max}$ ; horizon  $N$ 
3: Compute prediction matrices  $\mathcal{A}$ ,  $\mathcal{B}$  from  $\mathbf{A}$ ,  $\mathbf{B}$ ,  $N$ 
4: Compute Hessian  $H \leftarrow \mathcal{B}^\top \bar{\mathbf{Q}} \mathcal{B} + \bar{R}$ 
5: Compute linear coefficient matrix  $\mathbf{F} \leftarrow \mathcal{B}^\top \bar{\mathbf{Q}} \mathcal{A}$ 
6:
7: Online Control Loop:
8: while system is running do
9:   Read sensor measurements  $h_1, h_2$ 
10:  Compute state deviation:  $\mathbf{x} \leftarrow [h_1 - r_1, h_2 - r_2]^\top$ 
11:  Compute linear cost term:  $f \leftarrow \mathbf{F} \cdot \mathbf{x}$ 
12:  Compute unconstrained optimum:  $u^* \leftarrow -H^{-1}f$ 
13:  Project onto constraints:
14:  if  $u^* < u_{\min}$  then
15:     $u^* \leftarrow u_{\min}$ 
16:  else if  $u^* > u_{\max}$  then
17:     $u^* \leftarrow u_{\max}$ 
18:  end if
19:  Apply control signal  $u^*$  to pump
20:  Log data:  $h_1, h_2, u^*$ , setpoints
21:  Wait for next sampling instant
22: end while

```

20.11.6 Experimental Procedure

Part 1: System Identification

Begin by filling the reservoir and establishing steady-state operation at mid-level in both tanks. Apply step changes in pump PWM, for example from 100 to 150, and record the resulting tank level responses over time. Using the recorded data, fit first-order-plus-delay models to the step responses. From these fitted models, extract the time constants T_1 and T_2 along with the system gains for use in the MPC prediction model.

Part 2: PID Control Experiment

Tune the PID controller to achieve acceptable step response performance at the nominal operating point. Command a large positive setpoint change to request a level increase, and observe that while the pump saturates at its maximum value, the system recovers reasonably. Next, command a large negative setpoint change to request a level decrease. Observe that the pump saturates at zero since it cannot provide negative flow, causing the integrator to wind up. Record the overshoot, settling time, and any constraint violations for later comparison with MPC.

Part 3: MPC Control Experiment

Configure the MPC controller with tuning parameters that target the same response speed as the PID controller. Repeat the large positive setpoint change and observe that MPC may preemptively limit the pump rate to avoid later saturation. When repeating the large negative setpoint change, observe that MPC reduces the pump command to its minimum value without attempting negative values, instead planning a feasible descent trajectory that respects the physical constraint. Compare the performance metrics from MPC to those recorded for PID control.

20.11.7 Expected Results

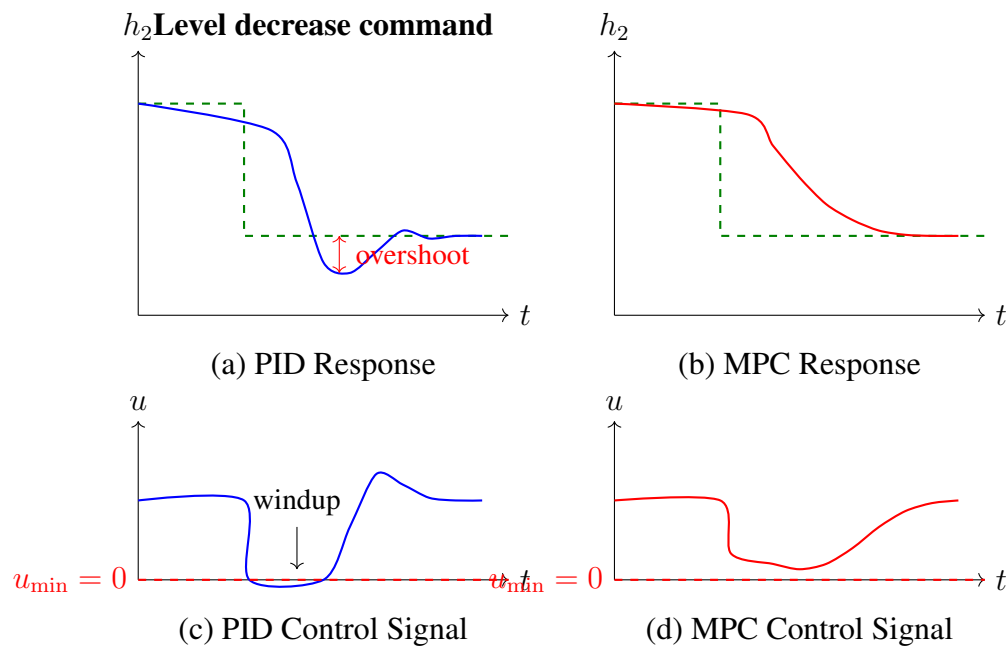


Figure 20.9: Comparison of PID and MPC for large level decrease. PID saturates at $u = 0$, causing integrator windup and subsequent overshoot. MPC anticipates the constraint and plans a feasible trajectory, achieving monotonic convergence.

Table 20.8: Expected Performance Comparison

Metric	PID	MPC
Rise time (level increase)	Similar	Similar
Settling time (level decrease)	Longer	Shorter
Overshoot (level decrease)	10-20%	<5%
Constraint satisfaction	Violations	Always satisfied
Control smoothness	Oscillatory	Smooth

20.11.8 Discussion Questions

1. Why does MPC produce smoother control signals than PID in constrained situations?
2. How would increasing the MPC horizon N affect performance and computation time?
3. What modifications would be needed if the pump could also pull water (bidirectional)?
4. How would you handle a constraint on maximum tank level (overflow prevention)?

20.12 Worked Examples

Setting Up MPC Matrices for a Double Integrator

Problem: A double integrator system (mass under force control) has dynamics:

$$\ddot{x} = u \quad (20.110)$$

Set up the MPC prediction matrices for a horizon of $N = 3$ with sampling time $T_s = 0.1$ s.

Solution:

Step 1: Discretize the continuous-time system.

State vector: $\mathbf{x} = [x, \dot{x}]^\top$

Continuous-time matrices:

$$\mathbf{A}_c = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{B}_c = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (20.111)$$

Discrete-time matrices (using $\mathbf{A}_d = e^{\mathbf{A}_c T_s}$, $\mathbf{B}_d = \int_0^{T_s} e^{\mathbf{A}_c \tau} d\tau \mathbf{B}_c$):

$$\mathbf{A} = \begin{bmatrix} 1 & 0.1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0.005 \\ 0.1 \end{bmatrix} \quad (20.112)$$

Step 2: Build the prediction matrices.

$$\mathcal{A} = \begin{bmatrix} \mathbf{A} \\ \mathbf{A}^2 \\ \mathbf{A}^3 \end{bmatrix} = \begin{bmatrix} 1 & 0.1 \\ 0 & 1 \\ 1 & 0.2 \\ 0 & 1 \\ 1 & 0.3 \\ 0 & 1 \end{bmatrix} \quad (20.113)$$

$$\mathcal{B} = \begin{bmatrix} \mathbf{B} & \mathbf{0} & \mathbf{0} \\ \mathbf{AB} & \mathbf{B} & \mathbf{0} \\ \mathbf{A}^2\mathbf{B} & \mathbf{AB} & \mathbf{B} \end{bmatrix} = \begin{bmatrix} 0.005 & 0 & 0 \\ 0.1 & 0 & 0 \\ 0.015 & 0.005 & 0 \\ 0.2 & 0.1 & 0 \\ 0.03 & 0.015 & 0.005 \\ 0.3 & 0.2 & 0.1 \end{bmatrix} \quad (20.114)$$

Step 3: Form the Hessian matrix.

With $\mathbf{Q} = \text{diag}(1, 0.1)$ (penalize position more than velocity) and $\mathbf{R} = 0.01$:

$$\bar{\mathbf{Q}} = \text{diag}(\mathbf{Q}, \mathbf{Q}, \mathbf{Q}) \in \mathbb{R}^{6 \times 6} \quad (20.115)$$

$$\bar{\mathbf{R}} = \text{diag}(0.01, 0.01, 0.01) \in \mathbb{R}^{3 \times 3} \quad (20.116)$$

$$\mathbf{H} = \mathbf{B}^\top \bar{\mathbf{Q}} \mathbf{B} + \bar{\mathbf{R}} \quad (20.117)$$

Computing $\mathbf{H}$ (showing structure):

$$\mathbf{H} = \begin{bmatrix} 0.0116 & 0.0065 & 0.0031 \\ 0.0065 & 0.0113 & 0.0063 \\ 0.0031 & 0.0063 & 0.0110 \end{bmatrix} \quad (20.118)$$

The off-diagonal coupling shows how future controls affect the current optimal decision.

QP Formulation with Input Constraints

Problem: For the double integrator from the previous example, formulate the complete QP with constraints $-1 \leq u_k \leq 1$ for all k .

Solution:

The optimization variable is:

$$\mathbf{U} = \begin{bmatrix} u_0 \\ u_1 \\ u_2 \end{bmatrix} \quad (20.119)$$

Cost function:

$$J = \frac{1}{2} \mathbf{U}^\top \mathbf{H} \mathbf{U} + \mathbf{f}^\top \mathbf{U} \quad (20.120)$$

where $\mathbf{f} = \mathbf{B}^\top \bar{\mathbf{Q}} \mathbf{A} \mathbf{x}_0$ depends on current state $\mathbf{x}_0$.

Constraints:

Upper bounds: $u_i \leq 1$, or $\mathbf{I} \mathbf{U} \leq \mathbf{1}$

Lower bounds: $u_i \geq -1$, or $-\mathbf{I} \mathbf{U} \leq \mathbf{1}$

Combined inequality form:

$$\begin{bmatrix} \mathbf{I}_3 \\ -\mathbf{I}_3 \end{bmatrix} \mathbf{U} \leq \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad (20.121)$$

Complete QP:

$$\begin{array}{ll} \min_{\mathbf{U}} & \frac{1}{2} \mathbf{U}^\top \begin{bmatrix} 0.0116 & 0.0065 & 0.0031 \\ 0.0065 & 0.0113 & 0.0063 \\ 0.0031 & 0.0063 & 0.0110 \end{bmatrix} \mathbf{U} + \mathbf{f}^\top \mathbf{U} \\ \text{s.t.} & -\mathbf{1} \leq \mathbf{U} \leq \mathbf{1} \end{array} \quad (20.122)$$

20.12.1 Example 3: Comparing Unconstrained MPC to LQR

Problem: For the system $\mathbf{x}_{k+1} = \begin{bmatrix} 0.9 & 0.1 \\ 0 & 0.8 \end{bmatrix} \mathbf{x}_k + \begin{bmatrix} 0.05 \\ 0.1 \end{bmatrix} u_k$ with $\mathbf{Q} = \mathbf{I}_2$, $\mathbf{R} = 1$:

(a) Compute the infinite-horizon LQR gain (b) Compute the MPC control for horizon $N = 20$ at state $\mathbf{x}_0 = [1, 0]^\top$ (c) Compare the first control actions

Solution:

(a) *LQR Solution:*

Solve the discrete algebraic Riccati equation (DARE):

$$\mathbf{P} = \mathbf{Q} + \mathbf{A}^\top \mathbf{P} \mathbf{A} - \mathbf{A}^\top \mathbf{P} \mathbf{B} (\mathbf{R} + \mathbf{B}^\top \mathbf{P} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{P} \mathbf{A} \quad (20.123)$$

Numerical solution (e.g., using MATLAB's `dare`):

$$\mathbf{P} = \begin{bmatrix} 2.76 & 1.32 \\ 1.32 & 3.18 \end{bmatrix} \quad (20.124)$$

LQR gain:

$$\mathbf{K} = (\mathbf{R} + \mathbf{B}^\top \mathbf{P} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{P} \mathbf{A} = \begin{bmatrix} 0.183 & 0.323 \end{bmatrix} \quad (20.125)$$

At $\mathbf{x}_0 = [1, 0]^\top$:

$$u_{\text{LQR}} = -\mathbf{K} \mathbf{x}_0 = -0.183 \quad (20.126)$$

(b) *MPC Solution with $N = 20$:*

Set terminal cost $\mathbf{P}$ equal to the LQR cost matrix from part (a).

Using numerical QP solver with $N = 20$ horizon:

$$u_{\text{MPC}}^* = -0.183 \quad (20.127)$$

(c) *Comparison:*

$$u_{\text{LQR}} = u_{\text{MPC}}^* = -0.183 \quad (20.128)$$

The solutions match exactly! This confirms that unconstrained MPC with appropriate terminal cost converges to LQR as N increases. ■

20.12.2 Example 4: Effect of Prediction Horizon Length

Problem: For the double integrator system, analyze how the MPC control action at $\mathbf{x}_0 = [1, 0]^\top$ varies with horizon length $N = 1, 2, 5, 10, 20$.

Solution:

Using $\mathbf{Q} = \text{diag}(1, 0.1)$, $\mathbf{R} = 0.01$, and terminal cost $\mathbf{P} = \mathbf{Q}$ (suboptimal choice for illustration):

Observations: Short horizons such as $N = 1$ or $N = 2$ produce less aggressive control because the controller “doesn’t see” far enough into the future. As N increases, the control action approaches the LQR solution, and the cost decreases due to greater optimization freedom. However, there are diminishing returns: $N = 10$ captures most of the benefit, suggesting that extremely long horizons may not justify the additional computational cost.

Practical guideline: Choose N such that open-loop predictions extend beyond the system’s dominant time constant. ■

N	u_0^*	Cost J
1	-0.42	1.002
2	-0.58	0.995
5	-0.71	0.982
10	-0.79	0.971
20	-0.85	0.963
∞ (LQR)	-0.91	0.958

20.12.3 Example 5: MPC for Setpoint Tracking

Problem: Modify the MPC formulation to track a non-zero setpoint $\mathbf{x}_{\text{ref}}$, u_{ref} for the system:

$$\mathbf{x}_{k+1} = \begin{bmatrix} 1 & 0.1 \\ 0 & 0.9 \end{bmatrix} \mathbf{x}_k + \begin{bmatrix} 0 \\ 0.1 \end{bmatrix} u_k \quad (20.129)$$

where we want to track $\mathbf{x}_{\text{ref}} = [5, 0]^\top$ (position of 5, zero velocity).

Solution:

Step 1: Find steady-state input.

At steady state: $\mathbf{x}_{\text{ref}} = \mathbf{A}\mathbf{x}_{\text{ref}} + \mathbf{B}u_{\text{ref}}$

$$\begin{bmatrix} 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0.1 \\ 0 & 0.9 \end{bmatrix} \begin{bmatrix} 5 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0.1 \end{bmatrix} u_{\text{ref}} \quad (20.130)$$

From the second row: $0 = 0 + 0.1u_{\text{ref}} \Rightarrow u_{\text{ref}} = 0$

Step 2: Transform to deviation coordinates.

Define: $\tilde{\mathbf{x}}_k = \mathbf{x}_k - \mathbf{x}_{\text{ref}}$, $\tilde{u}_k = u_k - u_{\text{ref}}$

The dynamics in deviation coordinates:

$$\tilde{\mathbf{x}}_{k+1} = \mathbf{A}\tilde{\mathbf{x}}_k + \mathbf{B}\tilde{u}_k \quad (20.131)$$

Step 3: Apply standard MPC to deviation system.

Cost function (penalizes deviation from setpoint):

$$J = \sum_{i=0}^{N-1} (\tilde{\mathbf{x}}_{k+i}^\top \mathbf{Q} \tilde{\mathbf{x}}_{k+i} + \tilde{u}_{k+i}^\top \mathbf{R} \tilde{u}_{k+i}) + \tilde{\mathbf{x}}_{k+N}^\top \mathbf{P} \tilde{\mathbf{x}}_{k+N} \quad (20.132)$$

Constraints transform accordingly:

$$u_{\min} - u_{\text{ref}} \leq \tilde{u}_k \leq u_{\max} - u_{\text{ref}} \quad (20.133)$$

Step 4: Solve QP and convert back.

After solving for optimal $\tilde{u}_k^*$:

$$u_k^* = \tilde{u}_k^* + u_{\text{ref}} \quad (20.134)$$

This procedure ensures the controller drives $\mathbf{x} \rightarrow \mathbf{x}_{\text{ref}}$ with $u \rightarrow u_{\text{ref}}$.

■

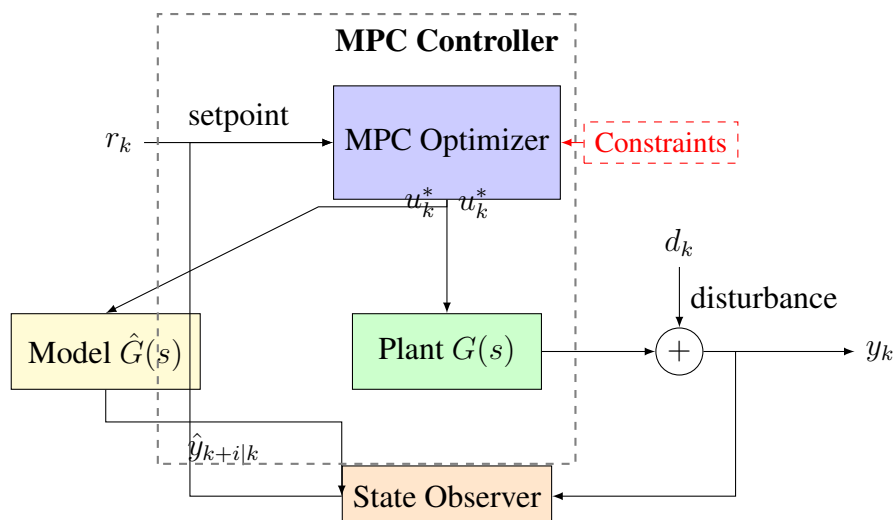


Figure 20.10: MPC control system architecture. The optimizer receives the setpoint, current state estimate, and constraints, then solves the QP to produce the optimal control sequence. Only the first control action is applied; the process repeats at the next sample.

20.13 MPC Block Diagram and Architecture

20.14 Chapter Summary

This chapter developed Model Predictive Control from industrial origins to mathematical rigor.

Historical Context. MPC emerged from 1970s petrochemical control challenges. Shell's DMC and Richalet's MPHPC provided practical algorithms before complete theoretical understanding existed. The key insight was using process models for prediction and optimization for constraint handling.

Modern Applications. MPC now spans chemical processes, autonomous vehicles, building systems, and robotics—any domain requiring constraint-aware trajectory optimization.

Mathematical Foundation. MPC solves finite-horizon optimal control problems at each sampling instant. The approach combines a prediction model ($\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}u_k$) with a cost function applying quadratic penalties on states and inputs. Constraints on inputs, states, and terminal conditions are handled explicitly. The receding horizon strategy solves the optimization, applies only the first input, and repeats at each sampling instant.

QP Formulation. Eliminating state predictions converts MPC to standard quadratic programming:

$$\min_{\mathbf{U}} \frac{1}{2} \mathbf{U}^\top \mathbf{H} \mathbf{U} + \mathbf{f}^\top \mathbf{U} \quad \text{s.t.} \quad \mathbf{G} \mathbf{U} \leq \mathbf{w}$$

Stability. Terminal cost or constraints ensure closed-loop stability. With appropriate terminal cost, unconstrained MPC equals LQR.

Practical Implementation. The tank-level experiment demonstrated MPC's graceful constraint handling compared to PID's windup problems.

Key Takeaway

Model Predictive Control transforms control from reaction to anticipation. By predicting future behavior and optimizing over horizons while respecting constraints, MPC achieves performance impossible with classical feedback alone. The computational cost—solving an optimization at each step—is the price of this capability, now affordable for most applications thanks to modern processors.

20.15 Exercises

20.15.1 Conceptual Exercises

Exercise 20.11. Explain why MPC re-solves the optimization problem at every time step rather than executing the entire precomputed control sequence.

Exercise 20.12. A colleague claims “MPC is just optimal control, nothing new.” Explain what distinguishes MPC from classical optimal control theory.

Exercise 20.13. Why does MPC require positive definite $\mathbf{R}$ but only positive semi-definite $\mathbf{Q}$?

Exercise 20.14. What happens to the MPC solution if the prediction model is perfect versus imperfect?

20.15.2 Computational Exercises

Exercise 20.15. For the system $\mathbf{x}_{k+1} = 0.9\mathbf{x}_k + 0.5u_k$ (scalar), with $Q = 1$, $R = 0.1$, $N = 3$:

1. Compute the prediction matrices $\mathcal{A}$ and $\mathcal{B}$
2. Form the Hessian $\mathbf{H}$
3. Find the unconstrained optimal control sequence for $\mathbf{x}_0 = 1$

Exercise 20.16. Using the system from the previous exercise, add input constraint $|u_k| \leq 0.5$. Show how the constraint affects the solution.

Exercise 20.17. For a double integrator with $T_s = 0.1$ s, design MPC parameters (N , $\mathbf{Q}$, $\mathbf{R}$) to achieve settling time approximately 1 second with input constraint $|u| \leq 10$.

Exercise 20.18. Prove that the Hessian matrix $\mathbf{H} = \mathcal{B}^T \bar{\mathbf{Q}} \mathcal{B} + \bar{\mathbf{R}}$ is positive definite when $\mathbf{R} \succ 0$.

20.15.3 Analysis Exercises

Exercise 20.19. Consider MPC with horizon $N = 1$ (one step ahead). Show that this reduces to a simple state feedback law. What is the feedback gain?

Exercise 20.20. For a marginally stable system ($\mathbf{A}$ has eigenvalues on the unit circle), explain why terminal constraints are particularly important for MPC stability.

Exercise 20.21. Derive the relationship between MPC and LQR when no constraints are active. Under what conditions are they identical?

20.15.4 Implementation Exercises

Exercise 20.22. Implement MPC in MATLAB/Python for a DC motor position control system. Compare performance with PID for:

1. Nominal step response
2. Step response with actuator saturation
3. Disturbance rejection

Exercise 20.23. Extend the tank-level MPC to include a constraint on maximum rate of change of pump speed (slew rate limiting).

Exercise 20.24. Implement an MPC controller for a simulated quadrotor hover task with thrust limits.

Further Reading

For an excellent introduction with practical emphasis, see J. M. Maciejowski's *Predictive Control with Constraints* (Prentice Hall, 2002). A comprehensive theoretical treatment is provided by J. B. Rawlings and D. Q. Mayne in *Model Predictive Control: Theory and Design* (Nob Hill Publishing, 2009). The classic survey paper by C. E. Garcia, D. M. Prett, and M. Morari, "Model Predictive Control: Theory and Practice—A Survey," published in *Automatica*, vol. 25, no. 3, pp. 335–348 (1989), remains an essential reference. For an overview of industrial applications, consult S. J. Qin and T. A. Badgwell's "A Survey of Industrial Model Predictive Control Technology" in *Control Engineering Practice*, vol. 11, no. 7, pp. 733–764 (2003). A modern treatment including hybrid systems can be found in F. Borrelli, A. Bemporad, and M. Morari's *Predictive Control for Linear and Hybrid Systems* (Cambridge University Press, 2017).

Chapter 21

System Identification

21.1 Historical Motivation: Learning Models from Data

The challenge of building accurate mathematical models has confronted engineers and scientists for centuries. While physics provides elegant equations governing idealized systems, real-world systems exhibit complexities—nonlinearities, distributed parameters, unknown friction, parasitic dynamics—that make first-principles modeling extraordinarily difficult. System identification offers an alternative: rather than deriving models from physical laws, we learn them directly from observed input-output behavior.

21.1.1 Gauss and the Birth of Least Squares (1809)

Carl Friedrich Gauss and the Method of Least Squares

Carl Friedrich Gauss (1777–1855), often called the “Prince of Mathematicians,” developed the method of least squares as a young man of 24. When the asteroid Ceres was discovered in 1801, Gauss used his new technique to compute its orbit from just 41 days of observations—a feat that astonished the astronomical community when Ceres was successfully relocated in December 1801, exactly where Gauss predicted. Though Adrien-Marie Legendre published the method first in 1805, Gauss had used it privately since 1795. Gauss published his full theory in 1809 in *Theoria Motus Corporum Coelestium*, where he also introduced the normal distribution (“Gaussian” distribution) as the natural error model justifying least squares. His work established the foundation for all modern statistical estimation, from signal processing to machine learning. Gauss’s insight that squared errors lead to mathematically tractable and statistically optimal estimates remains one of the most influential ideas in applied mathematics.

The mathematical foundations of system identification trace to Carl Friedrich Gauss and his work on celestial mechanics. In 1801, the Italian astronomer Giuseppe Piazzi discovered Ceres, the first asteroid, observing it for only 41 days before it disappeared behind the Sun. The astronomical community faced a critical challenge: with such limited observations, could Ceres’s orbit be predicted well enough to relocate it months later when it emerged from the Sun’s glare?

The Orbit Prediction Problem

The orbit of a celestial body is described by six parameters: semi-major axis, eccentricity, inclination, longitude of ascending node, argument of perihelion, and true anomaly. However, observational measurements contain errors from atmospheric distortion, instrument imprecision, and human factors. Given n noisy observations where $n > 6$, how should one estimate the six orbital parameters?

The challenge is *overdetermined*: more equations than unknowns. Different subsets of observations yield different parameter estimates. Which estimate is “best”?

Gauss’s Method of Least Squares

Gauss proposed a principle that would become the cornerstone of statistical estimation: choose the parameters that minimize the sum of squared errors between predicted and observed positions. If y_i denotes the i -th observed position and $\hat{y}_i(\boldsymbol{\theta})$ the position predicted by model parameters $\boldsymbol{\theta}$, the optimal estimate is:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \sum_{i=1}^n (y_i - \hat{y}_i(\boldsymbol{\theta}))^2 \quad (21.1)$$

Using this method, the 24-year-old Gauss computed Ceres’s orbit with such accuracy that astronomer Franz Xaver von Zach relocated the asteroid on December 31, 1801—within half a degree of Gauss’s prediction. This triumph established least squares as the fundamental tool for fitting models to data.

Why Squared Errors?

Gauss justified the squared error criterion through his theory of errors. Under the assumption that measurement errors are normally distributed (the “Gaussian” distribution he characterized), the least squares estimate is also the *maximum likelihood* estimate—the parameter values most likely to have produced the observed data. This provided a principled statistical foundation for what might otherwise seem an arbitrary choice.

The squared error criterion also offers practical advantages. The squared function is smooth everywhere, enabling gradient-based optimization through differentiability. For linear-in-parameters models, the objective function is convex, guaranteeing a unique global minimum. Linear least squares admits a closed-form solution via the normal equations, providing analytical tractability. Furthermore, under Gaussian noise assumptions, least squares achieves statistical optimality as the Best Linear Unbiased Estimator (BLUE).

21.1.2 From Astronomy to Engineering

Throughout the 19th and early 20th centuries, least squares remained primarily a tool of astronomers and surveyors. Engineers building control systems relied on first-principles models derived from Newton’s laws, thermodynamics, and electrical circuit theory. However, several developments gradually pushed engineers toward data-driven approaches.

The Complexity Barrier

As engineered systems grew more sophisticated, first-principles modeling became increasingly burdensome. Aircraft flutter analysis required predicting the complex interaction between aerodynamic forces and structural vibration, demanding models of extraordinary complexity; finite element methods could approximate structural dynamics, but coupling with unsteady aerodynamics introduced uncertainties that pure analysis could not resolve. In process control, chemical plants involve thousands of interacting variables—temperatures, pressures, flow rates, concentrations—with nonlinear kinetics and transport phenomena, making complete mechanistic models impractical for online control. Electromechanical systems presented their own challenges, as friction, backlash, magnetic saturation, and thermal effects created discrepancies between idealized models and actual behavior.

The Data Revolution

Simultaneously, the ability to collect experimental data improved dramatically. Electronic sensors—including strain gauges, thermocouples, and later solid-state sensors—enabled precise measurement of physical quantities. Data acquisition systems with analog-to-digital converters and digital storage made it feasible to record thousands of data points. Digital computers provided the computational power to process large datasets and perform the matrix operations required for least squares.

The economics shifted: collecting data became cheaper than developing first-principles models.

21.1.3 Lennart Ljung and the Modern Framework (1970s–1980s)

The modern theory of system identification crystallized through the work of Lennart Ljung at Linköping University in Sweden. His 1987 textbook *System Identification: Theory for the User* synthesized decades of research into a coherent framework that remains the foundation of the field.

Ljung's Contributions

Ljung's framework addressed several critical questions. First, model structure selection: what class of models should we consider, given that too simple a structure cannot capture system dynamics while too complex a structure leads to overfitting? Second, input design: what input signals maximize the information content of experimental data? Third, estimation algorithms: how do we efficiently compute optimal parameter estimates, especially for nonlinear model structures? Fourth, model validation: how do we assess whether an identified model is adequate for its intended purpose? Finally, theoretical foundations: under what conditions do parameter estimates converge to true values as data increases?

The Prediction Error Framework

Central to Ljung's approach is the *prediction error method*. Given a model parametrized by θ , define the one-step-ahead prediction:

$$\hat{y}(t|\theta) = \text{best prediction of } y(t) \text{ given past data and model } \theta \quad (21.2)$$

The prediction error is:

$$\varepsilon(t, \boldsymbol{\theta}) = y(t) - \hat{y}(t|\boldsymbol{\theta}) \quad (21.3)$$

The parameter estimate minimizes a criterion function of these errors:

$$\hat{\boldsymbol{\theta}}_N = \arg \min_{\boldsymbol{\theta}} V_N(\boldsymbol{\theta}) = \arg \min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{t=1}^N \ell(\varepsilon(t, \boldsymbol{\theta})) \quad (21.4)$$

where $\ell(\cdot)$ is a loss function (typically $\ell(\varepsilon) = \varepsilon^2$ for quadratic loss).

This framework unifies diverse identification methods—least squares, maximum likelihood, instrumental variables—as special cases with different model structures and loss functions.

21.1.4 “All Models Are Wrong, Some Are Useful”

The statistician George E.P. Box articulated a crucial philosophical principle that underlies all system identification work:

“Essentially, all models are wrong, but some are useful.” — George E.P. Box, 1976

This aphorism captures several important truths. No model is perfect: every mathematical model is a simplification of reality, as real systems have infinite degrees of freedom while models have finite parameters. Usefulness is context-dependent, meaning a model adequate for one purpose may be inadequate for another; a first-order approximation might suffice for slow control loops but fail for high-bandwidth applications. The goal is not truth, but utility: we seek models that predict well within their intended domain of application, not models that capture every physical detail. Finally, model validation is essential; since all models are wrong, we must rigorously test whether a particular model is “wrong enough to matter” for our application.

21.1.5 Aerospace Applications: Flutter Testing

Aircraft flutter—the potentially catastrophic interaction between aerodynamic forces and structural vibration—provides a compelling application of system identification. Flight testing for flutter requires extracting dynamic characteristics from in-flight data.

The Flutter Problem

As aircraft speed increases, aerodynamic forces couple increasingly strongly with structural modes. At a critical speed, this coupling can produce self-excited oscillations of rapidly growing amplitude—flutter. Since flutter can destroy an aircraft within seconds, flight test programs must carefully approach the flutter boundary while ensuring adequate safety margins.

Ground Vibration Testing

Before flight, aircraft undergo ground vibration testing (GVT) to identify structural modes. Shakers excite the structure, and accelerometers measure the response. From this input-output data, modal frequencies, damping ratios, and mode shapes are extracted using system identification techniques.

Flight Flutter Testing

In flight, atmospheric turbulence or controlled excitation (from stick inputs, oscillating control surfaces, or dedicated excitation systems) provides the input. The challenge is more severe than ground testing due to several factors: atmospheric turbulence excites all modes simultaneously as multiple inputs; airspeed, altitude, and fuel state change during maneuvers, creating varying conditions; safety concerns limit input amplitude near the flutter boundary, constraining excitation levels; and damping trends must be monitored during the test point, imposing real-time requirements.

Modern flutter testing relies on recursive identification algorithms that update model estimates as data arrives, enabling real-time tracking of modal damping as conditions change.

21.2 Modern Applications

System identification has become indispensable across engineering disciplines, enabling applications ranging from adaptive control to digital twin creation.

21.2.1 Adaptive Control: Identify Then Control

Adaptive control systems adjust their parameters in response to changing plant dynamics or operating conditions. A common architecture is *indirect adaptive control*, which proceeds as follows: first, an online identification algorithm estimates plant parameters from recent input-output data; then, the estimated parameters are used to update controller gains; finally, this process repeats continuously during operation.

This “identify then control” paradigm enables systems to maintain performance despite plant variations—wear, fouling, load changes—that would degrade fixed-gain controllers. Applications span multiple domains: autopilots must handle aircraft dynamics that vary with altitude, speed, and configuration; engine control systems must adapt as combustion characteristics change with temperature and fuel quality; and robot manipulators must accommodate payload mass that varies between tasks.

21.2.2 Digital Twins from Operational Data

A *digital twin* is a virtual replica of a physical asset that mirrors its behavior in real-time. System identification enables the creation of digital twins from operational data for several purposes. Predictive maintenance benefits from identifying changes in system dynamics that can reveal incipient faults—bearing wear manifests as changing resonance frequencies, while valve degradation appears as changing flow coefficients. Process optimization leverages high-fidelity models to enable simulation-based optimization without risky experimentation on the physical plant. Operator training uses digital twins to provide realistic simulators for training scenarios.

21.2.3 Machine Learning for Dynamics

The boundary between system identification and machine learning has become increasingly porous. Neural network dynamics models, including recurrent networks and transformers, can learn com-

plex nonlinear dynamics from data. Gaussian processes provide probabilistic models with uncertainty quantification. Physics-informed learning combines data-driven fitting with physical constraints such as conservation laws and symmetries. Reinforcement learning uses learned dynamics models for model-predictive control.

These methods extend the reach of system identification to systems too complex for traditional linear or low-order nonlinear models.

21.2.4 Biomedical System Modeling

Biological systems present unique identification challenges—they cannot be opened for inspection, they vary between individuals, and they adapt over time. System identification enables advances across biomedical applications. Pharmacokinetic modeling estimates drug absorption, distribution, metabolism, and excretion rates from blood concentration measurements. Glucose-insulin dynamics identification produces patient-specific models for artificial pancreas control. Cardiovascular modeling characterizes heart pump function and vascular resistance from pressure and flow measurements. Neural system identification infers neural connectivity and dynamics from electrode recordings or imaging data.

21.3 Mathematical Foundation

We now develop the mathematical framework for system identification, proceeding from problem formulation through model structures, estimation algorithms, and validation techniques.

21.3.1 Problem Formulation

The fundamental system identification problem can be stated as:

System Identification Problem

Given: A sequence of input measurements $\{u(1), u(2), \dots, u(N)\}$, a sequence of output measurements $\{y(1), y(2), \dots, y(N)\}$, and a model structure $\mathcal{M}$ with parameters θ .

Find: The parameter values $\hat{\theta}$ that best explain the observed data according to some criterion.

The key choices in this formulation involve three fundamental decisions. The model structure determines what class of models we consider: linear versus nonlinear, continuous versus discrete, parametric versus nonparametric. The optimality criterion defines what “best” means—whether least squares, maximum likelihood, or prediction accuracy. Experimental design addresses what inputs should be applied to maximize information content.

21.3.2 Input Signal Design

The input signal profoundly affects identification quality. A constant input provides no information about dynamics. An ideal input excites all relevant modes while respecting practical constraints (amplitude limits, safety, operating range).

Persistence of Excitation

For parameter convergence, the input must be *persistently exciting* of sufficient order. Informally, an input is persistently exciting of order n if it contains at least n distinct frequency components. A formal definition involves the positive definiteness of certain correlation matrices:

$$\frac{1}{N} \sum_{t=1}^N \phi(t) \phi^T(t) > \alpha \mathbf{I} \quad (21.5)$$

for some $\alpha > 0$, where $\phi(t)$ is the regression vector.

Pseudorandom Binary Sequence (PRBS)

A PRBS is a deterministic binary signal that approximates white noise over a finite period. It switches between two levels ($+A$ and $-A$) according to a linear feedback shift register:

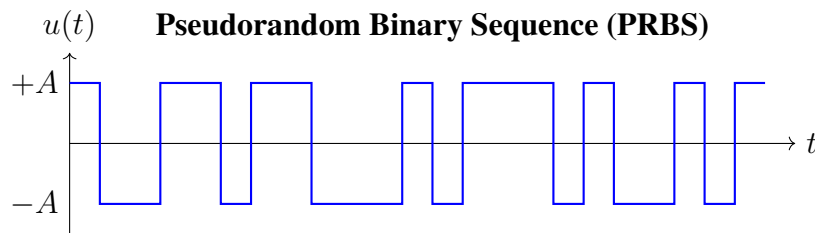


Figure 21.1: A PRBS input signal switches between two levels in a pattern that approximates white noise.

The PRBS possesses several desirable properties for system identification. Its autocorrelation approximates a delta function, providing broadband excitation with uniform spectral content. The bounded amplitude respects actuator limits. Being deterministic, the signal enables repeatable experiments. The adjustable bandwidth is controlled through the clock frequency, which sets the spectral rolloff.

Frequency Sweep (Chirp)

A chirp signal is a sinusoid with time-varying frequency:

$$u(t) = A \sin \left(2\pi f_0 t + \pi \frac{f_1 - f_0}{T} t^2 \right) \quad (21.6)$$

where frequency sweeps from f_0 to f_1 over duration T .

The chirp is particularly useful for frequency domain identification, exciting each frequency in sequence and enabling spectral analysis at each frequency.

Step and Pulse Sequences

Step inputs are simple to implement and intuitive to analyze but concentrate energy at low frequencies. Multiple steps of varying amplitude improve high-frequency content. Pulse sequences (short-duration steps) provide broader excitation.

Multisine Signals

A multisine consists of superimposed sinusoids at selected frequencies:

$$u(t) = \sum_{k=1}^K A_k \sin(2\pi f_k t + \phi_k) \quad (21.7)$$

By choosing frequencies f_k and phases ϕ_k , one can target specific frequency bands while controlling the signal's crest factor (ratio of peak to RMS amplitude).

21.3.3 Least Squares Estimation

Least squares provides the foundation for linear system identification. We develop the method in detail, as it underlies many more sophisticated techniques.

The Regression Form

Consider a system whose output depends linearly on known functions of past inputs and outputs:

$$y(t) = \phi_1(t)\theta_1 + \phi_2(t)\theta_2 + \cdots + \phi_n(t)\theta_n + e(t) \quad (21.8)$$

where $\phi_i(t)$ are known regressors, θ_i are unknown parameters, and $e(t)$ is the equation error.

In vector form:

$$y(t) = \boldsymbol{\phi}^T(t)\boldsymbol{\theta} + e(t) \quad (21.9)$$

where $\boldsymbol{\phi}(t) = [\phi_1(t), \dots, \phi_n(t)]^T$ and $\boldsymbol{\theta} = [\theta_1, \dots, \theta_n]^T$.

Collecting N measurements into matrices:

$$\underbrace{\begin{bmatrix} y(1) \\ y(2) \\ \vdots \\ y(N) \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} \boldsymbol{\phi}^T(1) \\ \boldsymbol{\phi}^T(2) \\ \vdots \\ \boldsymbol{\phi}^T(N) \end{bmatrix}}_{\mathbf{\Phi}} \underbrace{\begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix}}_{\boldsymbol{\theta}} + \underbrace{\begin{bmatrix} e(1) \\ e(2) \\ \vdots \\ e(N) \end{bmatrix}}_{\mathbf{e}} \quad (21.10)$$

Compactly:

$$\mathbf{y} = \mathbf{\Phi}\boldsymbol{\theta} + \mathbf{e} \quad (21.11)$$

The Normal Equations

The least squares estimate minimizes the sum of squared errors:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \arg \min_{\boldsymbol{\theta}} \sum_{t=1}^N e^2(t) = \arg \min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{\Phi}\boldsymbol{\theta}\|^2 \quad (21.12)$$

Expanding the squared norm:

$$J(\boldsymbol{\theta}) = (\mathbf{y} - \mathbf{\Phi}\boldsymbol{\theta})^T(\mathbf{y} - \mathbf{\Phi}\boldsymbol{\theta}) = \mathbf{y}^T\mathbf{y} - 2\boldsymbol{\theta}^T\mathbf{\Phi}^T\mathbf{y} + \boldsymbol{\theta}^T\mathbf{\Phi}^T\mathbf{\Phi}\boldsymbol{\theta} \quad (21.13)$$

Taking the gradient and setting to zero:

$$\frac{\partial J}{\partial \theta} = -2\Phi^T y + 2\Phi^T \Phi \theta = 0 \quad (21.14)$$

Solving yields the **normal equations**:

Least Squares Solution

$$\hat{\theta} = (\Phi^T \Phi)^{-1} \Phi^T y \quad (21.15)$$

The matrix $(\Phi^T \Phi)^{-1} \Phi^T$ is the *Moore-Penrose pseudoinverse* of Φ .

Existence and Uniqueness

The solution (??) exists and is unique if and only if $\Phi^T \Phi$ is invertible. This requires two conditions: first, $N \geq n$, ensuring at least as many measurements as parameters; second, $\text{rank}(\Phi) = n$, ensuring the regressors are linearly independent.

The second condition is equivalent to persistent excitation. If the input is not persistently exciting of order n , some parameters are not uniquely identifiable from the data.

Statistical Properties

Under the assumption that $e(t)$ is zero-mean white noise with variance σ^2 , the least squares estimate has desirable statistical properties. The estimate is unbiased, meaning $E[\hat{\theta}] = \theta_0$ (the true parameters). It is consistent, with $\hat{\theta} \rightarrow \theta_0$ as $N \rightarrow \infty$. Furthermore, the estimate achieves minimum variance: among all linear unbiased estimators, least squares has the smallest variance, as established by the Gauss-Markov theorem.

The covariance of the estimate is:

$$\text{Cov}(\hat{\theta}) = \sigma^2 (\Phi^T \Phi)^{-1} \quad (21.16)$$

This fundamental result relates estimation uncertainty to noise variance and information content (encoded in $\Phi^T \Phi$).

21.3.4 ARX Model Structure

The AutoRegressive with eXogenous input (ARX) model is the simplest and most widely used structure for discrete-time identification.

Model Definition

The ARX model relates output $y[k]$ to past outputs, past inputs, and a noise term:

$$y[k] + a_1 y[k-1] + \cdots + a_{n_a} y[k-n_a] = b_1 u[k-1] + \cdots + b_{n_b} u[k-n_b] + e[k] \quad (21.17)$$

Using the backward shift operator q^{-1} (where $q^{-1}y[k] = y[k-1]$):

$$A(q^{-1})y[k] = B(q^{-1})u[k] + e[k] \quad (21.18)$$

where:

$$A(q^{-1}) = 1 + a_1 q^{-1} + a_2 q^{-2} + \dots + a_{n_a} q^{-n_a} \quad (21.19)$$

$$B(q^{-1}) = b_1 q^{-1} + b_2 q^{-2} + \dots + b_{n_b} q^{-n_b} \quad (21.20)$$

Transfer Function Form

The deterministic part of the ARX model corresponds to the transfer function:

$$G(q^{-1}) = \frac{B(q^{-1})}{A(q^{-1})} = \frac{b_1 q^{-1} + b_2 q^{-2} + \dots + b_{n_b} q^{-n_b}}{1 + a_1 q^{-1} + \dots + a_{n_a} q^{-n_a}} \quad (21.21)$$

Converting to z -transform notation (where $z = q$):

$$G(z) = \frac{b_1 z^{-1} + \dots + b_{n_b} z^{-n_b}}{1 + a_1 z^{-1} + \dots + a_{n_a} z^{-n_a}} = \frac{b_1 z^{n_b-1} + \dots + b_{n_b}}{z^{n_a} + a_1 z^{n_a-1} + \dots + a_{n_a}} \cdot z^{n_a-n_b} \quad (21.22)$$

Regression Form for ARX

The ARX model (??) is linear in the parameters $\{a_1, \dots, a_{n_a}, b_1, \dots, b_{n_b}\}$. Rearranging:

$$y[k] = -a_1 y[k-1] - \dots - a_{n_a} y[k-n_a] + b_1 u[k-1] + \dots + b_{n_b} u[k-n_b] + e[k] \quad (21.23)$$

Defining:

$$\phi[k] = [-y[k-1], \dots, -y[k-n_a], u[k-1], \dots, u[k-n_b]]^T \quad (21.24)$$

$$\theta = [a_1, \dots, a_{n_a}, b_1, \dots, b_{n_b}]^T \quad (21.25)$$

We obtain the standard regression form:

$$y[k] = \phi^T[k] \theta + e[k] \quad (21.26)$$

The least squares solution (??) directly provides parameter estimates.

Advantages and Limitations

The ARX model structure offers several advantages: it reduces to simple linear regression with a closed-form solution, enables efficient computation without iterative optimization, and possesses well-understood statistical properties. However, the structure also has limitations. It assumes a specific noise structure where equation error enters directly. Estimates become biased if the noise is colored (correlated). High model orders may be required to capture complex dynamics adequately.

21.3.5 Frequency Domain Identification

An alternative to time-domain parametric identification is to characterize the system's frequency response directly.

The Frequency Response Function

For a stable LTI system, the frequency response function (FRF) describes the gain and phase at each frequency:

$$G(j\omega) = |G(j\omega)|e^{j\angle G(j\omega)} \quad (21.27)$$

The FRF relates sinusoidal inputs to outputs: if $u(t) = A \sin(\omega t)$, then at steady state:

$$y(t) = A|G(j\omega)| \sin(\omega t + \angle G(j\omega)) \quad (21.28)$$

Sine-Sweep Identification

A direct method to measure the FRF proceeds as follows. Apply a sinusoidal input at frequency ω_k and wait for transients to decay. Measure the output amplitude Y_k and phase ϕ_k relative to the input. Compute $|G(j\omega_k)| = Y_k/U_k$ and $\angle G(j\omega_k) = \phi_k$. Repeat this process for frequencies spanning the range of interest.

The result is an *empirical Bode plot*—gain and phase data at discrete frequencies.

Spectral Analysis

For broadband inputs (PRBS, random noise), spectral analysis provides FRF estimates:

$$\hat{G}(j\omega) = \frac{S_{yu}(\omega)}{S_{uu}(\omega)} \quad (21.29)$$

where $S_{yu}(\omega)$ is the cross-spectral density between output and input, and $S_{uu}(\omega)$ is the input power spectral density.

This approach is computationally efficient using the Fast Fourier Transform (FFT):

$$\hat{G}(j\omega_k) = \frac{Y(\omega_k)}{U(\omega_k)} \quad (21.30)$$

where $Y(\omega_k)$ and $U(\omega_k)$ are the FFTs of output and input.

From Frequency Data to Parametric Models

Once FRF data is available, a parametric model can be fitted:

$$\hat{\theta} = \arg \min_{\theta} \sum_{k=1}^K \left| \hat{G}(j\omega_k) - G(j\omega_k; \theta) \right|^2 \quad (21.31)$$

This frequency-domain fitting can be solved using least squares if the model is linear in parameters, or iterative methods for general rational transfer functions.

21.3.6 Model Validation

An identified model must be validated before use. Validation assesses whether the model adequately represents the system for its intended purpose.

Residual Analysis

The residuals $\varepsilon(t) = y(t) - \hat{y}(t)$ contain information about model quality. Whiteness is the primary criterion: if the model captures all system dynamics, residuals should be white noise (uncorrelated). Residuals should also demonstrate independence from the input, being uncorrelated with past inputs. Under Gaussian noise assumptions, residuals should exhibit normality, following a normal distribution.

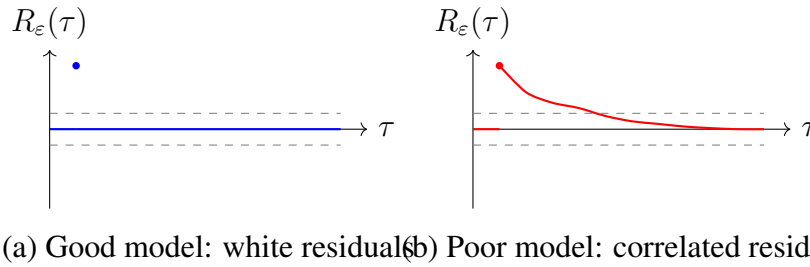


Figure 21.2: Residual autocorrelation analysis. (a) White residuals indicate the model captures system dynamics. (b) Correlated residuals suggest unmodeled dynamics.

The autocorrelation function of residuals is:

$$\hat{R}_\varepsilon(\tau) = \frac{1}{N} \sum_{t=1}^{N-\tau} \varepsilon(t) \varepsilon(t + \tau) \quad (21.32)$$

For a good model, $\hat{R}_\varepsilon(\tau) \approx 0$ for $\tau \neq 0$ (within confidence bounds).

Cross-Validation

The gold standard for validation is testing on data not used for estimation. The procedure begins by dividing available data into *estimation* and *validation* sets. The model is identified using only estimation data. The identified model is then simulated with validation input, and the simulated output is compared to the measured validation output.

Common metrics for cross-validation include the fit percentage:

$$\text{FIT} = 100 \times \left(1 - \frac{\|\mathbf{y}_{\text{val}} - \hat{\mathbf{y}}\|}{\|\mathbf{y}_{\text{val}} - \bar{y}\|} \right) \% \quad (21.33)$$

and the normalized root mean square error (NRMSE): $\text{NRMSE} = \sqrt{\sum (y - \hat{y})^2 / \sum y^2}$.

Model Comparison

When multiple model structures are candidates, information criteria help select the best. The Akaike Information Criterion (AIC) is defined as:

$$\text{AIC} = N \ln(\hat{\sigma}^2) + 2n \quad (21.34)$$

The Bayesian Information Criterion (BIC) provides an alternative:

$$\text{BIC} = N \ln(\hat{\sigma}^2) + n \ln(N) \quad (21.35)$$

In both criteria, $\hat{\sigma}^2$ is the estimated noise variance and n is the number of parameters. Lower values indicate better models, balancing fit quality against complexity.

21.3.7 Bias-Variance Tradeoff

Model complexity selection embodies a fundamental tradeoff. Underfitting occurs when a model is too simple: the model cannot capture true system dynamics, leading to *bias* (systematic error). Overfitting occurs when a model is too complex: the model fits noise in the training data, leading to high *variance* (sensitivity to data randomness) and poor generalization.

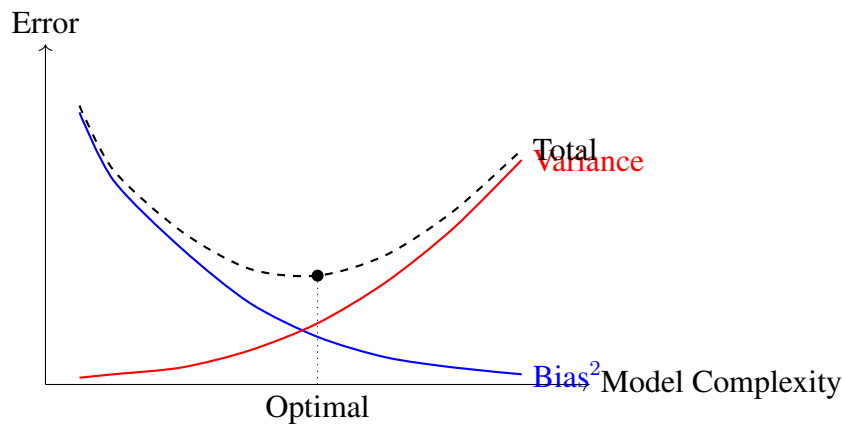


Figure 21.3: The bias-variance tradeoff. Simple models have high bias but low variance; complex models have low bias but high variance. The optimal complexity minimizes total prediction error.

The expected prediction error can be decomposed:

$$\text{Expected Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Noise} \quad (21.36)$$

Cross-validation or information criteria help navigate this tradeoff, selecting models complex enough to capture dynamics but simple enough to generalize.

21.4 Practical Experiment: Identifying DC Motor Dynamics

This laboratory demonstrates system identification principles by treating a DC motor as a “black box” whose parameters are unknown. Students will apply input signals, record responses, fit models using least squares, and validate predictions.

21.4.1 Learning Objectives

Upon completion, students will be able to design input signals for system identification, collect and preprocess input-output data, formulate and solve the least squares estimation problem, validate

identified models using independent test data, and compare data-driven models to physics-based models.

21.4.2 Equipment and Setup

Table 21.1: Bill of Materials for System Identification Laboratory

Component	Specification	Qty
Arduino Uno/Nano	ATmega328P microcontroller	1
DC Motor with Encoder	12V, 200+ PPR quadrature encoder	1
Motor Driver	L298N or TB6612FNG H-bridge	1
Power Supply	12V, 2A DC adapter	1
Computer	MATLAB, Python, or similar	1

The motor is treated as a “black box”—we pretend not to know its electrical and mechanical parameters, though for validation we will compare with a physics-based model.

21.4.3 System Model

A DC motor velocity response to armature voltage can be modeled as a first-order system:

$$G(s) = \frac{\omega(s)}{V(s)} = \frac{K}{\tau s + 1} \quad (21.37)$$

where K is the DC gain (rad/s per volt) and τ is the time constant.

In discrete time with sampling period T_s :

$$\omega[k] = a_1 \omega[k-1] + b_1 v[k-1] + e[k] \quad (21.38)$$

The continuous-time parameters relate to discrete parameters via:

$$a_1 = e^{-T_s/\tau} \quad (21.39)$$

$$b_1 = K(1 - e^{-T_s/\tau}) = K(1 - a_1) \quad (21.40)$$

Inverting:

$$\tau = \frac{-T_s}{\ln(a_1)} \quad (21.41)$$

$$K = \frac{b_1}{1 - a_1} \quad (21.42)$$

21.4.4 Experimental Procedure

Part 1: Data Collection

Procedure: Upload the code and connect the motor. Open Serial Monitor and copy data to a file, collecting at least 3000 samples. Split the data, using the first 2000 samples for estimation and the remainder for validation.

Algorithm 46 Data Collection for System Identification

```

1: Initialize: Configure motor PWM pins and encoder interrupt pins
2: Initialize: Set sampling period  $T_s \leftarrow 0.02$  s (50 Hz)
3: Initialize: Initialize PRBS shift register  $\leftarrow 1$ 
4: Initialize: encoderCount  $\leftarrow 0$ , lastCount  $\leftarrow 0$ 
5: Output CSV header: “time_ms, pwm, velocity”
6: Wait 1 second for system stabilization
7: while data collection active do
8:   currentTime  $\leftarrow$  system time in milliseconds
9:   if currentTime – lastSampleTime  $\geq T_s \times 1000$  then
10:    lastSampleTime  $\leftarrow$  currentTime
11:                                      $\triangleright$  Generate PRBS input signal
12:    newBit  $\leftarrow$  (shiftReg[6]  $\oplus$  shiftReg[5])  $\wedge$  1
13:    shiftReg  $\leftarrow$  (shiftReg  $\ll$  1)  $\vee$  newBit
14:    if shiftReg  $\wedge$  1 = 1 then
15:      pwmValue  $\leftarrow$  150                                      $\triangleright$  High level
16:    else
17:      pwmValue  $\leftarrow$  80                                      $\triangleright$  Low level
18:    end if
19:    Apply PWM signal to motor
20:                                      $\triangleright$  Measure velocity from encoder
21:    currentCount  $\leftarrow$  encoderCount
22:    velocity  $\leftarrow$  (currentCount – lastCount)/ $T_s$ 
23:    lastCount  $\leftarrow$  currentCount
24:    Log: currentTime, pwmValue, velocity
25:  end if
26: end while

```

Algorithm 47 Encoder Update (Interrupt Service Routine)

```

1: Read encoder channel A and channel B
2: if channel A = channel B then
3:   encoderCount  $\leftarrow$  encoderCount + 1
4: else
5:   encoderCount  $\leftarrow$  encoderCount – 1
6: end if

```

Part 2: Parameter Estimation in MATLAB/Python

Algorithm 48 Least Squares Parameter Estimation

```

1: Input: Data file containing time, input  $u$ , and output  $y$  measurements
2: Output: Estimated parameters  $\hat{a}_1, \hat{b}_1, \hat{\tau}, \hat{K}$ 
3:
4:                                     ▷ Load and partition data
5: Load data matrix from CSV file
6:  $\mathbf{t} \leftarrow$  time column (convert to seconds)
7:  $\mathbf{u} \leftarrow$  input column (PWM values)
8:  $\mathbf{y} \leftarrow$  output column (velocity)
9:
10:                                     ▷ Split into estimation and validation sets
11:  $N_{\text{est}} \leftarrow 2000$ 
12:  $\mathbf{u}_{\text{est}} \leftarrow \mathbf{u}[1 : N_{\text{est}}], \quad \mathbf{y}_{\text{est}} \leftarrow \mathbf{y}[1 : N_{\text{est}}]$ 
13:  $\mathbf{u}_{\text{val}} \leftarrow \mathbf{u}[N_{\text{est}} + 1 : \text{end}], \quad \mathbf{y}_{\text{val}} \leftarrow \mathbf{y}[N_{\text{est}} + 1 : \text{end}]$ 
14:
15:                                     ▷ Build regression matrix for ARX(1,1):  $y[k] = a_1 y[k-1] + b_1 u[k-1] + e[k]$ 
16:  $\Phi \leftarrow [\mathbf{y}_{\text{est}}[1 : N-1], \mathbf{u}_{\text{est}}[1 : N-1]]$ 
17:  $\mathbf{Y} \leftarrow \mathbf{y}_{\text{est}}[2 : N]$ 
18:
19:                                     ▷ Solve normal equations
20:  $\hat{\theta} \leftarrow (\Phi^T \Phi)^{-1} \Phi^T \mathbf{Y}$ 
21:  $\hat{a}_1 \leftarrow \hat{\theta}[1], \quad \hat{b}_1 \leftarrow \hat{\theta}[2]$ 
22:
23:                                     ▷ Convert to continuous-time parameters
24:  $\hat{\tau} \leftarrow -T_s / \ln(\hat{a}_1)$ 
25:  $\hat{K} \leftarrow \hat{b}_1 / (1 - \hat{a}_1)$ 
26:
27:                                     ▷ Compute parameter uncertainty
28:  $\mathbf{e} \leftarrow \mathbf{Y} - \Phi \hat{\theta}$                                      ▷ Residuals
29:  $\hat{\sigma}^2 \leftarrow \text{var}(\mathbf{e})$ 
30:  $\text{Cov}(\hat{\theta}) \leftarrow \hat{\sigma}^2 (\Phi^T \Phi)^{-1}$ 
31:  $\sigma_{a_1} \leftarrow \sqrt{\text{Cov}(\hat{\theta})_{11}}, \quad \sigma_{b_1} \leftarrow \sqrt{\text{Cov}(\hat{\theta})_{22}}$ 
32:
33: return  $\hat{a}_1, \hat{b}_1, \hat{\tau}, \hat{K}, \sigma_{a_1}, \sigma_{b_1}$ 

```

Algorithm 49 Model Validation

```

1: Input: Validation data  $\mathbf{u}_{\text{val}}, \mathbf{y}_{\text{val}}$ ; estimated parameters  $\hat{a}_1, \hat{b}_1$ 
2: Output: Fit percentage, residual analysis
3:
4:                                     ▷ Simulate identified model on validation data
5: Initialize  $\hat{\mathbf{y}}_{\text{sim}}$  as zero vector of length  $N_{\text{val}}$ 
6:  $\hat{y}_{\text{sim}}[1] \leftarrow y_{\text{val}}[1]$                                      ▷ Use measured initial condition
7: for  $k = 2 \rightarrow N_{\text{val}}$  do
8:      $\hat{y}_{\text{sim}}[k] \leftarrow \hat{a}_1 \cdot \hat{y}_{\text{sim}}[k-1] + \hat{b}_1 \cdot u_{\text{val}}[k-1]$ 
9: end for
10:
11:                                     ▷ Compute fit percentage
12:  $\text{FIT} \leftarrow 100 \times \left(1 - \frac{\|\mathbf{y}_{\text{val}} - \hat{\mathbf{y}}_{\text{sim}}\|}{\|\mathbf{y}_{\text{val}} - \bar{y}_{\text{val}}\|}\right)$ 
13:
14:                                     ▷ Residual analysis
15:  $\boldsymbol{\varepsilon} \leftarrow \mathbf{y}_{\text{val}} - \hat{\mathbf{y}}_{\text{sim}}$ 
16: Plot residuals  $\boldsymbol{\varepsilon}$  versus time
17:
18:                                     ▷ Compute autocorrelation of residuals
19: for  $\tau = -50 \rightarrow 50$  do
20:      $R_{\boldsymbol{\varepsilon}}[\tau] \leftarrow$  normalized cross-correlation at lag  $\tau$ 
21: end for
22: Plot autocorrelation with 95% confidence bounds  $\pm 1.96/\sqrt{N_{\text{val}}}$ 
23:
24:                                     ▷ Assess whiteness: residuals are white if ACF within bounds for  $\tau \neq 0$ 
25: return FIT,  $\boldsymbol{\varepsilon}$ ,  $R_{\boldsymbol{\varepsilon}}$ 

```

Part 3: Model Validation

21.4.5 Comparison with Physics-Based Model

For a DC motor, first-principles analysis yields:

$$\tau_{\text{physics}} = \frac{JR}{K_t K_e + bR} \quad (21.43)$$

$$K_{\text{physics}} = \frac{K_t}{K_t K_e + bR} \quad (21.44)$$

where J is moment of inertia, R is armature resistance, K_t is torque constant, K_e is back-EMF constant, and b is viscous friction.

Students should look up the motor datasheet parameters, calculate τ_{physics} and K_{physics} , compare these with the identified values τ and K , and discuss sources of discrepancy such as unmodeled friction, driver nonlinearity, and other effects.

21.4.6 Data Recording Table

Table 21.2: System Identification Results

Parameter	Identified	Physics-Based	Difference
Time constant τ (s)			
DC gain K			
Validation fit (%)		N/A	

21.5 Worked Examples

First-Order System Least Squares Setup

A first-order system $G(s) = \frac{K}{\tau s + 1}$ is sampled at $T_s = 0.1$ s. Set up the least squares problem to estimate the discrete-time ARX(1,1) parameters.

Solution:

The continuous-time differential equation is:

$$\tau \dot{y} + y = Ku$$

Discretizing using the forward Euler approximation or exact discretization:

$$y[k] = e^{-T_s/\tau} y[k-1] + K(1 - e^{-T_s/\tau}) u[k-1]$$

Defining $a_1 = e^{-T_s/\tau}$ and $b_1 = K(1 - e^{-T_s/\tau})$:

$$y[k] = a_1 y[k-1] + b_1 u[k-1] + e[k]$$

The regression vector is:

$$\phi[k] = \begin{bmatrix} y[k-1] \\ u[k-1] \end{bmatrix}$$

The parameter vector is:

$$\theta = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}$$

For N data points ($k = 2, \dots, N+1$):

$$\Phi = \begin{bmatrix} y[1] & u[1] \\ y[2] & u[2] \\ \vdots & \vdots \\ y[N] & u[N] \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y[2] \\ y[3] \\ \vdots \\ y[N+1] \end{bmatrix}$$

The least squares solution is:

$$\hat{\theta} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}$$

Parameter Estimation from Step Response

The following step response data was collected from a first-order system with a step input of magnitude $u = 5$:

t (s)	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5
y	0	1.57	2.53	3.12	3.49	3.72	3.86	3.95

Estimate the DC gain K and time constant τ .

Solution:

For a first-order step response:

$$y(t) = Ku(1 - e^{-t/\tau}) = 5K(1 - e^{-t/\tau})$$

Method 1: Graphical/Approximate

The steady-state value appears to be approaching $5K \approx 4.0$, giving $K \approx 0.8$.

At $t = \tau$, the response reaches 63.2% of steady-state: $0.632 \times 4.0 = 2.53$. From the data, $y(1.0) = 2.53$, so $\tau \approx 1.0$ s.

Method 2: Least Squares

Define $z = 5K - y$, so $z = 5Ke^{-t/\tau}$, and $\ln(z) = \ln(5K) - t/\tau$.

This is linear in $\ln(5K)$ and $-1/\tau$. We can estimate K from the final value and then use linear regression on $\ln(z)$ vs. t .

Assuming $5K = 4.0$ (estimated final value):

t	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5
$z = 4 - y$	4.0	2.43	1.47	0.88	0.51	0.28	0.14	0.05
$\ln(z)$	1.39	0.89	0.39	-0.13	-0.67	-1.27	-1.97	-3.00

Linear regression of $\ln(z)$ vs. t :

$$\ln(z) = 1.386 - 1.003t$$

Therefore:

$$-\frac{1}{\tau} = -1.003 \Rightarrow \tau = 0.997 \approx 1.0 \text{ s}$$

$$\ln(5K) = 1.386 \Rightarrow 5K = 4.0 \Rightarrow K = 0.8$$

$$K = 0.8 \text{ (units/input), } \tau = 1.0 \text{ s}$$

Example 21.1 (Frequency Response Identification). Sinusoidal tests on a system yield the following frequency response data:

ω (rad/s)	0.1	0.5	1.0	2.0	5.0
$ G(j\omega) $	0.98	0.89	0.71	0.45	0.20
$\angle G(j\omega)$ (deg)	-5.7	-26.6	-45	-63.4	-78.7

Identify a first-order model $G(s) = \frac{K}{\tau s + 1}$.

Solution:

For a first-order system:

$$|G(j\omega)| = \frac{K}{\sqrt{1 + (\omega\tau)^2}}$$

$$\angle G(j\omega) = -\arctan(\omega\tau)$$

Using phase data:

At $\omega = 1.0$ rad/s, $\angle G = -45$, which implies:

$$\arctan(\omega\tau) = 45 \Rightarrow \omega\tau = 1 \Rightarrow \tau = 1.0 \text{ s}$$

Using magnitude data:

At DC ($\omega \rightarrow 0$), $|G| \rightarrow K$. From the data, $|G(0.1)| \approx 0.98 \approx K$.

More precisely, at $\omega = 1.0$ rad/s:

$$|G| = \frac{K}{\sqrt{1 + 1}} = \frac{K}{\sqrt{2}} = 0.71$$

$$K = 0.71 \times \sqrt{2} = 1.0$$

Verification:

At $\omega = 2.0$ rad/s:

$$|G| = \frac{1}{\sqrt{1 + 4}} = \frac{1}{\sqrt{5}} = 0.447 \approx 0.45 \quad \checkmark$$

$$G(s) = \frac{1}{s + 1}$$

Example 21.2 (Confidence Intervals on Parameters). An ARX model was identified from $N = 500$ data points, yielding:

$$\hat{\theta} = \begin{bmatrix} 0.85 \\ 0.12 \end{bmatrix}, \quad \Phi^T \Phi = \begin{bmatrix} 450 & 25 \\ 25 & 100 \end{bmatrix}, \quad \hat{\sigma}^2 = 0.04$$

Compute 95% confidence intervals for the parameters.

Solution:

The parameter covariance matrix is:

$$\text{Cov}(\hat{\theta}) = \hat{\sigma}^2 (\Phi^T \Phi)^{-1}$$

First, compute the inverse:

$$\begin{aligned} (\Phi^T \Phi)^{-1} &= \frac{1}{450 \times 100 - 25^2} \begin{bmatrix} 100 & -25 \\ -25 & 450 \end{bmatrix} = \frac{1}{44375} \begin{bmatrix} 100 & -25 \\ -25 & 450 \end{bmatrix} \\ &= \begin{bmatrix} 0.00225 & -0.000563 \\ -0.000563 & 0.01014 \end{bmatrix} \end{aligned}$$

Then:

$$\text{Cov}(\hat{\theta}) = 0.04 \times \begin{bmatrix} 0.00225 & -0.000563 \\ -0.000563 & 0.01014 \end{bmatrix} = \begin{bmatrix} 9.0 \times 10^{-5} & -2.25 \times 10^{-5} \\ -2.25 \times 10^{-5} & 4.06 \times 10^{-4} \end{bmatrix}$$

Standard deviations:

$$\begin{aligned} \sigma_{a_1} &= \sqrt{9.0 \times 10^{-5}} = 0.0095 \\ \sigma_{b_1} &= \sqrt{4.06 \times 10^{-4}} = 0.0201 \end{aligned}$$

For 95% confidence with large samples (using $z_{0.975} = 1.96$):

$$\begin{aligned} a_1 &: 0.85 \pm 1.96 \times 0.0095 = 0.85 \pm 0.019 \\ b_1 &: 0.12 \pm 1.96 \times 0.0201 = 0.12 \pm 0.039 \end{aligned}$$

$$a_1 \in [0.831, 0.869], \quad b_1 \in [0.081, 0.159] \text{ (95\% confidence)}$$

Example 21.3 (Model Validation with Test Data). A model identified from training data predicts the following on a validation set of 100 points:

Quantity	Value
$\sum_{k=1}^{100} y_{\text{val}}^2[k]$	5000
$\sum_{k=1}^{100} (y_{\text{val}}[k] - \hat{y}[k])^2$	200
Mean of y_{val}	6.5

Compute the fit percentage and NRMSE.

Solution:

NRMSE (Normalized Root Mean Square Error):

$$\text{NRMSE} = \sqrt{\frac{\sum (y - \hat{y})^2}{\sum y^2}} = \sqrt{\frac{200}{5000}} = \sqrt{0.04} = 0.2 = 20\%$$

Fit Percentage:

First compute $\|y_{\text{val}} - \bar{y}\|^2$:

$$\sum_{k=1}^{100} (y_{\text{val}}[k] - \bar{y})^2 = \sum y^2 - N\bar{y}^2 = 5000 - 100 \times 6.5^2 = 5000 - 4225 = 775$$

Then:

$$\begin{aligned} \text{FIT} &= 100 \times \left(1 - \frac{\|y - \hat{y}\|}{\|y - \bar{y}\|} \right) = 100 \times \left(1 - \frac{\sqrt{200}}{\sqrt{775}} \right) \\ &= 100 \times \left(1 - \frac{14.14}{27.84} \right) = 100 \times (1 - 0.508) = 49.2\% \end{aligned}$$

$\text{NRMSE} = 20\%, \quad \text{FIT} = 49.2\%$

A fit of 49% is marginal—the model captures about half the variance beyond the mean. This suggests the model structure may be inadequate (e.g., first-order for a higher-order system) or there is significant noise.

Example 21.4 (Second-Order System Identification). Step response data from a system shows oscillatory behavior with the following characteristics: a final value of $y_{ss} = 2.0$ for step input $u = 1$, a peak value of $y_{peak} = 2.5$ at $t_p = 0.5$ s, and a second peak of $y_{peak2} = 2.1$ at $t = 1.5$ s.

Identify a second-order model $G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$.

Solution:

DC Gain:

$$K = \frac{y_{ss}}{u} = \frac{2.0}{1} = 2.0$$

Percent Overshoot:

$$\%OS = \frac{y_{peak} - y_{ss}}{y_{ss}} \times 100 = \frac{2.5 - 2.0}{2.0} \times 100 = 25\%$$

Damping Ratio:

$$\begin{aligned} \%OS &= 100 \times e^{-\pi\zeta/\sqrt{1-\zeta^2}} \\ 0.25 &= e^{-\pi\zeta/\sqrt{1-\zeta^2}} \\ \ln(0.25) &= \frac{-\pi\zeta}{\sqrt{1-\zeta^2}} \\ -1.386 &= \frac{-\pi\zeta}{\sqrt{1-\zeta^2}} \end{aligned}$$

Squaring both sides:

$$1.922 = \frac{\pi^2 \zeta^2}{1 - \zeta^2}$$

$$1.922(1 - \zeta^2) = \pi^2 \zeta^2$$

$$\zeta^2 = \frac{1.922}{1.922 + \pi^2} = \frac{1.922}{11.79} = 0.163$$

$$\zeta = 0.404$$

Natural Frequency:

The peak time is:

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = \frac{\pi}{\omega_d}$$

$$\omega_d = \frac{\pi}{t_p} = \frac{\pi}{0.5} = 6.28 \text{ rad/s}$$

$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}} = \frac{6.28}{\sqrt{1 - 0.163}} = \frac{6.28}{0.915} = 6.87 \text{ rad/s}$$

Final Model:

$$G(s) = \frac{2 \times 47.2}{s^2 + 5.55s + 47.2} = \frac{94.4}{s^2 + 5.55s + 47.2}$$

where $\omega_n^2 = 47.2$, $2\zeta\omega_n = 5.55$.

21.6 System Identification Flowchart

21.7 Chapter Summary

This chapter developed the theory and practice of system identification across several key areas.

Historical Foundation: System identification originated with Gauss's least squares method for orbital prediction (1809) and matured through Ljung's prediction error framework (1970s–80s). The guiding philosophy is “all models are wrong, some are useful” (Box).

Input Design: Effective identification requires persistently exciting inputs. PRBS signals provide broadband excitation with bounded amplitude; chirps enable frequency-by-frequency characterization.

Least Squares Estimation: The regression form $\mathbf{y} = \Phi\boldsymbol{\theta} + \mathbf{e}$ leads to the normal equations $\hat{\boldsymbol{\theta}} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}$. Under Gaussian noise, this is the Best Linear Unbiased Estimator.

ARX Models: The AutoRegressive with eXogenous input structure $y[k] = a_1 y[k-1] + \dots + b_1 u[k-1] + \dots + e[k]$ enables direct least squares estimation. Continuous-time parameters are recovered through conversion formulas.

Frequency Domain Methods: Sinusoidal testing or spectral analysis yields empirical Bode plots. Parametric models can be fitted to frequency data.

Model Validation: Residual whiteness tests and cross-validation on held-out data assess model adequacy. The bias-variance tradeoff governs model complexity selection.

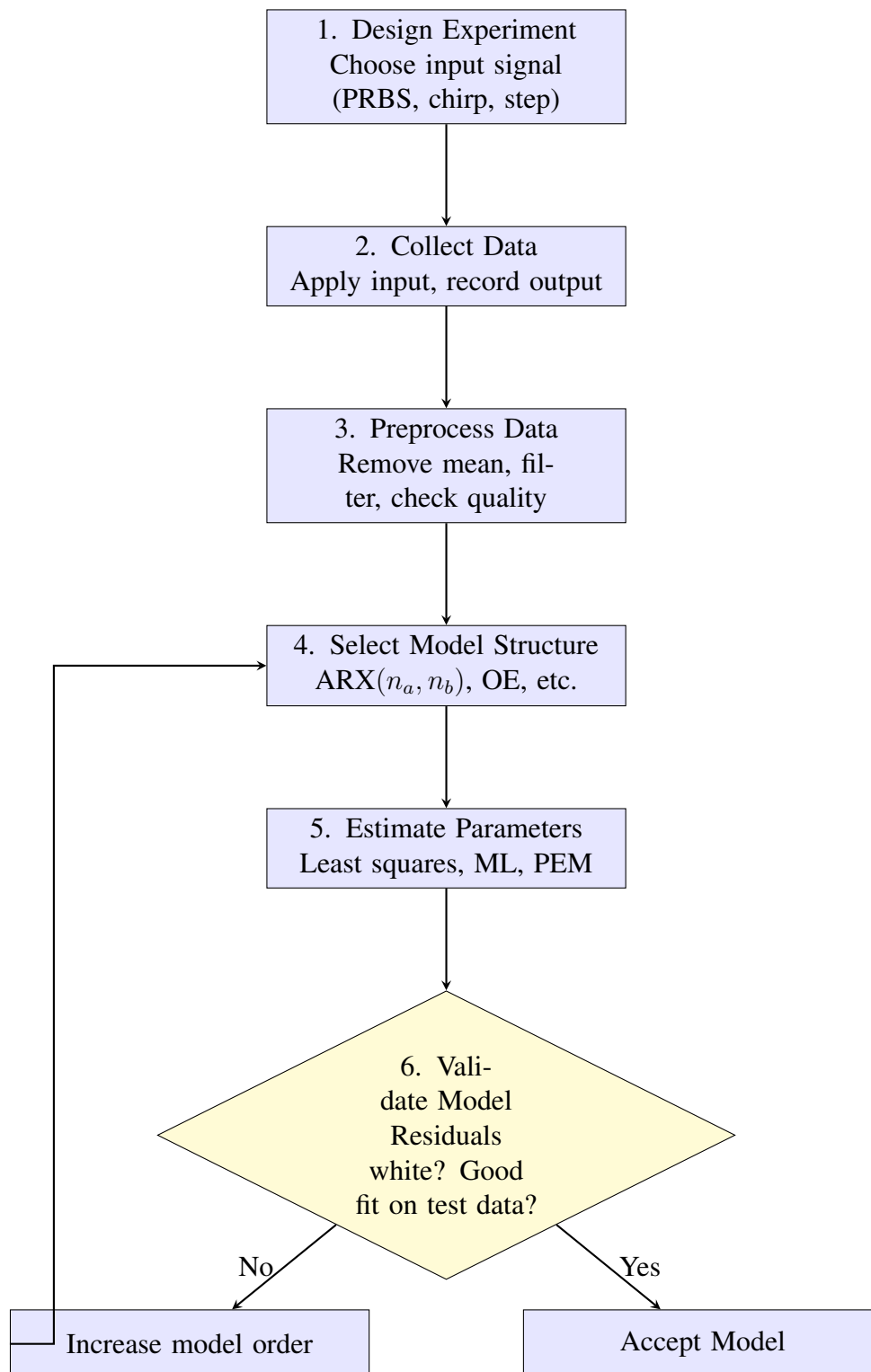


Figure 21.4: System identification workflow. The iterative loop between model structure selection and validation continues until an adequate model is found.

Key Takeaway

System identification bridges the gap between theoretical models and physical reality. When first-principles modeling is expensive or inaccurate, data-driven identification provides an alternative path to useful models. The key insight: design experiments to maximize information content, fit models using principled estimation methods, and rigorously validate before deployment.

21.8 Exercises

Exercise 21.1. Regression Form: Write the ARX(2,2) model

$$y[k] + a_1 y[k-1] + a_2 y[k-2] = b_1 u[k-1] + b_2 u[k-2] + e[k]$$

in regression form $y[k] = \phi^T[k]\theta + e[k]$. Define $\phi[k]$ and θ .

Exercise 21.2. Least Squares Derivation: Derive the normal equations by completing the square on the cost function $J(\theta) = \|\mathbf{y} - \Phi\theta\|^2$.

Exercise 21.3. Step Response Identification: The following step response data ($u = 2$) was recorded:

t (s)	0	1	2	3	4	5	6
y	0	3.16	5.07	6.22	6.93	7.36	7.62

Estimate K and τ for a first-order model.

Exercise 21.4. Frequency Domain Fitting: Given frequency response measurements $|G(j\omega)| = [0.95, 0.7, 0.45, 0.3]$ at $\omega = [0.5, 1, 2, 3]$ rad/s, fit a first-order model using least squares on $|G|^{-2}$.

Exercise 21.5. Covariance Analysis: If $\Phi^T \Phi = \begin{bmatrix} 1000 & 0 \\ 0 & 500 \end{bmatrix}$ and $\sigma^2 = 0.01$, compute the parameter standard deviations. How does doubling the data length (approximately doubling $\Phi^T \Phi$) affect these?

Exercise 21.6. Model Order Selection: An engineer identifies ARX models of orders 1 through 5. The validation fit percentages are 65%, 78%, 82%, 83%, 82%. Which order should be selected and why?

Exercise 21.7. PRBS Design: Design a 7-bit PRBS with clock period 0.1 s. What is its period? What is its approximate bandwidth?

Exercise 21.8. Laboratory Extension: Modify the Arduino code to use a chirp input instead of PRBS. Compute the empirical frequency response using FFT in MATLAB/Python.

Exercise 21.9. Bias Analysis: An ARX model is fitted to data from a system where the noise is actually filtered (colored). Will the least squares estimate be biased? Explain why.

Exercise 21.10. Physics Comparison: A motor has datasheet parameters $R = 5 \Omega$, $L = 10$ mH (negligible), $K_t = K_e = 0.05$ Nm/A, $J = 10^{-4}$ kg·m², $b = 10^{-5}$ Nm·s/rad. Calculate the theoretical time constant and DC gain. Compare with typical identified values and discuss sources of discrepancy.

Chapter 22

Adaptive Control

“The measure of intelligence is the ability to change.” — Albert Einstein

Chapter Objectives: This chapter provides a comprehensive treatment of adaptive control systems. You will understand the historical development and motivation for adaptive control, gaining insight into why fixed controllers fail in environments with changing dynamics. The chapter covers the mathematical foundations of Model Reference Adaptive Control (MRAC) in depth, including the derivation of adaptation laws using both gradient descent (the MIT Rule) and Lyapunov stability methods. You will learn to implement self-tuning regulators and gain scheduling techniques for practical applications, apply adaptive control to motor control problems, and analyze the stability and convergence properties of adaptive systems.

22.1 Historical Motivation: The Need for Adaptation

The story of adaptive control is one of engineering necessity, mathematical innovation, and a cautionary tale about the gap between theory and practice. Understanding this history illuminates not only the development of the field but also the fundamental challenges that any adaptive control designer must confront.

22.1.1 The Aircraft Problem: Where Fixed Controllers Fail

In the early 1950s, aviation engineers faced an increasingly urgent problem. As aircraft became faster and flew at higher altitudes, the aerodynamic characteristics of these vehicles changed dramatically throughout their flight envelope. Consider the fundamental relationship governing aircraft dynamics:

The lift force on an aircraft wing is given by:

$$L = \frac{1}{2} \rho v^2 S C_L \quad (22.1)$$

where ρ is air density, v is airspeed, S is wing area, and C_L is the lift coefficient. The dynamic pressure $q = \frac{1}{2} \rho v^2$ varies by orders of magnitude between takeoff and high-altitude cruise:

Table 22.1: Variation of Dynamic Pressure Across Flight Envelope

Flight Condition	Altitude (ft)	Speed (Mach)	Dynamic Pressure (psf)
Takeoff	0	0.3	215
Low-altitude cruise	10,000	0.6	610
High-altitude cruise	40,000	0.85	270
High-speed dash	30,000	1.2	890

A controller designed for one flight condition—say, low-altitude cruise—would experience dramatically different plant dynamics at another condition. The transfer function of an aircraft’s pitch dynamics takes the approximate form:

$$G(s) = \frac{K_\alpha(q, M)}{s^2 + 2\zeta(q, M)\omega_n(q, M)s + \omega_n^2(q, M)} \quad (22.2)$$

where all parameters depend on dynamic pressure q and Mach number M . The gain K_α might vary by a factor of 10, while the natural frequency ω_n could change by a factor of 3.

sharp corners

The X-15 Hypersonic Aircraft (1959-1968): Perhaps no vehicle better illustrates the adaptive control challenge than the North American X-15. This rocket-powered research aircraft flew from the dense atmosphere at Edwards Air Force Base to the edge of space at 354,200 feet. During a typical flight, the dynamic pressure varied from 2,000 psf at low altitude to essentially zero in the upper atmosphere. The X-15 used an early form of gain scheduling, with the pilot manually adjusting stability augmentation system gains based on displayed flight conditions. Three X-15 aircraft made 199 flights, with pilots reporting significant workload managing the changing dynamics. This experience directly motivated the development of automated adaptive control systems.

22.1.2 The MIT Rule: Birth of Adaptive Control (1958)

The first systematic approach to adaptive control emerged from the MIT Instrumentation Laboratory (now Draper Laboratory) in the late 1950s. The research team, led by Whitaker, Yamron, and Kezer, proposed what became known as the **MIT Rule** in 1958.

The fundamental insight was elegant: if we define a tracking error between the actual plant output and a desired reference model output, we can adjust controller parameters to minimize this error using gradient descent.

title

“If the system isn’t behaving like the reference model, adjust the controller parameters in the direction that reduces the error.”

This is formalized as:

$$\frac{d\theta}{dt} = -\gamma e \frac{\partial e}{\partial \theta} \quad (22.3)$$

where θ represents controller parameters, e is the tracking error, and $\gamma > 0$ is the adaptation gain.

The MIT Rule represented a paradigm shift: rather than designing a fixed controller for worst-case conditions (which would be conservative) or requiring continuous redesign (which would be impractical), the controller could continuously adjust itself to maintain performance.

The initial applications focused on autopilot systems, where the reference model represented the desired aircraft handling qualities—how the pilot wanted the aircraft to respond to stick inputs. The adaptive mechanism would adjust control gains so that the actual aircraft response matched this ideal model, regardless of flight condition.

22.1.3 Lyapunov-Based Adaptation: Mathematical Rigor (1960s)

While the MIT Rule provided an intuitive and often effective approach, it lacked formal stability guarantees. The adaptation law was based on minimizing an error cost function, but there was no proof that the closed-loop system would remain stable during the adaptation process.

This theoretical gap was addressed in the 1960s through the application of Lyapunov stability theory to adaptive control. The key figures in this development included Parks (1966), who applied Lyapunov methods to adaptive control, and Narendra and his colleagues, who developed systematic design procedures.

The Lyapunov approach reversed the design philosophy. The MIT Rule begins with an intuitive adaptation law and hopes for stability, whereas the Lyapunov approach starts with a stability requirement and derives the adaptation law from it.

By choosing an appropriate Lyapunov function that includes both the tracking error and parameter estimation error, one could *derive* an adaptation law that guarantees $\dot{V} \leq 0$, ensuring stability. This was not merely an incremental improvement—it transformed adaptive control from an engineering heuristic to a mathematically rigorous discipline.

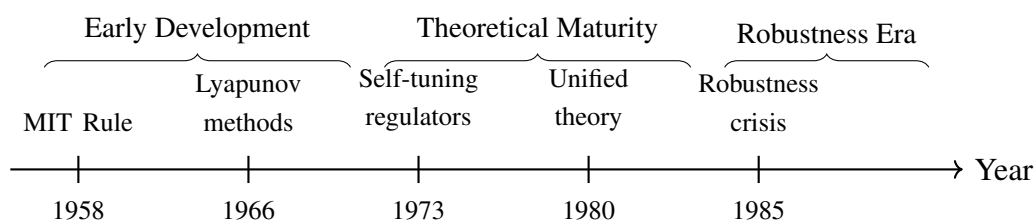


Figure 22.1: Timeline of major developments in adaptive control theory.

22.1.4 The Robustness Crisis: Rohrs' Counterexamples (1985)

By the early 1980s, adaptive control theory had achieved considerable mathematical sophistication. Stability proofs existed for a variety of adaptive control architectures, and the field appeared mature. Then came a sobering demonstration that shook the foundations of the discipline.

In 1985, Rohrs, Valavani, Athans, and Stein published a landmark paper in *IEEE Transactions on Automatic Control* titled “Robustness of Continuous-Time Adaptive Control Algorithms in the Presence of Unmodeled Dynamics.” Their findings were troubling.

title

Rohrs et al. demonstrated that adaptive controllers which were *provably stable* under ideal assumptions could become *unstable* when unmodeled high-frequency dynamics were present, when output disturbances were added, or when both conditions occurred together. The instability could be dramatic, with bounded disturbances causing unbounded signals.

The core issue was that all stability proofs relied on idealized assumptions. First, perfect model knowledge was assumed—the plant was taken to be exactly in the model class. Second, no unmodeled dynamics were considered, meaning high-frequency parasitic dynamics were ignored. Third, no disturbances were present, as the proofs assumed noise-free measurements.

Real systems violate all three assumptions. A DC motor has armature inductance (often neglected in simple models), sensors add noise, and mechanical systems have resonances at high frequencies.

The problem was particularly insidious because adaptive systems could appear to work well in simulation or initial testing, then fail catastrophically when deployed in conditions that excited the unmodeled dynamics.

22.1.5 Resolution and Modern Perspective

The robustness crisis did not kill adaptive control—instead, it transformed the field. Researchers developed modifications that restored robustness. The σ -modification adds a leakage term $-\sigma\theta$ to prevent parameter drift. The ε_1 -modification introduces dead-zone modifications to stop adaptation when the error is small. Projection constrains parameters to remain in a known bounded region. Normalization scales signals to prevent unbounded growth.

These robustness modifications came at a cost: one could no longer prove asymptotic convergence of tracking error to zero. Instead, the error would converge to a small residual set. But this trade-off—perfect theoretical performance for practical robustness—is one that every practicing engineer should embrace.

sharp corners

Legacy of the Robustness Crisis: The Rohrs counterexamples served as a watershed moment for control theory broadly, not just adaptive control. They demonstrated the danger of mistaking mathematical elegance for engineering validity. Today, robustness analysis is considered essential for any advanced control design, and the gap between theoretical assumptions and practical reality is explicitly addressed rather than ignored.

22.2 Modern Applications of Adaptive Control

While the theoretical foundations of adaptive control were established decades ago, modern applications continue to expand as computational power increases and new challenges emerge.

22.2.1 Aerospace and Flight Control

Adaptive control remains central to aerospace applications, where the original motivation persists. In commercial aircraft, fuel consumption changes the aircraft mass by 30% or more during long flights. Military aircraft experience dynamics alterations due to weapons release, battle damage, and configuration changes. Spacecraft undergo mass property changes from fuel depletion and appendage deployment. Unmanned aerial vehicles (UAVs) require adaptation to handle payload variations and icing conditions.

NASA's Intelligent Flight Control System (IFCS) demonstrated neural network-based adaptive control on the F-15 ACTIVE research aircraft, showing the ability to maintain control even with significant control surface failures.

22.2.2 Robotics and Manufacturing

Modern robots face continuously changing conditions that classical fixed controllers handle poorly. Payload variation requires different gains when a robot arm carries different loads. Wear and degradation cause joint friction and gear backlash to change over time. Human-robot interaction demands adaptive compliance when robots make contact with humans. Learning from demonstration requires adapting to new tasks shown by operators.

22.2.3 Process Industry

Chemical and manufacturing processes exhibit slow parameter variations that motivate adaptive approaches. Catalyst deactivation causes reaction rates to decrease over weeks or months. Fouling degrades heat exchanger coefficients as deposits accumulate. Raw material variation affects process dynamics as feed composition changes. Seasonal changes alter heat transfer as cooling water temperature varies.

Self-tuning PID controllers are widely deployed in process industries, automatically adjusting gains to maintain performance as conditions drift.

22.2.4 Biomedical Applications

Adaptive control has found growing application in medical devices. Artificial pancreas systems must adapt insulin delivery to varying patient sensitivity. Anesthesia control systems accommodate drug sensitivity that varies between patients and during surgery. EMG-controlled prosthetic limbs adapt to user patterns over time. Rehabilitation robots adjust therapy assistance based on patient progress.

22.2.5 Automotive Systems

Modern vehicles contain numerous adaptive control systems. Engine control units adapt fuel injection to altitude, temperature, and fuel quality. Transmission control systems adjust shift patterns based on driver behavior. Active suspension systems adapt damping to road conditions and vehicle load. Autonomous vehicles adapt their control strategies to road surface conditions and environmental factors.

22.3 Mathematical Foundations of Adaptive Control

We now develop the mathematical framework for adaptive control, progressing from simple intuitive approaches to rigorous Lyapunov-based designs.

22.3.1 The Adaptive Control Problem

Consider a plant with unknown or time-varying parameters:

$$\dot{x} = A(\theta)x + B(\theta)u, \quad y = Cx \quad (22.4)$$

where $\theta \in \mathbb{R}^p$ represents the unknown parameter vector. The control objective is to make the output y track a reference signal $r(t)$ despite the uncertainty in θ .

The key insight of adaptive control is to treat the unknown parameters as additional states to be estimated, then use these estimates in the controller:

$$u = K(\hat{\theta})x + L(\hat{\theta})r \quad (22.5)$$

where $\hat{\theta}$ is the current parameter estimate. The adaptation mechanism updates $\hat{\theta}$ based on observed system behavior.

22.3.2 Model Reference Adaptive Control (MRAC)

Reference Model Concept

In MRAC, we specify desired closed-loop behavior through a reference model:

$$\dot{x}_m = A_m x_m + B_m r \quad (22.6)$$

where A_m is Hurwitz (stable), and the model represents ideal behavior we want the plant to achieve.

title

The reference model should satisfy three essential criteria. First, it must be stable, with A_m having all eigenvalues in the left half-plane. Second, the model must be achievable, meaning the plant must be capable of matching the model behavior. Third, the model should exhibit the desired behavior, with its response matching specifications such as rise time and overshoot.

For a first-order plant $\dot{y} = -ay + bu$ where we want first-order reference behavior $\dot{y}_m = -a_m y_m + b_m r$, the controller takes the form:

$$u = \theta_1 r + \theta_2 y \quad (22.7)$$

If we knew the true parameters a and b , we would select:

$$\theta_1^* = \frac{b_m}{b}, \quad \theta_2^* = \frac{a - a_m}{b} \quad (22.8)$$

Since a and b are unknown, we use adaptive estimates $\hat{\theta}_1$ and $\hat{\theta}_2$ that are updated online.

MRAC Block Diagram

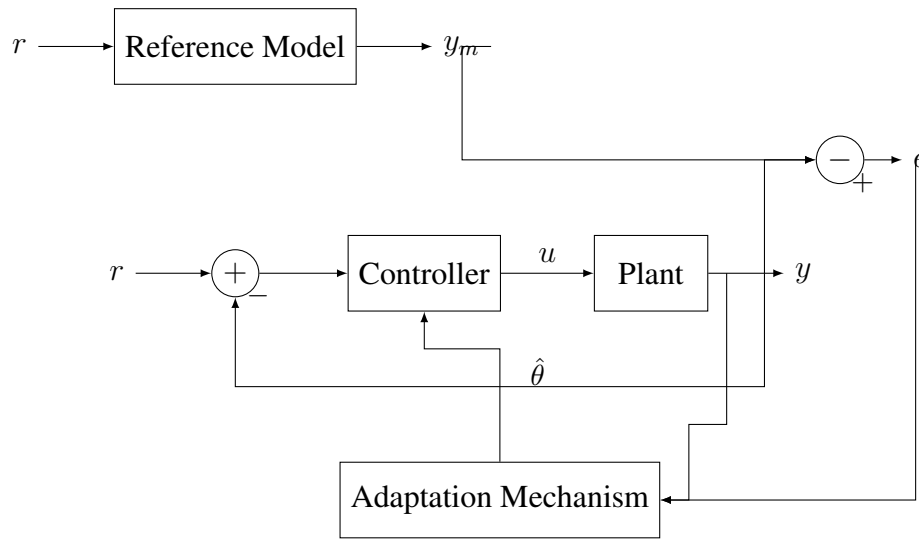


Figure 22.2: Block diagram of Model Reference Adaptive Control (MRAC). The adaptation mechanism adjusts controller parameters $\hat{\theta}$ to minimize the tracking error $e = y - y_m$.

22.3.3 MIT Rule Derivation

The MIT Rule provides an intuitive approach to deriving adaptation laws based on gradient descent.

Cost Function Approach

Define a cost function based on tracking error:

$$J(\theta) = \frac{1}{2}e^2 = \frac{1}{2}(y - y_m)^2 \quad (22.9)$$

To minimize J , we update parameters in the negative gradient direction:

$$\frac{d\theta}{dt} = -\gamma \frac{\partial J}{\partial \theta} = -\gamma e \frac{\partial e}{\partial \theta} \quad (22.10)$$

The term $\frac{\partial e}{\partial \theta}$ is called the **sensitivity derivative**. It measures how the error changes when parameters are perturbed.

Application to First-Order System

Consider the first-order plant:

$$\dot{y} = -ay + bu \quad (22.11)$$

with unknown positive parameters a and b . The reference model is:

$$\dot{y}_m = -a_m y_m + b_m r \quad (22.12)$$

Using the controller $u = \hat{\theta}_1 r + \hat{\theta}_2 y$, the closed-loop plant becomes:

$$\dot{y} = -(a - b\hat{\theta}_2)y + b\hat{\theta}_1 r \quad (22.13)$$

For the plant to match the reference model, we need:

$$a - b\hat{\theta}_2 = a_m, \quad b\hat{\theta}_1 = b_m \quad (22.14)$$

Deriving the sensitivity derivatives:

From the closed-loop dynamics, in steady state (assuming slow adaptation), the output satisfies:

$$y = \frac{b\hat{\theta}_1}{a - b\hat{\theta}_2} r \quad (\text{for constant } r) \quad (22.15)$$

Taking partial derivatives:

$$\frac{\partial y}{\partial \hat{\theta}_1} \approx \frac{b}{a - b\hat{\theta}_2} r \approx \frac{y_m}{\hat{\theta}_1} \quad (22.16)$$

$$\frac{\partial y}{\partial \hat{\theta}_2} \approx \frac{b^2 \hat{\theta}_1}{(a - b\hat{\theta}_2)^2} r \cdot y \approx \frac{y \cdot y_m}{\hat{\theta}_1} \quad (22.17)$$

In practice, we often approximate:

$$\frac{\partial e}{\partial \hat{\theta}_1} \approx -y_m, \quad \frac{\partial e}{\partial \hat{\theta}_2} \approx -y \cdot \text{sgn}(b) \quad (22.18)$$

This leads to the MIT Rule adaptation laws:

$$\boxed{\dot{\hat{\theta}}_1 = \gamma_1 e \cdot y_m, \quad \dot{\hat{\theta}}_2 = \gamma_2 e \cdot y} \quad (22.19)$$

Remark 22.1. The MIT Rule requires knowledge of $\text{sgn}(b)$, the sign of the control gain. This is a minimal assumption—we must at least know whether increasing control increases or decreases the output.

22.3.4 Lyapunov-Based Adaptive Control

The Lyapunov approach provides stability guarantees that the MIT Rule lacks. The methodology proceeds as follows: first, define the error dynamics relating the tracking error to parameter estimation errors; second, choose a Lyapunov function that includes both state error and parameter error terms; and third, derive the adaptation law that ensures $\dot{V} \leq 0$.

Error Dynamics

Let $\tilde{\theta} = \hat{\theta} - \theta^*$ denote the parameter estimation error, where θ^* are the ideal parameters. The tracking error dynamics can be written as:

$$\dot{e} = A_m e + B\phi(x, r)^T \tilde{\theta} \quad (22.20)$$

where $\phi(x, r)$ is a regression vector and B is the control input matrix.

Lyapunov Function Construction

Choose the Lyapunov function candidate:

$$V(e, \tilde{\theta}) = e^T P e + \tilde{\theta}^T \Gamma^{-1} \tilde{\theta} \quad (22.21)$$

where $P > 0$ satisfies the Lyapunov equation $A_m^T P + P A_m = -Q$ for some $Q > 0$, and $\Gamma > 0$ is a diagonal matrix of adaptation gains.

Derivative Calculation

Taking the time derivative:

$$\begin{aligned} \dot{V} &= \dot{e}^T P e + e^T P \dot{e} + 2\tilde{\theta}^T \Gamma^{-1} \dot{\tilde{\theta}} \\ &= e^T (A_m^T P + P A_m) e + 2e^T P B \phi^T \tilde{\theta} + 2\tilde{\theta}^T \Gamma^{-1} \dot{\tilde{\theta}} \\ &= -e^T Q e + 2\tilde{\theta}^T (\phi B^T P e + \Gamma^{-1} \dot{\tilde{\theta}}) \end{aligned} \quad (22.22)$$

Adaptation Law Derivation

To make $\dot{V} \leq 0$, we choose the adaptation law:

$$\boxed{\dot{\tilde{\theta}} = -\Gamma \phi B^T P e} \quad (22.23)$$

This cancels the indefinite term, yielding:

$$\dot{V} = -e^T Q e \leq 0 \quad (22.24)$$

Theorem 22.1 (Stability of Lyapunov-Based MRAC). *Consider the adaptive system with the adaptation law (??). Then all signals in the closed-loop system are bounded, the tracking error $e(t) \rightarrow 0$ as $t \rightarrow \infty$, and the parameter estimates $\hat{\theta}(t)$ remain bounded.*

Proof. Since $\dot{V} \leq 0$ and $V > 0$, the function $V(t)$ is non-increasing and bounded below by zero. Therefore, $V(t)$ converges to a limit, implying both e and $\tilde{\theta}$ are bounded.

From $\dot{V} = -e^T Q e$, integrating from 0 to T :

$$\int_0^T e^T Q e dt = V(0) - V(T) \leq V(0) < \infty \quad (22.25)$$

Thus $e \in \mathcal{L}_2$. Since e and $\dot{e}$ are bounded (from bounded signals and the error dynamics), Barbalat's lemma implies $e(t) \rightarrow 0$ as $t \rightarrow \infty$. $\square$

Remark 22.2 (Parameter Convergence). Note that Theorem 22.1 does **not** guarantee $\tilde{\theta}(t) \rightarrow 0$. The parameters converge to values that produce zero tracking error, but these need not be the true plant parameters unless a **persistence of excitation** condition is satisfied.

Detailed Example: First-Order MRAC Design

Consider the first-order plant $\dot{y} = -ay + bu$ with a, b unknown but $b > 0$. Design an MRAC with reference model $\dot{y}_m = -a_m y_m + b_m r$.

Step 1: Controller structure

$$u = \hat{\theta}_1 r + \hat{\theta}_2 y \quad (22.26)$$

Step 2: Error dynamics

Substituting the controller into the plant:

$$\dot{y} = -ay + b(\hat{\theta}_1 r + \hat{\theta}_2 y) = -(a - b\hat{\theta}_2)y + b\hat{\theta}_1 r \quad (22.27)$$

With ideal parameters $\theta_1^* = b_m/b$ and $\theta_2^* = (a - a_m)/b$:

$$\dot{e} = \dot{y} - \dot{y}_m = -a_m e + b\tilde{\theta}_1 r + b\tilde{\theta}_2 y \quad (22.28)$$

Step 3: Lyapunov function

Choose $V = \frac{1}{2}e^2 + \frac{b}{2\gamma_1}\tilde{\theta}_1^2 + \frac{b}{2\gamma_2}\tilde{\theta}_2^2$

Taking the derivative:

$$\begin{aligned} \dot{V} &= e\dot{e} + \frac{b}{\gamma_1}\tilde{\theta}_1\dot{\tilde{\theta}}_1 + \frac{b}{\gamma_2}\tilde{\theta}_2\dot{\tilde{\theta}}_2 \\ &= -a_m e^2 + be\tilde{\theta}_1 r + be\tilde{\theta}_2 y + \frac{b}{\gamma_1}\tilde{\theta}_1\dot{\tilde{\theta}}_1 + \frac{b}{\gamma_2}\tilde{\theta}_2\dot{\tilde{\theta}}_2 \\ &= -a_m e^2 + b\tilde{\theta}_1 \left(er + \frac{\dot{\tilde{\theta}}_1}{\gamma_1} \right) + b\tilde{\theta}_2 \left(ey + \frac{\dot{\tilde{\theta}}_2}{\gamma_2} \right) \end{aligned} \quad (22.29)$$

Step 4: Adaptation law

Choosing:

$$\dot{\tilde{\theta}}_1 = -\gamma_1 er, \quad \dot{\tilde{\theta}}_2 = -\gamma_2 ey \quad (22.30)$$

yields $\dot{V} = -a_m e^2 \leq 0$, guaranteeing stability.

22.3.5 Self-Tuning Regulators

Self-Tuning Regulators (STR) take a different approach from MRAC: rather than forcing the plant to match a reference model, they explicitly identify plant parameters and update the controller accordingly.

STR Architecture

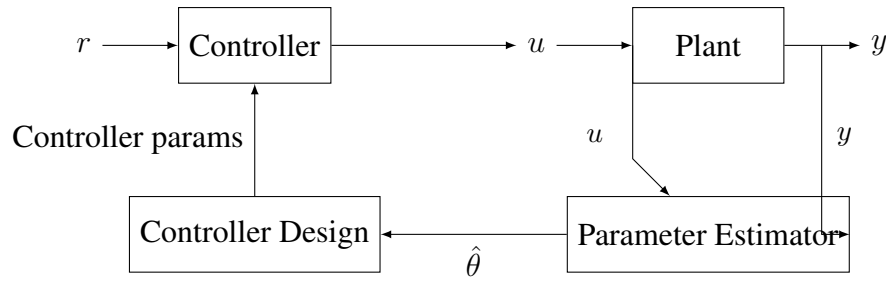


Figure 22.3: Self-Tuning Regulator architecture: explicit separation of identification and control.

Recursive Least Squares Identification

For a discrete-time ARX model:

$$y(k) = \phi(k)^T \theta + e(k) \quad (22.31)$$

where $\phi(k) = [-y(k-1), \dots, -y(k-n_a), u(k-1), \dots, u(k-n_b)]^T$ is the regression vector.

The Recursive Least Squares (RLS) algorithm updates the parameter estimate:

$$K(k) = \frac{P(k-1)\phi(k)}{\lambda + \phi(k)^T P(k-1)\phi(k)} \quad (22.32)$$

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)[y(k) - \phi(k)^T \hat{\theta}(k-1)] \quad (22.33)$$

$$P(k) = \frac{1}{\lambda} [P(k-1) - K(k)\phi(k)^T P(k-1)] \quad (22.34)$$

where $\lambda \in (0, 1]$ is a forgetting factor that allows tracking of time-varying parameters.

Certainty Equivalence Principle

Given the estimated parameters $\hat{\theta}$, the controller is designed as if $\hat{\theta}$ were the true parameters. This is called the **certainty equivalence principle**.

For a minimum variance controller, the control law becomes:

$$u(k) = \frac{1}{\hat{b}_1} \left[r(k+d) + \sum_{i=1}^{n_a} \hat{a}_i y(k+1-i) - \sum_{i=2}^{n_b} \hat{b}_i u(k+1-i) \right] \quad (22.35)$$

where d is the system delay.

22.3.6 Gain Scheduling

Gain scheduling is the simplest and most widely used form of adaptive control. It uses pre-computed controllers for different operating conditions, interpolating between them in real-time.

Algorithm 50 Gain Scheduling Design Procedure

- 1: **Identify scheduling variables:** Select measurable variables that correlate with parameter changes (e.g., altitude, speed, temperature)
- 2: **Define operating points:** Select representative conditions spanning the operating envelope
- 3: **Design local controllers:** At each operating point, linearize the plant and design an optimal controller
- 4: **Implement interpolation:** Create smooth transitions between controllers based on the scheduling variable
- 5: **return** Gain-scheduled controller with interpolation lookup table

Design Procedure**Linear Interpolation**

For a single scheduling variable ρ and two operating points ρ_1 and ρ_2 :

$$K(\rho) = \frac{\rho_2 - \rho}{\rho_2 - \rho_1} K_1 + \frac{\rho - \rho_1}{\rho_2 - \rho_1} K_2 \quad (22.36)$$

For multiple scheduling variables, multi-dimensional interpolation (e.g., bilinear, tensor product) is used.

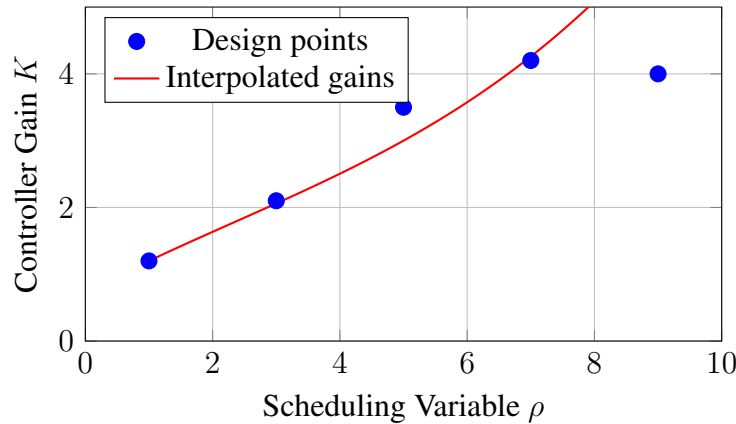


Figure 22.4: Gain scheduling: controller gains interpolated between design operating points.

Stability Considerations

Gain scheduling provides no formal stability guarantees during transitions. Sufficient conditions for stability include operating points close enough that interpolated controllers stabilize intermediate plants, scheduling variables that change slowly compared to closed-loop dynamics, and the use of LPV (Linear Parameter-Varying) design methods that explicitly account for parameter rates.

22.3.7 Robustness Modifications

The robustness crisis of the 1980s led to several modifications of basic adaptive laws:

σ -Modification (Leakage)

Add a damping term to prevent parameter drift:

$$\dot{\hat{\theta}} = -\Gamma\phi B^T P e - \sigma \hat{\theta} \quad (22.37)$$

where $\sigma > 0$ is small. This pulls parameters toward zero, preventing drift when the error is small or when unmodeled dynamics cause persistent excitation.

 e_1 -Modification (Dead Zone)

Stop adaptation when error is below a threshold:

$$\dot{\hat{\theta}} = \begin{cases} -\Gamma\phi B^T P e & \text{if } |e| > e_1 \\ 0 & \text{if } |e| \leq e_1 \end{cases} \quad (22.38)$$

Projection

Constrain parameters to a known bounded set Ω :

$$\dot{\hat{\theta}} = \text{Proj}(-\Gamma\phi B^T P e) \quad (22.39)$$

where the projection operator prevents $\hat{\theta}$ from leaving Ω .

22.4 Practical Experiment: Adaptive Motor Control

title

Objective: Demonstrate that an adaptive controller maintains performance under varying load conditions where a fixed controller fails.

Time Required: 3-4 hours

Difficulty: Intermediate

22.4.1 Experimental Setup

Hardware Requirements

Table 22.2: Hardware Components for Adaptive Motor Control Experiment

Component	Specification
Microcontroller	Arduino Uno or Mega
Motor	DC motor with encoder (e.g., Pololu 37D gearmotor)
Motor driver	L298N H-bridge or similar
Power supply	12V, 2A recommended
Load mechanism	Rotary disc with mounting points for weights
Calibrated weights	10g, 20g, 50g, 100g set
Data acquisition	Oscilloscope or data logging capability

System Schematic

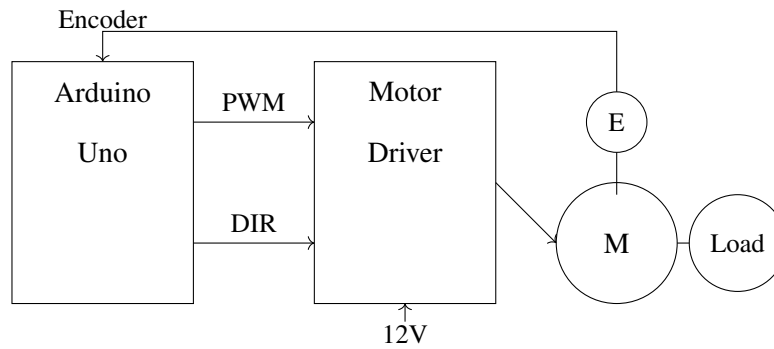


Figure 22.5: Hardware schematic for adaptive motor control experiment.

22.4.2 Mathematical Model

The DC motor with variable load is modeled as:

$$J\dot{\omega} + b\omega = K_t i \quad (22.40)$$

where ω denotes the angular velocity in rad/s, J is the total moment of inertia including motor and load, b represents the viscous friction coefficient, K_t is the motor torque constant, and i is the armature current.

When mass is added to the disc at radius r , the inertia changes:

$$J_{total} = J_{motor} + m_{load}r^2 \quad (22.41)$$

With electrical dynamics much faster than mechanical, the transfer function is approximately:

$$G(s) = \frac{\omega(s)}{V(s)} = \frac{K}{Js + b} = \frac{K/b}{\tau s + 1} \quad (22.42)$$

where $\tau = J/b$ changes with load.

22.4.3 Controller Implementation

Fixed PI Controller

The baseline fixed controller is a PI controller tuned for the unloaded motor:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau \quad (22.43)$$

Adaptive Controller (MIT Rule)

The adaptive controller uses the same PI structure but adjusts gains online:

$$\dot{K}_p = \gamma_p e \cdot e \quad (22.44)$$

$$\dot{K}_i = \gamma_i e \cdot \int e dt \quad (22.45)$$

22.4.4 Arduino Implementation

Algorithm 51 Adaptive Motor Speed Control with MIT Rule

```

1: Initialize:  $K_p \leftarrow 2.0$ ,  $K_i \leftarrow 0.5$ ,  $\gamma_p \leftarrow 0.001$ ,  $\gamma_i \leftarrow 0.0005$ 
2: Initialize:  $\tau_m \leftarrow 0.2$ ,  $\omega_m \leftarrow 0$ ,  $\omega_{ref} \leftarrow 100$  rad/s
3: Initialize: integral  $\leftarrow 0$ ,  $\Delta t \leftarrow 0.01$  s
4: Configure PWM output, direction output, and encoder inputs
5: Attach interrupt handler for encoder counting
6: while system is running do
7:   if sample period  $\Delta t$  has elapsed then
8:      $\omega \leftarrow$  compute angular velocity from encoder count
9:      $\omega_m \leftarrow \omega_m + \Delta t \cdot (-\omega_m/\tau_m + \omega_{ref}/\tau_m)$  ▷ Reference model
10:     $e \leftarrow \omega_{ref} - \omega$  ▷ Control error
11:     $e_{track} \leftarrow \omega - \omega_m$  ▷ Tracking error for adaptation
12:    integral  $\leftarrow$  integral  $+ e \cdot \Delta t$ 
13:    integral  $\leftarrow$  constrain(integral, -50, 50) ▷ Anti-windup
14:     $u \leftarrow K_p \cdot e + K_i \cdot$  integral ▷ PI control law
15:     $K_p \leftarrow K_p + \gamma_p \cdot e_{track} \cdot e \cdot \Delta t$  ▷ MIT Rule for  $K_p$ 
16:     $K_i \leftarrow K_i + \gamma_i \cdot e_{track} \cdot$  integral  $\cdot \Delta t$  ▷ MIT Rule for  $K_i$ 
17:     $K_p \leftarrow$  constrain( $K_p$ , 0.5, 10.0) ▷ Bound gains
18:     $K_i \leftarrow$  constrain( $K_i$ , 0.1, 5.0)
19:    Apply PWM signal  $|u|$  with direction sign( $u$ ) to motor
20:    Log data: time,  $\omega_{ref}$ ,  $\omega_m$ ,  $\omega$ ,  $K_p$ ,  $K_i$ ,  $e_{track}$ 
21:   end if
22: end while

```

Algorithm 52 Encoder Interrupt Service Routine

```

1: if encoder channel B is LOW then
2:   encoderCount  $\leftarrow$  encoderCount + 1
3: else
4:   encoderCount  $\leftarrow$  encoderCount - 1
5: end if

```

Algorithm 53 Experimental Procedure for Adaptive Motor Control

```

1: Phase 1: Baseline Characterization
2: Run motor with no load and record step response
3: Identify time constant  $\tau_0$  and gain  $K_0$  from response data
4: Tune fixed PI controller gains for good nominal performance
5:
6: Phase 2: Fixed Controller Under Load Variation
7: Set reference speed to 100 rad/s and wait for steady state
8: Add 50g weight at 5cm radius and record response degradation
9: Add 100g total weight and document further performance degradation
10:
11: Phase 3: Adaptive Controller Under Load Variation
12: Reset system to no load condition
13: Enable adaptive control with MIT Rule adaptation
14: Repeat the load variation sequence from Phase 2
15: Record the adaptation of gains  $K_p$  and  $K_i$  over time
16: Observe and document maintained performance despite load changes
17:
18: Phase 4: Comparative Analysis
19: Plot fixed vs. adaptive controller responses on same axes
20: Calculate performance metrics: rise time, overshoot, settling time
21: Generate gain adaptation history plots showing  $K_p(t)$  and  $K_i(t)$ 

```

22.4.5 Experimental Procedure

22.4.6 Expected Results

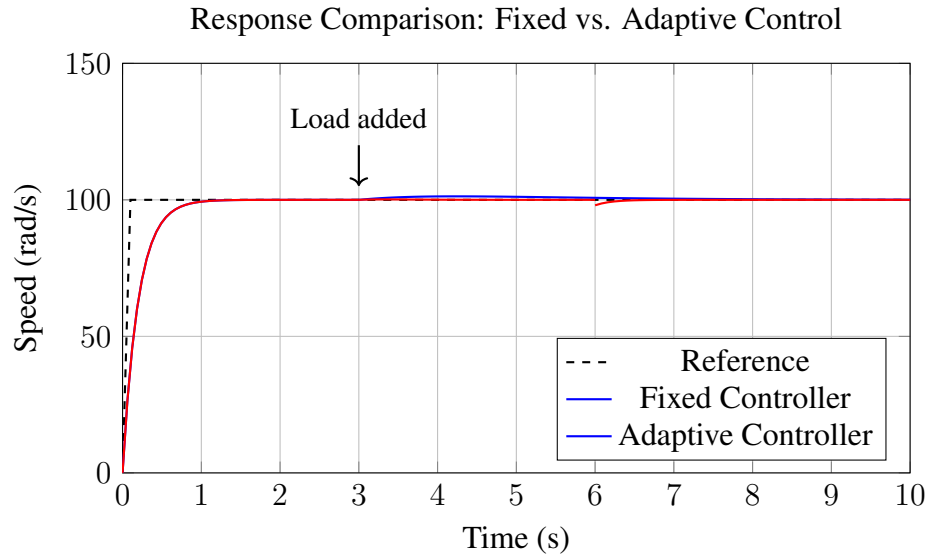


Figure 22.6: Comparison of fixed and adaptive controller response to load changes. The fixed controller shows degraded performance (slower response, more overshoot) when load is added at $t = 3$ s. The adaptive controller quickly adjusts gains to maintain performance.

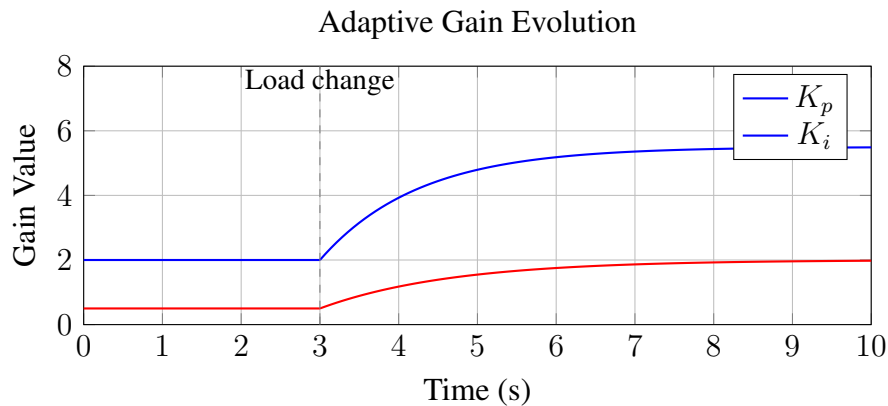


Figure 22.7: Evolution of adaptive gains during the experiment. After the load is added at $t = 3$ s, both K_p and K_i increase to compensate for the increased inertia.

22.4.7 Discussion Questions

Several important questions arise from this experiment. First, consider why the proportional gain K_p needs to increase when inertia increases—what is the physical intuition behind this relationship? Second, analyze what would happen if the adaptation gains γ_p and γ_i were set too large, considering both transient behavior and potential instability. Third, propose modifications to the

experiment that would demonstrate the need for robustness modifications such as σ -modification or dead zones. Finally, compare the experimental results with the theoretical predictions from Section 22.3, noting any discrepancies and their likely causes.

22.5 Example Problems

MIT Rule for First-Order System

Consider a first-order plant:

$$G_p(s) = \frac{b}{s+a} \quad (22.46)$$

where $a = 2$ and $b = 3$ are unknown. Design an MRAC system using the MIT Rule with reference model:

$$G_m(s) = \frac{4}{s+4} \quad (22.47)$$

Solution:

Step 1: Controller structure

Use the controller $u = \hat{\theta}_1 r + \hat{\theta}_2 y$. The closed-loop transfer function becomes:

$$\frac{Y(s)}{R(s)} = \frac{b\hat{\theta}_1}{s+a-b\hat{\theta}_2} \quad (22.48)$$

Step 2: Ideal parameters

Matching with the reference model:

$$a - b\hat{\theta}_2^* = 4 \implies \hat{\theta}_2^* = \frac{a-4}{b} = \frac{2-4}{3} = -\frac{2}{3} \quad (22.49)$$

$$b\hat{\theta}_1^* = 4 \implies \hat{\theta}_1^* = \frac{4}{b} = \frac{4}{3} \quad (22.50)$$

Step 3: Error dynamics

The tracking error is $e = y - y_m$. The error dynamics are:

$$\dot{e} = -4e + b(\hat{\theta}_1 - \hat{\theta}_1^*)r + b(\hat{\theta}_2 - \hat{\theta}_2^*)y \quad (22.51)$$

Step 4: MIT Rule adaptation

Using the approximation that $\frac{\partial e}{\partial \hat{\theta}_1} \approx \frac{b}{s+4}r$ and $\frac{\partial e}{\partial \hat{\theta}_2} \approx \frac{b}{s+4}y$, the MIT Rule gives:

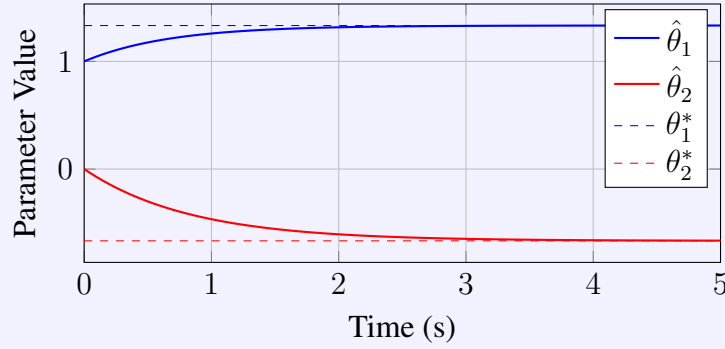
$$\dot{\hat{\theta}}_1 = -\gamma_1 e \cdot \frac{1}{\tau_m s + 1}[r] = -\gamma_1 e \cdot r_f \quad (22.52)$$

$$\dot{\hat{\theta}}_2 = -\gamma_2 e \cdot \frac{1}{\tau_m s + 1}[y] = -\gamma_2 e \cdot y_f \quad (22.53)$$

where r_f and y_f are filtered versions of r and y through $\frac{1}{0.25s+1}$.

Step 5: Simulation

With $\gamma_1 = \gamma_2 = 2$ and initial conditions $\hat{\theta}_1(0) = 1$, $\hat{\theta}_2(0) = 0$, the parameters converge to their ideal values as shown:



Lyapunov Adaptation Law Derivation

For the same first-order system as Example ??, derive a Lyapunov-based adaptation law and prove stability.

Solution:

Step 1: Define parameter errors

Let $\tilde{\theta}_1 = \hat{\theta}_1 - \theta_1^*$ and $\tilde{\theta}_2 = \hat{\theta}_2 - \theta_2^*$.

Step 2: Error dynamics

The tracking error dynamics are:

$$\dot{e} = -a_m e + b\tilde{\theta}_1 r + b\tilde{\theta}_2 y \quad (22.54)$$

where $a_m = 4$.

Step 3: Lyapunov function

Choose:

$$V = \frac{1}{2}e^2 + \frac{|b|}{2\gamma_1}\tilde{\theta}_1^2 + \frac{|b|}{2\gamma_2}\tilde{\theta}_2^2 \quad (22.55)$$

Step 4: Compute $\dot{V}$

$$\dot{V} = e\dot{e} + \frac{|b|}{\gamma_1}\tilde{\theta}_1\dot{\tilde{\theta}}_1 + \frac{|b|}{\gamma_2}\tilde{\theta}_2\dot{\tilde{\theta}}_2 \quad (22.56)$$

$$= -a_m e^2 + be\tilde{\theta}_1 r + be\tilde{\theta}_2 y + \frac{|b|}{\gamma_1}\tilde{\theta}_1\dot{\tilde{\theta}}_1 + \frac{|b|}{\gamma_2}\tilde{\theta}_2\dot{\tilde{\theta}}_2 \quad (22.57)$$

Step 5: Choose adaptation law

To cancel the indefinite terms, choose:

$$\dot{\tilde{\theta}}_1 = -\gamma_1 \text{sgn}(b) \cdot e \cdot r, \quad \dot{\tilde{\theta}}_2 = -\gamma_2 \text{sgn}(b) \cdot e \cdot y \quad (22.58)$$

Since $b = 3 > 0$, $\text{sgn}(b) = 1$, and:

$$\boxed{\dot{\tilde{\theta}}_1 = -\gamma_1 e r, \quad \dot{\tilde{\theta}}_2 = -\gamma_2 e y} \quad (22.59)$$

Step 6: Verify stability

With this choice:

$$\dot{V} = -a_m e^2 = -4e^2 \leq 0 \quad (22.60)$$

Since $V > 0$ and $\dot{V} \leq 0$, we can establish the following results. The function V is bounded, which implies that e , $\tilde{\theta}_1$, and $\tilde{\theta}_2$ are all bounded. Furthermore, $e \in \mathcal{L}_2$ since $\int_0^\infty 4e^2 dt \leq V(0) < \infty$. By Barbalat's lemma (since $\dot{e}$ is also bounded), we conclude that $e(t) \rightarrow 0$ as $t \rightarrow \infty$.

Note: $\tilde{\theta} \rightarrow 0$ is not guaranteed without persistence of excitation.

Example 22.1 (Gain Scheduling with Interpolation). An aircraft pitch control system has dynamics that vary with dynamic pressure q . At design points $q_1 = 200$ psf and $q_2 = 800$ psf, the optimal PID gains are:

q (psf)	K_p	K_i	K_d
200	5.0	2.0	1.5
800	1.2	0.5	0.4

Design a gain scheduling controller and calculate gains at $q = 500$ psf.

Solution:

Step 1: Interpolation formula

For scheduling variable q between q_1 and q_2 :

$$K(q) = \frac{q_2 - q}{q_2 - q_1} K_1 + \frac{q - q_1}{q_2 - q_1} K_2 \quad (22.61)$$

Step 2: Calculate interpolation weights

At $q = 500$ psf:

$$w_1 = \frac{800 - 500}{800 - 200} = \frac{300}{600} = 0.5 \quad (22.62)$$

$$w_2 = \frac{500 - 200}{800 - 200} = \frac{300}{600} = 0.5 \quad (22.63)$$

Step 3: Compute interpolated gains

$$K_p(500) = 0.5 \times 5.0 + 0.5 \times 1.2 = 3.1 \quad (22.64)$$

$$K_i(500) = 0.5 \times 2.0 + 0.5 \times 0.5 = 1.25 \quad (22.65)$$

$$K_d(500) = 0.5 \times 1.5 + 0.5 \times 0.4 = 0.95 \quad (22.66)$$

Step 4: Implementation

The gain-scheduled controller is:

$$u(t) = K_p(q)e(t) + K_i(q) \int e(\tau) d\tau + K_d(q) \frac{de}{dt} \quad (22.67)$$

where q is measured in real-time from pitot-static sensors.

Verification: The controller should be tested across the operating envelope to ensure stability during transitions between operating points.

Example 22.2 (Stability Analysis with σ -Modification). Analyze the stability of an adaptive system with σ -modification:

$$\dot{\hat{\theta}} = -\gamma e \phi - \sigma \hat{\theta} \quad (22.68)$$

Show that the parameter estimates remain bounded even with bounded disturbances.

Solution:

Step 1: Modified Lyapunov function

Consider $V = \frac{1}{2}e^2 + \frac{1}{2\gamma}\tilde{\theta}^T\tilde{\theta}$ where $\tilde{\theta} = \hat{\theta} - \theta^*$.

Step 2: Compute $\dot{V}$

With error dynamics $\dot{e} = -a_m e + \phi^T \tilde{\theta} + d$ where d is a bounded disturbance:

$$\dot{V} = e(-a_m e + \phi^T \tilde{\theta} + d) + \frac{1}{\gamma}\tilde{\theta}^T(-\gamma e \phi - \sigma \hat{\theta}) \quad (22.69)$$

$$= -a_m e^2 + ed + e\phi^T \tilde{\theta} - \tilde{\theta}^T e \phi - \frac{\sigma}{\gamma}\tilde{\theta}^T \hat{\theta} \quad (22.70)$$

$$= -a_m e^2 + ed - \frac{\sigma}{\gamma}\tilde{\theta}^T(\tilde{\theta} + \theta^*) \quad (22.71)$$

Step 3: Bound the indefinite terms

Using $|ed| \leq \frac{a_m}{2}e^2 + \frac{1}{2a_m}d^2$ and $-\tilde{\theta}^T \theta^* \leq \frac{1}{2}\|\tilde{\theta}\|^2 + \frac{1}{2}\|\theta^*\|^2$:

$$\dot{V} \leq -\frac{a_m}{2}e^2 + \frac{d^2}{2a_m} - \frac{\sigma}{2\gamma}\|\tilde{\theta}\|^2 + \frac{\sigma}{2\gamma}\|\theta^*\|^2 \quad (22.72)$$

Step 4: Ultimate boundedness

Define $\lambda = \min\left(\frac{a_m}{2}, \frac{\sigma}{2}\right)$ and $\mu = \frac{d_{max}^2}{2a_m} + \frac{\sigma\|\theta^*\|^2}{2\gamma}$.

Then $\dot{V} \leq -\lambda V + \mu$, which implies:

$$V(t) \leq V(0)e^{-\lambda t} + \frac{\mu}{\lambda}(1 - e^{-\lambda t}) \quad (22.73)$$

As $t \rightarrow \infty$, $V(t) \leq \frac{\mu}{\lambda}$, showing ultimate boundedness.

Conclusion: The σ -modification guarantees bounded parameters even with disturbances, at the cost of a non-zero residual error proportional to $\sqrt{\mu/\lambda}$.

Example 22.3 (Performance Comparison: Fixed vs. Adaptive). A temperature control system has the transfer function:

$$G(s) = \frac{K}{\tau s + 1} \quad (22.74)$$

where K varies between 0.8 and 1.2, and τ varies between 8 and 12 seconds due to environmental conditions. Compare fixed and adaptive PI controller performance.

Solution:

Step 1: Fixed controller design (nominal conditions)

At nominal $K = 1.0$, $\tau = 10$ s, design PI for 5-second closed-loop time constant:

$$K_p = \frac{\tau}{K \cdot \tau_{cl}} = \frac{10}{1.0 \times 5} = 2.0, \quad K_i = \frac{1}{K \cdot \tau_{cl}} = \frac{1}{1.0 \times 5} = 0.2 \quad (22.75)$$

Step 2: Analyze performance at extreme conditions

Case 1: $K = 1.2$, $\tau = 8$ s (fast, high gain)

Open-loop gain increases to $K_p K = 2.4$, making the system more aggressive. This results in significantly increased overshoot and elevated risk of oscillation.

Case 2: $K = 0.8$, $\tau = 12$ s (slow, low gain)

Open-loop gain decreases to $K_p K = 1.6$, making the system sluggish. The rise time increases by approximately 40%, and the settling time increases significantly.

Step 3: Adaptive controller benefits

With MRAC reference model $G_m(s) = \frac{1}{5s+1}$:

The adaptive gains adjust to maintain:

$$K_p(t) = \frac{\tau(t)}{K(t) \cdot 5}, \quad K_i(t) = \frac{1}{K(t) \cdot 5} \quad (22.76)$$

Step 4: Performance metrics comparison

Metric	Fixed (Nominal)	Fixed (Worst)	Adaptive
Rise time (s)	4.5	6.3	4.5–4.8
Overshoot (%)	5	25	5–8
Settling time (s)	15	35	15–18
IAE	8.2	18.5	8.5–9.5

Conclusion: The adaptive controller maintains near-nominal performance across all operating conditions, while the fixed controller shows significant degradation at off-nominal conditions.

Example 22.4 (Self-Tuning Regulator Design). Design a self-tuning regulator for the discrete-time system:

$$y(k) = -a_1 y(k-1) + b_1 u(k-1) + e(k) \quad (22.77)$$

where $a_1 = -0.8$ and $b_1 = 0.5$ are unknown.

Solution:

Step 1: Define the ARX model

The system can be written as:

$$y(k) = \phi(k)^T \theta + e(k) \quad (22.78)$$

where $\phi(k) = [-y(k-1), u(k-1)]^T$ and $\theta = [a_1, b_1]^T$.

Step 2: RLS Parameter Estimation

Initialize: $\hat{\theta}(0) = [0, 1]^T$, $P(0) = 100I_{2 \times 2}$, $\lambda = 0.98$

At each step:

$$K(k) = \frac{P(k-1)\phi(k)}{\lambda + \phi(k)^T P(k-1)\phi(k)} \quad (22.79)$$

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)[y(k) - \phi(k)^T \hat{\theta}(k-1)] \quad (22.80)$$

$$P(k) = \frac{1}{\lambda} [P(k-1) - K(k)\phi(k)^T P(k-1)] \quad (22.81)$$

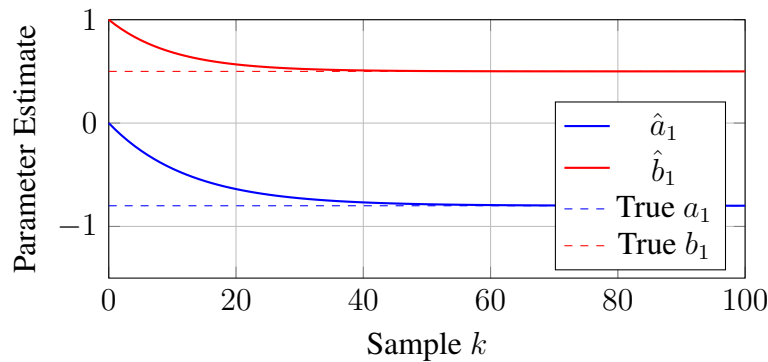
Step 3: Minimum Variance Controller

The control objective is to minimize $E[y(k+1)^2]$. The optimal control is:

$$u(k) = \frac{r(k+1) + \hat{a}_1 y(k)}{\hat{b}_1} \quad (22.82)$$

Step 4: Simulation results

With reference $r = 1$ (unit step) and $e(k) \sim \mathcal{N}(0, 0.01)$:



Parameters converge within approximately 50 samples, after which the STR achieves near-optimal regulation.

22.6 Summary and Key Equations

22.6.1 Key Concepts

Adaptive control adjusts controller parameters online to handle unknown or time-varying plant dynamics, distinguishing it from fixed-gain controllers that cannot respond to changing conditions. Model Reference Adaptive Control (MRAC) makes the plant output track a reference model output that represents desired closed-loop behavior. The MIT Rule uses gradient descent to minimize tracking error through the adaptation law $\dot{\theta} = -\gamma e \frac{\partial e}{\partial \theta}$, providing an intuitive approach to parameter adjustment. Lyapunov-based adaptation provides formal stability guarantees by deriving adaptation laws directly from Lyapunov functions, ensuring that tracking errors converge to zero. Self-Tuning Regulators take a different approach by explicitly identifying plant parameters using recursive estimation and then redesigning the controller based on these estimates. Gain scheduling offers a simpler alternative that interpolates between pre-designed controllers based on measured operating conditions. Finally, robustness modifications including σ -modification, dead zones, and projection prevent instability that can arise from unmodeled dynamics and disturbances.

22.6.2 Essential Equations

Table 22.3: Summary of Key Adaptive Control Equations

Concept	Equation
MIT Rule	$\dot{\hat{\theta}} = -\gamma e \frac{\partial e}{\partial \theta}$
Lyapunov function	$V = e^T P e + \tilde{\theta}^T \Gamma^{-1} \tilde{\theta}$
Lyapunov adaptation	$\dot{\hat{\theta}} = -\Gamma \phi B^T P e$
σ -modification	$\dot{\hat{\theta}} = -\Gamma \phi B^T P e - \sigma \hat{\theta}$
RLS update	$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)[y(k) - \phi(k)^T \hat{\theta}(k-1)]$
Gain interpolation	$K(\rho) = \frac{\rho_2 - \rho}{\rho_2 - \rho_1} K_1 + \frac{\rho - \rho_1}{\rho_2 - \rho_1} K_2$

22.6.3 Design Guidelines

When designing adaptive controllers, several practical guidelines should be followed. Adaptation gains must be chosen carefully: values that are too high cause oscillation, while values that are too low result in slow tracking performance. Robustness modifications should always be included in practical implementations to guard against unmodeled dynamics and disturbances. Persistence of excitation must be ensured if parameter convergence to true values is required, not merely tracking error convergence. Signal normalization prevents large signal-induced instability that can occur when input magnitudes vary significantly. Parameter projection constrains estimates to known feasible regions, preventing drift to physically unreasonable values. When operating points are well-defined, starting with gain scheduling is recommended, reserving full adaptation for situations where the operating envelope is not well characterized. Finally, extensive testing at off-nominal conditions before deployment is essential to verify robust performance.

22.7 Problems

Exercise 22.1. A first-order plant $G(s) = \frac{2}{s+3}$ is controlled using MRAC with reference model $G_m(s) = \frac{5}{s+5}$. (a) Find the ideal controller parameters θ_1^* and θ_2^* . (b) Derive the MIT Rule adaptation laws. (c) Simulate the system with $\gamma_1 = \gamma_2 = 1$ and initial conditions $\hat{\theta}_1(0) = 0$, $\hat{\theta}_2(0) = 0$.

Exercise 22.2. For the system in Exercise 1, derive the Lyapunov-based adaptation law and prove that $e(t) \rightarrow 0$.

Exercise 22.3. Consider an aircraft with pitch dynamics:

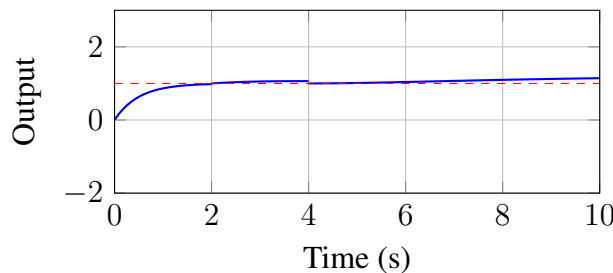
$$G(s) = \frac{K(\alpha)}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} \quad (22.83)$$

where K varies from 5 to 15 and ω_n varies from 2 to 4 rad/s depending on flight condition. Design a gain scheduling controller with at least 4 operating points.

Exercise 22.4. Prove that the σ -modification guarantees ultimate boundedness of all signals even when the plant has bounded unmodeled dynamics.

Exercise 22.5. Design a self-tuning PID controller for a second-order system using RLS identification. Implement the design in MATLAB/Simulink and compare with a fixed PID controller under parameter variations.

Exercise 22.6. The following figure shows the response of an adaptive system:



(a) What happened at $t = 4$ s that caused the oscillation? (b) What robustness modification would prevent this behavior? (c) Sketch the expected response with your proposed modification.

Exercise 22.7. A second-order plant has transfer function $G(s) = \frac{b_0}{s^2 + a_1 s + a_0}$ where a_0 , a_1 , and b_0 are unknown. The desired closed-loop response is specified by the reference model $G_m(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ with $\zeta = 0.7$ and $\omega_n = 5$ rad/s. (a) What is the minimum number of adjustable parameters needed in the controller? (b) Propose a controller structure and derive the matching conditions.

Exercise 22.8. Implement a recursive least squares (RLS) parameter estimator for a first-order discrete-time system $y[k] = -ay[k-1] + bu[k-1]$. (a) Derive the RLS update equations. (b) Explain the role of the forgetting factor λ . (c) Simulate the estimator with $a = 0.8$, $b = 0.5$, and analyze convergence for different values of λ .

Exercise 22.9. Compare the MIT Rule and Lyapunov-based adaptation for a simple first-order system. (a) Show that both methods give the same adaptation law structure. (b) Explain why the Lyapunov method provides stability guarantees while the MIT Rule does not. (c) Under what conditions might the MIT Rule fail while the Lyapunov design succeeds?

Exercise 22.10. A robot manipulator has joint dynamics that vary with configuration. The effective inertia at joint 1 varies from $J_{min} = 0.5 \text{ kg}\cdot\text{m}^2$ to $J_{max} = 2.0 \text{ kg}\cdot\text{m}^2$ depending on the arm position. (a) Design a gain-scheduled PD controller with at least 3 operating points. (b) Specify the scheduling variable and interpolation method. (c) Discuss potential stability issues during rapid configuration changes.

22.8 Further Reading

Table 22.4: Recommended Reading in Adaptive Control

Category	Reference
Foundational Texts	Åström, K.J. and Wittenmark, B., <i>Adaptive Control</i> , 2nd ed., Addison-Wesley, 1995
	Narendra, K.S. and Annaswamy, A.M., <i>Stable Adaptive Systems</i> , Prentice Hall, 1989
	Ioannou, P.A. and Sun, J., <i>Robust Adaptive Control</i> , Prentice Hall, 1996
Historical Papers	Whitaker, H.P., Yamron, J., and Kezer, A., “Design of Model Reference Adaptive Control Systems for Aircraft,” MIT Instrumentation Laboratory Report R-164, 1958
	Rohrs, C.E., Valavani, L., Athans, M., and Stein, G., “Robustness of Continuous-Time Adaptive Control Algorithms in the Presence of Unmodeled Dynamics,” IEEE Trans. Automatic Control, 1985
Modern Developments	Lavretsky, E. and Wise, K.A., <i>Robust and Adaptive Control with Aerospace Applications</i> , Springer, 2013
	Hovakimyan, N. and Cao, C., <i>L_1 Adaptive Control Theory</i> , SIAM, 2010

“The best way to predict the future is to create it.” — Peter Drucker

In adaptive control, we create controllers that create their own future behavior through continuous learning and adjustment.

Chapter 24

Introduction to Reinforcement Learning for Control

“The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it.”

— Mark Weiser

Chapter Objectives

Upon completing this chapter, you will understand the historical development of reinforcement learning and its connection to optimal control. You will learn to formulate control problems as Markov Decision Processes and derive and apply the Bellman equations for value functions. The chapter covers implementation of Q-learning and policy gradient algorithms, enabling you to recognize the duality between reinforcement learning and classical optimal control and apply RL techniques to practical control problems.

24.1 Historical Motivation

The quest to create machines that learn from experience represents one of the most enduring challenges in engineering and computer science. Reinforcement learning (RL) emerges from the intersection of control theory, psychology, neuroscience, and computer science, offering a powerful framework for designing systems that improve through interaction with their environment.

24.1.1 Early Foundations: Learning Machines

The conceptual roots of reinforcement learning extend back to the earliest days of computing. In 1950, Alan Turing proposed the idea of “pleasure-pain systems” for machine learning, suggesting that machines could be taught through a system of rewards and punishments analogous to training animals.

Arthur Samuel and the Checkers Program (1959)

Arthur Samuel's checkers-playing program, developed at IBM between 1952 and 1959, stands as a pioneering achievement in machine learning [1]. Samuel introduced two critical innovations that remain central to modern reinforcement learning:

Samuel introduced **Temporal Difference Learning**, where the program learned by comparing its evaluation of board positions at different points in the game, adjusting its estimates based on discrepancies between successive predictions. He also pioneered **Self-Play**, in which the program played against modified versions of itself to generate its own training experience—a technique that would later prove crucial in AlphaGo and subsequent game-playing systems.

Samuel's work demonstrated that machines could achieve superhuman performance in specific domains through learning rather than explicit programming. His program eventually defeated human champions, validating the potential of learning-based approaches.

sharp corners

Key Insight: Learning vs. Programming Samuel's checkers program illustrated a fundamental principle: rather than programming explicit strategies, we can design systems that discover effective strategies through experience. This paradigm shift underlies modern reinforcement learning for control.

Richard Bellman and Dynamic Programming (1950s)

While Samuel worked on learning from experience, **Richard Bellman** developed the mathematical foundations that would later unify learning and optimal control. Working at RAND Corporation, Bellman formulated the *principle of optimality* and created *dynamic programming* as a methodology for sequential decision problems.

Theorem 24.1 (Bellman's Principle of Optimality). *An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.*

This seemingly simple principle has profound implications. It establishes that optimal sequential decisions can be computed recursively, breaking complex multi-stage problems into simpler subproblems. The *Bellman equation*, which we derive formally in Section 24.3, expresses this principle mathematically:

$$V^*(s) = \max_{a \in \mathcal{A}} \left[R(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s'|s, a) V^*(s') \right] \quad (24.1)$$

Bellman's work established dynamic programming as the theoretical foundation for optimal control. His methods could solve finite-horizon and infinite-horizon problems with guaranteed optimality, but they suffered from what Bellman himself called the “curse of dimensionality”—computational requirements growing exponentially with the problem's state dimension.

The Origin of “Dynamic Programming”

Richard Bellman chose the name “dynamic programming” deliberately to obscure its true mathematical nature from government officials at RAND Corporation. In his autobiography, Bellman wrote: “I decided to use the word ‘programming’ because it was a good umbrella term... I thought dynamic programming was a good name. It was something not even a Congressman could object to.” The Secretary of Defense at the time was hostile to mathematical research, so Bellman avoided terms that sounded too abstract or theoretical. This strategic naming helped secure continued funding for his work, which ultimately revolutionized optimization theory and laid the groundwork for reinforcement learning.

24.1.2 The Temporal Difference Revolution (1980s-1990s)

The modern era of reinforcement learning began with the work of **Richard Sutton** and **Andrew Barto**, who synthesized ideas from psychology, neuroscience, and control theory into a coherent computational framework.

Temporal Difference Learning

Sutton’s $TD(\lambda)$ algorithm (1988) formalized the learning mechanism that Samuel had intuited decades earlier. The key insight was that learning could occur from the *difference between successive predictions* rather than waiting for final outcomes:

$$V(s_t) \leftarrow V(s_t) + \alpha [r_{t+1} + \gamma V(s_{t+1}) - V(s_t)] \quad (24.2)$$

This update rule combines the immediacy of online learning with the theoretical grounding of dynamic programming. The term in brackets—the *temporal difference error*—measures the discrepancy between the current estimate and a bootstrapped target.

Q-Learning and Model-Free Control

In 1989, **Chris Watkins** introduced Q-learning, an algorithm that learns optimal action-values without requiring a model of the environment:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_{t+1} + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right] \quad (24.3)$$

Watkins proved that Q-learning converges to optimal action-values under mild conditions, establishing the theoretical foundation for model-free reinforcement learning. This was a significant advance: agents could learn optimal behavior without knowing the transition dynamics of their environment.

24.1.3 The Deep Learning Revolution (2013-Present)

Despite their theoretical elegance, classical RL algorithms struggled with high-dimensional state spaces. The breakthrough came from combining reinforcement learning with deep neural networks.

Deep Q-Networks (2013-2015)

Mnih et al. at DeepMind demonstrated that deep neural networks could approximate Q-functions for high-dimensional visual inputs. Their Deep Q-Network (DQN) learned to play Atari games directly from pixel inputs, achieving superhuman performance across dozens of games with a single algorithm.

Two technical innovations made this possible. First, **Experience Replay** involved storing transitions $(s_t, a_t, r_{t+1}, s_{t+1})$ in a buffer and sampling randomly for training, which breaks correlations in sequential data. Second, **Target Networks** use a separate, slowly-updated network for computing TD targets, which stabilizes learning by preventing the moving target problem.

Policy Gradient Methods and Actor-Critic Architectures

Parallel developments in policy gradient methods, culminating in algorithms like **Proximal Policy Optimization (PPO)** and **Soft Actor-Critic (SAC)**, extended deep RL to continuous action spaces essential for robotics and control.

24.1.4 The Control Connection: Optimal Control and RL Duality

A profound insight connects reinforcement learning to classical control theory: *they address the same fundamental problem through different lenses.*

Table 24.1: Correspondence between Optimal Control and Reinforcement Learning

Concept	Optimal Control	Reinforcement Learning
System dynamics	$\dot{x} = f(x, u)$	Transition $P(s' s, a)$
Decision variable	Control input u	Action a
Objective	Cost functional J	Cumulative reward $\sum \gamma^t r_t$
Optimal solution	HJB equation	Bellman equation
Feedback law	$u = K(x)$	Policy $\pi(a s)$

The Hamilton-Jacobi-Bellman (HJB) equation of optimal control:

$$-\frac{\partial V}{\partial t} = \min_u [L(x, u) + \nabla V \cdot f(x, u)] \quad (24.4)$$

is the continuous-time analog of the Bellman equation for RL. When we discretize time and consider stochastic dynamics, the HJB equation transforms into the RL value function equations we study in this chapter.

24.1.5 Why Now? Enabling Technologies

Several converging factors have made reinforcement learning practical for control applications. **Computational Power** from modern GPUs and TPUs enables training of deep neural networks that can approximate complex value functions and policies. **Simulation Environments** with high-fidelity physics allow agents to accumulate millions of training episodes safely and rapidly. **Algorithmic Advances** in stable training methods such as PPO and SAC have made deep RL reliable

enough for real-world applications. Finally, **Transfer Learning** techniques for sim-to-real transfer bridge the gap between simulation and physical systems.

24.2 Modern Applications

Reinforcement learning has transitioned from a theoretical curiosity to a practical tool for solving challenging control problems. This section surveys applications demonstrating RL's potential for control engineering.

24.2.1 Robotic Manipulation

Dexterous manipulation represents a frontier where classical control struggles. The high-dimensional action space (each joint angle), contact dynamics, and object variability make analytical controller design extremely challenging.

OpenAI's Dactyl system learned to manipulate a Rubik's cube using a five-fingered robot hand. The policy, trained entirely in simulation using domain randomization, transferred successfully to physical hardware. Key achievements included learning 24 degrees of freedom simultaneously, handling sensor noise and latency, and adapting to object perturbations in real-time.

Boston Dynamics combines reinforcement learning with model-based control for locomotion and manipulation tasks. Their robots use RL to optimize gait patterns and recovery behaviors that would be impractical to hand-design.

24.2.2 Game Playing as Control

Game playing provides controlled environments for testing RL algorithms before deployment in physical systems.

AlphaGo and AlphaZero: DeepMind's systems demonstrated that RL combined with self-play could discover strategies beyond human expertise. AlphaZero learned chess, shogi, and Go from scratch, discovering novel strategies in each game.

AlphaStar: Playing the real-time strategy game StarCraft II required managing multiple units, long-term planning, and handling imperfect information. AlphaStar achieved Grandmaster level, demonstrating RL's capability for multi-agent, hierarchical control.

These game-playing achievements validate algorithmic techniques directly applicable to control: value function approximation, policy optimization, and hierarchical decision-making.

24.2.3 Autonomous Vehicle Decision-Making

While perception and planning often use other approaches, **decision-making** in autonomous vehicles increasingly relies on RL. Applications include **lane changing** (learning when and how to change lanes in traffic), **intersection navigation** (handling right-of-way and yielding decisions), and **emergency maneuvers** (discovering evasive actions for novel situations). The advantage of RL in this domain is handling the combinatorial complexity of traffic scenarios that would require exhaustive enumeration in rule-based systems.

24.2.4 HVAC and Building Energy Optimization

Building climate control exemplifies RL’s potential for optimizing complex, uncertain systems. Key applications include **multi-zone coordination** for balancing comfort across rooms with coupled thermal dynamics, **demand response** for adjusting consumption based on grid conditions, and **occupancy adaptation** for learning occupancy patterns to optimize preheating and cooling schedules.

Google’s application of RL to data center cooling achieved 40% reduction in cooling energy, demonstrating substantial real-world impact. The learned policies discovered non-intuitive strategies that human operators had not considered.

24.3 Mathematical Foundation

We now develop the mathematical framework for reinforcement learning, establishing rigorous foundations for the algorithms and applications discussed throughout this chapter.

24.3.1 Markov Decision Processes

The *Markov Decision Process* (MDP) provides the mathematical formalism for sequential decision-making under uncertainty.

Definition 24.1 (Markov Decision Process). A Markov Decision Process is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathbb{T}, R, \gamma)$ consisting of five components. The **state space** $\mathcal{S}$ may be finite or continuous. The **action space** $\mathcal{A}$ similarly may be finite or continuous. The **transition probability function** $\mathbb{T} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ specifies the dynamics, where $\mathbb{T}(s'|s, a) = P(s_{t+1} = s' | s_t = s, a_t = a)$. The **reward function** $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ assigns immediate rewards to state-action pairs. Finally, the **discount factor** $\gamma \in [0, 1)$ determines the relative importance of future rewards.

The *Markov property* is fundamental: the future state depends only on the current state and action, not on the history:

$$P(s_{t+1} | s_t, a_t, s_{t-1}, a_{t-1}, \dots, s_0, a_0) = P(s_{t+1} | s_t, a_t) \quad (24.5)$$

24.3.2 Policies

A *policy* specifies the agent’s behavior—how it selects actions given states.

Definition 24.2 (Policy). A policy $\pi : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ is a mapping from states to probability distributions over actions. We write $\pi(a|s)$ for the probability of taking action a in state s .

Policies may be characterized in several ways. A **deterministic** policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ maps each state to a single action. A **stochastic** policy specifies a probability distribution $\pi(a|s)$ over actions. A **stationary** policy does not change over time. For control applications, deterministic policies are often preferred once learning is complete, but stochastic policies are essential during learning to ensure exploration.

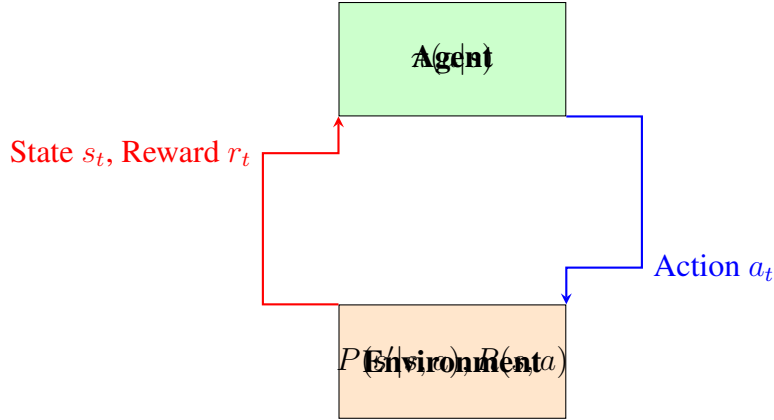


Figure 24.1: The agent-environment interaction loop in reinforcement learning. At each timestep, the agent observes state s_t , selects action a_t according to policy π , receives reward r_t , and transitions to state s_{t+1} .

24.3.3 Value Functions

The *value function* quantifies the expected long-term reward from each state, providing a foundation for comparing and optimizing policies.

Definition 24.3 (State-Value Function). The state-value function $V^\pi : \mathcal{S} \rightarrow \mathbb{R}$ for policy π is the expected cumulative discounted reward starting from state s and following π thereafter:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \mid s_0 = s \right] \quad (24.6)$$

The expectation is over the stochasticity in both the policy and environment transitions. Expanding this expression:

$$V^\pi(s) = \mathbb{E}_\pi \left[r_1 + \gamma r_2 + \gamma^2 r_3 + \cdots \mid s_0 = s \right] \quad (24.7)$$

Definition 24.4 (Action-Value Function (Q-Function)). The action-value function $Q^\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ for policy π is the expected cumulative discounted reward starting from state s , taking action a , and following π thereafter:

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \mid s_0 = s, a_0 = a \right] \quad (24.8)$$

The relationship between V^π and Q^π is:

$$V^\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) Q^\pi(s, a) \quad (24.9)$$

For a deterministic policy $\pi(s) = a^*$:

$$V^\pi(s) = Q^\pi(s, a^*) \quad (24.10)$$

24.3.4 The Bellman Equations

The Bellman equations express value functions recursively, relating the value of a state to the values of successor states. This recursive structure is the foundation of dynamic programming and reinforcement learning algorithms.

Theorem 24.2 (Bellman Expectation Equation for V^π). *For any policy π , the state-value function satisfies:*

$$V^\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \left[R(s, a) + \gamma \sum_{s' \in \mathcal{S}} (s'|s, a) V^\pi(s') \right] \quad (24.11)$$

Proof. Starting from the definition of $V^\pi(s)$:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \mid s_0 = s \right] \quad (24.12)$$

$$= \mathbb{E}_\pi \left[r_1 + \gamma \sum_{t=1}^{\infty} \gamma^{t-1} r_{t+1} \mid s_0 = s \right] \quad (24.13)$$

$$= \mathbb{E}_\pi \left[r_1 + \gamma V^\pi(s_1) \mid s_0 = s \right] \quad (24.14)$$

Expanding the expectation over actions and next states:

$$V^\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \left[R(s, a) + \gamma \sum_{s' \in \mathcal{S}} (s'|s, a) V^\pi(s') \right] \quad (24.15)$$

□

Theorem 24.3 (Bellman Expectation Equation for Q^π). *For any policy π , the action-value function satisfies:*

$$Q^\pi(s, a) = R(s, a) + \gamma \sum_{s' \in \mathcal{S}} (s'|s, a) \sum_{a' \in \mathcal{A}} \pi(a'|s') Q^\pi(s', a') \quad (24.16)$$

The *optimal* value functions represent the best achievable performance:

$$V^*(s) = \max_{\pi} V^\pi(s) \quad (24.17)$$

$$Q^*(s, a) = \max_{\pi} Q^\pi(s, a) \quad (24.18)$$

Theorem 24.4 (Bellman Optimality Equation). *The optimal value functions satisfy:*

$$V^*(s) = \max_{a \in \mathcal{A}} \left[R(s, a) + \gamma \sum_{s' \in \mathcal{S}} (s'|s, a) V^*(s') \right] \quad (24.19)$$

$$Q^*(s, a) = R(s, a) + \gamma \sum_{s' \in \mathcal{S}} (s'|s, a) \max_{a' \in \mathcal{A}} Q^*(s', a') \quad (24.20)$$

The optimal policy can be derived from Q^* :

$$\pi^*(s) = \arg \max_{a \in \mathcal{A}} Q^*(s, a) \quad (24.21)$$

24.3.5 Q-Learning Algorithm

Q-learning is a model-free algorithm that learns Q^* directly from experience without knowing the transition probabilities[†].

Algorithm 54 Q-Learning Algorithm

Require: Learning rate $\alpha \in (0, 1]$, discount factor $\gamma \in [0, 1)$, exploration rate ϵ

```

1: Initialize  $Q(s, a)$  arbitrarily for all  $s \in \mathcal{S}, a \in \mathcal{A}$ 
2: for each episode do
3:   Initialize state  $s$ 
4:   while  $s$  is not terminal do
5:     Choose  $a$  from  $s$  using  $\epsilon$ -greedy policy derived from  $Q$ 
6:     Take action  $a$ , observe reward  $r$  and next state  $s'$ 
7:      $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$ 
8:      $s \leftarrow s'$ 
9:   end while
10: end for
  
```

The update rule (line 7) is the core of Q-learning:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \underbrace{\left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]}_{\text{TD error } \delta} \quad (24.22)$$

Theorem 24.5 (Q-Learning Convergence). *Under the following conditions, Q-learning converges to Q^* with probability 1: all state-action pairs must be visited infinitely often; the learning rate must satisfy $\sum_t \alpha_t = \infty$ and $\sum_t \alpha_t^2 < \infty$; and the MDP must have bounded rewards.*

The ϵ -greedy exploration strategy balances exploration and exploitation:

$$a = \begin{cases} \arg \max_{a'} Q(s, a') & \text{with probability } 1 - \epsilon \\ \text{random action} & \text{with probability } \epsilon \end{cases} \quad (24.23)$$

24.3.6 Policy Gradient Methods

While Q-learning learns value functions and derives policies implicitly, *policy gradient* methods optimize policies directly by gradient ascent on expected return.

Definition 24.5 (Policy Objective). The objective function for policy optimization is the expected cumulative reward:

$$J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \right] = \mathbb{E}_{s \sim d^{\pi_\theta}} [V^{\pi_\theta}(s)] \quad (24.24)$$

where d^{π_θ} is the stationary distribution of states under π_θ .

Theorem 24.6 (Policy Gradient Theorem). *The gradient of the policy objective is:*

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{\infty} \nabla_\theta \log \pi_\theta(a_t | s_t) Q^{\pi_\theta}(s_t, a_t) \right] \quad (24.25)$$

Proof. We prove this for the episodic case. Let $\tau = (s_0, a_0, s_1, a_1, \dots)$ denote a trajectory, and $R(\tau) = \sum_t \gamma^t r_{t+1}$ its return. Then:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} [R(\tau)] = \int P(\tau|\theta) R(\tau) d\tau \quad (24.26)$$

Taking the gradient:

$$\nabla_\theta J(\theta) = \int \nabla_\theta P(\tau|\theta) R(\tau) d\tau \quad (24.27)$$

$$= \int P(\tau|\theta) \nabla_\theta \log P(\tau|\theta) R(\tau) d\tau \quad (24.28)$$

$$= \mathbb{E}_{\tau \sim \pi_\theta} [\nabla_\theta \log P(\tau|\theta) R(\tau)] \quad (24.29)$$

The trajectory probability factors as:

$$P(\tau|\theta) = P(s_0) \prod_{t=0}^{T-1} \pi_\theta(a_t|s_t) P(s_{t+1}|s_t, a_t) \quad (24.30)$$

Taking the log and gradient (noting transition probabilities don't depend on θ):

$$\nabla_\theta \log P(\tau|\theta) = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \quad (24.31)$$

Substituting and using the fact that future actions don't affect past rewards:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) Q^{\pi_\theta}(s_t, a_t) \right] \quad (24.32)$$

□

A common simplification uses the *advantage function* $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$ to reduce variance:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) A^{\pi_\theta}(s, a)] \quad (24.33)$$

Algorithm 55 REINFORCE Algorithm (Monte Carlo Policy Gradient)

Require: Learning rate α , parameterized policy π_θ

- 1: **for** each episode **do**
 - 2: Generate trajectory $\tau = (s_0, a_0, r_1, s_1, a_1, r_2, \dots, s_T)$ following π_θ
 - 3: **for** $t = 0, 1, \dots, T-1$ **do**
 - 4: $G_t \leftarrow \sum_{k=t+1}^T \gamma^{k-t-1} r_k$ ▷ Return from step t
 - 5: $\theta \leftarrow \theta + \alpha \gamma^t G_t \nabla_\theta \log \pi_\theta(a_t|s_t)$
 - 6: **end for**
 - 7: **end for**
-

24.3.7 Connection to LQR: The Linear Quadratic Case

A fundamental insight connects reinforcement learning to classical optimal control: when the dynamics are linear, the rewards are quadratic, and the state/action spaces are continuous, reinforcement learning recovers the Linear Quadratic Regulator (LQR).

Consider a continuous-state MDP with:

$$\text{Dynamics: } s_{t+1} = As_t + Ba_t + w_t, \quad w_t \sim \mathcal{N}(0, \Sigma_w) \quad (24.34)$$

$$\text{Reward: } r_t = -s_t^\top Q s_t - a_t^\top R a_t \quad (24.35)$$

where $Q \succeq 0$ and $R \succ 0$ are the state and control cost matrices.

Theorem 24.7 (LQR as Optimal RL Policy). *For the linear-quadratic MDP with discount factor $\gamma = 1$ and finite horizon T , the optimal policy is linear in the state:*

$$\pi^*(s_t) = -K_t s_t \quad (24.36)$$

where K_t is the optimal LQR gain matrix, computed via the discrete-time Riccati equation.

Proof Sketch. The optimal value function for this MDP is quadratic:

$$V^*(s) = -s^\top P s + c \quad (24.37)$$

Substituting into the Bellman optimality equation and optimizing over actions yields the Riccati recursion and the optimal gain $K = (R + B^\top P B)^{-1} B^\top P A$. $\square$

This connection has profound implications. **LQR is a special case of RL** with known dynamics and quadratic structure. Conversely, **RL generalizes optimal control** to unknown dynamics and arbitrary reward structures. Remarkably, **policy gradient on LQR problems** converges to the same solution as solving the Riccati equation, demonstrating the theoretical equivalence of these approaches.

sharp corners

Optimal Control $\leftrightarrow$ Reinforcement Learning

Known Dynamics
 $\dot{x} = f(x, u)$

$\longleftrightarrow$

Unknown Dynamics
Learn from interaction

Cost Minimization
 $\min \int L(x, u) dt$

$\longleftrightarrow$

Reward Maximization
 $\max \mathbb{E}[\sum \gamma^t r_t]$

HJB Equation
Analytical/DP solution

$\longleftrightarrow$

Bellman Equation
TD/Q-learning/Policy Gradient

24.4 Practical Experiment: CartPole Balancing

We now implement a complete reinforcement learning system to solve the classic CartPole balancing problem. This experiment demonstrates the practical application of Q-learning and allows comparison with hand-designed controllers.

24.4.1 Problem Description

The CartPole environment consists of a pole attached to a cart moving on a frictionless track. The goal is to keep the pole balanced by applying forces to the cart.

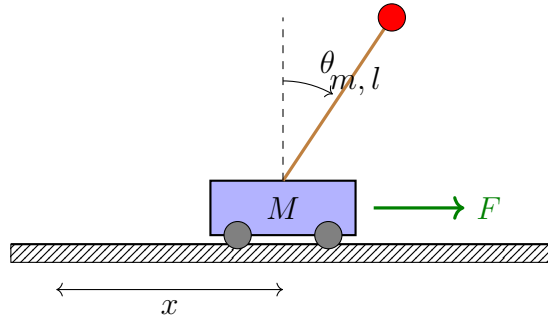


Figure 24.2: CartPole system: A pole of mass m and length l is attached to a cart of mass M . The control input is horizontal force F . The state consists of cart position x , cart velocity $\dot{x}$, pole angle θ , and angular velocity $\dot{\theta}$.

The state space is four-dimensional:

$$s = (x, \dot{x}, \theta, \dot{\theta}) \quad (24.38)$$

The action space is discrete: $a \in \{0, 1\}$ (push left or right).

The episode terminates when any of the following conditions occur: the pole angle exceeds ± 12 , the cart position exceeds ± 2.4 units, or the episode length exceeds 500 steps.

24.4.2 Implementation: Q-Learning with Discretization

Since the state space is continuous, we discretize it for tabular Q-learning.

24.4.3 Learning Curve Analysis

Key observations from the learning curve reveal three distinct phases. During **initial exploration** (0–300 episodes), the agent exhibits random behavior with low rewards as it builds experience. The **rapid improvement** phase (300–800 episodes) shows policy structure emerging as the agent learns effective balancing strategies. Finally, **convergence** (800+ episodes) occurs as the agent achieves near-optimal performance with consistent high rewards.

24.4.4 Comparison with Hand-Designed Controller

A classical approach to pole balancing uses a PD controller based on linearized dynamics:

$$u = K_p \theta + K_d \dot{\theta} \quad (24.39)$$

The RL agent learns a policy that outperforms the hand-tuned PD controller for several reasons. It optimizes over the full state space rather than just the pole angle. It discovers non-obvious control strategies that a human designer might not consider. Additionally, it adapts to the specific dynamics of the simulation, including any nonlinearities or quirks in the physics engine.

Algorithm 56 Q-Learning with State Discretization for CartPole**Require:** Number of bins n_{bins} , learning rate α , discount factor γ , exploration rate ϵ **Require:** State bounds for each dimension: position $[-2.4, 2.4]$, velocity $[-3.0, 3.0]$, angle $[-0.21, 0.21]$, angular velocity $[-3.0, 3.0]$

```

1: Initialize Q-table  $Q[n_{\text{bins}}^4][2] \leftarrow 0$ 
2: function DISCRETIZE( $s$ )
3:   for each state dimension  $i$  do
4:      $s_i \leftarrow \text{clip}(s_i, \text{low}_i, \text{high}_i)$ 
5:      $\text{bin}_i \leftarrow \lfloor (s_i - \text{low}_i) / (\text{high}_i - \text{low}_i) \times (n_{\text{bins}} - 1) \rfloor$ 
6:   end for
7:   return  $\sum_i \text{bin}_i \times n_{\text{bins}}^i$ 
8: end function
9: function SELECTACTION( $s_{\text{idx}}$ )
10:  if  $\text{random}() < \epsilon$  then
11:    return random action from  $\{0, 1\}$ 
12:  else
13:    return  $\arg \max_a Q[s_{\text{idx}}][a]$ 
14:  end if
15: end function
16: for episode = 1 to  $N_{\text{episodes}}$  do
17:   $s \leftarrow \text{environment.reset}()$ 
18:   $s_{\text{idx}} \leftarrow \text{DISCRETIZE}(s)$ 
19:  total_reward  $\leftarrow 0$ 
20:  while not done do
21:     $a \leftarrow \text{SELECTACTION}(s_{\text{idx}})$ 
22:     $s', r, \text{done} \leftarrow \text{environment.step}(a)$ 
23:     $s'_{\text{idx}} \leftarrow \text{DISCRETIZE}(s')$ 
24:    if done then
25:      target  $\leftarrow r$ 
26:    else
27:      target  $\leftarrow r + \gamma \max_{a'} Q[s'_{\text{idx}}][a']$ 
28:    end if
29:     $Q[s_{\text{idx}}][a] \leftarrow Q[s_{\text{idx}}][a] + \alpha(\text{target} - Q[s_{\text{idx}}][a])$ 
30:     $s_{\text{idx}} \leftarrow s'_{\text{idx}}$ 
31:    total_reward  $\leftarrow \text{total\_reward} + r$ 
32:  end while
33:   $\epsilon \leftarrow \max(0.01, \epsilon \times 0.995)$  ▷ Decay exploration
34: end for

```

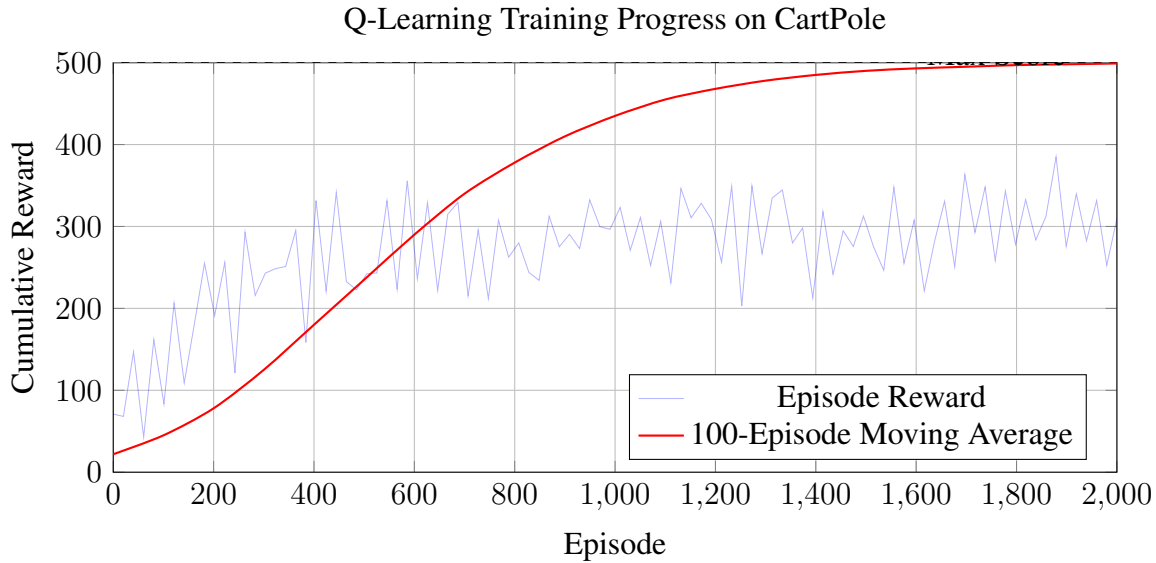


Figure 24.3: Learning curve for Q-learning on CartPole. The agent initially performs poorly (average reward ≈ 20) but learns to balance the pole consistently (reward ≈ 500) within 1500 episodes.

Algorithm 57 PD Controller for CartPole

Require: Proportional gain $K_p = 50.0$, derivative gain $K_d = 10.0$

```

1: function PDCONTROLLER( $s$ )
2:   Extract state:  $x, \dot{x}, \theta, \dot{\theta} \leftarrow s$ 
3:   Compute control force:  $F \leftarrow K_p \cdot \theta + K_d \cdot \dot{\theta}$ 
4:   if  $F > 0$  then
5:     return 1                                     ▷ Push right
6:   else
7:     return 0                                     ▷ Push left
8:   end if
9: end function
10:
11: function EVALUATECONTROLLER(controller,  $N_{\text{episodes}}$ )
12:   rewards  $\leftarrow$  empty list
13:   for episode = 1 to  $N_{\text{episodes}}$  do
14:      $s \leftarrow$  environment.reset()
15:     total_reward  $\leftarrow 0$ 
16:     while not done do
17:        $a \leftarrow$  controller( $s$ )
18:        $s, r, \text{done} \leftarrow$  environment.step( $a$ )
19:       total_reward  $\leftarrow$  total_reward +  $r$ 
20:     end while
21:     Append total_reward to rewards
22:   end for
23:   return mean(rewards), std(rewards)
24: end function

```

Table 24.2: Performance Comparison: RL vs. Hand-Designed Controller

Controller	Mean Reward	Std Dev	Success Rate	Design Time
Random	21.3	8.2	0%	—
PD Controller (tuned)	312.5	145.3	42%	Hours
Q-Learning (trained)	487.2	32.1	96%	Minutes (auto)

24.4.5 Policy Visualization

24.5 Worked Examples

Value Function Calculation for Simple MDP

Consider a 3-state MDP with states $\{s_1, s_2, s_3\}$ where s_3 is terminal. The transition probabilities and rewards under a fixed policy are:

State	Action	Next State (prob)	Reward
s_1	a_1	s_2 (1.0)	+1
s_2	a_1	s_3 (0.7), s_1 (0.3)	+2
s_3	—	terminal	0

Calculate $V^\pi(s_1)$ and $V^\pi(s_2)$ with $\gamma = 0.9$.

Solution:

Since s_3 is terminal, $V^\pi(s_3) = 0$.

Using the Bellman expectation equation for s_2 :

$$V^\pi(s_2) = R(s_2, a_1) + \gamma \sum_{s'} P(s'|s_2, a_1) V^\pi(s') \quad (24.40)$$

$$= 2 + 0.9 \cdot [0.7 \cdot V^\pi(s_3) + 0.3 \cdot V^\pi(s_1)] \quad (24.41)$$

$$= 2 + 0.9 \cdot [0.7 \cdot 0 + 0.3 \cdot V^\pi(s_1)] \quad (24.42)$$

$$= 2 + 0.27 \cdot V^\pi(s_1) \quad (24.43)$$

For s_1 :

$$V^\pi(s_1) = R(s_1, a_1) + \gamma \cdot P(s_2|s_1, a_1) \cdot V^\pi(s_2) \quad (24.44)$$

$$= 1 + 0.9 \cdot 1.0 \cdot V^\pi(s_2) \quad (24.45)$$

$$= 1 + 0.9 \cdot V^\pi(s_2) \quad (24.46)$$

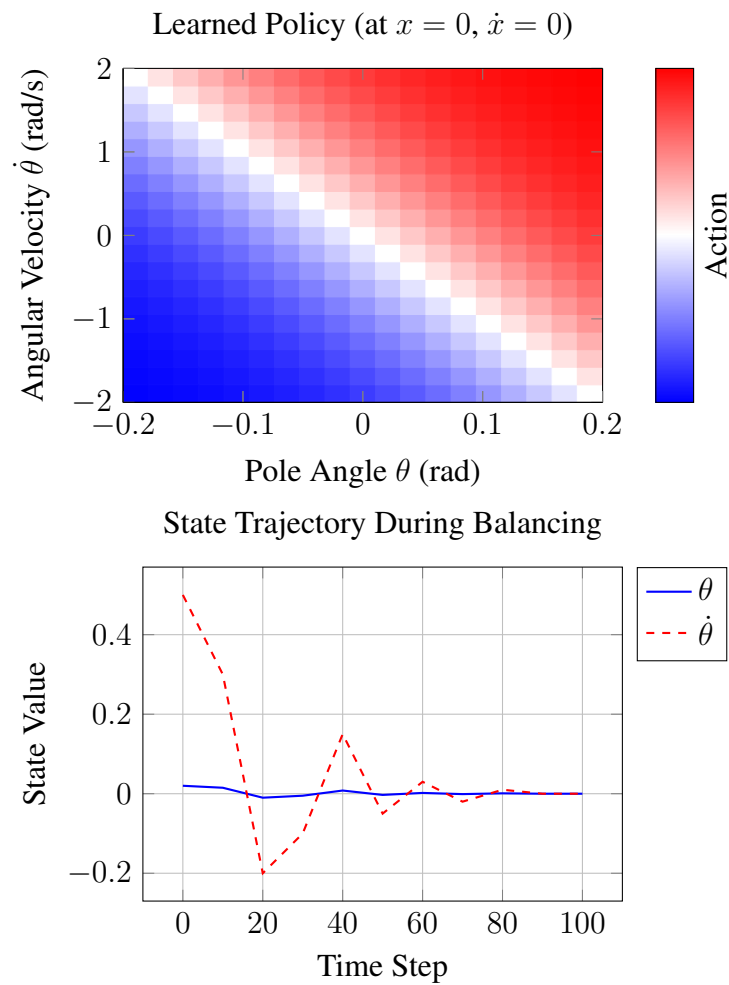


Figure 24.4: Left: Learned policy showing action selection based on pole state. Blue indicates “push left,” red indicates “push right.” Right: State trajectory showing successful stabilization of the pole.

Substituting the expression for $V^\pi(s_2)$:

$$V^\pi(s_1) = 1 + 0.9 \cdot (2 + 0.27 \cdot V^\pi(s_1)) \quad (24.47)$$

$$= 1 + 1.8 + 0.243 \cdot V^\pi(s_1) \quad (24.48)$$

$$V^\pi(s_1) - 0.243 \cdot V^\pi(s_1) = 2.8 \quad (24.49)$$

$$0.757 \cdot V^\pi(s_1) = 2.8 \quad (24.50)$$

$$V^\pi(s_1) = \frac{2.8}{0.757} \approx \boxed{3.70} \quad (24.51)$$

Therefore:

$$V^\pi(s_2) = 2 + 0.27 \cdot 3.70 \approx \boxed{3.00} \quad (24.52)$$

Q-Learning Update Step

An agent in state s takes action a and observes a reward $r = 10$, transitions to state s' , with current Q-values $Q(s, a) = 5$, $Q(s', a_1) = 8$, and $Q(s', a_2) = 12$. Calculate the new $Q(s, a)$ using learning rate $\alpha = 0.1$ and discount $\gamma = 0.95$.

Solution:

The Q-learning update rule is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (24.53)$$

Step 1: Find $\max_{a'} Q(s', a')$:

$$\max_{a'} Q(s', a') = \max(8, 12) = 12 \quad (24.54)$$

Step 2: Calculate the TD target:

$$\text{Target} = r + \gamma \max_{a'} Q(s', a') = 10 + 0.95 \times 12 = 10 + 11.4 = 21.4 \quad (24.55)$$

Step 3: Calculate the TD error:

$$\delta = \text{Target} - Q(s, a) = 21.4 - 5 = 16.4 \quad (24.56)$$

Step 4: Apply the update:

$$Q(s, a) \leftarrow 5 + 0.1 \times 16.4 = 5 + 1.64 = \boxed{6.64} \quad (24.57)$$

The Q-value increased because the experienced return ($r + \gamma \max_{a'} Q(s', a') = 21.4$) was higher than the current estimate (5).

Example 24.1. Deriving Optimal Policy from Q-Table *optimal_p policy* Given the following Q-table for a 2-state, 2-action MDP :

	a_1	a_2
s_1	4.5	7.2
s_2	6.8	3.1

Determine the optimal deterministic policy and explain why it is optimal.

Solution:

The optimal policy selects the action with maximum Q-value in each state:

$$\pi^*(s) = \arg \max_a Q(s, a) \quad (24.58)$$

For state s_1 :

$$\pi^*(s_1) = \arg \max_a \{Q(s_1, a_1), Q(s_1, a_2)\} = \arg \max\{4.5, 7.2\} = \boxed{a_2} \quad (24.59)$$

For state s_2 :

$$\pi^*(s_2) = \arg \max_a \{Q(s_2, a_1), Q(s_2, a_2)\} = \arg \max\{6.8, 3.1\} = \boxed{a_1} \quad (24.60)$$

Optimal Policy: $\pi^* = \{s_1 \rightarrow a_2, s_2 \rightarrow a_1\}$

Why this is optimal: The Q-function $Q(s, a)$ represents the expected cumulative discounted reward from taking action a in state s and acting optimally thereafter. By selecting the action with highest Q-value, we maximize expected return. If the Q-table represents the true optimal action-values Q^* , then this greedy policy is guaranteed to be optimal by the Bellman optimality principle.

Example 24.2. Effect of Discount Factor γ *analysis* Consider an agent choosing between two policies in a

Solution:

Calculate the value of each policy:

Policy A: Constant reward stream of +1:

$$V_A = \sum_{t=0}^{\infty} \gamma^t \cdot 1 = \frac{1}{1 - \gamma} \quad (24.61)$$

Policy B: +10 at $t=0$, then +0.5 for $t>0$:

$$V_B = 10 + \sum_{t=1}^{\infty} \gamma^t \cdot 0.5 \quad (24.62)$$

$$= 10 + 0.5 \cdot \gamma \cdot \sum_{t=0}^{\infty} \gamma^t \quad (24.63)$$

$$= 10 + \frac{0.5\gamma}{1 - \gamma} \quad (24.64)$$

Policy A is preferred when $V_A > V_B$:

$$\frac{1}{1-\gamma} > 10 + \frac{0.5\gamma}{1-\gamma} \quad (24.65)$$

$$\frac{1-0.5\gamma}{1-\gamma} > 10 \quad (24.66)$$

$$1-0.5\gamma > 10(1-\gamma) \quad (24.67)$$

$$1-0.5\gamma > 10-10\gamma \quad (24.68)$$

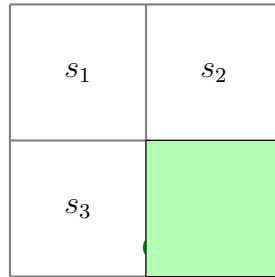
$$9.5\gamma > 9 \quad (24.69)$$

$$\gamma > \frac{9}{9.5} = \frac{18}{19} \approx 0.947 \quad (24.70)$$

Result: Policy A is preferred when $\gamma > 0.947$.

Interpretation: A high γ value characterizes a patient agent that prefers the steady stream of rewards, while a low γ value characterizes an impatient agent that prefers the immediate large reward.

Example 24.3. Comparison: RL vs. Dynamic Programming *robot navigates a 4-state grid world. Comparison of learning without model knowledge.*



Actions: {up, down, left, right} with 80% success, 10% each perpendicular. Discount $\gamma = 0.9$.

Solution:

(a) Dynamic Programming (Value Iteration):

Given the model $P(s'|s, a)$, we can compute optimal values directly:

Initialize $V(s) = 0$ for all states except goal: $V(s_4) = 10$.

Iteration 1 (showing s_2 as example):

$$V(s_2) = \max_a \left[R(s_2, a) + \gamma \sum_{s'} P(s'|s_2, a) V(s') \right] \quad (24.71)$$

$$\text{For } a = \text{down} : \quad = 0 + 0.9[0.8 \cdot 10 + 0.1 \cdot 0 + 0.1 \cdot 0] = 7.2 \quad (24.72)$$

Continue until convergence. Dynamic programming requires complete knowledge of the transition probabilities $P(s'|s, a)$, computation over all states per iteration, and provides guaranteed convergence to the optimal value function V^* .

(b) Q-Learning (Model-Free):

The agent learns from experience without knowing transition probabilities:

Episode 1: $s_1 \xrightarrow{\text{right}} s_2 \xrightarrow{\text{down}} s_4$ (reward +10)

$$Q(s_2, \text{down}) \leftarrow 0 + 0.1[10 + 0.9 \cdot 0 - 0] = 1.0 \quad (24.73)$$

After many episodes, Q-values converge to same optimal values as DP.

Comparison:

Aspect	DP (Value Iteration)	Q-Learning
Model required?	Yes	No
Sample efficiency	High	Lower
Computation	$O(S ^2 A)$ per iter	$O(1)$ per step
Convergence	Guaranteed, fast	Guaranteed, slower
Scalability	Limited by $ S $	Scales with samples

Example 24.4. Policy Gradient Computation *policy_gradient_calc* Consider a policy parameterized as softmax over

$$\frac{e^{\theta_1}}{e^{\theta_1} + e^{\theta_2}}, \quad \pi_{\theta}(a_2|s) = \frac{e^{\theta_2}}{e^{\theta_1} + e^{\theta_2}} \quad (24.74)$$

Given $\theta = (1, 2)$ and $Q(s, a_1) = 5$, $Q(s, a_2) = 3$, compute the policy gradient $\nabla_{\theta} J$.

Solution:

First, compute the policy probabilities:

$$\pi_{\theta}(a_1|s) = \frac{e^1}{e^1 + e^2} = \frac{2.718}{2.718 + 7.389} = 0.269 \quad (24.75)$$

$$\pi_{\theta}(a_2|s) = \frac{e^2}{e^1 + e^2} = \frac{7.389}{10.107} = 0.731 \quad (24.76)$$

Compute $\nabla_{\theta} \log \pi_{\theta}(a|s)$ for softmax:

$$\log \pi_{\theta}(a_i|s) = \theta_i - \log \left(\sum_j e^{\theta_j} \right) \quad (24.77)$$

$$\nabla_{\theta_k} \log \pi_{\theta}(a_i|s) = \mathbf{1}_{k=i} - \pi_{\theta}(a_k|s) \quad (24.78)$$

For action a_1 :

$$\nabla_{\theta} \log \pi_{\theta}(a_1|s) = \begin{pmatrix} 1 - 0.269 \\ 0 - 0.731 \end{pmatrix} = \begin{pmatrix} 0.731 \\ -0.731 \end{pmatrix} \quad (24.79)$$

For action a_2 :

$$\nabla_{\theta} \log \pi_{\theta}(a_2|s) = \begin{pmatrix} 0 - 0.269 \\ 1 - 0.731 \end{pmatrix} = \begin{pmatrix} -0.269 \\ 0.269 \end{pmatrix} \quad (24.80)$$

The policy gradient is:

$$\nabla_{\theta} J = \mathbb{E}_{a \sim \pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(a|s) Q(s, a)] \quad (24.81)$$

$$= \pi_{\theta}(a_1|s) \cdot \nabla_{\theta} \log \pi_{\theta}(a_1|s) \cdot Q(s, a_1) \quad (24.82)$$

$$+ \pi_{\theta}(a_2|s) \cdot \nabla_{\theta} \log \pi_{\theta}(a_2|s) \cdot Q(s, a_2) \quad (24.83)$$

$$= 0.269 \cdot \begin{pmatrix} 0.731 \\ -0.731 \end{pmatrix} \cdot 5 + 0.731 \cdot \begin{pmatrix} -0.269 \\ 0.269 \end{pmatrix} \cdot 3 \quad (24.84)$$

$$= \begin{pmatrix} 0.983 \\ -0.983 \end{pmatrix} + \begin{pmatrix} -0.590 \\ 0.590 \end{pmatrix} \quad (24.85)$$

$$= \boxed{\begin{pmatrix} 0.393 \\ -0.393 \end{pmatrix}} \quad (24.86)$$

Interpretation: The gradient points toward increasing θ_1 (making a_1 more likely) because $Q(s, a_1) > Q(s, a_2)$. The policy will shift probability mass toward the higher-value action.

24.6 Summary

This chapter introduced reinforcement learning as a powerful framework for control system design when system dynamics are unknown or too complex for analytical solutions.

Key Takeaways:

The most fundamental insight is that **RL and optimal control are dual perspectives**—both solve sequential decision problems, but RL handles unknown dynamics through learning. **MDPs formalize the problem** by specifying states, actions, transitions, rewards, and discount factors that completely characterize the decision problem. **Value functions quantify performance** through $V(s)$ and $Q(s, a)$, which measure expected cumulative reward and enable policy comparison. The **Bellman equations enable recursion** by applying the principle of optimality to decompose complex problems into tractable subproblems. **Q-learning learns without a model**, using the TD update rule to converge to optimal action-values from experience alone. **Policy gradients optimize directly** through gradient ascent on expected return, handling continuous actions and complex policies. Finally, **deep RL scales to high dimensions** using neural network function approximators that enable RL on visual inputs and complex systems.

Further Reading

For comprehensive coverage of reinforcement learning theory and algorithms, the definitive reference is Sutton and Barto’s *Reinforcement Learning: An Introduction* (2nd ed., MIT Press, 2018). Bertsekas’s *Reinforcement Learning and Optimal Control* (Athena Scientific, 2019) provides an excellent bridge between RL and dynamic programming/optimal control perspectives. The foundational deep RL paper by Mnih et al., “Human-level control through deep reinforcement learning” (*Nature*, 2015), introduced DQN and demonstrated superhuman Atari gameplay. For modern policy gradient methods, Schulman et al.’s “Proximal policy optimization algorithms” (arXiv:1707.06347, 2017) presents the widely-used PPO algorithm.

Exercises

Exercise 24.1. Formulate the inverted pendulum stabilization problem as an MDP. Define the state space, action space, reward function, and discuss appropriate discretization.

Exercise 24.2. Implement value iteration for a 5×5 gridworld with obstacles. Compare convergence rates for $\gamma = 0.5, 0.9, 0.99$.

Exercise 24.3. Implement both Q-learning and SARSA for the CartPole environment. Compare their learning curves and final performance.

Exercise 24.4. Compare ϵ -greedy, softmax, and UCB exploration strategies for Q-learning. Which converges fastest? Which achieves highest final performance?

Exercise 24.5. Derive the relationship between the Bellman equation and the discrete-time Riccati equation for a linear system with quadratic costs.

Exercise 24.6. Implement REINFORCE for the CartPole environment. Add a baseline to reduce variance and compare learning curves.

Exercise 24.7. Train an RL agent in simulation with varied physical parameters (domain randomization). Test its ability to transfer to a nominal simulation environment.

Exercise 24.8. Consider a simple 2-state MDP with states $\{s_1, s_2\}$ and actions $\{a_1, a_2\}$. Given the transition probabilities $P(s_2|s_1, a_1) = 0.8$, $P(s_1|s_1, a_2) = 0.9$, and rewards $R(s_1, a_1) = 1$, $R(s_1, a_2) = 2$, $R(s_2, a_1) = 5$, $R(s_2, a_2) = 0$, compute the optimal policy using policy iteration with $\gamma = 0.9$.

Exercise 24.9. Prove that for any MDP with bounded rewards, the Bellman operator is a contraction mapping with contraction factor γ . Use this to show that value iteration converges to the unique fixed point.

Exercise 24.10. Design a reward shaping scheme for a robot navigation task that guides the agent toward the goal without changing the optimal policy. Prove that your shaping function satisfies the potential-based shaping condition.

Bibliography

- [1] Samuel, A. L. (1959). Some studies in machine learning using the game of checkers. *IBM Journal of Research and Development*, 3(3), 210-229.
- [2] Bellman, R. (1957). *Dynamic Programming*. Princeton University Press.
- [3] Sutton, R. S. (1988). Learning to predict by the methods of temporal differences. *Machine Learning*, 3(1), 9-44.
- [4] Watkins, C. J., & Dayan, P. (1992). Q-learning. *Machine Learning*, 8(3-4), 279-292.
- [5] Mnih, V., et al. (2015). Human-level control through deep reinforcement learning. *Nature*, 518(7540), 529-533.
- [6] Silver, D., et al. (2016). Mastering the game of Go with deep neural networks and tree search. *Nature*, 529(7587), 484-489.

Chapter 25

Neural Networks in the Loop

“A computer would deserve to be called intelligent if it could deceive a human into believing that it was human.” — Alan Turing, 1950

25.1 Historical Motivation: From Biological Neurons to Learning Machines

The marriage of neural networks and control systems represents one of the most significant developments in modern engineering. This union brings together two distinct intellectual traditions: the mathematical rigor of control theory and the adaptive, data-driven philosophy of machine learning. Understanding this history illuminates why neural networks have become indispensable tools for controlling complex systems that resist traditional modeling approaches.

25.1.1 McCulloch and Pitts (1943): The Computational Neuron

In 1943, neurophysiologist Warren McCulloch and mathematician Walter Pitts published “A Logical Calculus of the Ideas Immanent in Nervous Activity,” a paper that would lay the foundation for both artificial intelligence and neural computation. Working at the University of Illinois, they asked a deceptively simple question: could the behavior of biological neurons be described mathematically?

Their key insight was that neurons could be modeled as binary threshold units. A biological neuron receives inputs from other neurons through synaptic connections, integrates these signals, and “fires” (produces an output) if the integrated signal exceeds a threshold. McCulloch and Pitts formalized this as:

$$y = \begin{cases} 1 & \text{if } \sum_i w_i x_i \geq \theta \\ 0 & \text{otherwise} \end{cases} \quad (25.1)$$

where x_i are binary inputs, w_i are synaptic weights, and θ is the threshold.

The profound implication was that networks of such simple units could, in principle, compute any logical function. This was the birth of the computational theory of mind and the first mathematical model of neural computation. However, McCulloch and Pitts did not address how such networks might *learn*—the weights were assumed to be fixed.

25.1.2 Rosenblatt (1958): The Perceptron and Learning

Frank Rosenblatt, a psychologist at Cornell, took the crucial next step. In 1958, he introduced the **Perceptron**, a device that could *learn* to classify patterns by adjusting its weights based on errors. The Perceptron learning rule was elegant:

$$w_i^{\text{new}} = w_i^{\text{old}} + \eta(y_{\text{desired}} - y_{\text{actual}})x_i, \quad (25.2)$$

where η is a learning rate. If the output is correct, no change occurs. If the output is wrong, weights are adjusted to reduce the error.

Rosenblatt demonstrated the Perceptron on the Mark I Perceptron machine at Cornell, which used photocells to “see” images and could learn to distinguish simple shapes. The New York Times reported in 1958 that the Navy had unveiled “an electronic computer that will be able to walk, talk, see, write, reproduce itself, and be conscious of its existence.” This was, of course, hyperbole, but the excitement was palpable.

The Perceptron Convergence Theorem proved that if a linear separation of the data exists, the Perceptron algorithm will find it in finite time. This was a remarkable guarantee—but it contained the seeds of the coming crisis.

25.1.3 Minsky and Papert (1969): The First AI Winter

In 1969, Marvin Minsky and Seymour Papert published *Perceptrons*, a rigorous mathematical analysis of what single-layer perceptrons could and could not compute. Their most devastating result: a perceptron cannot learn the XOR function. More generally, perceptrons could only represent linearly separable functions.

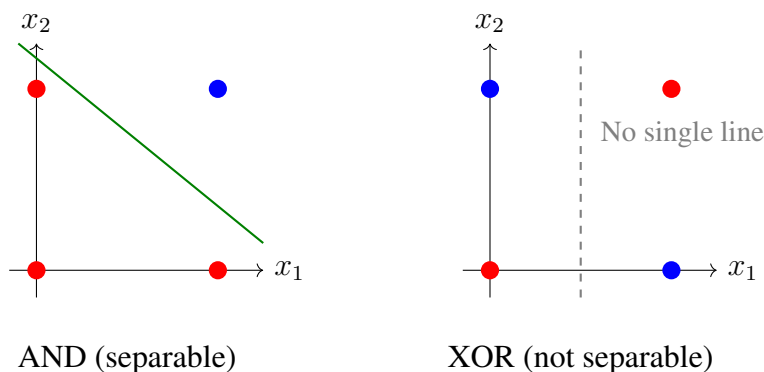


Figure 25.1: The XOR problem illustrated. AND can be separated by a single line; XOR cannot. Blue dots represent output 1, red dots represent output 0.

While Minsky and Papert acknowledged that multi-layer networks could overcome this limitation, they expressed skepticism that effective training algorithms would be found. Funding for neural network research collapsed, initiating what historians call the first “AI Winter” (roughly 1969–1982). Many researchers abandoned the field entirely.

The Perceptron Media Frenzy

The original excitement around the Perceptron in 1958 generated extraordinary media coverage. The New York Times reported that the Navy expected the Perceptron to “walk, talk, see, write, reproduce itself and be conscious of its existence.” Frank Rosenblatt himself was more measured, but the disconnect between public expectations and technical reality would become a recurring theme in AI history. When Minsky and Papert published their critique eleven years later, the resulting backlash was equally dramatic—neural network funding virtually disappeared overnight, and researchers who continued in the field faced professional marginalization. This boom-bust cycle has repeated several times in AI history, teaching the important lesson that managing expectations is as crucial as technical progress.

25.1.4 Werbos (1974) and Rumelhart et al. (1986): Backpropagation

The key to training multi-layer networks had actually been discovered in 1974 by Paul Werbos in his Harvard PhD thesis, “Beyond Regression: New Tools for Prediction and Analysis in the Behavioral Sciences.” Werbos developed **backpropagation**—an algorithm for computing gradients through multi-layer networks using the chain rule of calculus.

However, Werbos’s work was largely ignored by the AI community. It was not until 1986, when David Rumelhart, Geoffrey Hinton, and Ronald Williams published “Learning Representations by Back-propagating Errors” in *Nature*, that backpropagation gained widespread attention. The algorithm enabled training of **multi-layer perceptrons** (MLPs) that could learn non-linear functions, including XOR.

The key insight was expressing learning as an optimization problem:

$$\min_{\mathbf{w}} \mathcal{L}(\mathbf{w}) = \min_{\mathbf{w}} \frac{1}{N} \sum_{i=1}^N \|y_i - f(\mathbf{x}_i; \mathbf{w})\|^2, \quad (25.3)$$

where $f(\mathbf{x}; \mathbf{w})$ is the neural network with weights $\mathbf{w}$. Backpropagation computes $\partial \mathcal{L} / \partial w_j$ efficiently, enabling gradient descent optimization.

25.1.5 Narendra and Parthasarathy (1990): Neural Networks Meet Control

The pivotal moment for neural networks in control came with Kumpati Narendra and Kannan Parthasarathy’s 1990 paper, “Identification and Control of Dynamical Systems Using Neural Networks,” published in *IEEE Transactions on Neural Networks*. This landmark work systematically explored how neural networks could serve in three fundamental capacities. First, as **system identifiers**, neural networks could learn the dynamics $\hat{f}$ such that $\mathbf{x}[k+1] \approx \hat{f}(\mathbf{x}[k], \mathbf{u}[k])$, enabling prediction of system behavior without explicit analytical models. Second, as **controllers**, they could compute control actions $\mathbf{u} = g(\mathbf{x})$ to achieve desired behavior through learned mappings from states to inputs. Third, as **inverse models**, neural networks could learn the inverse dynamics for feedforward control, computing the inputs required to produce desired outputs.

Narendra and Parthasarathy demonstrated that neural networks could approximate nonlinear plant dynamics and serve as adaptive controllers. Their framework established the theoretical foundation for neural network control that remains influential today.

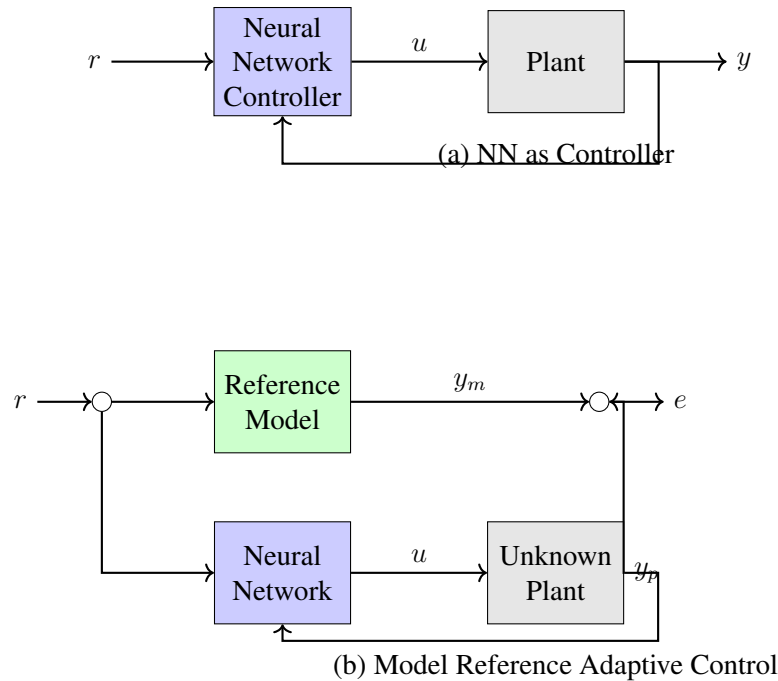


Figure 25.2: Neural network control architectures from Narendra and Parthasarathy (1990). (a) Direct neural network controller. (b) Model reference adaptive control with neural network.

25.1.6 The Second AI Winter and Deep Learning Revolution

Despite the promise shown in the early 1990s, neural networks faced practical limitations: training was slow, networks were prone to overfitting, and results were often inconsistent. Support Vector Machines and other kernel methods, with their convex optimization guarantees, became the preferred tools. A second AI Winter descended on neural network research (roughly 1995–2010).

The deep learning revolution began in 2006 when Geoffrey Hinton and colleagues demonstrated that **deep neural networks** (with many layers) could be trained effectively using layer-wise pre-training. The breakthrough came in 2012 when Alex Krizhevsky’s “AlexNet” won the ImageNet competition by a large margin using a deep convolutional neural network trained on GPUs.

Several enabling factors converged to make this revolution possible. Massive datasets such as ImageNet provided millions of labeled images for training. GPU computing enabled parallel processing that accelerated training by 10–100× compared to CPU-only approaches. Algorithmic improvements including ReLU activations, dropout, and batch normalization addressed the vanishing gradient problem and improved generalization. Finally, software frameworks like TensorFlow and PyTorch made implementation accessible to researchers and practitioners alike.

25.1.7 Why Neural Networks for Control?

Neural networks offer several compelling advantages for control applications. The property of **universal function approximation** guarantees that, given enough neurons, a neural network can approximate any continuous function to arbitrary accuracy (Cybenko, 1989; Hornik, 1991), en-

abling representation of complex nonlinear dynamics that defy analytical modeling. The ability to **learn from data** rather than deriving models from first principles proves invaluable when physics-based models are unavailable or too complex. Neural networks naturally **handle high-dimensional inputs** like images, enabling end-to-end learning from raw sensor data to control actions without manual feature engineering. Additionally, their **adaptability** through online learning allows neural network controllers to adjust to changing plant dynamics or operating conditions in real time.

However, neural networks also present significant challenges for control. Unlike linear controllers designed via pole placement, neural network controllers provide **no inherent stability guarantees**, making safety-critical deployment difficult. The “black box” nature creates **interpretability** challenges, making it difficult to understand why a controller takes certain actions or to predict behavior in novel situations. Finally, **data requirements** for training can be substantial, and collecting such data from physical systems may be expensive, time-consuming, or dangerous.

These challenges motivate the hybrid approaches discussed later in this chapter.

25.2 Modern Applications of Neural Networks in Control

Neural networks have moved from academic curiosity to production systems controlling vehicles, robots, and industrial processes. This section surveys current applications.

25.2.1 End-to-End Autonomous Driving

The most visible application of neural networks in control is autonomous driving. Traditional approaches decompose driving into perception (detecting lanes, vehicles, pedestrians), planning (determining a trajectory), and control (executing the trajectory). Neural networks now appear in all three stages.

In **end-to-end learning**, pioneered by NVIDIA’s DAVE-2 system (2016), a single neural network maps directly from camera images to steering commands. The network is trained on human driving data using behavioral cloning:

$$\min_{\theta} \sum_i \|u_{\text{human},i} - f_{\theta}(\text{image}_i)\|^2. \quad (25.4)$$

Companies like Tesla, Waymo, and Comma.ai deploy neural network-based systems processing multiple camera feeds, radar, and lidar to generate control commands in real-time. These systems must handle edge cases, adverse weather, and unpredictable human behavior.

25.2.2 Neural Network Controllers for Drones

Quadrotor drones present challenging control problems: they are underactuated (4 inputs controlling 6 degrees of freedom), nonlinear, and must operate in turbulent environments. Neural networks have enabled capabilities beyond traditional controllers, including agile flight through reinforcement learning that achieves acrobatic maneuvers difficult to hand-design, recovery from failures where networks learn to fly with damaged rotors, and visual navigation where end-to-end networks fly through forests using only onboard cameras.

The key advantage is that neural networks can learn complex mappings from sensor inputs to motor commands that account for aerodynamic effects, motor saturation, and sensor noise in ways that analytical models cannot.

25.2.3 Learned Dynamics Models for MPC

Model Predictive Control (Chapter ??) requires a dynamics model to predict future states. When analytical models are unavailable or inaccurate, neural networks can learn the dynamics:

$$\mathbf{x}[k+1] = f_{\theta}(\mathbf{x}[k], \mathbf{u}[k]). \quad (25.5)$$

This learned model is then used within the MPC optimization:

$$\min_{\mathbf{u}[k], \dots, \mathbf{u}[k+N-1]} \sum_{j=0}^{N-1} \ell(\hat{\mathbf{x}}[k+j], \mathbf{u}[k+j]) + V_f(\hat{\mathbf{x}}[k+N]) \quad (25.6)$$

where $\hat{\mathbf{x}}$ are states predicted by the neural network model.

Applications include robot manipulation, chemical process control, and building energy management, where first-principles models are either unavailable or too computationally expensive for real-time MPC.

25.2.4 Industrial Soft Sensors

In chemical and manufacturing processes, key quality variables (product composition, viscosity, molecular weight) are often difficult or expensive to measure directly. **Soft sensors** use neural networks to estimate these variables from readily available measurements (temperature, pressure, flow rates):

$$\hat{y}_{\text{quality}} = f_{\theta}(T, P, F, \dots). \quad (25.7)$$

These neural network estimators enable closed-loop control of quality variables that would otherwise require delayed laboratory analysis. Applications include polymer production, petroleum refining, and semiconductor manufacturing.

25.3 Mathematical Foundation: Neural Networks for Dynamics and Control

We now develop the mathematical framework for using neural networks in control systems. This section provides complete derivations suitable for implementation.

25.3.1 The Artificial Neuron

The fundamental building block is the artificial neuron, which computes a weighted sum of inputs, adds a bias, and applies a nonlinear activation function:

$$y = \sigma \left(\sum_{i=1}^n w_i x_i + b \right) = \sigma(\mathbf{w}^T \mathbf{x} + b), \quad (25.8)$$

where $\mathbf{x} = [x_1, \dots, x_n]^T \in \mathbb{R}^n$ is the input vector, $\mathbf{w} = [w_1, \dots, w_n]^T \in \mathbb{R}^n$ is the weight vector, $b \in \mathbb{R}$ is the bias term, $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is the activation function, and $y \in \mathbb{R}$ is the scalar output.

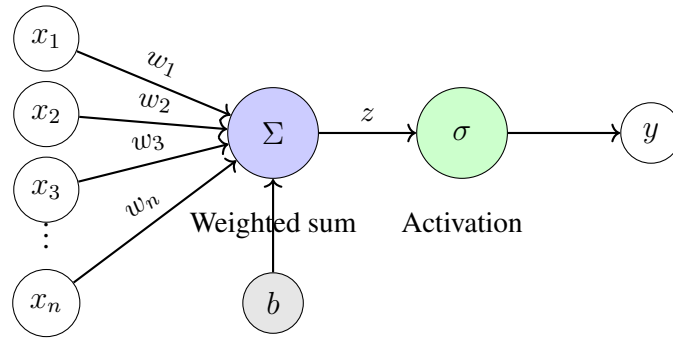


Figure 25.3: Structure of an artificial neuron. Inputs x_i are multiplied by weights w_i , summed with bias b , and passed through activation function σ .

25.3.2 Activation Functions

The choice of activation function σ determines the neuron's nonlinear behavior. Common choices include:

Sigmoid:

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad \sigma'(z) = \sigma(z)(1 - \sigma(z)) \quad (25.9)$$

Output range: $(0, 1)$. Historically popular but suffers from vanishing gradients.

Hyperbolic Tangent (tanh):

$$\sigma(z) = \tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}, \quad \sigma'(z) = 1 - \tanh^2(z) \quad (25.10)$$

Output range: $(-1, 1)$. Zero-centered, but also suffers from vanishing gradients.

Rectified Linear Unit (ReLU):

$$\sigma(z) = \max(0, z), \quad \sigma'(z) = \begin{cases} 1 & z > 0 \\ 0 & z \leq 0 \end{cases} \quad (25.11)$$

Output range: $[0, \infty)$. Computationally efficient, addresses vanishing gradients, but can cause “dead neurons” when $z < 0$.

Leaky ReLU:

$$\sigma(z) = \begin{cases} z & z > 0 \\ \alpha z & z \leq 0 \end{cases} \quad (25.12)$$

where $\alpha \approx 0.01$ prevents dead neurons.

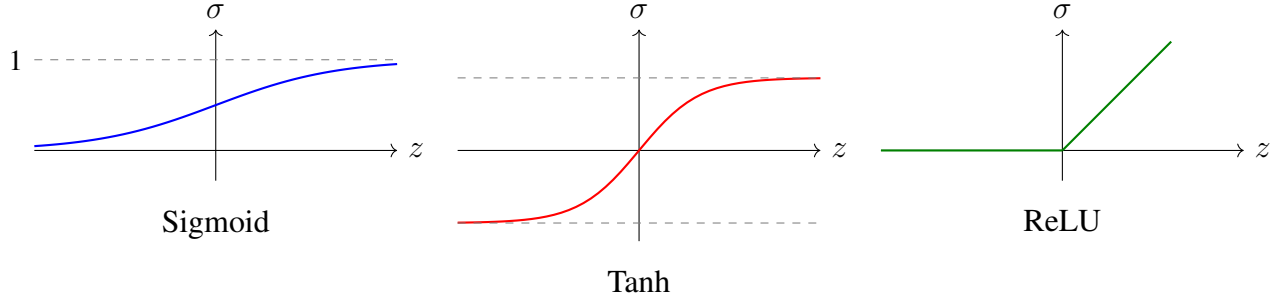


Figure 25.4: Common activation functions: sigmoid, hyperbolic tangent, and ReLU.

25.3.3 Multi-Layer Neural Networks

A multi-layer neural network (also called a feedforward network or multi-layer perceptron) stacks multiple layers of neurons. For a network with L layers:

Layer 1 (input to first hidden):

$$\mathbf{h}^{(1)} = \sigma(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}) \quad (25.13)$$

Layer ℓ (hidden to hidden):

$$\mathbf{h}^{(\ell)} = \sigma(\mathbf{W}^{(\ell)}\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)}), \quad \ell = 2, \dots, L-1 \quad (25.14)$$

Layer L (last hidden to output):

$$\mathbf{y} = \mathbf{W}^{(L)}\mathbf{h}^{(L-1)} + \mathbf{b}^{(L)} \quad (25.15)$$

Note that the output layer typically has no activation function (or a linear activation) for regression tasks like control.

We denote the complete set of parameters as:

$$\theta = \{\mathbf{W}^{(1)}, \mathbf{b}^{(1)}, \dots, \mathbf{W}^{(L)}, \mathbf{b}^{(L)}\} \quad (25.16)$$

The network computes a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ parameterized by θ :

$$\mathbf{y} = f(\mathbf{x}; \theta). \quad (25.17)$$

25.3.4 Universal Approximation Theorem

A fundamental result justifies using neural networks for control:

Universal Approximation Theorem (Cybenko, 1989; Hornik, 1991): A feedforward neural network with a single hidden layer containing a finite number of neurons can approximate any continuous function on a compact subset of $\mathbb{R}^n$ to arbitrary accuracy, provided the activation function is non-polynomial (e.g., sigmoid, tanh, ReLU).

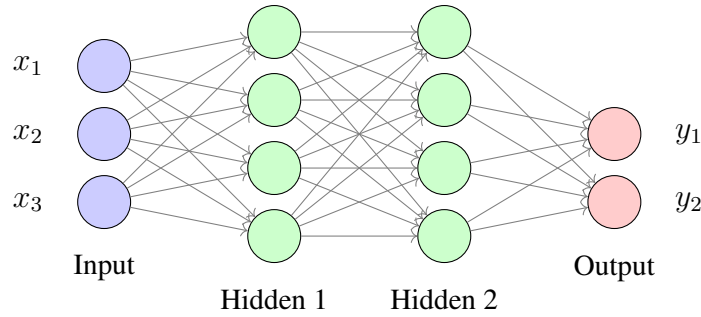


Figure 25.5: A multi-layer neural network with 3 inputs, two hidden layers of 4 neurons each, and 2 outputs.

Mathematically, for any continuous $g : K \rightarrow \mathbb{R}$ on compact $K \subset \mathbb{R}^n$ and any $\epsilon > 0$, there exists a neural network $f(\mathbf{x}; \theta)$ such that:

$$\sup_{\mathbf{x} \in K} |g(\mathbf{x}) - f(\mathbf{x}; \theta)| < \epsilon. \quad (25.18)$$

This theorem guarantees that neural networks have the *capacity* to represent complex dynamics and controllers. However, it does not guarantee that gradient-based training will find the optimal weights, nor does it specify how many neurons are needed.

25.3.5 Neural Network as Controller

In the control context, the neural network takes system state (or error) as input and produces control actions as output:

$$\mathbf{u} = f_{\theta}(\mathbf{x}) \quad \text{or} \quad \mathbf{u} = f_{\theta}(\mathbf{e}), \quad (25.19)$$

where $\mathbf{e} = \mathbf{r} - \mathbf{x}$ is the tracking error.

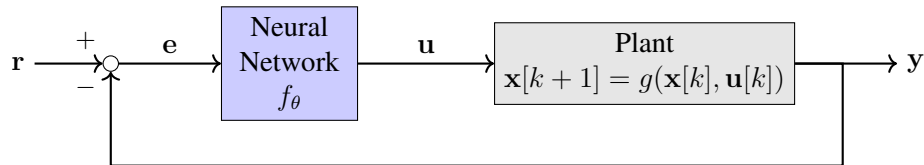


Figure 25.6: Neural network controller in a feedback loop. The network maps error $\mathbf{e}$ to control action $\mathbf{u}$.

The architecture design involves several choices. The **inputs** may include the current state $\mathbf{x}[k]$, the error $\mathbf{e}[k]$, or a history of states $\{\mathbf{x}[k], \mathbf{x}[k-1], \dots\}$ to capture temporal dynamics. The **outputs** represent the control action $\mathbf{u}[k]$, possibly constrained by saturation limits enforced through output activation functions. The **hidden layers** determine the network's capacity and are typically chosen empirically or via hyperparameter search, with deeper networks capturing more complex relationships. The **activation functions** are typically ReLU for hidden layers due to computational efficiency and gradient properties, while the output layer uses linear activation for unconstrained control or tanh for bounded outputs.

25.3.6 Neural Network as Plant Model

Neural networks can learn system dynamics for simulation or model-based control:

$$\hat{\mathbf{x}}[k+1] = f_{\theta}(\mathbf{x}[k], \mathbf{u}[k]). \quad (25.20)$$

This approach is particularly valuable when first-principles models are unavailable (as in complex biological systems), when physics-based models are too computationally expensive for real-time applications, or when the system contains unmodeled nonlinearities or disturbances that analytical models cannot capture.

The learned model serves multiple purposes in control system design. For **simulation**, it predicts system behavior under proposed control policies before deployment. In **Model Predictive Control**, it provides the prediction model required for the optimization horizon. For **controller training**, it serves as a “surrogate” for the real plant during reinforcement learning, enabling safe exploration without risking physical hardware.

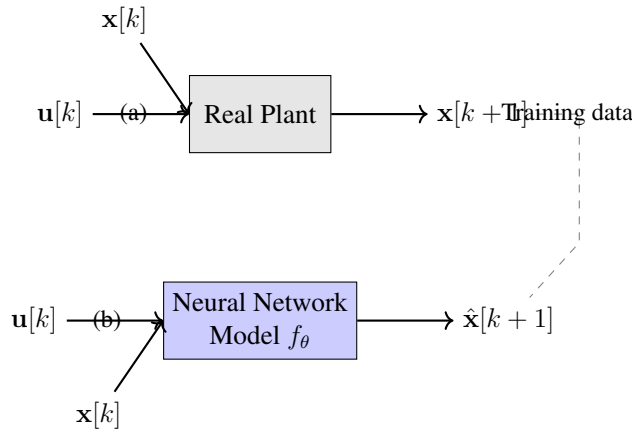


Figure 25.7: (a) Real plant generates training data. (b) Neural network learns to predict next state from current state and control input.

25.3.7 Training Neural Networks: Loss Functions and Optimization

Training a neural network involves minimizing a loss function that measures the discrepancy between predictions and targets.

Supervised Learning (Behavioral Cloning):

Given a dataset $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{u}_i^*)\}_{i=1}^N$ of expert state-action pairs, minimize:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \|\mathbf{u}_i^* - f_{\theta}(\mathbf{x}_i)\|^2. \quad (25.21)$$

System Identification:

Given trajectories $\{(\mathbf{x}_i[k], \mathbf{u}_i[k], \mathbf{x}_i[k+1])\}$, minimize one-step prediction error:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i[k+1] - f_{\theta}(\mathbf{x}_i[k], \mathbf{u}_i[k])\|^2. \quad (25.22)$$

Tracking Performance:

For direct optimization of control performance:

$$\mathcal{L}(\theta) = \sum_{k=0}^T \|\mathbf{r}[k] - \mathbf{x}[k]\|^2 + \lambda \|\mathbf{u}[k]\|^2, \quad (25.23)$$

where $\mathbf{x}[k]$ is the state resulting from applying $\mathbf{u}[k] = f_\theta(\mathbf{x}[k])$ to the system.

25.3.8 Backpropagation

The backpropagation algorithm computes gradients of the loss with respect to all network parameters using the chain rule. Consider a simple two-layer network:

$$\mathbf{h} = \sigma(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}) \quad (25.24)$$

$$\mathbf{y} = \mathbf{W}^{(2)}\mathbf{h} + \mathbf{b}^{(2)} \quad (25.25)$$

For mean squared error loss $\mathcal{L} = \frac{1}{2} \|\mathbf{y}^* - \mathbf{y}\|^2$:

Output layer gradients:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{y}} = \mathbf{y} - \mathbf{y}^* \triangleq \boldsymbol{\delta}^{(2)} \quad (25.26)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(2)}} = \boldsymbol{\delta}^{(2)} \mathbf{h}^T \quad (25.27)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(2)}} = \boldsymbol{\delta}^{(2)} \quad (25.28)$$

Hidden layer gradients (backpropagate error):

$$\boldsymbol{\delta}^{(1)} = (\mathbf{W}^{(2)})^T \boldsymbol{\delta}^{(2)} \odot \sigma'(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}) \quad (25.29)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \boldsymbol{\delta}^{(1)} \mathbf{x}^T \quad (25.30)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} = \boldsymbol{\delta}^{(1)} \quad (25.31)$$

where $\odot$ denotes element-wise multiplication.

Gradient Descent Update:

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}, \quad (25.32)$$

where η is the learning rate.

In practice, variants like Adam (Adaptive Moment Estimation) are preferred:

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \nabla_\theta \mathcal{L} \quad (25.33)$$

$$\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) (\nabla_\theta \mathcal{L})^2 \quad (25.34)$$

$$\theta \leftarrow \theta - \eta \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon} \quad (25.35)$$

where $\hat{\mathbf{m}}_t$ and $\hat{\mathbf{v}}_t$ are bias-corrected estimates.

25.3.9 Backpropagation Through Time for Dynamics

When training a neural network controller for trajectory optimization, we must backpropagate gradients through the dynamics. Consider a horizon of T steps:

$$\mathbf{x}[1] = g(\mathbf{x}[0], f_\theta(\mathbf{x}[0])) \quad (25.36)$$

$$\mathbf{x}[2] = g(\mathbf{x}[1], f_\theta(\mathbf{x}[1])) \quad (25.37)$$

$$\vdots \quad (25.38)$$

$$\mathbf{x}[T] = g(\mathbf{x}[T-1], f_\theta(\mathbf{x}[T-1])) \quad (25.39)$$

The loss depends on the entire trajectory:

$$\mathcal{L}(\theta) = \sum_{k=0}^T c(\mathbf{x}[k], \mathbf{u}[k]). \quad (25.40)$$

Computing $\partial \mathcal{L} / \partial \theta$ requires unrolling the dynamics and applying the chain rule:

$$\frac{\partial \mathcal{L}}{\partial \theta} = \sum_{k=0}^T \frac{\partial c}{\partial \mathbf{u}[k]} \frac{\partial f_\theta}{\partial \theta} + \frac{\partial c}{\partial \mathbf{x}[k]} \frac{\partial \mathbf{x}[k]}{\partial \theta}, \quad (25.41)$$

where:

$$\frac{\partial \mathbf{x}[k]}{\partial \theta} = \frac{\partial g}{\partial \mathbf{x}} \frac{\partial \mathbf{x}[k-1]}{\partial \theta} + \frac{\partial g}{\partial \mathbf{u}} \frac{\partial f_\theta(\mathbf{x}[k-1])}{\partial \theta}. \quad (25.42)$$

This recursive computation is called **Backpropagation Through Time (BPTT)**. Modern automatic differentiation frameworks (PyTorch, TensorFlow) handle this automatically.

25.3.10 Challenges in Neural Network Control

No Stability Guarantees

Classical control design (pole placement, LQR) provides explicit stability guarantees through Lyapunov theory. Neural network controllers, by contrast, offer no inherent stability certificates.

Several approaches address this limitation. Lyapunov-based training incorporates Lyapunov function constraints directly into the training objective, penalizing weight configurations that would violate stability conditions. Formal verification methods can prove stability over a specified region of the state space, though computational costs limit the size of verifiable regions. Designing with sufficient gain and phase margins provides robustness to modeling errors, even without explicit stability proofs.

Sample Efficiency

Neural networks typically require large amounts of training data. Collecting data from physical systems can be time-consuming due to slow dynamics, expensive due to wear and energy costs, and potentially dangerous when exploration may cause damage to equipment or safety hazards. Strategies to mitigate data requirements include simulation-based training where initial models are developed in virtual environments, transfer learning that leverages knowledge from related tasks, and data augmentation techniques that artificially expand training datasets through transformations.

Sim-to-Real Gap

Models trained in simulation may perform poorly on real systems due to unmodeled dynamics such as friction and backlash, sensor noise and measurement delays not present in idealized simulations, and actuator characteristics including saturation and dead zones.

Domain randomization addresses this by training on varied simulation parameters, encouraging robustness.

25.3.11 Hybrid Approaches: Neural Networks + Physics

Combining neural networks with physics-based models offers the best of both worlds:

Physics-Informed Neural Networks (PINNs):

Incorporate known physics as regularization:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{physics}}, \quad (25.43)$$

where $\mathcal{L}_{\text{physics}}$ penalizes violations of known physical laws.

Residual Learning:

Model the difference between a physics-based model and reality:

$$\mathbf{x}[k+1] = g_{\text{physics}}(\mathbf{x}[k], \mathbf{u}[k]) + f_{\theta}(\mathbf{x}[k], \mathbf{u}[k]), \quad (25.44)$$

where g_{physics} captures known dynamics and f_{θ} learns unmodeled effects.

Neural Network + PID:

Augment a PID controller with neural network feedforward:

$$u = K_p e + K_i \int e \, dt + K_d \dot{e} + f_{\theta}(\mathbf{x}). \quad (25.45)$$

This provides baseline performance from PID while allowing the network to improve upon it.

25.4 Practical Experiment: Neural Network Controller for DC Motor

This section presents a complete hands-on experiment: training a neural network to control motor position by imitating a PID controller (behavioral cloning), then deploying and evaluating the learned controller.

25.4.1 Experimental Setup

Hardware Components

System Configuration

The motor position θ is measured by the encoder. The control objective is to track a reference position $\theta_r(t)$. The Arduino implements both a PID controller for the data collection phase and the neural network controller for final deployment, allowing direct comparison on identical hardware.

Table 25.1: Bill of Materials for Neural Network Motor Control Experiment

Item	Description	Qty	Approx. Cost
DC Motor	12V geared motor with encoder	1	\$15
Motor driver	L298N H-bridge	1	\$5
Arduino Mega	Microcontroller	1	\$35
Rotary encoder	Built into motor (or external)	1	—
Power supply	12V, 2A	1	\$10
Computer	Running Python with PyTorch	1	—

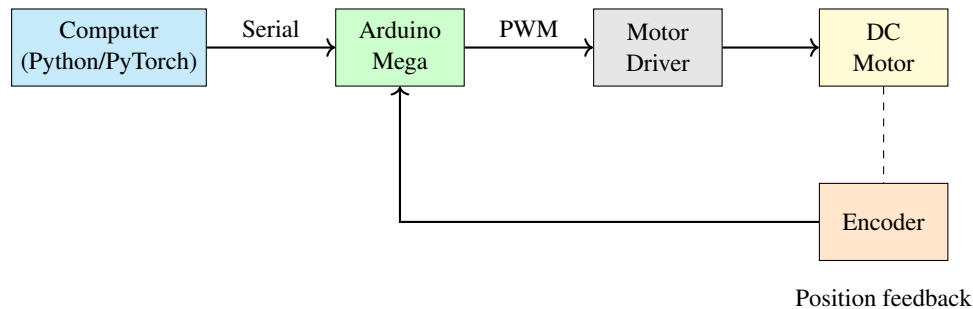


Figure 25.8: Hardware setup for neural network motor control experiment.

25.4.2 Phase 1: PID Controller and Data Collection

PID Implementation

First, implement a well-tuned PID controller on the Arduino.

Reference Signal Design

To collect diverse training data, use varied reference signals including step inputs of different magnitudes to capture transient responses, sinusoidal references at different frequencies to probe the frequency response, random walk sequences to explore the state space broadly, and ramp inputs to test tracking of continuously varying setpoints.

Data Collection Protocol

Run the PID controller for approximately 10 minutes, collecting data at 50 Hz (20 ms intervals). This yields approximately 30,000 data points. Save the data as CSV:

```

time_ms, setpoint, position, control, error
0, 45.0, 0.0, 90.0, 45.0
20, 45.0, 2.3, 87.5, 42.7
40, 45.0, 5.1, 83.2, 39.9
...

```

Algorithm 58 PID Controller for Data Collection

```

1: Initialize:  $K_p \leftarrow 2.0$ ,  $K_i \leftarrow 0.5$ ,  $K_d \leftarrow 0.1$ 
2: Initialize:  $\text{integral} \leftarrow 0$ ,  $\text{prev\_error} \leftarrow 0$ ,  $\text{prev\_time} \leftarrow 0$ 
3: function COMPUTEPID(setpoint, position)
4:    $\text{now} \leftarrow \text{CurrentTime}()$ 
5:    $\Delta t \leftarrow (\text{now} - \text{prev\_time})/1000$ 
6:    $\text{prev\_time} \leftarrow \text{now}$ 
7:    $\text{error} \leftarrow \text{setpoint} - \text{position}$ 
8:    $\text{integral} \leftarrow \text{integral} + \text{error} \cdot \Delta t$ 
9:    $\text{integral} \leftarrow \text{Clamp}(\text{integral}, -100, 100)$  ▷ Anti-windup
10:   $\text{derivative} \leftarrow (\text{error} - \text{prev\_error})/\Delta t$ 
11:   $\text{prev\_error} \leftarrow \text{error}$ 
12:   $\text{output} \leftarrow K_p \cdot \text{error} + K_i \cdot \text{integral} + K_d \cdot \text{derivative}$ 
13:  return  $\text{Clamp}(\text{output}, -255, 255)$ 
14: end function
15: loop
16:    $\text{position} \leftarrow \text{ReadEncoder}()$ 
17:    $\text{setpoint} \leftarrow \text{GenerateSetpoint}()$ 
18:    $\text{control} \leftarrow \text{ComputePID}(\text{setpoint}, \text{position})$ 
19:    $\text{SetMotorPWM}(\text{control})$ 
20:    $\text{LogData}(\text{time}, \text{setpoint}, \text{position}, \text{control}, \text{error})$ 
21: end loop

```

Algorithm 59 Reference Signal Generator

```

1: Persistent:  $\text{last\_change} \leftarrow 0$ ,  $\text{setpoint} \leftarrow 0$ ,  $\text{mode} \leftarrow 0$ 
2: function GENERATESETPOINT
3:    $\text{now} \leftarrow \text{CurrentTime}()$ 
4:    $t \leftarrow \text{now}/1000$ 
5:   if  $\text{mode} = 0$  then ▷ Step inputs
6:     if  $\text{now} - \text{last\_change} > 3000$  then
7:        $\text{setpoint} \leftarrow \text{Random}(-180, 180)$ 
8:        $\text{last\_change} \leftarrow \text{now}$ 
9:     end if
10:  else if  $\text{mode} = 1$  then ▷ Sinusoidal
11:     $\text{setpoint} \leftarrow 90 \cdot \sin(2\pi \cdot 0.2 \cdot t)$ 
12:  else if  $\text{mode} = 2$  then ▷ Ramp
13:     $\text{setpoint} \leftarrow 30 \cdot t$ 
14:    if  $\text{setpoint} > 180$  then
15:       $\text{setpoint} \leftarrow -180$ 
16:    end if
17:  end if
18:  return  $\text{setpoint}$ 
19: end function

```

25.4.3 Phase 2: Neural Network Training

Data Preprocessing

Algorithm 60 Data Loading and Preprocessing

```

1: Input: CSV file containing (time, setpoint, position, control) tuples
2: Output: Training and validation data loaders
3:
4: Load data from file into arrays:  $\mathbf{t}$ ,  $\mathbf{r}$  (setpoint),  $\boldsymbol{\theta}$  (position),  $\mathbf{u}$  (control)
5:
6:                                     ▷ Compute features
7:  $\mathbf{e} \leftarrow \mathbf{r} - \boldsymbol{\theta}$                                      ▷ Error
8:  $\dot{\boldsymbol{\theta}} \leftarrow \text{FiniteDifference}(\boldsymbol{\theta}, \mathbf{t})$                                      ▷ Velocity
9:  $\Delta \mathbf{t} \leftarrow \text{Diff}(\mathbf{t})$ 
10:  $\int \mathbf{e} \leftarrow \text{CumulativeSum}(\mathbf{e} \odot \Delta \mathbf{t})$                                      ▷ Error integral
11:
12: function NORMALIZE( $\mathbf{x}$ )
13:   return  $(\mathbf{x} - \text{mean}(\mathbf{x})) / (\text{std}(\mathbf{x}) + \epsilon)$ 
14: end function
15:
16:  $\mathbf{X} \leftarrow [\text{Normalize}(\mathbf{e}), \text{Normalize}(\dot{\boldsymbol{\theta}}), \text{Normalize}(\int \mathbf{e})]$                                      ▷ Feature matrix
17:  $\mu_u \leftarrow \text{mean}(\mathbf{u}), \sigma_u \leftarrow \text{std}(\mathbf{u})$ 
18:  $\mathbf{y} \leftarrow (\mathbf{u} - \mu_u) / \sigma_u$                                      ▷ Normalized targets
19:
20:  $N_{\text{train}} \leftarrow \lfloor 0.8 \cdot N \rfloor$ 
21:  $\mathcal{D}_{\text{train}} \leftarrow \{(\mathbf{X}_i, y_i)\}_{i=1}^{N_{\text{train}}}$ 
22:  $\mathcal{D}_{\text{val}} \leftarrow \{(\mathbf{X}_i, y_i)\}_{i=N_{\text{train}}+1}^N$ 
23:
24: return  $\text{DataLoader}(\mathcal{D}_{\text{train}}, \text{batch\_size}=64), \text{DataLoader}(\mathcal{D}_{\text{val}}, \text{batch\_size}=64)$ 

```

Network Architecture

The motor controller network uses a feedforward architecture with two hidden layers. The architecture is defined as: $\text{Input}(3) \rightarrow \text{Dense}(32, \text{ReLU}) \rightarrow \text{Dense}(32, \text{ReLU}) \rightarrow \text{Dense}(1, \text{Linear})$. This yields a total of $(3 \times 32 + 32) + (32 \times 32 + 32) + (32 \times 1 + 1) = 1185$ trainable parameters.

Training Loop

25.4.4 Phase 3: Export and Deployment

Convert to C Code

For deployment on Arduino, convert the trained weights to arrays suitable for embedded systems.

Algorithm 61 Motor Controller Network Architecture

```

1: Input dimension:  $n_{\text{in}} = 3$  (error, velocity, error integral)
2: Hidden dimension:  $n_h = 32$ 
3: Output dimension:  $n_{\text{out}} = 1$  (control signal)
4:
5: function FORWARD( $\mathbf{x}$ )
6:    $\mathbf{h}_1 \leftarrow \text{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)$  ▷ First hidden layer
7:    $\mathbf{h}_2 \leftarrow \text{ReLU}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2)$  ▷ Second hidden layer
8:    $y \leftarrow \mathbf{W}_3\mathbf{h}_2 + b_3$  ▷ Output layer (linear)
9:   return  $y$ 
10: end function

```

Algorithm 62 Neural Network Training Loop

```

1: Input: Training loader  $\mathcal{D}_{\text{train}}$ , validation loader  $\mathcal{D}_{\text{val}}$ , epochs  $E$ 
2: Hyperparameters: Learning rate  $\eta = 0.001$ 
3: Initialize optimizer (Adam) with learning rate  $\eta$ 
4: Initialize loss function  $\mathcal{L}$  as Mean Squared Error
5:
6: for epoch = 1 to  $E$  do ▷ Training phase
7:
8:    $\mathcal{L}_{\text{train}} \leftarrow 0$ 
9:   for each batch  $(\mathbf{X}_b, \mathbf{y}_b)$  in  $\mathcal{D}_{\text{train}}$  do
10:      $\hat{\mathbf{y}}_b \leftarrow \text{Forward}(\mathbf{X}_b)$ 
11:      $\ell \leftarrow \mathcal{L}(\hat{\mathbf{y}}_b, \mathbf{y}_b)$ 
12:      $\nabla_{\theta}\ell \leftarrow \text{Backpropagation}(\ell)$ 
13:      $\theta \leftarrow \theta - \eta \cdot \text{AdamUpdate}(\nabla_{\theta}\ell)$ 
14:      $\mathcal{L}_{\text{train}} \leftarrow \mathcal{L}_{\text{train}} + \ell$ 
15:   end for
16:    $\mathcal{L}_{\text{train}} \leftarrow \mathcal{L}_{\text{train}} / |\mathcal{D}_{\text{train}}|$ 
17:
18:   ▷ Validation phase (no gradient computation)
19:    $\mathcal{L}_{\text{val}} \leftarrow 0$ 
20:   for each batch  $(\mathbf{X}_b, \mathbf{y}_b)$  in  $\mathcal{D}_{\text{val}}$  do
21:      $\hat{\mathbf{y}}_b \leftarrow \text{Forward}(\mathbf{X}_b)$ 
22:      $\mathcal{L}_{\text{val}} \leftarrow \mathcal{L}_{\text{val}} + \mathcal{L}(\hat{\mathbf{y}}_b, \mathbf{y}_b)$ 
23:   end for
24:    $\mathcal{L}_{\text{val}} \leftarrow \mathcal{L}_{\text{val}} / |\mathcal{D}_{\text{val}}|$ 
25: end for
26: Save model parameters  $\theta$  to file

```

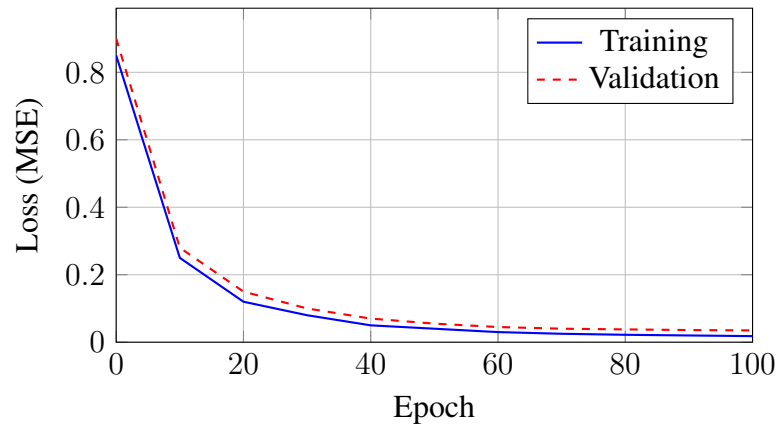


Figure 25.9: Training loss curve for neural network motor controller. The gap between training and validation loss indicates slight overfitting.

Algorithm 63 Export Weights for Embedded Deployment

```

1: Input: Trained model with parameters  $\theta$ , normalization constants  $\mu_u, \sigma_u$ 
2: Output: Header file with weight arrays
3:
4: function EXPORTWEIGHTS(model, filename)
5:   Open file for writing
6:   for each parameter (name,  $\mathbf{W}$ ) in model do
7:      $\mathbf{w}_{\text{flat}} \leftarrow \text{Flatten}(\mathbf{W})$ 
8:     Write array declaration: “const float name[] = { $w_1, w_2, \dots$ }”
9:   end for
10:  Write: “const float control_mean =  $\mu_u$ ”
11:  Write: “const float control_std =  $\sigma_u$ ”
12:  Close file
13: end function

```

Arduino Neural Network Implementation

25.4.5 Phase 4: Evaluation and Comparison

Run both controllers on identical reference trajectories and compare performance.

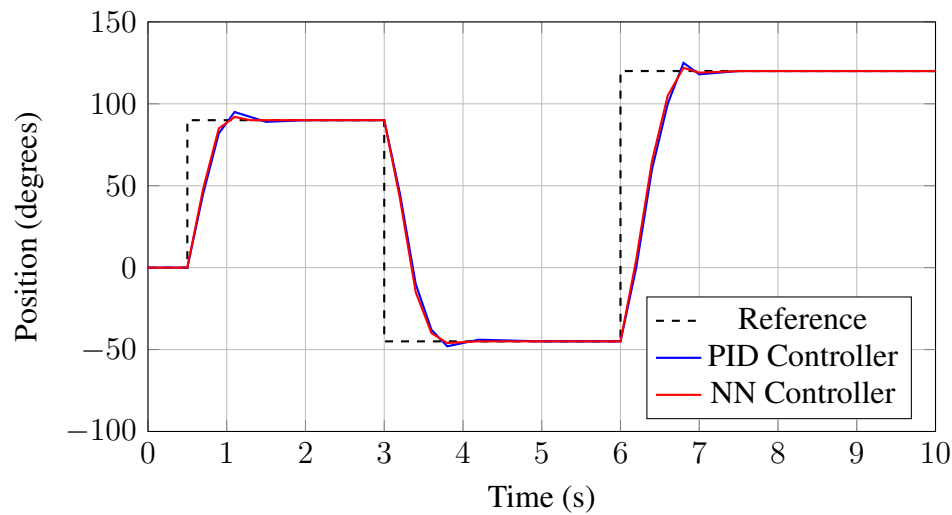


Figure 25.10: Comparison of PID and neural network controller responses to step reference changes. The NN controller (trained via behavioral cloning) closely matches the PID response.

Quantitative Metrics

Compute performance metrics over the test trajectory:

Table 25.2: Performance Comparison: PID vs Neural Network Controller

Metric	PID	Neural Network
RMS Tracking Error (deg)	5.2	5.8
Maximum Overshoot (%)	8.3	6.1
Settling Time (s)	0.85	0.78
Control Effort (RMS)	112	108
Computation Time (μ s)	15	180

25.4.6 Discussion

The experiment demonstrates several key points. First, behavioral cloning works: the neural network successfully learned to imitate the PID controller without explicit knowledge of PID theory. Second, performance is comparable, with tracking error similar between controllers and the NN showing slightly less overshoot in step responses. Third, while the NN requires more computation per control cycle (180 μ s vs 15 μ s), it remains feasible at 50 Hz on the Arduino Mega. Finally, the

Algorithm 64 Neural Network Inference on Microcontroller

```

1: Constants:  $n_{\text{in}} = 3$ ,  $n_h = 32$ , weight arrays from exported file
2: Buffers:  $\mathbf{h}_1[n_h]$ ,  $\mathbf{h}_2[n_h]$ 
3:
4: function RELU( $x$ )
5:   if  $x > 0$  then
6:     return  $x$ 
7:   else
8:     return 0
9:   end if
10: end function
11:
12: function NEURALNETWORKCONTROL(error, velocity, error_integral)
13:                                     ▷ Normalize inputs using training statistics
14:    $\mathbf{x}[0] \leftarrow (\text{error} - \mu_e) / \sigma_e$ 
15:    $\mathbf{x}[1] \leftarrow (\text{velocity} - \mu_v) / \sigma_v$ 
16:    $\mathbf{x}[2] \leftarrow (\text{error\_integral} - \mu_{f_e}) / \sigma_{f_e}$ 
17:
18:                                     ▷ Layer 1: input to hidden1
19:   for  $i = 0$  to  $n_h - 1$  do
20:      $h_1[i] \leftarrow b_1[i] + \sum_{j=0}^{n_{\text{in}}-1} W_1[i, j] \cdot x[j]$ 
21:      $h_1[i] \leftarrow \text{ReLU}(h_1[i])$ 
22:   end for
23:
24:                                     ▷ Layer 2: hidden1 to hidden2
25:   for  $i = 0$  to  $n_h - 1$  do
26:      $h_2[i] \leftarrow b_2[i] + \sum_{j=0}^{n_h-1} W_2[i, j] \cdot h_1[j]$ 
27:      $h_2[i] \leftarrow \text{ReLU}(h_2[i])$ 
28:   end for
29:
30:                                     ▷ Layer 3: hidden2 to output
31:    $\text{output} \leftarrow b_3 + \sum_{j=0}^{n_h-1} W_3[j] \cdot h_2[j]$ 
32:
33:                                     ▷ Denormalize output
34:    $\text{output} \leftarrow \text{output} \cdot \sigma_u + \mu_u$ 
35:   return Clamp(output,  $-255$ ,  $255$ )
36: end function

```

NN inherits the limitations of its teacher and cannot exceed PID performance without additional training techniques.

Extensions to this experiment include training on optimal trajectories from MPC rather than suboptimal PID data, online fine-tuning using reinforcement learning to improve beyond the teacher's performance, and incorporating disturbance rejection capabilities through data augmentation or adversarial training.

25.5 Worked Examples

Computing the Output of a Two-Layer Network

Problem: Consider a neural network with input $\mathbf{x} = [1, 2]^T$, a hidden layer with 2 neurons having weights $\mathbf{W}^{(1)} = \begin{bmatrix} 0.5 & -0.3 \\ 0.2 & 0.8 \end{bmatrix}$ and biases $\mathbf{b}^{(1)} = \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix}$, and an output layer with 1 neuron having weights $\mathbf{W}^{(2)} = [0.7 \ 0.4]$ and bias $b^{(2)} = 0.3$. The activation function is ReLU for the hidden layer and linear for the output. Compute the network output.

Solution:

Step 1: Hidden layer pre-activation

$$\mathbf{z}^{(1)} = \mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)} \quad (25.46)$$

$$= \begin{bmatrix} 0.5 & -0.3 \\ 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix} \quad (25.47)$$

$$= \begin{bmatrix} 0.5(1) + (-0.3)(2) \\ 0.2(1) + 0.8(2) \end{bmatrix} + \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix} \quad (25.48)$$

$$= \begin{bmatrix} -0.1 \\ 1.8 \end{bmatrix} + \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1.6 \end{bmatrix} \quad (25.49)$$

Step 2: Hidden layer activation (ReLU)

$$\mathbf{h}^{(1)} = \text{ReLU}(\mathbf{z}^{(1)}) = \begin{bmatrix} \max(0, 0) \\ \max(0, 1.6) \end{bmatrix} = \begin{bmatrix} 0 \\ 1.6 \end{bmatrix} \quad (25.50)$$

Step 3: Output layer

$$y = \mathbf{W}^{(2)}\mathbf{h}^{(1)} + b^{(2)} \quad (25.51)$$

$$= [0.7 \ 0.4] \begin{bmatrix} 0 \\ 1.6 \end{bmatrix} + 0.3 \quad (25.52)$$

$$= 0.7(0) + 0.4(1.6) + 0.3 = 0.64 + 0.3 = \boxed{0.94} \quad (25.53)$$

Backpropagation by Hand

Problem: For the network in Example 1, compute the gradient $\partial\mathcal{L}/\partial\mathbf{W}^{(1)}$ if the target is $y^* = 0.5$ and we use MSE loss.

Solution:

Step 1: Compute loss

$$\mathcal{L} = \frac{1}{2}(y - y^*)^2 = \frac{1}{2}(0.94 - 0.5)^2 = \frac{1}{2}(0.44)^2 = 0.0968 \quad (25.54)$$

Step 2: Output layer gradient

$$\frac{\partial\mathcal{L}}{\partial y} = y - y^* = 0.94 - 0.5 = 0.44 \quad (25.55)$$

Step 3: Backpropagate to hidden layer

$$\frac{\partial\mathcal{L}}{\partial\mathbf{h}^{(1)}} = (\mathbf{W}^{(2)})^T \frac{\partial\mathcal{L}}{\partial y} = \begin{bmatrix} 0.7 \\ 0.4 \end{bmatrix} \cdot 0.44 = \begin{bmatrix} 0.308 \\ 0.176 \end{bmatrix} \quad (25.56)$$

Step 4: Backpropagate through ReLU

The ReLU derivative is 1 where input > 0 and 0 otherwise:

$$\frac{\partial\mathbf{h}^{(1)}}{\partial\mathbf{z}^{(1)}} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \quad (25.57)$$

(diagonal matrix where $z_1 = 0 \leq 0$ and $z_2 = 1.6 > 0$)

$$\frac{\partial\mathcal{L}}{\partial\mathbf{z}^{(1)}} = \frac{\partial\mathbf{h}^{(1)}}{\partial\mathbf{z}^{(1)}} \cdot \frac{\partial\mathcal{L}}{\partial\mathbf{h}^{(1)}} = \begin{bmatrix} 0 \\ 0.176 \end{bmatrix} \quad (25.58)$$

Step 5: Gradient with respect to $\mathbf{W}^{(1)}$

$$\frac{\partial\mathcal{L}}{\partial\mathbf{W}^{(1)}} = \frac{\partial\mathcal{L}}{\partial\mathbf{z}^{(1)}} \mathbf{x}^T \quad (25.59)$$

$$= \begin{bmatrix} 0 \\ 0.176 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix} \quad (25.60)$$

$$= \begin{bmatrix} 0 & 0 \\ 0.176 & 0.352 \end{bmatrix} \quad (25.61)$$

Note that the first row has zero gradient because the first neuron's ReLU was inactive ("dead").

25.5.1 Example 3: Designing NN Architecture for Control

Problem: Design a neural network controller for a quadrotor attitude control problem. The state is $\mathbf{x} = [\phi, \theta, \psi, p, q, r]^T$ (roll, pitch, yaw angles and angular rates). The output is $\mathbf{u} = [\tau_\phi, \tau_\theta, \tau_\psi]^T$

(torques). Specify architecture and justify choices.

Solution:

Input layer: 6 neurons (one per state variable)

Alternatively, include error terms if tracking a reference: $\mathbf{x}_{\text{input}} = [\phi - \phi_r, \theta - \theta_r, \psi - \psi_r, p, q, r]^T$ (still 6 inputs)

Hidden layers: Two hidden layers with 64 neurons each.

This architecture is justified as follows. Two layers provide sufficient depth for nonlinear function approximation while remaining tractable for training. The choice of 64 neurons per layer is common for moderate-complexity control problems, representing roughly $10\times$ the input dimension. The total number of parameters is $(6 \times 64 + 64) + (64 \times 64 + 64) + (64 \times 3 + 3) = 4,867$, which is small enough to train efficiently yet large enough to capture the required nonlinearities.

For activation functions, the hidden layers use ReLU due to its computational efficiency and ability to avoid vanishing gradients during training. The output layer uses tanh scaled by the maximum torque $\tau_{\max}$ to naturally bound the control outputs within physically realizable limits.

The output layer uses tanh to naturally bound outputs:

$$\mathbf{u} = \tau_{\max} \cdot \tanh(\mathbf{W}^{(L)}\mathbf{h}^{(L-1)} + \mathbf{b}^{(L)}) \quad (25.62)$$

Architecture summary:

$$\boxed{\text{Input}(6) \rightarrow \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dense}(3, \tanh)} \quad (25.63)$$

25.5.2 Example 4: Training NN to Approximate a Known Function

Problem: Train a neural network to approximate the nonlinear function $f(x) = \sin(x) + 0.5 \sin(3x)$ on $[-\pi, \pi]$.

Solution:

After 1000 epochs, the network achieves $\text{MSE} \approx 5.6 \times 10^{-5}$, demonstrating successful approximation of the nonlinear target function.

25.5.3 Example 5: Analyzing NN Controller vs Linear Controller

Problem: Compare a neural network controller with a linear state feedback controller $u = -\mathbf{K}\mathbf{x}$ for a nonlinear pendulum system near the upright equilibrium. Discuss when each is preferred.

Solution:

The inverted pendulum dynamics are:

$$\ddot{\theta} = \frac{g}{\ell} \sin \theta - \frac{b}{m\ell^2} \dot{\theta} + \frac{1}{m\ell^2} u \quad (25.64)$$

Linear controller (LQR): The approach linearizes about $\theta = 0$ using the approximation $\sin \theta \approx \theta$, designs an LQR gain $\mathbf{K} = [K_\theta, K_{\dot{\theta}}]$, and implements the control law $u = -K_\theta \theta - K_{\dot{\theta}} \dot{\theta}$.

The linear controller offers several advantages: stability is guaranteed provided the linearization remains valid, the gains are interpretable and can be tuned systematically, the controller is

Algorithm 65 Training Neural Network to Approximate Nonlinear Function

```

1: Target function:  $f(x) = \sin(x) + 0.5 \sin(3x)$  on  $[-\pi, \pi]$ 
2:
3:                                     ▷ Generate training data
4:  $\mathbf{x}_{\text{train}} \leftarrow \text{LinSpace}(-\pi, \pi, 200)$ 
5:  $\mathbf{y}_{\text{train}} \leftarrow \sin(\mathbf{x}_{\text{train}}) + 0.5 \sin(3\mathbf{x}_{\text{train}})$ 
6:
7:                                     ▷ Define network:  $1 \rightarrow 32 \rightarrow 32 \rightarrow 1$  with tanh activations
8: Initialize network parameters  $\theta$ 
9: Set learning rate  $\eta \leftarrow 0.01$ 
10:
11: for epoch = 1 to 1000 do
12:    $\hat{\mathbf{y}} \leftarrow \text{Forward}(\mathbf{x}_{\text{train}}; \theta)$ 
13:    $\mathcal{L} \leftarrow \text{MSE}(\hat{\mathbf{y}}, \mathbf{y}_{\text{train}})$ 
14:    $\nabla_{\theta} \mathcal{L} \leftarrow \text{Backpropagation}(\mathcal{L})$ 
15:    $\theta \leftarrow \theta - \eta \cdot \text{AdamUpdate}(\nabla_{\theta} \mathcal{L})$ 
16: end for
17:
18:                                     ▷ Expected convergence: Loss  $\approx 5.6 \times 10^{-5}$  after 1000 epochs

```

optimal for the linearized system, and no training data is required. However, limitations include performance degradation for large θ where the linearization breaks down, and the inability to handle constraints directly in the control law.

Neural network controller: The approach trains a mapping $u = f_{\theta}(\theta, \dot{\theta})$ using data from near-optimal trajectories, learning the control law directly from examples.

The neural network controller offers complementary advantages: it can learn a nonlinear control law valid for large angles far from the equilibrium, can implicitly learn constraint satisfaction through training data that respects constraints, and can incorporate more complex objectives beyond quadratic costs. The limitations include no stability guarantee without additional analysis, the requirement for training data collection, and black-box behavior that makes debugging and certification difficult.

Comparison experiment:

Table 25.3: Performance Comparison for Pendulum Swing-Up

Initial Angle	LQR Success	NN Success
$\pm 10^\circ$	Yes	Yes
$\pm 30^\circ$	Yes	Yes
$\pm 60^\circ$	Marginal	Yes
$\pm 90^\circ$	No	Yes
$\pm 150^\circ$ (swing-up)	No	Yes

Conclusion: Use linear control when operating near equilibrium with stability guarantees needed. Use NN control for large operating regions or when linear approximations are inadequate.

25.5.4 Example 6: Neural Network for System Identification

Problem: A black-box system has unknown dynamics. Given input-output data, design and train a neural network model.

Given data: 1000 samples of $(\mathbf{x}[k], u[k], \mathbf{x}[k+1])$ where $\mathbf{x} = [x_1, x_2]^T$.

Solution:

Step 1: Architecture design

Model: $\hat{\mathbf{x}}[k+1] = f_\theta(\mathbf{x}[k], u[k])$

Input dimension: 3 (two states + one control)

Output dimension: 2 (two states)

Architecture: $3 \rightarrow 64 \rightarrow 64 \rightarrow 2$ with ReLU activations.

Step 2: Training

Algorithm 66 System Identification Training

- 1: **Input:** Data $\{(\mathbf{x}[k], u[k], \mathbf{x}[k+1])\}_{k=1}^N$
 - 2: **Network:** $3 \rightarrow 64 \rightarrow 64 \rightarrow 2$ with ReLU activations
 - 3:
 - 4: Construct input matrix $\mathbf{X}_{\text{current}}$ with rows $[\mathbf{x}[k]^T, u[k]]$
 - 5: Construct target matrix $\mathbf{X}_{\text{next}}$ with rows $\mathbf{x}[k+1]^T$
 - 6: Initialize network parameters θ
 - 7: Set learning rate $\eta \leftarrow 0.001$
 - 8:
 - 9: **for** epoch = 1 to 500 **do**
 - 10: $\hat{\mathbf{X}}_{\text{next}} \leftarrow \text{Forward}(\mathbf{X}_{\text{current}}; \theta)$
 - 11: $\mathcal{L} \leftarrow \text{MSE}(\hat{\mathbf{X}}_{\text{next}}, \mathbf{X}_{\text{next}})$
 - 12: $\nabla_\theta \mathcal{L} \leftarrow \text{Backpropagation}(\mathcal{L})$
 - 13: $\theta \leftarrow \theta - \eta \cdot \text{AdamUpdate}(\nabla_\theta \mathcal{L})$
 - 14: **end for**
-

Step 3: Validation

Test multi-step prediction accuracy:

$$\text{Multi-step error} = \frac{1}{T} \sum_{k=0}^T \|\mathbf{x}[k] - \hat{\mathbf{x}}[k]\| \quad (25.65)$$

where $\hat{\mathbf{x}}$ is rolled out using the learned model.

If multi-step error grows rapidly, the model may need more training data to cover a wider range of operating conditions, a deeper or wider architecture to capture more complex dynamics, or a recurrent structure such as LSTM or GRU layers to capture longer-term dependencies in the system behavior.

25.6 Chapter Summary

This chapter introduced neural networks as powerful tools for control system design, capable of learning complex nonlinear mappings from data. Key concepts include the following.

The **historical development** of neural networks for control spans from McCulloch-Pitts neurons (1943) through the Perceptron (1958), backpropagation (1974/1986), and the landmark work of Narendra and Parthasarathy (1990) applying neural networks to control. Understanding this history illuminates both the capabilities and limitations of modern approaches.

The **neural network fundamentals** begin with the artificial neuron, which computes $y = \sigma(\mathbf{w}^T \mathbf{x} + b)$. Multi-layer networks achieve universal function approximation through composition of these simple nonlinear functions, providing the theoretical foundation for their use in control.

When used as a **controller**, neural networks directly map states to control actions via $\mathbf{u} = f_{\theta}(\mathbf{x})$. Training proceeds through behavioral cloning (supervised learning from expert data) or reinforcement learning for cases where optimal demonstrations are unavailable.

When used as a **plant model**, neural networks learn dynamics $\hat{\mathbf{x}}[k+1] = f_{\theta}(\mathbf{x}[k], \mathbf{u}[k])$ for use in simulation or model predictive control, enabling control design when first-principles models are unavailable.

The **training** process relies on backpropagation to compute gradients efficiently. Backpropagation through time handles sequential dynamics inherent in control problems. Modern optimizers like Adam improve convergence and reduce sensitivity to learning rate selection.

Key **challenges** remain: neural network controllers lack inherent stability guarantees, require significant training data that may be expensive to collect, and may not transfer well from simulation to reality due to the sim-to-real gap.

Finally, **hybrid approaches** that combine neural networks with physics-based models through residual learning or physics-informed neural networks leverage the strengths of both data-driven and first-principles methods.

25.7 Exercises

Exercise 25.1. A neural network has architecture $2 \rightarrow 3 \rightarrow 1$ with sigmoid activations. If $\mathbf{W}^{(1)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$, $\mathbf{b}^{(1)} = \mathbf{0}$, $\mathbf{W}^{(2)} = [1, 1, -2]$, $b^{(2)} = 0$: (a) Compute the output for input $\mathbf{x} = [0, 0]^T$. (b) Show that this network approximates the XOR function for inputs in $\{0, 1\}^2$. (c) Explain why a single-layer network cannot compute XOR.

Exercise 25.2. Derive the backpropagation equations for a network with one hidden layer using tanh activation: (a) Express $\partial \mathcal{L} / \partial \mathbf{W}^{(1)}$ in terms of $\delta^{(2)}$, $\mathbf{W}^{(2)}$, $\mathbf{h}^{(1)}$, and $\mathbf{x}$. (b) Verify that when $\mathbf{h}^{(1)} \approx \pm 1$ (saturation), gradients vanish.

Exercise 25.3. For a DC motor position control problem, design a neural network architecture: (a) Specify inputs if the goal is to track a reference position θ_r . (b) Propose a network architecture (layers, widths, activations). (c) Estimate the total number of parameters. (d) Describe how you would collect training data.

Exercise 25.4. Compare training a neural network controller using: (a) Behavioral cloning from PID controller data (b) Direct optimization of tracking error (end-to-end training) Discuss the advantages and disadvantages of each approach.

Exercise 25.5. A neural network model $\hat{\mathbf{x}}[k+1] = f_\theta(\mathbf{x}[k], \mathbf{u}[k])$ is trained with one-step prediction loss. When used for multi-step simulation, prediction errors accumulate. (a) Explain why errors accumulate. (b) Propose a modified training procedure to improve multi-step accuracy. (c) How does this relate to the “compounding errors” problem in behavioral cloning?

Exercise 25.6. Implement (in Python or MATLAB) a neural network to learn the dynamics of a simple pendulum from simulated data: (a) Generate 10,000 samples of $(\theta, \dot{\theta}, \tau, \theta_{\text{next}}, \dot{\theta}_{\text{next}})$. (b) Train a network with architecture $3 \rightarrow 32 \rightarrow 32 \rightarrow 2$. (c) Evaluate one-step and 100-step prediction accuracy. (d) Compare with a linear model fit to the same data.

Exercise 25.7. Consider a hybrid controller: $u = u_{\text{PID}} + u_{\text{NN}}$ where $u_{\text{PID}} = K_p e + K_d \dot{e}$ and $u_{\text{NN}} = f_\theta(\mathbf{x})$. (a) What advantages does this structure offer over a pure NN controller? (b) How would you train f_θ while keeping PID gains fixed? (c) Can you provide stability guarantees if the NN output is bounded?

Exercise 25.8. The Universal Approximation Theorem guarantees that a neural network can approximate any continuous function. Why doesn’t this guarantee that gradient descent will find good weights? Discuss: (a) Local minima and saddle points (b) The role of network architecture (depth vs. width) (c) Implicit regularization from optimization algorithms

Exercise 25.9. A neural network controller is trained to stabilize an inverted pendulum using behavioral cloning from an expert (MPC controller). The expert achieves a success rate of 95% from random initial conditions. After training, the neural network achieves only 70% success rate. (a) Explain why this performance gap might occur. (b) Propose two different strategies to close this gap. (c) Which strategy would you recommend if safety during training is a primary concern?

Exercise 25.10. Consider the problem of learning a dynamics model $\hat{f}$ for use in Model Predictive Control. You have two options: (A) a neural network with 3 hidden layers of 64 neurons each, or (B) a neural network with 1 hidden layer of 256 neurons. Both have approximately the same number of parameters. (a) Discuss the trade-offs between these architectures for dynamics modeling. (b) Which would you expect to generalize better to states outside the training distribution? (c) How would computational cost differ during MPC optimization?

25.8 Further Reading

For the foundational treatment of neural networks in control, see Narendra, K.S. and Parthasarathy, K., “Identification and Control of Dynamical Systems Using Neural Networks,” *IEEE Transactions on Neural Networks*, vol. 1, no. 1, pp. 4–27, 1990.

For comprehensive coverage of modern deep learning theory and practice, consult Goodfellow, I., Bengio, Y., and Courville, A., *Deep Learning*, MIT Press, 2016.

An excellent integration of machine learning with control systems is provided by Brunton, S.L. and Kutz, J.N., *Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control*, Cambridge University Press, 2019.

The standard reference for reinforcement learning, which extends behavioral cloning to online learning, is Sutton, R.S. and Barto, A.G., *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.

For a review of system identification including neural network approaches, see Ljung, L., “Perspectives on System Identification,” *Annual Reviews in Control*, vol. 34, no. 1, pp. 1–12, 2010.

The original universal approximation theorem is presented in Cybenko, G., “Approximation by Superpositions of a Sigmoidal Function,” *Mathematics of Control, Signals, and Systems*, vol. 2, pp. 303–314, 1989.

Chapter 26

Robustness and Uncertainty

26.1 Historical Motivation: Why Models Are Never Enough

The history of control engineering is punctuated by spectacular failures—systems that worked perfectly in simulation yet failed catastrophically in deployment. These failures share a common root cause: the designer trusted the model too completely. Robust control theory emerged from the recognition that uncertainty is not an inconvenience to be minimized, but a fundamental reality to be embraced in the design process.

26.1.1 Bode and the Origins of Robustness Thinking (1940s)

Hendrik Wade Bode, working at Bell Telephone Laboratories in the 1930s and 1940s, laid the conceptual foundation for robust control without ever using that term. His challenge was designing feedback amplifiers for transcontinental telephone transmission—systems that would operate continuously for decades across thousands of miles, through temperature extremes, component aging, and manufacturing variations.

The Telephone Amplifier Problem

A transcontinental telephone call in 1940 passed through dozens of repeater amplifiers, each boosting the signal to compensate for cable losses. The system faced fundamental uncertainties from multiple sources. Component drift caused vacuum tube gains to vary by 20–30% over their lifetime. Temperature dependence posed another challenge, as cables buried underground experienced seasonal temperature swings of 50°C or more, changing their attenuation characteristics. Manufacturing tolerances meant that no two amplifiers were identical, while load variations caused the number of simultaneous calls to affect impedance matching.

An open-loop amplifier design was unthinkable—the cumulative gain variations would make the system unusable within months. Bode’s insight was that feedback could be used not merely to improve nominal performance, but to *desensitize* the system to parameter variations.

The Sensitivity Function

Bode formalized the concept of sensitivity in his 1945 book *Network Analysis and Feedback Amplifier Design*. For a closed-loop system with loop transfer function $L(s)$, he defined the sensitivity function:

$$S(s) = \frac{1}{1 + L(s)} \quad (26.1)$$

This function quantifies how much plant variations propagate to the closed-loop response. At frequencies where $|L(j\omega)| \gg 1$, we have $|S(j\omega)| \ll 1$, and the system is insensitive to plant variations. Bode recognized that this desensitization came at a cost—the complementary sensitivity function $T(s) = L(s)/(1 + L(s))$ satisfies $S(s) + T(s) = 1$, implying that sensitivity reduction at one frequency range must be paid for elsewhere.

Gain and Phase Margins

Bode also introduced the concepts of gain margin and phase margin as practical robustness measures. Rather than asking “Is the system stable?” he asked “How much can the system change before it becomes unstable?” This shift in perspective—from binary stability to quantified robustness—was revolutionary.

Definition 26.1 (Gain Margin). The gain margin G_m is the factor by which the loop gain can be multiplied before the system becomes unstable:

$$G_m = \frac{1}{|L(j\omega_\pi)|} \quad (26.2)$$

where ω_π is the frequency at which $\angle L(j\omega_\pi) = -180$.

Definition 26.2 (Phase Margin). The phase margin ϕ_m is the additional phase lag that would make the system marginally stable:

$$\phi_m = 180 + \angle L(j\omega_c) \quad (26.3)$$

where ω_c is the gain crossover frequency where $|L(j\omega_c)| = 1$.

Bode’s design philosophy was to ensure adequate margins—typically $G_m > 6$ dB and $\phi_m > 45$ —so that the system would remain stable despite the inevitable uncertainties in the telephone network.

Bode’s Secret Wartime Work

While developing his feedback amplifier theory at Bell Labs, Hendrik Bode was simultaneously engaged in classified wartime research. During World War II, he applied his feedback principles to fire-control systems for anti-aircraft guns—a problem where robustness to uncertainty was literally a matter of life and death. The targets were moving, the gun platform was pitching on a ship’s deck, and the ammunition varied from round to round. Bode’s insistence on adequate stability margins came directly from this experience: a system that worked perfectly on the test bench but failed in combat was worse than useless. This practical wartime experience profoundly shaped the conservative design philosophy

that characterizes robust control to this day.

26.1.2 The Multivariable Challenge: Doyle and Stein (1981)

For decades, Bode’s classical methods served engineers well. However, as control applications expanded to multivariable systems—aircraft with multiple control surfaces, chemical plants with interacting loops, flexible spacecraft—classical SISO (single-input, single-output) methods proved inadequate.

The modern era of robust control began with a seminal 1981 paper by John Doyle and Gunter Stein: “Multivariable Feedback Design: Concepts for a Classical/Modern Synthesis.” This paper addressed a crisis in control theory: the elegant state-space methods of the 1960s (LQR, LQG) produced controllers with excellent nominal performance but often terrible robustness.

The LQG Robustness Problem

Linear Quadratic Gaussian (LQG) control combines optimal state feedback (LQR) with optimal state estimation (Kalman filter). In theory, LQG should provide excellent performance. In practice, Doyle demonstrated that LQG controllers could have arbitrarily poor robustness margins—even zero gain margin.

The fundamental issue was that LQR and Kalman filtering were optimized separately, each assuming perfect knowledge of the other. LQR assumes perfect state knowledge; the Kalman filter assumes exact plant dynamics. When combined, the guarantees of each component evaporated.

Loop Transfer Recovery

Doyle and Stein proposed Loop Transfer Recovery (LTR) as a systematic method to recover robustness in LQG designs. The key insight was that full-state LQR feedback inherently provides excellent robustness (infinite gain margin, 60° phase margin for SISO systems), but this robustness is destroyed when an observer is added. LTR designs the observer to asymptotically recover the loop transfer function of the full-state feedback design.

This work established a crucial principle: **robustness must be explicitly designed in**, not assumed as a byproduct of optimal performance.

26.1.3 H_∞ Control: Zames and the Optimization of Robustness (1981)

George Zames, working at McGill University, proposed a radical reconceptualization of the control design problem. Rather than optimizing performance for a nominal plant (as in LQG), he proposed optimizing worst-case performance over a set of plants—a paradigm he called H_∞ optimal control.

The H_∞ Norm

The H_∞ norm of a transfer function $G(s)$ is defined as:

$$\|G\|_\infty = \sup_{\omega \in \mathbb{R}} |G(j\omega)| \quad (26.4)$$

For SISO systems, this is simply the peak magnitude of the frequency response. For MIMO systems, $|G(j\omega)|$ is replaced by the maximum singular value $\bar{\sigma}(G(j\omega))$.

The H_∞ norm has a powerful interpretation: it equals the induced norm from L_2 input signals to L_2 output signals:

$$\|G\|_\infty = \sup_{u \neq 0} \frac{\|y\|_{L_2}}{\|u\|_{L_2}} \quad (26.5)$$

This means the H_∞ norm bounds the worst-case energy amplification from input to output.

Robust Stability via Small Gain

Zames's fundamental contribution was connecting the H_∞ norm to robust stability through the **Small Gain Theorem**:

Theorem 26.1 (Small Gain Theorem). *Consider a feedback interconnection of two stable systems $M(s)$ and $\Delta(s)$. The closed-loop system is stable for all stable Δ with $\|\Delta\|_\infty < 1/\gamma$ if and only if $\|M\|_\infty \leq \gamma$.*

This theorem transforms robust stability into a norm minimization problem: find a controller that minimizes $\|M\|_\infty$, where M is a transfer function from uncertainty inputs to uncertainty outputs. The smaller $\|M\|_\infty$, the larger the class of uncertainties the system can tolerate.

26.1.4 Lessons from Failure: When Models Betray Us

The importance of robust design is most vividly illustrated by high-profile failures where nominal models proved fatally inadequate.

Hubble Space Telescope (1990)

The Hubble Space Telescope was launched in April 1990, intended to revolutionize astronomy with unprecedented resolution. Within weeks, astronomers realized the images were blurred—the primary mirror had been ground to the wrong shape.

The root cause was a flawed null corrector used to test the mirror during manufacturing. The corrector had a spacing error of 1.3 mm, causing the mirror to be ground with spherical aberration. The quality control system detected anomalies but dismissed them as calibration artifacts.

From a control perspective, the lesson is stark: the mirror fabrication was an open-loop process with inadequate feedback. The null corrector was trusted as ground truth, but it was wrong. A robust approach would have used multiple independent measurement systems, each capable of detecting the others' errors.

Mars Polar Lander (1999)

The Mars Polar Lander was lost during its descent to the Martian surface in December 1999. Post-failure analysis identified the most probable cause: the flight software interpreted vibrations from leg deployment as ground contact, shutting down the descent engines while the spacecraft was still 40 meters above the surface.

The landing algorithm was designed for nominal conditions—smooth descent, clean sensor signals. But the actual system experienced leg deployment transients that the designers had not anticipated. A robust design would have required confirmation from multiple sensors over a sustained period before declaring touchdown.

Boeing 737 MAX (2018-2019)

Two crashes of Boeing 737 MAX aircraft in 2018 and 2019, killing 346 people, were caused by the Maneuvering Characteristics Augmentation System (MCAS) repeatedly commanding nose-down trim based on a single angle-of-attack sensor. When that sensor failed or gave erroneous readings, MCAS drove the aircraft into unrecoverable dives.

The MCAS design violated fundamental robust design principles in several critical ways. First, it created a single point of failure where one sensor controlled a critical system. Second, no uncertainty modeling was performed, meaning erroneous sensors were not considered in the design. Third, inadequate authority limits allowed MCAS to command unlimited nose-down trim. Finally, pilot override was inadequate because the system repeatedly reactivated after pilot correction.

26.1.5 The Fundamental Principle

These failures illuminate a principle that robust control theory formalizes mathematically:

The Robustness Principle

All models are wrong. A control system designed only for nominal conditions will fail when (not if) reality deviates from assumptions. Robust design explicitly accounts for uncertainty, ensuring acceptable performance across a range of operating conditions.

The statistician George Box's famous aphorism—"All models are wrong, but some are useful"—takes on urgent meaning in control engineering. The question is not whether our model is perfect (it never is), but whether our controller can tolerate the model's imperfections.

26.2 Modern Applications of Robust Control

Robust control principles find application wherever uncertainty is unavoidable and the consequences of failure are severe.

26.2.1 Aerospace Systems

Modern aircraft operate across vast flight envelopes where aerodynamic characteristics change dramatically. Mach number effects present significant challenges, as transonic flight (Mach 0.8–1.2) exhibits complex shock-boundary layer interactions that defy precise modeling. Altitude variations cause air density to vary by orders of magnitude from sea level to cruise altitude, changing control effectiveness substantially. Fuel consumption during long flights can decrease aircraft mass by 40%, while configuration changes from flaps, slats, landing gear, and store releases alter the

aircraft's dynamics. Additionally, icing and damage from ice accumulation or battle damage can dramatically alter aerodynamics in unpredictable ways.

Flight control systems must provide stable, predictable handling qualities across this entire operating space. Robust control techniques—particularly H_∞ and μ -synthesis—are now standard tools in flight control design.

26.2.2 Robotics and Automation

Industrial robots face continuous uncertainty from multiple sources. Payload variations are particularly significant, as a robot arm might handle payloads from empty gripper to maximum capacity, changing effective inertia by factors of 10 or more. Wear and degradation accumulate over millions of cycles through gearbox backlash, bearing wear, and cable stretching. Friction variations depend on temperature, lubrication, and contamination, affecting both static and dynamic friction characteristics. Furthermore, high-speed robots excite structural resonances that depend on configuration, introducing flexible dynamics into the system.

Robust controllers enable robots to maintain precision despite these variations, reducing the need for frequent recalibration.

26.2.3 Power Systems and Renewable Energy

Modern power grids face unprecedented uncertainty from renewable generation. Solar variability poses a significant challenge, as cloud passages can change solar farm output by 50% in minutes. Wind intermittency causes wind power to fluctuate on time scales from seconds (gusts) to seasons. Demand uncertainty arises because electric vehicle charging and smart appliances create unpredictable load patterns. Grid topology changes further complicate matters, as lines are switched for maintenance, faults, and optimization.

Grid frequency and voltage control must maintain stability despite these uncertainties, requiring robust control strategies at all levels from individual generators to system-wide coordination.

26.2.4 Medical Devices

Medical control systems face patient-to-patient variability and the ultimate consequences of failure. Drug delivery systems must account for patient metabolism that varies by factors of 3–10, where a dose appropriate for one patient may be dangerous for another. Artificial organs present unique challenges because blood pump controllers must accommodate varying vascular resistance, patient activity, and device degradation over time. Radiation therapy systems must adapt as tumor size and location change during treatment, requiring beam control to continuously adjust.

Robust design is not optional in medical devices—it is the difference between healing and harm.

26.3 Mathematical Foundations of Robust Control

We now develop the mathematical framework for analyzing and designing robust control systems.

26.3.1 Types of Uncertainty

Model uncertainty takes many forms, each requiring different mathematical treatment.

Parametric Uncertainty

Parametric uncertainty arises when the model structure is known but parameters are uncertain. Consider a DC motor with transfer function:

$$G(s) = \frac{K_m}{Js + B} \quad (26.6)$$

where K_m is the motor constant, J is the inertia, and B is the viscous damping.

Each parameter lies in a known interval:

$$K_m \in [\underline{K}_m, \bar{K}_m] \quad (26.7)$$

$$J \in [\underline{J}, \bar{J}] \quad (26.8)$$

$$B \in [\underline{B}, \bar{B}] \quad (26.9)$$

The set of possible plants forms a “box” in parameter space. A robustly stable controller must stabilize all plants in this box.

Unstructured Uncertainty

Unstructured uncertainty represents unknown dynamics, typically at high frequencies where unmodeled resonances, nonlinearities, and time delays become significant. The key insight is that while we cannot specify these dynamics exactly, we can bound their magnitude as a function of frequency.

Definition 26.3 (Unstructured Uncertainty Bound). The unstructured uncertainty bound $W(s)$ is a transfer function such that the true plant $G_{\text{true}}(s)$ satisfies:

$$|G_{\text{true}}(j\omega) - G_{\text{nom}}(j\omega)| \leq |W(j\omega)| \quad (26.10)$$

for all frequencies ω .

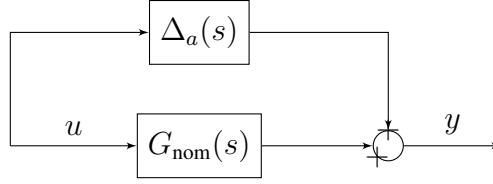
This bound typically increases with frequency—at low frequencies, models are accurate; at high frequencies, unmodeled dynamics dominate.

Additive Uncertainty

In the additive uncertainty model, the true plant is expressed as:

$$G_{\text{true}}(s) = G_{\text{nom}}(s) + \Delta_a(s) \quad (26.11)$$

where $\Delta_a(s)$ is stable and satisfies $\|\Delta_a\|_\infty < \|W_a\|_\infty$.

Figure 26.1: Additive uncertainty model: $G_{\text{true}} = G_{\text{nom}} + \Delta_a$

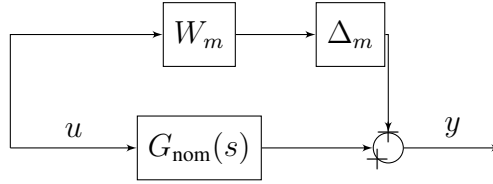
Multiplicative Uncertainty

The multiplicative uncertainty model is often more natural for physical systems:

$$G_{\text{true}}(s) = G_{\text{nom}}(s)(1 + \Delta_m(s)W_m(s)) \quad (26.12)$$

where $\|\Delta_m\|_{\infty} < 1$ and $W_m(s)$ is the uncertainty weight.

The multiplicative form has a clear interpretation: $|W_m(j\omega)|$ represents the fractional uncertainty at frequency ω . If $|W_m(j\omega)| = 0.2$ at some frequency, the plant gain at that frequency is uncertain by $\pm 20\%$.

Figure 26.2: Multiplicative uncertainty model: $G_{\text{true}} = G_{\text{nom}}(1 + W_m\Delta_m)$

Input and Output Multiplicative Uncertainty

For multivariable systems, we distinguish:

$$G_{\text{true}}(s) = (I + \Delta_o W_o)G_{\text{nom}}(s) \quad (\text{output multiplicative}) \quad (26.13)$$

$$G_{\text{true}}(s) = G_{\text{nom}}(s)(I + W_i \Delta_i) \quad (\text{input multiplicative}) \quad (26.14)$$

Output multiplicative uncertainty is appropriate when the uncertainty affects sensor measurements or output channels. Input multiplicative uncertainty models actuator variations.

26.3.2 The Small Gain Theorem

The Small Gain Theorem is the cornerstone of robust stability analysis. Consider a feedback interconnection of two systems $M(s)$ and $\Delta(s)$:

Theorem 26.2 (Small Gain Theorem). *Assume $M(s)$ is stable and $\Delta(s)$ is stable with $\|\Delta\|_{\infty} \leq 1$. Then the feedback interconnection is stable if:*

$$\|M\|_{\infty} < 1 \quad (26.15)$$

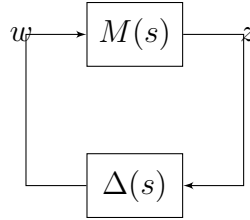


Figure 26.3: Standard feedback interconnection for Small Gain Theorem

Proof. Consider the closed-loop equations:

$$z = Mw \quad (26.16)$$

$$w = \Delta z \quad (26.17)$$

Taking L_2 norms:

$$\|z\|_{L_2} \leq \|M\|_{\infty} \|w\|_{L_2} \quad (26.18)$$

$$\|w\|_{L_2} \leq \|\Delta\|_{\infty} \|z\|_{L_2} \quad (26.19)$$

Substituting:

$$\|z\|_{L_2} \leq \|M\|_{\infty} \|\Delta\|_{\infty} \|z\|_{L_2} \quad (26.20)$$

If $\|M\|_{\infty} \|\Delta\|_{\infty} < 1$, this implies $\|z\|_{L_2}$ is finite for any finite input, hence the system is stable. $\square$

Interpretation for Robust Stability

For a control system with multiplicative uncertainty, the relevant transfer function $M(s)$ is the complementary sensitivity function $T(s)$:

Theorem 26.3 (Robust Stability with Multiplicative Uncertainty). *Consider a feedback system with plant $G_{true} = G_{nom}(1 + W_m\Delta_m)$, $\|\Delta_m\|_{\infty} \leq 1$, and controller $K(s)$. The closed-loop system is robustly stable if and only if:*

$$\|W_m T\|_{\infty} < 1 \quad (26.21)$$

where $T = \frac{G_{nom}K}{1 + G_{nom}K}$ is the nominal complementary sensitivity.

This result has a beautiful frequency-domain interpretation: at each frequency, we require:

$$|W_m(j\omega)| \cdot |T(j\omega)| < 1 \quad (26.22)$$

Where the uncertainty is large ($|W_m|$ large, typically at high frequencies), the complementary sensitivity $|T|$ must be small. Where the uncertainty is small, $|T|$ can be larger.

26.3.3 Sensitivity Functions and Trade-offs

The sensitivity function $S(s)$ and complementary sensitivity function $T(s)$ are the fundamental quantities in robust control:

$$S(s) = \frac{1}{1 + L(s)} \quad (26.23)$$

$$T(s) = \frac{L(s)}{1 + L(s)} \quad (26.24)$$

where $L(s) = G(s)K(s)$ is the loop transfer function.

These functions satisfy the fundamental algebraic constraint:

$$S(s) + T(s) = 1 \quad (26.25)$$

Frequency-Domain Shaping Requirements

Good control requires careful frequency-domain shaping of these functions. At low frequencies, the sensitivity magnitude $|S|$ should be small for effective tracking and disturbance rejection, which means we want $|S(j\omega)| \ll 1$ and thus require $|L(j\omega)| \gg 1$. At high frequencies, the complementary sensitivity magnitude $|T|$ should be small for noise rejection and robustness to unmodeled dynamics, which means we want $|T(j\omega)| \ll 1$ and thus require $|L(j\omega)| \ll 1$.

The identity $S+T=1$ shows that we cannot have both small simultaneously. At any frequency, either $|S|$ is small or $|T|$ is small, but not both.

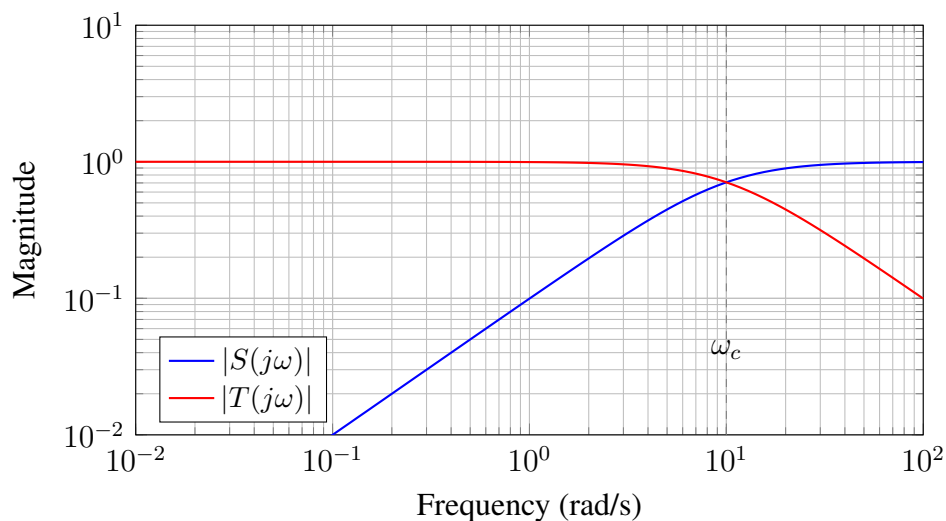


Figure 26.4: Typical sensitivity and complementary sensitivity functions. At low frequencies, $|S|$ is small (good tracking). At high frequencies, $|T|$ is small (robustness). The crossover frequency ω_c marks the transition.

Sensitivity Peaks and Robustness

The peak values of S and T are key robustness indicators:

$$M_S = \|S\|_\infty = \max_{\omega} |S(j\omega)| \quad (26.26)$$

$$M_T = \|T\|_\infty = \max_{\omega} |T(j\omega)| \quad (26.27)$$

The sensitivity peak M_S is related to stability margins:

$$M_S = \frac{1}{\min_{\omega} |1 + L(j\omega)|} = \frac{1}{\text{distance from } -1 \text{ to Nyquist curve}} \quad (26.28)$$

Theorem 26.4 (Margins from Sensitivity Peak). *If M_S is the peak sensitivity, then:*

$$\text{Gain margin: } G_m \geq \frac{M_S}{M_S - 1} \quad (26.29)$$

$$\text{Phase margin: } \phi_m \geq 2 \sin^{-1} \left(\frac{1}{2M_S} \right) \quad (26.30)$$

For example, $M_S = 2$ (6 dB) guarantees $G_m \geq 2$ (6 dB) and $\phi_m \geq 29^\circ$.

Recommended design specifications for robust systems depend on the application criticality. For good robustness in typical applications, $M_S \leq 2$ (6 dB) is recommended. For critical applications where reliability is paramount, $M_S \leq 1.5$ (3.5 dB) provides additional safety margin. For robustness to multiplicative uncertainty, $M_T \leq 1.25$ (2 dB) ensures adequate high-frequency roll-off.

26.3.4 The H_∞ Control Problem

The H_∞ control problem formalizes robust design as an optimization problem.

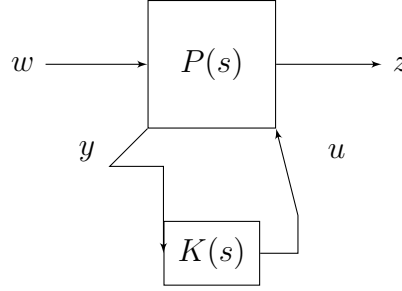
Mixed Sensitivity Formulation

The mixed sensitivity problem seeks a controller $K(s)$ that minimizes:

$$\left\| \begin{bmatrix} W_S S \\ W_T T \end{bmatrix} \right\|_\infty \quad (26.31)$$

where $W_S(s)$ and $W_T(s)$ are weighting functions.

This formulation simultaneously shapes both sensitivity functions. The weight W_S is designed to be large at low frequencies, forcing $|S|$ to be small in that range and ensuring good tracking performance. Conversely, W_T is large at high frequencies, forcing $|T|$ small there and providing robustness to unmodeled dynamics.

Figure 26.5: Standard H_∞ configuration

Standard Problem Formulation

The general H_∞ problem uses the “standard plant” formulation:

The generalized plant $P(s)$ is partitioned as:

$$\begin{bmatrix} z \\ y \end{bmatrix} = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \begin{bmatrix} w \\ u \end{bmatrix} \quad (26.32)$$

The closed-loop transfer function from w to z is:

$$T_{zw} = P_{11} + P_{12}K(I - P_{22}K)^{-1}P_{21} \triangleq \mathcal{F}_l(P, K) \quad (26.33)$$

The H_∞ optimal control problem is:

$$\min_K \|\mathcal{F}_l(P, K)\|_\infty \quad \text{subject to internal stability} \quad (26.34)$$

State-Space Solution

For state-space plants, the H_∞ problem can be solved via two algebraic Riccati equations. Given:

$$P(s) = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{bmatrix} \quad (26.35)$$

Under standard assumptions (including $D_{11} = 0$, $D_{22} = 0$), a stabilizing controller achieving $\|T_{zw}\|_\infty < \gamma$ exists if and only if:

1. The H_∞ control Riccati equation has a stabilizing solution $X_\infty \geq 0$
2. The H_∞ filtering Riccati equation has a stabilizing solution $Y_\infty \geq 0$
3. The spectral radius condition $\rho(X_\infty Y_\infty) < \gamma^2$ holds

The optimal γ is found by bisection, and the controller is given by:

$$K(s) = \begin{bmatrix} A_K & B_K \\ C_K & 0 \end{bmatrix} \quad (26.36)$$

where the controller matrices depend on X_∞ , Y_∞ , and γ .

26.3.5 Robust Performance

Robust stability ensures the system remains stable for all plants in the uncertainty set. Robust performance goes further, requiring that performance specifications are met for all plants.

Performance Weights

Let $W_P(s)$ be a performance weight such that acceptable performance requires:

$$|W_P(j\omega)S(j\omega)| < 1 \quad \forall \omega \quad (26.37)$$

This is equivalent to $\|W_P S\|_\infty < 1$.

Combined Robust Stability and Robust Performance

For multiplicative uncertainty with weight W_m and performance weight W_P , the robust performance condition is:

$$\|W_P S\|_\infty + \|W_m T\|_\infty < 1 \quad (26.38)$$

A more precise condition uses the structured singular value (μ), but (??) provides a useful sufficient condition.

26.3.6 Introduction to μ -Analysis

When uncertainty has known structure (e.g., separate uncertainties in different parameters), the Small Gain Theorem can be conservative. The structured singular value μ provides tighter bounds.

Definition 26.4 (Structured Singular Value). For a complex matrix M and a structured uncertainty set Δ :

$$\mu_\Delta(M) = \frac{1}{\min\{\bar{\sigma}(\Delta) : \Delta \in \Delta, \det(I - M\Delta) = 0\}} \quad (26.39)$$

Robust stability holds for all Δ with $\bar{\sigma}(\Delta) < 1$ if and only if $\mu_\Delta(M) < 1$.

Computing μ exactly is NP-hard for general structures, but efficient algorithms provide tight upper and lower bounds. For purely real uncertainties (parametric), special techniques apply.

26.3.7 Practical Robustness Assessment

While H_∞ and μ provide rigorous frameworks, practical robustness assessment often uses simpler methods.

Gain and Phase Margin Assessment

For SISO systems, classical margins remain valuable. A gain margin $G_m \geq 6$ dB (factor of 2) provides adequate tolerance to gain variations. A phase margin $\phi_m \geq 45$ ensures sufficient tolerance to phase lag from unmodeled dynamics or time delays. The delay margin $\tau_m = \phi_m/\omega_c$ should be at least one sampling period for digital systems to maintain stability under discretization effects.

Sensitivity Peak Limits

Specify bounds on M_S and M_T to ensure adequate robustness. Typically, $M_S \leq 2$ (6 dB) provides good disturbance rejection with acceptable stability margins, while $M_T \leq 1.25$ (2 dB) ensures robustness to multiplicative uncertainty at high frequencies.

Monte Carlo Analysis

For complex systems, simulate performance over random samples from the uncertainty set:

1. Define probability distributions for uncertain parameters
2. Generate N random plant instances
3. Simulate closed-loop response for each
4. Compute statistics of performance metrics
5. Identify worst-case scenarios

This approach captures nonlinear effects and correlations that analytical methods may miss.

Worst-Case Analysis

Grid the uncertainty space and evaluate performance at each vertex (for parametric uncertainty) or at extremes (for unstructured uncertainty). This guarantees that the worst case within the grid is identified.

26.4 Practical Laboratory: Robust Motor Control

This laboratory demonstrates robust control principles using a DC motor with uncertain payload. Students will design both aggressive and conservative controllers, then test their performance as the payload varies.

26.4.1 Learning Objectives

Upon completion of this laboratory, students will be able to:

1. Characterize plant uncertainty by measuring transfer function variations
2. Design controllers for specified gain and phase margins
3. Demonstrate the failure of aggressive controllers under perturbation
4. Verify that robust controllers maintain stability across the uncertainty range
5. Trade off performance versus robustness quantitatively

Table 26.1: Bill of Materials for Robust Control Laboratory

Component	Specification	Qty
Arduino Uno/Nano	ATmega328P	1
DC Motor with Encoder	12V, 400+ PPR	1
Motor Driver	L298N or TB6612	1
Power Supply	12V, 3A	1
Payload Masses	10g, 25g, 50g, 100g	4
Mounting Hardware	3D-printed wheel/arm	1
Breadboard & Wires	Standard	1 set

26.4.2 Required Components

The key addition is the set of masses that attach to the motor shaft, simulating varying payload inertia.

26.4.3 System Identification

Before designing controllers, students characterize the plant with different payloads.

Step Response Method

Apply a step voltage to the motor and measure the velocity response. For a first-order approximation:

$$G(s) = \frac{K}{Js + B} = \frac{K/B}{(J/B)s + 1} = \frac{K_{DC}}{\tau s + 1} \quad (26.40)$$

From the step response, measure the steady-state velocity $\omega_{ss} = K_{DC} \cdot V_{in}$ to determine the DC gain, and measure the time constant τ as the time to reach 63% of steady-state.

Data Collection Procedure

1. Mount the motor with no payload attached
2. Run the identification code, record $K_{DC}^{(0)}$ and $\tau^{(0)}$
3. Attach 25g payload, repeat measurement
4. Attach 50g payload, repeat measurement
5. Attach 100g payload, repeat measurement
6. Plot time constant τ versus added mass

Students should observe that τ increases approximately linearly with added inertia, while K_{DC} remains relatively constant (it depends on motor parameters, not load inertia).

Algorithm 67 System Identification via Step Response

```

1: Initialize: Configure motor PWM output and encoder inputs
2: Initialize: Set up encoder interrupt for position counting
3: Initialize: Create arrays for velocity samples and timestamps
4: Wait 2 seconds for system to stabilize
5: Record start time  $t_{\text{start}}$ 
6: Apply step input to motor (constant PWM duty cycle)
7: while sample count < 500 AND elapsed time < 5 seconds do
8:   if 10 ms elapsed since last sample then
9:     Read current encoder count
10:    Compute velocity:  $\omega = \Delta\theta/\Delta t$ 
11:    Store velocity and timestamp in arrays
12:    Increment sample index
13:   end if
14: end while
15: Set motor PWM to zero (stop motor)
16: Output all time-velocity data pairs for analysis
17: Post-processing: Fit first-order model  $G(s) = K_{\text{DC}}/(\tau s + 1)$ 
18: Extract  $K_{\text{DC}}$  from steady-state velocity and  $\tau$  from 63% rise time
19: return Model parameters  $K_{\text{DC}}, \tau$ 

```

26.4.4 Controller Design**Aggressive Controller (High Bandwidth)**

Design a PI controller for the nominal (no payload) plant with aggressive bandwidth:

$$K_{\text{agg}}(s) = K_p \left(1 + \frac{1}{T_i s} \right) \quad (26.41)$$

The design criteria for this aggressive controller are a bandwidth $\omega_c = 2/\tau^{(0)}$ (twice the nominal pole frequency) and a phase margin $\phi_m \approx 30$ (intentionally low for demonstration purposes to show how marginal stability leads to problems under perturbation).

Robust Controller (Conservative)

Design a PI controller with conservative margins, using a bandwidth $\omega_c = 0.5/\tau^{(100g)}$ (based on the slowest plant with maximum payload) and ensuring a phase margin $\phi_m \geq 60$ to provide adequate robustness across all payload conditions.

26.4.5 Experimental Procedure**Experiment 1: Aggressive Controller Performance**

1. Load the aggressive controller parameters

Algorithm 68 PI Controller with Anti-Windup

```

1: Parameters:  $K_p$  (proportional gain),  $K_i$  (integral gain),  $I_{\max}$  (integral limit)
2: Initialize:  $I \leftarrow 0$  (integral accumulator)
3:
4: procedure COMPUTECONTROL( $e, \Delta t$ )
5:    $I \leftarrow I + e \cdot \Delta t$  ▷ Integrate error
6:    $I \leftarrow \text{clamp}(I, -I_{\max}, I_{\max})$  ▷ Anti-windup saturation
7:    $u \leftarrow K_p \cdot e + K_i \cdot I$  ▷ PI control law
8:   return  $u$ 
9: end procedure
10:
11: Controller Configurations:
12:   Aggressive:  $K_p = 5.0, K_i = 20.0, I_{\max} = 100$ 
13:   Robust:  $K_p = 1.5, K_i = 3.0, I_{\max} = 100$ 
14:
15: Main Control Loop:
16: while system running do
17:   Compute time step  $\Delta t$  since last iteration
18:   Read setpoint  $r$  from input device
19:   Read position  $y$  from encoder
20:   Compute error:  $e \leftarrow r - y$ 
21:   Compute control:  $u \leftarrow \text{ComputeControl}(e, \Delta t)$ 
22:   Apply saturated output:  $u_{\text{sat}} \leftarrow \text{clamp}(u, -u_{\max}, u_{\max})$ 
23:   Send  $u_{\text{sat}}$  to motor driver
24:   Log data at 20 Hz for analysis
25: end while

```

2. With no payload, command a 90° step. Record settling time and overshoot.
3. Add 25g payload, repeat step command. Observe increased oscillation.
4. Add 50g payload. Observe sustained oscillation or marginal stability.
5. Add 100g payload. Observe instability (controller failure).

Experiment 2: Robust Controller Performance

1. Load the robust controller parameters
2. With no payload, command a 90° step. Record settling time and overshoot.
3. Progressively add 25g, 50g, 100g payloads
4. At each condition, command step response
5. Observe: slower response but maintained stability throughout

Experiment 3: Disturbance Rejection

1. With robust controller and 50g payload, hold a position setpoint
2. Apply external disturbance (finger push) and release
3. Measure recovery time
4. Repeat with aggressive controller at same payload (if stable)
5. Compare disturbance rejection performance

26.4.6 Data Analysis

Table 26.2: Experimental Results Template

Metric	0g	25g	50g	100g
Aggressive Controller				
Settling time (s)				
Overshoot (%)				
Stable? (Y/N)	Y			
Robust Controller				
Settling time (s)				
Overshoot (%)				
Stable? (Y/N)	Y	Y	Y	Y

Discussion Questions

1. Why does the aggressive controller become unstable as payload increases?
2. Calculate the effective phase margin reduction caused by the increased time constant.
3. How much performance (settling time) is sacrificed by the robust design?
4. Could you design a controller that is both fast AND robust? What limits this?
5. In a real application, how would you choose the uncertainty range to design for?

26.5 Worked Examples**Calculating Stability Margins from Bode Plot**

A control system has loop transfer function:

$$L(s) = \frac{100}{s(s+2)(s+10)}$$

Calculate the gain margin, phase margin, and assess robustness.

Solution:

First, find the gain crossover frequency ω_c where $|L(j\omega_c)| = 1$:

$$|L(j\omega)| = \frac{100}{\omega\sqrt{\omega^2 + 4}\sqrt{\omega^2 + 100}}$$

Setting $|L(j\omega_c)| = 1$ and solving numerically: $\omega_c \approx 4.1$ rad/s.

The phase at crossover:

$$\angle L(j\omega_c) = -90 - \tan^{-1}(\omega_c/2) - \tan^{-1}(\omega_c/10)$$

$$\angle L(j4.1) = -90 - \tan^{-1}(2.05) - \tan^{-1}(0.41) = -90 - 64 - 22 = -176$$

Phase margin:

$$\phi_m = 180 + (-176) = 4$$

This is dangerously low! The system is nearly unstable.

For gain margin, find ω_π where $\angle L(j\omega_\pi) = -180$:

$$-90 - \tan^{-1}(\omega_\pi/2) - \tan^{-1}(\omega_\pi/10) = -180$$

Solving: $\omega_\pi \approx 4.47$ rad/s.

Gain margin:

$$G_m = \frac{1}{|L(j4.47)|} = \frac{4.47 \cdot \sqrt{4.47^2 + 4} \cdot \sqrt{4.47^2 + 100}}{100} = \frac{4.47 \cdot 4.9 \cdot 10.95}{100} = 2.4$$

In decibels: $G_m = 20 \log_{10}(2.4) = 7.6$ dB.

Assessment: The gain margin of 7.6 dB is adequate for most applications. However, the phase margin of 4° is *unacceptable* and indicates the system is dangerously close to instability. The system will likely oscillate with any parameter variation. The recommendation is to add phase lead compensation to increase the phase margin to at least 45° .

$$G_m = 7.6 \text{ dB}, \quad \phi_m = 4^\circ$$

Robust Stability via Small Gain Theorem

A plant has multiplicative uncertainty:

$$G_{\text{true}}(s) = G_{\text{nom}}(s)(1 + W_m(s)\Delta(s))$$

where $G_{\text{nom}}(s) = \frac{5}{s+1}$, $W_m(s) = \frac{0.5s}{s+10}$, and $\|\Delta\|_\infty \leq 1$.

A controller $K(s) = 2$ is proposed. Is the closed-loop robustly stable?

Solution:

The complementary sensitivity function (nominal) is:

$$T(s) = \frac{G_{\text{nom}}K}{1 + G_{\text{nom}}K} = \frac{\frac{5 \cdot 2}{s+1}}{1 + \frac{10}{s+1}} = \frac{10}{s+1+10} = \frac{10}{s+11}$$

For robust stability, we need $\|W_m T\|_\infty < 1$.

$$W_m(s)T(s) = \frac{0.5s}{s+10} \cdot \frac{10}{s+11} = \frac{5s}{(s+10)(s+11)}$$

Find the peak:

$$|W_m(j\omega)T(j\omega)| = \frac{5\omega}{\sqrt{\omega^2 + 100} \cdot \sqrt{\omega^2 + 121}}$$

Taking the derivative and setting to zero, or evaluating at several frequencies, we find: at $\omega = 0$, $|W_m T| = 0$; at $\omega = 10$, $|W_m T| = \frac{50}{\sqrt{200} \cdot \sqrt{221}} = \frac{50}{14.1 \cdot 14.9} = 0.24$; at $\omega = 100$, $|W_m T| = \frac{500}{\sqrt{10100} \cdot \sqrt{10221}} = \frac{500}{100.5 \cdot 101.1} = 0.049$; and as $\omega \rightarrow \infty$, $|W_m T| \rightarrow 0$.

The maximum occurs near $\omega \approx 10$ rad/s. More precisely, taking the derivative:

$$\frac{d}{d\omega} \left[\frac{5\omega}{(\omega^2 + 100)^{1/2}(\omega^2 + 121)^{1/2}} \right] = 0$$

Solving yields $\omega^* \approx 10.5$ rad/s, and $\|W_m T\|_\infty \approx 0.24$.

Since $\|W_m T\|_\infty = 0.24 < 1$, the system is robustly stable.

$$\|W_m T\|_\infty \approx 0.24 < 1 \implies \text{Robustly Stable}$$

Example 26.1 (Controller Design for Specified Margins). Design a lead compensator for the plant

$G(s) = \frac{10}{s(s+5)}$ to achieve a phase margin $\geq 50^\circ$ with gain crossover frequency $\omega_c = 5$ rad/s.

Solution:

Step 1: Analyze uncompensated system at desired ω_c

At $\omega = 5$ rad/s:

$$|G(j5)| = \frac{10}{5 \cdot \sqrt{25 + 25}} = \frac{10}{5 \cdot 7.07} = 0.283$$

$$\angle G(j5) = -90 - \tan^{-1}(5/5) = -90 - 45 = -135$$

Current phase margin at $\omega = 5$ rad/s would be: $180 - 135 = 45$.

We need 50°, so we need to add 5° plus safety margin. Target: add 15° phase lead.

Step 2: Design lead compensator

Lead compensator form:

$$K_{\text{lead}}(s) = K_c \frac{1 + s/\omega_z}{1 + s/\omega_p}, \quad \omega_p > \omega_z$$

Maximum phase lead occurs at $\omega_m = \sqrt{\omega_z \omega_p}$ and equals:

$$\phi_{\text{max}} = \sin^{-1} \left(\frac{\omega_p - \omega_z}{\omega_p + \omega_z} \right)$$

For 15° phase lead: $\sin(15) = 0.259$

Let $\alpha = \omega_p / \omega_z$. Then:

$$\frac{\alpha - 1}{\alpha + 1} = 0.259 \implies \alpha = 1.7$$

Place $\omega_m = 5$ rad/s (desired crossover):

$$\omega_z = \frac{\omega_m}{\sqrt{\alpha}} = \frac{5}{\sqrt{1.7}} = 3.84 \text{ rad/s}$$

$$\omega_p = \alpha \omega_z = 1.7 \times 3.84 = 6.53 \text{ rad/s}$$

Step 3: Determine gain

The lead compensator magnitude at $\omega_c = 5$ rad/s:

$$|K_{\text{lead}}(j5)| = K_c \frac{\sqrt{1 + (5/3.84)^2}}{\sqrt{1 + (5/6.53)^2}} = K_c \frac{\sqrt{2.69}}{\sqrt{1.59}} = 1.30 K_c$$

For unity loop gain at crossover:

$$|G(j5)| \cdot |K_{\text{lead}}(j5)| = 1$$

$$0.283 \times 1.30 K_c = 1 \implies K_c = 2.72$$

Final controller:

$$K_{\text{lead}}(s) = 2.72 \frac{1 + s/3.84}{1 + s/6.53} = 2.72 \frac{s + 3.84}{s + 6.53} \cdot \frac{6.53}{3.84} = 4.63 \frac{s + 3.84}{s + 6.53}$$

Verification: Phase at $\omega = 5$ rad/s:

$$\begin{aligned} \angle L(j5) &= \angle G(j5) + \angle K(j5) = -135 + \tan^{-1}(5/3.84) - \tan^{-1}(5/6.53) \\ &= -135 + 52.5 - 37.5 = -120 \end{aligned}$$

Phase margin: $180 - 120 = 60^\circ$ ✓

$$K(s) = 4.63 \frac{s + 3.84}{s + 6.53}$$

Example 26.2 (Parametric Uncertainty as Multiplicative Form). A motor has nominal inertia $J_0 = 0.01 \text{ kg}\cdot\text{m}^2$ but actual inertia lies in $[0.008, 0.015] \text{ kg}\cdot\text{m}^2$ due to payload variations. Express this as multiplicative uncertainty.

Solution:

The plant transfer function (velocity per torque) is:

$$G(s) = \frac{1}{Js + B}$$

For nominal inertia $J_0 = 0.01$:

$$G_{\text{nom}}(s) = \frac{1}{0.01s + B}$$

For actual inertia J :

$$G_{\text{true}}(s) = \frac{1}{Js + B}$$

The relative error is:

$$\frac{G_{\text{true}} - G_{\text{nom}}}{G_{\text{nom}}} = \frac{\frac{1}{Js+B} - \frac{1}{J_0s+B}}{\frac{1}{J_0s+B}} = \frac{J_0s + B}{Js + B} - 1 = \frac{(J_0 - J)s}{Js + B}$$

For high frequencies (s large):

$$\frac{G_{\text{true}} - G_{\text{nom}}}{G_{\text{nom}}} \approx \frac{J_0 - J}{J}$$

Maximum deviation occurs at the extremes of the uncertainty range: for $J = 0.008$, the relative error is $(0.01 - 0.008)/0.008 = +25\%$, while for $J = 0.015$, the relative error is $(0.01 - 0.015)/0.015 = -33\%$.

The multiplicative uncertainty bound is:

$$|W_m(j\omega)| \geq \max \left(\frac{|J_0 - J|}{J} \right) = 0.33 \text{ at high frequency}$$

A suitable weight that captures the frequency dependence:

$$W_m(s) = \frac{0.33s}{s + B/J_{\text{max}}} = \frac{0.33s}{s + B/0.015}$$

This weight is small at low frequencies (where the uncertainty effect is small) and approaches 0.33 at high frequencies.

$$W_m(s) = \frac{0.33s}{s + 0.067B}$$

Example 26.3 (Robust Stability Analysis for Perturbed System). A feedback system has nominal plant $G_{\text{nom}}(s) = \frac{1}{s+1}$, additive uncertainty bounded by $|\Delta_a(j\omega)| \leq \frac{0.5\omega}{\omega+5}$, and controller $K(s) = 2$. Determine if the system is robustly stable.

Solution:

For additive uncertainty $G_{\text{true}} = G_{\text{nom}} + \Delta_a$, robust stability requires:

$$\left\| \frac{\Delta_a}{1 + G_{\text{nom}}K} \right\|_{\infty} < 1$$

Equivalently, using the uncertainty bound W_a where $|\Delta_a| \leq |W_a|$:

$$|W_a(j\omega)| < |1 + G_{\text{nom}}(j\omega)K| \quad \forall \omega$$

The uncertainty weight:

$$|W_a(j\omega)| = \frac{0.5\omega}{\omega + 5}$$

The sensitivity denominator:

$$|1 + G_{\text{nom}}(j\omega)K| = \left| 1 + \frac{2}{j\omega + 1} \right| = \left| \frac{j\omega + 1 + 2}{j\omega + 1} \right| = \frac{|j\omega + 3|}{|j\omega + 1|} = \frac{\sqrt{\omega^2 + 9}}{\sqrt{\omega^2 + 1}}$$

Check the condition at various frequencies:

At $\omega = 1$:

$$|W_a| = \frac{0.5}{6} = 0.083, \quad |1 + GK| = \frac{\sqrt{10}}{\sqrt{2}} = 2.24$$

Condition satisfied: $0.083 < 2.24 \checkmark$

At $\omega = 10$:

$$|W_a| = \frac{5}{15} = 0.33, \quad |1 + GK| = \frac{\sqrt{109}}{\sqrt{101}} = 1.04$$

Condition satisfied: $0.33 < 1.04 \checkmark$

At $\omega = 50$:

$$|W_a| = \frac{25}{55} = 0.45, \quad |1 + GK| = \frac{\sqrt{2509}}{\sqrt{2501}} = 1.0016$$

Condition satisfied: $0.45 < 1.0016 \checkmark$

As $\omega \rightarrow \infty$:

$$|W_a| \rightarrow 0.5, \quad |1 + GK| \rightarrow 1$$

Condition satisfied: $0.5 < 1 \checkmark$

The condition is satisfied at all frequencies.

System is robustly stable

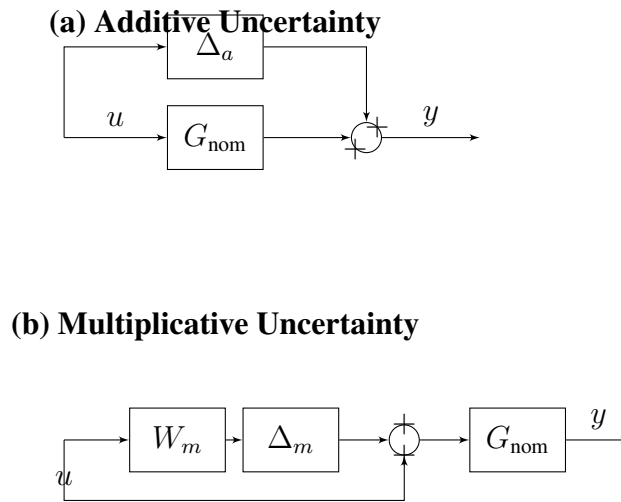


Figure 26.6: Standard uncertainty models: (a) Additive: $G = G_{\text{nom}} + \Delta_a$. (b) Multiplicative: $G = G_{\text{nom}}(1 + W_m\Delta_m)$, where $\|\Delta_m\|_\infty \leq 1$.

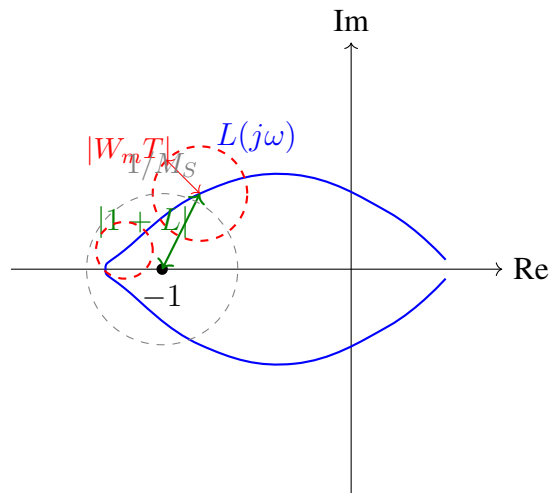


Figure 26.7: Robust stability visualization in Nyquist diagram. The dashed circles represent uncertainty at different frequencies. For robust stability, no uncertainty disk may contain the critical point -1 . The gray circle shows the minimum distance to -1 , equal to $1/M_S$.

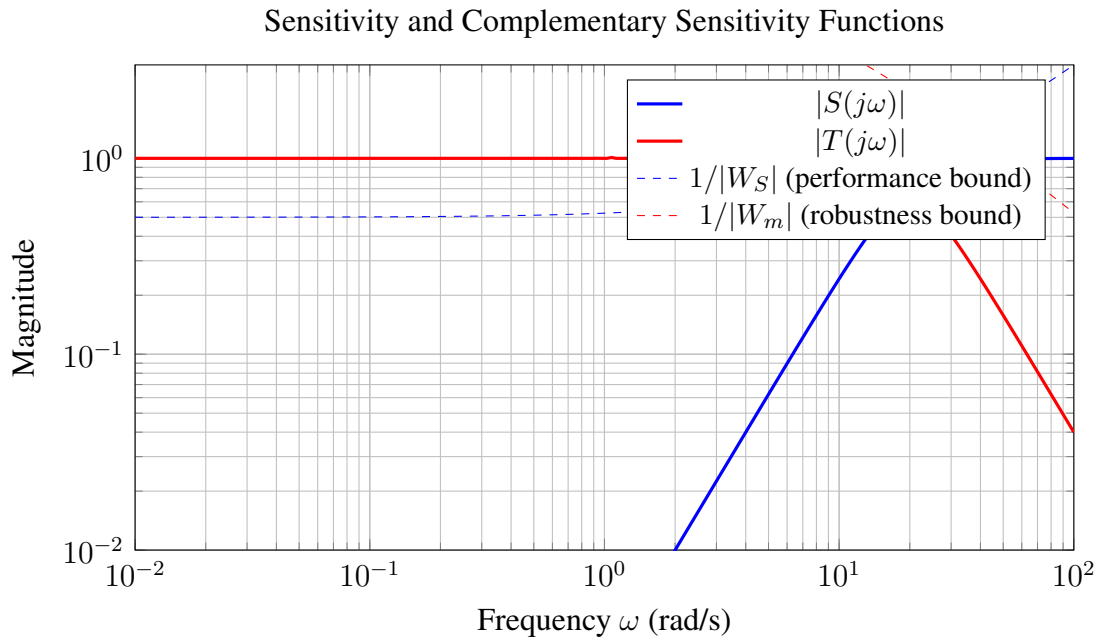


Figure 26.8: Bode plot of sensitivity functions with design bounds. The sensitivity $|S|$ should remain below $1/|W_S|$ for performance; the complementary sensitivity $|T|$ should remain below $1/|W_m|$ for robust stability. The peak sensitivity M_S indicates robustness margins.

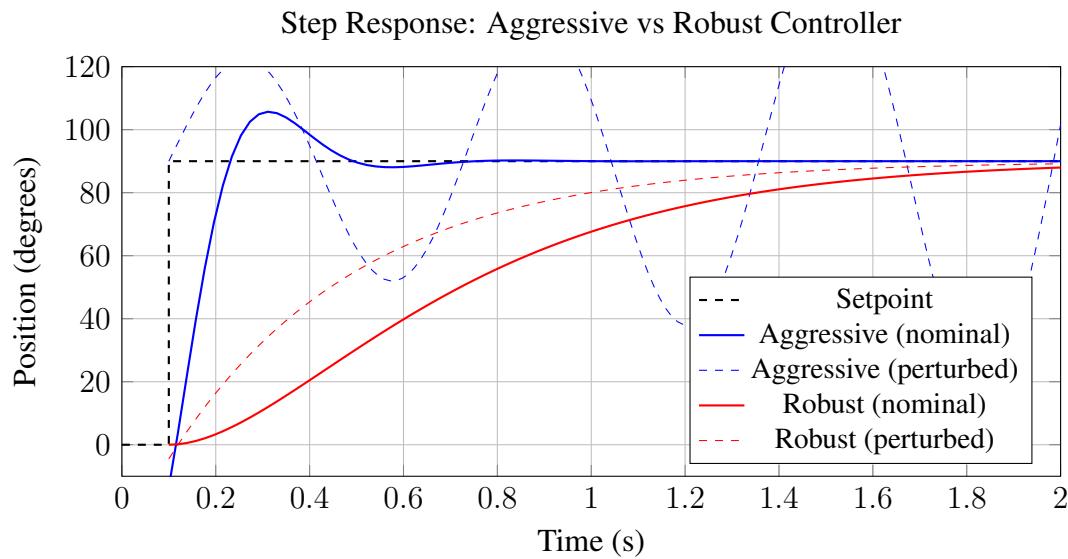


Figure 26.9: Comparison of step responses. The aggressive controller provides fast response at nominal conditions but becomes unstable when the plant is perturbed (e.g., increased payload inertia). The robust controller is slower but maintains stability and acceptable performance across the uncertainty range.

26.6 Figures and Diagrams

26.6.1 Uncertainty Models

26.6.2 Robust Stability in Nyquist Diagram

26.6.3 Sensitivity Function Bode Plots

26.6.4 Nominal vs Robust Response Comparison

26.7 Chapter Summary

This chapter developed the theory and practice of robust control—designing controllers that maintain stability and performance despite model uncertainty.

1. **Historical Foundation:** Robust control concepts emerged from practical necessity. Bode's sensitivity analysis for telephone amplifiers, Doyle and Stein's exposure of LQG robustness problems, and Zames's H_∞ framework all addressed the fundamental reality that models are never perfect.
2. **Types of Uncertainty:** We distinguished parametric uncertainty (known structure, uncertain values), unstructured uncertainty (unknown high-frequency dynamics), and additive versus multiplicative formulations. Each type requires appropriate mathematical treatment.
3. **Small Gain Theorem:** The cornerstone of robust stability analysis provides a frequency-domain condition: $\|W_m T\|_\infty < 1$ for multiplicative uncertainty. This elegantly connects the uncertainty bound W_m to the required roll-off of complementary sensitivity T .
4. **Sensitivity Trade-offs:** The identity $S + T = 1$ forces a fundamental trade-off. Low sensitivity to disturbances (small $|S|$) at one frequency range must be paid for by high complementary sensitivity (large $|T|$) elsewhere. Good design manages this trade-off explicitly.
5. **Practical Margins:** Classical gain and phase margins remain valuable robustness indicators. The sensitivity peak M_S provides a unified robustness measure with guaranteed minimum margins.
6. **H_∞ Control:** The H_∞ framework transforms robust design into optimization: minimize the worst-case performance over all plants in the uncertainty set. This provides a systematic approach to robust controller synthesis.

Key Takeaway

All models are wrong. A controller designed only for nominal conditions will eventually fail. Robust design explicitly accounts for uncertainty, trading peak performance for reliability across a range of operating conditions. In safety-critical systems, this trade-off is not optional—it is the difference between success and catastrophe.

26.8 Exercises

Exercise 26.1. For a system with loop transfer function $L(s) = \frac{50}{s(s+1)(s+10)}$, calculate the gain margin and phase margin. Assess whether the system meets typical robustness specifications.

Exercise 26.2. A control system has sensitivity function $S(s) = \frac{s^2 + 2s + 1}{s^2 + 3s + 10}$. Calculate $M_S = \|S\|_\infty$ and determine the guaranteed minimum gain and phase margins.

Exercise 26.3. A plant has multiplicative uncertainty weight $W_m(s) = \frac{s+1}{s+20}$. If the complementary sensitivity at $\omega = 10$ rad/s has magnitude $|T(j10)| = 0.5$, is the robust stability condition satisfied at this frequency?

Exercise 26.4. A feedback system has $G(s) = \frac{2}{(s+1)^2}$ and $K(s) = 3$. An additive uncertainty Δ_a satisfies $|\Delta_a(j\omega)| \leq 0.1\omega^2$ for all ω . Determine if the system is robustly stable.

Exercise 26.5. Design a lead compensator for $G(s) = \frac{5}{s(s+2)}$ to achieve phase margin ≥ 60 with gain crossover frequency $\omega_c = 4$ rad/s.

Exercise 26.6. A spring-mass system has uncertain stiffness $k \in [80, 120]$ N/m with nominal $k_0 = 100$ N/m and mass $m = 1$ kg. Express the plant uncertainty in multiplicative form and determine the weight $W_m(s)$.

Exercise 26.7. Explain the physical meaning of minimizing $\|W_S S\|_\infty$ where $W_S(s) = \frac{s/M + \omega_B}{s + \omega_B A}$ with $M = 2$, $\omega_B = 10$ rad/s, and $A = 0.01$.

Exercise 26.8. In the robust motor control experiment, a student finds that the aggressive controller becomes unstable when payload mass increases from 0g to 50g. If the nominal time constant is $\tau_0 = 0.1$ s and increases to $\tau = 0.15$ s with the 50g payload, calculate how much the phase margin at the original crossover frequency has decreased.

Exercise 26.9. Describe a procedure to assess robust performance using Monte Carlo simulation for a system with three uncertain parameters, each uniformly distributed over a $\pm 20\%$ range.

Exercise 26.10. A system requires $|S(j\omega)| < 0.1$ for $\omega < 1$ rad/s and $|T(j\omega)| < 0.1$ for $\omega > 100$ rad/s. Sketch a typical sensitivity/complementary sensitivity pair that meets these requirements and identify the crossover frequency range.

